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CHEMISTRY

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ATOMIC STRUCTURE 1

DISCOVERY OF NUCLEUS

In the year 1911, Lord Rutherford carried out an experiment to study penetration power of α-rays (stream of He^{2+} ions, i.e. helium nucleus consisting of two protons and two neutrons) by bombarding thin gold foil of 10^{-4} mm thickness with parallel beams of α-rays. He made following observations,

(i) Most of the α-particles had passed straight through the foil without any deflection.

(ii) Some of them got deflected by an angle less than 90°.

(iii) Only a few (one out of every 20000) were deflected back by an angle greater than 90°, i.e. they could not pass through the foil.

(iv) The scattering of α-particles $\propto \frac{1}{\sin^4(\theta/2)}$

The above observations led Rutherford to discard J.J. Thomson's watermelon model of atom and proposed his own model of atom. The main observation was that there is a positively charged central heavy mass situated at the centre of the atom called **nucleus**.

The following conclusions were drawn from the above observations:

(i) The size of various nuclei can be calculated from the below mentioned equation,

$r = (1.3\times10^{-13})m^{1/3}cm$

where m is the mass number and r is the radius of nucleus.

(ii) The density of nucleus, assumed to be spherical, may be expressed as

$$d = \frac{\text{mass}}{\text{volume}} = \frac{\text{Mass number}}{\text{Avogadro's number}} \times \frac{1}{\frac{4}{3}\times\pi r^3}$$

$$= \frac{1}{\frac{4}{3}\times3.14\times6.023\times10^{23}\times1.3\times1.3\times1.3\times10^{-39}}$$

$$= 1.80\times10^{14}\,gcm^{-1}$$

Thus, nuclear density is a constant independent of mass number of an element.

(iii) The dimensions of nucleus are of the order of 10^{-6} nm, i.e. 10^{-13} cm while dimensions of atom are of the order of 10^{-1} nm, i.e. 10^{-8} cm.

PLANCK'S QUANTUM THEORY

Radiant energy is emitted or absorbed in the integral multiple of some smallest unit or packet of energy, each packet of energy being called one **quantum** of energy. In Einstein's term, each packet of energy is **photon**, carrying energy equal to one quantum of radiation. There is a slight difference between Planck's concept and Einstein's theory. Planck thought that radiant energy is itself continuous but its absorption or emission takes place in the unit of quantum whereas Einstein made a step forward and thought that radiation is itself discrete and it has photon constitution, i.e. light is streams of photons.

The energy associated with one quantum of energy is expressed as

$$E = h\nu = h\frac{c}{\lambda} = hc\bar{\nu}$$

where, h = Planck's constant = 6.625×10^{-34} Js

$= 6.625\times10^{-27}$ erg·s

c = Velocity of light = $3\times10^{8}\,ms^{-1} = 3\times10^{10}\,cms^{-1}$

ν = Frequency of radiation (light) in cps or Hertz

λ = Wavelength of light in m or cm

$\bar{\nu} = \frac{1}{\lambda}$ = wave number in m^{-1} or cm^{-1}

From the above equation, it is clear that energy of photon is inversely proportional to the wavelength of radiation.

BOHR'S MODEL OF ATOM

Niels Bohr had proposed atomic model that could explain Planck's quantum theory, line spectra of atom or ions and stability of an atom. Rutherford's nuclear model of atom with electron revolving round the nucleus had a serious question mark on its stability. An electron being a charged particle and getting accelerated due to its circular motion is bound to radiate energy continuously. In doing so, the electron will move nearer and nearer to the nucleus. In other words, the electron will follow the spiral path and will ultimately fall into the nucleus, which means nothing but complete destruction of atom in no time. This means the destruction of whole universe as the universe is the creation of atoms. But it does not happen so, atom is very stable. Electron does not fall into the nucleus.

The main points of Bohr's model of atom are as follows:

1. Electrons revolve round the nucleus in some selected orbits and not just in any orbit. In other words, the electronic orbits are discrete rather than continuous.
2. Every selected orbit is associated with some definite amount of energy and so long as an electron revolves round the nucleus in any of the selected orbits it does not radiate energy, i.e. its energy remains constant. These selected orbits are called stationary orbits, closed orbits, shells or energy levels.

3. Permissible orbits are those in which the angular momentum of electron is $\frac{h}{2\pi}$ unit or an integral multiple of the same, i.e.

$$mvr = n\left(\frac{h}{2\pi}\right)$$

where n = 1, 2, 3...

The quantum number (n) is the serial number of the shells from the nucleus. Here, n = 1 means 1st, n = 2 means 2nd, n = 3 means 3rd shell and so on and these are denoted by the symbols K, L, M, ..., respectively.

4. The absorption or emission of energy occurs only when an electron jumps from lower energy orbit to higher energy orbit or falls down from higher to lower energy orbit, in the form of electromagnetic radiations, i.e.

$$\bar{\nu} = \frac{E_2 - E_1}{hc}$$

where $\bar{\nu}$ = wave number of radiation absorbed or emitted

E_1 and E_2 are the energies of the orbits within which electronic transition had taken place, $E_2 > E_1$.

The absorption or emission of radiation by any atom, molecule or ion must occur at the rate of one photon per atom, molecule or ion and this is called Bohr's quantum theory. Following expressions were derived by Niels Bohr for an atom or ion containing one electron.

(i) $r_n = \frac{n^2h^2}{4\pi^2 mkZe^2}$, for the 1st orbit of H atom

where r_n is called the Bohr's radius and hence $r_n = \frac{n^2}{Z}a_o$

where $a_o = 0.529\ \text{Å}$

(ii) (a) K.E. of electron $= \frac{kZe^2}{2r_n}$

(b) P.E. of electron $= kZe^2\int_{\infty}^{r_n}\frac{dr}{r^2} = -\frac{kZe^2}{r_n}$

Note that P.E. of electron is –ve and its numerical value is just the twice that of its K.E.

(c) Total energy of electron (E_n) = K.E. + P.E $= \frac{kZe^2}{2r_n} - \frac{kZe^2}{r_n} = -\frac{kZe^2}{2r_n}$

Putting the value of r_n, we get

$$E_n = -\frac{kZe^2}{2n^2h^2} \times 4\pi^2 mkZe^2$$

or $$E_n = -\frac{2\pi^2 mk^2Z^2e^4}{n^2h^2} = -13.6\left(\frac{Z^2}{n^2}\right)eV$$

where n = 1, 2, 3, ..., ∞ denoting 1st, 2nd, 3rd shells respectively.

ATOMIC SPECTRA

Atomic spectra are actually emission spectra and arise due to downward transition of electron from an excited state to a state of lower excitation or the ground state, e.g. M-shell ⟶ L-shell or M-shell ⟶ K-shell directly. The spectrum of an atom consists of a series of discrete lines and each line appear at different wave numbers characteristic of an element, hence also known as line spectra.

The wave number of the spectral line is given by the expression,

$$\bar{\nu} = \frac{1}{\lambda} = R_H \cdot Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

where n_1 and n_2 ($n_2 > n_1$) are the principal quantum numbers of electron in its final and initial state respectively.

$R_H = \frac{2\pi^2 me^4 k^2}{ch^3}$ = Rydberg constant for hydrogen = 109437 cm^{-1}

The experimental value for R_H is 109678 cm^{-1}, which is in close agreement with the value as calculated by Bohr and that is indicative of Bohr's success.

For, Lyman series: $n_1 = 1$ and $n_2 = 2, 3, 4, \ldots, \infty$ lines appear in UV region

Balmer series: $n_1 = 2$ and $n_2 = 3, 4, 5, \ldots, \infty$ lines appear in visible region

Paschen series: $n_1 = 3$ and $n_2 = 4, 5, 6, \ldots, \infty$ lines appear in IR region

Brackett series: $n_1 = 4$ and $n_2 = 5, 6, 7, \ldots, \infty$ lines appear in IR region

Pfund series: $n_1 = 5$ and $n_2 = 6, 7, 8, \ldots, \infty$ lines appear in IR region

Humphery series: $n_1 = 6$ and $n_2 = 7, 8 \ldots \infty$

The spectral line with lowest value of n_2 in any series corresponds to longest wavelength or minimum energy transition of that series. The shortest wavelength transition, i.e. the highest energy transition in any series is one for which $n_2 = \infty$. This is also called series limit.

The maximum number of spectral lines produced or the maximum number of radiations of different wavelengths produced by a sample of H atoms or hydrogen like species containing only one electron such as He^+, Li^{2+} etc., in which the maximum excited atom/ion contains electron in the shell for which the principal quantum number is n, as given below:

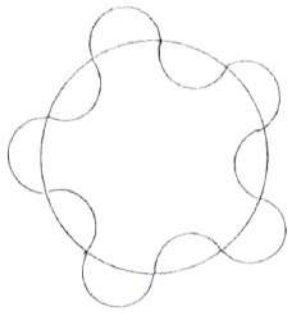
$$\text{No. of spectral lines} = \frac{n(n-1)}{2}$$

WAVE NATURE OF ELECTRON

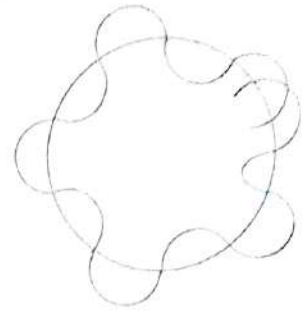
In order to explain quantization of the angular momentum of electron, de Broglie (1924) conceived the idea that electron is a particle and its motion is governed by some kind of wave. As per his theory, an electron revolves round the nucleus vibrating in simple harmonic motion and the orbit to be stable, its circumference must be integral multiple of the wavelength of electron. Otherwise, waves will not continually remain in phase and will vanish due to their destructive interference. Mathematically,

$2\pi r = n\lambda$

where $n = 1, 2, 3, \ldots, \infty$



Wave continually in phase: stable orbit or permitted orbit



Wave out of phase: unstable orbit

In analogy with particle wave dualism of photon and intimate relationship between wavelength of photon and its momentum, is expressed as

$$\lambda = \frac{h}{mc}$$

de Broglie suggested that a similar equation may also apply to all micromaterial particle like electron, photon, neutron, atom, ion, molecule etc. Thus, an electron of mass m moving with velocity v has a wave nature associated with it whose wavelength and momentum are interrelated as,

$$\lambda = \frac{h}{mv} \quad \ldots(1)$$

Equation (1) is known as de Broglie relationship.

From Eq.(1), following two important equations may be derived

$\lambda = \sqrt{\frac{150}{V}}$ Å (for electron only), and

$$\lambda = \frac{h}{\sqrt{2mE}}$$

where V = potential difference of the electric field in volt, through which electrons passes

E = K.E. of electron

Frequency of matter wave, $\nu = \frac{mv^2}{h} = \frac{2K.E.}{h}$

QUANTUM MECHANICAL MODEL OF ATOM

Schrodinger derived an equation for calculating energy of an electron possessing wave-particle dualism. The Schrodinger wave equation in three-dimension is as follows:

$$\frac{\partial^2\psi}{\partial x^2} + \frac{\partial^2\psi}{\partial y^2} + \frac{\partial^2\psi}{\partial z^2} + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$$

where m = mass of electron

h = Planck's constant

E = total energy of electron

V = P.E. of electron

Ψ = wave function of electron.

It can also be written as $\nabla^2\psi + \frac{8\pi^2 m}{h^2}(E - V)\psi = 0$

or $H\Psi = E\Psi$

Where $\nabla = \frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}$ = Laplace operator.

H = Hamiltonion operator.

The wave function (Ψ) is the amplitude function of the particle wave and $\Psi^2 dxdydz$ is the measure of the probability of finding an electron in a volume element dxdydz.

The model of atom based on this equation (Schrodinger wave equation) is known as quantum mechanical model of atom. The quantum mechanical model of atom has replaced orbit concept by orbital concept. An orbital is defined as the region in the three-dimensional space around the nucleus where there is high probability of finding an electron of a given energy.

As worded, the energy of electron in an orbital is fixed while its position is uncertain and thus it is in tune with Heisenberg uncertainty principle as discussed later in the chapter.

QUANTUM NUMBERS

There are four quantum numbers. The first three have been introduced by quantum mechanics while the fourth one has come from spectroscopy. These quantum numbers help us to determine energy of an electron in an atom and they introduce quantization in energy of an electron in an atom.

1. Principal quantum number (n)
2. Azimuthal quantum number (ℓ)
3. Magnetic quantum number (m)
4. Spin quantum number (s)

Principal Quantum Number (n)

(i) Permissible values; n = 1, 2, 3,...,∞, i.e. any integer except zero.

(ii) It furnishes following information:

(a) The principal shell in which electron is present.

(b) The average distance of electron from the nucleus.

(c) The size of the electron cloud, i.e. orbital. As n increases, size of orbital as well as average distance of electron from the nucleus increases.

(d) The energy of electron principally depends upon 'n' and if there are only one electron so that there is no scope of electron–electron repulsion, then energy of electron depends only on n as given by Bohr's theory of atom (Classical mechanical model of atom).

$$E_n = -\frac{Z^2}{n^2} \times 13.6 \text{ eV per atom}$$

Azimuthal Quantum Number (ℓ)

(i) Allowed values; ℓ may have any integral value from 0 to n – 1, i.e. $\ell = 0, 1, 2, \ldots, (n-1)$.

(ii) Azimuthal quantum number furnishes following information:
(a) The sub-shells (s, p, d, f) in which electron is present.
(b) Shapes of orbital

ℓ-value	Sub-shell	Shape of orbitals
$\ell = 0$	s	Spherical
$\ell = 1$	p	Dumb-bell
$\ell = 2$	d	Double dumb-bell
$\ell = 3$	f	Complicated

From $\ell = 4$ onward, the sub-shells are denoted by letters in alphabetical orders, e.g. $\ell = 4$, g sub-shell; $\ell = 5$, h sub-shell and so on.

(c) Orbital angular momentum of electron = $\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

The classical analogue of the same was, $n\left(\frac{h}{2\pi}\right)$.

Magnetic Quantum Number (m)

(i) Permissible values; m = all integral values from $-\ell$ to $+\ell$ including zero.

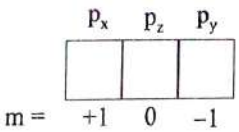
(ii) Magnetic quantum number furnishes following information:
(a) The orbital in which electron is present in a given sub-shell.
(b) Orientation of the orbital.
(c) z–component of orbital angular momentum of electron. The quantized value can be determined by the expression

$$L_z = m\left(\frac{h}{2\pi}\right)$$

where m = magnetic quantum numbers = $\sqrt{\ell(\ell+1)}\cos\theta$

where θ is the angle between orbital angular momentum and z-axis.

(d) For $\ell = 1$, m = +1, 0, –1. This means there are three p-orbitals of equal energy as shown below. In other words, p-orbital is triply degenerate or it is three-fold degenerate.



Spin Quantum Number

(i) Permissible values; $s = +\frac{1}{2}$ and $-\frac{1}{2}$

(ii) Spin quantum number furnishes following informations:

(a) Direction of spin of electron ("+" sign indicates clockwise spin and "–" sign indicates anti-clockwise spin.)

(b) It determines spin angular momentum of electron.

$$L_s = \sqrt{s(s+1)}\frac{h}{2\pi}$$

where L_s = spin angular momentum

s = total spin, i.e. summation of spin quantum numbers of all the electrons.

For electron pair $(\uparrow\downarrow)$, $s = +\frac{1}{2} - \frac{1}{2} = 0$

Total spin (s) $= \frac{1}{2} \times n$, where n = number of unpaired electron.

(c) It determines spin-only value of the magnetic moment of any atom, molecule or ion using the equation

$\mu_s = \sqrt{n(n+2)}$ B.M.

Bohr magneton (B.M.) $= 9.27 \times 10^{-24}$ JT^{-1}

NODES AND NODAL POINTS

The point where there is zero probability of finding an electron is called nodal point or node.

No. of radial or spherical nodes $= n - \ell - 1$

No. of angular nodes $= \ell$

Total no. of nodes $= n - 1$

No. of nodal planes = No. of angular nodes $= \ell$

PHOTOELECTRIC EFFECT

According to Einstein,

$$h\nu = h\nu_0 + \frac{1}{2}mv^2$$

where ν = Frequency of incident radiation ($\nu > \nu_0$)

ν_0 = Threshold frequency, characteristic of metal

$h\nu_0$ = Energy of photon used in knocking out the electron from the atom, i.e. work function (ω_0) of electron in the metal

$h\nu$ = Energy of incident photon

$h\nu > h\nu_0$ and $h(\nu - \nu_0)$ = Energy of photon converted into K.E.

The potential applied on the surface to reduce the velocity of photoelectron to zero is known as stopping potential V_0. Thus, K.E. = eV_0

$h\nu = \omega_0 + eV_0$

Rearranging the Einstein's equation, we get

K.E. = $h\nu - W$

Thus, a plot of K.E. vs frequency of incident radiations will be a straight line and the slope is equal to Planck's constant.

HEISENBERG'S UNCERTAINTY PRINCIPLE

This principle states,

It is impossible to determine both momentum and position of an electron or any micro particle simultaneously with accuracy.

Mathematically,

$$\Delta p \cdot \Delta x \geq \frac{h}{4\pi}$$

or $$m\Delta v \cdot \Delta x \geq \frac{h}{4\pi}$$

or $$\Delta v \cdot \Delta x \geq \frac{h}{4\pi m}$$

where $\Delta p = m\Delta v$ = uncertainty in momentum

Δv = uncertainty in velocity

Δx = uncertainty in position

m = mass of micro particle

AUFBAU PRINCIPLE

(i) Electrons prefer to occupy sub-shell with lower $(n + \ell)$ value to higher $(n + \ell)$ value.

(ii) $(n + \ell)$ value remaining the same, electron prefers to occupy the sub-shell with lower value of n, i.e. the one nearer to the nucleus.

It is to be noted that the energy of sub-shell depends upon its $(n + \ell)$ value. As $(n + \ell)$ value increases, energy increases. If $(n + \ell)$ value remains same, energy of sub-shell can be distinguished by 'n' value alone. However, lower the n value, lower is the energy. Thus, Aufbau principle can be stated as the electrons in a polyelectronic atom are filled one by one in order of increasing energy level.

PAULI'S EXCLUSION PRINCIPLE

According to this principle, no two electrons in the same atom can have all the four quantum numbers equal. In other words, an orbital can contain only a maximum of two electrons with opposite spin.

Based on this principle, following conclusions are drawn

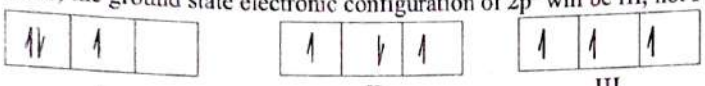
(i) As number of orbitals in any shell is n^2, maximum number of electrons in any shell is $2n^2$.

(ii) Maximum number of electrons in s, p, d and f-sub-shells is 2, 6, 10 and 14 respectively.

HUND'S RULE

This rule states that during the filling of electrons in degenerate orbitals or orbitals of equal energy, i.e. orbitals belonging to the same sub-shell, electrons tend to avoid pairing till each such orbital is filled singly with parallel spin.

Thus, the ground state electronic configuration of $2p^3$ will be III, not I or II.



I II III

ILLUSTRATIVE PROBLEMS

1. Calculate velocity of an electron placed in the third orbit of the hydrogen atom. Also calculate the number of revolutions per second that this electron makes around the nucleus.

Sol. Velocity of electron in nth Bohr's orbit is given by $V_n = V_o \frac{Z}{n}$

where, $V_o = 2.182 \times 10^6$ ms^{-1}

In this case, z = 1 and n = 3

$\therefore V_3 = \frac{2.182 \times 10^6}{3} = 7.273 \times 10^5$ ms^{-1}

$$x = \frac{1 \times 7.273 \times 10^5}{2 \times \frac{22}{7} \times 9 \times 0.529 \times 10^{-10}}$$

$= 2.4314 \times 10^{14}$ r.p.s.

2. What electron transition in a hydrogen atom, starting from the orbit n = 7, will produce infrared light of wavelength 2170 nm? ($R_H = 1.09677 \times 10^7$ m^{-1})

Sol. Electron transition from n_2 level is associated with wavelength λ given by

$$\frac{1}{\lambda} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

$\lambda = 2170$ nm $= 2170 \times 10^{-9}$ m

$R_H = 1.09677 \times 10^7$ m^{-1}

$n_1 = ?$, $n_2 = 7$

$$\therefore \frac{10^9}{2170} = 1.09677 \times 10^7 \left[\frac{1}{n_1^2} - \frac{1}{7^2}\right]$$

$$\frac{1}{n_1^2} - \frac{1}{7^2} = 0.0420$$

$$\frac{1}{n_1^2} = 0.0420 + \frac{1}{49} = 0.0624$$

$\therefore n_1 = 4$

Thus, electron transition from n = 7 to n = 4 will produce infrared light of wavelength 2170 nm.

3. (i) Estimate the difference in energy between the 1st and 2nd Bohr orbit for a hydrogen atom.

(ii) At what minimum atomic number, would a transition from n = 2 to n = 1 energy level result in the emission of X – rays with $\lambda = 3.0 \times 10^{-8}$ m?

(iii) Which hydrogen atom like species does this atomic number correspond to?

R_H (Rydberg constant) $= 1.09677 \times 10^7$ m^{-1}.

Sol. (i) $Z = 1, n_1 = 1, n_2 = 2$

$$\therefore \Delta E = 1.09677\times10^{7}\times1\times3\times10^{8}\times6.626\times10^{-34}\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$$

$$= 1.635\times10^{-18}\ J$$

(ii) Also, $\frac{1}{\lambda}=\frac{\Delta E}{hc}=R_H Z^2\left[\frac{1}{n_1^2}-\frac{1}{n_2^2}\right]$

For this case, $\lambda = 3\times10^{-8}$ m, $n_1 = 1$, $n_2 = 2$

$$\frac{1}{3\times10^{-8}} = 1.09677\times10^{7}\ Z^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right]$$

$Z^2 = 4.0523$

$Z = 2.013 \approx 2$ (∵ It should be a whole number)

(iii) This H like species with atomic number 2 is He^+.

4. One quantum is absorbed per molecule of gaseous iodine for converting into iodine atoms. If light absorbed has wavelength of 5000 Å, calculate energy required in kJ mol^{-1}.

Sol. $E = N_0 h\nu = \frac{N_0 hc}{\lambda}$

$$= \frac{6.02\times10^{23}\times6.626\times10^{-34}\times3\times10^{8}}{5000\times10^{-10}}$$

$= 239.3\times10^{3}$ J mol^{-1}

$= 239.3$ kJ mol^{-1}

5. A near ultraviolet photon of 300 nm is absorbed by a gas and then reemitted as two photons. One photon is red with wavelength 760 nm. What would be the wavelength of the second photon?

Sol. Wavelength of the first photon $= \lambda_1$

$$E_1 = \frac{hc}{\lambda_1}$$

and that of second photon $= \lambda_2$

$$\therefore E_2 = \frac{hc}{\lambda_2}$$

$E_{total} = E_1 + E_2$ = Emitted energy

$$\frac{hc}{\lambda} = \frac{hc}{\lambda_1}+\frac{hc}{\lambda_2}$$

$$\frac{1}{300} = \frac{1}{760}+\frac{1}{\lambda_2}$$

$$\frac{1}{\lambda_2} = \left(\frac{1}{300}-\frac{1}{760}\right)$$

$$\frac{1}{\lambda_2} = \frac{760-300}{760\times300}$$

$\therefore \lambda_2$ = wavelength of the second photon = 496 nm

6. Find the quantum numbers of excited state of electrons in He^+ ion which on transition to ground state and first excited state emit two photons of wavelengths, 30.4 nm and 108.5 nm respectively. ($R_H = 1.09678 \times 10^7\ m^{-1}$)

Sol. For transition to ground state, $n_1 = 1$, $n_2 = ?$

$$\frac{1}{30.4 \times 10^{-9}} = 1.09678 \times 10^7 \times 4\left[\frac{1}{1^2} - \frac{1}{n_2^2}\right]$$

This gives, $n_2 = 2$

For $n_1 = 2$ (first excited state), $n_2 = ?$

$$\frac{1}{108.5 \times 10^{-9}} = 1.09678 \times 10^7 \times 4\left[\frac{1}{2^2} - \frac{1}{n_2^2}\right]$$

This gives, $n_2 = 5$

7. An electron of a H – atom in its ground state absorbs 1.50 times as much as energy as the minimum required for it to escape (13.6 eV) from the atom. Calculate the wavelength of the emitted electron.

Sol. If $E_1 = 13.6$ eV

$$E_2 = 13.6 \times 1.50\,eV = 20.4\,eV$$

$$\Delta E = (E_2 - E_1) = 20.4 - 13.6 = 6.8\,eV$$

$$= 6.8 \times 1.6 \times 10^{-19}\ J$$

$$\lambda = \frac{h}{\sqrt{2m(K.E.)}}$$

$$= \frac{6.626 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 6.8 \times 1.6 \times 10^{-19}}} = 4.71 \times 10^{-10}\ m = 4.71\ \text{Å}$$

8. The Lyman series of the hydrogen spectrum can be represented by the equation in terms of wave number

$$\bar{\nu} = 1.09603 \times 10^7\ m^{-1}\left[\frac{1}{1^2} - \frac{1}{n^2}\right] (\text{where } n = 2, 3 \ldots\ldots\ldots\ldots)$$

(i) Calculate the maximum and minimum wavelength of lines in this series.

(ii) In what portion of the electromagnetic spectrum will this series be found?

Sol. (i) $\bar{\nu}_{min} = \frac{1}{\lambda_{max}} = 1.09603 \times 10^7\ m^{-1}\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right]$

$\lambda_{max} = 121.65$ nm

$\nu_{max} = \frac{1}{\lambda_{min}} = 1.09603 \times 10^7\ m^{-1}\left[\frac{1}{1^2} - \frac{1}{\infty}\right]$

$\lambda_{min} = 91.24$ nm

(ii) $\lambda_{max} = 121.65$ nm and $\lambda_{min} = 91.24$ nm

Both will be found in UV region.

9. When light with a wavelength of 400 nm falls on the surface of sodium, electrons with a kinetic energy of 1.05×10^5 J mol^{-1} are emitted.

(i) What is the minimum energy needed to remove an electron from sodium?

(ii) What is the maximum wavelength of light that will cause a photoelectron to be emitted?

Sol. (i) K.E. of the emitted electron $= 1.05\times10^5$ J mol^{-1}

$$= \frac{1.05\times10^5}{6.02\times10^{23}} \text{ J atom}^{-1} = 1.744\times10^{-19} \text{ J atom}^{-1}$$

Using photoelectric equation for the emission of electron

$h\nu = h\nu_0 + \text{K.E.}$

$h\nu_0 = h\nu - \text{K.E.}$

∴ Minimum energy required to remove an electron,

$$h\nu_0 = \frac{hc}{\lambda} - \text{K.E.}$$

$$= \left(\frac{6.626\times10^{-34}\times3\times10^8}{400\times10^{-9}} - 1.744\times10^{-19}\right)$$

$$= 3.2255\times10^{-19} \text{ J}$$

(ii) Wavelength of the photon $= \frac{hc}{E}$

$$= \frac{6.626\times10^{-34}\times3\times10^8}{3.2255\times10^{-19}}$$

$$= 616 \text{ nm}$$

10. Show that circumference of the Bohr orbit for hydrogen atom is an integral multiple of the de Broglie wavelength associated with the electron revolving about that orbit.

Sol. According to de Broglie:

$$\lambda = \frac{h}{mu} \quad \ldots \text{(i)}$$

Also, $mur = \frac{nh}{2\pi}$ … (ii)

By Eqs. (i) and (ii) $\lambda = \frac{h.2\pi r}{n.h}$

or $2\pi r = n\lambda$

11. (i) What is highest frequency photon that can be emitted from hydrogen atom? What is wavelength of this photon?

(ii) Calculate the longest wavelength transition in the Paschen series of He^+.

(iii) Calculate the ratio of the wavelength of first and the ultimate line of Balmer series of Li^{2+}?

Sol. (i) Highest frequency photon is emitted when electron comes from infinity to 1st energy level.

$$E = -\frac{13.6Z^2}{1^2} = -13.6 \text{ eV}$$

or, $13.6 \times 1.6 \times 10^{-19}$ Joule $= 2.176 \times 10^{-18}$ Joule

$E = h\nu$

$$\therefore \nu = \frac{E}{h} = \frac{2.176\times10^{-18} J}{6.626\times10^{-34} Js} = 0.328 \times 10^{16} \text{ Hz}$$

$$\nu = \frac{c}{\lambda} \quad \therefore \lambda = \frac{3\times10^{8}}{0.328\times10^{16}} = 9.146 \times 10^{-8} \text{ m}$$

(ii) $\bar{\upsilon} = R_H \times Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

For He; Z = 2; For Paschen series $n_1 = 3$

For longest wavelength $n_2 = 4$

$$\frac{1}{\lambda} = 109678\times(2)^2 \times\left[\frac{1}{3^2} - \frac{1}{4^2}\right] = 109678\times4\times\left[\frac{1}{9} - \frac{1}{16}\right] = 109678 \times4\times\frac{7}{144}$$

$\lambda = 4689$ Å

(iii) Wave number of first line of Balmer,

$$\bar{\nu}_1 = RZ^2\left[\frac{1}{2^2} - \frac{1}{3^2}\right] = \frac{5\times9R}{36} = \frac{5R}{4}$$

$\therefore$ Wavelength of first line of Balmer $= \frac{4}{5R}$

Wave number of ultimate line of Balmer, $\bar{\nu}_2 = RZ^2\left[\frac{1}{2^2} - \frac{1}{\infty}\right] = \frac{9R}{4}$

$\therefore$ Wavelength of ultimate line of Balmer $= \frac{4}{9R}$

$\therefore$ **Ratio** $= \frac{9}{5}$

12. The kinetic energy of an electron in H like atom is 6.04 eV. Find the area of the third Bohr orbit to which this electron belongs. Also report the atom.

Sol. K.E. = 6.04 in 3rd orbit

$E_{total} = K.E + P.E.$

$= K.E - 2 \times K.E.$

$\Rightarrow -K.E = -6.04$ eV

E_1 for H = – 13.6 eV and not for any orbit E = – 6.04 eV for H atom. Thus, atom for which K.E. is given is other than H

E_n H like atom $= E_{nH} \times Z^2$

$$\frac{E_1}{n^2} \times z^2 \Rightarrow 6.04 = \frac{13.6}{3^2} \times z^2$$

$Z^2 = 3.99 \approx 4 \Rightarrow Z = 2$

$\therefore$ The atom is $He^+ \Rightarrow r_n = 0.529 \times \frac{n^2}{Z} = 0.529 \times \frac{3^2}{2} = 2.3805$ Å

Area, $\pi r^2 = \frac{22}{7} \times \left(2.3805 \times 10^{-8}\right)^2$

$= 17.8 \times 10^{-16}\ cm^2$

13. The Lyman series of the hydrogen spectrum can be represented by the equation.

$$\nu = 3.2881 \times 10^{15}\ s^{-1} \left[\frac{1}{(1)^2} - \frac{1}{(n)^2}\right]$$

(where n = 2, 3,....)

Calculate the maximum and minimum wavelength of lines in this series.

Sol. $\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c} = \frac{3.2881 \times 10^{15}}{3 \times 10^8}\ m^{-1} \left[\frac{1}{(1)^2} - \frac{1}{n^2}\right]$

Wavelength is maximum ($\bar{\nu}_{min}$) when n is minimum so that $\frac{1}{n^2}$ is maximum

$\therefore\ \bar{\nu}_{min} = \frac{1}{\lambda_{max}} = \frac{3.2881 \times 10^{15}}{3 \times 10^8} \left[\frac{1}{(1)^2} - \frac{1}{(2)^2}\right]$

$\therefore\ \lambda_{max} = \frac{3 \times 10^8}{3.2881 \times 10^{15}} \times \frac{4}{3}$

$= 1.2165 \times 10^{-7}\ m = 121.67\ nm$

Wavelength is minimum ($\bar{\nu}_{max}$) when n is ∞

i.e. series converge

$\therefore\ \nu_{max} = \frac{1}{\lambda_{min}} = \frac{3.2881 \times 10^{15}}{3 \times 10^8}$

$\therefore\ \lambda_{min} = 0.9124 \times 10^{-7}\ m\ \ 91.24\ nm$

14. Calculate the accelerating potential that must be imparted to a proton beam to give it an effective wavelength 0.005 m.

Sol. For proton $\nu = \frac{h}{m\lambda} = \frac{6.6 \times 10^{-34}}{1.6 \times 10^{-27} \times 0.005 \times 10^{-9}}$

$= 7.94 \times 10^4\ m\,sec^{-1}$

$$\text{velocity} = v = \sqrt{\frac{2eV}{m}} = \sqrt{\frac{2 \times 1.6 \times 10^{-9} \times V}{1.67 \times 10^{-27}}}$$

$V = 32.85$ volt

SOLVED PROBLEMS

SUBJECTIVE QUESTIONS

Prob. 1. The dissociation energy of H_2 is 150 kJ mol^{-1}. If H_2 is exposed to light energy of wavelength 500 nm, what % of light energy will be converted into kinetic energy ?

Sol. 37.12 %

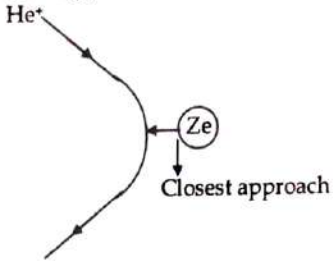
Dissocation energy H_2 = 150 kJ/mol $= 24.90 \times 10^{-20}$ J $= 2.49 \times 10^{-19}$ J

H_2 exposed to light energy = 3.96×10^{-19} Joule

% of light energy converted into kinetic energy = 37.12%

Prob. 2. With what velocity should an α-particle travel towards the nucleus of a Cu-atom so as to arrive at a distance 10^{-13} metre from the nucleus of the Cu-atom?
(Cu = 29, e = 1.6×10^{-19} C, $\varepsilon_0 = 8.85 \times 10^{-12}$ J^{-1} C^2 m^{-1}, $m_e = 9.1 \times 10^{-31}$ kg)

Sol. Total energy of α particle in the beginning = its K.E. = $\frac{1}{2}mv^2$



Total energy of α particle at the point of closest approach
= its electrostatical potential energy
(as its velocity is zero there)

$= \frac{Z \times 4e^2}{4\pi\varepsilon_0 d}$

From the law of conservation of energy
We have

$\frac{Z \times 4e^2}{4\pi\varepsilon_0 d} = \frac{1}{2}mv^2$

$\frac{1}{4\pi\varepsilon_0} = K = 9 \times 10^9$ MKS units ε_0 = permittivity of vacuum.

Substituting $Z = 29$, and $d = 10^{-13}$ m the given values in the above equation, we get $v = 6.3 \times 10^6$ m/s^{-1}

Prob. 3. Iodine molecule dissociates into atoms after absorbing light of 4500 Å. If one quantum of radiation is absorbed by each molecule. Calculate the kinetic energy of iodine atoms. (Bond energy per molecule of I_2 = 240 kJ mol^{-1})

Sol. Bond energy per molecule of $I_2 = \frac{240 \times 1000}{6.022 \times 10^{23}}$ J $= 3.984 \times 10^{-19}$ J

Energy absorbed $= \frac{hc}{\lambda} = \frac{6.626 \times 10^{-34} \times 3 \times 10^8}{4500 \times 10^{10}} = 4.417 \times 10^{-19}$ J

∴ The part of the absorbed energy which will convert into K. E. of one I_2 molecule
$= 4.417 \times 10^{-19} - 3.984 \times 10^{-19}$ J

$= 4.33 \times 10^{-20}$ J. KE of one Iodine atom $= \frac{4.33 \times 10^{-20}}{2} = 2.165 \times 10^{-20}$ J.

Prob. 4. If a 1g body is traveling along the x-axis with an uncertainty in velocity of 1 cm/sec. What is theoretical uncertainty in its position? (a) if an electron is traveling with uncertainty in velocity of 1 m/sec. (b) What is the theoretical uncertainty in its position?

Sol. According to uncertainty

(a) $\Delta x \times m.\Delta u = \frac{h}{4\pi}$

$\Delta x = \frac{h}{4\pi m \Delta u} = \frac{6.6 \times 10^{-34}}{4 \times 3.14 \times 10^{-3} \times 10^{-2}} = 3 \times 10^{-30}$ metre

(b) $\Delta x = \dfrac{6.6\times10^{-34}}{4\times3.14\times9.1\times10^{-31}\times1} = 3\times10^{-5}$ meter $= 30\ \mu m.$

Prob. 5. Calculate the energy required to excite one litre of hydrogen gas at 1 atm and 298 K to the first excited state of atomic hydrogen. The energy for the dissociation of H–H bond is 436 kJ mol^{-1}. Also calculate the minimum frequency of photon to break this bond.

Sol. Mole of H_2 present in one litre = PV/RT $= \dfrac{1\times1}{0.0821\times298} = 0.0409$ mol

Thus, energy needed to break H–H bonds in 0.0409 mole of H_2
$= 0.0409 \times 436 = 17.83$ kJ

Also energy needed to excite one H-atom from 1st to 2nd energy level,

$$\Delta E = E_2 - E_1 = \frac{E_1}{4} - E_1 \qquad (E_1 = -13.6\ eV)$$

$$= 13.6\left(1-\frac{1}{4}\right) eV = 10.2\ eV \quad = 10.2\times1.6\times10^{-19}\ J/atom$$

One atom absorbs one quantum of energy

$\therefore$ Energy needed to excite $0.0409 \times 2 \times 6.02 \times 10^{23}$ atoms of H
$= 10.2 \times 1.6 \times 10^{-19} \times 0.0409 \times 2 \times 6.02 \times 10^{23}$ J $= 80.36$ kJ

Thus, total energy needed = 17.83 + 80.36 = **98.19 kJ**

Energy required to break (H–H) bond $= \dfrac{436\times10^3}{6.023\times10^{23}}$ joule

$\because$ $E = h\nu$ $\quad\therefore\quad \dfrac{436\times10^3}{6.023\times10^{23}} = 6.623\times10^{-34}\times\nu$

$\therefore$ ν = **10.93×10^{14} sec^{-1} or Hz.**

Prob. 6. Estimate the difference in energy between 1st and 2nd Bohr orbit for a H-atom. At what minimum atomic no., a transition from n = 2 to n = 1 energy level would result in the emission of X-ray with $\lambda = 3.0\times10^{-8}$ m ? Which hydrogen atom like species does this atomic no. corresponds to ?

Sol. E_1 for H = –13.6 eV

$\therefore$ E_2 for H = $(-13.6/2^2) = -13.6/4 = -3.4$ eV

$\therefore$ $E_2 - E_1 = -3.4-(-13.6) = +10.2$ eV

Also for transition of H like atom; $\lambda = 3.0\times10^{-8}$ m

$$\frac{1}{\lambda} = R_H Z^2\left[\frac{1}{1^2}-\frac{1}{2^2}\right] \qquad \therefore\ \frac{1}{3\times10^{-8}} = 1.09\times10^7\times Z^2\times\frac{3}{4}$$

$\therefore$ $Z^2 = 4$ and $Z = 2$ Thus, the H like atom is He^+

Prob. 7. The photoelectric emission requires a threshold frequency ν_0. For a certain metal $\lambda_1 = 2200$ Å and $\lambda_2 = 1900$ Å produce electrons with a maximum kinetic energy KE_1 and KE_2, If $KE_2 = 2KE_1$, calculate ν_0 and corresponding λ_0.

Sol. $h\nu_1 = KE_1 + h\nu_0$ ...(i)

$H\nu_2 = KE_2 + h\nu_0$ ...(ii)

Multiplying (i) by 2 ad subtracting (ii) from it

$2h\nu_1 - h\nu_2 = h\nu_0$

$2\nu_1 - \nu_2 = \nu_0$

$$\nu_0 = \left[\frac{2c}{\lambda_1}-\frac{c}{\lambda_2}\right] \qquad \nu_0 = 1.1483\times10^{15}\ sec^{-1}$$

$$\nu_0 = \frac{c}{\nu_0} = \frac{3\times10^8}{1.1483\times10^{15}} = 2.612\times10^{-7}\ m$$

OBJECTIVE QUESTIONS

Prob. 1. The principal quantum number of an atom is related to the
(A) Size of the orbitals (B) Spin angular momentum
(C) Orbital angular momentum (D) Orientation of the orbitals in space

Sol. **(A)** Principal quantum number is related to size and energy level of major shell.

Prob. 2. Among KO_2, $KAlO_2$, CaO_2 and NO_2^+, unpaired electron is present in:
(A) NO_2^+ and CaO_2 (B) KO_2 and $KAlO_2$
(C) KO_2 only (D) CaO_2 only

Sol. **(C)** KO_2 has $K^+O_2^-$ structure having one unpaired electron.

Prob. 3. Mercurous ion (Hg_2^{2+}) has magnetic moment of
(A) Zero (B) 1.732 BM (C) 2.89 BM (D) 5.92 BM

Sol. **(A)** $\mu = \sqrt{n(n+2)}$ B.M. (Here n = no. of unpaired e^-)

Prob. 4. The velocity of electron in the second orbit of He^+ will be
(A) 2.18×10^6 m/s (B) 1.09×10^6 m/s (C) 4.36×10^6 m/s (D) None

Sol. **(A)** Velocity of electron in the nth orbit of H–like atom is given by

$$= 2.18\times10^6 \times \frac{Z}{n} m/s$$

For second orbit of He^{2+} ion
$Z = 2, n = 2$
$\therefore$ Velocity of electron = 2.18×10^6 m/s

Prob. 5. With a certain radiation (exciting) of a particular frequencyv, to which hydrogen atoms are exposed, the maximum number of spectral lines obtainable in the emission is 15. The uppermost energy level to which the e^- is excited is n =
(A) 4 (B) 5 (C) 6 (D) 7

Sol. **(C)** Number of spectral lines, $\frac{n(n-1)}{2} = 15 \Rightarrow n = 6$

Prob. 6. The minimum no. of orbitals possible for a shell containing g – sub shell is
(A) 25 (B) 9 (C) 36 (D) none of these

Sol. **(A)** For g–sub–shell value of 'n' is 5. So number of orbitals i.e. n^2. Thus $(5)^2 = 25$.

Prob. 7. An e^-, a proton and an alpha particle have K.E. of 16 E, 4 E and E respectively. What's the qualitative order of their Broglie wavelengths?
(A) $\lambda_e > \lambda_P > \lambda_a$ (B) $\lambda_P = \lambda_a > \lambda_e$ (C) $\lambda_P < \lambda_e < \lambda_a$ (D) $\lambda_a < \lambda_e = \lambda_P$

Sol. **(B)** $\lambda = \frac{h}{\sqrt{2mK.E.}}$

Prob. 8. The ratio of the radius difference between 4^{th} and 3^{rd} orbit of H atom and that of Li^{2+} ion is
(A) 1:1 (B) 3:1 (C) 3:4 (D) 1:3

Sol. **(B)** For H, $r_4 - r_3 = \left(0.529\times\frac{n^2}{Z}\right) = \frac{0.529}{1}[16-9] = 0.529\times7$

For Li^{2+}, $r_4 - r_3 = \frac{0.529}{3}[16-9] = \frac{0.529}{3}\times7 \Rightarrow \frac{\Delta r_H}{\Delta r_{H^+}} = 3:1$

Prob. 9. Pick out the orbital(s) with maximum number of nodal planes.
(A) $3d_{xy}$ (B) $4d_{z^2}$ (C) $4d_{xy}$ (D) $2p_x$

Sol. **(A, B, C)**

Prob. 10. For 1.Rutherford's experiment, which established the nuclear model of the atom, which statement is/are false
(A) He used a beam of β–particles, which impinged on a metal foil and got absorbed.
(B) He used a beam of γ–rays, which impinged on a metal foil and ejected electrons.
(C) He used helium atoms, which impinged on a metal foil and got scattered.
(D) He used helium nuclei, which impinged on a metal foil and got scattered.

Sol. **(A, B, C)**

Prob. 11. Which statements concerning Bohr's model are true?
(A) Predicts that probability of electron near nucleus is more
(B) Angular momentum of electron in H-atom $= \frac{nh}{2\pi}$
(C) Introduces the idea of stationary states
(D) Explains line spectrum of hydrogen

Sol. **(B, C, D)**

Prob. 12. Choose the correct relations on the basis of Bohr's theory
(A) Velocity of electron $\propto \frac{1}{n}$ (B) Frequency of revolution $\propto \frac{1}{n^3}$
(C) Radius of orbit $\propto n^2Z$ (D) Force of electron $\propto \frac{1}{n^4}$

Sol. **(A, B, D)**

Prob.13. Which of the following are the limitation of Bohr's model ?
(i) It could not explain the intensities or the fine spectrum of the spectral lines.
(ii) No justification was given for the principle of quantisation of angular momentum.
(iii) It could not explain why atoms should combine to form bond.
(iv) It could not successfully applied to single electron atoms.
(A) (i), (ii), (iv) (B) (ii), (iii) (C) (i), (iii), (iv) (D) (i), (ii), (iii)

Sol. **(D)**
Bohr's model was successfully applied to single electron.

Prob.14. Select the correct statement:
(A) Spin quantum number was proposed by Uhlenbeck and Gold-Smith
(B) Spin angular momentum is given by $\sqrt{S(S+1)}\cdot\frac{h}{2\pi}$, where S stands for absolute value of spin quantum number i.e., without sign
(C) Each fine line of atomic spectrum was found to consist a doublet pair of lines.
(D) The five *d*-orbitals energetically not identical

Sol. **(A, B, C)**

Prob.15. IE of a H-like ion A is greater than that of another H-like ion B. Let r, u, E and L represent the radius of orbit, speed of electron, energy of atom and orbital angular momentum of electron respectively in ground state:
(A) $r_A > r_B$ (B) $u_A > u_B$ (C) $E_A > E_B$ (D) $L_A > L_B$

Sol. **(B, C)**

COMPREHENSION – I

In the Rutherford's experiment, α-particles were bombarded towards the copper atoms so as to arrive at a distance of 10^{-13} metre form the nucleus of copper and then getting either deflected or traversing back. The α-particles did not move further closer.

Prob. 1. The velocity of the α-particles must be

(A) 8.32×10^8 cm/sec (B) 6.32×10^8 cm/sec
(C) 6.32×10^8 m/sec (D) 6.32×10^8 km/sec

Sol. **(B)** At the distance of closest approach $v^2 = \frac{2Ze2e}{4\pi\varepsilon_0 \times r} \times \frac{1}{m}$

Hence, substituting the value of m = 4, Z = 29, e = 1.66×10^{-19}, $r_0 = 10^{-13}$ mts
Hence, choice **(B)** is correct while **(A), (C)** and **(D)** are incorrect as from above calculation.

Prob. 2. From Rutherford's α-particle scattering, it can be concluded that

(A) $N\alpha \text{Sin}^4 \frac{\theta}{2}$ (B) $N\alpha \frac{1}{\sin^4 \theta}$ (C) $N\alpha \frac{1}{\sin^4 \theta/2}$ (D) $N = \sin \frac{\theta}{2}$

Sol. **(C)** It is observed that N = number of particles scattering $\propto \frac{1}{\sin^4(\theta/2)}$.

Hence, choice **(C)** is correct while choice **(A), (B)** and **(D)** are incorrect.

Prob. 3. It can also be concluded that the electrostatic potential energy is equal to

(A) $\frac{1}{4\pi\varepsilon_0}\frac{q_1q_2}{r}$ (B) $\frac{1}{4\pi\varepsilon_0}\frac{5ze^2}{mv^2}$ (C) $\frac{ze^2}{r}$ (D) mv^2

Sol. **(A)** The electrostatic potential $= \frac{1}{4\pi\varepsilon_0} \times \frac{q_1q_2}{r}$

Hence, choice **(A)** is correct while **(B), (C)** and **(D)** are incorrect.

COMPREHENSION – II

Read the following paragraph and answer the questions given below:

Subshells are filled in increasing order of their energy. Lower is the (n + *l*) value of sub shells, lower will be energy of that sub shells and hence electron will first enters in that subshells.
During the filling of electrons in the orbitals of same energy level, first the spin becomes maximum so that repulsion between electrons will be lesser and then pairing becomes easier.
It is not possible that two electrons of same atom will have the same sets of all four quantum numbers.

1. The numbers of elements in second period, if spin quantum number could have value of $-\frac{1}{2}, 0, \frac{1}{2}$ are
(A) 8 (B) 10
(C) 12 (D) 18

Sol. **C**

2. Which of the following ions are coloured and paramagnetic in aqueous medium?
(A) Fe^{3+} Ti^{3+} Co^{3+} (B) Cu^+ Ti^{4+} Sc^{3+}
(C) Fe^{3+} Ni^{2+} V^{5+} (D) Cu^+ Cu^{2+} Fe^{2+}

Sol. **A**

3. A vanadium compound has magnetic moment of 1.73 BM. Electronic configuration of Vanadium ion is
(A) [Ar] $4s^0\ 3d^1$ (B) [Ar] $4s^2\ 3d^3$
(C) [Ar] $4s^1\ 3d^0$ (D) [Ar] $4s^0\ 3d^5$

Sol. **A**

MATRIX – MATCH TYPE

1. Match **Column I** with **Column II**.

Column – I (Quantum numbers)		Column – II (Relation)	
(A)	n	(p)	Stark effect
(B)	ℓ	(q)	Schrodinger's equation
(C)	m	(r)	Bohr's Atomic model
(D)	s	(s)	Pauli's exclusivity
		(t)	Afbau sequencing

Sol. **((A → q, r, s, t) (B → q, s, t) (C → p, q, s) (D → s))**

2. Match the entries in **Column I** with the correctly related quantum number(s) in **Column II.** Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix given in the ORS.

Column I		Column II
(A)	Orbital angular momentum of the electron in a hydrogen-like atomic orbital	(p) Principal quantum number
(B)	A hydrogen-like one-electron wave function obeying Pauli principle	(q) Azimuthal quantum number
(C)	Shape, size and orientation of hydrogen-like atomic orbitals	(r) Magnetic quantum number
(D)	Probability density of electron at the nucleus in hydrogen-like atom	(s) Electron spin quantum number

Sol. **(A) → (q,r); (B) → (p,q,r,s); (C) → (p,q,r); (D) → (p,q)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1 Acid		Column – 2 pKa		Column – 3 O, S	
(I)	1s	(i)	0	(P)	5
(II)	$4d_{xy}$	(ii)	2	(Q)	4
(III)	6f	(iii)	1	(R)	3
(IV)	5p	(iv)	3	(S)	0

3. Which of the following combination is correct ?
[A] (I) (iii) (S) [B] (II) (ii) (Q) [C] (III) (ii) (P) [D] (IV) (i) (S)

4. Which of the following combination is incorrect ?
[A] (I) (i) (S) [B] (II) (iii) (Q) [C] (III) (ii) (P) [D] (IV) (ii) (Q)

5. Which of the following combination is correct for 5p ?
[A] (IV) (iv) (Q) [B] (IV) (i) (P) [C] (IV) (iii) (R) [D] (IV) (iv) (S)

Sol.
3. **C**
4. **B**
5. **A**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

Prob.1. STATEMENT-1: The plot of atomic number (y-axis) versus number of neutrons (x-axis) for stable nuclei shows a curvature towards x-axis from the line of 45° slope as the atomic number is increased.
and
STATEMENT-2: Proton-proton electrostatic repulsions begin to overcome attractive forces involving protons and neutrons in heavier nuclides.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(A)**

Prob. 2. STATEMENT-1: Spectral lines would not be seen for $2p_x - 2p_z$ transition.
and
STATEMENT-2: p-orbitals are degenerate orbitals.

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. (A)

Prob. 3. STATEMENT-1: All s-orbital in H-atom corresponds to a non-zero probability density at nucleus.
and
STATEMENT-2: The probability density is given by : $\Psi \propto e^{\frac{-Zr}{2a_0}}$

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(A)**

$\Psi \propto e^{\frac{-Zr}{2a_0}}$ at $r = 0$; $\Psi^2 \propto^{-0} \propto 1$

SINGLE INTEGER TYPE QUESTIONS

Prob. 1. Find the quantum no. 'n' corresponding to the excited state of He^+ ion if on transition to the ground state that ion emits two photons in succession with wavelengths 108.5 and 30.4 nm.

Sol. **(5)**

Prob. 2. The velocity of electron in a certain Bohr's orbit of H – atom bears the ratio 1 : 275 to the velocity of light. What is the number of orbit?

Sol. **(2)**

Velocity of electron in certain Bohr's orbit $(U_x) = \frac{1}{275} \times$ velocity of light.

$= \frac{1}{275} \times 3 \times 10^{10} = 1.09 \times 10^8 \text{ cms}^{-1}$

Also, $U_x = \frac{2\pi e^2}{nh}$

$\Rightarrow 1.09 \times 10^8 = \frac{2 \times 3.14 \times \left(4.803 \times 10^{-10}\right)^2}{6.626 \times 10^{-27} \times n}$

$\therefore n = 2.006 \approx 2$

Prob. 3. The wave function of atomic orbital of H like species is given by: $\Psi_{2s} = \frac{1}{4\sqrt{2}\pi} Z^{3/2} (2 - zr) e^{-zr/2}$

The radius for nodal surface of He^+ ion in Å is .....

Sol. **(1)**

Prob. 4. 1 mole of photons, each of frequency 250 sec^{-1} would have approximately a total energy (in erg).

Sol. **(1)**

Total energy = $h\nu N_A$

$= 6.6 \times 10^{-34} \times 250 \times 6.023 \times 10^{23}$

$= 0.99 \times 10^{-7} \text{ J} \approx 1 \text{ erg.}$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. If x is the total number of radial and angular nodes present in of orbital. Then find the value of $\frac{x}{5}$

Sol. **0.8**

Prob. 2. Calculate the value of A if $A = \frac{E_{1,2}}{2E_{2,1}} \times 1.9$

if $E_{n,2}$ = energy of e^- in n^{th} orbit

z = atomic number of hydrogen like species.

Sol. **15.2**

Prob. 3. If y is the difference in the number of angular node and number of radial nodes in the orbital to which last e^- of Cr is present. Then $\frac{x}{3} = ?$

Sol. **0.66**

Prob. 4. If $y \times 10^4$ k is the temperature at which translation kinetic energy of H atom equal to that for H-atom for then the first line Lyman transition ($N_A = 6 \times 10^{23}$). Then y = ?

Sol. **7.84**

STOICHIOMETRY 2

Gravimetric Analysis

(i) It is based on simple mole concept and balanced chemical equations.
(ii) The law of limiting reagent is important.
(iii) Generally, it deals with
  (a) Mass–mass relationships
  (b) Mass–volume relationships
  (c) Volume–volume relationships

Volumetric Analysis

(i) It is based on mole concept and concentration terms.
(ii) The law of equivalence is important.
(iii) Generally, it deals with
  (a) Simple titration
  (b) Back titration
  (c) Double titration
  (d) Precipitation titration

$$\text{Equivalent weight} = \frac{\text{Molecular weight}}{\text{n-factor}}$$

n-factor for redox reactions:

n = no. of moles of electrons lost or gained per mole of the reactant salt.

n-factor for non-redox reactions:

(i) n = basicity (replaceable H^+ ions), for example, n = 2 for H_3PO_3
(ii) n = acidity (replaceable OH^- ions), for example, n = 3 for $Al(OH)_3$
(iii) n = valency
(iv) n = no. of charge on cation in one molecule, for example, n = 6 for $Ca_3(PO_4)_2$

Law of Equivalents

According to the stoichiometry of the reaction, the stage in a titration when the reactants and products are present in equivalent amounts.

BACK TITRATION

Back titration is used to calculate the % purity of a sample.

Let us assume that we are given a solid substance Na_2CO_3 (C) and two solutions HCl (A) and NaOH (B).

Let, the gram equivalent of HCl is x and that of NaOH is y and given that x > y.

Therefore, out of x equivalent of HCl, y equivalent of HCl (excess equivalent of HCl w.r.t. Na_2CO_3) reacts with NaOH.

$\therefore$ Equivalents of HCl remaining for reaction with $Na_2CO_3 = x - y$

$\therefore$ Equivalents of $Na_2CO_3 = x - y$

For the reaction,

$Na_2CO_3 + 2HCl \longrightarrow 2NaCl + CO_2 + H_2O$

n-factor of $Na_2CO_3 = 2$

$\therefore$ Moles of $Na_2CO_3 = \frac{x-y}{n} = \frac{x-y}{2}$

$\therefore$ Weight of pure $Na_2CO_3 = \left(\frac{x-y}{2}\right) \times 106$ g

Let, the weight of sample of Na_2CO_3 = w g

$\therefore$ % purity of $Na_2CO_3 = \left(\frac{x-y}{2}\right) \times \frac{106}{w} \times 100$

DOUBLE TITRATION

This method involves two indicators namely phenolphthalein and methyl orange. These indicators are weak organic acids or bases which shows completion of reaction by changing its colour. Phenolphthalein acts as an indicator in weakly basic medium, but not in acidic medium. But methyl orange acts as an indicator both in acidic medium as well as in basic medium.

Extent of reaction of different bases with acid (HCl) using these two indicators is summarized as below:

Bases	Indicators	
	Phenolphthalein	Methyl orange
NaOH	100% reaction is indicated	100% reaction is indicated
Na_2CO_3	50% of reaction up to $NaHCO_3$ stage is indicated	100% reaction is indicated
$NaHCO_3$	No reaction is indicated	100% reaction is indicated $NaHCO_3 + HCl \longrightarrow NaCl + H_2O + CO_2$

Suppose, the volume of given standard acid solution (HCl) is required,

for complete reaction of NaOH = V_a ml

for complete reaction of $Na_2CO_3 = V_b$ ml

for complete reaction of $NaHCO_3 = V_c$ ml

There may be different combinations of mixture of bases are possible. We will opt the following two methods,

Method 1: We carry two titrations separately with two different indicators.

Method 2: We carry single titration but adding second indicator after first end point is reached.

Results with two indicators:

Mixture	Method – I		Method – II	
	Volume of HCl used with indicator		Volume of HCl used with indicator	
	Phenolphthalein	Methyl orange	Phenolphthalein	Methyl orange added after the first end point is reached
$NaOH + Na_2CO_3$	$V_a + \frac{V_b}{2}$	$V_a + V_b$	$V_a + \frac{V_b}{2}$	$\frac{V_b}{2}$
$NaOH + NaHCO_3$	V_a	$V_a + V_c$	V_a	V_c
$Na_2CO_3 + NaHCO_3$	$\frac{V_b}{2}$	$V_a + V_c$	$\frac{V_b}{2}$	$\frac{V_b}{2} + V_c$

[**Note:** If mixture contains NaOH, Na_2CO_3 and $NaHCO_3$, then these are the following cases with the indicators.

Case 1: The titre readings of HCl, using phenolphthalein from beginning $= V_a + \frac{V_b}{2}$

Case 2: If methyl orange is added after the first end point, then the titre readings of HCl $= \frac{V_b}{2} + V_c$

Case 3: If methyl orange is added from the very beginning, the total titre readings of HCl

$= V_a + V_b + V_c$

Volume Strength of H_2O_2

'x' volume strength of H_2O_2 means 1 litre of H_2O_2 gives 'x' litre of O_2 at NTP

Normality of $H_2O_2 = \frac{x}{5.6}$

Molarity of $H_2O_2 = \frac{x}{11.2}$

ILLUSTRATIVE PROBLEMS

1. **30 ml of a solution containing 9.15 gm/litre of an oxalate $K_xH_y(C_2O_4)_z.nH_2O$ are required for titrating 27 ml of 0.12 N NaOH and 36 ml of 0.12 N $KMnO_4$ separately. Calculate x, y, z and n. Assume all H atoms (except H_2O) are replaceable and x, y, z are in the simple ratio of gm atoms.**

Sol. Let, molecular weight of oxalate salt is M

(i) n-factor in acid–base reaction = 2

(ii) n-factor in redox titration = $2 \times z$

($C_2O_4^{2-} \longrightarrow 2CO_2 + 2e^-$)

$\therefore$ Meq. of acid in 30 ml = Meq. of NaOH used

$$30 \times \frac{9.15}{M} \times y = 27 \times 0.12 \quad \ldots (1)$$

$$\text{Also, } 30 \times \frac{9.15}{M} \times (2z) = 36 \times 0.12 \quad \ldots (2)$$

$$\text{From equations (1) and (2) } \frac{y}{2z} = \frac{27}{36} \Rightarrow \frac{y}{z} = \frac{3}{2} \quad \ldots (3)$$

Also, total cationic charge = total anionic charge

$$\therefore x + y = 2z \quad \ldots (4)$$

By equations (3) and (4)

$x : y : z :: 1 : 3 : 2$

These are in simplest ratio and molecular formula is $KH_3(C_2O_4)_2.nH_2O$

Molecular weight of salt = $39 + 3 + 176 + 18n = 218 + 18n$

$$\text{From equation (1), } M = \frac{30 \times 9.15 \times 3}{27 \times 0.12} = 254.16$$

$\therefore 218 + 18n = 254.15$

$\therefore n = 2$

$\therefore$ **Oxalate salt is $KH_3(C_2O_4)_2.2H_2O$**

2. **The calcium contained in a solution of 1.048 gm of the substance being analysed was precipitated with 25 ml of $H_2C_2O_4$. The excess of $C_2O_4^{2-}$ in one fourth of the filtrate was back titrated with 5 ml of 0.1025 N $KMnO_4$ solution. To determine the concentration of the $H_2C_2O_4$ solution taken, it was diluted four fold, the titration of 25 ml of the dilute solution used up 24.1 ml of $KMnO_4$ solution. What is percentage of CaO in the substance being analysed.**

Sol. Equivalents of $KMnO_4$ used up with one fourth of the filtrate containing excess $H_2C_2O_4$

$$= \frac{5 \times 0.1025}{1000} = 5.125 \times 10^{-4}$$

$\therefore$ Excess equivalents of $H_2C_2O_4 = 5.125 \times 10^{-4} \times 4 = 2.05 \times 10^{-3}$

Equivalents of $KMnO_4$ reacting in the second titration $= \frac{24.1}{1000} \times 0.1025 = 2.47 \times 10^{-3}$

Equivalents of $H_2C_2O_4$ in the original solution = $2.47 \times 10^{-3} \times 4 = 9.88 \times 10^{-3}$

$\therefore$ Equivalents $H_2C_2O_4$ consumed by CaO = $9.88 \times 10^{-3} - 2.05 \times 10^{-3} = 7.83 \times 10^{-3}$

$\therefore$ Equivalents of CaO = 7.83×10^{-3}

Moles of CaO = $\frac{7.83 \times 10^{-3}}{2} = 3.915 \times 10^{-3}$.

Mass of CaO = $3.915 \times 10^{-3} \times 56 = 0.22$ gm

% CaO = $\frac{0.22}{1.048} \times 100 =$ **20.99%**

3. If 1 g of $H_2(g)$ reacts with 1 g of $Cl_2(g)$ in a closed container, then how many grams of HCl will be produced during the reaction?

Sol. No. of moles of $H_2(g)$ taken = $\frac{1}{2}$

No. of moles of $Cl_2(g)$ taken = $\frac{1}{71}$

Now, $H_2(g)$ and $Cl_2(g)$ will react accordingly.

	$H_2(g)$	+	$Cl_2(g)$	$\longrightarrow$	$2HCl(g)$
At t = 0	$\frac{1}{2}$ mole		$\frac{1}{71}$ mole		0
t = t	$\frac{1}{2} - \frac{1}{71}$		$-\frac{1}{71}$ $\frac{}{0}$		$2 \times \frac{1}{71}$ mole $\frac{}{\frac{2}{71}}$ mole

Here, $Cl_2(g)$ is the limiting reagent which is completely consumed.

$\therefore$ Molar ratio of Cl_2 : HCl = 1 : 2

$\because$ No. of moles of Cl_2 consumed = $\frac{1}{71}$

$\therefore$ No. of moles of HCl produced = $\frac{2}{71}$

$\therefore$ Weight of HCl produced = $\frac{2}{71} \times 36.5$ g = 1.038 g

4. A mixture of CO and CO_2 when treated with I_2O_5, gives I_2 vapour according to following equation:

$$5CO + I_2O_5 \longrightarrow 5CO_2 + I_2$$

I_2 vapour was separated and treated with $HClO_4$ and resultant HIO_4 required 0.001 moles of glycerol ($CH_2OHCHOHCH_2OH$) for complete oxidation.
After treatment with I_2O_5 and removal of I_2, the mixture was treated with excess of 0.1 N NaOH solution. Finally, this solution required 20 ml of 1 N HCl to reach the end point using phenolphthalein as indicator followed by using methyl orange as indicator after 1st end point and 10 ml of further HCl was consumed. Calculate the volume of NaOH used and % mole of CO_2 in original mixture.

Sol.

$5CO + I_2O_5 \longrightarrow 5CO_2 + I_2$

$I_2 + 2HClO_4 \longrightarrow 2HIO_4 + Cl_2$

$2HIO_4 + CH_2OHCHOHCH_2OH \longrightarrow 2HCHO + HCOOH$

Number of moles of HIO_4 required $= 2 \times 0.001 = 0.002$

Number of moles of CO present initially $= 0.005$

Number of moles of $CO_2 = 5 + n$

$CO_2 + 2NaOH \longrightarrow Na_2CO_3 + H_2O$

Moles of Na_2CO_3 formed $= 5 + n$

Let moles of NaOH $= b$

Moles of NaOH left $= b - 2(n+5)$

When phenolphthalein is used,

$b - 2(n + 5) + (5 + n) = 20$

$b - (n + 5) = 20$ ... (1)

When methyl orange is used,

$(5 + n) = 10$ ... (2)

from Eq. (1) and (2),

$n = 5$

$\therefore b = 30$

Thus, moles of NaOH taken $= 30$

Moles of $CO_2 = 5$

Moles of $CO = 5$

Volume of NaOH,

$30 = 0.1 \times v$

$\therefore v = 300$ ml

5. Calculate the amount of SeO_3^{-2} in solution on the basis of following data. 20 ml of M/60 solution of $KBrO_3$ was added to a definite volume of SeO_3^{-2} solution. The bromine evolved was removed by boiling and excess of $KBrO_3$ was back titrated with 5.1 ml of M/25 solution of $NaAsO_2$. The reactions are given below:

(a) $SeO_3^{-2} + BrO_3^- + H^+ \rightarrow SeO_4^{-2} + Br_2 + H_2O$

(b) $BrO_3^- + AsO_2^- + H_2O \rightarrow Br^- + AsO_4^{-3} + H^+$

Sol. Meq of $KBrO_3$ (originally) $= 20 \times \frac{1}{60} \times 5 = 1.66$

Meq of $KBrO_3$ (left) = Meq of $NaAsO_2 = 5.1 \times \frac{1}{25} \times 2 = 0.408$

$\therefore$ Meq of $KBrO_3$ consumed by $SeO_3^{2-} = 1.66 - \frac{0.408}{6} \times 5 = 1.66 - 0.34 = 1.32$

Meq of $SeO_3^{2-} = 1.32$

$\therefore$ wt. of SeO_3^{2-} in the given solution $= \frac{1.32}{2} \times 127 \times 10^{-3}$

$= \mathbf{0.0838\ g}$

6. 12 g of impure cyanogen undergo hydrolysis by two different pathways:

(i) $(CN)_2 + 4H_2O \longrightarrow (NH_4)_2C_2O_4$

(ii) $(CN)_2 + 2H_2O \longrightarrow NH_2CONH_2$

The exact amount of urea was obtained when 11.52 g of pure ammonium carbonate was heated. If 20 ml of 1.6 M acidic $KMnO_4$ solution was required to completely oxidize $(NH_4)_2C_2O_4$, calculate

(a) % purity of cyanogen.

(b) % progress of reaction in two different pathways.

(c) In which reaction carbon is oxidized and in which it is reduced?

Sol. $(NH_4)_2CO_3 \longrightarrow NH_2-\overset{\overset{O}{\|}}{C}-NH_2 + 2H_2O$

Moles of urea $= \frac{11.52}{96} = 0.12$

$$2MnO_4^- + 5(NH_4)_2C_2O_4 \longrightarrow 2Mn^{2+} + 10CO_2 + 10NH_4^+$$

Moles of $(NH_4)_2C_2O_4 = 20 \times 1.6 \times \frac{5}{2} \times 10^{-3} = 0.08$

(i) $(CN)_2 + 4H_2O \longrightarrow (NH_4)_2C_2O_4$

a a

(ii) $(CN)_2 + 2H_2O \longrightarrow NH_2CONH_2$

b b

$a = 0.08$

$b = 0.12$

Total moles of $(CN)_2 = 0.20$

Weight of $(CN)_2 = 0.2 \times 52 = 10.4$

(a) % purity of $(CN)_2 = \frac{10.4}{12} \times 100 = 86.67\%$

(b) % progress in case (i)

$= \frac{0.08}{0.20} \times 100 = 40\%$

% progress in case (ii)

$= \frac{0.12}{0.20} \times 100 = 60\%$

In case (i), oxidation state remains same.

In case (ii), it is oxidized.

7. **A fused mixture of rubidium fluoride and uranium (IV) fluoride can be oxidized with fluorine to produce a uranium compound in which the uranium is mainly but not entirely in the +5 oxidation states. The product is found to contact 54.93% uranium. A 1.0357 gm sample of the product immersed in 100 ml of 10.07 centimolar acidified potassium iodide solution, reaction accordingly as following equation.**

$$2I^- + 2UF_6^- \longrightarrow 2UF_4 + I_2 + 4F^-$$

The iodine produced was titrated with 14.80 ml of 14.94 centimolar sodium thiosulphate solution. What % of the original mixture oxidized to +5 oxidation state?

Sol. Since

$2I^- + 2UF_6^- \longrightarrow 2UF_4 + I_2 + 4F^-$

$I_2 + 2S_2O_3^{2-} \longrightarrow 2I^- + S_4O_6^{2-}$

$\therefore 2S_2O_3^{2-} \equiv I_2 \equiv 2UF_6^-$

$\because$ 14.80 ml of 14.98 $\left(\frac{M}{100}\right)$ $Na_2S_2O_3$ solution

$= 14.80 \times 0.1494$ milli mole of solution

$= 2.211$ milli mole of $Na_2S_2O_3$ solution

$= \frac{2.211}{2}$ milli mole of I_2

$= 2.211$ milli mole UF_6^-

$= 2.211 \times 10^{-3} \times 238$ gm

Uranium (V)

$= 0.5626$ U (V)

The mass of uranium in 1.037 gm product

$= 1.0357 \times 0.5443$ gm $= 0.564$ gm

The percent of uranium that oxidized to + 5 O.S.

$= \frac{0.5626}{0.564} \times 100 =$ **93.3%**

8. **1.249 g of a sample of pure $BaCO_3$ and impure $CaCO_3$ containing some CaO was treated with dil. HCl and it evolved 168 ml of CO_2 at N.T.P. From this solution, $BaCrO_4$ was precipitated filtered and washed. The precipitate was dissolved in dil. H_2SO_4 and diluted to 100 ml. 10 ml of this solution when treated with KI solution liberated iodine which required exactly 20 ml of 0.05 N $Na_2S_2O_3$. Calculate the percentage of CaO in the sample.**

Sol. Let, weight of $BaCO_3$, $CaCO_3$ and CaO are x, y and z respectively.

$\therefore x + y + z = 1.249$

$BaCO_3 \longrightarrow BaCrO_4$

Redox change $\quad Cr^{6+} + 3e \longrightarrow Cr^{3+}$

$2I^- \longrightarrow I_2 + 2e^-$

Meq. of $BaCO_3$ = Meq. of $BaCrO_4$ = Meq. of I_2

$$\therefore \frac{x}{197/3} \times 1000 = 20 \times 0.05 \times \frac{100}{10} \quad \ldots (1)$$

$\therefore x = 0.657$ gm

The equivalent weight of $BaCrO_4$ is M/3, therefore for $BaCO_3$, it should be M/3 also because mole ratio of $BaCO_3$ and $BaCrO_4$ is 1 : 1.

Applying POAC for C atom,

Moles of C in $BaCO_3$ + Moles of C in $CaCO_3$ = Moles of C in CO_2

$$\therefore \frac{x}{197} + \frac{y}{100} = \frac{168}{22400}$$

$\Rightarrow 200x + 294y = 295.5 \quad \ldots (2)$

From equation (1) and (2)

$y = 0.416$ gm

$\therefore 0.0657 + 0.416 + z = 1.249$

$z = 0.176$

$$\% \text{ of CaO} = \frac{0.176}{1.249} = \mathbf{14.09\%}$$

9. Chile salt peter a source of $NaNO_3$ also contains $NaIO_3$. The $NaIO_3$ can be used as a source of iodine produced in the following reactions:

Step 1: $IO_3^- + 3HSO_3^- \longrightarrow I^- + 3H^+ + 3SO_4^{2-}$

Step 2: $5I^- + IO_3^- + 6H^+ \longrightarrow 3I_2 + 3H_2O$

One litre of chile salt peter solution containing 5.80 gm $NaIO_3$, is treated with stoichiometric quantity of $NaHSO_3$. Now additional amount of same solution is added to the reaction mixture to bring about the second reaction. How many grams of $NaHSO_3$ are required in step 1 and what additional volume of chile salt peter must be added in step 2 to bring in complete conversion of I^- to I_2?

Sol. Meq. of $NaHSO_3$ = Meq. of $NaIO_3 = N \times V = \frac{5.8}{198} \times 6 \times 1000$

$I^{5+} + 6e^- \longrightarrow I^-$ $\quad$ (For KIO_3, n-factor = 6)

$\therefore$ Meq. of $NaHSO_3$ = 175.76

$$\frac{W_{NaHSO_3}}{M/2} \times 1000 = 175.76$$

$\Rightarrow W_{NaHSO_3} = \frac{175.75 \times 104}{2 \times 1000} = \mathbf{9.14\ gm}$

Also, Meq. of I^- formed using n-factor 6 = 175.76

In step 2, n-factor of I^- is 1 and n-factor for IO_3^- is 5

Meq. of I^- formed using n-factor 1 = $\frac{175.76}{6}$

Meq. of $NaIO_3$ used in 2nd step = $\frac{175.76}{6}$

$\Rightarrow \frac{5.8}{198} \times 5 \times V = \frac{175.76}{6}$

$\therefore V_{NaIO_3} = \mathbf{20\ ml}$

10. Calculate the amount of calcium oxide required when it reacts with 852 g of P_4O_{10}.

Sol. $6CaO + P_4O_{10} \longrightarrow 2Ca_3(PO_4)_2$

Base Acid Salt

Mole ratio of reactant = $\frac{CaO}{P_4O_{10}} = \frac{6}{1}$

Mol. wt. of $P_4O_{10} = 284$

Mole of CaO required = 6 × mole of $P_4O_{10} = 6 \times \frac{852}{284}$

Weight of CaO required = $\frac{6 \times 852}{284} \times 56$

= **1008 g**

11. The formula weight of an acid is 82.0. 100 cm^3 of a solution of this acid containing 39.0 g of the acid per litre were completely neutralized by 95.0 cm^3 of aqueous NaOH containing 40.0 g of NaOH per litre. What is the basicity of the acid ?

Sol. Normality of acid = $\frac{39}{82/n \times 1}$

Normality of NaOH = $\frac{40}{40} \times \frac{1000}{1000} = 1$

Now, Meq. of acid = Meq. of NaOH

$\frac{39n}{82} \times 100 = 1 \times 95$

$\therefore$ n = 2; i.e., acid is dibasic.

SOLVED PROBLEMS

Subjective Question

Prob. 1. 10 gm of a sample of SO_2Cl_2 containing insoluble impurities was dissolved in 200 ml of water (d = 1 gm/cc). The resulting solution was titrated against Barium chloride solution and the end point of white precipitate was formed. Precipitate was filtered off and filtrate was diluted to 5 times. 50 cc of diluted solution was required 0.04N, 100cc NaOH for half neutralization. What is percentage purity of the original solution. What is mass of this white ppt?

Sol. $\underset{x\ mole}{SO_2Cl_2} + BaCl_2 + 2H_2O \longrightarrow BaSO_4(s) + 4HCl$

molarity of HCl after 5 times dilution

(i.e. 1 litre solution) $= \frac{4x}{1} = 4x$

$\underset{4x}{HCl} + \underset{4x}{NaOH} \longrightarrow NaCl + H_2O$

Complete neutralization NaOH required

$4x \times 50 = 0.04 \times 100 \times 2$

$4x = 0.04 \times 4$

$x = 0.04$

wt. of white ppt $= 0.04 \times 233 = 9.32$ gm

wt. of $SO_2Cl_2 = 0.04 \times 135 = 5.4$ gm

% purity of sample $= \frac{5.4}{10} \times 100 = 54\%$.

Objective Questions

Prob. 1. In basic medium, CrO_4^{2-} oxidizes $S_2O_3^{2-}$ to form SO_4^{2-} and itself changes to $Cr(OH)_4^-$. How many ml of 0.154 M CrO_4^{2-} are required to react with 400 ml of 0.246 M $S_2O_3^{2-}$?

(A) 200 ml (B) 156.4 ml

(C) 170.4 ml (D) 190.4 ml

Sol. $40 \times 0.246 \times 8 = V \times 0.154 \times 3$ (Meq. of $S_2O_3^{2-}$ = Meq. of CrO_4^{2-})

$\therefore V = 170.4$ ml

Hence, **(C)** is the correct answer.

Prob. 2. 20 ml of 0.2 M $Al_2(SO_4)_3$ is mixed with 20 ml of 0.6 M $BaCl_2$. Concentration of Al^{3+} ion in the solution will be

(A) 0.2 M (B) 10.3 M

(C) 0.1 M (D) 0.25 M

Sol.

	$Al_2(SO_4)_3$	+ $3BaCl_2$	$\longrightarrow$	$3BaSO_4\downarrow$	+ $2AlCl_3$
Initial Meq.	$20 \times 0.2 \times 6 = 24$	$20 \times 0.6 \times 2 = 24$		0	0
Final Meq.	0	0		24	24

$$[Al^{3+}] = \frac{24}{40 \times 3} = 0.2 \text{ M}$$

Hence, **(A)** is the correct answer.

Prob. 3. The number of moles of $KMnO_4$ that will be required to react with one mol of ferrous oxalate is

(A) $\frac{3}{5}$ (B) $\frac{2}{5}$

(C) $\frac{4}{5}$ (D) 1

Sol. **(A),** $Mn^{7+} + 5e^- \rightarrow Mn^{2+}] \times 3$

$$\left.\begin{matrix} Fe^{2+} \rightarrow Fe^{3+} + e^- \\ C_2O_4^{2-} \rightarrow 2CO_2 + 2e^- \end{matrix}\right] \times 5$$

3 moles of $KMnO_4$ = 5 moles of FeC_2O_4

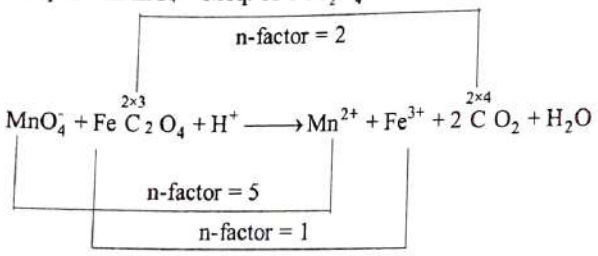
$\therefore$ 1 mol of ferrous oxalate $\equiv \frac{3}{5}$ mol of $KMnO_4$

Prob. 4. What volume of 0.1M $KMnO_4$ is needed to oxidize 100 mg of FeC_2O_4 in acid solution?

(A) 4.1 ml (B) 8.2 ml

(C) 10.2 ml (D) 4.6 ml

Sol. Meq. of $KMnO_4$ = Meq. of FeC_2O_4



$\therefore V = 4.1$ml

Hence, **(A)** is the correct answer.

Prob. 5. Which of the following is/are redox reaction(s)?

(A) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$ (B) $2BaO + O_2 \rightarrow 2BaO_2$

(C) $2KClO_3 \rightarrow 2KCl + 3O_2$ (D) $SO_2 + 2H_2S \rightarrow 2H_2O + 3S$

Sol. **(B), (C)** and **(D).**

Prob. 6. In the reaction,

$3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3^-$

(A) bromine is oxidized and carbonate is reduced

(B) bromine is oxidized

(C) bromine is reduced

(D) it is disproportionation reaction or autoredox change

Sol. **(B), (C)** and **(D).**

Prob. 7. During the titration of a mixture of Na_2CO_3 and $NaHCO_3$ against HCl,

(A) phenolphthalein is used to detect the first end point

(B) phenolphthalein is used to detect the second end point

(C) methyl orange is used to detect the second end point

(D) methyl red is used to detect the first end point

Sol. **(A, C)**

COMPREHENSION

Read the paragraph carefully and answer the following questions:

The process of addition of the known solution from the burette to the measured volume of solution of the substance to be estimated until the reaction between the two is just complete is termed as titration.

Titration is classified in following two categories:
(a) Acid–base titration
(b) Redox titration

The acid-base titration is further classified in following three categories:
(a) Simple titration
(b) Back titration
(c) Double titration

In simple titration, we can find the concentration of substance with the help of another substance with known concentration, which can react with it.
Back titration is used to calculate % purity of a sample.
In double titration mainly two indicators, i.e. phenolphthalein (HPh) and methyl orange (MeOH) are used. Suppose, for complete neutralization of Na_2CO_3, $NaHCO_3$ and NaOH, the amount of standard HCl required is a, b and c ml. The titration of the mixture may be carried by two different methods as summarized below:

Mixture	Volume of HCl used with		Volume of HCl	
	HPh from beginning	MeOH from beginning	HPh from beginning	MeOH orange after first end point
$NaOH + Na_2CO_3$	$c + \frac{a}{2}$	(a + c)	$c + \frac{a}{2}$	$\frac{a}{2}$ (For remaining 50% Na_2CO_3)
$NaOH + NaHCO_3$	c + 0	(b + c)	C + 0	B (For 100% remaining $NaHCO_3$)
$Na_2CO_3 + NaHCO_3$	$\left(\frac{a}{2} + 0\right)$	(a + b)	$\frac{a}{2} + 0$	$\left(\frac{a}{2} + b\right)$ for remaining 50% of Na_2CO_3 and 100% $NaHCO_3$ are indicated

1. 150 ml of N/10 HCl is required to react completely with 1.0 g of a sample of limestone. Calculate the percentage purity of $CaCO_3$.
   (A) 65% (B) 25%
   (C) 75% (D) 85%

2. An aqueous solution of 6.3 g of oxalic acid dihydrate is made upto 250 ml. The volume of 0.1 N NaOH required to completely neutralize 10 ml of this solution is
   (A) 40 ml (B) 20 ml
   (C) 10 ml (D) 4 ml

3. A solution contained Na_2CO_3 and $NaHCO_3$. 15 ml of the solution required 5 ml of N/10 HCl for neutralization using phenolphthalein as an indicator. Addition of methyl orange required a further 15 ml of the acid for neutralization. The amount of Na_2CO_3 present in the solution is
   (A) 21.2 g (B) 2.12 g
   (C) 0.212 g (D) 4.24 g

Answers

1. C 2. A 3. B

MATRIX-MATCH TYPE

1. Match the following

Column – I (Acid)		Column – II (Basicity, Equivalent wt.)	
(a)	H_3PO_3	(p)	Dibasic oxidizing agent
(b)	H_2SO_4	(q)	Dibasic reducing agent
(c)	H_2S	(r)	Eq. wt. $= \frac{\text{mol. mass}}{2}$
(d)	$\begin{matrix}COOH\\ \vert\\ COOH\end{matrix}$	(s)	Eq. wt = mol. mass

Sol. **(A → q, r, s), (B → p, r, s), (C → q, r, s), (D → q, r, s)**

Answer 2, 3 and 4 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	$P_2H_4 \longrightarrow PH_3 + P_4H_2$	(i)	Disproportionation	(P)	$E = \frac{3M}{4}$
(II)	$I_2 \longrightarrow I^- + IO_3^-$	(ii)	Intramolecular redox	(Q)	$E = \frac{3M}{5}$
(III)	$MnO_4^- + Mn^{2+} + H_2O \longrightarrow Mn_3O_4 + H^+$	(iii)	Intermolecular redox	(R)	$E = \frac{15M}{26}$
(IV)	$H_3PO_2 \longrightarrow PH_3 + H_3PO_3$	(iv)	Non-redox reaction	(S)	$E = \frac{5M}{6}$

2. The only correct combination is
[A] (I) (i) (S) [B] (I) (iv) (S) [C] (I) (i) (Q) [D] (III) (ii) (P)

3. The only incorrect combination is
[A] (I) (i) (S) [B] (II) (i) (Q) [C] (IV) (i) (P) [D] (III) (ii) (R)

4. The only correct combination in which element in their respective elementary state is taken.
[A] (III) (iii) (R) [B] (IV) (i) (P) [C] (II) (i) (Q) [D] (I) (i) (S)

Sol.
2. **A**
3. **D**
4. **C**

SINGLE INTEGER ANSWER TYPE

Prob. 1. What mass of H_2SO_4 could be produced (in kg) in the process given below if 5.52 kg of FeS_2 is used

$$FeS_2(s) + O_2(g) \longrightarrow Fe_2O_3(s) + SO_2(g)$$

$$SO_2(g) + O_2(g) \longrightarrow SO_3(g)$$

$$SO_3(g) + H_2SO_4(l) \longrightarrow H_2s_2O_7(l)$$

$$H_2S_2O_7(l) + H_2O(l) \longrightarrow H_2SO_4(aq)$$

(At. wt. of Fe = 56, S = 32)

Sol. **(9)**

$FeS_2 \rightarrow 2SO_2 \rightarrow 2SO_3 \rightarrow H_2S_2O_7 \rightarrow 2H_2SO_4$

1 mol of $FeS_2 \equiv 2$ mol of $H_2SO_4 = 2 \times 98$ gm H_2SO_4

$\therefore$ Moles of $FeS_2 = \dfrac{5.52 \times 1000}{120}$

H_2SO_4 from $\dfrac{5.52 \times 1000}{120}$ moles of $Fes_2 = \dfrac{5.52 \times 1000}{120} \times 2 \times 98$

$= 9$ kg of H_2SO_4

Prob. 2. 0.45 g of the acid of molecular mass 90 was neutralized by 20 mL of 0.5 N caustic potash. What is basicity of the acid?

Sol. **(2)**

Prob. 3. 2 mole of $Fe_2C_2O_4$ is oxidized by x mole of $Cr_2O_7^{2-}$ in acidic medium, x is

Sol. **(1)**

Prob. 4. The volume (in litre) of Cl_2 at STP react with 10 g Fe to produce $FeCl_3$ will be

Sol. **(6)**

Prob. 5. Find the weight of NaOH in its 50 milli equivalents.

Sol. **(2)**

$\because \quad \text{Meq.} = \dfrac{w}{E} \times 1000$

$\therefore \quad 50 = \dfrac{w}{40} \times 1000$

$\therefore \quad w = \mathbf{2}$

Prob. 6. How many millilitre of 0.5 M H_2SO_4 are needed to dissolve 0.5 g of copper II carbonate ?

Sol. **(8)**

Meg. of H_2SO_4 = Meq. of $CuCO_3 \qquad \left(\text{Eq. wt. of } CuCO_3 = \dfrac{M}{2}\right)$

$$0.5 \times 2 \times V = \frac{0.5 \times 2 \times 1000}{123.5}$$

$\therefore \quad V = \mathbf{8.097\ mL}$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. 50 litres of water containing $Ca(HCO_3)_2$ when converted into soft water required 22.2 g $Ca(OH)_2$. Calculate the amount of $Ca(HCO_3)_2$ present per litre of hard water.

Sol. **0.97 gL^{-1}**

Reaction

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2CaCO_3 + 2H_2O$$

162 g 74 g

74 g $Ca(OH)_2$ reacts with 162 g of $Ca(HCO_3)_2$

22.2 g $Ca(OH)_2$ will react with $\frac{162 \times 22.2}{74}$

$= 48.6$ g $Ca(HCO_3)_2$

50 L water contains = 48.6 g $Ca(HCO_3)_2$

1 L water contains $= \frac{48.6}{50} = 0.972\ gL^{-1}$

Prob. 2. 1.2 g of a commercial sample of oxalic acid was dissolved in 200 ml of water. 10 ml of this sample required 8.5 ml of N/10 $KMnO_4$. Calculate % purity of sample.

Sol. **98.25**

Redox changes are :

For $H_2C_2O_4$ $\quad (C^{3+})_2 \longrightarrow 2C^{4+} + 2e^-$

For $KMnO_4$ $\quad 5e^- + Mn^{7+} \longrightarrow Mn^{2+}$

Meq. of oxalic acid in 10 ml solution = Meq. of $KMnO_4$ used for it $= 8.5 \times \frac{1}{10}$

$\therefore$ Meq. of oxalic acid in 200 ml solution

$$= 8.5 \times \frac{1}{10} \times \frac{200}{10} = 17$$

$$\therefore \quad \frac{W}{E} \times 1000 = 17$$

Formula of oxalic acid is $H_2C_2O_4.2H_2O$

$\therefore$ Molar mass = 126

$$\therefore \quad \frac{w}{126/2} \times 1000 = 17$$

$$\therefore \quad \%\ \text{purity of oxalic acid} = \frac{1.071}{1.2} \times 100 = 89.25\ \%$$

Prob. 3. How many millilitres of 0.5M H_2SO_4 are needed to dissolve 0.5 g of copper(II) carbonate ?

Sol. **8.09**

According to normality equation,

$N_1V_1 = N_2V_2$

N_1 = Normality of $H_2SO_4 = 0.5 \times 2 = 1$ N

V_1 = Volume of H_2SO_4

N_2 = Normality of copper(II) carbonate

$$= \frac{0.5}{123.5/2} = \frac{0.5 \times 2}{123.5}\ \text{N}$$

V_2 = Volume of copper(II) carbonate = 1000 ml

So, $1.0 \times V_1 = \frac{0.5 \times 2}{123.5} \times 1000$

Therefore, $V_1 = 8.09$ ml

3

GASEOUS STATE

GAS LAWS

Boyle's Law

The volume of a gas is inversely proportional to the pressure of the gas at constant temperature.

$V \propto \frac{1}{P}$, PV = constant

$P_1V_1 = P_2V_2$ at constant T

Charle's Law

The volume of a gas is directly proportional to the temperature of the gas at constant pressure.

$V \propto T$, $\frac{V_1}{V_2} = \frac{T_1}{T_2}$ at constant P

Gay Lussac's Law

The pressure of a gas is directly proportional to the temperature of the gas at constant volume.

$P \propto T$, $\frac{P_1}{P_2} = \frac{T_1}{T_2}$ at constant T

Avogadro's Law

The volume of a gas increases with increase in the number of moles at constant P and T.

$V \propto n$, at constant P and T

$\frac{V_1}{V_2} = \frac{n_1}{n_2}$

DALTON'S LAW OF PARTIAL PRESSURE

$P = P_A + P_B + P_C$

where P is the total pressure of the mixture of gases

P_A, P_B and P_C are the partial pressures of gases A, B and C

$P_A = X_AP$

where X_A = mole fraction of A

$P_{dry\ gas} = P_{moist\ gas}$ – aqueous tension

Aqueous tension = $P_{water\ vapours}$

GRAHAM'S LAW OF DIFFUSION/EFFUSION

It states that the rate of diffusion/effusion of a gas is inversely proportional to the square root of its density.

$$r_1 \propto \frac{1}{\sqrt{d_1}}$$

$$\frac{r_1}{r_2} = \frac{V_1/t_1}{V_2/t_2} = \sqrt{\frac{d_2}{d_1}} = \sqrt{\frac{VD_2}{VD_1}} = \sqrt{\frac{M_2}{M_1}}$$

where M_1 = Molecular mass of gas 1
M_2 = Molecular mass of gas 2
VD_1 = Vapour density of gas 1
VD_2 = Vapour density of gas 2

$$r \propto \frac{P}{\sqrt{M}}$$

IDEAL GAS EQUATION

The ideal gas equation is obtained by combining Boyle's and Charle's law.

$PV = nRT$

where R = Gas constant

The above relation can be also represented as

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2};\ PV = \frac{W}{M}RT;\ P = \frac{dRT}{M}\ (d = \text{gm/litre})$$

KINETIC THEORY OF GAS

The postulates of the theory are used to derive kinetic gas equation, i.e.

$$PV = \frac{1}{3}mnc^2$$

where P = Pressure of the gas

V = Volume of the gas

M = Mass of a molecule

n = Number of molecules in the given amount of gas

c = Root mean square velocity

For one gram mole of gas,

$n = N_A$ and $m \times N_A = M$ [N_A = Avogadro's number]

$$\therefore\ PV = \frac{1}{3}Mc^2$$

$$\text{K.E. per mole} = \frac{3}{2}RT = \frac{3}{2}PV$$

$$\text{K.E. per molecule} = \frac{3}{2}\frac{RT}{N_A} = \frac{3}{2}kT$$

where k = Boltzmann constant = 1.38×10^{-23} J K^{-1} $molecule^{-1}$

TYPES OF MOLECULAR VELOCITIES

(i) **RMS velocity:**

$$C_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3PV}{M}} = \sqrt{\frac{3P}{d}} \text{ (rms velocity)}$$

(ii) **Average velocity:**

$$C_{av} = \sqrt{\frac{8RT}{\pi M}} \text{ (average velocity)}$$

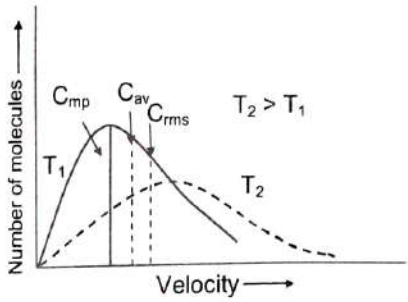
(iii) **Most probable velocity:**

$$C_{mp} = \sqrt{\frac{2RT}{M}} \text{ (most probable velocity)}$$

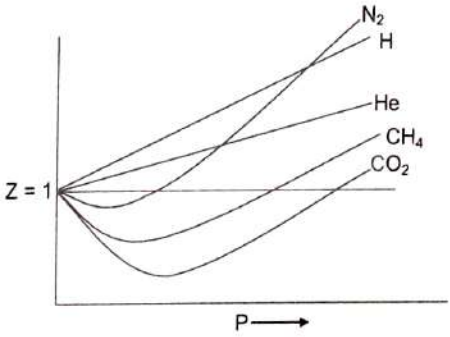
$$C_{rms} : C_{av} : C_{mp} = 1 : 0.9213 : 0.816$$

$$C_{mp} : C_{av} : C_{rms} = 1 : 1.128 : 1.224$$

Maxwell's Distribution Curve for Velocities



Compressibility Factor (Z)



PV = ZnRT

For ideal gases, Z = 1

So, PV = nRT

If Z > 1, then compressibility is less than expected, Repulsive forces are dominating.

If Z < 1, then compressibility is more than expected, Attractive forces are dominating.

$$Z = \frac{\text{Volume occupied by real gas}}{\text{Volume occupied by ideal gas}}$$

VAN DER WAAL'S GAS EQUATION FOR REAL GASES

The following assumptions of Kinetic theory of gas are not followed by the real gases:

1. The volume of a gas molecule is negligible in comparison to the volume occupied by whole gas, which is practically not correct at all pressures. This leads to volume correction.
2. There are no intermolecular forces of attraction, which is practically not observed in real gases. This leads to pressure correction.

(i) **Volume correction**

$V_{actual} = V - nb$ for n molecules

where b = effective volume per molecule $= \frac{4}{3}\pi r^3 \times 4$

Units of b = ltr mol^{-1}

(ii) **Pressure correction**

$P_{actual} = P + P'$

where, $P' \propto d^2 \propto \frac{n^2}{V^2}$

So, $P' = \frac{an^2}{V^2}$

Units of a = atm $ltr^2\ mol^{-2}$

$= Nm^4\ mol^{-2}$

So, ideal gas equation becomes

$$\left(P + \frac{an^2}{V^2}\right)(V - nb) = nRT$$

(a) At very low pressure & high temperature for 1 mole gas

P′ and nb can be neglected.

$PV = RT$

(b) At moderate pressure, nb can be neglected.

$$\left(P + \frac{a}{V^2}\right)(V) = RT$$

$Z = \frac{PV}{RT}$ so $Z = 1 - \frac{a}{RTV}$

$Z < 1$

(c) At high pressure

P′ can be ignored but b cannot be ignored.

So, $P(V - b) = RT$

$Z = \frac{PV}{RT} = 1 + \frac{Pb}{RT}$

$Z > 1$

In case of H and He due to their small size, we always ignore the internal pressure $\left(P' \text{ or } \frac{an^2}{V^2}\right)$.

So, they always show $Z > 1$ at moderate and high pressures.

Boyle Temperature

It is a particular temperature for a real gas above which the gas behaves as an ideal gas over a long range of pressure.

$$T_b = \frac{a}{bR}$$

where a and b are van der Waal's constants.

CRITICAL PARAMETERS

(a) **Critical temperature:** The temperature above which a gas cannot be liquefied by applying any extent of pressure.

(b) **Critical pressure:** The pressure required to apply on a gas to liquefy at critical temperature.

(c) **Critical volume:** The volume attained by a gas at critical temperature and pressure.

$$T_c = \frac{8a}{27Rb},\ P_c = \frac{a}{27b^2},\ V_c = 3b$$

Molar Heat Constants

$$C_V = \Delta K.E. = \frac{3}{2}R$$

$$C_P = \Delta K.E. + P\Delta V = \frac{3}{2}R + R = \frac{5}{2}R$$

$C_P - C_V = R$

Specific heat (at constant V) × Molecular weight = C_V

Specific heat (at constant P) × Molecular weight = C_P

ILLUSTRATIVE PROBLEMS

1. A graph is plotted between PV_m along Y-axis and P along X-axis, where V_m is the molar volume of a real gas. Find the intercept on Y-axis.

Sol. $\therefore \left(P + \frac{a}{V_m^2}\right)(V_m - b) = RT$

$$\left(P + \frac{aP^2}{(PV_m)^2}\right)\left(\frac{PV_m}{P} - b\right) = RT$$

$[(PV_m)^2 P + aP^2][(PV_m) - b)] = P(PV_m)^2RT$

$\Rightarrow P[(PV_m)^2 + aP]\,(PV_m - bP) = P(PV_m)^2RT$

Put $P = 0$

$\Rightarrow (PV_m)^3 = (PV_m)^2\,RT$

Intercept $= RT$

2. 50 ml. of a mixture of NH_3 and H_2 was completely decomposed by sparking into nitrogen and hydrogen. 40 ml of oxygen was then added and the mixture was sparked again. After cooling to the room temperature the mixture was shaken with alkaline pyro-gallol and a contraction of 6 ml. was observed. Calculate the % of NH_3 in the original mixture.

Sol. Let the volume of NH_3 be x ml

$\therefore$ Volume of $H_2 = (50 - x)$ ml

On sparking,

$$NH_3 \longrightarrow \frac{1}{2}N_2 + \frac{3}{2}H_2$$

Since 40 ml of O_2 is added and sparked, it must have reacted with H_2 to form liquid water. Moreover since 6 ml contraction occurs with alkaline pyrogallol, 34 ml is the volume of O_2 used up.

$\therefore$ Total volume of H_2 is 68 ($\because 2H_2 + O_2 \longrightarrow 2H_2O$)

$$\therefore (50 - x) + \frac{3}{2}x = 68$$

$$50 + \frac{x}{2} = 68; \quad x = 36$$

$\therefore$ **$NH_3 = 72\%$**

3. **The density of the vapour of a substance at 1 atm pressure and 500 K is 0.36 kg m^{-3}. The vapour effuses through a small hole at a rate of 1.33 times faster than oxygen under the same condition.**

(a) Determine

(i) molecular, weight,

(ii) molar volume,

(iii) compression factor (Z) of the vapour and

(iv) which forces among the gas molecules are dominating, the attractive or the repulsive?

(B) If the vapour behaves ideally at 1000 K, determine the average translational kinetic energy of a molecule.

Sol. (a) $d = 0.36 \text{ kg m}^{-3} = 0.36 \text{g/L}$

(i) From Graham's Law of diffusion

$$\frac{r_v}{r_{O_2}} = \sqrt{\frac{M_{O_2}}{M_v}} ; 1.33 = \sqrt{\frac{32}{M_v}} \therefore M_v = \frac{32}{(1.33)^2} = 18.09; \text{ where } M_v = \text{MW of the vapour}$$

(ii) Thus, $0.36\text{g} = \frac{0.36}{18.09}$ mol

$\frac{0.36}{18.09}$ mol occupies 1 L volume

so 1 mol occupies $\frac{18.09}{0.36}$ L

Thus, molar volume of vapour = 50.25 L

Assuming ideal behaviour the volume of the vapour would be

$$\frac{V_1}{T_1} = \frac{V_2}{T_2} \Rightarrow V_2 = 22.4 \times \frac{500}{273} = 41.025 \text{ L}$$

(iii) Compressibility factor (Z) $= \frac{(PV)_{obs}}{(PV)_{ideal}} = \frac{1 \times 50.25}{1 \times 41.025} = 1.224$

(iv) Z being greater than unity, it is the short range repulsive force that would dominate.

(b) $E = \frac{3}{2} k_B T = \frac{3}{2} \times 1.38 \times 10^{-23} \times 1000 = 2.07 \times 10^{-20}$ J per molecule.

4. **Derive expression relating to change in E_{trans} for a gas for small temperature change and calculate change in E_{trans} for a sample of Ne(g) as the temperature is increased from 100 K to 101 K ?**

Sol. $E_{trans} = \frac{3}{2} RT$

$$\frac{dE_{trans}}{dT} = \frac{3}{2} \times R = \frac{3}{2} RT\left(\frac{1}{T}\right) = \frac{E_{trans}}{T}$$

$$dE_{trans} = E_{trans} \cdot \frac{dT}{T}$$

$$\Delta E_{trans} = E_{trans} \ln\left(\frac{T_2}{T_1}\right)$$

$$= \frac{3}{2} \times 8.314 \times 100 \ln \frac{101}{100}$$

$$= 12.4 \text{ J mol}^{-1}$$

5. Two bulbs A and B of equal capacity are filled with He and SO_2 respectively at the same temperature

(i) If the pressure in two bulbs is same, calculate the ratio of U_{rms} for them.

(ii) At what temperature velocity of SO_2 becomes half of the speed of He molecules at $27^\circ C$.

(iii) How will the speeds be affected if volume of B becomes four times.

(iv) How will the speeds be affected if half of the molecules of SO_2 are removed from B.

Sol. Let the U_{rms} of He and SO_2 be U_1 and U_2 respectively

(i) For He $u_1 = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3RT}{4}}$

For SO_2 $u_2 = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3RT}{64}}$ {Temp is same}

$$\frac{u_1}{u_2} = 4$$

(ii) Given u_{rms} of $SO_2 = \frac{1}{2} u_{rms}$ of He

$$\sqrt{\frac{3RT}{64}} = \frac{1}{2}\sqrt{\frac{3R(300)}{4}}$$ (T = 300 K for He)

T = 1200 K

or, 927°C

(iii) Since speed is independent of P and V terms and thus no effect on speed by changing volume.

(iv) Since u_{rms} is independent of no. of molecules and thus no effect on speed by changing no. of molecules.

6. Using Vander waal's equation of state, calculate the pressure correction factor for two moles of a gas confined in a four litre flask that exerts a pressure of 11 atm at 300 K. (b = 0.05 L mol^{-1})

Sol. $(P + K)(V - nb) = nRT$

$(11 + K)(4 - 2 \times 0.05) = 2 \times 0.082 \times 300$

$K = 1.61 \text{ atm L}^2 \text{ mol}^{-2}$

7. A definite amount of a gaseous hydrocarbon having less than 5 carbon atoms was burned with sufficient amount of O_2 at a very high temperature. The volume of all reactants was 600 ml, after the explosion the volume of the products was found to be 700 ml under the similar conditions. What is the molecular formula of compound?

Sol. $C_xH_y(g) + \left(x+\frac{y}{4}\right)O_2(g) \longrightarrow xCO_2(g) + \frac{y}{2}H_2O(\ell)$

A ml	$A\left(x+\frac{y}{4}\right)$ ml	0	0
0	0	Ax	$\frac{Ay}{2}$

$A + Ax + \frac{Ay}{4} = 600$ ...(1)

$Ax + \frac{Ay}{2} = 700$ ...(2)

$\Rightarrow \frac{x}{y} = \frac{3}{8}$

Since, $x < 5$

$\Rightarrow x = 3, y = 8$

$\Rightarrow A = 100$ ml

So, the hydrocarbon is C_3H_8 and the volume taken was 100 ml.

8. The pressure in bulb dropped from 2000 to 1500 mm Hg in 47 mins. when the O_2 present in the bulb leaked through a small hole. The bulb was then completely evacuated. A mixture of O_2 and another gas of molecular weight of 79 in the molar ratio 1:1 at a total pressure of 4000 mm Hg was introduced. Find the molar ratio of two gases remaining in the bulb after a period of 74 mins.

Sol. 2000 mm pressure of $O_2 \xrightarrow{t = 47 \text{ min}} 1500$ mm

4000 mm pressure of mixture $\xrightarrow{t = 74 \text{ min}}$ 1:1 (O_2 + gas)

For pure O_2, $\frac{P_1}{P_2} = \frac{n_1}{n_2}$

where n_1 and n_2 are original no. of moles of O_2 and moles of O_2 after 47 minutes.

$\therefore \frac{n_1}{n_2} = \frac{2000}{1500}$

$\therefore n_2 = 3/4n_1$

or moles of O_2 diffused in 47 minutes $= n_1 - \frac{3n_1}{4} = \frac{n_1}{4}$

or moles of O_2 diffused in 74 minutes $= \frac{n_1 \times 74}{47 \times 4}$

$$= \frac{74}{188}n_1 = \frac{74}{188} \text{ (if } n_1 = 1) = 0.3936$$

Since, diffusion of O_2 in mixture also occurs at partial pressure of 2000 mm (The ratio of gas and O_2 being 1:1).

Now, gas and O_2 both diffusing in the form of mixture through same orifice at partial pressure of 2000 mm each.

$\therefore \frac{n_{O_2}}{74} \times \frac{74}{n_g} = \sqrt{\left(\frac{79}{32}\right)}$

$\therefore n_g = n_{O_2} \times \sqrt{\left(\frac{32}{79}\right)} = \frac{74}{188} \times \sqrt{\left(\frac{32}{79}\right)} = 0.249$

$\therefore$ Moles of O_2 left after 74 minutes $= 1 - 0.3936$
$= 0.6064$

Moles of gas after 74 minutes $= 1 - 0.249$
$= 0.7510$

$\therefore O_2$: gas $= 0.6064 : 0.7510$
$\therefore 1 : 1.236$

9. Calculate the pressure of one mole of ammonia gas which is in a 100 ml vessel, at 500 K, given the second and third virial coefficients are $B = -68.3 \frac{cc}{mol}$ and $C = 2397 \frac{cm^6}{mol^2}$

Sol. $B = -68.3 \frac{cm^3}{mol} = b - \frac{a}{RT}$ ...(i)

$C = b^2$ ...(ii)

Solving (i) and (ii) at 500 K

$a = 4.8$ atm L^2 mol^{-2}

$b = 0.049$ L mol^{-1}

Now for 1 mol of Vander waal gas

$$p = \frac{RT}{V-b} - \frac{a}{V^2} = \frac{0.082 \times 500}{0.1 - 0.049} - \frac{4.8}{0.01}$$

$= \mathbf{323.92}$ **atm**

10. Calculate the average volume available to a molecule in a sample of nitrogen gas at STP. What is the average distance between neighbouring molecules if nitrogen molecules are spherical in nature?

Sol. 6.023×10^{23} molecules of N_2 occupy 22400 cm^3

$\therefore$ one molecules of N_2 occupies $= \frac{22400}{6.023 \times 10^{23}} cm^3$

or volume of one molecule of $N_2 = 3.72 \times 10^{-20}$ cm^3

Also, the average distance between two molecules = 2r

and $\frac{4}{3}\pi r^3 = 3.72 \times 10^{-20}$

$\therefore r^3 = \frac{3.72 \times 10^{-20} \times 3 \times 7}{4 \times 22}$

$\therefore r = 20.7 \times 10^{-8}$ cm

Thus, average distance $= 2 \times 20.7 \times 10^{-8}$

$= 41.4 \times 10^{-8}$ cm

11. The compression factor (compressibility factor) for one mole of a van der Waals' gas at 0°C and 100 atmosphere pressure is found to be 0.5. Assuming that the volume of a gas molecule is negligible, calculate the van der Waal's constant 'a'.

Sol. $Z = \frac{PV}{RT} = 0.5$

$\therefore \frac{100 \times V}{0.082 \times 273} = 0.5$

$\therefore V = 0.112$ litre

Now, using van der Waals' equation

$\left[P + \frac{a}{V^2}\right][V] = RT$

or $\left[P + \frac{a}{V^2}\right] = \frac{RT}{V}$ ($\because$ b is negligible)

$\therefore \left[100 + \frac{a}{(0.112)^2}\right] = \frac{0.082 \times 273}{0.112}$

$= 199.88$

$\therefore \frac{a}{(0.112)^2} = 99.88$

$\therefore a = 1.253$ litre^2 mole^{-2} atm.

12. Two containers A and B have the same volume. Container A contains 5 moles of O_2 gas. Container B contains 3 moles of He and 2 moles of N_2. Both the containers are separately kept in vacuum at the same temperature. Both the containers have very small orifices of the same area through which the gases leak out. Compare the rate of effusion of O2 with that of He gas mixture.

Sol. Since both the containers are in the same conditions of P, V and T,

$\frac{r_{O_2}}{r_{mix}} = \sqrt{\frac{M_{mix}}{M_{O_2}}}$

As the mixture contains three moles of He and 2 moles of N_2, the effective molecular weight of the mixture would be.

$\frac{3}{5} \times 4 + \frac{2}{5} \times 28 = 13.6$

$\therefore \frac{r_{O_2}}{r_{mix}} = \sqrt{\frac{13.6}{32}} = 0.652$

Though this solution looks OK, there is one big flaw in it.

The error is that we have assumed that He and N_2 from vessel B would effuse out with the same rate. This assumption was made in because we have taken the composition of the gas mixture coming out of the vessel to be same as that of the mixture that was inside the vessel. It should be duly noted that the two mixtures (inside and the are that effused out) have different compositions. Therefore first we must find the composition of the gas mixture coming out of the vessel B.

$\frac{r_{N_2}}{r_{He}} = \frac{2}{3}\sqrt{\frac{4}{28}} = \frac{2}{3}\sqrt{\frac{1}{7}} = 0.252$

This means that initially the ratio of moles of N_2 to the moles of He coming out of the vessel are in the molar ratio of 0.252 and not $\frac{2}{3}$.

Let moles of the He coming out to be x

$\therefore$ Moles of N_2 coming out is $0.252\,x$

$\therefore \frac{n_{N_2}}{n_{total}} = \frac{0.252x}{1.252x} = 0.2$

$\frac{n_{He}}{n_{total}} = 0.8 \Rightarrow M_{mix} = 0.2 \times 28 + 0.8 \times 4 = 8.8$

$\therefore \frac{r_{O_2}}{r_{mix}} = \sqrt{\frac{8.8}{32}} = \mathbf{0.52}$

SOLVED PROBLEMS

Subjective Question

Prob. 1. A sample of Ne is originally 10% by mole in Ne^{20} isotope and remaining are Ne^{22} isotope. In how many steps of effusion, 25% enrichment of Ne^{20} can be achieved.

Sol. n = 24 steps

Initial ratio $\frac{Ne^{20}}{Ne^{22}} = \frac{10}{90} = \frac{1}{9}$

Desired ratio $\frac{Ne^{20}}{Ne^{22}} = \frac{25}{75} = \frac{1}{3}$

Then $\left(\frac{n_A}{n_B}\right)_{desired} = \left(\frac{n_A}{n_B}\right)_{initial}\left(\frac{M_B}{M_A}\right)^{\frac{n-1}{2}}$

$\Rightarrow \frac{1}{3} = \frac{1}{9}\left(\frac{22}{20}\right)^{\frac{n-1}{2}}$

$\Rightarrow n = 24$

Objective Questions

Prob. 1. The rate of diffusion of two gases x and y is in the ratio of 1 : 5 and that of y and z in the ratio of 1 : 6. The ratio of the rate of diffusion of z with respect to x is

(A) 5 : 6 (B) 1 : 30

(C) 6 : 5 (D) 30 : 1

Sol. $\frac{r_x}{r_y} = \frac{1}{5}$ and $\frac{r_y}{r_z} = \frac{1}{6}$

$\frac{r_z}{r_x} = \frac{30}{1}$

Hence, **(D)** is the correct answer.

Prob. 2. The circulation of blood in human body supplies O_2 and releases CO_2. the concentration of O_2 and CO_2 is variable but on an average, 100 ml blood contains 0.02 g of O_2 and 0.08 g of CO_2. The volume of O_2 and CO_2 at 1 atm and at body temperature 37°C, assuming 10 lt blood in human body, is

(A) 2 lt, 4 lt (B) 1.5 lt, 4.5 lt

(C) 1.59 lt, 4.62 lt (D) 3.82 lt, 4.62 lt

Sol. $\because$ 100 ml blood has 0.02 g P_2 and 0.08 g CO_2

$\therefore$ 10,000 ml blood has 2 g O_2 and 8 g CO_2

Using PV = nRT

For O_2, $1 \times V_{O_2} = \frac{2}{32} \times 0.0821 \times 310 \Rightarrow V_{O_2} = 1.59$ litre

For CO_2, $1 \times V_{CO_2} = \frac{8}{44} \times 0.0821 \times 310 \Rightarrow V_{CO_2} = 4.62$ litre

Hence, **(C)** is the correct answer.

Prob. 3. If for two gases of molecular weights M_A and M_B at temperature T_A and T_B, $T_AM_B = T_BM_A$, then which property has the same magnitude for both the gases.

(A) Density
(B) Pressure
(C) KE per mol
(D) V_{rms}

Sol. (i) density of a gas $(\rho) = \dfrac{PM}{RT}$

Since $\dfrac{M_B}{T_B} = \dfrac{M_A}{T_A}$, $\therefore$ at the same pressure $\rho_A = \rho_B$.

But if pressure is different then $\rho_A \neq \rho_B$.

(ii) Pressure of the gases would be equal if their densities are equal other wise not.

KE per mol $= \dfrac{3}{2}RT$

$\therefore$ It will be different for the two gases.

(iii) $V_{rms} = \sqrt{\dfrac{3RT}{M}}$, since $\dfrac{T_A}{M_A} = \dfrac{T_B}{M_B}$; V_{rms} of A = V_{rms} of B

Hence, **(D)** is the correct answer.

Prob. 4. Ammonia gas at 76 cm Hg pressure was connected to a manometer. After sparking in the flask, ammonia is partially dissociated as follows.

$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$

The level in the mercury column of the manometer was found to show the difference of 18 cm. What is the partial pressure of $H_2(g)$ at equilibrium ?

(A) 18 cm Hg
(B) 9 cm Hg
(C) 27 cm Hg
(D) 24 cm Hg

Sol.

	$2NH_3(g)$	$\rightleftharpoons N_2(g)$	$+ 3H_2(g)$
t = 0	76	0	0
at equilibrium	76 – 2x	x	3x

Total pressure = 76 – 2x + x + 3x = 76 + 2x
Increase in pressure = 2x = 18
x = 9
Partial pressure of H_2 = 3x = **27 cm**
Hence, **(C)** is the correct answer.

Prob. 5. Gaseous benzene reacts with hydrogen gas in presence of a nickel catalyst to form gaseous cyclohexane according to the reaction:

$C_6H_{6(g)} + 3H_{2(g)} \longrightarrow C_6H_{12(g)}$

A mixture of C_6H_6 and excess H_2 has a pressure of 60 mm of Hg in an unknown volume. After the gas has been passed over a nickel catalyst and all the benzene converted to cyclohexane, the pressure of the gas was 30 mm of Hg in the same volume and temperature. The fraction of C_6H_6 (by volume) present in the original volume is:

(A) $\frac{1}{3}$
(B) $\frac{1}{4}$
(C) $\frac{1}{5}$
(D) $\frac{1}{6}$

Sol. Suppose initially, pressure of $C_6H_6(g) = P_1$ mm and that of $H_2(g) = P_2$ mm.

$\therefore P_1 + P_2 = 60\,mm$ ... (1)

After the reaction

Pressure of $C_6H_6(g) = 0$

Pressure of $H_2(g) = P_2 - 3P_1$

Pressure of $C_6H_{12}(g) = P_1$

Total pressure $= P_2 - 3P_1 + P_1 = 30\,mm$ ... (2)

Or, $P_2 - 2P_1 = 30\,mm$

Solving (1) & (2) we get

$P_1 = 10\,mm$

$P_2 = 50\,mm$

Fraction of $C_6H_6(g)$ by volume = Fraction of moles = Fraction of pressures

$= \frac{10}{60} = \frac{1}{6}$ Hence, **(D)** is the correct answer.

Prob. 6. CH_4 diffuses two times faster than a gas X. The number of molecules present in 32 g of gas X is : (N is Avogadro number)

(A) N (B) $\frac{N}{2}$

(C) $\frac{N}{4}$ (D) $\frac{N}{16}$

Sol. $\frac{r_{CH_4}}{r_{gas}} = \sqrt{\frac{m_{gas}}{m_{CH_4}}}$

$m_{gas} = 64$

Number of molecule $= \frac{w}{\text{molar mass}} \times N$

$= \frac{32}{64} \times N = \frac{N}{2}$

Hence, **(B)** is the correct answer.

Prob. 7. 50 mililitres of CO is mixed with 20 mL of oxygen and sparked. After reaction, the mixture is treated with an aqueous KOH solution. Choose the correct option.

(A) The volume of the CO that reacts = 40 mL

(B) The volume of the CO_2 formed = 40 mL

(C) The volume of the CO that remains after treatment with KOH = 10 mL

(D) The volume of the CO that remains after treatment with KOH = 20 mL

Sol. **(A, B, C)**

Prob. 8. The Vander Waal's parameters of two gases are given as

Gas	$a\left(L^2\ atm\ mol^{-2}\right)$	$b\left(L\ mol^{-1}\right)$
A	7	8.33×10^{-3}
B	21	1.66×10^{-2}

Then, which of the following statement are correct?

(A) Critical volume of A > critical volume of B

(B) Critical pressure of A < critical pressure of B

(C) Critical volume of A < critical volume of B

(D) Boyle temperature of A < Boyle temperature of B

Sol. **(C, D)**

$$P_c = \frac{a}{27b^2},\ V_c = 3b,\ T_b = \frac{a}{Rb}$$

Reasoning Type

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Rate of diffusion will increase if an inert gas is added to the gaseous mixture and allowed to diffuse through a pin hole.
and
STATEMENT-2: Addition of extra gas will increase the inside pressure
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D), Assertion**→ False
Reason → True but not the reason.

2. **STATEMENT-1:** Pressure exerted by a real gas is lesser than that exerted by an ideal gas under similar conditions.
and
STATEMENT-2: At high pressure repulsive forces dominate in real gases.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

3. **STATEMENT-1:** Considering the Vander Waal's equation of state $\left(P+\frac{a}{V^2}\right)(V-b)=RT$ for ammonia (NH_3) and nitrogen (N_2). The value of a for NH_3 is larger than that for N_2.
and
STATEMENT-2: Ammonia has a lower molecular weight than nitrogen.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

4. **STATEMENT-1:** Pressure exerted by a real gas is lesser than that exerted by an ideal gas under similar conditions.
and
STATEMENT-2: At high pressure repulsive forces dominate in real gases.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(D)**

5. STATEMENT-1: Helium shows only +ve deviation from ideal behaviour at room temperature.
and
STATEMENT-2: Helium is inert gas.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

MATRIX – MATCH TYPE QUESTION

1. Match the following column – I and Column – II

Column – I		Column – II	
(A)	Density of a gas	(p)	directly proportional to pressure at constant T and V.
(B)	Kinetic energy of a gas	(q)	Inversely proportional to square root of molar mass of the gas
(C)	Rate of diffusion of a gas	(r)	directly proportional to temperature (K) at constant pressure
(D)	Most probable speed of a gas	(s)	directly proportional to square root of temperature
		(t)	Inversely proportional to square root of density of gas at constant temperature and pressure

Sol. **(A → p), (B → p, r), (C → p, q, s, t), (D → q, s, t)**

2. Match the following column – I and Column – II

Column – I		Column – II	
(A)	Equal rate of diffusions	(p)	Negative deviation (Cl_2 & NH_3)
(B)	Equal KE at 298 K	(q)	CO_2 & N_2O
(C)	$Z < 1$	(r)	Positive deviation (H_2 & He)
(D)	$Z > 1$	(s)	SO_2, CO_2

Sol. **A → q; B → s; C → p; D → r**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Boyle's Law	(i)	$P_{total} = P_1^l + P_2^l + P_3^l + \ldots$	(P)	Constant n, P
(II)	Charles Law	(ii)	$P \propto \frac{1}{V}$	(Q)	Constant T, V, n
(III)	Dalton's Law	(iii)	$V \propto T$	(R)	Constant P, T
(IV)	Avogadro's Law	(iv)	$V \propto n$	(S)	Constant n, T

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (II) (iii) (R) [C] (III) (i) (Q) [D] (I) (ii) (S)

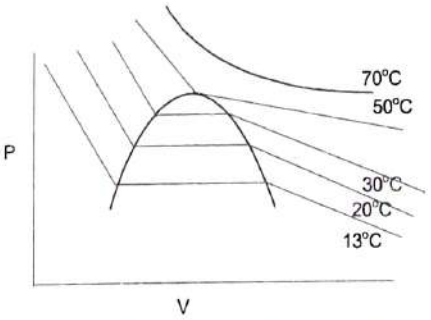
4. Which of the following combination is incorrect ?
[A] (I) (ii) (S) [B] (III) (ii) (Q) [C] (IV) (iv) (R) [D] (II) (iii) (P)

5. Which of the following combination is correct ?
[A] (IV) (i) (Q) [B] (IV) (iv) (R) [C] (II) (i) (Q) [D] (II) (i) (P)

Sol.
3. **C**
4. **B**
5. **B**

COMPREHENSION

Thomas Andrews plotted the P.V. curve for CO_2. He plotted the isotherms of CO_2 at various temperatures. Later on it was found that real gases behave in the same manner as CO_2.
Andrews noticed that at high temp isotherms look like that of an ideal gas and cannot be liquified even at very high pressure. As temp is lowered, shape of curve changes from ideal behaviour.



1. The critical temp. of gases A, B, C and D are 126 K, 154.3 K, 304.15 K & 405 K respectively which gas shown greater deviation from ideal behaviour with increase in pressure:
(A) A (B) B
(C) C (D) D

Sol. **(D)**

2. The compressibility factor of a van der Waal's gas at its critical point is:
(A) 1 (B) 0.375
(C) 2.67 (D) 0.48

Sol. **(B)**

SINGLE INTEGER ANSWER TYPE

1. The C_{rms} of He is 5.3×10^2 ms^{-1} and the distance per collision is 6.18×10^{-8} metre collision^{-1}. The collision frequency is $x \times 10^9$ collision sec^{-1}. What will be the value of x?

Sol. (7)

$$\text{Collision frequency} = \frac{\text{Most probable speed}}{\text{mean free path}}$$

Mean free path = distance per collision

$$C_{mp} = \frac{C_{rms}}{1.224}$$

$$= \frac{5.3\times10^2}{1.224}$$

$= 4.33 \times 10^2$ m/s

$$\text{Collision frequency} = \frac{4.33\times10^2}{6.18\times10^{-8}}$$

$= 7 \times 10^9$

So x = 7

2. Solid phosphorus melts and vapourizes at high temperature. Gaseous phosphorus effuses at a rate that is 0.567 times that of Ne in the same apparatus under the same conditions. How many atoms are in a molecule of gaseous phosphorus.

Sol. **(2)**

r_1 = rate of effusion of phosphorus

r_2 = rate of effusion of Ne

$$\frac{r_1}{r_2} = 0.567 = \sqrt{\frac{M_2}{M_1}}$$

$0.567 = \sqrt{\frac{20}{x\times31}}$ x = no. of P atoms

squaring both side

$$0.32 = \frac{20}{31x}$$

$$x = \frac{20}{31\times0.32} = 2$$

Hence, no. of atoms present in phosphorus molecule = 2

3. Vander Waal's constant 'b' of a gas is 125.57 centilitre/mol. How near can be the centres of the two molecules approach each other?

Sol. 1.6×10^{-7} cm

$b = N_A\times4\times4/3\pi r^3$

$125.57\times10^{-2}\times1000 = 6.023\times10^{23}\times4\times4/3\times3.14r^3$

$$r^3 = \frac{125.57\times10\times3}{6.023\times10^{23}\times4\times4\times3.14}$$

$= 125\times10^{-24}$

$r = 5\times10^{-8}$ cm

So collision diameter (nearest possible distance) = 1.0×10^{-7} cm

4. If an ideal gas at 100 K is heated to 109 K in a rigid container, the pressure increases by X°%. What is the value of X ?

Sol. (9)

$$\frac{P_f}{P_i} = \frac{109}{100}$$

$$\Delta P = P_f - P_i = \frac{9}{100} P_i$$

$$\%\text{ increase in pressure} = \frac{\Delta P}{P_i} \times 100 = \frac{9P_i}{100P_i} \times 100$$

$$= 9$$

5. Calculate the ratio of the rate of diffusion of gas A to B if their pressures are in the ratio 2: 1 and their molecules if travel with same rms speed then ratio of their kinetic energies is 1: 9

Sol. (6)

$$\frac{kE_A}{kE_B} = \frac{1}{9} = \frac{m_A}{m_B} (\text{as } V_A = V_B)$$

$$\frac{P_A}{P_B} = \frac{2}{1} \text{ so } \frac{r_A}{r_B} = \frac{2}{1}\sqrt{\frac{9}{1}} = 6:1$$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. An evacuated glass vessel weighs 50.0 gm when empty, 148.0 gm when filled with a liquid of density 0.98 g/ml and 50.5 g when filled with an ideal gas at 1 atm of 'T K'. The molecular weight of gas is found to be "xRT gm". Find the value of $\frac{x}{3}$. [Where R is universal gas constant.]

Sol. **1.66**

Prob. 2. Two separate bulbs contain ideal gases A and B. The density of gas A is twice that of gas B. The molecular mass of A is half that of gas B. The two gases are at the same temperature. The ratio of the pressure of A to that of gas B is x : 1. Then value of $\frac{x}{3}$ is

Sol. **1.33**

Prob. 3. 8 g of O_2 has the same number of molecules as in x g of CO. What is the value of $\frac{x}{4}$?

Sol. **1.75**

Prob. 4. Two bulbs of volume 5 litre and 10 litre containing and ideal gas at 9 atm and 6 atm pressure respectively are joined together. The final pressure (in atm) in the two bulbs is x, if the temperature remains the same. Then value of $\frac{x}{4}$

Sol. **1.75**

Prob. 5. Under identical condition of temperature and pressure, 1 litre of CH_4 weighted 1.2 gram while 2 litre of another gaseous hydrocarbon, C_nH_{2n-2} weighted 8.1 gram. What is the value of $\frac{n}{3}$?

Sol. **1.33**

Prob. 6. Two grams of hydrogen diffuse from a container in 10 minutes. x grams of oxygen would diffuse through the same container in the same time under similar conditions. Then value of $\frac{x}{3}$ is

Sol. **2.666**

Prob. 7. A mixture of 20 ml of CO, CH_4 and N_2 was burnt in an excess of oxygen resulting in the reduction of 13 ml of volume. The residual gas was treated with KOH solution when there was further 14 ml reduction in volume. The volume (ml) of CH_4 in the given mixture x. Then value of $\frac{x}{3}$

Sol. **1.13**

Prob. 8. When 2 g of a gaseous substance A is introduced into an initially evacuated flask at 25°C, the pressure is found to be 1 atm. 3 g of another gas B is then added to it at the same temperature and pressure, the final pressure is found to be 1.5 atm. x times the molecular weights of B is heavier than A, assuming ideal gas behaviour, then value of $\frac{x}{2}$ is

Sol. **1.5**

Prob. 9. If a gas contains only three molecules that move with velocity of 100, 200, 500 m/s. What is the rms velocity of that gas in m/s ?

Sol. **316.228 m/sec**

Prob.10. If the temperature of 1 mole of gas is increased by 50°C, calculate the change in kinetic energy of system.

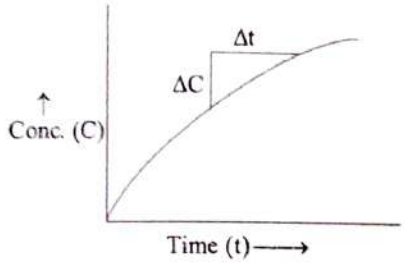
Sol. **623.253**

CHEMICAL KINETICS 4

RATE OF REACTION

(i) Instantaneous rate of reaction $= \frac{dc}{dt}$

(ii) Average rate of reaction $= \frac{\Delta C}{\Delta t}$



RATE LAW

$aA + bB \longrightarrow cC + dD$

According to differential rate law,

$$-\frac{1}{a}\frac{d[A]}{dt} = -\frac{1}{b}\frac{d[B]}{dt} = +\frac{1}{c}\frac{d[C]}{dt} = +\frac{1}{d}\frac{d[D]}{dt} = K[A]^m[B]^n$$

where $\frac{-d[A]}{dt}$ = rate of disappearance of A

$+\frac{d[C]}{dt}$ = rate of appearance of C

The exponents *'m'* and *'n'* are known as the order of reaction with respect to A and B respectively. *(m + n)* gives the overall order of the reaction. Order of reaction may be fractional, zero or negative. If reaction is multi-step, then slowest step is rate determining step.

Order of Reaction

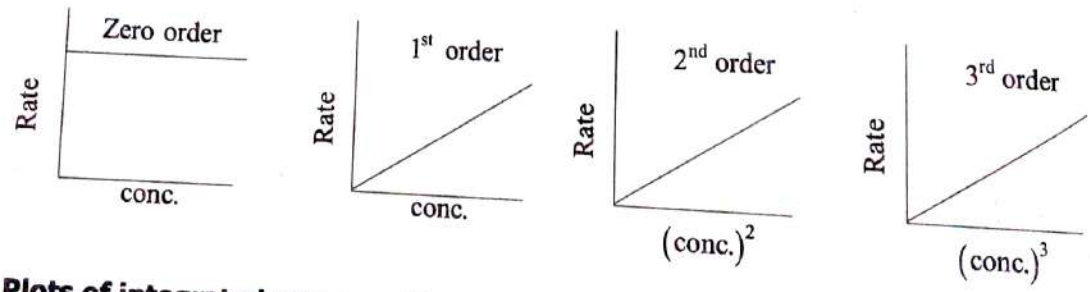
S. No	Order of reaction	Example	Rate law	Rate constant	Half-life	Unit of k
1	Zero order	$2NH_3 \rightarrow N_2+3H_2$	$r = k(NH_3)^0$	$k = \frac{x}{t}$	$t_{1/2} = \frac{a}{2k}$	$mol\ L^{-1}\ sec^{-1}$
2.	1st order	$A \rightarrow$ Product	$r = k(a-x)$	$k = \frac{2.303}{t}\log\frac{a}{a-x}$	$t_{1/2} = \frac{0.693}{k}$	sec^{-1}
		$2A \longrightarrow$ Product	$r = k(a-x)^2$	$kt = \left(\frac{1}{a-x} - \frac{1}{a}\right)$	$t_{1/2} = \frac{1}{k \times a}$	$sec^{-1}\ mol^{-1}\ L$
3.	2nd order					
		$A+B \longrightarrow$ Product	$r = k(a-x)(b-x)$	$kt = \frac{2.303}{(a-b)}\log\frac{b(a-x)}{a(b-x)}$	$t_{1/2} = \frac{2.303}{k(a-b)}$	$sec^{-1}\ mol^{-1}\ L$
4.	nth order	$nA \longrightarrow$ Product	$r = k(a-x)^n$	$kt = \frac{1}{(n-1)}\left(\frac{1}{(a-x)^{n-1}} - \frac{1}{a^{n-1}}\right)$	$kt_{1/2} = \frac{(1-2^{n-1})}{\frac{1}{2}(n-1)a^{n-1}}$	$mol^{1-n}\ L^{n-1}\ sec^{-1}$

Determination of order of reaction by $t_{1/2}$ methods:

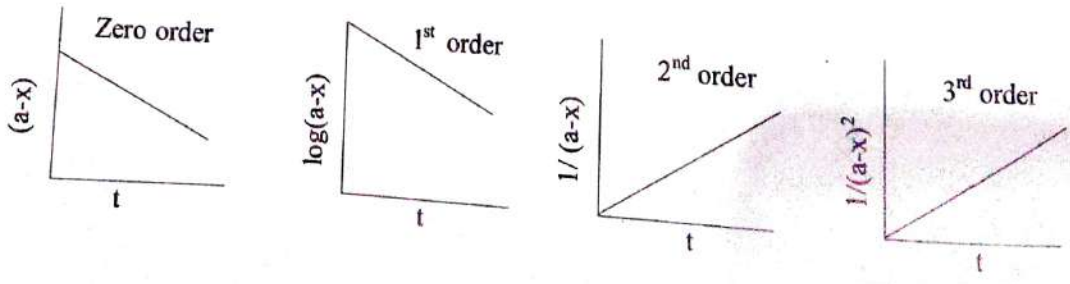
$$n = 1 + \frac{\log(t_{1/2})_1 - \log(t_{1/2})_2}{\log(A_0)_2 - \log(A_0)_1}\ ;\ \text{where } t_{1/2} \propto \frac{1}{[A_0]^{n-1}}$$

Determination of order of reaction by graphical methods:

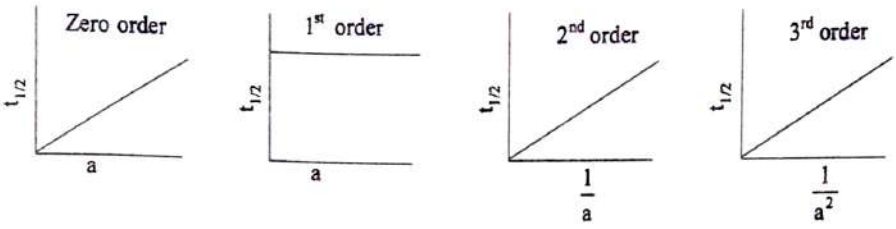
(a) **Plots of rate v/s concentration**



(b) **Plots of integrated rate equation**



(c) **Plots of $t_{1/2}$ v/s initial concentration**



General expression for the time taken for n^{th} fraction of a reaction of 1st order to complete

$$t_{1/n} = \frac{2.303}{k}\log\frac{n}{n-1}$$

General expression for time taken for the reactant to reduce to n^{th} fraction

$$t = \frac{2.303}{k}\log n$$

Amount of substance left after n half lives = $\frac{A_0}{2^n}$

Decomposition of Hydrogen Peroxide

$$H_2O_2 \longrightarrow H_2O + \tfrac{1}{2} O_2$$

$$k = \frac{2.303}{t}\log\frac{V_0}{V_t}$$

where V_0 = Volume of $KMnO_4$ solution used before the commencement of reaction, i.e. at zero time

V_t = Volume of $KMnO_4$ solution used at any instant of time

Pseudo first order reaction

(i) **Hydrolysis of ethyl acetate:**

$$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$$

$$k = \frac{2.303}{t}\log\frac{V_\infty - V_0}{V_\infty - V_t}$$

where V_0 = volume of NaOH solution used in the beginning (t = 0)

V_t = volume of NaOH solution used at any instant of time t

V_∞ = volume of NaOH solution used after the reaction at infinite time ($t = \infty$)

(ii) **Inversion of cane sugar (sucrose):**

$$C_{12}H_{22}O_{11} + H_2O \xrightarrow{H^+} \underset{\text{Glucose}\ +52.5^\circ}{C_6H_{12}O_6} + \underset{\text{Fructose}\ -92^\circ}{C_6H_{12}O_6}$$

$$k = \frac{2.303}{t}\log\frac{r_0 - r_\infty}{r_t - r_\infty}$$

where r_0 = reading of the polarimeter at zero time

r_t = reading of the polarimeter at any time

r_∞ = reading of the polarimeter at infinite time

Effect of temperature on rate of reaction (Arrhenius equation)

$$k = Ae^{-E_a/RT}$$

$$\log k = \frac{-E_a}{2.303\,R}\frac{1}{T} + \log A$$

$$\text{or } \log\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303\,R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

If $T_2 > T_1 \Rightarrow k_2 > k_1$

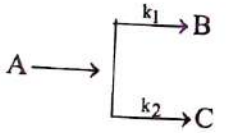
where E_a – Activation energy

A – Frequency factor

If $T \to \infty \Rightarrow k = A$

$$e^{-E_a/RT} = \frac{\text{No. of effective collisions per unit time}}{\text{Total no. of collisions per unit time}}$$

Parallel reactions:



$$\frac{dx}{dt} = k_1(a-x) + k_2(a-x) = (k_1 + k_2)(a-x)$$

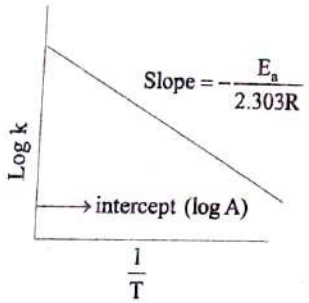
$$k_1 + k_2 = \frac{1}{t}\ln\left(\frac{a}{a-x}\right)$$

Rate of formation (B), $\frac{d(B)}{dt} = k_1(a-x)$

Rate of formation (C), $\frac{d(C)}{dt} = k_2(a-x)$

where 'a' is initial concentration of A and x is the amount decomposed after time 't' then

Plot log k v/s 1/T



ILLUSTRATIVE PROBLEMS

1. **A substance reacts according to 1st order kinetics. The rate constant for the reaction is 10^{-2} sec^{-1}. If its initial concentration is 1 M**

(a) What is initial rate?

(b) What is the rate after 1 minute?

Sol. (a) Initial rate = $k(A)^1 = 10^{-2}(1) = 10^{-2}$ mol $litre^{-1}$ sec^{-1}

(b) $k = \frac{2.303}{t}\log\frac{a}{a-x}$

$10^{-2} = \frac{2.303}{1\times 60}\log\frac{1}{1-x}$

$(1 - x) = 0.549$

Rate after 1 minute = $k[1 - x]^1$

$= 10^{-2}(0.549)$

$= 5.49\times10^{-3}$ mol $litre^{-1}$ sec^{-1}

2. **The reaction given below is observed to be first order with rate constant 7.48×10^{-3} sec^{-1}. Calculate the time required for the total pressure in a system containing A at an initial pressure of 0.1 atm to rise to 0.145 atm and also find the total pressure after 100 sec.**

$2A(g) \rightarrow \quad 4B(g) + C(g)$

Sol.

	$2A(g)$	$\longrightarrow$	$4B(g)$	+	$C(g)$
Initial	P_o		0		0
At time t	$P_o - P'$		$2P'$		$P'/2$

$P_{total} = P_o - P' + 2P' + P'/2 = P_o + \frac{3P'}{2}$

$P' = \frac{2}{3}(0.145 - 0.1) = \quad 0.03$ atm

$k = \frac{2.303}{t}\log\frac{P_o}{P_o - P'}$

$t = \frac{2.303}{7.48\times10^{-3}}\log\left(\frac{0.1}{0.07}\right) = \mathbf{47.7\ sec}$

Also, $k = \frac{2.303}{t}\log\left(\frac{0.1}{P_0 - P'}\right)$

$7.48 \times 10^{-3} = \frac{2.303}{100}\log\left(\frac{0.1}{0.1 - P'}\right)$

$0.1 - P' = 0.047$

$P' = 0.053$

$P_{total} = \quad 0.1 + \frac{3}{2}(0.053) \approx \mathbf{0.180\ atm.}$

3. **The time required for 10% completion of first order reaction at 298 K is equal to that required for its 25% completion at 308 K. If the pre-exponential factor for the reaction is $3.56 \times 10^9\ s^{-1}$, calculate its rate constant at 318 K and also the energy of activation.**

Sol. For first order reactions,

$$t = \frac{2.303}{k}\log\frac{N_0}{N_t}$$

$$\text{At 298 K; } t = \frac{2.303}{k_{298}}\log\frac{100}{90}$$

$$\text{At 308 k; } t = \frac{2.303}{k_{308}}\log\frac{100}{75}$$

Since time is same

$$\frac{2.303}{k_{298}}\log\frac{100}{90} = \frac{2.303}{k_{308}}\log\frac{100}{75}$$

$$\text{or } \frac{0.0458}{k_{298}} = \frac{0.1249}{k_{308}}$$

$$\text{or } \frac{k_{308}}{k_{298}} = \frac{0.1249}{0.0458} = 2.73$$

According to Arrhenius equation,

$$2.303 \log\frac{k_{308}}{k_{298}} = \frac{E_a}{8.314}\left[\frac{1}{298} - \frac{1}{308}\right]$$

$$\text{or } 2.303 \log 2.73 = \frac{E_a}{8.314}\left[\frac{10}{298\times 308}\right]$$

$E_a = 76.65$ kJ/mol

and $K_{318} = \mathbf{9.036 \times 10^{-4}\ s^{-1}}$

4. **The rate of a certain reaction depends on concentration according to equation $-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]}$**

What will be the order of reaction when (i) concentration is very high (ii) very low?

Sol. Given, $-\frac{d[A]}{dt} = \frac{k_1[A]}{1+k_2[A]}$

$$\Rightarrow \frac{-d[A]}{dt} = \frac{k_1}{\frac{1}{[A]} + k_2}$$

(i) When [A] is very high $\frac{1}{[A]}$ is very small, and thus negligible

$$\therefore -\frac{d[A]}{dt} = \frac{k_1}{k_2} = \text{constant}$$

Thus, order of reaction is zero.

(ii) When [A] is very low.

$[1 + k_2 [A] = k'$

$$\therefore -\frac{d[A]}{dt} = \frac{k_1[A]}{k'} = k''[A]$$

Thus, order of reaction is one.

5. A catalyst decreases the energy of activation of the reaction: A → B, ΔH = 37.012 kcals, by 2%. At 27°C 4×10^{-8}% of the product molecules i.e. B is able to cross over the potential energy barrier in absence of catalyst. How many times the catalyst will increase the rate of forward reaction at 27°C?

Sol. $e^{-Ea/RT}$ = fraction of the total number of molecules having energy equal to or greater than Ea

At 27° for backward reaction we have as per question:

$$e^{-Ea/2 \times 300} = \frac{4 \times 10^{-8}}{100} = 4 \times 10^{-10} \quad (R = 1.987 \approx 2 \text{ cals } K^{-1} \text{ mole}^{-1}$$

$$-\frac{Ea}{600} = 2.303(\log 4 - 10)$$

$$Ea = -2.303 \times 600\ (0.6 - 10)$$

$$= 12.988 \text{ kcals}$$

$$\Delta H = Ea(FR) - Ea(BR) \Rightarrow Ea(FR) = \Delta H + Ea(BR)$$

$$= 37.012 + 12.988 = 50 \text{ kcals mole}^{-1}$$

Catalyst will decrease the Ea by 1 kcal mol^{-1} which is 2% of 50 kcals $mole^{-1}$.

$$\frac{K^{cat}}{K} = e^{\Delta Ea/RT} = e^{1/2 \times 10^{-3} \times 300} = 5.29$$

Thus a catalyst will increase the rate by 5.29 times.

6. The first order reaction: Sucrose → Glucose + Fructose takes place at 308 K in 0.5 N HCl. At time zero the initial total rotation of the mixture is 32.4°. After 10 minutes the total rotation is 28.8°. If the rotation of sucrose per mol is 85°, that of glucose is 74° and fructose is −86.04°. Calculate the half life of the reaction.

Sol. Let r_1, r_2, $-r_3$ be rotation per mole of sucrose, glucose and fructose respectively.

Sucrose →	Glucose +	Fructose	
a	0	0	initial
a − x	x	x	after time t

$$a\,r_1 = 32.4 \quad \Rightarrow \quad a = \frac{32.4}{85} = 0.38$$

$$(a - x)\,r_1 + xr_2 - xr_3 = 28.8$$

$$x\,(r_2 - r_3 - r_1) = 28.8 - 32.4 = -3.6$$

$$x(7.4 - 86.04 - 85) = -3.6$$

$$x = 0.022$$

$$k = \frac{2.303}{10} \log\left(\frac{0.38}{0.38 - 0.022}\right)$$

$$k = 5.96 \times 10^{-3} \text{ min}^{-1}$$

$$\therefore\ t_{½} = \mathbf{116.2 \text{ min}}$$

SOLVED PROBLEMS

Subjective Question

Prob. 1. The decomposition of dinitrogen pentoxide (N_2O_5) follows first order rate law. Calculate the rate constant from the following data. $\left(\log\frac{1.45}{0.88}=0.2169\right)$

$t = 800$ sec $\quad [N_2O_5] = 1.45$ mol L^{-1}
$t = 1600$ sec $\quad [N_2O_5] = 0.88$ mol L^{-1}

Sol. $$K = \frac{2.303}{t_2 - t_1}\log\frac{[A_1]}{[A_2]}$$

$$= \frac{2.303}{1600-800}\log\frac{1.45}{0.88}$$

$$= \frac{2.303}{800}\times 0.2169 = 6.24\times10^{-4}$$

Prob. 2. For the reaction, $A \rightleftharpoons B$, ΔE is -33.0 kJ mol^{-1} calculate.

(A) Equilibrium constant K_c for the reaction at 300K

(B) If E_f and E_b are in the ratio 20 : 31

Calculate E_f, E_b at 300K

Assuming pre exponential factor same for forward and backward reaction.

Sol (A) $\Delta H = E_f - E_b = -33$ kJ

$$K_f = Ae^{\frac{-E_f}{RT}}$$

$$K_b = Ae^{\frac{-E_b}{RT}}$$

$$K_c = \frac{K_f}{K_b} = e^{(E_b - E_f)/RT} = 5.572\times10^5 \text{ at } 300\text{ K}$$

(B) $\frac{E_f}{E_b} = \frac{20}{31}$ and $E_f - E_b = -33$ kJ

$$E_f - \frac{31}{20}E_f = -33$$

$$E_f = \frac{33\times20}{11} = 60\text{ kJ}$$

$$\therefore\ E_b = 93\text{ KJ}$$

Objective Questions

Prob. 1. Half life of a reaction becomes half when initial concentration of reactants are made double. The order of the reaction will be:

(A) 1 (B) 2

(C) 0 (D) 3

Sol. $t_{1/2} \propto \frac{1}{a^{n-1}}$

$t_{1/2} \propto \frac{1}{a}$

where n = order of reaction for second order reaction.

Hence, **(B)** is the correct answer.

Prob. 2. The reaction $A \xrightarrow{k}$ Product, is zero order while the reaction $B \xrightarrow{k}$ Product, is first order reaction. For what initial concentration of A are the half lives of the two reactions equal? (Rate constant value are same for both two reaction).

(A) $(\log_e 4)$ M (B) 2 M

(C) log 2 M (D) ln 2 M

Sol. For zero order reaction, x = kt

$\therefore \frac{a}{2} = k \times t_{1/2}$, i.e. $t_{1/2} = \frac{a}{2k}$ ... (i)

For first order reaction, $t_{1/2} = \frac{\log_e 2}{k}$ ... (ii)

From (i) and (ii), $\frac{a}{2k} = \frac{\log_e 2}{k}$

$a = \log_e 4$ M

Hence, **(A)** is the correct answer.

Prob. 3. The inversion of cane sugar proceeds with constant half-life of 50 minutes at pH = 5 for any concentration of sugar. However, if pH = 6, the half-life changes to 500 minutes. The rate law expression for the sugar inversion can be written as

(A) $r = k[sugar]^2 [H^+]^6$ (B) $r = k[sugar]^1 [H^+]^0$

(C) $r = k[sugar]^1 [H^+]^1$ (D) $r = k[sugar]^0 [H^+]^1$

Sol. At pH = 5, the half–life is 50 minutes for all concentrations of sugar that is $t_{1/2} \propto$ [sugar].

The reaction is first order with respect to sugar.

Let, rate = $r = k[H^+]^x$ [sugar]

k' [sugar] $k' = k[H^+]^x$

$$\frac{0.693}{t_{1/2}} = k\left[H^+\right]^x \qquad \left[H^+\right]^x\left[t_{1/2}\right] = \frac{0.693}{k}$$

$$50\times\left[10^{-5}\right]^x = 500\times\left(10^{-6}\right)^x \qquad 10^x = 10^1$$

$x = 1$

Hence, **(C)** is the correct answer.

Prob. 4. Two substances A and B are present such that $[A_0] = 4[B_0]$ and half-life of A is 5 minutes and that of B is 15 minutes. If they start decaying at the same time following first order kinetics how much time later will the concentration of both of them would be same?

(A) 15 minutes (B) 10 minutes

(C) 5 minutes (D) 12 minutes

Sol. Amount of A left in n_1 half-lives $= \left(\frac{1}{2}\right)^{n_1}[A_0]$

Amount of B left n_2 half-lives $= \left(\frac{1}{2}\right)^{n_2}[B_0]$

At the end, $\frac{[A_0]}{2^{n_1}} = \frac{[B_0]}{2^{n_2}}$

$\Rightarrow \frac{4}{2^{n_1}} = \frac{1}{2^{n_2}}\{[A_0] = 4[B_0]\}$

$\therefore \frac{2^{n_1}}{2^{n_2}} = 4 \Rightarrow 2^{n_1-n_2} = (2)^2$

$\therefore n_1 - n_2 = 2$

$\therefore n_2 = (n_1 - 2)$ ... (1)

Also, $t = n_1 \times t_{1/2(A)}$ $t = n_2 \times t_{1/2(B)}$

(Let, concentration of both become equal after time t)

$\therefore \frac{n_1 \times t_{1/2(A)}}{n_2 \times t_{1/2(B)}} = 1 \Rightarrow \frac{n_1 \times 5}{n_2 \times 15} = 1 \Rightarrow \frac{n_1}{n_2} = 3$ ... (2)

From Eqs. (1) and (2), we get

$n_1 = 3, n_2 = 1$

$t = 3 \times 5 = 15$ minutes

Hence, **(A)** is the correct answer.

Prob. 5. van't Hoff equation is

(A) $\frac{d(\ln K)}{dt} = \left(-\Delta E / RT^2\right)$ (B) $(d/dT)\ln K = +\left(E/RT^2\right)$

(C) $(d/dT)\ln K = -(\Delta E/RT)$ (D) $K = Ae^{-E_a/RT}$

Sol. **(B)** and **(D)** are the correct answers.

Prob. 6. Which of the following are examples of unimolecular reactions?

(A) $O_3 \longrightarrow O_2 + O$

(B)

(C) $NO + O_3 \longrightarrow NO_2 + O_2$

(D) $O + NO + N_2 \longrightarrow NO_2 + N_2$

Sol. **(A, B)**

Prob. 7. If a reaction, $A + B \rightarrow C$, is exothermic to the extent of 30 kJ/mol and the forward reaction has an activation energy 70 kJ/mol, the activation energy for the reverse reaction is

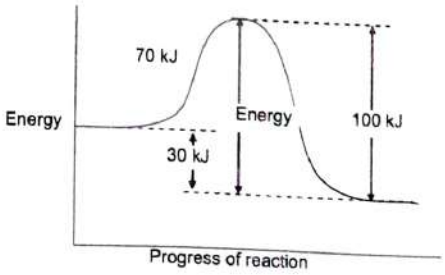
(A) 30 kJ/mol

(B) 40k J/mol

(C) 70 kJ/mol

(D) 100 kJ/mol

Sol.



Activation energy for backward reaction = 100 kJ

Hence, **(D)** is the correct answer.

COMPREHENSION

Read the following passage and answer the questions:

When the temperature is increased, heat is supplied which increases the kinetic energy of the reacting molecules. This shall increase the number of collisions and ultimately the rate of reaction shall be enhanced. Arrhenius suggested a equation which describes K as a function of temperature, i.e.

$k = Ae^{-E_a/RT}$

where

k = rate constant

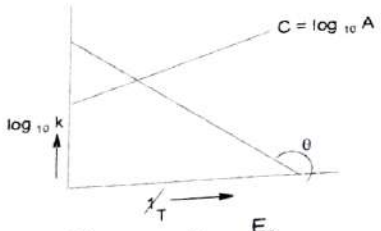
A = a constant (frequency factor)

E_a = energy of activation

$$\log_{10} k = \log_{10} A - \frac{E_a}{2.303R}\left[\frac{1}{T}\right]$$

$Y = C + MX$

It is the equation of straight line with negative slope. On plotting $\log_{10} k$ against $\left[\frac{1}{T}\right]$ we get a straight line as shown below:



$$\text{Slope} = \tan\theta = \frac{E_a}{2.303R}$$

The slope gives activation energy and intercept gives frequency factor.

Also $\log\frac{k_2}{k_1} = \frac{E_a}{2.303}\left[\frac{T_2 - T_1}{T_1T_2}\right]$

1. The rate constant of a reaction is increased by 5% when its temperature is raised from 27°C to 28°C. The activation energy of the reaction in kJ/mol is
   (A) 36.6 (B) 16.6
   (C) 46.6 (D) 26.6

Ans. **(A)**

2. The activation energy of a reaction is 9 kcal/mol. The increase in rate constant when its temperature is raised from 295 to 300 K is approximately
   (A) 10% (B) 50%
   (C) 100 % (D) 255%

Ans. **(D)**

3. A first order reaction is 50% complete in 30 minutes at 27°C and in 10 minutes at 47°C. The energy of activation of the reaction is
   (A) 43.84 kJ/mol (B) 34.84 kJ/mol
   (C) 84.00 kJ/mol (D) 30.00 kJ/mol

Ans. **(A)**

4. The rate of chemical reaction doubles for every 10°C rise in temperature. If the temperature is increased by 60°C, the rate of reaction increases by about
   (A) 20 times (B) 32 times
   (C) 164 times (D) 128 times

Ans. **(C)**

5. The rate of certain reaction increases by 2.5 times when the temperature is raised from 300 K to 310 K. If k is the rate constant at 300 K then rate constant at 310 K will be equal to
   (A) k (B) 2 k
   (C) 2.5 k (D) 13 k

Ans. **(C)**

MATRIX-MATCH TYPE QUESTION

1. Match the following:

Column I:		Column II:	
(A)	$CH_3COOC_2H_5 + H_2O \xrightarrow{H^+} CH_3COOH + C_2H_5OH$	(p)	Saponification
(B)	Radioactive decay	(q)	Unit of rate constant K = sec^{-1}
(C)	$CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$	(r)	1^{st} order
(D)	Decomposition of H_2O_2	(s)	Pseudo 1^{st} order

Sol. **A → s, q; B → r, q; C → p; D → r, q**

2. Match the following column.

Column – I		Column – II	
(A)	Reaction of zero order	(p)	Rate constant has the units of L $mol^{-1}S^{-1}$
(B)	Reaction of 1^{st} order	(q)	Half – life period is directly proportional to initial concentration
(C)	Reaction of 2^{nd} order	(r)	Example is reaction of acetic anhydride with excess of ethanol
(D)	Pseudo unimolecular reactions	(s)	Percent dissociation of the reactant at time t = $\left(1-e^{-kt}\right)\times 100$
		(t)	Radioactive decay is an example

Sol. **(A) → q, (B) → (r, t) (C) → p, (D) → r**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1 Linear Plot		Column – 2 Slope		Column – 3 Order	
(I)	$\ln\left(\frac{dx}{dt}\right)$ vs ln[A]	(i)	–k	(P)	One
(II)	$[A]_t$ vs time	(ii)	k	(Q)	Zero
(III)	$\frac{1}{[A]_t}$ vs time	(iii)	O	(R)	n
(IV)	$\log\frac{1}{[A]_t}$ vs time	(iv)	n	(S)	Two

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (I) (ii) (S) [C] (I) (iv) (R) [D] (I) (ii) (S)

4. Which of the following combination is incorrect ?
[A] (I) (ii) (R) [B] (III) (ii) (S) [C] (IV) (i) (P) [D] (I) (iv) (R)

5. Which of the following combination is correct ?
[A] (III) (iii) (P) [B] (IV) (i) (P) [C] (III) (iv) (Q) [D] (III) (i) (R)

Sol.
3. **C**
4. **A**
5. **B**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: R.M.S velocity of a gas sample can be expressed as $C_{rms} = \sqrt{\frac{3PV}{M}}$

and

STATEMENT-2: The R.M.S velocity of a gas sample decreases if its volume is decreased.

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1

(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT – 1 is True, STATEMENT-2 is False

(D) STATEMENT – 1 is False, STATEMENT-2 is True

Ans. **(C)**

2. STATEMENT-1: $K \propto e^{-E_a/RT}$, the Arrhenius equation represents the dependence of rate constant with temperature.

and

STATEMENT-2: Plot of log K against $\frac{1}{T}$ is linear and the activation energy can be calculated with this plot.

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1

(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT – 1 is True, STATEMENT-2 is False

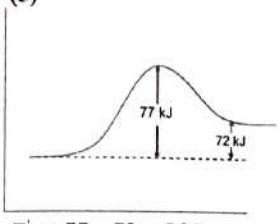
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Ans. **(A)**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. There is formation of H_2O_2 in the upper atmosphere.
$H_2O + O \longrightarrow 2\ OH \longrightarrow H_2O_2$
$\Delta H = 72\ kJ\ mol^{-1}$, $E_a = 77\ kJ\ mol^{-1}$
What will be the $E_a^{'}$ for bimolecular recombination of two OH radicals to form H_2O and O in kJ.

Sol. **(5)**



$E_a^{'} = 77 - 72 = 5\ kJ$

2. If in the fermentation of sugar in an enzymatic solution that is 0.12 m the concentration of the sugar is reduced to 0.06 m in 10 h and to 0.03 m in 20 h. What is the order of reaction.

Sol. **(1)**
0.12 m $\longrightarrow$ 0.06 m, 50%, Change $t_{1/2} = 10$ h.
0.12 m $\longrightarrow$ 0.03 m, 75%, change $t_{3/4} = 20$ h
Since $t_{3/4} = 2 \times t_{1/2}$. hence order is 1.

NUMERICAL VALUE ANSWER TYPE

Prob. 1. A reaction has rate constant 1.15×10^{-2} sec^{-1}. How long will 5 gm of this reactant take to reduce to 3g?
Sol. **4.4**

Prob. 2. The rate of fist order reaction is 0.04 mol $L^{-1}s^{-1}$ at 10 minutes and 0.03 mol $L^{-1}s^{-1}$ at 20 after initial. The half life of reaction in minutes is $x \times 10$ min. Calculate x
Sol. **2.51**

Prob. 3. A catalyst lowers the activation energy for a certain reaction from 75 to 20 KJ mol^{-1}, find $K_{catalysed}/K_{uncatalysed}$.
Sol. **6.3**

Prob. 4. In the decomposition of H_2O_2 at 300K, the activation energy is found to be 18 Kcal/mol, while it decreased to 6 Kcal/mol when carried out in presence of catalyst. The catalysed reaction is $x \times 10^8$ times faster than uncatalysed one. Calculate x
Sol. **4.83**

Prob. 5. Consider the following first order completing reactions $X \xrightarrow{k_1} A + B$ and $Y \xrightarrow{k_2} C + D$. If 50% of reaction of X was completed when 96% of the reaction of Y was completed. Calculate the ratio of their rate constants (k_2/k_1).
Sol. **4.65**

Prob. 6. For a reaction rate = $K[A]^2[B]$ by what factor the initial rate of reaction will increase if initial concentration of A is taken 1.5 times and B is tripled ?
Sol. **6.75**

Prob. 7. A 0.50 g sample of rock was found to have 2.5×10^{-6} mol of $^{40}_{19}K(t_{1/2} = 1.3 \times 10^9 \text{ yr})$ and 7.5×10^{-6} mol of $^{40}_{20}Ca$. The age of rock is $x \times 10^9$ year. Find x
Sol. **2.6**

Prob. 8. Rate constant of reaction is 2.303 min^{-1}. The initial concentration of reaction is 1 mol/litre then rate of reaction after 1 minute is $x \times 10^{-1}$. Find x
Sol. **2.3**

Prob. 9. $(CH_3)_2N_2 \longrightarrow C_2H_6 + N_2$

The above reaction is first order reaction. After 40 minutes from start, the total pressure developed is found to be 350 mm Hg in place of initial pressure 200 mm Hg of azomethane. The value of rate constant k is $P \times 10^{-4}$ sec^{-1}. Calculate P
Sol. **5.77**

Prob.10. An analysis of rock shows the relative number of Sr^{87} and Rb^{87} ($t_{1/2} = 4.7 \times 10^{10}$ year) atoms is 0.05. The age of rock is $x \times 10^9$ year. Find x
Sol. **3.28**

5

CHEMICAL EQUILIBRIUM

Chemical reaction can be classified as

(i) **Irreversible reactions:** The reactions which move only in one direction, i.e. forward direction.

(ii) **Reversible reactions:** The reactions which move in both directions under similar conditions and do not reach to completion.

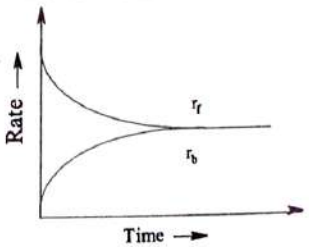
Chemical equilibrium is an important characteristic property of reversible reactions which can be defined as the state at which both forward and backward reactions occurs at the same speed.

$aA + bB \rightleftharpoons cC + dD$

$r_f = k_f[A]^a[B]^b$

$r_b = k_b[C]^c[D]^d$

Equilibrium constant, $K_c = \frac{k_f}{k_b}$



FACTORS AFFECTING EQUILIBRIUM CONSTANT

Methods of representing the equation:

For a reversible reaction,

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

Consider the backward reaction, then

$$K_c' = \frac{[N_2][H_2]^3}{[NH_3]^2}$$

$$\therefore K_c' = \frac{1}{K_c}$$

Stoichiometry representation:

When a reversible reaction can be represented by different stoichiometric equations, then the value of equilibrium constant will be numerically different.

(a) $2NO_2(g) \rightleftharpoons N_2(g) + 2O_2(g)$

$$K_c = \frac{[N_2][O_2]^2}{[NO_2]^2}$$

(b) $NO_2(g) \rightleftharpoons \frac{1}{2}N_2(g) + O_2(g)$

$$K_c' = \frac{[N_2]^{1/2}[O_2]}{[NO_2]}$$

The relation is

$K_c' = \sqrt{K_c}$

Use of partial pressure:

$aA + bB \rightleftharpoons cC + dD$

$$K_p = \frac{(P_C)^c (P_D)^d}{(P_A)^a (P_B)^b}$$

$K_p = K_c(RT)^{\Delta n}$

where $\Delta n = n_p - n_r$

Temperature:

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R}\left[\frac{T_2 - T_1}{T_1 T_2}\right]$$

DEGREE OF DISSOCIATION (Dissociated amount by 1 mole)

The degree of dissociation is dependent on density measurement.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Initial moles	a	0	0
At equilibrium	$a(1-\alpha)$	$a\alpha$	$a\alpha$

Since total mass is constant, $M \times \frac{1}{m}$

Hence, $\frac{a(1+\alpha)}{a} = \frac{M}{m}$

or $1 + \alpha = \frac{M}{m} = \frac{D}{d}$

D = V.D. of undissociated PCl_5

d = V.D of gaseous mixture at equilibrium

Reaction Quotient (Q_c)

For the reaction;

$aA + bB \rightleftharpoons cC + dD$

$$Q_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

(i) $Q_c > K_c$ – backward reaction occurs.

(ii) $Q_c < K_c$ – Forward reaction occurs.

(iii) $Q_c = K_c$ – Reaction mixture is at equilibrium.

LE CHATELIER'S PRINCIPLE

It describes the effect of change in concentration, pressure and temperature on equilibrium in a reversible system. If any system at equilibrium is subjected to a change of any one of the factors mentioned above, the system adjusts itself in such a way so as to nullify the effect of that change.

Conclusions:

(i) When temperature is increased,
if $\Delta H > 0$, reaction moves in forward direction.
if $\Delta H < 0$, reaction moves in backward direction.

(ii) When pressure is increased,
if $\Delta n > 0$, reaction moves in backward direction.
if $\Delta n < 0$, reaction moves in forward direction.

(iii) When volume is increased

if $\Delta n > 0$, reaction moves in forward direction.

if $\Delta n < 0$, reaction moves in backward direction.

(iv) If concentration of only one of the species of the reaction is increased, then reaction occurs in that direction in which concentration gets decreased.

(v) Addition of an inert gas

(a) At constant volume: no effect on equilibrium constant but total pressure is increased.

(b) At constant pressure,

if $\Delta n > 0$, reaction moves in forward direction.

if $\Delta n < 0$, reaction moves in backward direction.

(vi) Addition of catalyst: Equilibrium constant remains unaffected but rate of reaction increases.

Standard Free Energy & Equilibrium Constant

$\Delta G^\circ = -RT\ln K_{eq} = -2.303\ RT \log K_{eq}$

(i) ΔG° = –ve or $K_{eq} > 1$; forward reaction is feasible.

(ii) ΔG° = +ve or $K_{eq} < 1$; reverse reaction is feasible.

(iii) $\Delta G^\circ = 0$ or $K_{eq} = 0$; reaction is at equilibrium.

ILLUSTRATIVE PROBLEMS

1. For the dissociation reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

derive the expression for the degree of dissociation in terms of K_p and total pressure P.

Sol.

$$\begin{array}{lll} & N_2O_4(g) \rightleftharpoons & 2NO_2(g) \\ t \to 0 & 1 & 0 \\ t \to t_{eq} & (1-\alpha) & 2\alpha \end{array}$$

$$K_p = \frac{(n_{NO_2})^2}{n_{N_2O_4}}\left[\frac{P}{\sum n}\right]^{\Delta n_g}$$

$$= \frac{(2\alpha)^2}{(1-\alpha)}\left[\frac{P}{(1-\alpha+2\alpha)}\right]^{(2-1)}$$

$$= \frac{4\alpha^2}{(1-\alpha)} \times \left(\frac{P}{(1+\alpha)}\right)^1 = \frac{4\alpha^2}{(1-\alpha^2)} \times P$$

$$\Rightarrow \frac{4P}{K_P} = \frac{(1-\alpha^2)}{\alpha^2} = \frac{1}{\alpha^2} - 1$$

$$\Rightarrow \frac{1}{\alpha^2} = \left(1 + \frac{4P}{K_P}\right) = \frac{4P + K_p}{K_p}$$

$$\Rightarrow \alpha^2 = \frac{K_p}{4P + K_p}$$

$$\alpha = \sqrt{\frac{K_p}{4P + K_p}}$$

2. At a certain temperature and total pressure of 10^5 Pa iodine vapour contain 40% by volume of I atoms $I_2(g) \rightleftharpoons 2I_2(g)$ calculate Kp for equilibrium.

Sol. Partial pressure of I atom is

$$P_I = \frac{40}{100} \times P_{total}$$

$$= \frac{40}{100} \times 10^5 = 4 \times 10^4 \text{ Pa}$$

Partial pressure of I_2 molecule is

$$P_{I_2} = \frac{60}{100} \times P_{total}$$

$$= \frac{60}{100} \times 10^5 = 6 \times 10^4 \text{ Pa}$$

For given reaction

$$K_p = \frac{(P_I)^2}{(P_{I_2})} = 2.67 \times 10^4 \text{ Pa}$$

3. A 250 ml flask, and 100 ml flask are separated by a stop cock. At 350 K, the nitric oxide in the larger flask exerts a pressure of 0.46 atm, and the smaller one contains oxygen at 0.86 atm. The gases are mixed by opening the stop cock. The reaction occurring are $2NO + O_2 \rightarrow 2NO_2 \rightleftharpoons N_2O_4$.

The first reaction is complete while the second one is at equilibrium. Assuming all the gases to behave ideally, calculate the K_p for the equilibrium reaction if the final total pressure is 0.37 atm.

Sol. Moles of nitric oxide $= \frac{0.46 \times 250 \times 10^{-3}}{0.082 \times 350} = 4 \times 10^{-3}$

Moles of oxygen $= \frac{0.86 \times 100 \times 10^{-3}}{0.082 \times 350} = 3 \times 10^{-3}$

After the completion of first reaction,

Moles of oxygen left $= 10^{-3}$, Moles of NO_2 left $= 4 \times 10^{-3}$

	$2NO_2(g)$	$\rightleftharpoons$	$N_2O_4(g)$
initial	4		0
equilibrium	$4\times10^{-3}(1-x)$		$4\times10^{-3}(x/2)$

Total moles at equilibrium $= 4\times10^{-3}[(1-x)+(x/2)] + 10^{-3}$

$0.37 \times 350 \times 10^{-3} = 10^{-3}(4 - 4x + 2x + 1) \times 0.082 \times 350$

On solving we get, $x = 0.2439$

Partial pressure of $NO_2 = \frac{3 \times 0.37}{4.5} = 0.25$ and Partial pressure of $N_2O_4 = \frac{0.5 \times 0.37}{4.5} = 0.04$

$\therefore K_P = \frac{0.04}{0.25 \times 0.25} = \mathbf{0.64\ atm^{-1}}$

4. Following equilibria are established on mixing two gases A_2 and C.

(i) $3A_2(g) \rightleftharpoons A_6(g)$; $K_p = 1.6\ atm^{-2}$

(ii) $A_2(g) + C(g) \rightleftharpoons A_2C(g)$

If A_2 and C are mixed in 2 : 1 molar ratio, calculate equilibrium partial pressures of A_2, C and A_2C and K_p for equation (ii). The total pressure is 1.4 atm and partial pressure of A_6 is 0.2 atm.

Sol. $3A_2(g) \rightleftharpoons A_6(g)$

$K_p = 1.6 = \frac{0.2}{P_{A_2}^3} \Rightarrow \mathbf{P_{A_2} = 0.5\ atm}$

Let the partial pressure of A_2 and C before the start of reaction be a and a/2 atm respectively

$A_2(g) + C(g) \rightleftharpoons A_2C(g)$

If x moles of A_2 reacts to form x moles of A_2C, then

$1.4 = 0.5 + 0.2 + p_{A_2C} + P_C \Rightarrow P_{A_2C} + P_C = 0.7$ ...(i)

$a/2 - x + x = 0.7 \Rightarrow \mathbf{a = 1.4\ atm}$

Now, partial pressure of A_2 consumed in Ist reaction $= 3 \times 0.2 = 0.6$ atm

Partial pressure of A_2 consumed in IInd reaction $= 1.4 - 3 \times 0.2 - 0.5$

$= 0.3$ atm $\Rightarrow \mathbf{P_{A_2C} = 0.3\ atm}$

$P_c = 0.7 - 0.3 = 0.4$ atm

$K_p = \frac{0.3}{0.4 \times 0.5} = \mathbf{1.5\ atm^{-1}}$

5. **The degree of dissociation of HI at a particular temperature is 0.8 find the volume of 1.5 M $Na_2S_2O_3$ solution required to react completely with the iodine present at equilibrium in acidic conditions when 0.135 mole of each H_2 and I_2 are heated at 440 K in a closed vessel of capacity 2L.**

$2HI(g) \rightleftharpoons H_2(g) + I_2(g)$

1

1–0.8 0.4 0.4

$K_c = K_p = \frac{0.4 \times 0.4}{(0.2)^2} = 4$

$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ $\quad K_p = \frac{1}{4}$

0.135 0.135

0.135–x 0.135–x 2x

$K_c = \frac{1}{4} = \frac{(2x)^2}{(0.135-x)^2}$

$x = 0.045$

moles of I_2 at equivalence point = 0.09

I_2 Vs hypo

Equivalents of I_2 = hypo equivalents

$0.09 \times 2 = 1.5 \times V \times 1$

$V =$ **144 ml**

6. **A container whose volume is V, contains an equilibrium mixture that consists of 2 moles each of PCl_5, PCl_3 and Cl_2 (all gases). The pressure is 30.3975 kPa and temperature is T. A certain amount of Cl_2 (g) is now introduced keeping the pressure and temperature constant until the equilibrium volume changes to 2V. Calculate the amount of Cl_2 that was added and the value of K_p.**

Sol.

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

At equilibrium 2mol 2mol 2mol

Total pressure = 30.3975 kPa = 3 atm = P(say)

$K_p = \frac{P_{PCl_3} \times P_{Cl_2}}{P_{PCl_5}} = \frac{\frac{P}{3} \times \frac{P}{3}}{\frac{P}{3}}$

$K_p = \frac{P}{3} = 1$ ... (1)

When chlorine is added to the system. The system will try to nullify the effect and hence formation of PCl_5 will be preferred.

Since, P and T are constant

$\frac{V_1}{V_2} = \frac{n_1}{n_2} = n_1 = 6$ $\quad [\because V_1 = V, V_2 = 2V]$

$\therefore n_2 = 12$ moles

Say, 'a' moles of Cl_2 were added

$PCl_5 \rightleftharpoons PCl_3 + Cl_2$

Initial	2	2	2
Final	$2+x$	$2-x$	$2+a-x$

$n_T - 12 = 6 + a - x$

$\therefore a - x = 6$

$$K_p = \frac{\frac{2+ax-x}{12} \times \frac{2-x}{12}}{\frac{2+x}{12}} \times P = \frac{(2+a-x)(2-x)}{4(2+x)} = 1 \qquad \ldots (2)$$

Solving equations (1) and (2), we get

$a = \frac{20}{3}$ moles

Hence, $\frac{20}{3}$ moles of Cl_2 was added.

7. PCl_5 dissociates into PCl_3 and Cl_2 thus

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

If the total pressure of the system in equilibrium is P at a density ρ and temperature T, show that Degree of dissociation $\alpha = \frac{PM}{\rho RT} - 1$

where M is the relative molar mass of PCl_5. If the vapour density of the gas mixture at equilibrium has the value of 62 when the temperature is 230°C, what is the value of P/ρ?

Sol. $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Since $PV = nRT = \frac{w}{M}RT$

$$M = \frac{wRT}{PV} = \frac{\rho RT}{P}$$

$$\therefore \text{V.D. (d)} = \frac{\rho RT}{2P}$$

	$PCl_5(g)$	$\rightleftharpoons$	$PCl_3(g)$	+	$Cl_2(g)$
Initial	C		0		0
At eqm.	$C(1-\alpha)$		$C\alpha$		$C\alpha$

$$\frac{\text{Total moles at equilibrium}}{\text{Total initial moles}} = \frac{\text{Initial vapour density}}{\text{Vapour density at equilibrium}}$$

$$\frac{C(1+\alpha)}{C} = \frac{D}{d}$$

$$\Rightarrow 1 + \alpha = \frac{M/2}{\rho RT/2P} = \frac{PM}{\rho RT}$$

$$\therefore \alpha = \frac{PM}{\rho RT} - 1$$

$$\text{If } d = 62, \frac{M}{2} = \frac{208.5}{2} = 104.25$$

$$\alpha = \frac{104.25 - 62}{62} = 0.68$$

$$\therefore \frac{P}{\rho} = \frac{(1+\alpha)RT}{M} = 0.3327 \text{ atm/(gm/L)}$$

8. **When 1-pentyne (A) is treated with 4 N alcoholic KOH at $175°C$, it is converted slowly into an equilibrium mixture of 1.3% 1-pentyne (A), 95.2% 2-pentyne (B) and 3.5% of 1, 2-pentadiene (C). The equilibrium was maintained at $175°C$. Calculate $\Delta G°$ for the following equilibria.**

$B \rightleftharpoons A \qquad \Delta G_1^\circ = ?$

$B \rightleftharpoons C \qquad \Delta G_2^\circ = ?$

From the calculated value of ΔG_1° and ΔG_2° indicate the order of stability of (A), (B) and (C). Write a reasonable reaction mechanism showing all intermediates leading to (A), (B) and (C).

Sol.

$$\text{Pentyne} \xrightleftharpoons{KOH} 2-\text{Pentyne} + 1,2-\text{Pentadiene}$$

$$\underset{}{} \quad (A) \qquad (B) \qquad (C)$$

At eq. m% 1.3 95.5 3.5

$$K_c = \frac{[B][C]}{[A]} = \frac{95.2 \times 3.5}{1.3} = 256.31 \qquad \ldots (i)$$

For eqm. $B \rightleftharpoons A$

$$K_1 = \frac{[A]}{[B]}$$

From Eqs. (i) and (ii), $K_1 = \dfrac{[C]}{K_c}$

$$= \frac{3.5}{256.31} = 0.013$$

$$\Delta G^\circ = -2.303RT \log_{10} K$$

$$= -2.303 \times 8.314 \times 448 \log_{10} 0.013$$

$$= 161787$$

$$= 16.178 \text{ kJ}$$

Stability order for A and B is B>A

Similarly, $B \rightleftharpoons C$

$$K_2 = \frac{[C]}{[B]}$$

$$= \frac{K_c \times [A]}{[B]^2}$$

$$= \frac{256.31 \times 1.3}{95.2 \times 95.2}$$

$$\therefore \Delta G_2^\circ = -2.303RT \log_{10} K$$

$$= -2.303 \times 8.314 \times 448 \log_{10} 0.037$$

$$= 12282 \text{ J}$$

$$= 12.282 \text{ kJ}$$

Thus, stability order for B and C is B>C

Total order of stability is B>C>A.

SOLVED PROBLEMS

Subjective Question

Prob. 1. A theoretically completed equilibrium constant for the polymerization of formaldehyde to glucose in aqueous solution is 6×10^{22}

$6HCHO \rightleftharpoons C_6H_{12}O_6$

If a 1.0 M solution of glucose were to reach dissociation equilibrium with respect to above equilibrium what would be the concentration of formaldehyde in the solution?

Sol. $K_c = \dfrac{[C_6H_{12}O_6]}{[HCHO]^6}$

$6\times10^{22} = \dfrac{1}{[HCHO]^6}$

$[HCHO] = 1.6 \times 10^{-4}$ M

Objective Questions

Prob. 1. The equilibrium, $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$, is attained at 25°C in a closed container and an inert gas He is introduced. Which of the following statements are correct?

(A) Concentration of PCl_5, PCl_3 and Cl_2 are changed (B) More Cl_2 is formed

(C) Concentration of PCl_3 is reduced (D) Nothing happens

Sol. Since decomposition of PCl_5 is carried out in a closed container. When helium gas is introduced the volume remains constant and there is no change in the partial pressures of PCl_5, PCl_3 and Cl_2. Therefore, nothing happens on introduction of helium gas and constant volume.

Hence, **(D)** is the correct answer.

Prob. 2. 40% of a mixture of 0.2 mol of N_2 and 0.6 mol of H_2 react to give NH_3 according to the equation, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$, at constant temperature and pressure. Then the ratio of the final volume to the initial volume of gases are

(A) 4:5 (B) 5:4

(C) 7:10 (D) 8:5

Sol.

	N_2 +	$3H_2$	$\rightleftharpoons$	$2NH_3$
t = 0	0.2	0.6		0
at t = t	0.2 – x	0.6–3x		2x

40% of $N_2 = 0.2 \times 0.4 = 0.08$

40% of $H_2 = 0.6 \times 0.4 = 0.24$

$\therefore$ Number of moles of N_2 remaining = 0.2 – 0.08 = 0.12

Number of moles of H_2 remaining = 0.6 – 0.24 = 0.36

Number of moles of NH_3 formed = 0.16

Total number of moles = 0.12 + 0.36 + 0.16 = 0.64

$\therefore \dfrac{\text{Final volume}}{\text{Initial volume}} = \dfrac{\text{Final moles}}{\text{Initial moles}} = \dfrac{0.64}{0.80} = \dfrac{4}{5}$

Hence, **(A)** is the correct answer.

Prob. 3. In the following equilibrium, the example of phase equilibria is

(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ (B) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(C) $H_2O(s) \rightleftharpoons H_2O(l)$ (D) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

Sol. **(C)**

Prob. 4. In a system, $A(s) \rightleftharpoons 2B(g) + 3C(g)$

If the conc. of C at equilibrium is increased by a factor of 2, it will cause the equilibrium concentration of B to change to

(A) two times the original value
(B) one half of its original value
(C) $2\sqrt{2}$ times the original value
(D) $1/2\sqrt{2}$ times the original value

Sol. $A(s) \rightleftharpoons 2B(g) + 3C(g)$

Let, x and y be the concentrations of B and C at equilibrium respectively.

$\therefore K_c = x^2y^3$ .... (1)

Now, the concentration of C is changed from y to y' such that $y' = 2y$. If x' is the new concentration of B

$\therefore K_c = (x')^2 (y')^3 = (x')^2(2y)^3$ ....(2)

From Eqs. (1) and (2)

$(x')^2(8y^3) = x^2y^3$

$\therefore x' = \sqrt{\frac{x^2}{8}} = \frac{x}{2\sqrt{2}}$

$\therefore$ Equilibrium concentration of B changes to $\frac{1}{2\sqrt{2}}$ times the original value.

Hence, **(D)** is the correct answer.

Prob. 5. K_p for a reaction at 25°C is 10 atm. The activation energy for forward and reverse reactions are 12 and 20 kJ/ mol respectively. The K_c for the reaction at 40°C will be

(A) 4.33×10^{-1} M
(B) 3.33×10^{-2} M
(C) 3.33×10^{-1} M
(D) 4.33×10^{-2} M

Sol. Enthalpy changed of a reaction is given by

$\Delta H = E_{a(f)} - E_{a(b)}$

where $E_{a(f)}$ and $E_{a(b)}$ are energies of activation for the forward and backward reactions.

$\Delta H = 12 - 20 = -8$ kJ/mol

K_p for the reaction at 25°C = 10 atm. Since K_p is expressed in atmosphere, $\Delta n = +1$

$\because K_p = K_c (RT)^{\Delta n}$, $K_c = \frac{10}{0.0821 \times 298} = 0.4$ M

K_c at 40°C is given by

$$\log \frac{(K_c)_{40}}{(K_c)_{25}} = \frac{\Delta H}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right] = \frac{-8 \times 1000}{2.303 \times 8.314} \times \frac{15}{298 \times 313} = -0.06719$$

$(K_c)_{40}/(K_c)_{25} = 0.85$

$(K_c)_{40} = 0.85 \times 0.4 = 0.34$ M

Hence, **(C)** is the correct answer.

Prob. 6. The dissociation of phosgene, which occurs according to the reaction

$$COCl_2(g) \rightleftharpoons CO(g) + Cl_2(g)$$

is an endothermic process. Which of the following will increase the degree of dissociation of $COCl_2$?

(A) Adding Cl_2 to the system
(B) Adding helium to the system at constant pressure
(C) Decreasing the temperature of the system
(D) Reducing the total pressure

Sol. **(B, D)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** K_P can be equal to or less than or greater than K_C.
and
STATEMENT-2: $K_P = K_C(RT)^{\Delta ng}$, thus, K_P and K_C depend upon temperature.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(B)**

2. **STATEMENT-1:** The concentration term of pure liquids and solids can be ignored while writing equilibrium constant expression.
and
STATEMENT-2: The concentration of pure liquids and solids donot remain constant.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(C)**

COMPREHENSION

Read the following passage and answer the questions:
The dissociation constant is related to the standard free energy difference between products and reactants in the following way. If K_a is the dissociation constant, then standard free energy dissociation is related as $K_a = 10^{-\Delta G^\circ/2.3\,RT}$.

If ΔG° is negative, this means that product P is more stable than starting material and vice–versa. Following is a type of association reaction, where standard entropies of A, B and C are 191, 130 and 192 J K^{-1} mol^{-1} respectively at 25°C.

$A_{(g)} + 3B_{(g)} \rightleftharpoons 2C_{(g)}$; $\Delta H^\circ = -46$ kJ

1. Value of ΔS° for above reaction is:
(A) -137 J K^{-1} mol^{-1} (B) -197 J K^{-1} mol^{-1} (C) $+127$ J K^{-1} mol^{-1} (D) $+227$ J K^{-1} mol^{-1}

Sol. **(B)**

2. ΔG° for above reaction is
(A) -3.3×10^4 J (B) -3.98×10^4 J (C) -1.3×10^4 J (D) -1.0×10^5 J

Sol. **(A)**

3. log K_p for the above reaction is
(A) 5.84 (B) 5.98 (C) 3.85 (D) 4.84

Sol. **(A)**

4. Value of K_p for the reverse reaction, $2C \rightleftharpoons A + 2B$ is:
(A) 1.2×10^{-3} atm^2 (B) 1.4×10^{-2} atm^2 (C) 1.44×10^{-6} atm^2 (D) 1.4×10^{-5} atm^2

Sol. **(C)**

MATRIX – MATCH TYPE QUESTION

1. Match the following:

Column I:		Column II:	
(A)	$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$	(p)	$K_P = K_c$
(B)	$NH_2COONH_4(s) \rightleftharpoons 2NH_3 + CO_2(g)$	(q)	$K_P < K_c$
(C)	$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	(r)	$K_P > K_c$
(D)	$2SO_2(g) + O_2 \rightleftharpoons 2SO_3(g)$	(s)	Addition of inert gas at constant pressure shift the equilibrium in the product side
		(t)	No effect of increase of pressure.

Sol. **A → p, t; B → r, s; C → p t; D → q**

2. Match the following:

Column I:		Column II:	
(A)	$NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$	(p)	Reaction is fast in forward direction
(B)	$Q < K$	(q)	$K_P = K_c (RT)^2$
(C)	Melting of ice	(r)	Affected by temperature
(D)	Equilibrium constant	(s)	Affected by pressure

Sol. **A → q; B → p; C → r, s; D → r**

3. Match the following:

Column I:		Column II:	
(A)	$\Delta G^\circ = 0$	(p)	Reaction is at equilibrium
(B)	For a heterogeneous endothermic equitation with equilibrium constant K	(q)	Equilibrium constant decrease as term is increased
(C)	$A(g) \rightleftharpoons B(g) + C(g) + \text{heat}$ equilibrium constant K	(r)	The equilibrium is unaffected

Sol. **A → p; B → r; C → q**

Answer 4, 5 and 6 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	$3O_2(g) \rightleftharpoons 2O_3(g)$	(i)	no unit	(P)	$K_p > K_c$
(II)	$SO_2 + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$	(ii)	$atm^{-1/2}$	(Q)	$K_p < K_c$
(III)	$2HF(g) \rightleftharpoons H_2(g) + F_2(g)$	(iii)	atm^{-1}	(R)	$K_p = K_c$
(IV)	$CO(g) + 3H_2(g) \rightleftharpoons CH_4(g) + H_2O(g)$	(iv)	atm^{-2}	(S)	$K_p \geq K_c$

4. Which of the following combination is correct ?
[A] (I) (ii) (S) [B] (II) (iii) (P) [C] (I) (iii) (Q) [D] (IV) (i) (P)

5. In the given column-III, only correct combination for (R) $K_p = K_c$?
[A] (IV) (i) (P) [B] (II) (ii) (R) [C] (III) (iii) (Q) [D] (III) (i) (R)

6. In the given column-II, which combination is correct for (iv) atm^{-2} ?
[A] (I) (ii) (S) [B] (IV) (iv) (Q) [C] (I) (iii) (Q) [D] (IV) (ii) (P)

Sol. 4. **C** ; 5. **D**; 6. **B**

SINGLE INTEGER ANSWER TYPE

1. $2SO_2 + O_2 \rightleftharpoons 2SO_3$

Starting with 2 mol of SO_2 and 1 mol of O_2 in 1-L flask, mixture required 0.4 mol of MnO_4^- in acidic medium. Find the K_c value.

Sol. **(2)**

$5SO_2 + 2MnO_4^- \longrightarrow 2Mn^{2+} + 5SO_4^{2-}$

2 mol of $MnO_4^- \equiv 5$ mole of SO_2.

$\therefore$ 0.4 ml of $MnO_4^- \equiv 1$ mol SO_2.

$2SO_2 + O_2 \rightleftharpoons 2SO_3$

2 1 0

$(2-2x)$ $(1-x)$ $2x$

$(2 - 2x) = 1$ mol (as determined by MnO_4^-)

$\therefore \left[SO_3\right] = \frac{2x}{V} = 1M$, $\left[SO_2^-\right] = \frac{2-2x}{V} = 1M$

$\left[O_2^-\right] = \frac{(1-x)}{V} = 0.5M$, $K_C = \frac{\left[SO_3\right]^2}{\left[SO_2\right]^2\left[O_2\right]} = 2$

2. For the equilibrium
$NH_2CONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

$p_{CO_2} = 1atm$ at 100^0 C. What will be the equilibrium constant at 100^0C in atm.

Sol. **(4)**

$NH_4CONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$

1 0 0

$(1-x)$ $2x$ x

Let p_{total} be p

$\therefore p_{CO_2} = \frac{x}{(2x+x)} \times p = \frac{p}{3} = 1$ $\quad p_{NH_3} = \frac{2x}{(2x+x)} \times p = \frac{2}{3} \times 3 = 3$ atm.

NUMERICAL VALUE ANSWER TYPE

Prob. 1. The equilibrium constant, K_P for the reaction $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ is 4.0 atm^{-1} at 1000 K. What would be the partial pressure of O_2 if at equilibrium the amount of SO_2 and SO_3 is the same ?

Sol. **0.25 atm**

Prob. 2. The equilibrium $SO_2(g) + \frac{1}{2}O_2(g) \rightleftharpoons SO_3(g)$ is established in a container of 4L at a particular temperature. If the number of moles of SO_2, O_2 and SO_3 at equilibrium are 2, 1 and 4 respectively then find the value of equilibrium constant.

Sol. **4.00**

Prob. 3. The equilibrium constant of the reaction $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$ is 64. If the volume of the container is reduced to one fourth of its original volume, the value of the equilibrium constant will be

Sol. **64.00**

Prob. 4. In the following reaction started only with A_8,

$2A_8(g) \rightleftharpoons 2A_3(g) + 3A_2(g) + A_4(g)$

Mole fraction of A_2 is found to 0.36 at a total pressure of 100 atm at equilibrium. The mole fraction of $A_8(g)$ at equilibrium is

Sol. **0.28**

Prob. 5. Two solid compounds X and Y dissociates at a certain temperature as follows

$X(s) \rightleftharpoons A(g) + 2B(g)$; $K_{p_1} = 9\times10^{-3}$ atm^3

$Y(s) \rightleftharpoons 2B(g) + C(g)$; $K_{p_2} = 4.5\times10^{-3}$ atm^3

The total pressure of gases over a mixture of X and Y is

Sol. **0.45 atm**

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CHEMISTRY

IONIC EQUILIBRIUM 6

IONIC PRODUCT OF WATER

Water is amphiprotic and gives H^+ and OH^- on self ionization.
Ionic product of water, $K_w = [H^+]\,[OH^-] = 10^{-14}$ at 25°C.
It depends only on temperature. It increases with increase in temperature.

pH value:

$pH = -\log [H_3O^+]$
or $[H_3O^+] = 10^{-pH}$
Similarly, $pOH = -\log [OH^-]$
$[OH^-] = 10^{-pOH}$
And, $pH + pOH = pK_w = 14$ at 25°C
For conjugate acid-base pair in aq. solution, i.e. HA and A^-

$$HA + H_2O \rightleftharpoons H_3O^+ + A^- \quad K_a$$

$$\frac{A^- + H_2O \rightleftharpoons HA + OH^-}{H_2O \rightleftharpoons H^+ + OH^-} \quad K_b$$

Thus, $K_w = K_a \times K_b$
And, $pK_a + pK_b = pK_w = 14$ at 25°C
Higher the pK_a value, lower is the acid strength and higher is the basic strength.

OSTWALD'S DILUTION LAW

It deals with the effect of dilution on weak electrolytes (weak acid or weak base). Degree of dissociation of weak electrolytes is proportional to the square root of dilution.

$$K_a = \frac{C\alpha^2}{1-\alpha}$$

where K_a = dissociation constant of weak electrolyte
α = degree of dissociation
C = initial concentration of weak electrolyte

If $\alpha \ll 1$, then $\alpha = \sqrt{\frac{K_a}{C}}$

$$[H_3O^+] = C\alpha = \sqrt{K_a C}$$

In case of weak monobasic acid:

$$pH = -\log [H_3O^+] = -\log \sqrt{K_a C}$$

$$pH = \frac{1}{2}[pK_a - \log C] \quad \ldots (1)$$

In case of weak monoacidic base:

$$pOH = \frac{1}{2}[pK_b - \log C]$$

$$\text{or } pH = 14 - pOH = 14 - \frac{1}{2}[pK_a - \log C] \quad \ldots (2)$$

Eqs. (1) and (2) are applied only when $\alpha \ll 1$ and the value of C must be high.

Total $[H_3O^+]$ in mixture of two weak acids:

$$[H_3O^+] = \sqrt{K_{a_1} C_1 + K_{a_2} C_2}$$

Total $[OH^-]$ in a mixture of two weak bases:

$$[OH^-] = \sqrt{K_{b_1}C_1 + K_{b_2}C_2}$$

Total $[H_3O^+]$ in a mixture of weak acid and a strong acid:

$$[H_3O^+] = \frac{C_2 + \sqrt{C_2^2 + 4K_1C_1}}{2}$$

Here, C_1 is concentration of weak acid having dissociation constant K_1 and C_2 is the concentration of strong acid.

Common Ion Effect

The degree of dissociation of a weak electrolyte is suppressed by the addition of another strong electrolyte containing common ion. It is called common ion effect.

In case of weak electrolyte,

$$XY \rightleftharpoons X^+ + Y^-$$

$$K = \frac{[X^+][Y^-]}{[XY]}$$

In case any other strong electrolyte containing X^+ or Y^- ions is added, the equilibrium tends to shift to left in order to keep K constant at given temperature.

(i) At half neutralization point of weak acid
where $[HA] = [A^-]$
$pH = pK_a$
$[H^+] = K_a$

(ii) At half neutralization point of weak base
where $[BOH] = [B^+]$
$\therefore pOH = pK_b$
$[OH^-] = K_b$
$\therefore pH = 14 - pK_b$

(iii) In case of weak dibasic acid (e.g. H_2CO_3), pH at half neutralization point is

$$pH = \frac{pK_{a_1} + pK_{a_2}}{2}$$

Salt Hydrolysis

S. No.	Salt	Nature	Hydrolysis constant	Degree of hydrolysis	$[H_3O^+]$	pH
1.	Salt of strong acid and strong base, e.g. NaCl, KCl	Neutral	–	No hydrolysis	–	$\frac{pK_w}{2}$
2.	Salt of weak acid and strong base, e.g. CH_3COONa	Basic	$K_h = \frac{K_w}{K_a}$	$h = \sqrt{\frac{K_w}{K_aC}}$	$\sqrt{\frac{K_wK_a}{C}}$	$pH = \frac{pK_w + pK_a + \log C}{2}$
3.	Salt of weak base and strong acid, e.g. NH_4Cl	Acid	$K_h = \frac{K_w}{K_b}$	$h = \sqrt{\frac{K_w}{K_aC}}$	$\sqrt{\frac{K_wC}{K_b}}$	$pH = \frac{pK_w - pK_b - \log C}{2}$
4.	Salt of weak acid and weak base	May be acidic, basic or neutral	$K_h = \frac{K_a}{K_a \times K_b}$	$h = \sqrt{\frac{K_w}{K_a \times K_b}}$	$\sqrt{\frac{K_w \times K_a}{K_b}}$	$pH = \frac{pK_w + pK_a - pK_b}{2}$

In case of salt of weak acid and weak base, nature of medium will be

(i) Neutral, if $K_a = K_b$
(ii) Basic, if $K_b > K_a$
(iii) Acidic, if $K_a > K_b$

BUFFER SOLUTIONS

A solution which resists change of pH on the addition of small amount of acid or alkali.

Acidic Buffer

These are prepared by mixing weak acid and its salt with strong base or it contains a weak acid and its conjugate base, e.g. CH_3COOH and CH_3COONa.

$$pH = pK_a + \log \frac{[\text{conjugate base}]}{[\text{weak acid}]}$$

Basic Buffer

These are prepared by mixing weak base and its salt with strong acid or it contains a weak base and its conjugate acid, e.g. NH_4OH and NH_4Cl.

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

Salt Buffer

It is a salt of weak acid and weak base.

$$[H^+] = \sqrt{\frac{K_w \times K_a}{K_b}}$$

Buffer Range

(i) An acidic buffer is effective within pH range $(pK_a - 1)$ to $(pK_a + 1)$.
(ii) A basic buffer is effective in pH range $(pK_w - pK_b \pm 1)$.

Buffer Capacity

$$\text{Buffer capacity} = \frac{\text{No. of moles of acid / base added per litre}}{\text{change in pH}}$$

It is maximum when

For acidic buffer, $\frac{[\text{salt}]}{[\text{acid}]} = 1$; $pH = pK_a$

For basic buffer, $\frac{[\text{salt}]}{[\text{base}]} = 1$; $pOH = pK_b$

SOLUBILITY PRODUCT (K_{sp})

K_{sp} of an electrolyte is the product of the molar concentration of its ions in a saturated solution with each concentration term raised to the power equal to the number of ions produced on dissociation of one molecule of that electrolyte.

K_{sp} value for a given salt is constant at a given temperature, e.g. K_{sp} for solute X_aY_b is

$$X_aY_b \rightleftharpoons aX^{b+} + bY^{a-}$$

$$aS \quad bS \quad (S \text{ is solubility})$$

$K_{sp} = [X^{b+}]^a [Y^{a-}]^b$

or $K_{sp} = [aS]^a [bS]^b$

or Solubility $(S) = \left(\frac{K_{sp}}{a^a b^b}\right)^{\frac{1}{a+b}}$

(i) Solubility of a solute in the presence of common ion, e.g. solubility of AgCl in presence of xM $AgNO_3$.

$$S = \frac{K_{sp}(AgCl)}{x}$$

(ii) Solubility of salt of weak acid and strong base, e.g. CH_3COONa in water.

$$CH_3COONa(s) \rightleftharpoons CH_3COO^- + Na^+$$

$$\underset{(S-x)}{CH_3COO^-} + H_2O \rightleftharpoons \underset{S}{CH_3COOH} + \underset{x}{OH^-}$$

$(S - x)$ x x

Thus, $(S - x)S = K_{sp}$ ...(1)

and $\frac{x^2}{(S-x)} = K_h = \frac{K_w}{K_a}$ ...(2)

(iii) Solubility of salt of weak acid and strong base (CH_3COONa) in an acid buffer of given pH strength (buffer does not have any common ion) is

$$CH_3COONa \rightleftharpoons CH_3COO^- + Na^+$$

$$\underset{S-x}{CH_3COO^-} + H^+ \rightleftharpoons \underset{S}{CH_3COOH}$$

$S - x$ x

$K_{sp} = S(S - x)$ ...(1)

$\frac{x}{[H^+][S-x]} = \frac{1}{K_a}$ ...(2)

(iv) Solubility of salt of weak acid and strong base in a basic buffer of given strength y is

$$CH_3COONa \rightleftharpoons CH_3COO^- + Na^+$$

$$\underset{(S-x)}{CH_3COO^-} + H_2O \rightleftharpoons \underset{S}{CH_3COOH} + OH^-$$

$(S - x)$ x y (Mainly due to solution basicity)

$S(S - x) = K_{sp}$ ...(1)

$\frac{x \times y}{(S-x)} = \frac{K_w}{K_a}$ ...(2)

Precipitation of Salts

(i) If ionic product (IP) > K_{sp}, precipitation takes place till IP equals K_{sp}.
(ii) If IP < K_{sp}, no precipitate will be formed.
(iii) If IP = K_{sp}, precipitate will not be formed, but the solution is at saturation stage.

THEORY OF INDICATOR

Ostwald's Theory

Indicators are generally weak organic acids or bases. The unionized and ionized form of indicators have different colours. They change colours within certain pH range and are used for the determination of end point in the titrations.

$$HIn + H_2O \rightleftharpoons H_3O^+ + In^-$$

Colour A Colour B

$$K_{in} \text{ (indicator constant)} = \frac{[H_3O^+][In^-]}{[HIn]}$$

K_{in} depends only on temperature.

$$[H_3O^+] = K_{in}\frac{[HIn]}{[In^-]}$$

$$\text{Thus, } pH = pK_{in} + \log\frac{[In^-]}{[HIn]}$$

$pH \leq pK_{in} - 1$, the solution has a characteristic colour of HIn.

$pH \geq pK_{in} + 1$, the solution has a characteristic colour of In^-.

Thus, transition of colour takes place in pH range, $pK_{in} - 1$ to $pK_{in} + 1$

Equivalence point

At equivalence point, the pH value of the solution, that is undergoing titration, changes steeply and pH range of this steep change depends on the nature of acid–base pair being titrated.

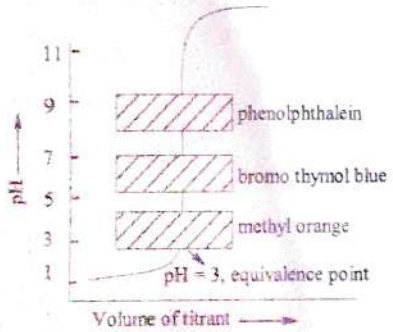
At equivalence point, $[In^-] = [HIn]$

$pH = pK_{in}$ or $[H^+] = K_{in}$

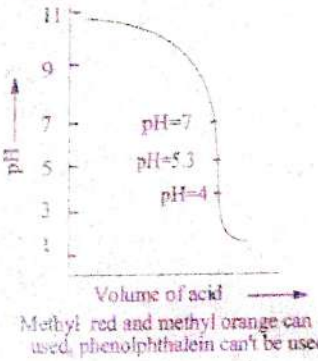
Selection of Suitable Indicator

The indicator should offer a colour change at equivalence point or the pH transition range of the indicator must coincide with the steep portion of the titration curve.

Titration of strong acid (HCl) and strong base (NaOH):

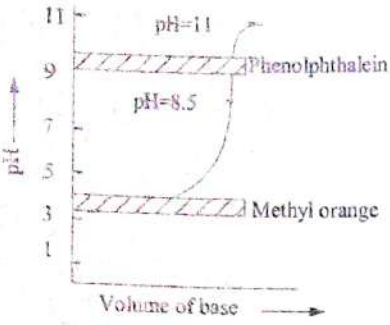


Titration of weak base (NH_4OH) and strong acid (HCl):



Methyl red and methyl orange can be used, phenolphthalein can't be used.

Titration of weak acid (CH_3COOH) and strong base (NaOH):



Phenolphthalein is a suitable indicator.

ILLUSTRATIVE PROBLEMS

1. A weak acid HA after treatment with 12 mL of 0.1 M strong base BOH has a pH of 5. At the end point, the volume of same base required is 26.6 mL. Calculate K_a of acid.

Sol. For neutralization:

Total Meq. of acid = Meq. of base = $26.6 \times 0.1 = 2.66$

Now for partial neutralization of acid

	HA	+	BOH	→	BA	+	H_2O
Meq. before reaction	2.66		1.2		0		0
Meq. after reaction	1.46		0		1.2		1.2

The resultant mixture acts as a buffer and [HA] and [BA] may be placed in terms of Meq. since volume of mixture is constant.

$$\therefore \quad pH = -\log K_a + \log \frac{[Salt]}{[Acid]}$$

$$\text{or} \quad 5 = -\log K_a + \log \frac{[1.2]}{[1.46]}$$

$$\therefore \quad K_a = 8.219 \times 10^{-6}$$

2. What is the concentrations of H^+, HSO_4^-, SO_4^{2-} and H_2SO_4 in a 0.2 M solution of sulphuric acid?

Give $H_2SO_4 \longrightarrow H^+ + HSO_4^-$ strong

$HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ $K_2 = 1.3 \times 10^{-2} M$

Sol. From first dissociation $[H^+]$ is 0.20 M

Let x be amount of H^+ from second dissociation

Hence $[H^+]_{total} = 0.20\ M + x$

This in equilibrium gives

HSO_4^-	$\rightleftharpoons$	H^+	+	SO_4^{2-}
$0.20M - x$		$0.20M + x$		x

$$K_2 = \frac{(x)(0.20M + x)}{0.20M - x} = 1.3 \times 10^{-2} M$$

$$x^2 + (0.20 + 1.3 \times 10^{-2})x - (1.3 \times 10^{-2})(0.20) = 0$$

$x = 0.0116M$

$[H^+]_{total} = (0.2 + x) = 0.2116\ M$

$[HSO_4^-] = 0.2 - x = 0.1884\ M$

$[SO_4^{2-}] = x = 0.0116\ M$

$[H_2SO_4] = 0$

3. Should $Mg(OH)_2$ precipitate from a solution that is 0.01 M $MgCl_2$ if the solution is also made 0.10 M in NH_3 [$K_{sp[Mg(OH)_2]} = 1.8 \times 10^{-11}$, $K_{b(NH_4OH)} = 1.8 \times 10^{-5}$]

Sol. $[OH^-]$ in 0.10 M $NH_4OH = \sqrt{K_b C}$ (Ostwald's dilution law)

$$= \sqrt{1.8\times10^{-5}\times0.010} = 1.34\times10^{-3}\ M$$

$\Rightarrow [Mg^{2+}] = 0.01$ M

Ionic product $= [Mg^{2+}][OH^-]^2 = (0.01)\times(1.34\times10^{-3})^2$

$= \mathbf{1.8\times10^{-8} > K_{sp}}$

As. Ionic product is greater than K_{sp} of $Mg(OH)_2$, hence precipitation should occur

4. **For the indicator thymol blue, the value of pH is 2.0 when half the indicator is present in unionised form calculate the percentage of indicator in the unionised form in a solution of 4×10^{-3}M $[H^+]$ concentration.**

Sol. Since at pH = 2 half of inductor is present in unionised form therefore

$[HIn] = [In^-]$

$$pH = pK_{In} + \log\frac{[In^-]}{[HIn]}$$

$pK_{In} = pH = 2$

pH of solution containing 4.0×10^{-3}M of H^+ is

$pH = -\log(4\times10^{-3}) = 2.4$

$$\log\left(\frac{[In^-]}{[HIn]}\right) = pH - pK_{In} = 2.4 - 2.0 = 0.4$$

$$\frac{[In^-]}{[HIn]} = 2.5$$

Adding 1 on both sides

$$\frac{[In^-]+[HIn]}{[HIn]} = 3.5$$

$$\frac{[HIn]}{[In^-]+[HIn]} = \frac{1}{3.5} = 0.286$$

% of indicator in unionised from = 28.6

5. **Given $[Ag(NH_3)_2]^+ \rightleftharpoons Ag^+ + 2NH_3$, $K_c = 6.2\times10^{-8}$ and $K_{sp(AgCl)} = 1.8\times10^{-10}$ at 298 K. Calculate the concentration of the complex in 1.0 M aqueous ammonia.**

Sol. $Ag(NH_3)_2^+(aq) \rightleftharpoons Ag^+(aq) + 2NH_3(aq)$

$x + y \qquad 2x$

$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

$(x+y) \qquad y$

In case of simultaneous solubility, Ag^+ remains same in both the equilibrium

$$K_c = \frac{(x+y)\times(2x)}{[Ag(NH_3)_2^+]} \quad \ldots(1)$$

$$K_{sp} = (x+y)\times y \quad \ldots(2)$$

$$\therefore \frac{K_c}{K_{sp}} = \frac{2x}{[Ag(NH_3)_2^+]\times y}$$

Given, $[NH_3] = 2x = 1M$

$[Ag(NH_3)_2^+] = [Cl^-] = y$ because Ag^+ obtained from AgCl passes in $[Ag(NH_3)_2]^+$ state.

$$\frac{K_c}{K_{sp}} = \frac{1}{y \times y}$$

$$\Rightarrow y^2 = \frac{1.8 \times 10^{-10}}{6.2 \times 10^{-8}} = 0.29 \times 10^{-2}$$

$\therefore y = 0.539 \times 10^{-1} = 0.0539$

i.e. $[Ag(NH_3)_2] = 0.539$.

6. **A solution has 0.05 M Mg^{2+} and 0.05 NH_3. Calculate the concentration of NH_4Cl required to prevent the formation of $Mg(OH)_2$ in solution.**
$K_{sp[Mg(OH)_2]} = 9.0 \times 10^{-12}$ and ionization constant of NH_3 is 1.8×10^{-5}.

Sol. The minimum $[OH^-]$ at which there will be no precipitation of $Mg(OH)_2$ can be obtained by

$K_{sp} = [Mg^{2+}][OH^-]^2$

$\Rightarrow 9.0 \times 10^{-12} = (0.05) \times [OH^-]^2$

$\therefore [OH^-] = 1.34 \times 10^{-5}$ M

Thus, solution having $[OH^-] = 1.34 \times 10^{-5}$ M will not show precipitation of $Mg(OH)_2$ in 0.05 M Mg^{2+}. These hydroxyl ions are to be derived by basic buffer of NH_4Cl and NH_4OH.

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$$pOH = pK_b + \log\frac{[NH_4^+]}{[NH_4OH]}$$

$NH_4OH \rightleftharpoons NH_4^+ + OH^-$

In presence of $[NH_4Cl]$, all the NH_4^+ are provided by NH_4Cl as due to common ion effect, dissociation of NH_4OH will decrease.

$$-\log[OH^-] = -\log 1.8 \times 10^{-5} + \log\frac{[NH_4^+]}{[0.05]}$$

$\therefore NH_4^+ = 0.067$ M or $[NH_4Cl] = 0.067$ M

7. **A solution contains a mixture of Ag^+ (0.10 M) and Hg^{2+} (0.10 M) which are to be separated by selective precipitation. Calculate the maximum concentration of iodide ion at which one of them get precipitated completely. What percentage of metal ion is precipitated ?**
K_{sp} AgI = 8.5×10^{-17}, K_{sp} $HgI_2 = 2.5 \times 10^{-26}$

Sol. $K_{sp}(AgI) = 8.5 \times 10^{-17} = [Ag^+][I^-]$

$[I^-]$ required to start precipitated of AgI

$$= \frac{8.5 \times 10^{-17}}{0.10} = 8.5 \times 10^{-16} M$$

$K_{sp}(HgI_2) = 2.5 \times 10^{-26} = [Hg^{2+}][I^-]^2$

$$[I^-] = \sqrt{\frac{2.5\times10^{-26}}{0.10}} = 5\times10^{-13}M$$

When $[I^-]$ reaches to 5×10^{-13}, AgI gets precipitated almost completely

When HgI_2 starts precipitating

$$[Ag^+] = \frac{8.5\times10^{-17}}{5\times10^{-13}} = 1.7\times10^{-4}M$$

$$\%\ Ag^+ \text{ remaining} = \frac{1.70\times10^{-14}\times100}{0.10} = 0.17$$

$$\%\ Ag^+ \text{ precipitated} = 100 - 0.17 = 99.83$$

8. **For the reaction, $[Ag(CN)_2] \rightleftharpoons Ag^+ + 2CN^-$, the equilibrium constant, K_c at 25°C is 4.0×10^{-19}. Calculate the silver ion concentration in a solution which is originally 0.10 M in KCN and 0.03 M in $AgNO_3$.**

Sol. $Ag^+ + 2CN^- \rightleftharpoons Ag(CN)_2^-$

$$K_c = \frac{[Ag(CN)_2^-]}{[Ag^+][CN^-]^2} = \frac{1}{K_c} = 2.5\times10^{20} \quad \ldots (i)$$

Very high value of K_c show that complex forming equilibrium is spontaneous and almost all the Ag^+ ion would have reacted leaving x M in Solution:

	Ag^+	+ $2CN^-$	$\rightleftharpoons$	$[Ag(CN)_2^-]$
Initial	0.03M	0.1M		0
At eqm.	x M	$(0.1-0.03\times2)$M		0.03M

$$K_c = 2.5\times10^{20} = \frac{0.03}{x(0.1-0.03\times2)^2}$$

$$\therefore x = [Ag^+] = 7.5\times10^{-20}M$$

9. **HI_n is an acidic indicator ($K_{In} = 10^{-7}$) which dissociates into aqueous acidic solution of 30 ml of 0.15 M H_3PO_4 ($K_1 = 1\times10^{-3}$, $K_2 = 1\times10^{-7}$, $K_3 = 1\times10^{-13}$) then find out the concentration ratio of I_n^- to HI_n. Letting that HI_n and I_n^- posses colour P and Q respectively and colour P predominates over colour Q, when concentration of HI_n is 120 times than that of I_n^- and colour Q predominates over P, and when concentration of I_n^- is 127 times of HI_n. What is the pH range of the indicator? If this solution is titrated with 30 ml of KOH solution, then what molarity of KOH is needed to reach the equivalence point with indicator?**

Sol.

$$H_3PO_4 \overset{K_1=10^{-3}}{\rightleftharpoons} H^+ + H_2PO_4^-$$

	H_3PO_4		H^+	$H_2PO_4^-$
t = 0	C		0	0
t = t	$C(1-\alpha)$		$C\alpha$	$C\alpha$

$$K_1 = \frac{\alpha^2 C}{1-\alpha}$$

$$1\times10^{-3} = \frac{\alpha^2 C}{1-\alpha} = \frac{\alpha^2\times0.15}{1-\alpha}$$

Neglecting "α" w.r.t. 1

$\therefore 1 \times 10^{-3} = \alpha^2 \times 0.15$

$\therefore \alpha = \sqrt{\frac{1\times10^{-3}}{0.15}} = \sqrt{\frac{1}{5}\times10^{-3}} = \sqrt{0.66\times10^{-6}} = 0.816 \times 10^{-3} = 0.82 \times 10^{-3}$

$\therefore [H^+] = \alpha C = 0.82 \times 10^{-3} \times 0.15 = 0.123 \times 10^{-3}$

$[H^+] = 1.23 \times 10^{-4}$

$pH = -\log [H^+] = 3.91$

For an acidic indicator,

$HI_n \overset{K_{in}}{\rightleftharpoons} H^+ + I_n^-$

$\therefore pH = pK_{in} + \log\frac{[I_n^-]}{[HI_n]}$

$3.91 = 7 + \log\frac{[I_n^-]}{[HI_n]}$ ($\because K_{in} = 10^{-7}$)

$\log\frac{[I_n^-]}{[HI_n]} = -3.09$

$\Rightarrow \frac{[I_n^-]}{[HI_n]} = 8.128 \times 10^{-4}$

Now, when P colour predominates over Q colour, then

$pH = pK_{in} + \log 120$

$pH = 7 + \log 120 = 9.0279$

When Q colour predominates over P colour, then

$pH = pK_{in} + \log\frac{1}{127} = 4.896$

For colour P,

$pH \leq 7 + \log\frac{1}{121} = 4.896$

For colour Q.

$pH \geq 7 + \log 120 = 9.0279$

$\therefore$ pH range of the indicator is 4.896 - 9.0279

At 1st neutralization point, the acid H_3PO_4 will be convert into KH_2PO_4, here pH of the solution

$= \frac{pK_1 + pK_2}{2} = \frac{3+7}{2} = 5$

At the 2nd step of neutralization, the acid H_3PO_4 will be converted into K_2HPO_4,

here the pH of the solution $= \frac{pK_2 + pK_3}{2} = \frac{7+13}{2} = 10$

The equivalence point will be obtained when acid H_3PO_4 will change into K_2HPO_4

$\therefore$ n-factor of the acid = 2

$\therefore$ normality of KOH $= \frac{30\times0.15\times2}{30}$ N

Molarity of KOH = **0.3 M**

SOLVED PROBLEMS

Subjective Question

Prob. 1. Calculate the pH of a buffer solution that is 0.250 M in formic acid and 0.100 M in sodium formate. pK_a of formic acid is 3.74.

Sol. $$pH = pK_a + \log\frac{[Salt]}{[Acid]}$$

$$pH = 3.74 + \log\frac{0.1}{0.25} = 3.34$$

Prob. 2. Calculate the pH of 0.1 M solution of $NaHCO_3$

given $CO_2 + H_2O \rightleftharpoons HCO_3^- + H^+$ $\quad K_1 = 4.2\times10^{-7}\,M$

$HCO_3^- \rightleftharpoons H^+ + CO_3^{2-}$ $\quad K_1 = 4.2\times10^{-7}\,M$

Sol. Since in solution of salt containing an ampholeric onion, pH of solution is

$$pH = \frac{1}{2}[pK_1 + pK_2]$$

So pH of $NaHCO_3$ solution

$$pH = \frac{1}{2}[-\log(4.2\times10^{-7}) - \log(4.8\times10^{-11})]$$

$$= 8.35$$

Objective Questions

Prob. 1. The K_{sp} of $Mg(OH)_2$ is 1×10^{-12}. 0.01 M $Mg(OH)_2$ will precipitate at the limiting pH

(A) 3 (B) 9

(C) 5 (D) 8

Sol. $[Mg^{2+}]\,[OH^-]^2 = 10^{-12}$

$0.01 \times [OH^-]^2 = 10^{-12}$

$[OH^-]^2 = 10^{-10}$

$[OH^-] = 10^{-5}$ M

$[H^+] = 10^{-9}$ M, and pH = -log $(10^{-9}) = 9$

Hence, (B) is the correct answer.

Prob. 2. A certain ion B^- has Arrhenius constant for basic character, (e.g. constant 2.8×10^{-7}. The equilibrium constant for Lowry – Bronsted basic character is

(A) 2.8×10^{-7} (B) 3.57×10^{-8}

(C) 3.57×10^{8} (D) 2.8×10^{7}

Sol. $B^- + H^+ \rightleftharpoons HB$, $\quad K_b = \dfrac{[HB]}{[B^-][H^+]}$ ... (i)

Bronsted base

$B^- + H_2O \rightleftharpoons HB + OH^-$, $\quad K_a = \dfrac{[HB][OH^-]}{[B^-]}$ ... (ii)

Arrhenius base

By equation (i) and (ii)

$$\frac{K_a}{K_b} = \frac{[HB][OH^-]}{[B^-]} \times \frac{[B^-][H^+]}{[HB]} = [H^+][OH^-] = K_w$$

$$\therefore K_b = \frac{K_a}{K_w} = \frac{2.8\times10^{-17}}{10^{-14}} = 2.8\times10^{-3}$$

Hence, **(D)** is the correct answer.

Prob. 3. The pOH of 10^{-8} M HCl is

(A) 8 (B) 6

(C) 6 and 7 (D) 7 and 8

Sol. **(C)**

Prob. 4. 2.5 mL of $\frac{2}{5}$ M weak monoacidic base $(K_b = 1\times10^{-12}$ at $25^\circ C)$ is titrated with $\frac{2}{15}$ M HCl in water at $25^\circ C$. The concentration of H^+ at equivalence point is $(K_w = 1\times10^{-14}$ at $25^\circ C)$

(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M

(C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

Sol. $BOH + HCl \longrightarrow BCl + H_2O$

C

$B^+ + H_2O \rightleftharpoons BOH + H^+$

$C(1-h)$ Ch Ch

Volume of HCl used = $= \frac{2.5\times\frac{2}{5}}{2/15} = 7.5$ ml

Concentration of Salt, $C = \frac{2.5\times\frac{2}{5}}{10} = 0.1$ M

$$\therefore \frac{Ch^2}{1-h} = \frac{K_w}{K_b}$$

Solving h = 0.27 (hence h cannot be ignored with respect to 1)

$[H^+] = Ch = 0.1\times0.27 = 2.7\times10^{-2}$ M

Hence, **(D)** is the correct answer.

Prob. 5. What in general may be the general criteria of choosing a suitable indicator for a given titration?

(A) The indicator should have a broad pH range

(B) pH at the end point of titration should be close to neutral point of indicator

(C) Indicator should have neutral at pH = 7

(D) The indicator must show a sharp change in colour near the equivalence point of titration point

Sol. **(B, D)**

Prob. 6. An aqueous solution of 1.0 M ammonium formate assuming complete dissociation and pK_a CHCOOH = 3.8. pK_b (NH_3) = 4.8. With the help of above data find out correct statement out of the following.

(A) pH of solution = 6.5 (B) solution is basic in nature

(C) pH of the solution = 7.5 (D) solution is acidic in nature

Sol. **(A, D)**

By equation (i) and (ii)

$$\frac{K_a}{K_b} = \frac{[HB][OH^-]}{[B^-]} \times \frac{[B^-][H^+]}{[HB]} = [H^+][OH^-] = K_w$$

$$\therefore K_b = \frac{K_a}{K_w} = \frac{2.8\times10^{-17}}{10^{-14}} = 2.8\times10^{-7}$$

Hence, **(D)** is the correct answer.

Prob. 3. The pOH of 10^{-8} M HCl is

(A) 8 (B) 6

(C) 6 and 7 (D) 7 and 8

Sol. **(C)**

Prob. 4. 2.5 mL of $\frac{2}{5}$M weak monoacidic base $\left(K_b = 1\times10^{-12} \text{ at } 25^\circ C\right)$ is titrated with $\frac{2}{15}$M HCl in water at 25°C. The concentration of H^+ at equivalence point is $\left(K_w = 1\times10^{-14} \text{at } 25^\circ C\right)$

(A) 3.7×10^{-13}M (B) 3.2×10^{-7}M

(C) 3.2×10^{-2}M (D) 2.7×10^{-2}M

Sol.

$$BOH + HCl \longrightarrow BCl + H_2O$$

C

$$B^+ + H_2O \rightleftharpoons BOH + H^+$$

$C(1-h)$ Ch Ch

Volume of HCl used = $= \frac{2.5\times\frac{2}{5}}{2/15} = 7.5$ ml

Concentration of Salt, $C = \frac{2.5\times\frac{2}{5}}{10} = 0.1$ M

$$\therefore \frac{Ch^2}{1-h} = \frac{K_w}{K_b}$$

Solving h = 0.27 (hence h cannot be ignored with respect to 1)

$[H^+] = Ch = 0.1\times0.27 = 2.7\times10^{-2}$ M

Hence, **(D)** is the correct answer.

Prob. 5. What in general may be the general criteria of choosing a suitable indicator for a given titration?

(A) The indicator should have a broad pH range

(B) pH at the end point of titration should be close to neutral point of indicator

(C) Indicator should have neutral at pH = 7

(D) The indicator must show a sharp change in colour near the equivalence point of titration point

Sol. **(B, D)**

Prob. 6. An aqueous solution of 1.0 M ammonium formate assuming complete dissociation and pK_a CHCOOH = 3.8. pK_b (NH_3) = 4.8. With the help of above data find out correct statement out of the following

(A) pH of solution = 6.5 (B) solution is basic in nature

(C) pH of the solution = 7.5 (D) solution is acidic in nature

Sol. **(A, D)**

Prob. 7. Which of the following compounds/ions can act as Bronsted acid as well as a Bronsted base?

(A) HCO_3^- (B) K_2CO_3

(C) H_2SO_4 (D) $H_2PO_4^-$

Sol. (A, D)

Prob. 8. If S is solubility of a sparingly soluble salt M_2X_5, its solubility product is equal to

(A) $9\ S^4$ (B) $27\ S^3$

(C) $6\ S^2$ (D) $108\ S^5$

Sol. (D)

Reasoning Type

This section contains Reasoning Type Questions. Each question contains STATEMENT-1 (Assertion) and STATEMENT-2 (Reason). Each question has 4 choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** A solution of 2 : 1 molar ratio of CH_3COONa and HCl results in acidic buffer.
and
STATEMENT-2: An acidic buffer is a mixture of weak acid and its salt with strong base.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

2. Which of the following statement (s) is (are) correct?
(A) The pH of 1.0×10^{-8} M HCl solution is 8
(B) The conjugate base of $H_2PO_4^-$ is HPO_4^{-2}
(C) Auto protolysis constant of water increases with temperature
(D) When a solution of a weak monoprotic acid is titrated against a strong base at half neutralization point, pH = ½ pK_a

Sol. Only (B) and (C) are the correct answers.
Hence **(B)** and **(C)** is the correct answer.

3. Choose the correct statement (s) out of the following
(A) In CH_3COOH/CH_3COONa buffer, the reserve acidity of the solution is due to CH_3COO^- ions
(B) It is not necessary that an acidic substance must contain hydrogen ions
(C) HCO_3^- is both Bronsted base and Bronsted acid
(D) Sodium carbonate does not contain OH^- ions and thus is not a base

Sol. (A), (B), (C) are the correct answers but (D) is wrong.
Hence **(A)**, **(B)** and **(C)** are the correct answer.

COMPREHENSION

Read the following passage and answer the questions:

The product of the concentrations of the ions of an electrolyte raised to power of their coefficients in the balanced chemical equation in the solution at any concentration. Its value is not constant and varies with change in concentration. Ionic product of the saturated solution is called solubility product K_{sp}.

(i) When $K_{ip} = K_{sp}$, the solution is just saturated and no precipitation takes place.

(ii) When $K_{ip} < K_{sp}$, the solution is unsaturated and precipitation will not take place.

(iii) When $K_{ip} > K_{sp}$, the solution is supersaturated and precipitation takes place.

Answer the following questions:

Prob. 1. The solubility product, K_{sp}, of sparingly soluble salt MX at 25°C is 2.5×10^{-9}. The solubility of the salt in mol L^{-1} at this temperature is:

(A) 1×10^{-14} (B) 5×10^{-8}

(C) 1.25×10^{-9} (D) 5×10^{-5}

Sol. **(D)**

Prob. 2. Which of the following is most soluble?

(A) Bi_2S_3 ($K_{sp} = 1 \times 10^{-70}$) (B) MnS ($K_{sp} = 7 \times 10^{-16}$)

(C) CuS ($K_{sp} = 8 \times 10^{-37}$) (D) Ag_2S ($K_{sp} = 6 \times 10^{-51}$)

Sol. **(B)**

Prob. 3. The concentration of Ag^+ ions in a given saturated solution of AgCl at 25°C is 1.06×10^{-5} g ion per litre. The solubility product of AgCl is:

(A) 0.353×10^{-10} (B) 0.530×10^{-10}

(C) 1.12×10^{-10} (D) 2.12×10^{-10}

Sol. **(D)**

MATRIX-MATCH TYPE QUESTION

1. Match the following:

Column – I		Column – II	
(A)	CH_3COOH (pK_a = 4.74, 0.1 M) + CH_3COONa (0.1 M)	(p)	Acidic buffer at it's maximum capacity
(B)	CH_3COOH (0.1 M) + HCl (0.1 M)	(q)	Buffer solution
(C)	CH_3COOH(pK_a = 4.74, 0.1 M) + NH_4OH (pK_b = 4.74, 0.1 M)	(r)	pH < 7 at 25°C
(D)	CH_3COONa (300 ml of 0.1 M) + HCl (100 ml of 0.1 M)	(s)	pH = 7 at 25°C

Sol. **A → (p, q, r), B → (r), C → (q, s), D → (q, r)**

2. Match the following.

Column – I (Mixture)		Column – II (pH)
(A)	50 ml of 0.1 M NaOH + 50 ml of 0.05 M CH_3COOH	(p) 4.74
(B)	50 ml of 0.05 M NaOH + 50 ml of 0.1 M CH_3COOH	(q) 8.72
(C)	50 ml of 0.1 M NaOH + 50 ml of 0.10 M CH_3COOH	(r) 7
(D)	50 ml of 0.05 M NH_4OH + 50 ml of 0.05 M CH_3COOH	(s) 12.39
		(t) 10.2

{Given log 0.333 = 0.48, pK_a for CH_3COOH = 4.74, pK_b for NH_4OH = 4.74)

Sol. **(A) → (s) (B) → (p) (C) → (q) (D) → (r)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column - 1		Column - 2		Column - 3	
(I)	Salt of weak acid and weak base	(i)	$pH=1/2[pK_w+pK_a+\log C]$	(P)	$HCl + NH_4OH \longrightarrow NH_4Cl + H_2O$
(II)	Salt of weak acid and strong base	(ii)	$pH=1/2[pK_w+pK_a-pK_b]$	(Q)	$\begin{matrix}COOH\\ \vert\\ COOH\end{matrix} + 2NaOH \longrightarrow \begin{pmatrix}COO\\ \vert\\ COO\end{pmatrix} Na_2 + 2H_2O$
(III)	Salt of strong acid and strong base	(iii)	$pH=1/2[pK_w-pK_b-\log C]$	(R)	$CH_3-\overset{O}{\overset{\Vert}{C}}-OH + Fe(OH)_3 \longrightarrow \left(CH_3-\overset{O}{\overset{\Vert}{C}}-O\right)_3 Fe + 3H_2O$
(IV)	Salt of strong acid and weak base	(iv)	$pH=1/2[pK_w]$	(S)	$HCl + NaOH \longrightarrow NaCl + H_2O$

3. Which of the following combination is correct ?
[A] (I) (ii) (R) [B] (III) (i) (P) [C] (III) (ii) (Q) [D] (IV) (ii) (S)

4. Which of the following combination is correct for salt of strong acid and strong base ?
[A] (III) (i) (P) [B] (III) (iii) (S) [C] (I) (iii) (Q) [D] (III) (iv) (S)

5. For the given salt in Column-I which the incorrect combination for salt of strong acid and weak base ?
[A] (III) (iv) (Q) [B] (II) (ii) (S) [C] (IV) (iii) (S) [D] (I) (i) (R)

Sol.

3. **A**
4. **D**
5. **C**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. Calculate the percentage dissociation of 0.5 M NH_3 at 25°C in a solution of pH = 12.

Sol. **(2)**

$$NH_4OH_{(c)} \rightleftharpoons NH_4^+ + OH^-$$

$$C \qquad 0 \qquad 0$$

$$C(1-\alpha) \qquad C\alpha \qquad C\alpha$$

pH = 12, $[H^+] = 10^{-12}$, or $[OH^-] = 10^{-2}$

$[OH^-] = C\alpha = 10^{-2}$

$$\alpha = \frac{10^{-2}}{c} = \frac{10^{-2}}{0.5}$$

2×10^{-2} or 2%

2. 0.1 M HA is titrated against 0.1 M NaOH. Find the pH at the end point. Dissociation constant for the acid HA is 5×10^{-6} and degree of hydrolysis, h < 1.

Sol. **(9)**

$$HA + NaOH \rightleftharpoons NaA + H_2O$$

(weak acid)

In the resultant solution, $[NaA] = \frac{0.1}{2} = 0.05$ M

As NaA is a salt of weak acid and strong base it hydrolysis as

$$A^- + H_2O \rightleftharpoons HA + OH^-$$

$pH = 7 + \frac{1}{2}[pKa + \log C]$

$= 7 + \frac{1}{2}[-\log(5 \times 10^{-6}) + \log 0.05]$

$= 7 + \frac{1}{2}[6 - 0.6990 + 0.6990 - 2] = 9$

3. The dissociation constant of a substituted benzoic acid at 25°C is 1.0×10^{-4}. The pH of a 0.01 M solution of its sodium salt is

Sol. **(8)**

$K_a (C_6H_5COOH) = 1 \times 10^{-4}$

pH of 0.01 M C_6H_5COONa

$$\underset{0.01(1-h)}{C_6H_5COO^-} + H_2O \rightleftharpoons \underset{0.01\,h}{C_6H_5COOH} + \underset{0.01\,h}{OH^-}$$

$$K_h = \frac{K_w}{K_a} = \frac{0.01\,h^2}{1-h}$$

$$\frac{10^{-14}}{10^{-4}} = \frac{10^{-2}h^2}{1-h} \quad (1-h=1)$$

$[OH^-] = 0.01\,h = 0.01 \times 10^{-4} = 10^{-6}$

$[H^+] = 10^{-8}$

pH = 8

4. 0.15 mole of pyridimium chloride has been added into 500 cm^3 of 0.2 M pyridine solution. Calculate pH of the resulting solution assuming no change in volume. ($K_{b\,(pyridium)} = 1.5 \times 10^{-9}$ M)

Sol. **(5)**

$$[\text{Pyridium chloride}] = \frac{0.15}{500} \times 1000 = 0.3 \text{ M}$$

Pyridine = 0.2 M

A mixture of pyridine and salt pyridinium chloride forms a basic buffer.

$$\therefore pOH = pK_b + \log\frac{[Salt]}{[Base]}$$

$$\Rightarrow pOH = -\log 1.5 \times 10^{-9} + \log\frac{[0.30]}{[0.20]}$$

$$= -\log 1.5 + 9\log_{10}^{10} + \log 1.5 = 9$$

$\therefore [OH^-] = 10^{-9}$

$[H^+] = 10^{-5}$

so, pH = 5.

5. 0.1 millimole of $CdSO_4$ are presence in 10 mL acid solution of 0.08 N HCl. Now H_2S is passed to precipitate all the Cd^{2+} ions. What would be the pH of solution after filtering of precipitate boiling of H_2S and making the solution 100 mL by adding H_2O?

Sol. **(2)**

$$CdSO_4 + HCl + H_2S \longrightarrow CdS + H_2SO_4$$

Millimole added	0.1	$10 \times 0.08 = 8$		
After reaction	0	0.8	0.1	0.1

Millimole of H^+ left = 0.8 (From HCl) + $0.1 \times 2 = 1.0$ (From H_2SO_4)

Total voume = 100 mL

$[H^+] = \frac{1}{100} = 10^{-2}$M, pH = 2.

6. K_a for butyric acid is 2×10^{-5}. Calculate pH in 0.2 M aqueous solution of sodium butyrate.

Sol. **(9)**

$$NaBu + H_2O \rightleftharpoons NaOH + BuH$$

	1	0	0
	(1 – h)	h	h

$$[OH^-] = Ch = C\sqrt{\frac{Kh}{C}} = \sqrt{K_n C} = \sqrt{\frac{K_w . C}{K_a}}$$

$$= \sqrt{\frac{10^{-14} \times 0.2}{2 \times 10^{-5}}} = \sqrt{10^{-10}} = 10^{-5}$$

$[H^+] = 10^{-9}$

$\therefore$ pH = 9

7. A certain buffer solution contains equal concentration of X^- and HX (K_b for X^- is 10^{-10}) what is pH of buffer ?

Sol. **(4)**

For a conjugate acid –base pair we have

$K_a(HX) \times K_b(X^-) = 10^{-4}$

$$K_a = \frac{10^{-14}}{10^{-10}} = 10^{-4}$$

$[HX^-] = [X^-]$

$$\text{So } pH = pK_a + \log\frac{[salt]}{[acid]}$$

= 4

8. 50 mL of 0.05 M Na_2CO_3 is titrated against 0.1 M HCl. On adding 40 mL of HCl, pH of the solution will be [Given : For H_2CO_3, $pK_{a_1} = 6.35$, $pK_{a_2} = 10.33$; log3 = 0.477, log2 = 0.30]. (Express answer in nearest integer).

Sol. (6)

	CO_3^{2-}	+ H^+	⟶	HCO_3^-
Initial milli moles	50×0.05	40×0.1		–
Final	–	1.5		2.5

	HCO_3^-	+ H^+	⟶	H_2CO_3
Initial milli moles	2.5	1.5	–	
Final	1	1.5		1.5

$$pH = pK_{a_1} + \log\frac{[HCO_3^-]}{[H_2CO_3]} = 6.173 \approx 6$$

9. In a baffer solution, conc. of $(NH_4)_2SO_4$ and NH_4OH are 0.2 M and 0.4 M, calculate pH of solution If $K_{NH_4^+} = 10^{-9}$

Sol. (9)

$$pOH = pK_b + \log\frac{[salt]}{[base]}$$

$(pK_b = -\log K_b = -\log K_{NH_4OH})$

$K_{NH_4^+} = 10^{-9}$ $\therefore$ $NH_4^+ \times K_{NH_4OH} = K_w$

$$K_{NH_4OH} = \frac{10^{-14}}{10^{-9}} = 10^{-5}$$

$$pOH = -\log 10^{-5} + \log\frac{0.4}{0.4} = 5$$

$$pH = 10 - pOH = 14 - 5 = 9$$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. The total number of different kind of buffers obtained during the titration of H_3PO_4 with NaOH are
Sol. **3.00**

Prob. 2. What is the pH of an aqueous solution of ammonium acetate ($K_a = K_b = 1.8 \times 10^{-5}$) ?
Sol. **7.0**

Prob. 3. What is the solubility of $Cd(OH)_2$ in a buffer solution having pH = 8 ? [$K_{sp}(Cd(OH)_2) = 2.5 \times 10^{-14}$]
Sol. **0.025**

Prob. 4. If the equilibrium constant for the reaction of weak acid HA with a strong base is 10^9, then determine pH of 0.1 M NaA solution.
Sol. **9.00**

Prob. 5. A small quantity of phenolphthalein (an acid indicator) is added to a decimolar solution of Sodium butyrate. Calculate the ratio of the coloured to the colourless form of the indicator. K_a for butyric acid $= 1.5 \times 10^{-5}$, K for the indicator 3.075×10^{-10} and $K_w = 10^{-14}$. Take log 1.23 = 0.09
Sol. **0.25**

ELECTROCHEMISTRY 7

TYPES OF CELLS

Electrolytic Cell

It is a device in which conversion of electrical energy into chemical energy is done.

Electrochemical Cell

It is also known as Galvanic or Voltaic cell. It is a device in which chemical energy is converted into electrical energy.

FARADAY'S LAWS OF ELECTROLYSIS

First Law

Weight of substance deposited (W) $\propto$ quantity of electricity passed

$W \propto Q$

or $W = ZQ$

or $W = Zit$

where Z = electrochemical equivalent,

i = current in amperes

t = time in seconds

Second Law

$\frac{W}{E}$ = constant

where E = equivalent weight

$\therefore \frac{W_1}{W_2} = \frac{E_1}{E_2}$

One Faraday (F) is the quantity of charge carried by one mole of electrons.

1 F = 96,487 coulombs $\cong$ 96500 coulombs.

By combining first and second law, we get,

$\frac{W}{E} = \frac{i \times t}{96500}$

ELECTROLYTIC CONDUCTANCE

$$\text{Conductance (C)} = \frac{1}{\text{Resistance(R)}}$$

The unit of conductance is ohm^{-1} or mho.

The electrolytic conductance of a solution depends upon the following factors:

(i) Number of ions
(ii) Mobility of ions
(iii) Magnitude of charge on each ion

SPECIFIC RESISTANCE AND SPECIFIC CONDUCTANCE

If 'ℓ' is the distance between two electrodes, 'a ' is the area of cross-section of the electrode.

R is resistance

$$R \propto \frac{\ell}{a}$$

$$R = \rho \times \frac{\ell}{a} \qquad \left(\frac{\ell}{a} \text{ is called cell constant}\right)$$

Where ρ specific resistance or resistivity.

$$\frac{1}{\rho} = \frac{1}{R} \times \left(\frac{\ell}{a}\right)$$

Specific conductance $(\kappa) = C \times \frac{\ell}{a}$

$\frac{1}{\rho}$ = (specific conductance or conductivity)

The unit of κ is Ω^{-1} cm^{-1} or S m^{-1}.

κ is also defined as conductance per cc of electrolytes.

Equivalent Conductance $(\wedge_e)$

$\wedge_e = \kappa \times V$

where V = Volume of solution in cc containing 1 gram-equivalent of the electrolytes

or $\wedge_e = \kappa \times \frac{1000}{C(gm.eq/litre)} = \kappa \times \frac{1000}{N}$

where N = normality

Molar Conductance $(\wedge_m)$

$\wedge_m$ = Specific conductance × Volume in cc containing one mole of the electrolyte

$$\wedge_m = \kappa \times \frac{1000}{C(mole/lit)} = \kappa \times \frac{1000}{M}$$

where M = molarity

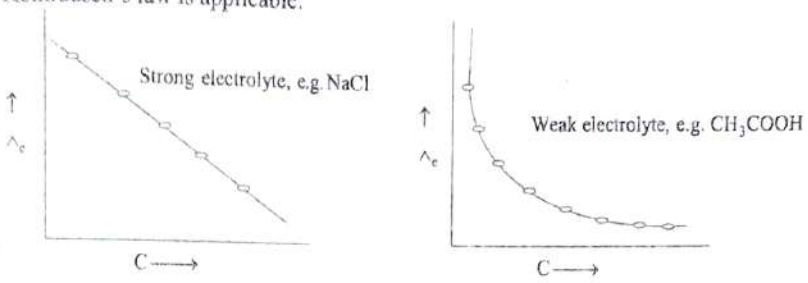
The unit of specific conductance is Ω^{-1} cm^2 mol^{-1} and in SI system, the unit is Sm^2 mol^{-1}.

Thus, $\wedge_m = n \wedge_e$

where n = n-factor of electrolyte = $\frac{M}{E}$

Effect of Dilution on Conductance

The value of Λ_c and Λ_m increases and attains a limiting value at infinite dilution, i.e. Λ_e^{∞} and Λ_m^{∞}. In case of weak electrolytes, it is mainly due to increase in number of ions (due to increase in degree of dissociation), upon dilution. But in case of strong electrolytes, it is due to increase in mobility of ions.
The value of Λ_e^{∞} and Λ_m^{∞} of strong electrolytes can be extrapolated experimentally while for weak electrolytes Kohlrausch's law is applicable.



KOHLRAUSCH'S LAW

At infinite dilution, molar conductivity is given by the sum of equivalent conductivities of contributing ions.
For example, for A_xB_y

$$\Lambda_m^{\infty} = x\Lambda_A^{\infty} + y\Lambda_B^{\infty}$$

Applications

(i) Determination of degree of dissociation (α)

$$\alpha = \frac{\Lambda_m}{\Lambda_m^{\infty}}$$

(ii) Determination of solubility of sparingly soluble salts

$$\Lambda_m^{\infty} = \frac{1000\kappa}{C}$$

Here, C is molarity of solution or solubility of solute

(iii) Calculation of dissociation constant of weak electrolyte

$$\alpha = \frac{\Lambda_m}{\Lambda_m^{\infty}} \text{ and } K_a = \frac{C\alpha^2}{1-\alpha} \approx C\alpha^2 \text{ when } \alpha << 1$$

Similarly, K_b can be calculated for weak base.

ELECTROCHEMICAL CELLS

(i) It consists of usually two half-cells, which are linked by an electrical connection and salt bridge. Salt bridge is made up of inert electrolyte and its function is to maintain electrical neutrality in solutions of half-cell.

(ii) The electrode at which oxidation takes place is called anode (negatively charged) and the electrode at which reduction takes place is called cathode (positively charged).

(iii) Electrode potential is defined as the tendency of an electrode to lose (oxidation potential) or gain (reduction potential) electrons. The relative value of electrode potential is measured by the reference electrode (standard hydrogen electrode or SHE).

(iv) IUPAC cell representation.

$$Zn \mid \underset{C_1}{ZnSO_4} \parallel \underset{C_2}{CuSO_4} \mid Cu$$

Anode Cathode

C_1 and C_2 are concentrations in mole/ litre of electrolyte solutions.

Two parallel vertical lines (||) represents salt bridge.

In case of a gas, the gas is indicated after the electrode in case of anode and before the electrode in case of cathode.

$$Pt, H_2 \mid H^+ \text{ or } H^+ \mid H_2, Pt$$

Anode Cathode

(v) EMF or $E_{cell} = E_{rp(cathode)} - E_{rp(anode)}$

$E_{cell} = E_{op(anode)} + E_{rp(cathode)}$

where rp = reduction potential

op = oxidation potential

or $E^\circ_{cell} = E^\circ_{rp(cathode)} - E^\circ_{rp(anode)}$

For spontaneous cell reaction, E_{cell} should be positive.

(vi) Nernst equation

$$E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF}\log K$$

or $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n}\log K$ (when T 298 K, R = 8.314 J mol^{-1} K^{-1}, F = 96500 C)

At equilibrium $E_{cell} = 0$

$$\therefore E^\circ_{cell} = \frac{0.0591}{n}\log Keq$$

Applications of EMF Measurement

(i) Determination of equilibrium constant.

(ii) Determination of pH of the solution.

$Pt(H_2)|H^+(pH=?) \parallel Cl^-(1M) \mid Hg_2Cl_2$, Hg (Calomel electrode or reference electrode)

Cell reaction:

$$H_2(g) \longrightarrow 2H^+ + e^- \qquad E^\circ_o = 0.0 \text{ volt}$$

$$Hg_2Cl_2(s) + 2e^- \longrightarrow 2Hg(l) + 2Cl^-(aq) \qquad E^\circ_{red} = 0.2676 \text{ V}$$

$$H_2(g) + Hg_2Cl_2(s) \longrightarrow 2H^+(aq) + 2Hg(l) + 2Cl^-(aq) \qquad E^\circ_{cell} = 0.2676 \text{ V}$$

$E_{cell} = E^\circ_{cell} - \frac{0.0591}{2}\log\left[H^+\right]^2$ as a activities of all other reagents and products can be taken.

$$\text{or } pH = \frac{E_{cell} - E^\circ_{cell}}{0.0591}$$

(iii) Determination of K_{sp} of sparingly soluble salts.

Cell reaction:

$$Ag(s) \longrightarrow Ag^+ + e^- \qquad E^\circ_o = E^\circ_{Ag/Ag^+}$$

$$AgX(s) + e^- \longrightarrow Ag(s) + X^- \qquad E^\circ_{Ag^+/Ag}$$

$$AgX(s) \longrightarrow Ag^+ + X^- \qquad E^\circ_{cell} = E^\circ_o + E^\circ_{red}$$

$$K = \frac{[Ag^+][X^-]}{[AgX]} = K_{sp}$$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{1}\log K_{sp}$$

At equilibrium, $E_{cell} = 0$. Thus, $E^{\circ}_{cell} = 0.0591 \log K_{sp}$

(iv) Determination of ionization constant of weak acid or weak base.

For cell, $Pt(H_2)|H^+, HA(C_1) \| Cu^{+2}(C_2) | Cu$

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{2}\log\frac{\left[H^+\right]^2}{\left[Cu^{2+}\right]}$$

$$\text{or } E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{2}\log\frac{\left(\sqrt{K_a C_1}\right)^2}{C_2}$$

(v) Determination of thermodynamic data.

$$\Delta G^{\circ} = \Delta H^{\circ} + T\left[\frac{d(\Delta G^{\circ})}{dT}\right]_p \text{ (Gibbs–Helmholtz equation)}$$

Applying $\Delta G^{\circ} = -nFE^{\circ}_{cell}$

$$\left(\frac{dE^{\circ}_{cell}}{dT}\right)_p = -\frac{\Delta H^{\circ}}{nFT} + \frac{E^{\circ}_{cell}}{T} \text{ (Temperature coefficient of the emf of the cell)}$$

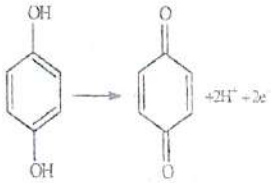
On comparing Gibbs–Helmholtz equation with $\Delta G^{\circ} = \Delta H^{\circ} - T\Delta S^{\circ}$, we get

$$\Delta S = -\left[\frac{d(\Delta G)}{dT}\right]_p$$

Applying $\Delta G = -nFE_{cell}$

$$\Delta S = nF\left(\frac{dE_{cell}}{dT}\right)_p$$

TYPES OF HALF-CELL

S. No	Type	Example	Half-cell reaction
1.	Metal-metal ions half-cell	$Ag\|Ag^+(aq)$	$Ag(s) \longrightarrow Ag^+ + e^-$
2.	Gas ion half-cell	$Pt(H_2)\|H^+(aq)$	$H_2 \longrightarrow 2H^+ + 2e^-$
3.	Metal insoluble salt anion half-cell	$Hg, Hg_2Cl_2\|Cl^-(aq)$ calomel electrode	$2Hg(l) + 2Cl^-(aq) \longrightarrow Hg_2Cl_2 + 2e^-$
4.	Oxidation/ reduction half-cell	$Pt\|Fe^{2+}, Fe^{3+}$	$Fe^{2+} \longrightarrow Fe^{3+} + e^-$
5.	Quinohydrone	Pt |Quinohydrone| $H^+(aq)$	
6.	Metal, metal oxide hydroxide half-cell	$Hg/HgO\|OH^-(aq)$	$Hg(l)+2OH^-(aq) \rightarrow HgO(s)+H_2O(l)+2e^-$

CONCENTRATION CELLS

Electrolyte Concentration Cell

Electrodes are identical and immersed in solution of the same electrolyte of different concentrations. EMF generated due to tendency of electrolyte to diffuse from higher concentration to lower concentration. As the concentrations becomes equal, the EMF tends to become zero.

$Zn|Zn^{2+}(C_1) \,||\, Zn^{2+}(C_2)|Zn$

$E_{cell} = E_{cathode} - E_{anode}$

$$E_{cell} = \frac{0.0591}{n}\log\frac{C_2}{C_1}$$

Electrode Concentration Cell

Potential difference between identical electrodes at different activities, dipped in same electrolytic solution.

$Pt(H_2) \mid HCl\ (1M) \mid Pt(H_2)$

$P_1 \qquad\qquad P_2$

$$\frac{1}{2}H_2(P_1) \longrightarrow H^+(aq) + e^-$$

$$\frac{H^+(1M) + e^- \longrightarrow \frac{1}{2}H_2(P_2)}{\frac{1}{2}H_2(P_1) \longrightarrow \frac{1}{2}H_2(P_2)} \quad E^\circ_{cell} = 0.0$$

$$E_{cell} = \frac{-0.0591}{2}\log\frac{P_2}{P_1}$$

$$E_{cell} = \frac{0.0591}{2}\log\frac{P_1}{P_2}$$

ILLUSTRATIVE PROBLEMS

1. Calculate the potential of an indicator electrode versus the standard hydrogen electrode which originally was 0.1M MnO_4^- and 0.8M H^+ and which has been treated with 90% of the Fe^{2+} necessary to reduce all the MnO_4^- to Mn^{2+} by the reaction.
$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ $E^\circ = 1.51$ V

Sol. The Galvanic cell under consideration is

$Pt \mid H_2(g)1bar \mid H^+(a = 1) \parallel MnO_4^-, Mn^{2+}, H^+ \mid Pt$

Anode half cell : $H^+(a = 1) \longrightarrow \frac{1}{2}H_2(1bar) + e$

Cathode half cell:	MnO_4^-	+ $8H^+$	+ $5e$	$\longrightarrow$ Mn^{+2}	+ $4H_2O$
Initial Conc.:	0.1	0.8		0	0
After reaction with Fe^{+2}	(0.01)	(0.08)		(0.09)	

So, electrode potential of indicator electrode

$$E_{MnO_4^-/Mn^{+2}} = E^\circ_{MnO_4^-/Mn^{+2}} - \frac{0.0591}{5}\log\frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8} \quad (E^\circ_{SHE} = 0)$$

$$= 1.51 - \frac{0.0591}{5}\log\frac{(0.09)}{(0.01)(0.08)^8}$$

$$= 1.51 - \frac{0.0591}{5}\log\frac{9}{1.67\times10^{-9}}$$

$$= 1.51 - \frac{0.0591}{5}\log(5.36\times10^9)$$

$$= 1.51 - 0.1149$$

$$= \mathbf{1.395\ V}$$

Thus, potential of an indicator electrode versus the SHE is 1.395 V.

2. In an electrolysis experiment, a constant current was passed for 5 hours through two cells connected in series. The first cell contains a solution of gold salt and the second cell contains copper sulphate solution. 9.85 g of gold was deposited in the first cell. If the oxidation number of gold is +3, find the amount of copper deposited on the cathode in the second cell. Also calculate the magnitude of the current in ampere.

Sol. $$\frac{\text{Mass Au deposited}}{\text{Mass of Cu deposited}} = \frac{\text{Eq mass of Au}}{\text{Eq mass of Cu}}$$

$$\text{Eq mass of Au} = \frac{197}{3}$$

$$\text{Eq mass of Cu} = \frac{63.5}{2}$$

$$\therefore \frac{9.85}{\text{Mass of Cu deposited}} = \frac{\frac{193}{3}}{\frac{63.5}{2}}$$

$\therefore$ Mass of Cu deposited $= 9.85 \times \frac{63.5}{2} \times \frac{3}{197} = 4.7625$ g.

Applying $w = \frac{Zit}{96500}$

$\Rightarrow 4.7625 = \frac{63.5/2 \times i \times 5 \times 60 \times 60}{96500}$

$\therefore i = \frac{4.7625 \times 2 \times 96500}{63.5 \times 5 \times 3600} = 0.804$ A

3. Estimate the cell potential of a Daniel cell having 1.0 M Zn^{++} and originally having 1.0 M Cu^{+2} after sufficient ammonia has been added to the cathode compartment to make the NH_3 concentration 2.0 M. Given $E^0_{Zn/Zn^{+2}}$ and $E^0_{Cu^{+2}/Cu}$ are 0.76 and + 0.34 volt respectively. Also equilibrium constant for the $[Cu(NH_3)_4]^{+2}$ formation is 1×10^{12}.

Sol. $Cu^{+2} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{+2}$

Since due to high value of K_f almost all of the Cu^{+2} ioins are converted to $[Cu(NH_3)_4]^{+2}$ ion.

Since originally $[Cu^{++}] = 1M$

i.e. $[Cu(NH_3)_4]^{+2} \cong 1$ M

$\because K_f = \frac{[Cu(NH_3)_4]^{+2}}{[Cu^{++}][NH_3]^4}$

or, $1 \times 10^{12} = \frac{1}{x \times (2.0)^4}$

$\therefore x = 6.25 \times 10^{-14}$ M

From Nernst equation, for a Daniel cell

$E_{cell} = E^0_{cell} - \frac{0.0593}{n} \log \frac{[Zn^{++}]}{[Cu^{+2}]}$ (Since activities of Cu and Zn are unity)

$= E^\circ_{Cu^{+2}/Cu} - E^\circ_{Zn^{+2}/Zn} - \frac{0.0593}{2} \log \frac{1}{6 \times 10^{-14}}$

$= 0.34 + 0.76 - \frac{0.0593}{2}[14 - \log 6] = 0.71$ V

4. Determine the degree of hydrolysis and hydrolysis constant of aniline hydrochioride in M/32 solution of salt at 298 K from the following cell data also at 298 K.
Pt | H_2 (1 bar) | H^+ (1M) || M/32 $C_6H_5NH_2HCl$ (aq) | H_2 (1 bar) | Pt, $E_{cell} = -0.188$ V

Sol. We can consider the given cell as a combination of two hydrogen electrodes

$\Rightarrow E_{cell} = -0.0591 \log \frac{1}{[H^+]_R}$

$\Rightarrow [H^+] = 6.63 \times 10^{-4}$ moldm^{-3}

$C_6H_5NH_3^+ + H_2O \rightleftharpoons C_6H_5NH_2 + H_3O^+$

$[H_3O^+] = Ch$

$h = \frac{6.63 \times 10^{-4}}{1/32} = 2.118 \times 10^{-2}$

$K_h = \frac{[C_6H_5NH_2][H_3O^+]}{[C_6H_5NH_3^+]} = \frac{(ch)(ch)}{c(1-h)} = 1.43 \times 10^{-5}$ moldm^{-3}

5. Chromium metal can be plated out from an acidic solution containing CrO_3 according to the following equation:

$$CrO_3(aq) + 6H^+ + 6e^- \longrightarrow Cr(s) + 3H_2O$$

Calculate,

(i) how many grams of chromium will be plated out by 24000 coulombs?

(ii) how long will it take to plate out 1.5 gm of Cr using 12.5 ampere current?

Sol. (i) Eq. wt. of Cr $= \dfrac{\text{Atomic weight}}{\text{Change in oxidation number per mole}} = \dfrac{52}{6}$

$\because$ 96500 coulomb deposit $= \dfrac{52}{6}$ gm (Cr)

$\therefore$ 24000 coulomb will deposit $= \dfrac{52}{6} \times \dfrac{24000}{96500}$ gm $= 2.1554$ gm of Cr

(ii) $W_{Cr} = 1.5$gm, $i = 12.5$ ampere

$E_{Cr} = \dfrac{52}{6}$, $t = ?$

$W = \dfrac{Eit}{96500} \Rightarrow 1.5 = \dfrac{52 \times 12.5 \times t}{6 \times 96500}$

$\therefore t = 1336.15$ sec.

6. Calculate the quantity of electricity delivered by a Daniel cell initially containing 1 L each of 1M Cu^{2+} ion and 1M Zn^{2+}, which is operated until its potential drops to 1 V. (Given: $E^\circ_{Zn^{2+}/Zn} = -0.76$ V; $E^\circ_{Cu^{2+}/Cu} = +0.34$ V)

Sol. At anode : $Zn \rightarrow Zn^{2+} + 2e^-$

At cathode : $Cu^{2+} + 2e^- \rightarrow Cu$

Cell reaction : $Zn + Cu^{2+} \rightarrow Zn^{2+} + Cu$

As electricity is withdrawn from the cell, the concentration of Cu^{2+} decreases while that of Zn^{2+} increases

$$E_{cell} = E^\circ_{cell} - \frac{0.059}{n} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$1 = 1.103 - \frac{0.059}{2} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$\therefore \frac{[Zn^{2+}]}{[Cu^{2+}]} = 3041$$

If x mole/lit is the amount of Cu^{2+} that is converted to Cu when the cell potential drops to 1 V then

$$\frac{[Zn^{2+}]}{[Cu^{2+}]} = \frac{1+x}{1-x} = 3041$$

$$x = \frac{3040}{3042} = 0.9993$$

To change Cu^{2+} to Cu and Zn to Zn^{2+}, 2 electrons ion are involved, hence quantity of charge drawn from the cell $= 2 \times 0.9993 \times 96500 = 1.92 \times 10^5$ C.

7. To find the standard potential of M^{3+}/M electrode, the following cell is constituted:
$Pt | M | M^{3+}(0.0018\ mol^{-1}L) \| Ag^{+}(0.01\ mol^{-1}L) | Ag$
The emf of this cell is found to be 0.42 volt. Calculate the standard potential of the half reaction
$M^{3+} + 3e^{-} \longrightarrow M$. $E^{\circ}_{Ag^{+}/Ag} = 0.80$ volt

Sol. The cell reaction is $M + 3\,Ag^{+} \longrightarrow 3\,Ag + M^{3+}$

Applying Nerst equation,

$$E_{Cell} = E^{\circ}_{Cell} - \frac{0.0591}{3}\log\frac{[M^{3+}]}{[Ag^{+}]^{3}}$$

$$0.42 = E^{\circ}_{Cell} - \frac{0.0591}{3}\log\frac{(0.0018)}{(0.01)^{3}} = E^{\circ}_{Cell} - 0.064$$

$$E^{\circ}_{Cell} = (0.42 + 0.064) = 0.484 \text{ volt}$$

$$E^{\circ}_{Cell} = E^{\circ}_{Cathode} - E^{\circ}_{Anode}$$

or $E^{\circ}_{Anode} = E^{\circ}_{Cathode} - E^{\circ}_{Cell}$

$=(0.80 - 0.484) = 0.32$ volt

8. During the discharge of a lead storage battery, density of H_2SO_4 fell from 1.294 to 1.139 g/mL. Sulphuric acid of density 1.294 is 39% H_2SO_4 by weight and that of density 1.139 g/mL is 20% H_2SO_4 by weight. The battery holds 3.5 litre of the acid and volume remains practically constant during discharge. Calculate ampere-hour of which the battery must have been used. The charging and discharging reactions are:

$Pb + SO_4^{2-} \longrightarrow PbSO_4 + 2e^{-}$ (charging)

$PbO_2 + 4H^{+} + SO_4^{2-} + 2e^{-} \longrightarrow PbSO_4 + 2H_2O$ (discharging)

Sol. Weight of solution before discharge = $3500 \times 1.294 = 4529$ g

Weight of H_2SO_4 before discharge = $\frac{39}{100} \times 4529 = 1766.31$ g

Weight of solution after discharge = $3500 \times 1.139 = 3986.5$ g

Weight of H_2SO_4 after discharging = $\frac{20}{100} \times 3986.5 = 797.3$ g

Loss in mass of H_2SO_4 during discharge = 1766.31 - 797.3 = 969.01 g

Now from first law of electrolysis, $W = \frac{Q \times E}{96500}$

$$969.01 = \frac{Q \times 98}{96500}$$

$Q = 954178.21$ coulomb

$$\text{Ampere} - \text{hour} = \frac{\text{Coulomb}}{3600} = \frac{954178.21}{3600}$$

$= 265.04$ ampere-hour.

9. The chlorate ion can disproportionate in basic solution according to reaction:
$2ClO_3^- \rightleftharpoons ClO_2^- + ClO_4^-$
What is the equilibrium concentration of perchlorate ions from a solution initially at 0.1 M in chlorate ions at 298 K? (Express answer in units of 10^{-3}).
Given : $E^\circ_{ClO_4^-/ClO_3^-} = 0.36$ V and $E^\circ_{ClO_3^-/ClO_2^-} = 0.33$ V at 298 K

Sol. 19

$$H_2O(\ell) + ClO_3^-(aq) \longrightarrow ClO_4^-(aq) + 2H^+(aq) + 2e^-$$
$$2H^+(aq) + ClO_3^-(aq) + 2e^- \longrightarrow 2ClO_2(aq) + H_2O(\ell)$$
$$2ClO_3^-(aq) \longrightarrow ClO_2^-(aq) + ClO_4^-(aq)$$

$E^\circ_{cell} = 0.33 - 0.03$; $E^\circ_{cell} = \frac{RT}{2F}\ln K$

$-0.03 = \frac{0.06}{2}\log K$ or $K = 0.1$

$2ClO_3^- \rightleftharpoons ClO_4^- + ClO_2^-$

$0.1 - 2x \qquad x \qquad x$

$\frac{x^2}{(0.1-2x)^2} = \frac{1}{10}$; $3.16x = 0.1 - 2x$

$5.16x = 0.1$

$x = 0.1/5.16 = 0.0193 \cong 19 \times 10^{-3}$

SOLVED PROBLEMS

Subjective Questions

Prob. 1. Calculate the emf of the cell.

$$Mg(s)\left|Mg^{2+}(0.2M)\right\|Ag^{+}\left(1\times10^{-3}M\right)\left|Ag\right.$$

$E^{\circ}_{Ag^{+}/Ag} = +0.8$ volt, $E^{\circ}_{Mg^{2+}/Mg} = -2.37$ volt

What will be the effect on emf if concentration of Mg^{2+} ion is decreased to 0.1 M?

Sol. $E^{\circ}_{Cell} = E^{\circ}_{Cathode} - E^{\circ}_{Anonde}$

$= 0.80 - (-2.37) = 3.17$ volt

Cell reaction. $Mg + 2\,Ag^{+} \longrightarrow 2\,Ag + Mg^{2+}$

$$E_{Cell} = E^{\circ}_{Cell} - \frac{0.0591}{n}\log\frac{Mg^{2+}}{\left[Ag^{+}\right]^{2}}$$

$$= 3.17 - \frac{0.0591}{2}\log\frac{0.2}{\left[1\times10^{-3}\right]^{2}}$$

$= 3.17 - 0.1566 = 3.0134$ volt

When $Mg^{2+} = 0.1$ M

$$E_{Cell} = E^{\circ}_{Cell} - \frac{0.0591}{2}\log\frac{0.1}{\left(1\times10^{-3}\right)^{2}}$$

$= (3.17 - 0.1477)$ volt

$=$ **3.0223 Volt.**

Objective Questions

Prob. 1. The standard electromotive force of the cell

$Fe|Fe^{2+}{}_{(aq)} \| Cd^{2+}{}_{(aq)}| Cd$

If the temperature co-efficient of emf is -0.125 V K^{-1}, the value of ΔS° at 25°C would be

(A) -20. 125 KJ K^{-1} (B) -24.125 KJ K^{-1}

(C) -26.125 KJ K^{-1} (D) -22.125 KJ K^{-1}

Sol. $$\frac{dE^{\circ}}{dT} = \frac{\Delta S^{\circ}}{nF}$$

$$\Rightarrow \Delta S^{\circ} = \left(\frac{dE^{\circ}}{dT}\right)\times nF$$

$= -0.125\times2\times96500$

$= -24.125\times$ KJ K^{-1}

Hence, **(B)** is the correct answer.

Prob. 2. The solution of $CuSO_4$ in which copper rod is immersed is diluted to 10 times, the reduction electrode potential will

(A) Increase by 0.030 V (B) Decrease by 0.030 V

(C) Increase by 0.059 V (D) Decrease by 0.059 V

Sol. $Cu^{2+} + 2e \longrightarrow Cu$

$E_1 = E^{\circ}_{Cu^{+2}/Cu} - \frac{0.0591}{2}\log\left(\frac{1}{Cu^{2+}}\right)_1$ as $E^{\circ}_{SHE} = 0$ and $^aH_2(g)$ and $^aH_3O^+$ in SHE are 1.

$E_2 = E^{\circ}_{Cu^{+2}/Cu} - \frac{0.0591}{2}\log\left(\frac{1}{Cu^{2+}}\right)_2$ as $E^{\circ}_{SHE} = 0$ and $^aH_2(g)$ and $^aH_3O^+$ in SHE are 1.

$E_2 = E^{\circ}_{Cu^{+2}/Cu} - \frac{0.0591}{2}\log\frac{10}{(Cu^{2+})_1}$

$= E^{\circ}_{Cu^{+2}/Cu} - \frac{0.0591}{2}\log\frac{1}{(Cu^{2+})_1} - \frac{0.0591}{2}\log 10$

$= E_1 - 0.030$ V

Hence, **(B)** is the correct answer.

Prob. 3. If $E^{\circ}_{Fe^{+2}/Fe} = -0.44$ V and $E^{\circ}_{Fe^{+3}/Fe^{+2}} = 0.77$ V. What is $E^{\circ}_{Fe^{+3}/Fe}$?

(A) - 0.036 V
(B) - 0.33 V
(C) + 0.33 V
(D) +0.0367 V

Sol. $Fe^{+2}_{aq} + H_2(g) + 2H_2O(aq) \longrightarrow Fe(s) + 2H_3O^+(aq)$; $\Delta G^{\circ}_1 = -n_1FE^{\circ}_1$

$Fe^{+3}(aq) + \frac{1}{2}H_2(g) + H_2O(aq) \longrightarrow Fe^{+2}(aq) + H_3O^+$; $\Delta G^{\circ}_2 = -n_2FE^{\circ}_2$

$Fe^{+3}(aq) + \frac{3}{2}H_2(g) + 3H_2O(aq) \longrightarrow Fe(s) + 3H_3O^+(aq)$; $\Delta G^{\circ}_3 = \Delta G^{\circ}_1 + \Delta G^{\circ}_2$

$\Rightarrow n_3FE^{\circ}_3 = n_1FE^{\circ}_1 + n_2FE^{\circ}_2$

$\Rightarrow E^{\circ}_3 = \frac{n_2E^{\circ}_2 + n_1E^{\circ}_1}{n_1 + n_2} = -0.036$ V

Hence, **(A)** is the correct answer.

Prob.4. Electrolysis of dil. H_2SO_4 generates $H_2S_2O_8$ (per disulphuric acid). What current is needed to produce $H_2S_2O_8$ at the rate of 1 mole per hour assuming 75% charge efficiency.

(A) 71.48 amp.
(B) 17.48 amp.
(C) 35.74 amp.
(D) 53.74 amp.

Sol. $2HSO_4^- \longrightarrow 2e^- + \underset{(1\ mole/hr)}{H_2S_2O_8}$

2 mole/hr electrons need to be released = 2F

$= \frac{8}{3}$F (assuming 75% charge efficiency)

$= \frac{8}{3} \times 96500\ C = i \times 60 \times 60$

$i = 71.48$ amp.

Hence, **(A)** is the correct answer.

Prob. 5. Calculate the standard Gibbs free energy change in kJ for the reaction:

$Cu^+(aq) + I^-(aq) \longrightarrow CuI(or)$

Given:

$CuI(or) + e \longrightarrow Cu(or) + I^-(aq)$ $E^\circ = -0.17V$

$Cu^+(aq) + e \longrightarrow Cu(or)$ $E^\circ = 0.53V$

(A) –67.55 (B) 135.1

(C) 1.78V (D) –1.75V

Sol. $E_1^\circ = -0.17V$

$CuI(or) + e \longrightarrow Cu(or) + I^-(aq)$

$\Delta G_1^\circ = -1 \times F \times (-0.17V)$

$Cu^+(aq) + e \longrightarrow Cu(or)$ $E_2^\circ = 0.53V$

$\Delta G_2^\circ = -1 \times F \times (0.53V) = -0.53F$

The ΔG° for the reaction: $Cu^+(aq) + I^-(aq) \longrightarrow CuI(or)$

$\Delta G^\circ = \Delta G_2^\circ - \Delta G_1^\circ$

$= -0.53F - (0.17F) = -0.70F$

$= -0.70 \times 96500J = -67550J$

$= -67.55kJ$

Hence, **(A)** is the correct answer.

Prob. 6. Standard free energy of $Cu^+ | Cu$ half cell, if SRP of $Cu^{2+} | Cu$ and SRP of $Cu^{2+} | Cu^+$ are 0.34 and 0.16 V respectively, will be : (where 'F' is Faraday constant).

(A) - 0. 42 F (B) + 0.52 F

(C) + 0.42 F (D) - 0.52 F

Sol. **(D)**

(I) : $Cu^{2+} + 2e \longrightarrow Cu$; $\Delta G_1 = -2F \times 0.34$ $\therefore Cu^+ + e^- \longrightarrow Cu$, $\Delta G_3 = \Delta G_1 + \Delta G_2$

(II) : $Cu^+ \longrightarrow Cu^{2+} + e^-$; $\Delta G_2 = F \times 0.16$.

Prob.7. Which of the following is/are correct for a lead acid battery supplying electrical energy?

(A) $PbSO_4$ is formed at one electrode only (B) $Pb + H_2SO_4 \longrightarrow PbSO_4 + H_2$

(C) Weight of electrolyte decreases (D) PbO_2 accepts electrons as well as protons

Sol. **(C, D)**

Reactions:

Cathode: $PbO_2 + 2H_2SO_4 \xrightarrow{+2e} PbSO_4 + 2H_2O + SO_4^{-2}$

Anode : $Pb + SO_4^{-2} \xrightarrow{-2e} PbSO_4$

Prob. 8. $Pb(s) | \underset{\text{Saturated solution}}{PbSO_4} \,||\, \underset{\text{saturated solution}}{PbI_2} | Pb(s)$

Which of the following expressions represent the emf of the above cell at 25°C?

(A) $E = \frac{0.0592}{2} \log \frac{\left(a_{Pb^{2+}}\right)_R}{\left(a_{Pb^{2+}}\right)_L}$ (B) $E = \frac{0.0592}{2} \log \frac{\left(a_{Pb^{2+}}\right)_1}{\left(a_{Pb^{2+}}\right)_2}$

(C) $E = \frac{0.0592}{2} \log \frac{\left[K_{sp}(PbI_2)\right]^{1/3}}{\left[K_{sp}(PbSO_4)\right]^{1/2}}$ (D) $E = \frac{0.0592}{2} \log \frac{K_{sp}(PbI_2)}{K_{sp}(PbSO_4)}$

Sol. (A)

Prob. 9. Which of the following statements is/are correct?

(A) The conductance of one cm^3 of a solution is called specific conductance

(B) Specific conductance increases while molar conductance decreases on progressive dilution

(C) The limiting equivalent conductivity of weak electrolyte cannot be determined exactly by extrapolating the plot of Λ_{eq} against $\sqrt{C}$ (C = concentration of the electrolyte)

(D) The conductance of metals is due to the movement of free electrons

Sol. **(A, C, D)**

Prob. 10. If the half cell reaction $A + e^- \longrightarrow A^-$; has a large negative reduction potentials, then which is/are not correct?

(A) 'A' is readily reduced

(B) 'A' is readily oxidised

(C) A^- is readily reduced

(D) A^- is readily oxidised

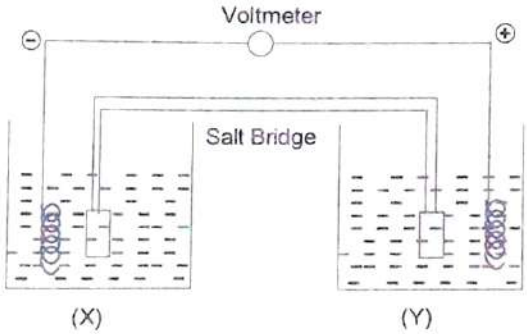
Sol. **(A, B, C)**

Greater the reduction potential, the more easily that species can be reduced.

COMPREHENSION

The cell potential for the unbalanced chemical reaction:

$Hg_2^{2+} + NO_3^- + H_3O^+ \longrightarrow Hg^{2+} + HNO_2 + H_2O + e^-$



Is measured under standard conditions in the electrochemical cell shown in the diagram were 0.02 V. Answer the following questions:

1. Choose the correct statement for the given cell diagram

(A) Compartment (X) has less pH than compartment (Y)

(B) (Y) compartment has acidic solution

(C) Current will flow from Y → X through internal supply

(D) ΔG° for the above cell reaction is less than 1 kJ at the equilibrium

Sol. **(B)**

2. Equilibrium constant for cell reaction is:

(A) $10^{3/2}$

(B) $e^{-\frac{3}{2}}$

(C) $e^{-\frac{2}{3}}$

(D) $10^{\frac{2}{3}}$

Sol. (D)

$$E^\circ = \frac{0.0591}{n}\log K_c$$

$$\log K_c = \frac{2\times 0.02}{0.06} = \frac{2}{3}$$

$$= 10^{\frac{2}{3}}$$

3. If same amount of charge (which is required for formation of 0.1 mol of HNO_2 in the above given cell) used for electrolysis of 0.1 M, 1 L aqueous solution of $CuSO_4$. Then the volume of gases liberated at STP will be:

(A) 2.24 litre
(B) 5.6 litre
(C) 22.4 litre
(D) 1.12 litre

Sol. (D)

Amount of electricity required to form 0.1 mole of HNO_2 = 0.2 F

At anode : $H_2O \longrightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e$

So, moles of O_2 produced at anode $= \frac{1}{20}$

$\therefore$ Volume of O_2(g) produced at STP $= \frac{1}{20}\times 22.4 = 1.12$ litres

MATRIX-MATCH TYPE QUESTION

1. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Specific conductance, κ	(p)	$\Lambda_m^c / \Lambda_m^\circ$
(B)	Molar conductance, Λ_m	(q)	Decreases with dilution
(C)	Resistance of electrolyte (R)	(r)	Increases with dilution
(D)	Degree of ionization of weak electrolyte	(s)	Increases with increase in the distance between parallel plates
		(t)	$ohm^{-1}m^{-1}$

Sol. A → (q, t); B → (r); C → (r, s); D → (p, r)

Answer 2, 3 and 4 by appropriately matching the information given in the three columns of the following table.

Column – 1 Cell		Column – 2 Cell reaction		Column – 3 No. of e⁻ transferred for cell reaction	
(I)	Dry cell	(i)	$Zn(Hg) + HgO(s) \rightarrow ZnO(s) + Hg(l)$	(P)	1
(II)	Lead battery	(ii)	$2H_2 + O_2 \rightarrow 2H_2O$	(Q)	2
(III)	Mercury cell	(iii)	$2Pb + 2PbO_2 + 4H_2SO_4 \rightarrow 4PbSO_4 + 4H_2O$	(R)	3
(IV)	Fuel cell	(iv)	$Zn + 2MnO_2 + 2NH_4Cl \rightarrow ZnCl_2 + Mn_2O_3 + 2NH_3 + H_2O$	(S)	4

2. Which of the following combination is incorrect ?
[A] (I) (ii) (Q) [B] (II) (iii) (S) [C] (III) (i) (Q) [D] (IV) (ii) (S)

3. Which of the following combination is correct ?
[A] (I) (iii) (R) [B] (I) (i) (Q) [C] (IV) (iv) (P) [D] (I) (iv) (Q)

4. Which of the following combination is correct ?
[A] (I) (iii) (Q) [B] (II) (i) (P) [C] (III) (i) (Q) [D] (IV) (i) (S)

Sol. 2. **A**
3. **D**
4. **C**

Answer 5, 6 and 7 by appropriately matching the information given in the three columns of the following table.

Column – 1 CELL		Column – 2 E°_{CELL}		Column – 3 Feasibility at standard conditions	
(I)	$Cu(s)\|Cu^{2+}(aq)\|\|Zn^{2+}(aq)\|Zn(s)$	(i)	0 V	(P)	Spontaneous
(II)	$Ag(s)\|Ag^{\oplus}(aq)\|\|Cu^{2+}(aq)\|Cu(s)$	(ii)	0.76 V	(Q)	Non-spontaneous
(III)	$Zn(s)\|Zn^{2+}(aq)\|\|H^{\oplus}(aq)\|H_2(g)\|Pt(s)$	(iii)	-1.1 V	(R)	At equilibrium
(IV)	$Pt(s)\|H_2(g)\|H^{\oplus}(aq.)\|\|H^{\oplus}(aq)\|H_2(g)\|Pt(s)$	(iv)	-0.46 V	(S)	Can't be predicted

5. Which of the following combination is correct ?
[A] (I) (ii) (Q) [B] (II) (i) (P) [C] (II) (iv) (Q) [D] (IV) (ii) (S)

6. Which of the following combination is incorrect ?
[A] (I) (iii) (Q) [B] (IV) (i) (R) [C] (III) (ii) (P) [D] (II) (iii) (S)

7. Which of the following combination is correct ?
[A] (I) (iii) (Q) [B] (III) (i) (P) [C] (III) (iii) (R) [D] (IV) (iv) (S)

Sol. 5. **C**
6. **D**
7. **A**

SINGLE INTEGER ANSWER TYPE

1. Find the current in amperes required to deposit 2.977 g of Pd by the electrolysis of an aqueous solution of Pd^{4+} salt per 1 hour (At. Wt. of Pd = 106.4)

Sol. (3)

$$Pd^{4+} + 4e \longrightarrow Pd$$

$$\frac{W}{E} = \frac{it}{96500}$$

$$\frac{2.977}{(106.4/4)} = \frac{i \times 1 \times 60 \times 60}{96500}$$

$\therefore i = 3$ ampere

2. If the equilibrium constant for the reaction,
$Cd^{2+}(aq) + 4NH_3(aq) \rightleftharpoons \left[Cd(NH_3)_4\right]^{2+}(aq)$; is 10^x, then find the value of 'x' ?

$\left[E^{\circ}_{Cd^{2+}|Cd} = -0.40V \text{ \& } E^{\circ}_{Cd(NH_3)_4^{2+}|Cd} = -0.61\ V\right]$

Sol. (7)

$$E^{\circ}_{cell} = \frac{0.0591}{2}\log K_{eq}$$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Consider the following table of standard reduction potentials:

Reduction half reaction:	$E^{\circ}_{R.P.}$ (in V)
$X^{3+} + 2e^- \longrightarrow X^+$	1.47
$Y^{2+} + 2e^- \longrightarrow Y$	0.60
$Z^{2+} + 2e^- \longrightarrow Z$	−0.21
$W^{2+} + e^- \longrightarrow W$	−1.38

For a balanced equation of the overall cell reaction that delivers the hightest voltage, calculate E°_{cell} for the reaction

Sol. **2.85**

Used the two half reactions that have the most positive and the most negative standard reduction potentials, respectively.

Anode: $2W \longrightarrow 2W^+ + 2e^-$ (Oxidation)

Cathode: $X^{3+} + 2e^- \longrightarrow X^+$ (Reduction)

Net cell reaction: $2W + X^{3+} \rightarrow 2W^+ + X^+$

$E^{\circ}_{cell} = E^{\circ}_{X^{3+}/X^+} - E^{\circ}_{W^+/W}$

$= 1.47 - (-1.38) = \mathbf{2.85\ V}$

Prob. 2. If for the half cell reactions

$Cu^{2+} + 2e^- \longrightarrow Cu; \quad E^{\circ} = 0.34V$

$Cu^{2+} + e^- \longrightarrow Cu^+; \quad E^{\circ} = 0.15V$

Calculate ΔG° of $2Cu^+ \longrightarrow Cu^+ + Cu$.

Sol. **−36.67**

Anode: $2Cu^+ \longrightarrow 2Cu^{2+} + 2e^-$

Cathode: $Cu^{2+} + 2e^- \longrightarrow Cu$

Net cell reaction: $2Cu^+ + Cu^{2+} \rightarrow Cu, \quad n = 2$

$E^{\circ}_{cell} = (E^{\circ}_{R.P})_{cathode} - (E^{\circ}_{R.P})_{anode}$

$= (0.34 - 0.15)\ V = 0.19\ V$

$\Delta G^{\circ} = -nFE^{\circ}_{cell} = -2 \times 96500 \times 0.19\ J$

$= \dfrac{-2 \times 96500 \times 0.19}{1000} kJ = -36.67\ kJ$

Prob. 3. A solution containing 4.5 mM $Cr_2O_7^{2-}$ of and mM of Cr^{3+} shows a pH of 2. Calculate the potential of half cell. Standard potential of the reaction: $Cr_2O_7^{2-} \longrightarrow Cr^{3+}$ is 1.33 V

Sol. **1.07**

The balanced half cell reaction is

$Cr_2O_{7(aq)}^{2-} + 14H_{(aq)}^+ + 6e^- \longrightarrow 2Cr_{(aq)}^{3+} + 7H_2O; \quad (n = 6)$

$Q = \dfrac{(15 \times 10^{-3})}{(4.5 \times 10^{-3})(10^{-2})^{14}}$

$$= \frac{225\times10^{-6}}{4.5\times10^{-31}} = 5\times10^{25}$$

$$E_{R.P} = E^{\circ}_{R.P} - \frac{0.059}{n}\log Q$$

$$= 1.33 - \frac{0.059}{6}\log(5\times10^{25})$$

$$= 1.33 - \frac{0.056}{6}\times 26.70$$

$= (1.33 - 0.26)$ volt = **1.07 volt**

Prob. 4. Calculate the EMF of the following cell: $Zn \mid Zn^{2+}(0.01M) \parallel Zn^{2+}(0.1M) \mid Zn$ at 298K

Sol. **0.0295**

The reactions at the two half-cell are:

At anode: $Zn \longrightarrow Zn^{2+}(0.01M) + 2e^-$

At cathode: $Zn^{2+}(0.1M) + 2e^- \longrightarrow Zn$

Net cell reaction: $Zn^{2+}(0.1M) \rightarrow Zn^{2+}(0.01M)$; $(n = 2)$

$$Q = \frac{[Zn^{2+}]_{Anode}}{[Zn^{2+}]_{Cathode}} = \frac{0.01}{0.1} = 10^{-1}$$

$$E_{cell} = 0 - \frac{0.059}{2}\log(10^{-1}) \quad (\because E^{\circ}_{cell} = 0 \text{ for concentration cell})$$

$$\therefore \quad E_{cell} = -\frac{0.059}{2}(-1)\log 10 = 0.0295 \text{ V}$$

Prob. 5. The emf of the cell: $Ag \mid AgI(s), 0.05(M)\ KI \parallel 0.05\,M\ AgNO_3 \mid Ag$ is 0.788 V. Calculate the solubility product of AgI.

Sol. **–13.36**

The given cell is:

Common ion

$Ag \mid AgI, 0.05(M)\ KI \parallel 0.05\,M\ AgNO_3 \mid Ag$

Anode: $Ag_{(s)} \longrightarrow Ag^+(C = ?) + e^-$

Cathode: $Ag^+(0.05M) + e^- \longrightarrow Ag_{(s)}$

Net cell reaction: $Ag^+(0.05M) \rightarrow Ag^+(C = ?)$; $(n = 1)$

$$E_{cell} = 0 - \frac{0.059}{n}\log\frac{[Ag^+]_{Anode}}{[Ag^+]_{Cathode}}$$

Substituting the value of $[Ag^+]_{Anode}$ in Nernst equation, we have

$$0.788 = -\frac{0.059}{1}\log\left(\frac{K_{sp}}{0.05}\right)$$

$$\therefore \log(400K_{sp}) = \left(-\frac{0.788}{0.059}\right) = -13.36$$

SOLID STATE 8

Solid state is one of the physical states of matter in which particles (molecules, ions or atoms) are closely packed and are held together by strong intermolecular forces.

TYPES OF SOLIDS

Solids are broadly divided into two types.

Crystalline solids: The constituent particles are arranged in a regular fashion.

Amorphous solids: The constituent particles are not arranged in a proper fashion.

The differences between crystalline and amorphous solids.

Crystalline	Amorphous
They have sharp melting points.	They melt over a range of temperature.
They are anisotropic.	They are isotropic.
They undergo clean cleavage.	They undergo irregular cleavage.

Types of Crystalline Solids

Depending upon the constituent particles, there are four types of crystalline solids.

Crystal type	Constituent particles	Binding force	Examples
Ionic	Positive and negative ions	Electrostatic forces of attraction	NaCl, KNO_3, LiF $BaSO_4$ etc.
Covalent	Atoms	Covalent bonds	Diamond, silicon, quartz
Molecular	Molecules	van der Waal's forces/Hydrogen bonds	Solid CO_2, I_2, S_8, ice
Metallic	Positive ions and mobile electrons	Metallic bonds	All metals and alloys

LATTICE AND UNIT CELL

Crystal Lattice: A regular arrangement of the constituent particles of a crystal in three-dimensional space. It is also termed as space lattice.

Unit Cell: The smallest three-dimensional portion of the space lattice which when repeated over and again in different directions produces the complete space lattice.

Types of Unit Cells

Simple: Particles are present at all corners.
Face-centred cubic (FCC): In addition to the particles on the corners, there is one particle present on the centre of every face.
End-face centred: In addition to the particles in corners, one particle in the centre of one pair of two opposite faces.
Body centred cubic (BCC): In addition to the particles in the corners, there is one particle present within the body of the unit cell.

Crystal Systems

Types of crystal systems, their axial ratios, angles and types of unit cells are given below:

Crystal system	Axial ratios	Axial angles	Unit cells/Bravais lattices
Cubic	$a = b = c$	$\alpha = \beta = \gamma = 90^\circ$	Simple, BCC, FCC
Tetragonal	$a = b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Simple, BCC
Orthorhombic	$a \neq b \neq c$	$\alpha = \beta = \gamma = 90^\circ$	Simple, FCC, end-face centred, BCC
Monoclinic	$a \neq b \neq c$	$\alpha = \beta = 90^\circ \neq \gamma$	Simple, end-face centred
Triclinic	$a \neq b \neq c$	$\alpha \neq \beta \neq \gamma \neq 90^\circ$	Simple
Rhombohedral	$a = b = c$	$\alpha = \beta = \gamma \neq 90^\circ$	Simple
Hexagonal	$a = b \neq c$	$\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$	Simple

In all, there are seven crystal systems and fourteen Bravais lattices.

Law of constancy of interfacial angles: The angles between the corresponding faces of a crystal (called interfacial angles) are always same irrespective of the size and external shape of the crystal.
The interfacial angles are measured with the help of an instrument called goniometer.

Cubic Systems

Number of Particles per Unit Cell (Cubic)

Contribution of each atom present on the corner $= \frac{1}{8}$

Contribution of each atom present on the face $= \frac{1}{2}$

Contribution of an atom present on the edge $= \frac{1}{4}$

Contribution of each atom within the body $= 1$

No. of atoms present in simple unit cell $= \frac{1}{8} \times 8 = 1$

No. of atoms present in FCC $= \frac{1}{8} \times 8$ (corner atoms) $+ \frac{1}{2} \times 6$ (face)

$= 4$

No. of atoms present in BCC $= \frac{1}{8} \times 8$ (corners) $+ 1$ (body centre)

$= 2$

In case of hexagonal close packing (hcp), no. of atoms per unit cell = 6
FCC arrangement and cubic close packing (ccp) arrangement are same and it has closest packing in cubic crystal system (packing fraction = 0.74, i.e. 74% of unit cell is occupied by atom).

Coordination Number

Coordination number is the number of its closest neighbours (oppositely charged ions in case of ionic crystal) that an atom has in a unit cell.

For elemental solids

(i) Coordination number of simple cubic structure = 6

(ii) Coordination number of FCC structure = 12

(iii) Coordination number of BCC structure = 8

Structure of Ionic Crystals

Common types of structures possessed by simple ionic compounds (AB type) are

1. Rock salt (NaCl) type
2. Caesium chloride (CsCl) type
3. Zinc blende (ZnS) type

Main features of NaCl type structure:

(i) It has FCC arrangement of ions.

(ii) Na^+ and Cl^- ions have 6 : 6 coordination.

(iii) Each unit cell has four NaCl units.

(iv) Examples: Halides of alkali metals (except those of Cs) and those of NH_4^+, oxides and sulphides of alkaline earth metals (except BeS), halides of Ag (except AgI).

Main features of CsCl type structure:

(i) It has BCC arrangement of ions.

(ii) Cs^+ and Cl^- ions have 8 : 8 coordination.

(iii) Each unit cell has one CsCl unit.

(iv) Examples: CsBr, CsI, CsCN, TiBr, TiCl etc.

Main features of ZnS type structure:

(i) It has FCC arrangement of S^{2-} ions.

(ii) Zn^{2+} and S^{2-} ions have 4 : 4 coordination (tetrahedrally arranged).

(iii) Examples: CuCl, CuBr, AgI

Structure of ionic compounds of AB_2 type

Ionic solids	Main features
Calcium fluorite (CaF_2) type	(i) It has ccp arrangement of Ca^{2+} ions and F^- ions in all tetrahedral voids. (ii) Ca^{2+} and F^- ions have 8 : 4 coordination.
Anti-fluorite structure e.g. Na_2O	(i) Anions in ccp arrangement and cations in all tetrahedral voids. (ii) Coordination number ratio of cation : anion is 4 : 8.

Types of Close Packing

The arrangement of atoms to form close packed structure can be done by two different ways.

(i) Hexagonal close packing (hcp): The pattern of arrangement is ABABAB ... type and the space occupied is 74%.

(ii) Cubic close packing (ccp): The pattern of arrangement is ABCABC ... type and the space occupied is 74%.

Tetrahedral void: Each sphere is surrounded by four spheres arranged tetrahedrally around it.
Octahedral void: Each sphere is surrounded by six spheres.
The radii of the voids in terms of radii of packing spheres (R) are
Radius (r) of tetrahedral void = 0.225R (When no distortion is there)
Radius (r) of octahedral void = 0.414R(When no distortion is there)
The number of voids in terms of number of atoms are
No. of octahedral voids = No. of atoms in the close packed arrangement
No. of tetrahedral voids = 2 × No. of atoms or octahedral voids
No. of octahedral voids in hcp = 6
No. of tetrahedral voids = 12
No. of octahedral voids in ccp = 4
No. of tetrahedral voids ccp = 8
All noble gases have ccp arrangement except He which has hcp arrangement.

The relationship between radius ratio $\left(\frac{r^+}{r^-}\right)$, coordination number and structural arrangements are tabulated below:

Radius ratio	Coordination number	Structural arrangement
$0.155 \le \left(\frac{r^+}{r^-}\right) < 0.225$	3	Triangular
$0.225 \le \left(\frac{r^+}{r^-}\right) < 0.4141$	4	Tetrahedral
$0.414 \le \left(\frac{r^+}{r^-}\right) < 0.732$	6	Octahedral
$0.732 \le \left(\frac{r^+}{r^-}\right) < 1$	8	Body centred cubic

The relationship between atomic radius (r) and the edge (a) of the unit cell is given below for elemental crystals and non distorted ionic crystals:

Simple	Body centred	Face centred
$r = \frac{a}{2}$	$r = \frac{\sqrt{3}}{4}a$	$r = \frac{a}{2\sqrt{2}}$

Packing Fraction (Undistorted Crystals)

$$\text{Packing Fraction} = \frac{\text{Volume occupied by spheres of unit cell}}{\text{Volume of unit cell}}$$

Type of unit cell	Packing Fraction
Simple unit cell	$\frac{\frac{4}{3}\pi r^3}{(2r)^3} \approx 0.52$
Body centred unit cell	$\frac{2\times\frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{3}}\right)^3} \approx 0.68$
Face centred unit cell	$\frac{4\times\frac{4}{3}\pi r^3}{\left(\frac{4r}{\sqrt{2}}\right)^3} \approx 0.74$

Theoretically, the density of a cubic unit cell can be calculated by following equation

$$\rho = \frac{Z \times M}{a^3 \times N_0}$$

where ρ - density of unit cell

Z - no. of formula units present in one unit cell of ionic compounds

M - formula mass of the ionic compound

a - edge length of the unit cell

N_0 – Avogadro's number

DEFECTS IN CRYSTALS

Metal Excess Defects

Formation of non-stoichiometric compound due to excess of any of the ions forming the compound. For example, NaCl heated in Na vapour turns yellow.

Metal Deficiency Defects

It is a defect in which if any one of the positive ions is missing from its lattice site, then its extra negative charge is balanced by some near by metal ion acquiring two charges instead of one. This type of defect is found in compounds of transition metals.

Schottky Defect

The defect in which equal number of cations and anions are missing from their lattice sites. As a result, the density of the crystal decreases.

Frenkel Defect

The defect in which cations are missing from their lattice sites and occupy interstitial sites. As a result, the density of the crystal remains unchanged.

ILLUSTRATIVE PROBLEMS

1. If the radius of the bromide ion is 0.182 nm, how large a cation can fit in each of the tetrahedral hole ?

Sol. For tetrahedral void, $\frac{r^+}{r^-} = 0.414$

$r^+ = 0.414 \times r^- = 7.5 \times 10^{-2}$ nm

2. The composition of a sample of wurstite is $Fe_{0.93}O_{1.0}$. What percentage of iron is present in the form of Fe(III)?

Sol. $Fe_{0.93}O_{1.0}$ is a non-stoichiometric compound. It is a mixture of Fe^{2+} and Fe^{3+} ions
Let, x atoms of Fe^{3+} ions are present in the compound for every O^{-2}.
So, (0.93 – x) atoms of Fe^{2+} ions are present in the compound
In the compound,
Total positive charge + Total negative charge = 0
$x \times (+3) + (0.93 - x) \times (+2) - 2 = 0$
or $3x + 0.93 \times 2 - 2x = 2$ or $x + 2 \times 0.93 = 2$
$x = 0.14$

$\therefore$ % of Fe^{3+} ions present $= \frac{0.14}{0.93} \times 100 = 15.053$

3. The edge length of unit cell of a metal having molecular weight 75 g/mol is 5 Å which crystallizes in cubic lattice. If the density is 2 g/cc then find the radius of metal atom. ($N_A = 6 \times 10^{23}$). Give the answer in pm.

Sol. $\rho = \frac{ZA}{N_o V}$

$\rho = \frac{ZA}{N_o V} \Rightarrow Z = \frac{\rho N_o V}{A}$

$= \frac{2 \times 6 \times 10^{23} \times (5 \times 10^{-8})^3}{75}$

$= 2 \Rightarrow$ Unit cell is BCC. $\Rightarrow a\sqrt{3} = 4r \Rightarrow r = \frac{a\sqrt{3}}{4}$

$= 500 \times \sqrt{3/4}$ pm $= 216$ pm

4. An element crystallizes as face centred cubic lattice with density 5.20 g/cm^3 and edge length of unit cell as 300 pm. Calculate the mass of the element which contains 3.01×10^{24} atoms.

Sol. $Z = 4$, $d = 5.2$ g/cm^3, $M = ?$

$a = 300$ pm $= 3 \times 10^{-12}$ m $= 3 \times 10^{-10}$ cm

$a^3 = 27 \times 10^{-24}$ cm^3

$N_o = 6.023 \times 10^{23}$

$$M = \frac{\rho N_0 a^3}{Z} = \frac{5.2 \times 6.023 \times 10^{23} \times 27 \times 10^{-24}}{4}$$

$= 21.13$ g/mol

6.023×10^{23} atoms have $= 21.13$ g

$\therefore 3.01 \times 10^{24}$ atoms have $= \frac{21.13 \times 3.01 \times 10^{24}}{6.02 \times 10^{23}}$

$= 105.65$ g

5. The length of the unit cell edge of a BCC crystal is 352 pm. Calculate the radius of an atom in the crystal lattice.

Sol. $\sqrt{3}\,a = 4r$

$$r = \frac{\sqrt{r} \times 352}{4} \text{ pm}$$

$= \mathbf{152.4\ pm}$

6. Copper has fcc crystal structure. Assuming an atomic radius of 130 pm for copper atom (Atomic wt. of Cu = 63.54 g/mole):
(a) What is the length of unit cell of Cu?
(b) What is the volume of the unit cell?
(c) How many atoms belong to the unit cell?
(d) Find the density of Cu.

Sol. As we know

(a) For FCC structure

$4r = \sqrt{2}\,a$

$a = 2\sqrt{2}\ r$

$= 2\sqrt{2} \times 130$ pm $= \mathbf{367.64\ pm}$

(b) Volume of unit cell $= a^3 = \frac{n \times M_m}{N_0 \times \rho} = (3.67 \times 10^{-8})^3 = \mathbf{4.94 \times 10^{-23}\ cm^3}$

(c) $n = 4$

(d) $\rho = \frac{4 \times 63.54}{6.023 \times 10^{23} \times (3.67 \times 10^{-8})^3} = \mathbf{8.54\ gm/cm^3}$

7. A spinel is an important class of oxides consisting of two types of metal ions with the oxide ions arranged in ccp layers. The normal spinel has 1/8th of the tetrahedral void occupied by one type of metal and one half of the octahedral voids occupied by another type of metal ions. Such a spinel is formed by Zn^{2+}, Al^{3+} and O^{2-} with Zn^{2+} in tetrahedral void. Give the simplest formula of the spinel.

Sol. Let the no. of tetrahedral voids = n

No. of octahedral voids $= \frac{n}{2}$

$\frac{1}{8}$th tetrahedral voids is occupied by Zn^{2+}

$\therefore$ No. of $Zn^{2+} = \frac{n}{8}$

$\frac{1}{2}$ of octahedral voids occupied by Al^{3+}

$\therefore$ No. of $Al^{3+} = \frac{n}{2\times 2} = \frac{n}{4}$

Ratio of $Zn^{2+} : Al^{3+}$

1 : 2

For 1 mole Zn^{2+}, $Al^{3+} = 2$

Total +ve charge = $2 + 3\times 2 = 8$

$\therefore$ No. of $O^{2-} = 4$

$\therefore$ The simplest formula will be $ZnAl_2O_4$

8. A metal crystallizes in two cubic phases, face centred cubic (fcc) and body centred cubic (bcc) whose unit cell length are 3.5 and 3.0 Å respectively. Calculate the ratio of density of fcc and bcc.

Sol. $\rho = \frac{n\times M_m}{N_A \times a^3}$

For face centred cubic cell $n = 4$, $a = 3.5$ Å

$\therefore \rho_{(fcc)} = \frac{4\times M_m}{N_A \times (3.5)^3}$ ... (i)

For bcc lattice $n = 2$, $a = 3.0$ Å

$\rho_{bcc} = \frac{2\times M_m}{N_A \times (3.0)^3}$ ... (ii)

From equations (i) and (ii)

$\therefore \frac{\rho_{(fcc)}}{\rho_{(bcc)}} = \frac{4}{2}\times\frac{3^3}{(3.5)^3} = \frac{4\times 3\times 3\times 3}{2\times 3.5\times 3.5\times 3.5} = 1.259$

9. A compound formed by elements X and Y, crystallizes in the cubic structure, where X is at the corners of the cube and Y is at the six face centres. What is the formula of the compound? If side length is 5 Å, estimate the density of the solid assuming atomic weight of X and Y as 60 and 90 respectively.

Sol. From eight corner atoms, one atom (X) contributes to one unit cell.
From six face centres, three atoms (Y) contribute to one unit cell.
So, the formula of the compound is XY_3.
As we know that,

Mass per unit cell = $60 + 3\times 90 = \frac{330 \text{ gm}}{N_0}$

$\rho = \frac{1\times 330}{6.023\times 10^{23}\times (5\times 10^{-8})^3}$ gm/cm^3 ($a = 5$Å $= 5\times 10^{-8}$ cm)

$= \frac{330}{6.023\times 10^{23}\times 125\times 10^{-24}}$ gm/cm^3 $= 4.38$ gm/cm^3

SOLVED PROBLEMS

Subjective Question

Prob. 1. Iron crystallizes in several modifications. At about 910°C, the body centred cubic α-form undergoes a transition to the face centred cubic γ-form. Assuming that the distance between nearest neighbours is same in the two forms at the transition temperature. Calculate the ratio of density of γ-iron to that of α-iron at the transition temperature.

Sol. In bcc structure (α form) of iron

$4r = \sqrt{3}a$

n = 2 atoms per unit cell from (one atom from 8 corners, one atom from the body centre)

So, ρ (γ-form) $= \dfrac{n \times M_m}{a^3 \times N_o} = \dfrac{2 \times M_m}{\left(\dfrac{4}{\sqrt{3}}r\right)^3 \times N_o}$ ... (i)

In fcc structure (r form) of iron

$4r = \sqrt{2}\,a$

$\therefore a = 2\sqrt{2}\,r$

n = 4 atoms per unit cell

$\therefore \rho$ (γ form) $= \dfrac{4 \times M_m}{(2\sqrt{2}r)^3 \times N_o}$ ... (ii)

Dividing Eq. (ii) by Eq. (i)

$$\frac{\rho(\gamma \text{ form})}{\rho(\alpha \text{ form})} = \frac{2}{\left(\frac{4}{\sqrt{3}}r\right)^3} \times \frac{\left(2\sqrt{2}r\right)^3}{4} = 1.0887$$

Objective Questions

Prob. 1. In a solid 'AB' having NaCl structure atoms B^- occupy the corners of the unit cell. If all the face centred atoms along one of the axis are removed, then the resultant stoichiometry of the solid is

(A) AB_2 (B) A_2B

(C) A_4B_3 (D) A_3B_4

Sol. In one NaCl type unit cell of solid, B^- are present at corners and face centres and A^+ are present at edge centres and body centre. Thus, a unit cell will contain $4A^+$ and $4B^-$ ions. In this unit two face centred ions B^- lies along one axis and are removed.

Contribution of two face centred B^- ions $= 2 \times \dfrac{1}{2} = 1$

In the resultant unit cell;

$\therefore$ Number of A^+ ions present per unit cell = 4

$\therefore$ Number of B^- ions present per unit cell = 4 – 1 = 3

$\therefore$ Stoichiometry = A_4B_3. Hence, **(C)** is the correct answer.

Prob. 2. CsBr has BCC structure. The length of its one side is 4.3 Å. The minimum distance between Cs^+ and Br^- ion will be

(A) 0.897 Å (B) 3.72 Å

(C) 1.749 Å (D) None of the above

Sol. For bcc structure

$2(r^+ + r^-) = \sqrt{3}a$ (For minimum distance)

or $2(r^+ + r^-) = \sqrt{3} \times 4.3$

or $(r^+ + r^-) = \sqrt{3} \times \frac{4.3}{2} = 3.72\ \text{Å}$

Hence, the distance between Cs^+ and Br^- ions will be 3.72 Å

Hence, **(B)** is the correct answer.

Prob. 3. The hcp and ccp structure for a given element will not have

(A) same density
(B) the same fraction of unoccupied space
(C) coordination number
(D) same edge length

Sol. **(D)**

Prob. 4. An ionic compound AB has ZnS type of structure, if the radius A^+ is 22.5 pm, then the ideal radius of B^- so as not to cause any distortion is:

(A) 54.35 pm
(B) 100 pm
(C) 145.16 pm
(D) None of the above

Sol. Since ionic compound AB has ZnS type of structure, therefore it has tetrahedral holes, for which

$$\frac{\text{Radius of cation}}{\text{Radius of anion}} = 0.225$$

$$\frac{r^+}{r^-} = 0.225$$

$$\frac{22.5}{r^-} = 0.225$$

Hence, $r^- = 100$ pm

Hence, **(B)** is the correct answer.

Prob. 5. Select the correct statements:

(A) For CsCl unit cell (edge-length = a), $r_c + r_a = \frac{\sqrt{3}}{2}a$

(B) For NaCl unit cell (edge-length = ℓ), $r_c + r_a = \frac{\ell}{2}$

(C) The void space in a bcc unit cell is 0.68

(D) The void space % in a face centred unit cell is 26%

Sol. In bcc structure, $r^+ + r^- = \frac{\sqrt{3}}{2}a$

Hence, for CsCl, $r_c + r_a = \frac{\sqrt{3}}{2}a$

$\therefore$ (A) is correct.

Since, NaCl crystallizes in fcc structure

$\therefore 2r_c + 2r_a =$ edge length of the unit cell

Hence, $r_c + r_a = \frac{l}{2}$

$\therefore$ (B) is correct.

Since packing fraction of a bcc unit cell is 0.68

$\therefore$ Void space = 1 - 0.68 = 0.32

$\therefore$ (C) is not correct.

In fcc unit cell, PF = 74%

$\therefore$ VF = 100 - 74 = 26%

$\therefore$ (D) is correct.

Hence, **(A), (B)** and **(D)** are the correct answers.

Prob. 6. A mineral having the formula AB_2 crystallises in the cubic close packed lattice, with A-atoms occupying the lattice points. Now, select the correct statement(s) of the following:

(A) Co-ordination number of A-atoms is 8

(B) Co-ordination number of B-atoms is 4

(C) Fraction of octahedral sites occupied by B-atoms is 50%

(D) Fraction of the tetrahedral sites occupied by B-atoms is 100%

Sol. **(A, B, D)**

In antifluorite structure anion B^- are present in all octahedral voids and cation A^{2+} forms ccp lattice.

Hence, A, B, and D are the correct answers.

Prob. 7. Which of the following is true for an ionic compound A_2B crystallizes in anti-fluorite type

(A) Cation is occupying tetrahedral void

(B) Cation is occupying octahedral void

(C) If ions present at one body diagonal are removed the new formula will be A_8B_5

(D) The ratio of radius of A^+/B^- is in between 0.414 to 0.732

Sol. **(A, C)**

In antifluorite type structure cation occupies all tetrahedral void.

When ions of one body diagonal will be removed then two corner B^- will be removed and $2A^+$ from tetrahedral voids will be removed. New effective formula = $A_6B_{3.75}$ so A_8B_5.

For coordination number 4: 4 in tetrahedral geometry radian ration lies between 0.224 to 0.414

REASONING TYPE QUESTIONS

1. **Assertion:** In rock salt structure, all the octahedral voids in the close packing of anions are occupied by cations.

   Reason: In rock salt structure, the distance of closest approach between two anions equal to half the face diagonal of the unit cell.

Sol. **Assertion** → True

Reason → True but not correct explanation.

COMPREHENSION

Read the paragraph carefully and answer the following questions:

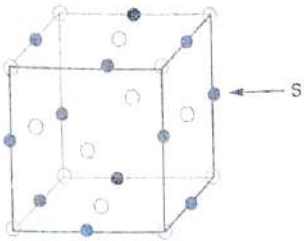
In HCP or CCP, constituent particles occupy 74% of the available space. The remaining space (26%) in between the spheres remains unoccupied and is called interstitial voids or holes.

Considering the close packing arrangement, each sphere in the second layer rests on the hollow space of the first layer, touching each other. Tetrahedral and octahedral voids are there.

If R is the radius of the spheres in the close packed arrangement then,

R (radius of tetrahedral void) = 0.225 R

Prob. 1. In the figure given below, the site marked as S is a



(A) tetrahedral void
(B) cubic void
(C) octahedral void
(D) None of the above

Prob. 2. If the anions (A) form hexagonal close packing and cations (B) occupy only 2/3rd octahedral voids in it, then the general formula of the compound is

(A) AB
(B) A_3B_2
(C) A_2B
(D) AB_3

Prob. 3. In the spinel structure, oxide ions are cubic close packed whereas 1/8th of tetrahedral voids are occupied by A^{2+} cations and 1/2 of octahedral voids are occupied by B^{3+} cations. The general formula of the compound having spinel structure is

(A) $A_2B_2O_4$
(B) AB_2O_4
(C) $A_2B_4O_2$
(D) $A_4B_2O_2$

Prob. 4. Mark the false statement:

(A) CsCl crystal always shows Schottky defect.
(B) Crystals having F–centres are coloured and paramagnetic.
(C) Photosensitivity of AgBr is due to the presence of Frenkel defect in this crystal.
(D) None of the above

Prob. 5. In Schottky defect

(A) cations are missing from the lattice sites and occupy the interstitial sites
(B) equal number of cations and anions are missing
(C) anions are missing and electrons are present in their place
(D) equal number of extra cations and electrons are present in the interstitial sites

Answers

1.	C	2.	B	3.	B	4.	D
5.	B						

MATRIX – MATCH TYPE

1. Match the Column – I with the Column – II.

Column – I	Column – II
(A) Equal number of cations and anions are missing	(p) spinel structure
(B) Metal excess non-stoichiometry in NaCl	(q) density of crystal decreases
(C) Occupancy of tetrahedral and octahedral voids	(r) Schottky defect
(D) Anion vacancy with trapped electron	(s) F-centre
	(t) Frenkel defect

Sol. **(A) → (q, r); (B) → (s); (C) → (p); (D) → (s)**

2. Match the type of defects in Column – I with their corresponding characteristics in Column – II:

Column – I (Defects)		Column – II (Characteristics)	
(A)	Frankel defect	(p)	Stoichiometric defect
(B)	Schottky defect	(q)	Non-stoichiometric defect
(C)	Metal excess defect	(r)	Density of the crystal is decreased
(D)	Metal deficiency defect	(s)	KCl crystal doped with $SrCl_2$
		(t)	Density of the crystal is increased

Sol. **A → (p); B → (p, r); C → (q, r, t); D → (q, r, s)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

	Column – 1		Column – 2		Column – 3
(I)	Cubic	(i)	$a = b \neq c$	(P)	$\alpha = \beta = \gamma = 90^\circ$
(II)	Tetragonal	(ii)	$a \neq b \neq c$	(Q)	$\alpha = \gamma = 90^\circ \neq \beta$
(III)	Orthorhombic	(iii)	$a = b = c$	(R)	$\alpha \neq \beta \neq \gamma \neq 90^\circ$
(IV)	Monoclinic	(iv)	$a = b \neq c$	(S)	$\alpha = \beta = 90^\circ; \gamma = 120^\circ$

3. Which of the following combination is correct ?
[A] (I) (iii) (P) [B] (II) (ii) (R) [C] (III) (iii) (Q) [D] (IV) (i) (S)

4. Which of the following combination is incorrect ?
[A] (II) (i) (P) [B] (III) (ii) (P) [C] (IV) (ii) (Q) [D] (IV) (iv) (R)

5. Which of the following combination is correct ?
[A] (IV) (iii) (R) [B] (II) (i) (P) [C] (I) (iv) (Q) [D] (II) (iii) (P)

Sol. 3. **A**; 4. **D**; 5. **B**

SINGLE INTEGER ANSWER TYPE

1. An ionic crystal A^+B^- has NaCl type structure, such that lattice points are occupied by B^- and A^+ are in voids. If all the face centred ions are removed, the number of A^+ ions left in one unit cell of AB is.

Sol. **(4)**
All corners as well as face centres are occupied by B^- ions only one unit cell of AB has 4 A^+ ions.

2. A solid is made up of two elements A (at. Wt. = 56) and B(at. Wt. = 32). A-atoms are present at each of the corner of cubic unit cell, while B-atoms are present at alternate face-centres as well as at alternate edge centres. If 0.24 kg of this solid contains $x \times N_A$ (where N_A is Avogadro's number) unit cells, then the value of 'x' is. ..

Sol. **(2)**
Formula of the solid is AB$_2$
Formula Wt. of the solid = 120 g
Now, 120 g solid contains = N_A unit cell
∴ 240 g solid contains = $2N_A$ unit cell
Hence, x = 2

3. Metal A crystallizes in a lattice with coordination no. 8. What will be total no. of atoms attached to a single unit if atoms lying on two body diagonals are removed?

Sol. (4)

CN = 8 so metal crystallizes in BCC, so total 5 atoms will be lying on 2 body diagonals.

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Radius ratio (ideal) for octahedral geometry is :

Sol. **0.414**

Prob. 2. Ratio of atoms at face placed by corner in fcc is ?

Sol. **0.75**

$$\frac{6}{8} = \frac{\text{face}}{\text{corner}}$$

Prob. 3. In ccp of atoms A(r_a), the radius of B that can fit in tetrahedral void is x r_B. x is ?

Sol. **0.225**

Prob. 4. Packing fraction of simple cubic is :

Sol. **0.524**

Prob. 5. Ratio of Z of bcc to fcc is :

Sol. **0.50**

$$\frac{2}{4} = 0.50$$

Prob. 6. Ratio of octahedral to tetrahedral voids in ccp is :

Sol. **0.5**

$$\frac{O}{Td} = \frac{4}{8}$$

Prob. 7. Ratio of Z in simple cubic (scp) packing by HCP is :

Sol. **0.16**

$$\frac{1}{6} = 0.16$$

Prob. 8. Ratio of nearest neighbours to next nearest neighbours in scp is :

Sol. **0.50**

$$\frac{6}{12} = 0.50$$

Prob. 9. The length of body diagonal of a cube in x a where a = edge length, then value of x in decimal is :

Sol. **1.73**

$\sqrt{3}a$

Prob.10. Void space fraction of fcc is :

Sol. **0.26**

$1 - 0.74 = 0.26$

LIQUID SOLUTION 9

A binary solution is a mixture of two components, one is called solute and the other one is called solvent. The concentration of solute can be expressed in various ways mentioned below.

Molarity

The number of moles of solute present in 1 litre of the solution is called as molarity (M) of the solution.

$$M = \frac{\text{No. of moles of solute}}{\text{Volume of solution in litre}}$$

Normality

The number of equivalents of solute present in 1 litre of the solution is called as normality (N) of the solution.

$$N = \frac{\text{No. of equivalents of solute}}{\text{Volume of solution in litre}}$$

Mole Fraction

The ratio of number of moles of solute to total number of moles is called mole fraction. The sum of mole fractions of all the species in a solution is equal to 1.

$$X = \frac{n}{N+n}$$

where n = number of moles of solute

N = number of moles of solvent

Per cent by Weight

The weight of solute present in 100 g of solution is called as per cent by weight of solute.

$$\% \text{ by weight of solute} = \frac{\text{Weight of solute}}{\text{Weight of solution}} \times 100$$

Per cent by Volume

This is expressed when solute and solvent both are in liquid state. The volume of solute in 100 ml of the solution is called per cent by volume.

$$\% \text{ by volume} = \frac{\text{Volume of solute}}{\text{Volume of solution}} \times 100$$

Molality

The number of moles of solute present in 1 kg of solvent is called as molality (m) of the solution.

$$m = \frac{\text{No. of moles of solute}}{\text{Weight of solvent in kg}}$$

TYPES OF SOLUTIONS

Solutions can be broadly classified into two types, i.e. ideal solution and non-ideal solution.

Ideal Solution

Suppose, A is dissolved in the solvent B to get a solution. The solution will be ideal if force of attraction among the A -- A molecules and B – B molecules are same that of A – B molecules. Some of the conditions followed by ideal solution are

$\Delta H_{mix} = 0$

$\Delta V_{mix} = 0$ assuming pressure and temperature are held constant.

The solution should obey Raoult's law.

$P_T = \left(X_A P_A^o + X_B P_B^o\right)$

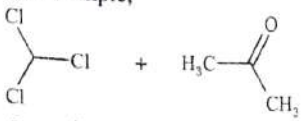
Non-ideal Solution

The solution which doesn't satisfy the above conditions are called non-ideal solution. These are further divided into two types, (a) which shows positive deviation from Raoult's law and (b) which shows negative deviation from Raoult's law.

$P_T \Rightarrow \left(X_A P_A^o + X_B P_B^o\right)$

When ΔH_{mixing} = + ve and ΔV_{mixing} = +ve, the solution shows +ve deviation from Raoult's law.

For example,



$P_T \Rightarrow \left(X_A P_A^o + X_B P_B^o\right)$

When ΔH_{mixing} = (–) ve and ΔV_{mixing} = (–) ve, the solution shows negative deviation from Raoult's law. For example, C_2H_5OH + Benzene.

Raoult's Law (For The Solution of Volatile Liquids)

Suppose, any solution is a mixture of two volatile liquids A and B. Then it has been found that partial vapour pressure of A at a given temperature is equal to the product of the vapour pressure of pure A at that temperature and the mole fraction of A in the solution. We can say that

$P'_A = X_A P_A^0$

where P'_A is the partial vapour pressure of component A in vapour and P_A^0 is the vapour pressure of pure A at given temperature.

Similarly,

$P'_B = X_B P_B^0$

and the total vapour pressure

$P_T = P'_A + P'_B$

$P_T = X_A P_A^0 + X_B P_B^0$

where $X_A + X_B = 1$

With the help of above relation, the mole fraction of A and B in vapour phase can be calculated.

$$X'_A = \frac{X_A P_A^0}{P_T}$$

$$X'_B = \frac{X_B P_B^0}{P_T} \quad (\text{when } X'_A + X'_B = 1)$$

COLLIGATIVE PROPERTIES OF SOLUTIONS

The properties which simply depends upon the number of particles and not on their nature with respect to their structure, are called colligative properties. They are

(i) Relative lowering of vapour pressure
(ii) Elevation of boiling point
(iii) Depression in freezing point
(iv) Osmotic pressure

Relative Lowering of Vapour Pressure

The vapour pressure of the solution is less than that of pure solvent in presence of a non-volatile solute. The decrease in vapour pressure with respect to the vapour pressure of pure solvent is called relative lowering of vapour pressure, i.e. $\left(\frac{P - P_s}{P}\right)$, where P is the vapour pressure of pure solvent and P_s is the vapour pressure of solution at same temperature.

Raoult gave a relationship between relative lowering of vapour pressure and composition of solution. If n is the number of moles of solute and N is the number of moles of solvent, then

$$\frac{P - P_s}{P} = \frac{n}{n + N}$$

If $n \ll N$

$$\frac{P - P_s}{P} = \frac{n}{N} \text{ or } \frac{w / M_1}{W / M_2}$$

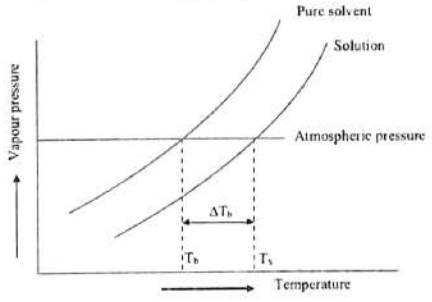
where w is the weight of solute, W is the weight of solvent, M_1 and M_2 are the molecular weight of solute and solvent respectively.

Limitations of Raoult's law:

(i) It is applicable to dilute solutions.
(ii) It is applicable to the non-volatile solute only.
(iii) It is not applicable to the solute which dissociates or associates in the solution.

Elevation of Boiling Point

The boiling point of any liquid is the temperature at which the vapour pressure of the liquid becomes equal to that of surrounding. On adding non-volatile solute, the vapour pressure of the solution decreases and it is obvious that now the solution will boil at higher temperature as compared to pure solvent. If T_b is the boiling point of pure solvent and T_s is the boiling point of solution, then $T_s > T_b$ and $(T_s - T_b) = \Delta T_b$, is called elevation in boiling point. It can be represented by the following graph.



$\Delta T_b \propto m$ (molality of solute)

$\Delta T_b = K_b \times m$

where K_b is called molal elevation constant.

If $m = 1$, then $\Delta T_b = K_b$.

Hence, molal elevation constant of a solvent is the rise in boiling temperature if the solution is 1 molal with respect to solute.

Depression in Freezing Point

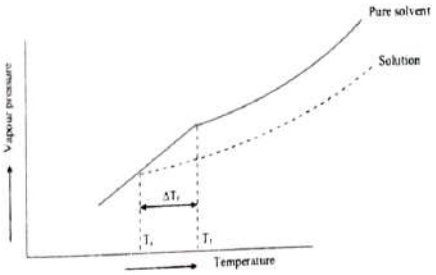
The freezing point of a pure liquid is the temperature at which it begins to freezes. At this temperature solid and liquid forms are in equilibrium, i.e.

liquid $\rightleftharpoons$ solid

At equilibrium, the vapour pressure of liquid will be equal to vapour pressure of solid. On addition of a non-volatile solute, the vapour pressure lowers and a solid-liquid equilibrium in solution will exist at a lower temperature than in case of pure solvent. There will be a decrease in freezing point and this decrease is called depression in freezing point.

$\Delta T_f = T_f - T_s$

where ΔT_f is depression in freezing point, T_f and T_s are the freezing point of pure solvent and solution respectively.



$\Delta T_f \propto m$

$\Delta T_f = K_f \times m$

where K_f is called molal depression constant.

If $m = 1$, then $\Delta T_f = K_f$.

All those arguments are valid for triple point and not for the freezing point.

K_f can be defined as depression in freezing point of the solution if the solution is 1 molal with respect to solute.

Relation of K_b, K_f with latent heat of vapourization and latent heat of fusion:

$$K_f = \frac{RT_f^2}{1000\ell_f}$$

where T_f is the freezing point of pure solvent

ℓ_f is the latent heat of fusion/g of solvent

$$K_b = \frac{RT_b^2}{1000\ell_v}$$

where T_b is the boiling point of pure solvent

ℓ_v is the latent heat of vapourization/g of solvent

Abnormal Colligative Properties

As we know, colligative properties of a solution depends upon the number of particles present in solution, in case there is an ionization or association in solution, the number of particles increases or decreases respectively and the value of the colligative properties will also be changed. For this, van't Hoff introduced a factor 'i' called van't Hoff factor and is defined as

$$i = \frac{\text{Abnormal colligative property}}{\text{Normal colligative property}}$$

$$i = \frac{P_{obs}}{P_{normal}} = \frac{\Delta P_{obs}}{\Delta P_{normal}} = \frac{(\Delta T_b)_{observed}}{(\Delta T_b)_{normal}} = \frac{(\Delta T_f)_{observed}}{(\Delta T_f)_{normal}}$$

or $$i = \frac{\text{Actual no. of particles in solution}}{\text{No. of particles taken}}$$

Since all the colligative properties are inversely proportional to molecular mass, m

$$i = \frac{\text{Normal molecular mass}}{\text{Observed molecular mass}}$$

ILLUSTRATIVE PROBLEMS

1. A mixture which contains .550 g camphor and 0.045 g of an organic solute freezes at 157°C. The solute contains 93.46% of C and 6.54% of H by weight. What is the molecular formula of the compound?
(Freezing point of camphor = 178.4°C and $K_f = 37.7°$)

Sol. $\text{Molality} = \frac{\Delta T_f}{k_f} = \frac{178.4 - 157}{37.7°} = \frac{20.4}{37.7°}$

$\text{Molality} = \frac{0.45/M}{.550} \times 1000 = \frac{4500}{55M}$ (M is mol. wt.)

$\frac{4500}{55M} = \frac{20.4}{37.7°} = M = 154$

From given weight % of C and H we can calculate

Moles of C $= \frac{93.46}{12} = 7.79$

Moles of H $= \frac{6.54}{1} = 6.54$

Molar ratio of C and H is 12 : 10

Empirical formula is $(C_{12}H_{10})$

M.F = $C_{12}H_{10}$

2. Benzene and toluene form nearly ideal solution. If at 300K $P^0_{benzene}$ = 103.01 mm and $P^0_{toluene}$ = 32.06 mm.
(i) Calculate the vapour pressure of a solution containing 0.6 mol fraction of toluene in liquid phase.
(ii) Calculate the mole fraction of toluene in the vapour form for this composition of liquid.

Sol. $P_{total} = P^0_T X_T + P^0_B X_B = P^0_T X_T + P^0_B (1 - X_T)$

$= 32.06 \times 0.6 + 103.01 \times 0.4$

$= 60.44$ mn.

$\frac{1}{60.44} = \frac{X'_T}{32.06} + \frac{1 - X'_T}{103.01}$ (X′ Refers to mol fraction in vapour phase)

$60.44X'_T = 0.6 \times 32.06 \Rightarrow X'_T = 0.3182$

3. A 0.025 m solution of mono basic acid had a freezing point of -0.06°C. Calculate K_a for the acid at the freezing point. K_f for H_2O = 1.86 K Molality^{-1}. Assume molality equal to molarity.

Sol. For mono basic acid HA

$HA \rightleftharpoons H^+ + A^-$

1 0 0

$(1-\alpha)$ α α $\quad i = (1 + \alpha)$

Assuming $K_a = \frac{c\alpha^2}{(1-\alpha)}$ ..... (i)

$\Delta T_f = K_f \times i$

$\Rightarrow 0.06 = 1.86 \times 0.025 \times (1+\alpha)$

$\therefore (1+\alpha) = \dfrac{0.06}{0.0465}$

$\therefore \alpha = 0.029, \quad C = 0.025M$

$K_a = \dfrac{c\alpha^2}{(1-\alpha)}$

From equation (i), $K_a = \dfrac{0.025 \times (0.029)^2}{(1-0.029)} = \mathbf{2.16 \times 10^{-5}}$

4. The freezing point of a 0.08 molal aqueous solution of $NaHSO_4$ is −0.372°C. Calculate the dissociation constant for the reaction at the given temperature.
$HSO_4^- \rightleftharpoons H^+ + SO_4^{-2}$
(K_f of water = 1.86 k kg/mole)

Sol.

$NaHSO_4 \longrightarrow Na^+ + HSO_4^-$

0.08 0 0

(0.08) (0.08)

Suppose the degree of dissociation of HSO_4^- is α

$HSO_4^- \rightleftharpoons H^+ + SO_4^{-2}$

At Eqb. $C(1-\alpha)$ $C\alpha$ $C\alpha$

$\Rightarrow [H^+] = 0.08\,\alpha$

$[SO_4^{-2}] = 0.08\,\alpha$

$[HSO_4^-] = 0.08\,(1-\alpha)$

Total ions present $= 0.08 + 0.08(1-\alpha) + 0.08\,\alpha + 0.08\,\alpha$

$= 0.16 + 0.08\,\alpha$

$i = \dfrac{0.16 + 0.08\alpha}{0.08} = (2 + \alpha)$

$\Delta T_f = K_f \times m \times i$

$0.372 = (2 + \alpha) \times 1.86 \times 0.08$

$\Rightarrow \alpha = \mathbf{0.5}.$

5. Calculate the boiling point of a solution containing 0.60 g of benzoic acid in 50 g of $CS_2(\ell)$ assuming 90% association of the acid. The boiling point and K_b CS_2 are 46.2°C and 2.44 K Kg mol^{-1} respectively.

Sol.

$\Delta T = i \times K_b \times \text{molality}$

$\Delta T = i \times K_b \times \dfrac{W_{C_6H_5COOH} \times 100}{m \times W_{CS_2}}$

If out of every mole of solute 2α moles have associated, α moles of dimer are present in the solution

$\Rightarrow$ Total moles of solute $= n(1-\alpha) \Rightarrow i = 1-\alpha$

$i = 0.55$

$\Delta T_b = iK_b m = 0.55 \times 2.44 \times \dfrac{0.60 \times 1000}{122 \times 50}$

$= 0.132\ K = 0.132°C$

$\Rightarrow$ Final boiling point $= 0.132 + 46.2 = 46.33°C$

6. The molar volume of liquid benzene (density = 0.877 gml^{-1}) increases by a factor of 2750 as it vapourizes at 20°C. At 27°C when a non-volatile solute (that does not dissociate) is dissolved in 54.6 cm^3 of benzene, vapour pressure of this solution, is found to be 98.88 mm Hg. Calculate the freezing point of the solution.
Given: Enthalpy of vapourization of benzene (*l*) = 394.57 Jg^{-1}.
Enthalpy of fusion of benzene (*l*) = 10.06 $kJmol^{-1}$.
Molal depression constant for benzene = 5.0 K $kgmol^{-1}$.

Sol. Let, the moles of benzene vapourizes at 20°C = n_1

Volume of n_1 mole of benzene (*l*) = $\frac{78n_1}{0.877}$

Volume of n_1 mole of benzene (g) = $2750\left(\frac{78n_1}{0.877}\right)$

$PV = nRT$

$$P^{\circ}\left(\frac{78n_1}{0.877}\right)\left(\frac{2750}{1000}\right) = n_1 \times 0.082 \times 293$$

$P^{\circ}_{Benzene} = 0.0982$ atm. = 74.63 mm Hg

$P^{\circ}_{Benzene}$ at 27° C can be calculated as

$$\log\left(\frac{P_2}{P_1}\right) = \frac{\Delta H_{vap}}{2.303R}\left[\frac{T_2 - T_1}{T_1T_2}\right]$$

$$\log\left(\frac{P_2}{74.63}\right) = \frac{394.57 \times 78}{2.303 \times 8.314}\left[\frac{7}{300 \times 293}\right]$$

$P^{\circ}_{Benzene}$ (at 27°C) = 100.2 mm Hg

$$\frac{\text{moles of solute} \times 78}{54.6 \times 0.877} = \frac{100.2 - 98.88}{98.88}$$

$\Rightarrow$ Moles of solute = 8.20×10^{-3}

$$M = \text{molality of solution} = \frac{8.20 \times 1000}{54.6 \times 0.877}$$

M = 0.168

Freezing point change = $5 \times 0.168 = 0.85$

Freezing point of pure benzene can be calculated using $K_f = \frac{RT_f^2 MW_{solvent}}{1000\Delta Hf} \Rightarrow T_{f_{benzene}} = 278.5\,K$

$\Rightarrow$ New freezing point = 278.5 – 0.85 = 277.65

7. The K_{sp}(25°C) of sparingly soluble salt $XY_2(s)$ is 3.56×10^{-5} (mol $lit^{-1})^3$ and at 30°C the vapour pressure of its saturated solution in water is 31.78 mm Hg. Calculate the enthalpy change of the reaction.
$XY_2(s) \rightleftharpoons X^{2+}(aq) + 2Y^-(aq)$
Given: Vapour pressure of pure water = 31.82 mm Hg

Sol. Let, the solubility of $AB_2(s)$ at 30°C be 's' molLit^{-1}

$AB_2(s) \rightleftharpoons A^{2+}(aq) + 2B^-(aq)$

s 2s

$$\frac{P^\circ - P_s}{P_s} = \frac{n}{N}$$

$$\frac{31.82 - 31.78}{31.78} = \frac{3s}{55.56}$$

$s = 0.0233$ mol/lit.

K_{sp} (at 30°C) $= (0.0233)(2 \times 0.0233)^2$

$= 5.05 \times 10^{-5}$

K_{sp} (at 25°C) $= 3.56 \times 10^{-5}$

Now, $\log \frac{K_{sp}(30^\circ C)}{K_{sp}(25^\circ C)} = \frac{\Delta H}{2.303\ R}\left[\frac{T_2 - T_1}{T_1\ T_2}\right]$

$$\log\left(\frac{5.05\times10^{-5}}{3.56\times10^{-5}}\right) = \frac{\Delta H}{2.303\times8.314}\left(\frac{5}{303\times298}\right)$$

$\Delta H^\circ = 48.802$ J/mole.

8. The freezing point of an aqueous saturated solution of I_2 is – 0.0024°C. More than this can dissolve in a KI solution because of the following equilibrium:

$I_2(aq) + I^-(aq) \longrightarrow I_3^-(aq)$

0.1 M KI solution dissolves 12.5 g/L of I_2. Calculate the equilibrium constant K_c for the above equilibrium. Also calculate the freezing point of the resulting solution. Assume (i) molarity to be equal to molality (ii) assume that conc. of I_2 in all saturated solutions is same.

[K_f for water = 1.86 K kg mol^{-1}]. (iii) the equilibrium constant of the above mentioned reaction remain unaffected by temperature change.

Sol. $\Delta T_f = K_f m$

$$m = \frac{0.0024}{1.86} = 0.0013 \text{ mole/kg}$$

12.5 g of I_2 =

$I_2(aq)$	+ $I^-(aq)$ $\rightleftharpoons$	$I_3^-(aq)$	
0.0492	0.1	0	Initial
(0.0492–x)	(0.1–x)	x	Equilibrium

Now, 0.0492 – x = 0.0013, because of conc. of I_2 in all saturated solutions is same.

x = 0.0479

$$K_c = \frac{[I_3^-]}{[I_2][I^-]}$$

$$K_c = \frac{(0.0479)}{(0.0013)(0.0521)}$$

$K_c = 707.2$

$\Delta T_f = 1.86\ (0.0479 + 0.0013 + 0.0521 + 0.1)$

$\Delta T_f = 0.3744$ K

∴ Freezing point of the resulting solution = 272.77 K

SOLVED PROBLEMS

Subjective Question

Prob. 1. Calculate the entropy change for vapourisation of water if latent heat of vapourisation for water is 540 cal/g. Given K_b for H_2O = 0.51 K/molality.

Sol. $$K_b = \frac{RT_b^2.M}{1000\Delta H_v} = \frac{RT_bM}{1000\Delta S_v}\left(\because \Delta S_v = \frac{\Delta H_v}{T_b}\right)$$

$$\therefore \Delta S_v = \frac{RT_b.M}{1000K_b} = \frac{2\times 373\times 18}{1000\times 0.51}$$

$= 26.33$ cal/mol

Prob. 2. If boiling point of an aqueous solution is 100.1°C. What is its freezing point? Given latent heat of fusion and vaporization of water are 80 cal g^{-1} and 540 cal g^{-1} respectively.

Sol. For a given aqueous solution

$\Delta T_b = K_b' \times$ molality

$\Delta T_f = K_f' \times$ molality

$$\therefore \frac{\Delta T_b}{\Delta T_f} = \frac{K_b'}{K_f'} = \frac{RT_b^2}{1000\ell_v} \times \frac{1000\ell_f}{RT_f^2}$$

$$\frac{\Delta T_b}{\Delta T_f} = \frac{T_b^2 \times \ell_f}{T_f^2 \times \ell_v}$$

$\because$ $T_b = 100 + 273 = 373$ K

$T_f = 0 + 273 = 273$ K

$\ell_f = 80$ cal g^{-1}

$\ell_v = 540$ cal g^{-1}

$$\therefore \frac{0.1}{\Delta T_f} = \frac{373\times 373\times 80}{273\times 273\times 540}$$

$\therefore$ $\Delta T_f = 0.362$

$\therefore$ $T_f = 0.0 - 0.362 = -0.362$°C

Prob. 3. The molar volume of liquid benzene (density = 0.877 gml^{-1}) increases by a factor of 2750 as it vapourizes at 20°C. At 27°C when a non-volatile solute (that does not dissociate) is dissolved in 54.6 cm^3 of benzene, vapour pressure of this solution, is found to be 98.88 mm Hg. Calculate the freezing point of the solution.

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$P^{\circ}_{\text{Benzene}} = 0.0982$ atm. $= 74.63$ mm Hg

$P^{\circ}_{\text{Benzene}}$ at 27° C can be calculated as

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$$\log\left(\frac{P_2}{74.63}\right) = \frac{394.57 \times 78}{2.303 \times 8.314}\left[\frac{7}{300 \times 293}\right]$$

$P^{\circ}_{\text{Benzene}}$ (at 27°C) = 100.2 mm Hg

$$\text{Molality of the solution} = \frac{X_{\text{solute}} \times 1000}{X_{\text{solvent}} \times 78} = \frac{\left(\frac{100.2 - 98.88}{98.88}\right) \times 1000}{0.98 \times 78} = 0.17$$

$\Delta T_f = K_f m$

$= 5 \times 0.17 = 0.85$

We know, that, $K_f = \left[\frac{RT_f^2}{1000\ \Delta H_f}\right] M$

$$5 = \left(\frac{8.314 \times T_f^2}{1000 \times 10060}\right) 78$$

$T_f = 278.5$ K

∴ Freezing point of solution = 278.5 – 0.85 = **277.65 K**

Prob. 4. 100 ml aqueous solution of glucose with osmotic pressure 1.2 atm at 25°C is mixed with 300 ml aqueous solution of urea at 2.4 atm at 25°C. Calculate the osmotic pressure of mixture.

Sol. $\pi_1 V_1 + \pi_2 V_2 = \pi_R (V_1 + V_2)$

$$\pi_R = \frac{\pi_1 V_1 + \pi_2 V_2}{V_1 + V_2}$$

$$= \frac{1.2 \times 100 + 2.4 \times 300}{500}$$

$$= \mathbf{1.86\ atm}$$

Objective Questions

Prob. 1. The relative lowering of vapour pressure of an aqueous solution of a non-volatile solute of molecular weight 60 (which neither dissociates nor associates in the solution) is 0.018. If K_f of water is $1.86°m^{-1}$, the depression in freezing point will be

(A) 0.93°C (B) 3.72°C

(C) 1.86°C (D) 0.018°C

Sol. (C)

0.018 = mole fraction of solute

$$0.018 = \frac{x}{x + 55.5}$$ where x is the no. of moles of solute in 1 litre water.

$$\Rightarrow \quad x = \frac{0.999}{1-0.018}$$

$$x = 0.99$$

or $\quad \simeq 1 = m$

$$\Delta T_f = K_f \times m$$

$$\therefore \Delta T_f = 1.86 \times 1 = 1.86$$

Prob. 2. Which of the following salt will have the same value of Van't Hoff's factor (i) as that of $K_4[Fe(CN)_6]$

(A) $Al_2(SO_4)_3$ (B) NaCl

(C) $Al(NO_3)_3$ (D) Na_2SO_4

Sol. **(A)**

On dissociation both $K_4(Fe(CN)_6]$ and $Al_2(SO_4)_3$ give five ions each.

Prob. 3. The boiling point of an aqueous solution of a non-volatile solute is 100.15°C. What is the freezing point of an aqueous solution obtained by diluting the above solution with an equal volume of water? The value of K_b and K_f for water are 0.512°C and 1.86°C molality^{-1} respectively.

(A) –0.544°C (B) –0.52°C

(C) –0.272°C (D) –1.86°C

Sol. Fro the same solution

$$\frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b} \Rightarrow \Delta T_f = \frac{K_f}{K_b} \times \Delta T_b$$

$$\therefore \Delta T_f = \frac{0.15 \times 1.86}{0.512} = 0.545$$

On diluting the solution to double

$$\Delta T_f \propto \frac{1}{\text{Wt. of solvent}}$$

$$\Delta T_f = \frac{0.545}{2} = 0.272$$

$\therefore$ Freezing point = **–0.272**

Hence, **(C)** is the correct answer.

Prob. 4. Acetic acid exists in benzene solution in the dimeric form, in an actual experiment the Van't Hoff factor was found to be 0.52. Then the degree of association of acetic acid is

(A) 0.48 (B) 0.88

(C) 0.96 (D) 0.52

Sol. $2CH_3COOH \rightleftharpoons (CH_3COOH)_2$

1	0
(1-α)	α/2

$i = 1-\alpha/2$

or, $\alpha = 2\times(1-i)$

$= 2(1-0.52) =$ **0.96**

Hence, **(C)** is the correct answer.

Prob. 5. The van't Hoff factor for 0.1M. $Ba(NO_3)_2$ solution is 2.74. The percentage of dissociation is:

(A) 91.3%
(B) 87%
(C) 100%
(D) 74%

Sol. $Ba(NO_3)_2 \rightleftharpoons Ba^{2+} + 2NO_3^-$

1	0	0
$(1-\alpha)$	α	2α

$i = 1-\alpha + \alpha + 2\alpha = (1 + 2\alpha)$

$i = (1 + 2\alpha)$

$$\therefore \alpha = \frac{(i-1)}{2} = \frac{2.74-1}{2} = 0.87$$

$= \mathbf{87\%}$

Hence, **(B)** is the correct answer

Prob. 6. The wt. of solute (M= 60) required to be added in (180g of water to reduce the vapour pressure $4/5^{th}$ of pure water is :-

(A) 140 g
(B) 134 g
(C) 150 g
(D) 240 g

Sol.
$$\frac{P^0 - P_s}{P_s} = \frac{w_1 \times m_2}{m_1 \times w_2}$$

$$\Rightarrow \frac{P^0 - \frac{4}{5} \times P^0}{\frac{4P^0}{5}} = \frac{w_1 \times 18}{60 \times 180}$$

$$\therefore w_1 = \frac{60 \times 180}{4 \times 18} = \mathbf{150g}$$

Hence, **(C)** is the correct answer.

Prob. 7. The vapour pressure of an aqueous solution of glucose is 750 mm of Hg at 373 K. the molality is

(A) 0.64
(B) 0.741
(C) 0.68
(D) 0.94

Sol. $\frac{P^0 - P_s}{P_s} = \frac{w_1 \times m_2}{m_1 \times w_2}$ (Assuming the solution to be dilute and remembering that at 373 K the equilibrium vapour pressure of water is 760 mm of Hg).

$$\Rightarrow \frac{P^0 - P_s}{P_s} = \frac{(w_1 \times 1000)}{(m_1 \times w_2)} \times \frac{m_2}{1000} = M \times \frac{m_2}{1000}$$

(m = molality, m_2 = mol wt of solvent)

$$\Rightarrow \frac{760 - 750}{750} = M \times \frac{18}{1000}$$

$$m = \frac{10 \times 1000}{750 \times 18} = 0.741 \text{mol/Kg of solvent}$$

Hence, **(B)** is the correct answer.

Prob. 8. The freezing point of solution containing 50 cm^3 of non volatile liquid (A) in 50g of water is -34^0C. Assuming ideal behaviour, the density of glycol in g/cm^3 is:

(A) 1.24
(B) 1.43
(C) 1.133
(D) 1.23

Sol. $V = 50\ cm^3$ (for A)

Let density of A be d g/cc.

Wt. of A ethylene glycol $(w_1) = 50 \times d$

Wt. of water $w_2 = 50g$.

$$\Delta T_f = \frac{K_f \times w_1 \times 1000}{m_1 \times w_2}$$

$$\Rightarrow 34 = \frac{1000 \times 1.86 \times 50 \times d}{62 \times 50}$$

$\therefore$ **d = 1.133g/cm^3.**

Prob. 9. The amount of ice that will separate out on cooling a solution containing 50g of non volatile solute glycol in 200g of water to -9.3^0C. (K_f for water = 1.86 Kmol^{-1}Kg)

(A) 48.2g

(B) 38.71g

(C) 37.1g

(D) 4.2g

Sol. $$\Delta T_f = \frac{K_f \times w_1 \times 1000}{m_1 \times w_2}$$

$$\Rightarrow 9.3 = \frac{1000 \times 1.86 \times 50}{62 \times w_2}$$

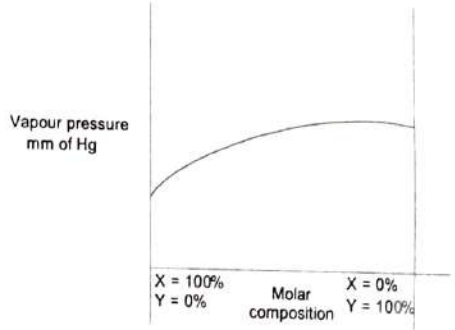
$w_2 = 161.2g$

Wt. of ice separated = (200-161.2)g

= **38.71g.**

Hence, **(B)** is the correct answer.

Prob. 10. The following graph shows the variation in vapour pressure of a solution of two liquids X and Y as a function of molar composition.



It can be deduced that

(A) pairs of liquids which give rise to this slope of curve evolve heat on mixing

(B) the curve shows positive deviation from Raoult's law

(C) Y has higher boiling point than X

(D) the two liquids could be ethanol and cyclohexane

Sol. **(B, D)**

Graph shown is of a non-ideal solution with positive deviation from Raoult's law, $\Delta H_{mix} > 0$.

Vapour pressure of X < vapour pressure of Y at T

Boiling point of X > boiling point of Y

Prob. 11. Among 0.1 M solution of urea and Na_3PO_4 and aluminium sulphate

(A) the vapour pressure and freezing point are the highest for urea

(B) the vapour pressure and freezing point are the lowest for urea

(C) the elevation in boiling point is the highest for $Al_2(SO_4)_3$

(D) the depression in freezing point is the highest for $Al_2(SO_4)_3$

Sol. **(B, C, D)**

Prob. 12. When a non-volatile solute is added to a pure solvent, the

(A) V.P. of solution becomes lower than the V.P. of pure solvent

(B) rate of evaporation of solvent is reduced

(C) solute does not affect the rate of condensation

(D) rate of evaporation of solvent = rate of condensation of the solvent at equilibrium

Sol. **(A, B, C, D)**

Prob. 13. Consider the following solutions kept at the same temperature of 300 K

(I) 0.01 M aq. solution of $K_4[Fe(CN)_6]$

(II) 0.1 M aq. solution of CaCl$_2$

(III) 0.2 M aq. solution of glucose

(IV) 0.1 M aq. solution of CH_3COOH ($K_a = 10^{-5}$)

Assuming molarity = molality, select the correct statements of the following:

(A) Correct order of freezing points : (I) > (IV) > (III) > (II)

(B) Correct order of boiling points is : (I) < (IV) < (III) < (II)

(C) Solution (III) and (IV) are isotonic

(D) V.P. of (II) is highest among all

Sol. **(A, B)**

	(I)	(II)	(III)	(IV)
$\Delta T_f = i \times K_f \times m$:	$0.05\ K_f$	$0.3\ K_f$	$0.2\ K_f$	$0.101\ K_f$
$\Delta T_b = i \times K_b \times m$:	$0.05\ K_b$	$0.3\ K_b$	$0.2\ K_b$	$0.101\ K_b$
$\pi = iCRT$:	$0.05\ RT$	$0.3\ RT$	$0.2\ RT$	$0.101\ RT$

$$1-\frac{P_s}{P^\circ}=\frac{\frac{in_B}{n_A}}{\frac{in_B}{n_A}+1}=\frac{\frac{0.05}{1000}\times\frac{18}{1}}{\frac{0.05}{1000}\times\frac{18}{1}+1} \quad \frac{\frac{0.3}{1000}\times\frac{18}{1}}{\frac{0.3}{1000}\times\frac{18}{1}+1} \quad \frac{\frac{0.2}{1000}\times\frac{18}{1}}{\frac{0.2}{1000}\times\frac{18}{1}+1} \quad \frac{\frac{0.101}{1000}\times\frac{18}{1}}{\frac{0.101}{1000}\times\frac{18}{1}+1}$$

Hence, A and B is the correct answer.

Prob.14. The plots of $\frac{1}{X_A}$ vs. $\frac{1}{Y_A}$ (where, X_A and Y_A are the mole fraction of liquid A in liquid and vapour phase respectively) is linear with slope and intercepts respectively:

(A) $\frac{P_A^0}{P_B^0}$ and $\frac{\left(P_A^0 - P_B^0\right)}{P_B^0}$

(B) $\frac{P_A^0}{P_B^0}$ and $\frac{\left(P_B^0 - P_A^0\right)}{P_B^0}$

(C) $\frac{P_B^0}{P_A^0}$ and $\frac{\left(P_A^0 - P_B^0\right)}{P_B^0}$

(D) $\frac{P_B^0}{P_A^0}$ and $\frac{\left(P_B^0 - P_A^0\right)}{P_B^0}$

Sol. **(B)**

$P'_A = P^0_A X_A$ and $P'_A = P^0_A X_A$

$\therefore$ and $P'_B = P_M Y_B$

$\therefore \quad \frac{P'_A}{Y_A} = \frac{P'_B}{Y_B}$

$\Rightarrow \frac{P^0_A X_A}{Y_A} = \frac{P^0_B X_B}{Y_B} = \frac{P^0_B (1 - X_A)}{(1 - Y_B)}$

$\Rightarrow \frac{1}{X_A} = \frac{1}{Y_A}\frac{P^0_A}{P^0_B} + \frac{(P^0_B - P^0_A)}{P^0_B}$

$\Rightarrow Y = mx + C$

$\therefore$ Slope $= m = \frac{P^0_A}{P^0_B}$ and intercept $C = \frac{(P^0_B - P^0_A)}{P^0_B}$

Prob.15. At 40°C the vapour pressure in torr, of methyl alcohol – ethyl alcohol solutions is represented by the equation. $P = 119X_A + 135$; where , X_A is mole fraction of methyl alcohol, then the value of $\lim_{X_{A\to 1}} \frac{P_A}{X_A}$ is

(A) 254 torr
(B) 135 torr
(C) 119 torr
(D) 140 torr

Sol. **(A)**

$P = 119\,X_A + 135$; $\lim_{X_A \to 1} \frac{P_A}{X_A} = 119 + 135 = 254$ torr

Prob.16. Y g of non-volatile organic substance of molecular mass M is dissolved in 250 g benzene. Molal elevation constant of benzene is K_b. Elevation in its boiling point is given by:

(A) $\frac{M}{K_b Y}$
(B) $\frac{4K_b Y}{M}$
(C) $\frac{K_b Y}{4M}$
(D) $\frac{K_b Y}{M}$

Sol. **(B)**

$$\Delta T = \frac{100 \times K_b \times Y}{250 \times M} = \frac{4K_b Y}{M}$$

Prob.17. The boiling point T_b^* of a solvent becomes T_b on addition of X_1 mole fraction of solute. Heat of vaporization of solvent is ΔH_{vap}. The relation between elevation in boiling point $\Delta T_b \left(= T_b - T_b^*\right)$ can be given by : $\frac{\Delta T_b}{T_b - T_b^*} = -\frac{R \ln X_1}{\Delta H_{vap}}$. A graph plotted between $\log_{10} X_1$ vs. $\frac{1}{T_b}$ gives:

(A) A straight line with slope $\frac{R}{\Delta H_{vap}}$ and intercept T_b^*

(B) A straight line with slope $\frac{2.303R}{\Delta H_{vap}}$ and intercept $\frac{1}{T_b}$

(C) A straight line with slope $\frac{\Delta H_{vap}}{2.303R}$ and intercept $\frac{1}{T_b^* \times 2.303}$

(D) A straight line with slope $\frac{\Delta H_{vap}}{2.303R}$ and intercept $-\frac{\Delta H_{vap}}{2.303RT_b^*}$

Sol. **(D)**

$$\frac{T_b - T_b^*}{T_b \times T_b^*} = -\frac{2.303R \log_{10} X_1}{\Delta H_{vap}} \Rightarrow \frac{1}{T_b^*} - \frac{1}{T_b} = -\frac{2.303R \log_{10} X_1}{\Delta H_{vap}}$$

$$\frac{1}{T_b} - \frac{1}{T_b^*} = \frac{2.303R \log_{10} X_1}{\Delta H_{vap}}$$

$$\log_{10} X_1 = \frac{\Delta H_{vap}}{2.303RT_b} - \frac{\Delta H_{vap}}{2.303RT_b^*}$$

Prob.18. The plots of ratio of osmotic pressure (in atm) and density of solution (π / ρ) vs ρ (in g cm^{-3}) on extrapolation to y axis gives the value of

(A) $\frac{M}{RT}$

(B) $\frac{RT}{M}$

(C) Zero

(D) RTM

Sol. **(B)**

$$\pi = \frac{\rho RT}{M}; \left(\frac{\pi}{\rho}\right)_{\rho \to 0} = \frac{RT}{M}$$

Prob. 19. Which statement is in correct about osmotic pressure (p), volume (v) and temperature (T) ?

(A) $p \propto \frac{1}{v}$ if T is constant

(B) $p \propto T$ if v is constant

(C) $p \propto v$ if T is constant

(D) pv is constant if T is constant

Sol. **(C)**

Prob. 20. For a non volatile solute :

(A) vapour pressure of solute is zero

(B) vapour pressure of solution = vapour pressure of pure solvent

(C) vapour pressure of solution = vapour pressure of solvent in solution

(D) all of the above

Sol. **(A, C)**

COMPREHENSION

Read the paragraph carefully and answer the following questions:

The vapour pressure of a solvent in a solution is lower than that of pure solvent, at the same temperature. A higher temperature is needed to raise the vapour pressure up to the atmospheric pressure, when boiling begins. However, increase is small, like 0.1 $molkg^{-1}$ aqueous sucrose solution boils at 100.05°C.
Sea water, an aqueous solution, which is rich in Na^+ and Cl^- ions, freezes about 1°C lower than frozen water. At the freezing point of a pure solvent, the rates at which two molecules stick together to form the solid and leave it to return to liquid state are equal when solute is present. Fewer solvent molecules are in contact with surface of solid. However, the rate at which the solvent molecules leave the surface of solid remains unchanged. That is why temperature is lowered to restore the equilibrium. The freezing point depression in an ideal solution is proportional to molality of the solute.

1. An aqueous solution of 0.1 $molkg^{-1}$ concentration of sucrose should have freezing point of ($K_f = 1.86$)
   (A) –0.2°C
   (B) 0.2°C
   (C) 2°C
   (D) –2°C

2. When 250 mg of eugenol is added to 100 gm of camphor ($K_f = 39.7$), it lowered the freezing point by 0.62°C. The molar mass of eugenol is
   (A) 1.6×10^2 gm/mole
   (B) 1.6×10^4 gm/mole
   (C) 1.6×10^3 gm/mole
   (D) 200 gm/mole

3. The freezing point of a 5% by mass CH_3COOH (aq.) solution is –1.576°C. The van't Hoff factor is (K_f of water = 1.86)
   (A) 0.996
   (B) 2
   (C) 0.5
   (D) None of the above

4. The freezing point of benzene solution was 5.4°C. The osmotic pressure at 10°C is (boiling point of benzene = 5.5°C)
   (A) 0.2 atm
   (B) 0.4 atm
   (C) 0.6 atm
   (D) 0.8 atm

Answers

1.	A	2.	A	3.	A	4.	B

REASONING TYPE QUESTIONS

1. **STATEMENT-1:** *The freezing point of 0.05 M urea solution is different from that of 0.05 M sodium chloride solution.*
and
STATEMENT-2: *The depression in freezing point is directly proportional to the number of species present in the solution.*
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

2. **STATEMENT-1:** *If one component obeyed Raoult's law over a certain range of composition, the other components would not obey Henry's law in that range.*
and
STATEMENT-2: *Raoult's law is a special case of Henry's law.*
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(B)**

3. **STATEMENT-1:** *The boiling point of 0.1 M urea solution is less than that of 0.1 M KCl solution.*
and
STATEMENT-2: *Elevation of boiling point is directly proportional to the number of species present in the solution.*
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

4. **STATEMENT-1:** *Acetic acid has a molecular weight of 120 in benzene solution.*
and
STATEMENT-2: *There is dimmer formation by H-bonding.*
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

5. **STATEMENT-1:** *The vapour pressure of 0.1 M sugar solution is more than that of 0.1 M potassium chloride solution*
and
STATEMENT-2: *Lowering of vapour pressure is directly proportional to the number of species present in the solution*
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

MATRIX – MATCH TYPE QUESTIONS

2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	CH_3COOH in H_2O	(p)	Neither association nor dissociation
(B)	CH_3COOH in benzene	(q)	When a non-volatile solute is added.
(C)	Polymer in water	(r)	Molecular mass observed greater than molecular mass of AcOH
(D)	Vapour pressure of a liquid decrease	(s)	$\Delta T_{b(obs)} > \Delta T_{b(calc.)}$

Sol. **(A) → (s); (B) → (r); (C) → (p); (D) → (q)**

2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	0.1 M $Ca_3(PO_4)_2$	(p)	Solution with highest boiling point
(B)	0.1 M NaCl	(q)	Solution with van't Hoff factor greater than 1
(C)	0.1 M glucose	(r)	Solution with lowest osmotic pressure
(D)	0.1 M $CaCl_2$	(s)	Solution with lowest freezing point
		(t)	Van't Hoff factors $(1+\alpha)$

Sol. **A → (p, q, s) B → (q) C → (r) D → (q, t)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Lyophilics	(i)	Gold number	(P)	Change is independent of pH
(II)	Lyophobics	(ii)	Organic substances	(Q)	Change depends on pH
(III)	Protection	(iii)	Blue tinge	(R)	Tyndall effect
(IV)	Smoke	(iv)	Inorganic substances	(S)	Gelatin

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q)
[B] (II) (iii) (R)
[C] (III) (iii) (S)
[D] (I) (ii) (Q)

4. Which of the following combination is correct ?
[A] (I) (iii) (Q)
[B] (III) (i) (S)
[C] (I) (ii) (Q)
[D] (IV) (iii) (R)

5. Which of the following combination is correct ?
[A] (IV) (i) (Q)
[B] (I) (i) (R)
[C] (II) (iv) (P)
[D] (II) (i) (P)

Sol.

3. **D**
4. **A**
5. **C**

SINGLE INTEGER TYPE QUESTIONS

1. A 0.0020 m aqueous solution of an ionic compound $Co(NH_3)_5NO_2$ freezes at – 0.00732°C. What will be the number of moles of ions which 1 mol of ionic compound produces (K_f = - 1.86° c/m).

Sol. **(2)**

$\Delta T_f = iK_f m$

$\therefore i = \dfrac{\Delta T_f}{K_f \times m} = \dfrac{0.00732}{1.86 \times 0.002} = 2$

2. What will be the mass of glucose that should be dissolved in 50 g of water in order to produce the same lowering of vapour pressure as produced by dissolving 1 g of urea in the same quantity of water?

Sol. **(3)**

$\dfrac{p^\circ - p_s}{p^\circ} = \dfrac{n_2}{n_1} = \dfrac{W_2M_1}{W_1M_2}$

As $\dfrac{p^\circ - p_s}{p^\circ} = \dfrac{n_2}{n_1}$ same in the two cases.

$\left(\dfrac{W_2M_2}{W_1M_2}\right)_{glucose} = \left(\dfrac{W_2M_1}{W_1M_2}\right)_{urea}$

$\Rightarrow \dfrac{W_2 \times 18}{50 \times 180} = \dfrac{1 \times 18}{50 \times 60} \Rightarrow W_2 = 3g$

3. A $\dfrac{M}{10}$ solution of potassium ferrocyanide is 46% dissociated at 18°C. What will be its osmotic pressure?

Sol. **(6)**

Normal osmotic pressure = $\dfrac{w}{m \times V} \times S \times T$

(when no dissociation has taken place)

$\dfrac{w}{m} = 0.1$, V = 1 litre, S = 0.0821, T = 18 + 273 = 291 K

Normal osmotic pressure = $\dfrac{0.1}{1} \times 0.0821 \times 291 = 2.389$ atm

Potassium ferrocyanide is an electrolyte. It dissociates as:

$$\underset{(1-\alpha)}{K_4Fe(CN)_6} \rightleftharpoons \underset{4\alpha}{4K^+} + \underset{\alpha}{[Fe(CN)_6]^{4-}}$$

Total number of particles = $1 - \alpha + 4\alpha + \alpha = 1 + 4\alpha$

$\alpha = 0.46$; so $1 + 4\alpha = 1 + 4 \times 0.46 = 2.84$

$\dfrac{\text{Observed osmotic pressure}}{\text{Normal osmotic pressure}} = \dfrac{2.84}{1}$

Observed osmotic pressure = 2.84 × 2.389

= 6.785 atm

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Calculate molality of the solution obtained by dissolving 11.7 g NaCl in 500 gm water.

Sol. **0.4**

Prob. 2. At 27°C, calculate osmotic pressure of M/10 solution of urea.

Sol. **2.46**

Prob. 3. At 24°C, osmotic pressure of cane sugar solution is 2.5 atm. Calculate concentration of the solution in mol/litre.

Sol. **0.1025**

Prob. 4. How much urea should be dissolved in 100 gm of water in order to reduce its vapour pressure by 25%.

Sol. **111.11**

Prob. 5. Vapour pressure of an aqueous solution of glucose at 373 K is 750 mm Hg. Calculate mole fraction of glucose.

Sol. **0.013**

Prob. 6. An aqueous solution of glucose is prepared at 303 K by dissolving 10 g glucose in 90 gm water. If vapour pressure of water is 32.8 mm Hg. What is the vapour pressure of solution ?

Sol. **32.44**

Prob. 7. On dissolving 6 gm glucose in 50 g water, 0.34°C elevation is boiling point was recorded. Calculate molal elevation constant of water.

Sol. **0.51**

Prob. 8. Calculate Kb for water if its latent heat of vaporization is 540 cal/g.

Sol. **0.515**

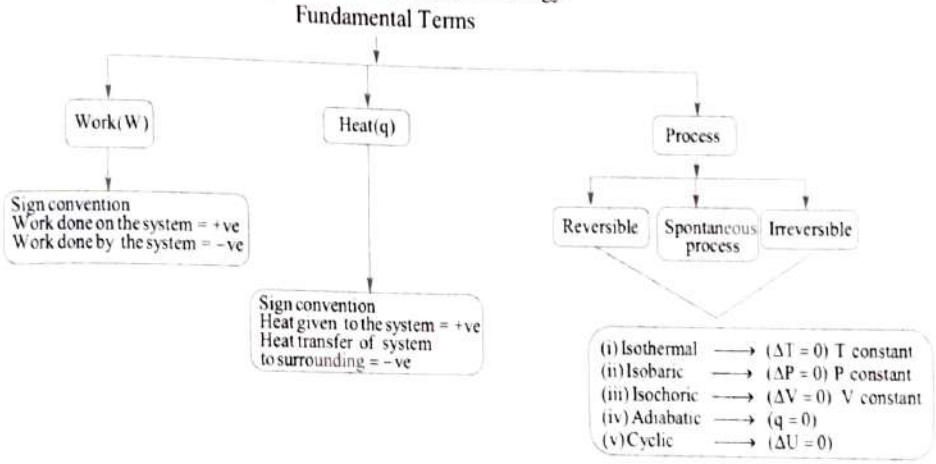
ENERGETICS 10

Energetics can be divided into two branches of science, viz.
1. Thermodynamics
2. Thermochemistry

THERMODYNAMICS

Thermodynamics is the branch of science which mainly deals with
(i) interconversion of different forms of energy
(ii) temperature and thermal manifestation.

The subject matter of thermodynamics is based on two fundamental postulates called two laws of thermodynamics (I and II law), dealing with interconversion of energy.



First Law of Thermodynamics

In any process energy can be changed from one form to another, but it is neither created nor destroyed.

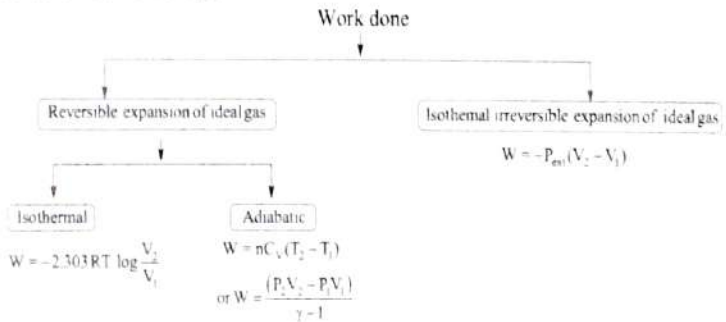
$dq = dE + (-dW) = dE + (PdV)$

For a finite change,

$q = \Delta E - W = \Delta E + P\Delta V$

Work Done in Thermodynamics

(i) Expansion of ideal or real gas
(ii) Compression of ideal or real gas



Work done in an irreversible process:

$$W = -P_{ext} \times \Delta V = -P_{ext}(V_2 - V_1) = -P_{ext}\, R\left(\frac{P_1T_2 - P_2T_1}{P_1P_2}\right)$$

Work done in reversible process, i.e. maximum work

Isothermal conditions: $W_{rev} = -2.303\ n\,RT\ \log_{10}\left(\frac{V_2}{V_1}\right)$

$$W_{rev} = -2.303\ n\,RT\ \log_{10}\left(\frac{P_1}{P_2}\right)$$

Adiabatic conditions: $W_{rev} = \left(\frac{nR}{\gamma - 1}\right)(T_2 - T_1)$

For adiabatic process, following conditions hold good.

PV^{γ} = constant

$T^{\gamma}P^{1-\gamma}$ = constant

$V^{\gamma-1}T$ = constant

Heat Capacities

At constant pressure, $C_p = \left(\frac{\delta H}{\delta T}\right)_p$

At constant volume, $C_v = \left(\frac{\delta E}{\delta T}\right)_v$

$$\frac{C_p}{C_v} = \gamma$$

$C_p - C_v = R$ (Mayor's rule)

Second Law of Thermodynamics

Heat cannot be completely converted to an equivalent amount of work in a cycle process. In other words, the entropy of the universe cannot decrease, i.e. $\Delta S_{univ} >$ zero. (Clausius inequality)

$\Delta S_{univ} = \Delta S_{system} + \Delta S_{surrounding}$

i.e. $\Delta S_{system} + \Delta S_{surrounding} >$ zero

Third Law of Thermodynamics

According to third law of thermodynamics, the entropy of a perfectly crystalline substance is zero at zero degree Kelvin (0°K).

Entropy

(i) Entropy is a state function and an extensive property.

(ii) For any reversible process, $\Delta S_{system} + \Delta S_{surrounding} = 0$

(iii) For irreversible process, $\Delta S_{universe} > 0$

$\Delta S = \sum S_{products} - \sum S_{reactants}$

$\Delta S = \frac{q_{rev}}{T} = 2.303\ nR \log_{10}\left(\frac{V_2}{V_1}\right) = 2.303\ nR \log_{10}\left(\frac{P_1}{P_2}\right)$ (For reversible isothermal expansion of a gas following $PV = nRT$)

$\Delta S_{fusion} = \frac{\Delta H_{fusion}}{T}$

$\Delta S_{vap} = \frac{\Delta H_{vap}}{T}$

Free Energy

$G = H - TS$

$\Rightarrow \Delta G = \Delta H - T\Delta S$ (at constant temperature T)

At equilibrium, $\Delta G = 0$

Heat supplied (at constant pressure) $= \Delta H$

Heat supplied (at constant volume) $= \Delta E$

$\Delta H = \Delta E + P\Delta V$ (at constant pressure)

$\Delta H = \Delta E + \Delta nRT$

Δn = Number of moles of product – Number of moles of reactants (only for gases)

Criteria for spontaneous process:

Consider the equation, $\Delta G = \Delta H - T\Delta S$

ΔS	ΔH	ΔG	T
+ve	Exothermic (–ve)	–ve	At all temperatures
+ve	Endothermic (+ve)	–ve	At high temperature ($T\Delta S > \Delta H$)
–ve	Exothermic (–ve)	–ve	At low temperature
–ve	Endothermic (+ve)	+ve	At all temperatures

Kirchhoff's Equation (For variation of ΔH_r at constant pressure and assumption that Cp_s do not change with temperature)

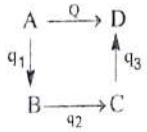
$\Delta H_2 - \Delta H_1 = \Delta C_p(T_2 - T_1)$

$\Delta C_p = \sum C_{p(products)} - \sum C_{p(reactants)}$

THERMOCHEMISTRY

Laws of Thermochemistry

Hess's law of constant heat summation (Constant pressure)



$Q = q_1 + q_2 + q_3$

Certain Terms in Thermochemistry

(i) Heat or enthalpy of formation

(ii) Enthalpy of combustion

(iii) Enthalpy of phase change

(a) Vapourization: $H_2O(l) \longrightarrow H_2O(g)$ $\Delta H = 540$ cal/gm

(b) Fusion: $H_2O(s) \longrightarrow H_2O(l)$ $\Delta H = 1.44$ Kcal/mole

(c) Transition: C(graphite) $\longrightarrow$ C(diamond) $\Delta H = 450$ cal/ gm

(iv) Enthalpy of neutralization

(v) Enthalpy of solution (heat change takes place when a substance is dissolved in a solvent like water)

(vi) Bond energy

(vii) Resonance energy

ILLUSTRATIVE PROBLEMS

1. Calculate heat of neutralization from following data.

200 ml of 1 M HCl is mixed with 400 ml of 0.5 M NaOH and the temparature rise in calorimeter was found to be 4.4°C. Water equivalent of calorimeter is 12 g and specific heat is 1 cal ml^{-1} $degree^{-1}$ for solution.

Sol. Meq. of acid and base = 200

i.e. 200 Meq. of HCl react with 200 Meq. of NaOH to produce heat = ΔH

$\therefore$ 1000 Meq. of HCl when react with 1000 Meq. of NaOH will give heat

= $5 \times \Delta H$ = Heat of neutralization

Now, heat produced during neutralisation of 200 Meq. of acid and base

= heat taken up by calorimeter + solution

= $M_1 \times S_1\Delta T + M_2 \times S_2\Delta T$

= $12 \times 1 \times 4.4 + 600 \times 1 \times 4.4 = 2692.8$ cal

$\therefore$ Heat of neutralization = 5×2692.8 cal

= **– 13.464 kcals**

2. Calculate the entropy change accompanying the conversion of 1 mole of ice at 273.1 K and 1 atm pressure into steam at 373.1 K and 1 atm pressure, given that at 273.1 K, the molar heat of fusion of ice, ΔH_f, is 6.00 kJ mol^{-1} and at 373.1K, the molar heat of vaporisation of water, ΔH_v, is 40.6 kJ mol^{-1}. Also assume that the molar heat capacities, C_p, in the temperature range 373.1–273.1 remains constant. Given that $C_p = 75.25$ $mol^{-1}K^{-1}$ and log 13.66 = 1.1354.

Sol. Let ΔS_1 = entropy change due to fusion of ice.

$$\therefore \Delta S_1 = \frac{\Delta H_f}{T_f} = \frac{6000 J mol^{-1}}{273.1K}$$

$$= 21.96 J mol^{-1} K^{-1}$$

ΔS_2 = Entropy change for heating 1 mole of water from 273.1K to 373.1 K at constant pressure.

$$\text{Then } \Delta S_2 = \int_{T_1}^{T_2} \frac{C_p}{T} dT$$

$$= C_p \ln \frac{T_2}{T_1}$$

$$= C_p \times 2.303 \log \frac{T_2}{T_1}$$

$$= 75.2 \times 2.303 \log \frac{373.1}{273.1}$$

$$= 75.2 \times 2.303 \times 0.1354$$

$$= 23.45 \text{ J mol}^{-1} \text{ K}^{-1}$$

ΔS_3 = Entropy change due to conversion of water into steam.

or entropy change when water is in equilibrium with vapour.

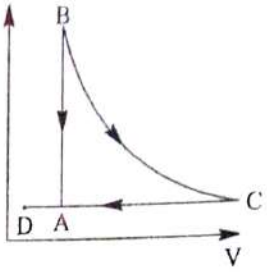
$$w_{BC} = -1 \times RX\ 2T_1 \ln\left(\frac{2V_1}{V_1}\right) = -840 \text{ cal}$$

For isobaric change (CD)

$$W_{CD} = -P\Delta V = -R\Delta T = -P\left[\frac{T_1}{2} - 2T_1\right]$$

$$= 900 \text{ cal}$$

W Total = $-840 + 900 =$ **60 cal**



5. Diborane is a potential rocket fuel which undergoes combustion according to the reaction

$B_2H_6(g) + 3O_2(g) \longrightarrow B_2O_3(s) + 3H_2O(g)$

From the following data, calculate the standard enthalpy change for the combustion of diborane.

$2B(s) + \frac{3}{2}O_2(g) \longrightarrow B_2O_3(g)$ $\quad \Delta H^\circ = -1273\,kJ$ $\quad$ ... (1)

$H_2(g) + \frac{1}{2}O_2(g) \longrightarrow H_2O(l)$ $\quad \Delta H^\circ = -286\,kJ$ $\quad$ ... (2)

$H_2O(l) \longrightarrow H_2O(g)$ $\quad \Delta H^\circ = 44\,kJ$ $\quad$ ... (3)

$2B(s) + 3H_2(g) \longrightarrow B_2H_6(g)$ $\quad \Delta H^\circ = 36\,kJ$ $\quad$ ... (4)

Sol. We have to get standard enthalpy change for $B_2H_6(g) + 3O_2(g) \rightarrow B_2O_3(s) + 3H_2O(g)$

Multiplying equation (2) and (3), by 3, add equation (1), (2), (3) and substract (4) from that

(i) $2B(s) + 3/2\ O_2(g) \rightarrow B_2O_3(s)$; $\quad \Delta H^\circ = -1273$ kJ

(ii) $3H_2(g) + 3/2\ O_2(g) \rightarrow 3H_2O\ (l)$; $\quad \Delta H^\circ = -286 \times 3$ kJ

(iii) $3H_2O\ (l) \rightarrow 3H_2O\ (g)$; $\quad \Delta H^\circ = 44 \times 3$ kJ

(iv) $B_2H_6(g) \rightarrow 2B(s) + 3H_2(g)$; $\quad \Delta H^\circ = -36$ kJ

$B_2H_6(g) + 3O_2(g) \rightarrow B_2O_3(s) + 3H_2O(g)$; $\quad \Delta H^\circ = -2035$ kJ

6. For a certain equilibrium, over the temperature range of 500 K to 700 K, the equilibrium constant, K_p conforms to the equation $\log K_p = 30.1 - \frac{6.36 \times 10^3}{T} - 5.7 \log T$. Calculate the magnitude of ΔH^0 and ΔS^0 at 600 K. Derive an expression for ΔH^0 in the form $\Delta H^0 = A + BT$ where A and B are constants. What are the values of A and B?

Sol. $$\log K = 30.1 - \frac{6.36 \times 10^3}{T} - 5.7 \log T$$

$$\ell nK = 2.303 \times 30.1 - \frac{6.36 \times 2.303 \times 10^3}{T} - 5.7 \ell nT$$

$$\frac{d\ell nK_p}{dT} = \frac{\Delta H^0}{RT^2} = \frac{6.36 \times 2.303 \times 10^3}{T^2} - \frac{5.7}{T}$$

$\therefore \Delta H^0 = [6.36 \times 2.303 \times 10^3 \times R] - (5.7\ R)T$ of the form $A + BT$

where $A = (6.36 \times 2.303 \times 10^3 \times R) =$ **1464 7 R**; $B =$ **– 5.7 R**

$(\Delta H^0)_{600K} = 29.103 \times 10^3 - 6795.54 =$ **22307.46 cal**

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Delta G^\circ = -\ RT\ lnK$

$$\ln K \text{ at } 600\ K = 2.303 \times 30.1 - \frac{6.36 \times 2.303 \times 10^3}{600} - 5.7 \ln 600$$

$= 69.32 - 24.41 - 36.46$

$= 8.45$

$\Delta G^\circ = -1.987 \times 600 \times 8.45 = -10.074$ cal.

$$\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = (22307 + 10074)/600 = 53.96 \text{ cal/K}$$

7. Calculate the standard enthalpy of combustion of benzene (*l*) on the basis of the following data:

i) Resonance energy of benzene = 152 kJ/mol

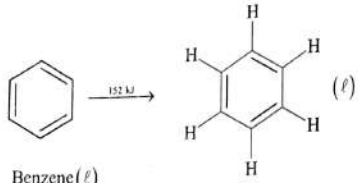
ii) Standard enthalpy of hydrogenation of cyclohexene(*l*) = -119 kJ/mol

iii) $\Delta H_f^0\ C_6H_{12}(l)$ = −156 kJ/mol

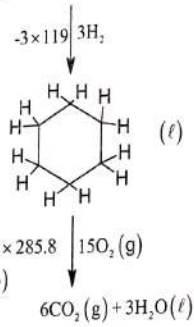
iv) ΔH°_f of H_2O (*l*) = −285.8 kJ/mol

v) ΔH°_f of CO_2(g) = −393.5 kJ/mol

Sol.



Benzene(ℓ) Cyclohexatriene no resonance assumed



= −3267.4 kJ

8. Using the data (all values are in kilocalories per mole at 25°C) given below calculate the bond energy of C–C and C–H bonds

$C(s) \longrightarrow C(g)$ $\Delta H = 172$ kcal

$H_2(g) \longrightarrow 2H(g)$ $\Delta H = 104$ kcal

$H_2(g) + 1/2\ O_2(g) \longrightarrow H_2O(l)$ $\Delta H = -68$ kcal

$C(s) + O_2(g) \longrightarrow CO_2(g)$ $\Delta H = -94.0$ kcal

Heat of combustion of $C_2H_6 = -372$ kcal

Heat of combustion of $C_3H_6 = -530$ kcal

Sol. Let x kcal be C–C bond energy and y kcal be C–H bond energy per mole

$2C(\text{graphite}) + 3H_2(g) \longrightarrow C_2H_6(g)$

$\Delta H = -2 \times 94 - 3 \times 68 + 372 = -20$ kcal

$-20 \text{ kcal} = 2 \times 172 + 3 \times 104 - BE\ (C_2H_6)$

$\Delta H^\circ = -3 \times 94 - 4 \times 68 + 530 = -24$ kcal

$BE\ (C_3H_8) = 956$ kcal

$BE(C_2H_6) = 676 \text{ kcal} = x + 6y$ ...(1)

$BE(C_3H_8) = 956 \text{ kcal} = 2x + 8y$ ...(2)

Solving (1) and (2) gives

$y = 99$ kcal (C–H)BE

$x = 82$ kcal (C–C)BE

SOLVED PROBLEMS

Subjective Question

1. Enthalpy for the reaction $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$ is – 84.54 kJ. Magnitude of enthalpy of formation of $Ag^+(aq)$ and $Br^-(aq)$ are in the ratio 8 : 9. Formation of $Ag^+(aq)$ is an endothermic process whereas formation of Br^- is an exothermic process. Enthalpy of formation of AgBr is – 99.54 kJ/mol. Calculate the enthalpy of formation of $Ag^+(aq)$ in kJ/mol.

Sol. **120**

(i) $Ag^+(aq) + Br^-(aq) \rightarrow AgBr(s)$; $\Delta H = -84.54$ kJ

(ii) $Ag(s) + aq \rightarrow Ag^+(aq)$; $\Delta H = 8x$ kJ

(iii) $\frac{1}{2}Br_2(l) + aq \rightarrow Br^-(aq)$; $\Delta H = -9x$ kJ

(iv) $Ag(s) + \frac{1}{2}Br_2(l) \rightarrow AgBr(s)$; $\Delta H = -99.54$ kJ

From (i) + (ii) + (iii) – (iv),

$(-84.54) + 8x + (-9x) - (-99.54) = 0$

$\Delta \quad x = 15$

$\Delta \; \Delta_f H_{Ag^+(aq)} = 8x = 120$ kJ /mol

Objective Questions

Prob. 1. The temperature of a definite amount of an ideal monoatomic gas becomes four times in an isoentropic process. Which of the following is correct relation between the final and initial parameters of gas?

(A) $V_f = 8V_i$ (B) $V_f = 16\,V_i$

(C) $P_f = 32\,P_i$ (D) $P_f = \frac{1}{16}P_i$

Sol. **(C)**

$$\left(\frac{V_1}{V_2}\right) = \left(\frac{T_2}{T_1}\right)^{\frac{1}{\gamma-1}} = (4)^{3/2} = 8$$

$$\frac{P_1}{P_2} = \left(\frac{T_2}{T_1}\right)^{\frac{\gamma}{1-\gamma}} = (4)^{\frac{5/3}{-2/3}} = (4)^{-5/2} = \frac{1}{32}$$

Prob. 2. For a liquid normal boiling point is – 173°C then at 2 atm pressure its boiling point should be nearly.
($\because \Delta H_{vap.} = 200$ cal/mole, R= 2 cal/mol-K, ln 2 = 0.7)

(A) –73°C (B) 333°C

(C) 60°C (D) 103°C

Sol. **(C)**

$$\ln\frac{P_2}{P_1} = \frac{\Delta H}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\ln\frac{2}{1} = \frac{200}{2}\left[\frac{1}{100} - \frac{1}{T_2}\right]$$

$$T_2 = \frac{100}{0.3} = 333.33K = 60.33°C$$

Prob. 3. A 3 mol sample of a gas having $\gamma = 1.33$ at 300 K is allowed to expand under reversible adiabatic condition from 5 L to 40 L. The value of ΔH is

(A) -12.46 KJ
(B) -14.950 KJ
(C) - 24.62 KJ
(D) - 10.24 KJ

Sol. $T_2 = T_1\left(\frac{V_1}{V_2}\right)^{\gamma-1}$

$= 300\times\left(\frac{5}{40}\right)^{1.33-1} = 300\times\left(\frac{1}{8}\right)^{1/3} = 300\times 1/2 = 150$

$\gamma = \frac{Cp}{Cv} = 1+\frac{R}{Cv} = 1.33$

$\Rightarrow Cv = 3R \Rightarrow Cp = 4R$

$\Delta H = nCp\,\Delta T = 3\times 4R\times\Delta T = 3\times 4\times 8.314\times(-150) = -14.950\ KJ.$

Hence, **(B)** is the correct answer.

Prob. 4. The difference between the heats of reaction at constant pressure and constant volume for reaction $2C_6H_6(l) + 15O_2 \longrightarrow 12CO_2(g) + 6H_2O(l)$ at 25°C in kJ is

(A) –7.43
(B) 3.72
(C) –3.72
(D) 7.43

Sol. **(A)**

$\Delta H = \Delta E + \Delta n_g RT$

$\Delta H - \Delta E = -3RT$

$= -3\times 8314\times 298$

$= -7.43$ kJ

Prob. 5. When a solid melt there will be

(A) an increase in enthalpy
(B) a decrease in free enthalpy
(C) no change in enthalpy
(D) a decrease in internal energy

Sol. **(A), (B)**

Prob. 6. At 0°C, ice and water are in equilibrium. The values of ΔG and ΔS for the conversion of ice to liquid water are $\left(\Delta H^{\circ}_{\text{fusion for ice}} = 6000\ J/\text{mole}\right)$.

(A) 0 and 21.98 JK^{-1} $mole^{-1}$
(B) –ve and zero
(C) +ve and 21.98 JK^{-1} $mole^{-1}$
(D) zero and zero

Sol. Since the given process is in equilibrium i.e. $\Delta G = 0$

$\Delta G = \Delta H - T\,\Delta S$

or $0 = \Delta H - T\Delta S$ or $\Delta H = T\Delta S$

or $\Delta S = \frac{\Delta H}{T} = \frac{6}{273}\times 10^3\ JK^{-1}\ mole^{-1}$

$= 21.98\ JK^{-1}\ mole^{-1}$. Hence, **(A)** is the correct answer.

Prob. 7. An adiabatic process is one in which

(A) all energy is transferred as heat
(B) no energy is transferred as heat
(C) the temperature of a gas decreases in a reversible adiabatic expansion
(D) None of the above

Sol. **(B, C)**

Prob. 8. Which of the following statements is/are correct?

(A) Absolute value of internal energy cannot be determined
(B) Absolute value of heat content can be determined
(C) Absolute value of entropy can be determined
(D) All the three E, H and S are extensive properties.

Sol. **(A), (C)** and **(D)** are correct answers but (B) is wrong.

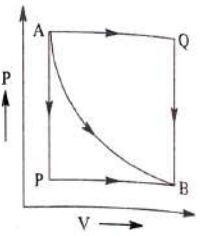
Prob. 9. During an adiabatic reversible expansion of an ideal gas

(A) Internal energy of the system decreases
(B) Temperature of the system decreases
(C) Pressure remains constant
(D) Pressure increases

Sol. The gas does work on the expense of it's internal energy thereby causing fall in temperature. Increasing volume and falling temperature causes fall in the pressure.
Hence (A) and (B) are the correct answer.

Prob. 10. For the fixed amount of an ideal gas, three process are carried out via AB, APB & AQB. Select correct options:

(A) ΔS_{AB} does not depend upon the type of process
(B) If AB path is reversible isothermal process, then $\Delta S_{AB} = \frac{Q_{AB}}{T}$, where Q_{AB} is heat involved in process AB & T is the temperature for the isothermal process in K
(C) ΔS_{AB} via AQB > ΔS_{AB} via APB
(D) ΔS_{AB} via AQB < ΔS_{AB} via APB



Sol. **(A, B)**

Prob. 11. Consider 2 mole of an ideal diatomic gas at 300 K and 0.507 MPa to be expanded adiabatically to a final pressure of 0.203 MPa against a constant pressure of 0.101 MPa calculate final pressure.

(A) 160 K
(B) 230 K
(C) 270 K
(D) 310 K

Sol. **(C)**

$$W = -P_{ext}\left(\frac{nRT_2}{P_2} - \frac{nRT_1}{P_1}\right)$$

For adiabatic process $\Delta U = W$

$$\Delta U = W = nC_v(T_2 - T_1)$$

For diatonic molecule $C_v = \frac{5}{2}R$

$$nC_v(T_2 - T_1) = P_{ext}nR\left(\frac{T_1}{P_1} - \frac{T_2}{P_2}\right)$$

$$\left(\frac{5}{2}R\right)(T_2 - T_1) = (0.101)\left[\frac{300}{0.507} - \frac{T_2}{0.203}\right]$$

$$T_2 = 270\ K$$

Prob. 12. Spontaneous adsorption of a gas on solid surface is an exothermic process because

(A) ΔH increases for system
(B) ΔS increases for gas
(C) ΔS decreases for gas
(D) ΔG increases for gas

Sol. **(C)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: $\Delta G = -T\Delta S_{Total}$ is valid for only isothermal reversible process.

and

STATEMENT-2: $dG = VdP - SdT$ is valid for only reversible process.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

2. **STATEMENT-1:** The heat absorbed during the isothermal expansion of an ideal gas against vacuum is zero.

and

STATEMENT-2: The volume occupied by the molecules of a real gas is negligible. Compared to gaseous volumes at high pressure.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(C)**

3. **STATEMENT-1:** ΔH° of a reaction is independent of temperature.

and

STATEMENT-2: ΔH° of a reaction $= \sum H^\circ(\text{products}) - \sum H^\circ(\text{reactants})$

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(D)**

4. **STATEMENT-1:** There is a natural asymmetry between converting work to heat and converting heat to work.

and

STATEMENT-2: No process is possible in which the sole result is the absorption of heat from a reservoir and its complete conversion into work.

(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1

(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT-1 is True, STATEMENT-2 is False

(D) STATEMENT-1 is False, STATEMENT-2 is True

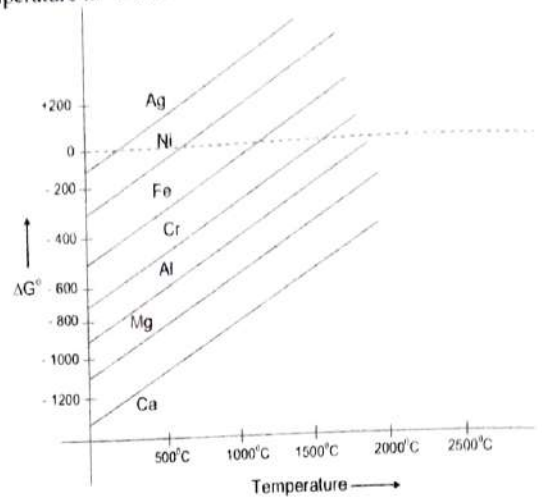
Sol. **(B)**

COMPREHENSION

Read the following passage and answer the questions:

Many principles, we are using to extract a metal from its ore, thermodynamics plays an important role to direct us in the reduction of a metal by using a suitable reducing agent, the following principle is used to explain the spontaneity of a reaction.

If ΔG = -ve the reaction is spontaneous, here the graph is given for oxidation of a metal to metal oxide taking ΔG° in kJ on y-axis and temperature in ^0C on x-axis.



1. Iron can be reduced from Iron oxide by using.
   (A) Ag (B) Hg
   (C) Cr (D) Ni

Sol. **(C)**
By using given graph ΔG° for formation of Cr_2O_3 is more than ΔG° for the formation of Fe_2O_3.

2. Pick out incorrect statement form the following.
   (A) Al_2O_3 is easily reduced with Cr at very high temperature.
   (B) 'Ca' is better reducing agent than 'Al'.
   (C) Cr_2O_3 decomposes easily than Al_2O_3.
   (D) Fe_2O_3 requires less temperature for decomposition, than Al_2O_3.

Sol. **(A)**
Al_2O_3 can't be reduced with 'Cr' as the reaction in between Al_2O_3 and Cr; ΔG° is positive (Non-spontaneous)

3. Ag_2O oxide can be reduced by using.
   (A) Mg (B) Cr
   (C) Fe (D) All

Sol. **(D)**
All metals Mg, Cr, Fe can reduce Ag_2O to Ag.

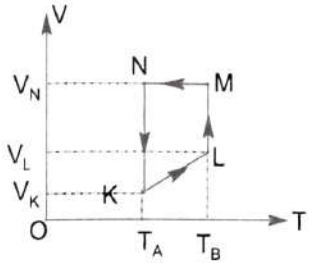
MATRIX – MATCH TYPE QUESTIONS

1. From the following data, mark the option(s) where ΔH is correctly written for the given reaction:
Given: $H^+(aq) + OH^-(aq) \longrightarrow H_2O(\ell)$ $\Delta H = -57.3$ kJ
$\Delta H_{solution}$ of HA (g) = - 70.7 kJ/mol
$\Delta H_{solution}$ of HA = 15 kJ/mol and BOH is strong base.
$\Delta H_{solution}$ of BOH(g) = 20 kJ/mol
$\Delta H_{ionisation}$ of HA = 15 kJ/mol

	Column - I (Reaction)		Column - II ΔH_R (kJ / mol)
(A)	$HA(aq) + BOH(aq) \longrightarrow BA(aq) + H_2O$	(p)	- 42.3
(B)	$HA(g) + BOH(g) \longrightarrow BA(aq) + H_2O$	(q)	-93
(C)	$HA(g) \longrightarrow H^+(aq) + A^-(aq)$	(r)	- 55.7
(D)	$B^+(aq) + OH^-(aq) \longrightarrow BOH(aq)$	(s)	0
		(t)	None of these

Sol. **A → p; B → q; C → r; D → s**

2. Consider the following figure of cyclic process with two moles of a mono atomic gas:



$\frac{V_L}{V_K} = 2$; $\frac{V_N}{V_K} = 4$

Based on this, match the process in column – I with its characteristics

	Column - I		Column – II
(A)	K → L	(p)	Expansion
(B)	L → M	(q)	Compression
(C)	N → K	(r)	W < 0
(D)	K L M N K	(s)	Isobaric
		(t)	Isothermal

Sol. **A → (p, r, s) B → (p, r, t) C → (p, t) D → (r)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1 Thermodynamic Process		Column – 2 Thermodynamic Condition		Column – 3 Thermodynamic change	
(I)	Isothermal	(i)	$\Delta T \neq 0$	(P)	$dU \neq 0$
(II)	Isobaric	(ii)	$\Delta V = 0$	(Q)	$dU = 0$
(III)	Isochoric	(iii)	$\Delta P = 0$	(R)	$dq = 0$
(IV)	Adiabatic	(iv)	$\Delta T = 0$	(S)	$dw = 0$

3. Which of the following combination is incorrect ?
[A] (I) (iv) (Q) [B] (II) (iii) (P) [C] (III) (i) (Q) [D] (IV) (i) (R)

4. Which of the following combination is correct ?
[A] (I) (iii) (R) [B] (I) (i) (Q) [C] (IV) (iv) (P) [D] (I) (iv) (Q)

5. Which of the following combination is correct ?
[A] (I) (iii) (Q) [B] (II) (i) (P) [C] (III) (i) (Q) [D] (IV) (i) (S)

Sol. 3. **C**, 4. **D**, 5. **B**

SINGLE INTEGER ANSWER TYPE

Prob. 1. If a monoatomic gas PV^x = constant. It possesses molar heat capacity as $C_p = \frac{R}{\gamma - 1} + \frac{R}{1-x}$ how much heat in calories is required to raise the temperature of one mole of that by 2°C if it obeys $\frac{P}{V}$ = constant ?

Sol. **8**

$$dq = nC_v dT = \left[\left(\frac{R}{0.66}\right) + \frac{R}{2}\right]2 = 8\,\text{cal.}$$

Prob. 2. For a liquid, enthalpy of fusion is 1.435 kcal mol^{-1} and molar entropy change is 5.26 cal $mol^{-1}K^{-1}$. The freezing point of liquid (in °C) will be:

Sol. **0°C**

$$\Delta S = \frac{\Delta H}{T}$$

$$T = \frac{1.435 \times 10^3}{5.26}$$

$$= 273\ \text{K or } 0°\text{C}$$

Prob. 3. When 3.0 g ethane gas is placed in bomb-calorimeter at 300 K and combusted with excess of oxygen, the temperature of the calorimeter raised to 301 K. If the heat capacity of bomb calorimeter is 2 kcal/K, then the enthalpy of combustion is found to be – 4.3x kcal mol^{-1}. Find the value of x.

Sol. **5**

Heat absorbed by calorimeter = + S × Δt = 2 × (301 – 300) = 2 kcal

So, heat produced by the combustion of 3.0 g of C_2H_6 = - 2 kcal

∴ heat produced by the combustion of 30 g (1 mole) of C_2H_6 = - 20 kcal mol^{-1} = ΔU

So, $C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(l)$

So, $\Delta H = \Delta U + \Delta n_g RT$

$= (-20) + (-2.5) \times (2 \times 10^{-3}) \times 300$

$= -20 - 1.5$

$= -21.5$ kcal mol^{-1}

$= -4.3x$ (given)

$\therefore x = 5$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. 100 g of N_2 gas at 300 K are held by a piston under 30 atm. The pressure is suddenly released to 10 atm and gas is allowed to expand adiabatically. If $C_V = 20.8\ JK^{-1}$. Calculate ΔS_{system}.

Sol. **$10.71\ JK^{-1}$**

$$T_2 = \frac{\left[20.8 + \frac{10}{30} \times 8.314\right]}{20.8 + 8.314} \times 300 = 243\ K$$

$$\Delta S = \frac{100}{28}(29.114)\ln\frac{243}{300} + \frac{100}{28} \times \ln\frac{30}{10}$$

$$= 10.71\ JK^{-1}$$

Prob. 2. 4 moles of an ideal gas initially at 1.5 atm and 300 K is heated to 600 K where pressure is 4.5 atm. (Given $C_{vm} = a + bT$; $a = 25\ JK^{-1}mol^{-1}$; $b = 0.03\ JK^{-2}\ mol^{-1}$) Determine ΔH

Sol. **56.17 kJ**

$$\Delta H = n\int_{T_1}^{T_2} C_p.dT = n\left[(a+R)\Delta T + \frac{b}{2}(T_2^2 - T_1^2)\right]$$

$$= 4\left[33.3 \times 300 + \frac{0.03}{2}(36-9) \times 10^4\right]$$

$$= 56.17\ kJ$$

Prob. 3. 2 moles of an ideal gas $C_{vm} = \frac{7}{2}R$, are initially at a pressure of 4 bar and temperature of 500 K. They are expanded adiabatically and reversibly until the volume has doubled. Calculate ΔE for this process.

Sol. **–5254.2 J**

$$T_2 = T_1\left(\frac{v_1}{v_2}\right)^{\gamma-1} = 500\left(\frac{1}{2}\right)^{2/7} = 410\ K$$

As process sis adiabatic and reversible $q = 0$,

$$W = nC_v\Delta T = -2 \times \frac{7}{2} \times 8.314 \times 90 = -5254.2\ J$$

$$\Delta E = W = -5254.2\ J$$

Prob. 4. Phosphine, often burns on contact with air to form P_4O_{10} and H_2O. Given the following thermal information

$\Delta H_f^0\ H_3PO_4(s) = -1279$ KJ/mol; $\Delta H_f^0\ P_4O_{10}(s) = -2984$ KJ/mol; $\Delta H_f^0\ H_2O(\ell) = -286$ KJ/mol and

$4PH_3(g) + 8O_2(g) \longrightarrow P_4O_{10}(s) + 6H_2O(\ell)$

$\Delta H° = -1177.7$ KJ/mol of $PH_3(g)$

Determine $\Delta H°$ for the following reaction

$PH_3(g) + 2O_2(g) \longrightarrow H_3PO_4(s)$

Sol. **–1281.7 kJ**

$4PH_3 + 8O_2 \longrightarrow P_4O_{10}(s) + 6H_2O(\ell)$; $\Delta H^0 = -1177.7 \times 4$

$\Rightarrow -1177.7 \times 4 = -2984 - 6 \times 286 - 4\ \ \Delta H_f^0(PH_3)$

$\Rightarrow \Delta H_f^0\ (PH_3) = 2.7\ kJ/mol$

$PH_3(g) + 2O_2(g) \longrightarrow H_3PO_4\ (s)$

$\Delta H° = -1279 - 2.7 = -1281.7$ kJ

Prob. 5. Calculate ΔH for the following homogeneous reaction

$$CH_3COCH_3(g) + 2O_2(g) \longrightarrow CH_3COOH(g) + CO_2(g) + H_2O(\ell)$$

From the following data

Bond energies
- C – H → 414.49 kJ/mol
- C = O → 732 kJ/mol
- C – C → 347.92 kJ/mol
- O = O → 494.0 kJ/mol
- C – O → 351.0 kJ/mol
- O – H → 462.0 kJ/mol

Sol. **–621.61 kJ**

$$CH_3-\overset{\overset{O}{\|}}{C}-CH_3 + 2O_2 \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-O-H + CO_2 + H_2O$$

$$\Delta H^\circ = [3 \times 414.49 + 347.92 + 2 \times 494] - [2 \times 732 + 351 - 3 \times 462]$$
$$= -621.61 \text{ kJ}$$

CHEMICAL BONDING 11

Types of Chemical Bonds

(i) **Primary bonds**
  (a) Ionic bond
  (b) Covalent bond
  (c) Metallic bond

(ii) **Secondary bonds**
  (a) Hydrogen bond
  (b) van der Waal's force of attraction

Ionic bond: The bond formed by complete transfer of one or more electrons from an atom of electropositive element to a highly electronegative bonding partner.

Covalent bond: The bond formed by sharing of electrons between two atoms of electronegative elements.

Octet Rule

According to this rule, during the bond formation atoms have tendency to attain ns^2np^6 type of configuration.

Deviation from the octet rule:

(i) Incomplete octet: $BeCl_2$, BF_3.

(ii) Expansion of octet: PCl_5, ClF_3, ICl_3, SF_6, IF_7, BrF_5 and OsO_4 etc.

VALENCE BOND THEORY (ORBITAL OVERLAP CONCEPT OF COVALENCY)

According to this theory, formation of a covalent bond between two atoms is necessarily accompanied by overlapping of half filled atomic orbitals of the valence shell of the atoms of bonding partners.

Types of covalent bond

σ - bond	π - bond
(i) Formed by overlapping of orbitals along their axes (end to end or head to head overlap)	(i) Formed by overlapping of orbitals along a line perpendicular to the molecular axis
(ii) Formed by s – s, s – p_z and p_z – p_z overlapping and also by hybridised orbitals.	(ii) Formed by p_x – p_x and d_{xy} – p_y overlapping etc.
(iii) Stronger	(iii) Weaker than σ–bond
(iv) These bonds are highly directional, forms the structure of a molecule.	(iv) No primary effect on the direction of the bond, but shortens the internuclear distance
(v) Almost free rotation is possible	(v) No free rotation

Polarity in covalent bonds

Non-polar covalent bond	Polar covalent bond (have some ionic character)
Covalent bond formed between two identical or similar atoms, e.g. H_2, Cl_2 etc.	Covalent bond formed between two unlike atoms having different electronegativity, e.g. $\overset{\delta^+}{H}\overset{\delta-}{Cl}$, $\overset{\delta^+}{H}-\overset{\delta-}{F}$ etc, or two like atoms having different neighbours, e.g. $\overset{\delta^+}{CH_3}-\overset{\delta-}{CCl_3}$.

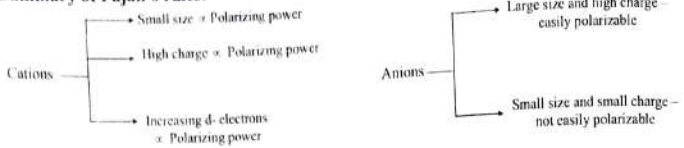
Polarization of Ions

Deformation of the electron cloud of another by the force of attraction of cation is called polarization. Polarization takes place when cation is very small and anion is large.

The figure depicts the polarization of an anion (A^-) by a cation (C^+).

Important Facts

Summary of Fajan's rules:



Ionic character of a covalent bond:

The existence of a 100% ionic or covalent bond a simple the ideal case. In reality no bond is either completely covalent or ionic. Polarity in a bond is expressed in terms of dipole moment (μ). It is given by the product of actual charge separated (δ) due to polarization and the internuclear distance. The magnitude of (δ) depends on the extent of ionic character developed in the bond. One unit of dipole moment is debye (D).

$1D = 10^{-18}$ esu cm (c.g.s)

S.I unit is coulomb meter (col. m)

$1D = 3.3356 \times 10^{-30}$ (col. m)

The overall dipole moment of a molecule is obtained by vectorial addition of the different bond moments present together with the dipole moments of the lone pairs (if any).

Dipole moment and per cent ionic character:

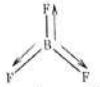
$$\% \text{ ionic character} = \frac{\text{Actual dipole moment of the bond (obs)}}{\text{Dipole moment of a pure ionic bond (cal)}} \times 100$$

Dipole moment and structure of molecules:

A molecule may be non-polar even though it has polar bonds:

For example,

CO_2: $\overleftarrow{O}=C=\overrightarrow{O}$ $\mu = 0$; Non-polar

BF_3:  $\mu = 0$; Non-polar

Other examples are CH_4, CCl_4, CS_2 etc.

Hybridization

Combination of atomic orbitals having similar energies in order to produce equal number of orbitals capable of formation of stronger bond is called hybridization.

Rules for determining the hybridization of an atom (when the bonding partners are not bonded to any other atom):

(i) Count the valence electrons of the concerned atom and the bonded atoms.

(ii) Divide the above value by 8 (2 in case of compounds with hydrogen).
Quotient gives the number of σ-bonds and the remainder gives the number of non-bonded electrons.

$$\therefore \text{ No of lone-pairs} = \frac{\text{Number of non-bonded electrons}}{2}$$

(iii) Total number of hybrid orbitals needed = Number of σ-bonds + number of lone pairs

VSEPR THEORY

For maximum stability of molecule, the hybrid orbitals occupied by electrons in the valency shell of the central atom are so arranged in space that they lie as far away from one another as possible in order to minimise the repulsion.

Important Facts

(i) Lone pair – lone pair repulsion > lone pair – bond pair repulsion > bond pair – bond pair repulsion.
(ii) Whenever there are lone pair in trigonal bipyramidal geometry, they should be placed in equatorial position so that the repulsion is minimum.
(iii) The more electronegative substituents prefer hybrid orbitals having less s–character and more electropositive substituents prefer hybrid orbitals having more s–character (Bent's rule).

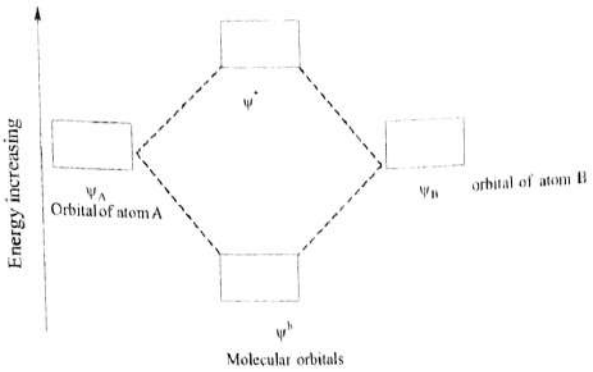
Types of molecule	No. of bond pair of electrons	No. of lone pair of electrons	Total number of electron pair	Hybridization	Geometry	Examples
AB_2	2	0	2	sp	Linear	BeF_2, C_2H_2 $[Ag(NH_3)_2]^+$
AB_3	3	0	3	sp^2	Trigonal planar	BCl_3, BH_3
AB_2L	2	1	3	sp^2	V-shape	SO_2, $SnCl_2$, $PbCl_2$
AB_4	4	0	4	dsp^2	Square planar	$[Ni(CN)_4]^{2-}$
AB_4	4	0	4	sp^3	Tetrahedral	CH_4, NH_4^+
AB_3L	3	1	4	sp^3	Pyramidal	NH_3, PX_3
AB_2L_2	2	2	4	sp^3	V – shaped	H_2O, NH_2^-, $SeCl_2$
AB_5	5	0	5	sp^3d	Trigonal bipyramidal	PF_5
AB_4L	4	1	5	sp^3d	See-saw	SF_4, $TeBr_4$
AB_3L_2	3	2	5	sp^3d	T – shaped	ClF_3
AB_2L_3	2	3	5	sp^3d	Linear	XeF_2, I_3^-
AB_6	6	0	6	sp^3d^2	Octahedral	SF_6, $[SbF_6]^-$
AB_5L	5	1	6	sp^3d^2	Square pyramidal	IF_5
AB_4L_2	4	2	6	sp^3d^2	Square planar	XeF_4, ICl_4^-
AB_7	7	0	7	sp^3d^3	Pentagonal bipyramidal	IF_7

MOT (Molecular Orbital Theory)

The molecular orbitals are obtained from the atomic orbitals by the principle of linear combination of atomic orbitals. In general, whenever atomic orbitals Ψ_A and Ψ_B of two atoms combine, two molecular orbitals Ψ and Ψ^* are formed by the linear combination of these orbitals, provided

(i) Ψ_A and Ψ_B are of same symmetry
(ii) Ψ_A and Ψ_B are of similar energies
(iii) Ψ_A and Ψ_B overlap

Ψ is wave function describing molecular orbital known as bonding molecular orbital (B.M.O).
Ψ^* is wave function describing the molecular orbital known as anti-bonding molecular orbital (A.B.M.O).
A simple diagram showing the formation of molecular orbitals from atomic orbitals:



Energy order, $\psi <$ atomic orbitals (ψ_A and ψ_B) $< \psi^*$

The main points of the theory are

(i) The electrons are filled in the molecular orbitals in the same way as they are filled in the atomic orbitals, following the Aufbau principle, Pauli's exclusion principle and Hund's rule of maximum multiplicity.

(ii) Bond order $= \frac{\text{Number of electrons in B.M.O} - \text{Number of electrons in A.B.M.O}}{2}$

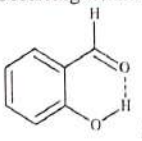
(iii) Greater the bond-order, shorter is the bond-length and higher is the bond-dissociation energy.

HYDROGEN BONDING

Whenever an H atom is attached to strongly electro negative element like F, O or N, it is single electron is pulled away strongly and nucleus is exposed. This strongly attracts a lone pair or any other molecule (inter molecular hydrogen bonding or within the same molecule.

(intramolecular hydrogen bonding); intramolecular hydrogen bonding is generally preferred (it possible).

Types of bonding

Intermolecular	Intramolecular
(i) Occurring between two or more similar or different molecules, e.g. H_2O, NH_3, HF etc.	(i) Occurring within a single molecule.  (Salicylaldehyde)
(ii) It causes association of molecules due to which melting point and boiling point increases.	(ii) No association of molecules, lower melting point and boiling point.
(iii) It accounts for the solubility of compound in water.	(iii) Less solubility in water.

Some examples of the effect of hydrogen bonding:

(i) State of H_2O and H_2S.

(ii) Melting point and boiling point of the compounds of N, O and F.

(iii) Ice has less density than water.

(iv) Contraction in water when warmed up to 4°C.

Intermolecular Forces

Intermolecular forces act between molecules or between molecules and ions. Generally, these forces are much weaker than the bonding forces. Dipole–dipole forces and ion–dipole forces attract molecules with dipole moments towards other polar molecules or ions. London forces are the result of temporary dipoles induced in the ordinary non-polar molecules. The term van der Waal's forces refers to dipole–dipole, dipole–induced dipole and london forces.

The strength of the van der Waal's forces increases as the size of the units linked increases. When other factors are absent, this can be applied to explain the melting or boiling points of similar compounds in a group, e.g. $PbH_4 > SnH_4 > GeH_4 > SiH_4 > CH_4$.

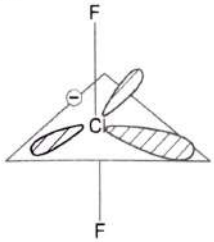
ILLUSTRATIVE PROBLEMS

1. Predict the shapes of the following ions:
(a) BeI_3^-
(b) BF_4^-
(c) IF_4^-
(d) IBr_2^-

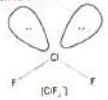
Sol. (a) Triangular
(b) Tetrahedral
(c) Square planar
(d) Linear

2. Explain when ClF_2^- is linear but ClF_2^+ is a bent?

Sol. Chlorine atom lies in sp^3d hybrid state. Three lone pairs are oriented along the corners of triangular plane.



In $\overset{\oplus}{Cl}F_2$ Chlorine atom lies in sp^3 hybrid state. Two lone pairs are oriented along two corners of tetrahedral.



3. (a) CO_2 is a gas whereas SiO_2 is solid at room temperature. Explain.
(b) Write down the structure and hybridization of the following compounds.
(i) XeO_3
(ii) XeO_6^{4-}
(iii) $XeOF_2$

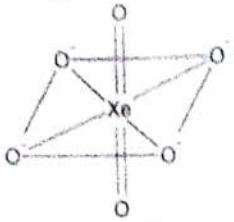
Sol. (a) O = C = O
In CO_2, $p\pi$-$p\pi$ overlap is possible due to shorter C – O bond length and similarity in energy between the orbitals of carbon and oxygen. But in SiO_2 π-overlap is not possible and thereby to complete the octet Si has to go for 4-sigma bonds with oxygen, it becomes a macromolecule and has high melting point.

(b) (i)

Shape: Pyramidal

Hybridization: sp^3

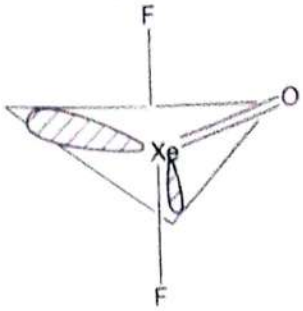
(ii)



Shape: Octahedral

Hybridization: sp^3d^2

(iii)



Shape: T-shape

Hybridization: sp^3d

4. (i) Predict the hybridization and draw the structure of the following compounds:

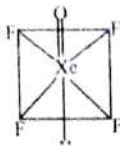
(a) $XeOF_4$, (b) PCl_3Br_2, (c) PCl_2Br_3, (d) XeF_4

(ii) Arrange the following as indicated:

(a) NH_3, PH_3, AsH_3, SbH_3 (decreasing bond angle)

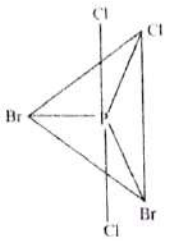
(b) $MgCO_3$, $BaCO_3$, $CaCO_3$, $BeCO_3$ (increasing stability)

Sol. (i) (a) sp^3d^2



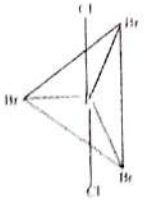
Square pyramidal

(b) sp^3d



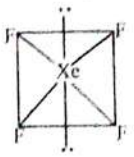
TBP

(c) sp^3d



TBP

(d) sp^3d



Square planar

(ii) (a) $NH_3 > PH_3 > AsH_3 > SbH_3$

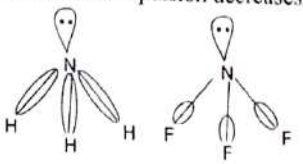
(b) $BeCO_3 < MgCO_3 < CaCO_3 < BaCO_3$

5. (a) Why is the bond energy of P – Cl bond different in PCl_3 and PCl_5?
(b) Out of CH_3Cl and $CHCl_3$, which one will have higher dipole moment and why?
(c) Bond angle in NH_3 is greater than NF_3 whereas the bond angle in PH_3 is lesser than PF_3.

Sol. (a) In PCl_3, P exhibits sp^3 hybridization and molecule is triangular pyramidal. But in PCl_5, P exhibits sp^3d hybridization. Therefore, it is trigonal bipyramidal. Three P – Cl bonds are on the plane while two P – Cl bonds are above and below the plane which are different from remaining three P – Cl bonds.

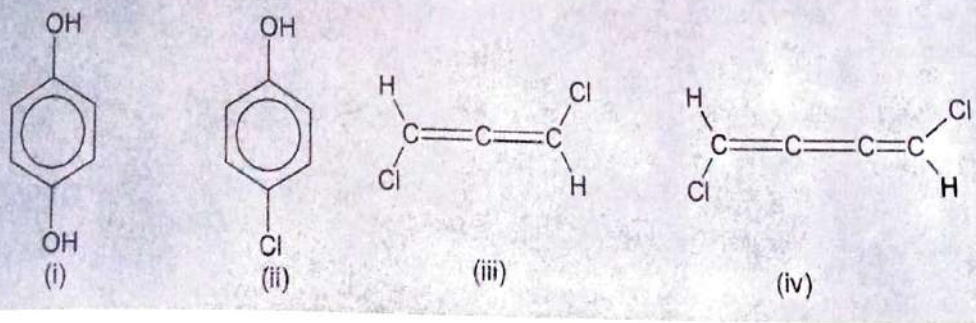
(b) CH_3Cl is having higher dipole moment in comparison to $CHCl_3$. Because in $CHCl_3$, each C – Cl bond dipole induces an opposite dipole on the other two C – Cl bonds. Each C – Cl moment is thereby reduced, ultimately lowering the overall dipole moment of the molecule.

(c) In case of NF_3, F atoms are highly electronegative because of which the charge cloud is displaced further away from the central atom and repulsion decreases, thereby reducing the bond angle.



In case of PF_3, the valence shell of phosphorus atom is incompletely filled. P–F bonds in the PF_3 gains some double bond character due to p_π–d_π back bonding. Thus, repulsion between P–F bond increases, thus, bond angle also increases. This is not possible in case of NF_3 because of non-availability of vacant d-orbitals on nitrogen.

6. Which of the following compounds have zero dipole moment:



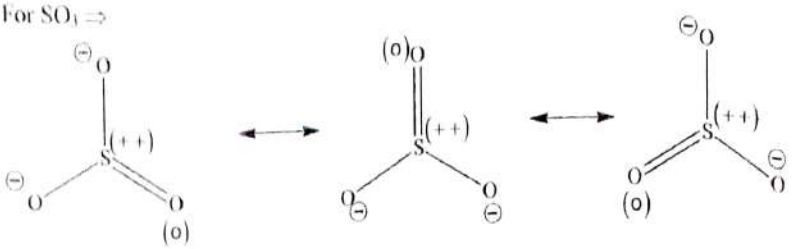
Sol. (iv)

SOLVED PROBLEMS

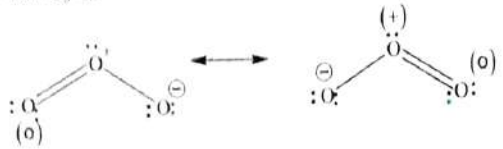
Subjective Question

Prob. 1. Write resonance structures, including formal charges, for SO_3, O_3, NO_2.

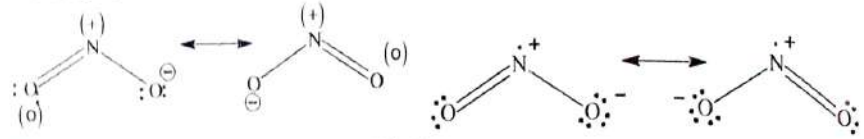
Sol. For SO_3 ⇒



For O_3 ⇒



For NO_2 ⇒

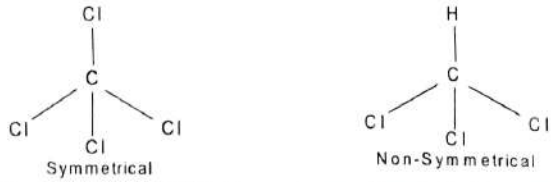


Prob. 2. PF_5 is known but PH_5 is non existent. Why ?

Sol. If P is bonded to F, a high partial positive charge density is developed on P atom and contraction of 3d orbital takes place, so that its energy becomes comparable to the 3s & 3p orbitals and it can take part in hybridization. On the other hand, in PH_5, this sort of contraction of d orbital can not take place, thus, 3d orbital can not take part in hybridization.

Prob. 3. Dipole moment of CCl_4 is zero while that of $CHCl_3$ is non zero.

Sol. Both CCl_4 & $CHCl_3$ have tetrahedral structure but CCl_4 is symmetrical while $CHCl_3$ is non-symmetrical.



Due to the symmetrical structure of CCl_4 the resultant of bond dipoles comes out to be zero. But in case of $CHCl_3$ it is not possible as the presence of hydrogen introduces some dissymmetry.

Prob. 4. Why ClF_3 exists but FCl_3 does not ?

Sol. Cl has $3s^2 3p^5$ configuration. One of the paired *p*-orbital is unpaired during excitation of its electron in *3d*-orbital to show a covalence of three.

On the other hand F does not possess d orbital in 2nd and thus it shows only one covalence.

Prob. 5. Which one of the *p*-orbital and *sp*-hybrid orbital has a greater directional orientation ?

Sol. The *p*-orbital has equal sized lobes. *sp*-hybrid orbital has one larger lobe and most of the sp-probability density is on one side. This makes the latter more directional in character.

Objective Questions

Prob 1. Which of the following pairs have different bond order ?

(A) CO and N_2

(B) O_2^- and F_2

(C) NO and CN^-

(D) H_2 and He_2^+

Sol. NO is having the bond order = 2
but CN^- is having bond order = 3
Hence, **(C)** is the correct answer.

Prob 2. The pair of species having identical shape is

(A) CF_4, SF_4

(B) PCl_3, BF_3

(C) XeF_2, CO_2

(D) PF_5, IF_5

Sol. Both are linear molecules.
Hence, **(C)** is the correct answer.

Prob. 3. For H_2O_2, H_2S, H_2O and HF, the correct order of increasing extent of hydrogen bonding is :

(A) $H_2O > HF > H_2O_2 > H_2S$

(B) $H_2O > HF > H_2S > H_2O_2$

(C) $HF > H_2O > H_2O_2 > H_2S$

(D) $H_2O_2 > H_2O > HF > H_2S$

Sol. **(C)**

Prob. 4. Which of the following is having diamagnetic property ?

(A) O_2

(B) NO_2

(C) KO_2

(D) Na_2O_2

Sol. $Na_2O_2 \rightarrow 2Na^+ + O_2^{2-}$
O_2^{2-} has no unpaired electron.
Hence, **(D)** is the correct answer.

Prob. 5. Which of the d orbitals of I participates in hybridization in IF_7 ?

(A) $5d_{z^2}$, $5d_{xy}$, $5d_{\infty}$

(B) $5d_{z^2}$, $5d_{xy}$, $5d_{x^2-y^2}$

(C) $5d_{xy}$ $5d_{yz}$, $5d_{\infty}$

(D) $5d_{x^2-y^2}$, $5d_{xy}$, $5d_{yz}$

Sol. $5d_{z^2}$ hybridizes with $5p_z$ to form axial bond, whereas $5d_{x^2-y^2}$ and $5d_{xy}$ mix up with 5s, $5p_x$, $5p_y$ and $5p_z$ to form equatorial bonds.
Hence, **(B)** is the correct answer.

Prob. 6. The melting point of AlF_3 is 104° *C* and that of SiF_4 is - 77° *C* (it sublimes) because :

(A) there is a very large difference in the ionic character of the Al – F and Si – F bonds

(B) in AlF_3, Al^{3+} interacts very strongly with the neighbouring F^- ions to give a three dimensional structure but in SiF_4, no interaction is possible

(C) the silicon ion in the tetrahedral SiF_4 molecule is not shielded effectively from the fluoride ions whereas in AlF_3, Al^{3+} the ion is shielded on all sides

(D) the attractive forces between the SiF_4 molecules are strong whereas those between the AlF_3 molecules are weak

Sol. **(B)**

Prob. 7. Which of the following has been arranged in order of decreasing dipole moment ?

(A) $CH_3Cl > CH_3F > CH_3Br > CH_3I$

(B) $CH_3F > CH_3Cl > CH_3Br > CH_3I$

(C) $CH_3Cl > CH_3Br > CH_3I > CH_3F$

(D) $CH_3F > CH_3Cl > CH_3I > CH_3Br$

Sol. **(A)**

Prob. 8. Which ion has the highest polarizing power?

(A) Mg^{2+} (B) Al^{3+}

(C) Ca^{2+} (D) Na^{+}

Sol. Small cation and large size has more polarizing power.

Hence, **(B)** is the correct answer.

Prob. 9. Which of the following statements are not correct?

(A) Sodium dissolved in liquid ammonia conducts electricity because of the presence of solvated electrons

(B) The value of Debye unit of dipole moment in CGS unit is 10^{-8} esu cm

(C) The value of Debye unit of dipole moment in SI units is 3.34×10^{-30} cm

(D) The bond angle in BH_4^- is about 90°

Sol. **(B, D)**

Prob. 10. Which of the following pairs are isostructural?

(A) $\left[NF_3, NO_3\right]$ (B) $\left[NF_3, H_3O^+\right]$

(C) $\left[NO_3^-, BF_3\right]$ (D) $[NF_3, HN_3]$

Sol. **(B, C)**

Prob. 11. In which of the following compounds vacant orbital takes part in bonding.

(A) B_2H_6 (B) Al_2Cl_6

(C) C_2H_5Cl (D) $BeCl_2$(solid)

Sol. **(A), (B)** and **(D)** are the correct answers.

Prob. 12. A hydrazine molecule is split in NH_2^+ and NH_2^- ions. Which of the following statements is/are correct?

(A) NH_2^+ shows sp^2 – hybridisation whereas NH_2^- shows sp^3 – hybridisation

(B) $Al(OH)_4^-$ has a regular tetrahedral geometry

(C) sp^2 – hybridized orbitals have equal *s*- and *p*- character

(D) Hybridized orbitals always form σ - bonds

Sol. **(A, B, D)**

Prob. 13. Which one is most ionic ?

(A) P_2O_5 (B) CrO_3

(C) MnO (D) Mn_2O_7

Sol. **(C)**

Prob. 14. Which one of the following conversions involve, change in both hybridisation and shape ?

(A) $CH_4 \longrightarrow C_2H_6$ (B) $NH_3 \longrightarrow NH_4^+$

(C) $BF_3 \longrightarrow BF_4^-$ (D) $H_2O \longrightarrow H_3O^+$

Sol. **(C)**

Prob. 15. Which has higher bond energy and stronger bond ?

(A) F_2 (B) Cl_2

(C) Br_2 (D) I_2

Sol. **(B)**

COMPREHENSION

Lewis concept of covalency of an element involved octet rule. Later on it was found that many elements in their compounds e.g. BeF_2, BF_3 etc. have incomplete octet whereas PCl_5, SF_6 etc. have expanded octet. This classical concept also failed in predicting the geometry of molecules. Modern concept of covalence was proposed in terms of valence bond theory. Hybridization concept along with V.B. Theory successfully explained the geometry of various molecules but failed in many molecules. The geometry of such molecules was explained by VSEPR concept. Finally molecular orbital theory was proposed to explain many other molecules.

1. Which are true statements among the following?
(I) I_3^- has bent structure
(II) $p_\pi - p_\pi$ bonds are present in SO_2
(III) SeF_4 and CH_4 has same shape
(IV) XeF_2 and CO_2 has same shape
(V) SF_4 is sea-saw structure whereas ICl_3 is T-shaped.
(A) I, II, IV, V
(B) I, II, III, IV
(C) II, III, IV, V
(D) I, III, IV, V

Sol. **(A)**
SeF_4 is sea-saw shaped whereas CH_4 is tetrahedral.

2. The shape of MnO_4^- ion and hybridization of Mn in MnO_4^- is
(A) tetrahedral sp^3
(B) tetrahedral d^1s
(C) square planar dsp^2
(D) square planar sp^2d

Sol. **(B)**

3. Which statements are correct for CO^+ and N_2^- according to M.O. Theory:
(I) Both have same configuration
(II) Bond order for CO^+ and N_2^- are 3.5 and 2.5
(III) Bond order of CO^+ and N_2^- are same
(IV) During the formation of N_2^- from N_2 bond length increases.
(V) During the formation of CO^+ from CO, the bond length decreases.
(A) II, IV, V
(B) I, III, IV, V
(C) I, III
(D) I, II, III

Sol. **(A)**

$$CO : (\sigma_{1s})^2 (\sigma_{2s}^*)^2 (\sigma_{2s})^2 (\sigma_{2p_z})^2 (\pi_{2p_x})^2 (\pi_{2p_y})^2 (\sigma_{2s}^*)^2$$

$$N_2 : (\sigma_{1s})^2 (\sigma_{1s}^*)^2 (\sigma_{2s})^2 (\sigma_{2s}^*)^2 (\pi_{2p_z})^2 (\pi_{2p_x})^2 (\sigma_{2p_z})^2$$

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** If $d_{x^2-y^2}$ and p_y orbitals come close together along z axis, then they can form π bond by sideways overlapping.
and
STATEMENT-2: Both orbitals do not have electron density along z axis.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

Sol. **(D)**
P_y and $d_{x^2-y^2}$ cannot form π bond as lateral overlapping between them is not possible.

2. **STATEMENT-1:** PCl_5 conducts current in solid state.
and
STATEMENT-2: PCl_5 exists as $[PCl_4]^+$ and $[PCl_6]^-$ ions.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

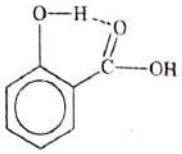
Sol. **(D)**
Solid ionic compounds conduct current only in fused state. PCl_5 in solid state exists as $[PCl_6]^-[PCl_4]^+$.

3. **STATEMENT-1:** $BiCl_5$ does not exist.
and
STATEMENT-2: In Bi, inert pair effect is most predominant due to the presence of 14 poorly shielding d electrons in inner core.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

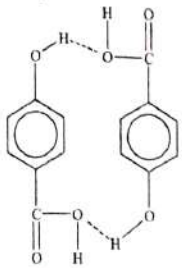
Sol. **(A)**

4. STATEMENT-1: p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
because
STATEMENT-2: o-Hydroxybenzoic acid has intramolecular hydrogen bonding.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol.



More stabilized by intramolecular hydrogen bonding



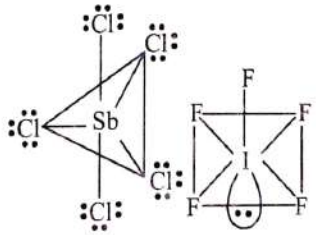
More stronger intermolecular forces increases the boiling point.

Hence, **(D)** is the correct answer.

5. STATEMENT-1: $SbCl_5$ and IF_5 are not isostructural.
and

STATEMENT-2: These two molecules are isosteric to each other.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**



sp^3d sp^3d^2

Statement-1 is true, statement-2 is false.
Isosteric means having same number of atoms and same number of electrons.

MATRIX – MATCH TYPE

1. Match the following question

Column – I		Column - II	
(A)	Chloral hydrate	(p)	Bond angle 120° – 140°
(B)	Solid hydrogen fluoride	(q)	Two dimensional sheet with almost hexagonal symmetry
(C)	Crystalline boric acid	(r)	Syrupy
(D)	Liquid glycerol	(s)	Intramolecular H-bond
		(t)	Intermolecular H-bond

Sol. **A → s; B → p, t; C → q, t; D → r, t**

2. Column –I lists oxidation/reduction of O_2 and N_2. Column – II lists some of the properties associated with these oxidation/reduction reactions. Match the entries of column-I with those listed in column-II

Column – I		Column – II	
(A)	$N_2 \longrightarrow N_2^-$	(p)	Bond order increases
(B)	$N_2 \longrightarrow N_2^-$	(q)	Bond order decreases
(C)	$O_2 \longrightarrow O_2^-$	(r)	Paramagnetism increases
(D)	$O_2 \longrightarrow O_2^-$	(s)	Paramagnetism decreases
		(t)	No change in bond order

Sol. **A → (q, r); B → (q, r); C → (p, s); D → (q, s)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1 Compound		Column – 2 Hybridisation		Column – 3 Geometry	
(I)	SF_4	(i)	sp^2	(P)	Square planar
(II)	NH_3	(ii)	sp^3d^2	(Q)	v-shape
(III)	XeF_4	(iii)	sp^3d	(R)	Pyramidal
(IV)	SO_2	(iv)	sp^3	(S)	See-saw

3. Which of the following combination is correct ?
[A] (I) (ii) (R) [B] (I) (iii) (S)
[C] (II) (i) (R) [D] (IV) (iii) (Q)

4. Which of the following combination is incorrect ?
[A] (I) (iii) (S) [B] (II) (iv) (R)
[C] (III) (ii) (P) [D] (IV) (iv) (Q)

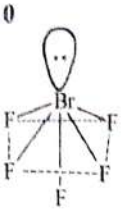
5. Which of the combination in column 1 is isostructural with PF_3 ?
[A] SF_4 [B] NH_3
[C] XeF_4 [D] SO_3

Sol.
3. **B**
4. **D**
5. **B**

SINGLE INTEGER ANSWER TYPE

1. Based on VSEPR theory, the number of 90 degree F–Br–F angles in BrF_5 is

Sol. **0**



All four planar bonds (F–Br–F) will reduce from 90° to 84.8° after $\ell p - bp$ repulsion.

2. The number of σ-bonds in $C(CN)_4$ ....

Sol. **8**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. If I–I distance is x (Å) for a given compound H_2CCl_2 if C–I bond length is 2.35 Å ($\sin 60^\circ = 0.866$). Find the value of x

Sol. **4.07 Å**

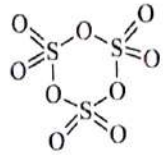
Prob. 2. If Hund's rule is not applicable, then if number of diamagnetic species among the following is x (H_2, O_2, N_2^-, C_2, No, OF, N_2^-, BN). Then find $\frac{x}{2}$

Sol. **2.5** as x = 5

Prob. 3. If x is the maximum number of equal P–O bonds in $P_2O_7^{4-}$ ion. Then $\frac{x}{4} = ?$

Sol. **1.5**

Prob. 4. Consider the following molecule



Calculate the value of $\frac{p}{q}$ is

Sol. **1.5**

Here p & q are total number of dπ–pπ bonds and total number of sp^3 hybridised atoms in the given molecule.

Prob. 5. Bond order of N_2^+ is

Sol. **2.5**

12 GENERAL ORGANIC CHEMISTRY & ISOMERISM

In general, organic reactions can be represented as,

Substrate + Reagent $\longrightarrow$ Intermediate $\longrightarrow$ Product (s)

The mechanism of organic reactions can be explained on the basis of three factors,

(i) Electronic effect
(ii) Fission of covalent bonds
(iii) Nature of attacking reagents

ELECTRONIC EFFECTS

The displacement of electrons in a covalent bond due to presence of some atom or group in the molecule or under the influence of attacking reagent, which creates centres of different electron densities in the substrate can be explained by following effects.

Inductive Effect

The polarity produced in the molecule due to electronegativity difference among the bonded atoms. There are two types of inductive effects, i.e. +I effect and –I effect.

+I effect : Groups (electron releasing)

$(CH_3)_3C- > (CH_3)_2CH- > CH_3CH_2- > CH_3- > H$

–I effect : Groups (electron withdrawing)

$\overset{+}{N}(CH_3)_3 > NO_2 > CN > F > Cl > Br > I > OH > OCH_3 > C_6H_5$

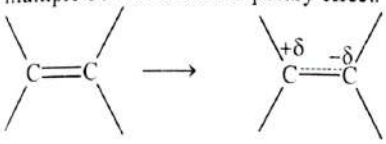
Mesomeric Effect

The polarity produced in the molecule due to interaction between two π bonds or a π bond and lone pair of electrons.

$$R-\overset{+}{C}=\ddot{\underset{\ddot{}}{O}} \longleftrightarrow R-C\equiv\overset{+}{\underset{\ddot{}}{O}}$$

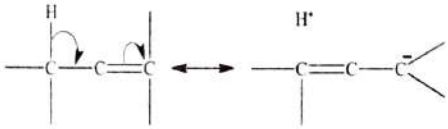
Electromeric Effect

Under the influence of attacking reagent, there is a transfer of shared pair of electrons to one of atoms joined by multiple bonds. It is a temporary effect.



Hyperconjugation Effect

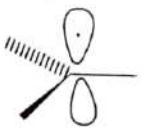
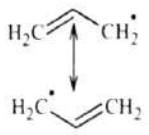
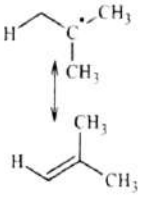
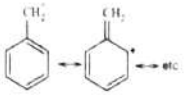
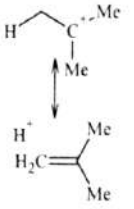
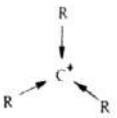
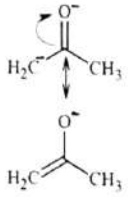
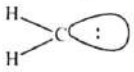
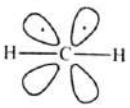
The effect of transfer of electrons from occupied orbital or double bond to an unoccupied orbital.



FISSION OF BONDS (BREAKING)

(i) Homolytic fission: Generation of free radicals.
(ii) Heterolytic fission: Generation of carbocations and carbanions.

REACTION INTERMEDIATES

Intermediate	Shape	Order of stability	Stability factors			
			Resonance	Hyperconjugation	Inductive effect	Steric hindrance
Free radicals	sp^2 – planar 	1° < 2° < 3° < allylic			–	✓
Carbocation	sp^2 – planar	1° < 2° < 3° < allylic			+ I effect 	✓
Carbanion	sp^3- pyramidal	1° > 2° > 3°		–	- I effect	
Carbene	Singlet sp^2  Triplet sp 	–	–	–	–	–

ORGANIC REACTIONS

Substitution Reactions

(i) **Free radical substitution:**

$$CH_4 + Cl_2 \xrightarrow[\Delta]{h\nu} CH_3Cl + HCl$$

(ii) **Nucleophilic substitution:**

(a) S_N1 mechanism (intermediate–carbocation)

$$(CH_3)_3CBr + Nu^- \longrightarrow (CH_3)_3CNu + Br^-$$

(b) S_N2 mechanism (no transition state formation)

$$CH_3 - Cl + Nu^- \longrightarrow CH_3 - Nu + Cl^-$$

(c) S_N mechanism (intermediate–carbocation)

$C_4H_9O_2Br + OH^-(aq) \longrightarrow C_4H_8O_2$

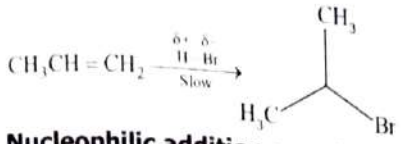
(iii) **Electrophilic substitution:**

$C_6H_6 + Cl_2 \xrightarrow[(anhyd.)]{AlCl_3} C_6H_5Cl + HCl + AlCl_3$

Addition Reactions

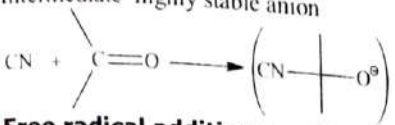
(i) **Electrophilic addition reaction:**

Intermediate–carbocation



(ii) **Nucleophilic addition reaction:**

Intermediate–highly stable anion



(iii) **Free radical addition reaction:**

$CH_3CH = CH_2 \xrightarrow{H\ Br} CH_3CH_2CH_2Br$

Elimination Reactions

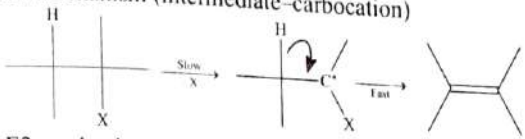
(i) **α - elimination :**

Intermediate–carbene

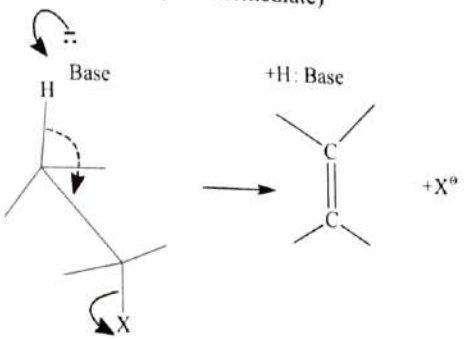
$CHCl_3 \xrightarrow[Slow]{OH^-} CCl_3^- \xrightarrow[-Cl^-]{Fast} CCl_2$

(ii) **β - elimination :**

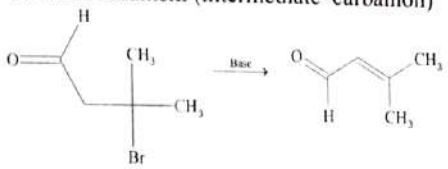
(a) E1 mechanism (intermediate–carbocation)



(b) E2 mechanism (no intermediate)



(c) E1-cb mechanism (intermediate–carbanion)



Rearrangement Reactions

(i) **H^- shifting:** 1, 2-hydride shift in alkyl halide
(ii) **Alkyl group shifting:** 1, 2-methyl shift in diol
(iii) **Bond shifting:** Beckmann rearrangement

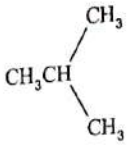
ISOMERISM

Isomerism is a property by virtue of which compounds having same molecular formula show different structural formula and properties.

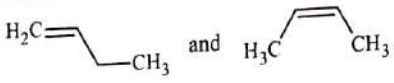
TYPES OF ISOMERISM

Structural Isomerism

(i) **Chain:**

$CH_3CH_2CH_2CH_3$ and 

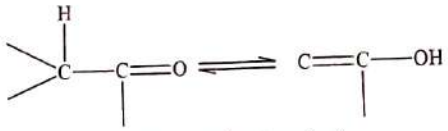
(ii) **Position:**



(iii) **Functional:**

(a) CH_3CH_2OH and CH_3OCH_3
(b) Alkene and cycloalkane (C_nH_{2n})
(c) Aldehyde, ketone and cyclic ether ($C_nH_{2n}O$)
(d) Carboxylic acid, ester and hydroxyaldehyde ($C_nH_{2n}O_2$)

(iv) **Tautomerism:**



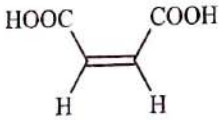
(keto-enol tautomerism)

(v) **Metamerism:**

$CH_3CH_2OCH_2CH_3$ and $CH_3OCH_2CH_2CH_3$

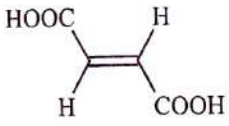
Stereoisomerism

(i) **Geometrical:**



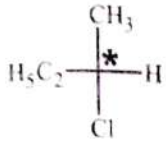
cis

and



trans

(ii) **Optical:**



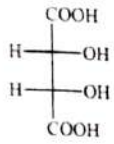
PROPERTIES OF CIS AND TRANS FORMS

(i) **Physical properties:**

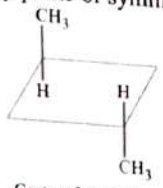
(a) Dipole moment, cis > trans

(b) Melting point, trans > cis

(ii) **Optical activity:** If any molecule is not superimposable on its mirror image, its bulk sample rotates the plane of plane polarized light. Such molecules do not have any symmetry element like point of symmetry line of symmetry plane of symmetry, etc.



Plane of symmetry (optically inactive)



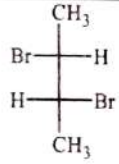
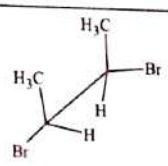
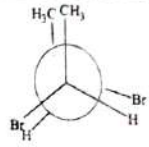
Centre of symmetry (optically inactive)

Total isomers:

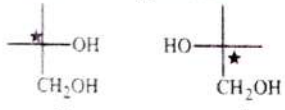
Optical isomers having 'n' number of chiral centre	No. of d and l forms (a)	No. of Mesoforms (m)	Total no. of optical isomers (a + m)
When molecule has no symmetry	2^n	0	2^n
When molecule has symmetry	If 'n' is even	$2^{(n/2)-1}$	2^n
	If 'n' is odd	$2^{(n-1)/2}$	$2^{3n-3/2}$

Representation of stereoisomers:

For example, 1, 2-dibromobutane

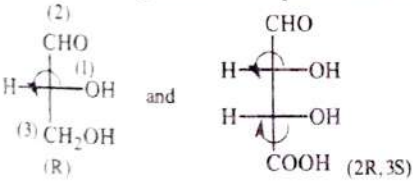
Fischer projection	Sawhorse projection	Newman projection
		

D and L configuration



D L

R and S configuration: The priorities are to be decided using Chann – Ingold and Prelog sequence rules.

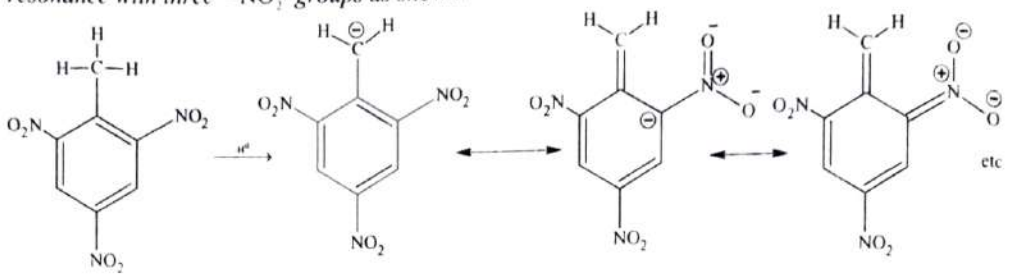


(R) and (2R, 3S)

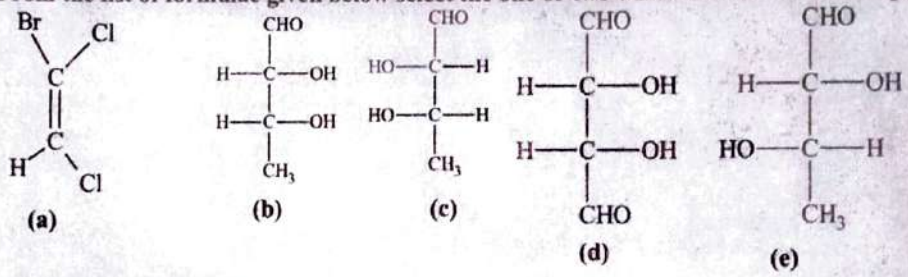
ILLUSTRATIVE PROBLEMS

1. 2, 4, 6-trinitrotoluene is relatively acidic compound. Explain why this proton is easily removed from the methyl group?

Sol. *The proton is removed easily because the anion formed after removal of the proton is stabilized by resonance with three $-NO_2$ groups as shown:*



2. From the list of formulae given below select the one or two which best fit each description.

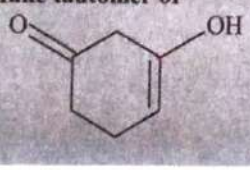


(a) **(b)** **(c)** **(d)** **(e)**

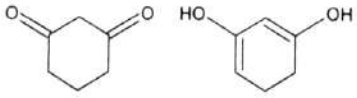
(i) A cis stereoisomer
(ii) A pair of enantiomers
(iii) Meso form
(iv) A pair of diastereoisomer
(v) Pair of compounds which might form a racemic mixture.

Sol. (i) (a)(ii) (b),(c)
(iii) (d) (iv) (c), (e)
(v) (b), (c)

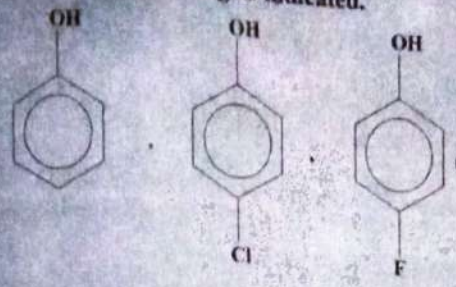
3. Make tautomer of

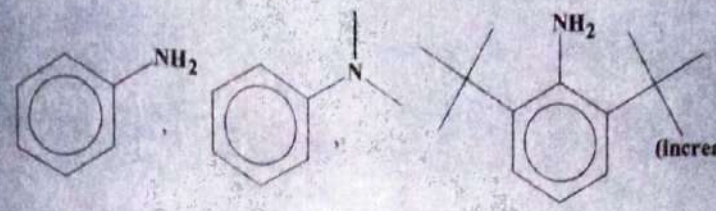


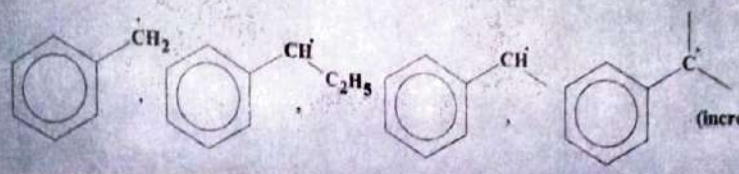
Sol.

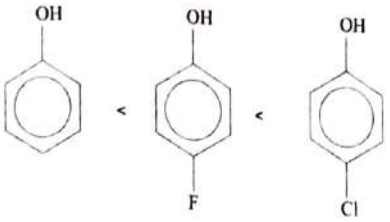


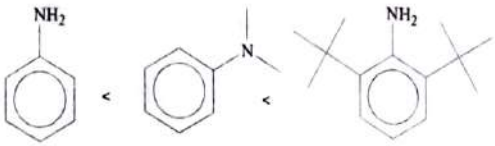
4. Arrange the following as indicated.

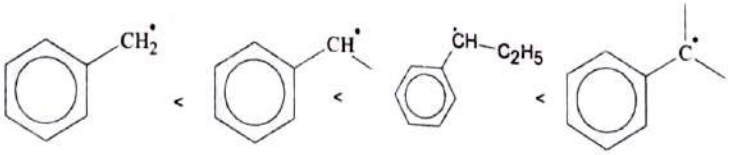
(a)  (increasing acidic strength)

(b)  (increasing basic strength)

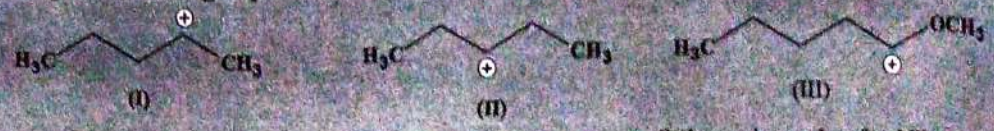
(c)  (increasing order of stability)

Sol. (a) 

(b) 

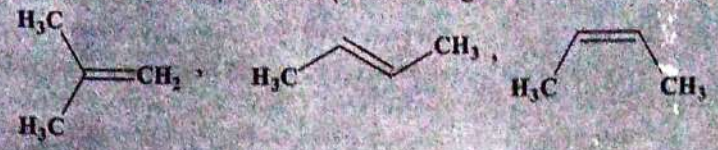
(c) 

5. Arrange the following as per the instructions given in bracket:

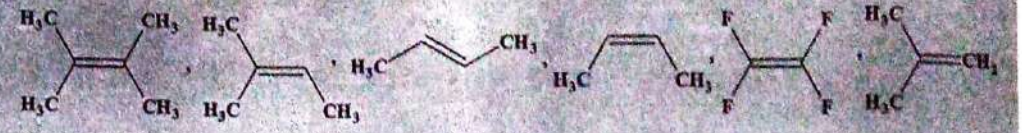
(a) 

(in increasing order of stability)

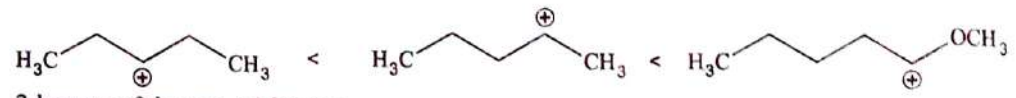
(b) 1-hexene, 2-hexene, 3-hexene (in increasing order of heat released on hydrogenation)

(c) 

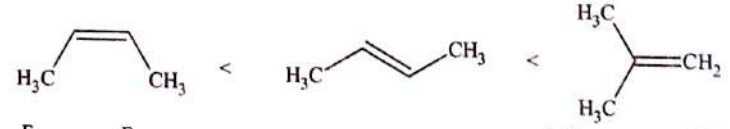
(in increasing order of stability)

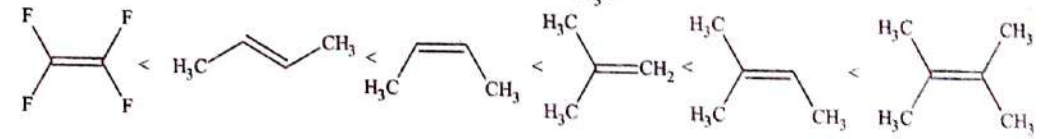
(d) 

(in increasing order of reactivity towards E^+ addition)

Sol. (a) 

(b) 2-hexene < 3-hexene < 1-hexene

(c) 

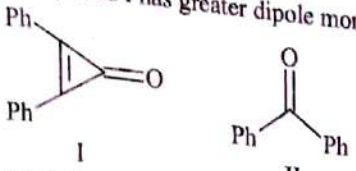
(d) 

SOLVED PROBLEMS

Subjective Question

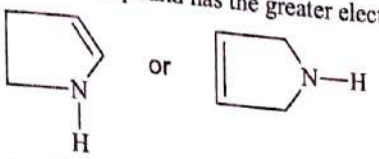
Prob. 1. Give proper explanations of the following:

(i) Compound I has greater dipole moment than compound II



I II

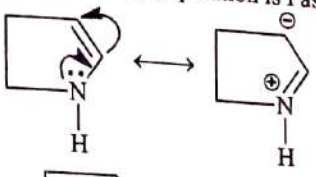
(ii) Which compound has the greater electron denci5ty on its nitrogen atom ?

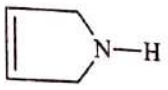


(iii) Species I is more stuble than II

Sol. (i) Greater charge separation is I assign it a higher dipole moment.

(ii)



in  No such resonance is possible.

Hence e^- density is more over on N atom.

Objective Questions

Prob. 1. The type of isomerism due to different types of alkyl group on either side of functional groups in the molecule of compounds $CH_3CH_2OCH_2CH_3$ and $CH_3OC_3H_7$ is referred as

(A) metamerism
(B) chain isomerism
(C) functional isomerism
(D) position isomerism

Sol. It is definition of metamerism.

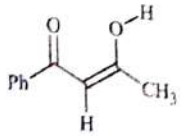
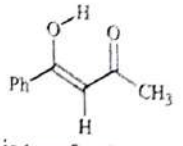
Hence, **(A)** is the correct answer.

Prob. 2. Number of isomers possible for C_4H_8 are _____

(A) 4
(B) 3
(C) 2
(D) 5

Sol. +M effect of NH_2 >+ M of – OH group disperse the charge of carbocation, hence increases the stability $-CH_3$ group shows +I effect, so it will disperse the charge less than $-NH_2$, –OH group. Whereas $-NO_2$ group shows – M/ – I effect due to which the positive charge on the carbocation increases, hence stability decreases.

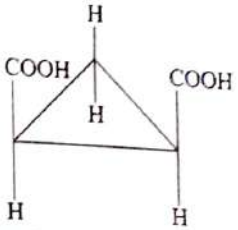
Hence, **(B)** is the correct answer.

Prob. 6.  and 

Which of the following statement is true for the above two compounds?

(A) Above pairs are resonance contributors.
(B) They are structural isomers.
(C) They are group isomers.
(D) They are tautomers.

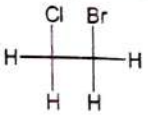
Sol. **(B, D)**

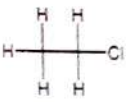
Prob. 7. For, 

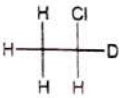
(A) cis form is more stable due to H-bonding
(B) cis form is less stable due to steric hindrance
(C) trans form is more stable due to lesser crowding
(D) trans form is less stable in the absence of H–bonding

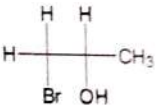
Sol. **(B, C)**

Prob. 8. Which has asymmetric carbon atoms?

(A) 

(B) 

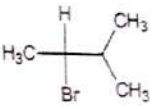
(C) 

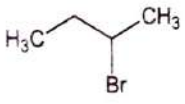
(D) 

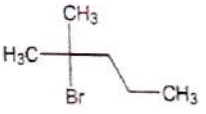
Sol. **(C, D)**

Prob. 9. Which of the following has an asymmetric carbon atom?

(A)

(B) 

(C) 

(D) 

Sol. **(B, C)**

COMPREHENSION

Read the following paragraph and answer the questions given below:

Organic molecules show chemical reaction on account of hemolytic bond fission or heterolytic bond fission in addition to inductive effect, electromeric effect, resonance and hyperconjugation. The usual reactions shown by organic molecules are substitution (S_N1 and S_N2) addition (syn and anti) and elimination (E_1 and E_2) reactions alongwith rearrangement reactions. In addition to these, various oxidation and reduction reactions are also given. The species which attacks a +ve centre is called nucleophile and the species capable of attacking –ve centres are called electrophile.

1. The most stable carbocation among the following is

(A) $C_6H_5^+$

(B) $C_6H_5\underset{\displaystyle CH_3}{\underset{|}{C}}H\overset{+}{C}H_2$

(C) $C_6H_5-\overset{+}{C}(CH_3)_2$

(D) $C_6H_5\dot{C}H_2$

Sol. **(C)**

2. During nitration of an organic compound, HNO_3 acts as

(A) acid

(B) base

(C) neutral

(D) salt

Sol. **(B)**

3. Which of the following is neither a nucleophile or electrophile?

(A) NO_2^+

(B) Cl^+

(C) CN^-

(D) $(CH_3)_4N^+$

Sol. **(D)**

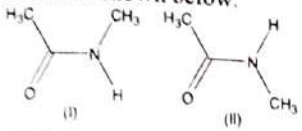
REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: Heat of neutralization of CH_3COOH Vs NaOH is numerically less than 57.1 KJ/eq.
and
STATEMENT-2: Some heat is used to ionize weak acetic acid.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

2. **STATEMENT-1:** There is evidence for existence of N–methyl acetamide in two structural forms I and II as shown below:



and

STATEMENT-2: Rotation about C–N bond is restricted due to resonance.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

3. **STATEMENT-1:** CH_3^- adds to >C = O group irreversibly but CN^- ion adds reversibly.
and
STATEMENT-2: CH_3^- ion is much stronger nucleophile than CN^- ion.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

4. **STATEMENT-1:** Grignard's reagent and dialkylcadmium both react with acid chlorides to form tert-alcohols.
and
STATEMENT-2: Grignard reagents are less reactive than dialkylcadmium.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(D)**

5. **STATEMENT-1:** Allyl carbocation is more stable than vinyl carbocation.
and
STATEMENT-2: Vinyl carbocation is destabilized by resonance.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

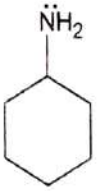
Sol. **(C)**

6. **STATEMENT-1:** Phenol gives Friedal craft alkylation reaction faster than benzene.
and
STATEMENT-2: -OH group is activating group for electrophilic aromatic substitution reaction when present on benzene.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

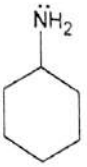
Sol. **(A)**

7. **STATEMENT-1:** Acidic strength boron halide is as $BF_3 < BCl_3 < BBr_3 < BI_3$
and
STATEMENT-2: Fluorine is more electronegative & creates greater electron deficiency on boron.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. (C)

8. **STATEMENT-1:** is less basic than 

and

STATEMENT-2: Lone pair of  is delocalized

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1

(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1

(C) STATEMENT – 1 is True, STATEMENT-2 is False

(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(C)**

9. **STATEMENT-1:** Carboxylic acid can not be converted to amide directly by heating with NH_3

and

STATEMENT-2: NH_3 reacts with acid to give carboxylate ion which is inactive for nucleophilic attack.

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1

(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1

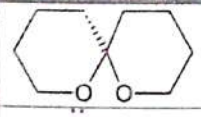
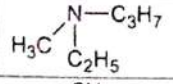
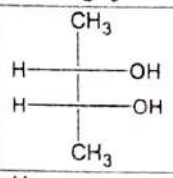
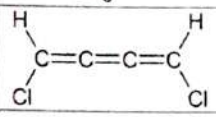
(C) STATEMENT – 1 is True, STATEMENT-2 is False

(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(D)**

MATRIX – MATCH TYPE QUESTIONS

1. Match the following:

	Column – I		Column – II
(A)		(p)	Chiral molecule
(B)		(q)	Achiral molecule
(C)		(r)	Optically active
(D)		(s)	Optically inactive
		(t)	Plane of symmetry

Sol. **A → (p, r) B → (p, s) C → (q, s, t) D → (q, s, t)**

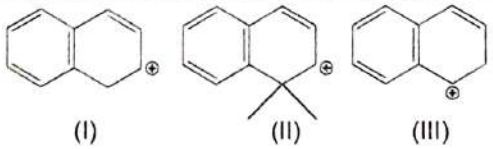
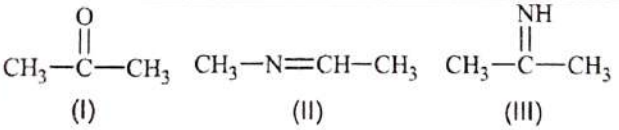
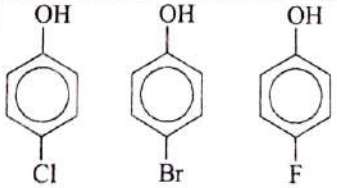
A is chiral, and optically active.

B is chiral but optically inactive due to pyramidal inversion.

C is achiral because of plane of symmetry.

D has a plane of symmetry.

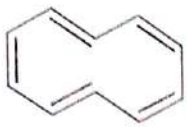
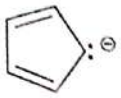
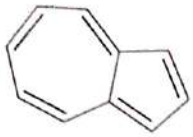
2. Match the following Column.

	Column-I Compounds		Column-II Order
(A)	 (I) (II) (III) decreasing order of carbocationstability	(p)	III > I > II
(B)	 (I) (II) (III) decreasing order of P_{ka}	(q)	I > II > III
(C)	 (I) (II) (III) Increasing order of P_{ka}	(r)	III > II > I
(D)	H_2O_2 O_2F_2 O_2Cl_2 (I) (II) (III) Decreasing order of (O—O) bond length	(s)	II > III >I

Sol. **(A) → q; (B) → r; (C) → q; (D) → p**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Antiaromaticity is a characteristic of a cyclic molecule with a π electron system that has higher energy due to the presence of 4n electrons in it. Unlike aromatic compounds, which follow Huckel's rule [(4n + 2)π electrons] and are highly stable, antiaromatic compounds are highly unstable and highly reactive. To avoid the instability of antiaromaticity, molecules may change shape, becoming non-planar and therefore breaking some of the π interactions.

Column – 1		Column – 2		Column – 3	
(I)		(i)	10π electrons	(P)	Aromatic
(II)		(ii)	Non-planar	(Q)	Huckel number of π e
(III)		(iii)	2 equivalent RS	(R)	Non-aromatic
(IV)		(iv)	5 equivalent RS	(S)	Dipolar

3. For the given species in column I, the only correct combination is
[A] (I) (ii) (S)
[B] (I) (ii) (P)
[C] (I) (i) (Q)
[D] (III) (i) (R)

4. For aromaticity, the only incorrect combination is
[A] (II) (iv) (P)
[B] (III) (iii) (P)
[C] (IV) (iii) (S)
[D] (I) (ii) (R)

5. Which of the following options is the only correct combination ?
[A] (I) (ii) (P)
[B] (II) (iv) (P)
[C] (III) (ii) (R)
[D] (IV) (ii) (S)

Sol.

3. **C**

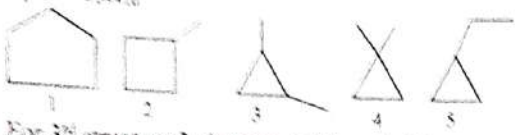
4. **C**

5. **B**

SINGLE INTEGER TYPE QUESTION

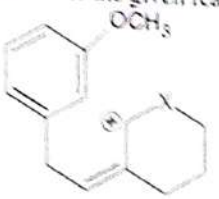
Prob. 1. The total number of cyclic structural as well as stereo isomers possible for a compound with the molecular formula C_5H_{10} is

Sol. Cyclic C_5H_{10}



For 3rd structure 2 cis-trans and 1 optical isomer are possible.
Total 7 isomers.

Prob. 2. Consider the given reaction

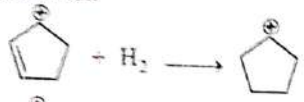


In the given reaction X may be:
(1) O (2) NH (3) S (4) $-CH_2-$
The reactivity of which compounds for the above reaction will be minimum.

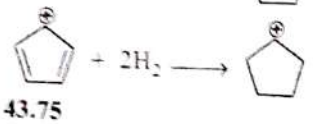
Sol. **(4)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. From the following, calculate the –ve value of resonance energy of the cyclopentadienyl cation, in k.cal/mol.



$\Delta H = -14.58$ k.cal / mol



$\Delta H = -72.91$ k.cal / mol

Sol. **43.75**

On hydrogenation of the allyl cation liberates –14.58 k.cal/mol. So two should liberate –29.16 k.cal/mol. Hydrogenation actually gains –72.91 k.cal/mol making the cation higher in energy by 43.75 k.cal/mol. The resonance energy is negative : –43.75 k.cal/mol

Prob. 2. Pure natural lysergic acid has $[\alpha]_D = +40°$. A certain (dl) mixture of lysergic acid has $[\alpha]_D = +10°$. Optical purity of the mixture is

Sol. **0.25**

$$\text{Optical purity} = \frac{[\alpha]_{net}}{[\alpha]_{mix}} = \frac{10°}{40°} = 0.25°$$

Prob. 3. The equilibrium constant for the ring-flip of fluorocy clohexane is 1.2 at 25°C. Calculate the percentage of the axial conformer at that temperature.

Sol. **45.45**

$$k_{eq} = \frac{[eq.]}{[ax]} = 1.2 \qquad \% \text{ axial} = \frac{[ax]}{[eq.]+[ax]} \times 100$$

$$= \frac{1}{[1.2]+1} \times 100 \qquad = \frac{1}{2.2} \times 100 = 45.45$$

ELECTROPHILIC AROMATIC SUBSTITUTION

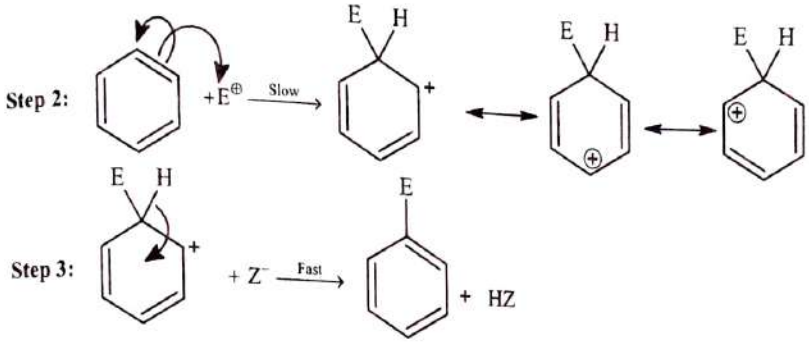
13

AROMATIC SUBSTITUTION REACTION

The replacement of an atom or group on an aromatic ring is known as aromatic substitution reaction. Some of the important substitution reactions are halogenation, nitration, sulphonation, chloromethylation, formylation and Friedel–Crafts reaction.

General mechanism:

Step 1: $E-Z \xrightarrow{\text{Catalyst}} E^+ + :Z^-$



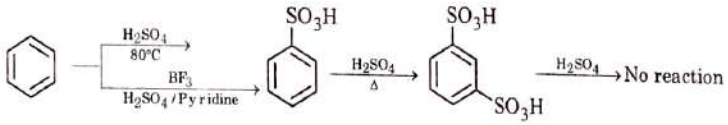
Halogenation

$E^+ \longrightarrow X^+ \left(Cl^+, Br^+, I^+\right)$

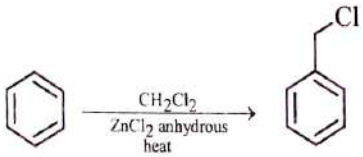
Nitration

$E^+ \longrightarrow NO_2^+$

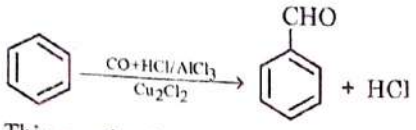
Sulphonation



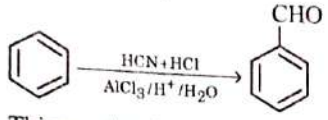
Chloromethylation



Formylation

(a) 

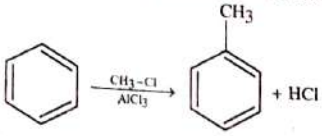
This reaction is known as Gattermann–Koch aldehyde synthesis. (This is not valid for phenols and phenolic ethers).

(b) 

This reaction is known as Gattermann aldehyde reaction.

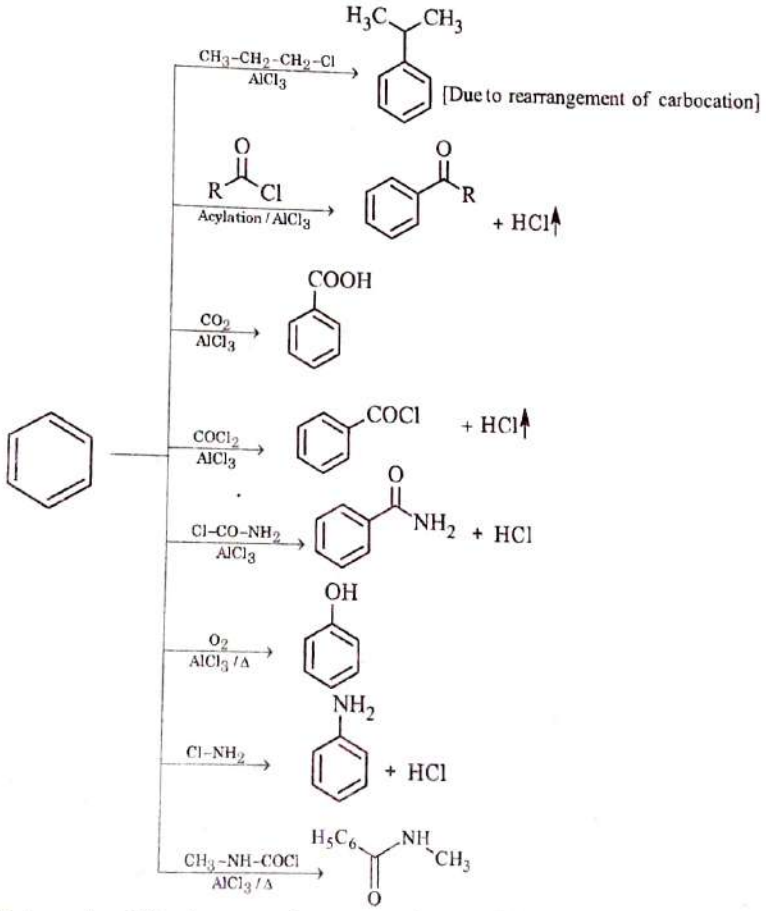
Friedel–Crafts Alkylation

Alkylation is possible with alkene, methanol and dialkyl sulphate.



Note: In general alkylation can be carried out using alcohols/alkene in presence of protic/Lewis acids.

Friedel–Crafts Reactions



[Note: 1. Nitrobenzene does not undergoes Friedel–Crafts reaction due to deactivating nature of NO_2 group.
2. In aniline due to acid base reaction between NH_2 group and $AlCl_3$ a bond forms causing loss of lone pair on $-NH_2$ thereby removing the +M effect and highly increasing – I effect.]

DIRECTIVE INFLUENCE OF SUBSTITUENTS

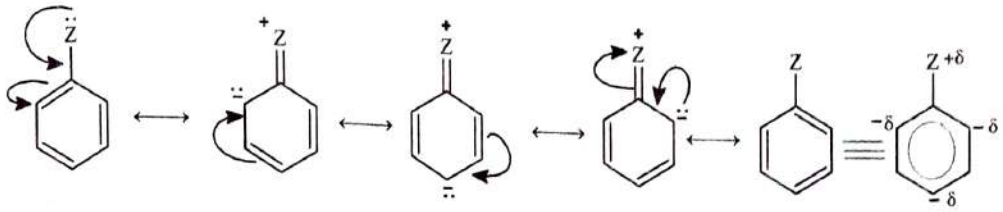
The group already present in a mono substituted product exerts a great influence on the position to be taken by the second incoming group. This is known as directive influence of the group.

Effect of substituents on electrophilic aromatic substitution

Ortho and para directors	Meta directors
Strongly activating $-NH_2$, $-NHR$, $-NR_2$, $-OH$, $-O^-$	Moderately deactivating $-C \equiv N$, $-SO_3H$, $-COOH$, $-COOR$, $-CHO$, $-COR$
Moderately activating $-NHCOCH_3$, $-NHCOR$, $-OCH_3$, $-OR$	Strongly deactivating $-NO_2$, $-NR_3^+$, $-CF_3$, $-CCl_3$
Weakly activating $-CH_3$, $-C_2H_5$, $-R$, $-C_6H_5$	
Weakly deactivating $-F$, $-Cl$, $-Br$, $-I$	

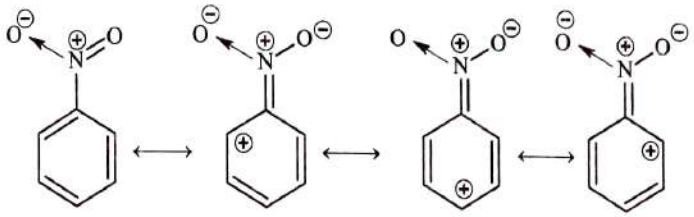
Effect of Activating Group

This may take place due to +I and +M, - I and –M (with +M > - I) preferably. The +M effect can be understood by considering the following structures.



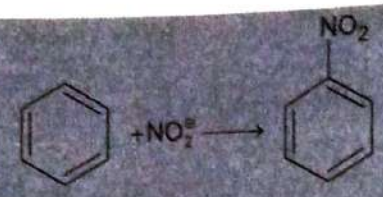
Effect of Deactivating Group

Could be due to – I only, - I, - M, +I and – M (with – M>+I). The – M effect can be understood by considering the following resonating structures for nitrobenzene.

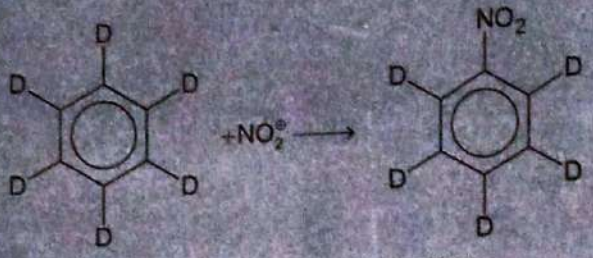


ILLUSTRATIVE PROBLEMS

1. (i)



(ii)



Which reaction have fast rate at the same temperature.

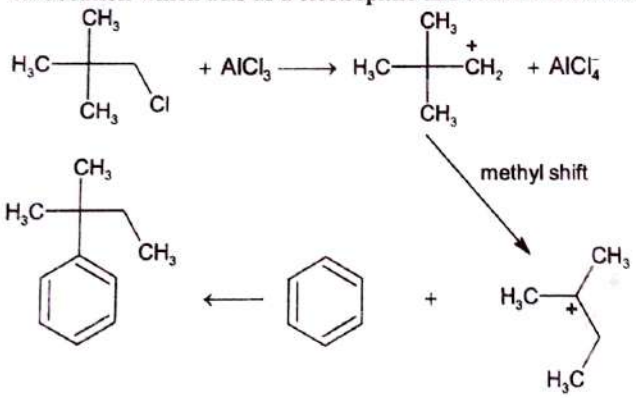
Sol. Both reaction have same rate because C – H or C – D bond breaking will not participate in rate determining step.

2. Friedel-Crafts acylation requires an excess of the catalyst but Friedel-Crafts alkylation requires only a catalytic amount. Why?

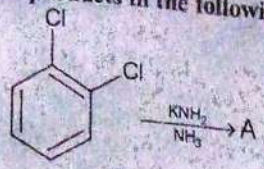
Sol. The product of acylation co-ordinates with is a carbonyl compound which can bind the Lewis acid catalyst. For this reason, Friedel-Crafts acylation requires an excess of the catalyst. The product of alkylation can act as Lewis base catalyst. So, all the catalyst is always available for alkylation.

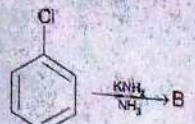
3. During reaction of benzene with neopentyl chloride in the presence of $AlCl_3$, the major product is 2-methyl-2-phenylbutane and not neopentylbenzene. Explain.

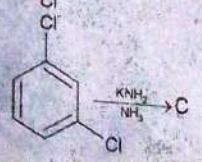
Sol. The initially formed primary carbocation rearranges by methyl migration to form a more stable tertiary carbocation which acts as a electrophile and attack the benzene to give the major product.

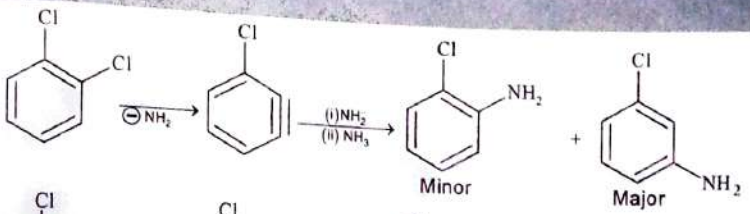


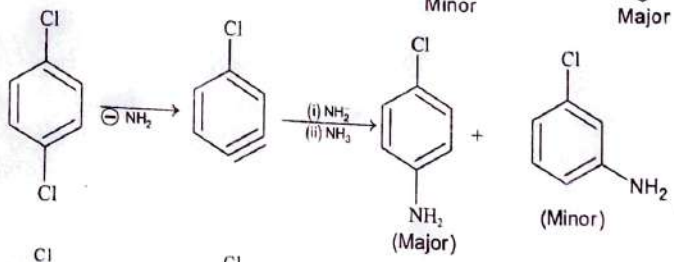
4. **Predict the products in the following reactions.**

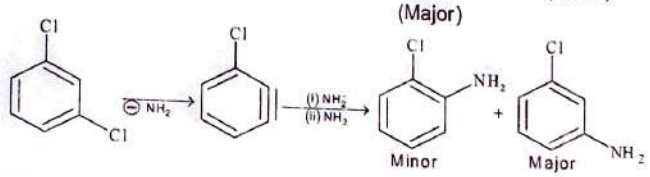
(a) 

(b) 

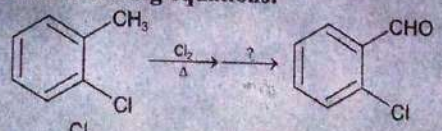
(c) 

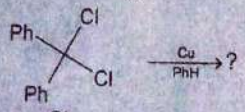
Sol. (a) 

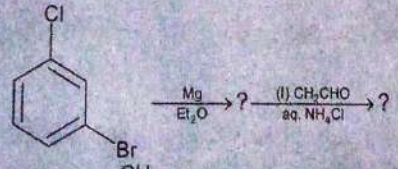
(b) 

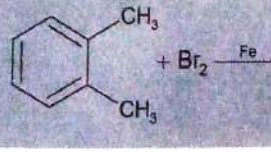
(c) 

5. **Complete the following equations:**

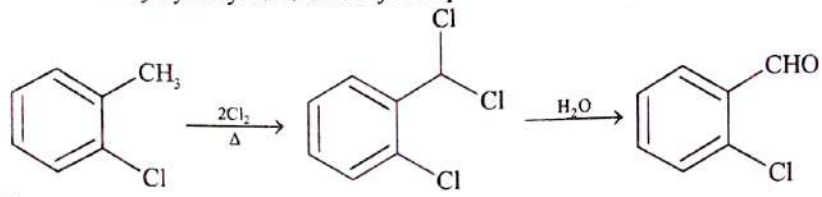
(a) 

(b) 

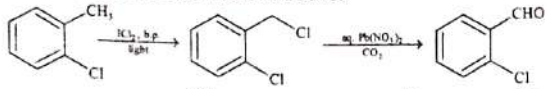
(c) 

(d) 

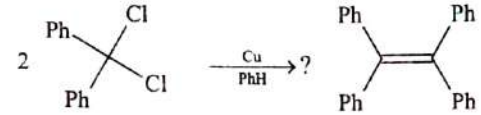
Sol. (a) Since the final product is an aldehyde, one precursor could be the 1,1-dichloride; these compounds are readily hydrolysed to carbonyl compounds.



There is also an alternative route.

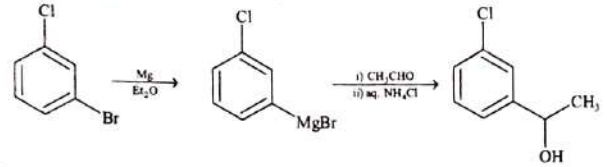


(b)



The use of two molecules of a 1,1-dichloride and copper suggests the formation of an alkene. This is an extension of the Wurtz reaction.

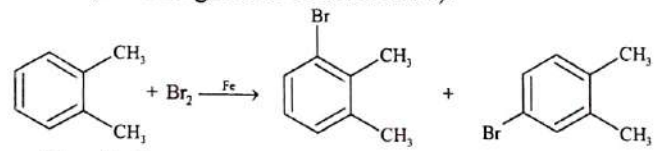
(c)



The given reagents clearly indicates that a Grignard reagent is involved. However, aryl chlorides do not form Grignard reagents in ether; THF is necessarily a solvent. On the other hand, aryl bromides (and iodides) readily react in ether solution.

The use of aqueous ammonium chloride is a safety precaution to prevent the possibility of dehydration of the alcohol (which might occur if acid is used).

(d)



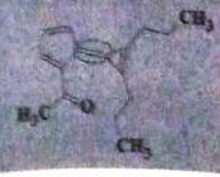
The use of iron indicates nuclear substitution. Me is o/p directing, and there is always the possibility of a steric effect at the o-position. So, we can suggest that 4-Br product will predominate. Experimental work would have to be done to find out what are the actual results. From the literature, it appears that the 4-Br compound is formed exclusively.

6. The $-\overset{+}{N}(CH_3)_3$ and -CN substituents are both meta directors. Still nitration of the corresponding benzene derivative does give a minor amount of the ortho and para isomers of the product. Which of the two substituents should lead to a greater ortho /para ratio? Why?

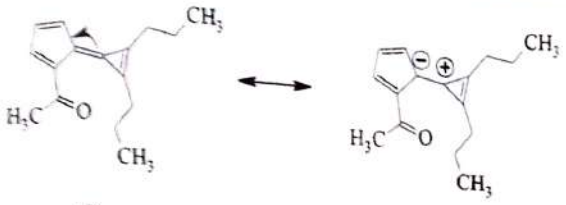
Sol. The -CN group, being a linear two-atom group, is much less sterically demanding than the t-butyl- like - $-\overset{+}{N}(CH_3)_3$ group. So, the CN group should give the greater ortho/para ratio. The actual data is mentioned in the table below:

Group	% Meta	% Ortho	% Para	Ortho / para ratio
$-\overset{+}{N}(CH_3)_3$	87	2	11	0.18
-CN	81	17	2	8.5

7. Consider the following reaction sequence:
Ordinarily the barrier to rotation about a carbon–carbon double bond is high. But in the compound below the carbon–carbon double bond (circled) was observed to have a poor rotational barrier. Explain.



Sol.

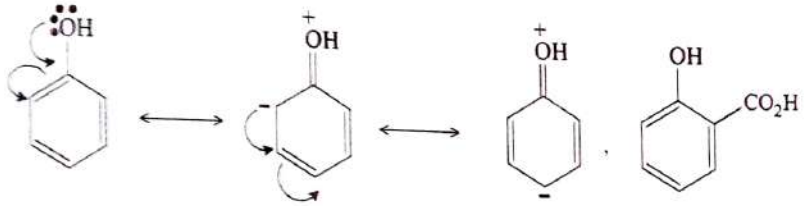


(I) (II)

The molecule will exist in more stabilized resonance form II as in this form both the rings gain aromaticity. Now, we don't have a double bond between the ring and hence poor rotational barrier.

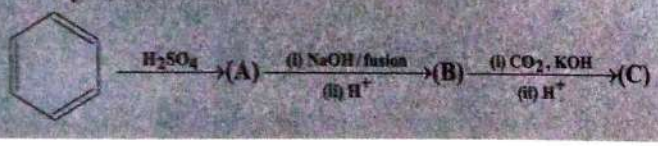
8. Phenol undergoes electrophilic substitution at a faster rate in alkaline medium. Why?

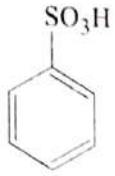
Sol. Phenol is ortho and para directing due to the +M effect of its $-\ddot{O}H$ group.

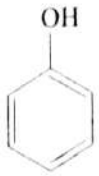


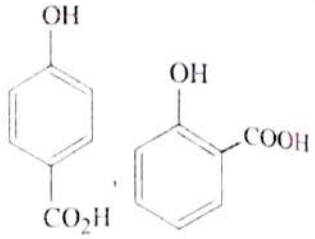
Also $-\ddot{O}H$ has –I effect. But in alkaline medium, phenol is converted to phenoxide ion which has both +M and –I effect. Hence, ring is more activated and hence electrophilic substitution occurs more readily.

9. Identify A, B and C.



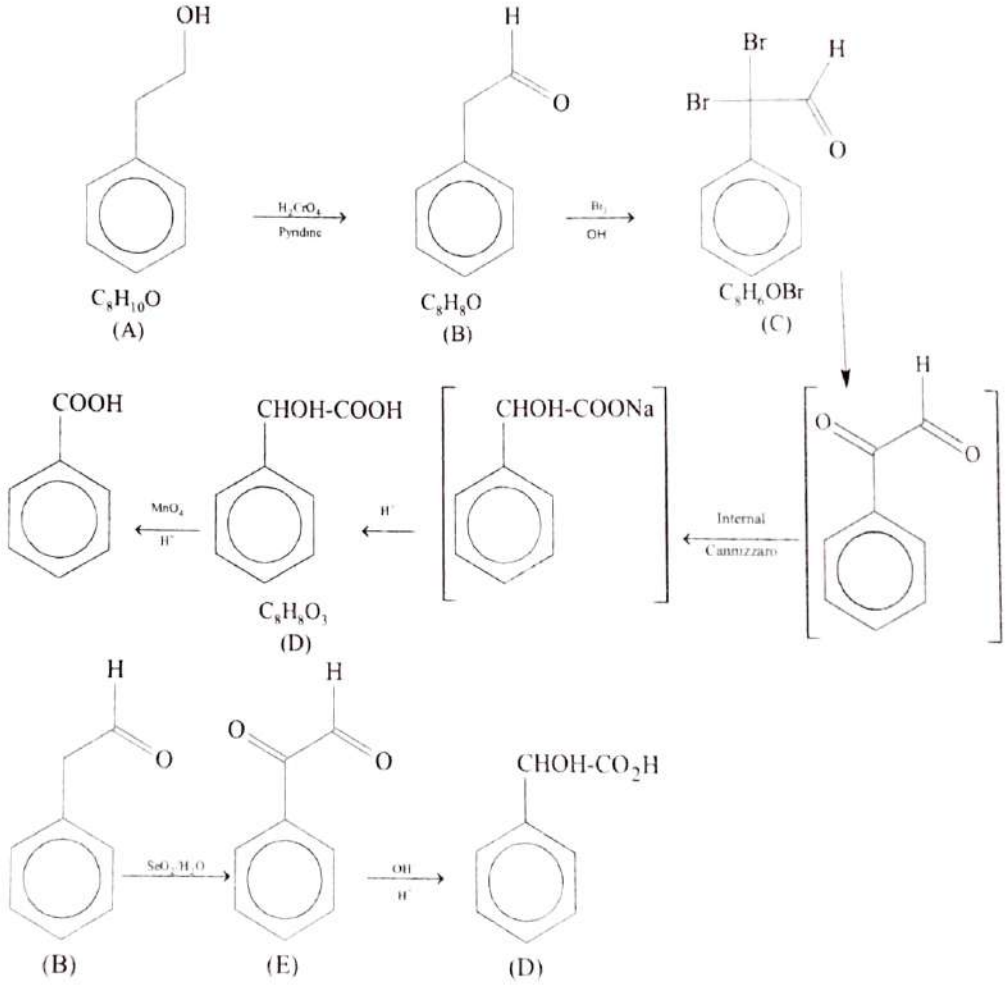
Sol. A = 

B = 

C = 

10. **Treatment of compound A ($C_8H_{10}O$) with chromic acid / pyridine gives B (C_8H_8O) which reacts with alkaline bromine yielding C ($C_8H_6OBr_2$). C generates D ($C_8H_8O_3$) upon treatment with alkali and subsequent acidification. Oxidation of D with alkaline $KMnO_4$ and subsequent acidification gives benzoic acid. Treatment of (B) with SeO_2/H_2O gives E ($C_8H_6O_2$) which also gives D when treated with alkali followed by acidification. Identify A, B, C, D and E with proper reasoning.**

Sol.

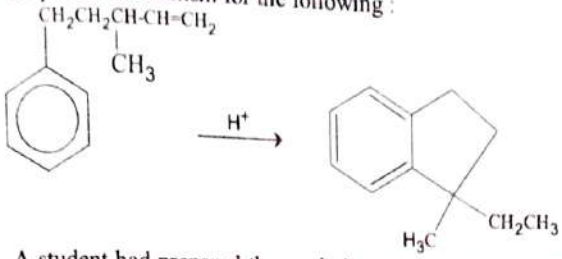


SOLVED PROBLEMS

Subjective Questions

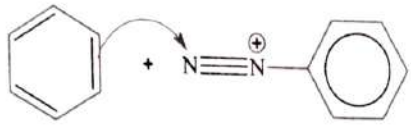
Prob. 1. a) In the mechanism for electrophilic substitution using a diazonium ion as the electrophile, why does nucleophilic attack occur on the terminal nitrogen atom of the diazonium ion rather than on the nitrogen atom bonded to the benzene ring?

b) Propose a mechanism for the following :



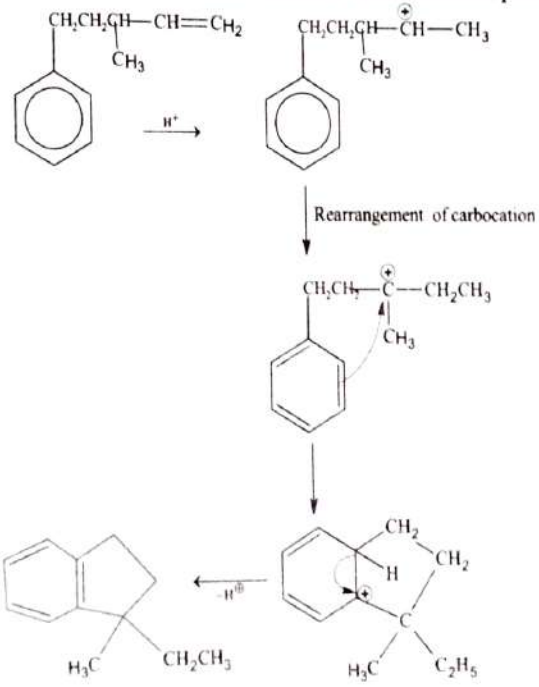
c) A student had prepared three ethyl – substituted benzaldehydes X, Y and Z, but had neglected to label them. The premed student at the next bench said that they could be identified by brominating a sample of each and determining how many bromo – substituted product were formed. Is the premed student's advice sound?

Sol. a)

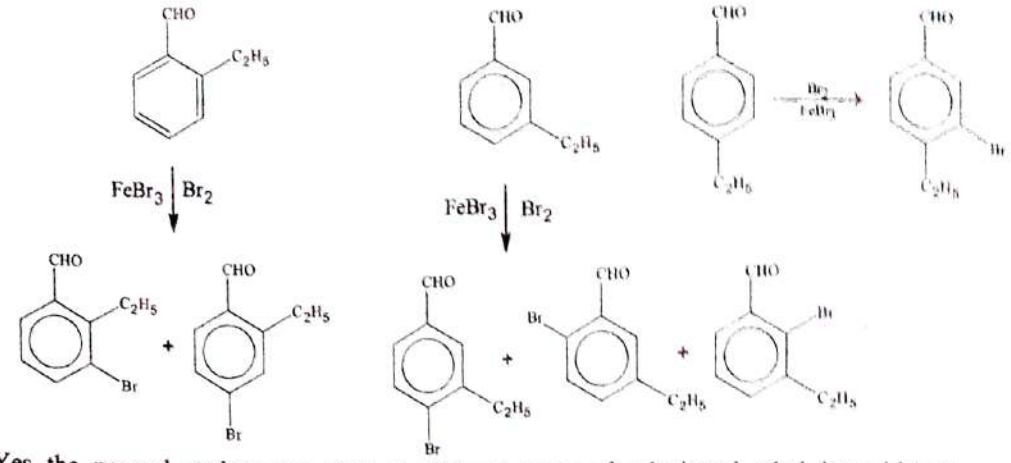


Because the nitrogen bonded to benzene can not become pentavalent due to lack of d – orbitals.

b)

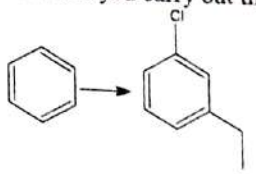


c)

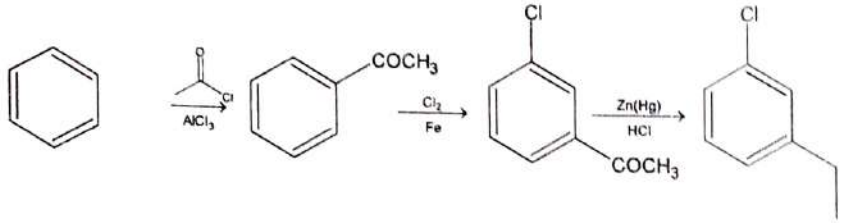


Yes the premed student was right. 3 different types of substituted ethyl benzaldehyde on bromination will give 3 different number of products.

Prob. 2. How can you carry out the following conversion?



Sol.



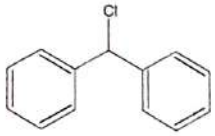
Objective Questions

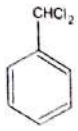
Prob. 1. In aniline, the $-NH_2$ group

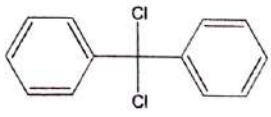
(A) activates the benzene ring via both inductive and resonance effects.
(B) deactivates the benzene ring via both inductive and resonance effects.
(C) activates the benzene ring via resonance effect and deactivates it via. inductive effect
(D) activates the benzene ring via inductive effect and deactivates it via resonance effect.

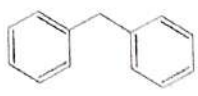
Sol. Due to lone pair on N atom, it activates the ring. On the other hand, electronegativity of N is more than C so it deactivates the benzene ring via inductive effect.
Hence, **(C)** is the correct answer.

Prob. 2. Which of the following structures correspond to the product expected, when excess of C_6H_6 reacts with CH_2Cl_2 in the presence of anhydrous $AlCl_3$?

(A) 

(B) 

(C) 

(D) 

Sol. **(D)**

Prob. 3. The most basic compound among the following is

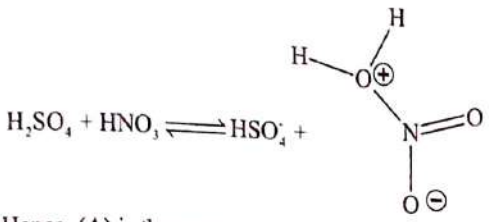
(A) benzylamine
(B) aniline
(C) acetanilide
(D) p-nitroaniline

Sol. Except benzylamine, all show less tendency to donate electron pair due to resonance.
Hence, **(A)** is the correct answer.

Prob. 4. Nitrobenzene can be prepared from benzene by using a mixture of conc. HNO_3 and conc. H_2SO_4. In the nitrating mixture HNO_3 acts as a

(A) base
(B) acid
(C) reducing agent
(D) catalyst

Sol. HNO_3 accepts a proton from H_2SO_4.



Hence, **(A)** is the correct answer.

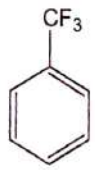
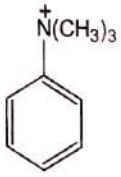
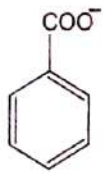
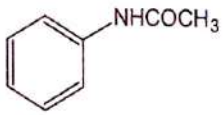
Prob. 5. Which of the following is characteristic of a carbocation?

(i) It can rearrange to a more stable carbocation.
(ii) It can eliminate a hydride ion to form an alkene.
(iii) It can add to an alkene to form a new carbocation.
(iv) It can abstract a hydride ion from an alkane.

(A) (i)
(B) (iii)
(C) (iv)
(D) (ii)

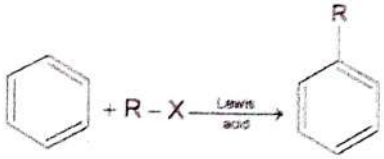
Sol. **(A, B, C, D)**

Prob. 6. Which will lead to S_E reactions at meta position?

(A) 
(B) 
(C) 
(D) 

Sol. **(A, B)**

COMPREHENSION

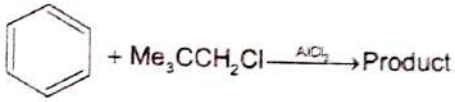
Reaction  + HX is an example of Friedel-Craft alkylation reaction.

In number of cases of Friedel-Crafts alkylation, the final product is found to contain a rearrange alkyl group. Generally with stronger Lewis acid product is rearranged due to enough polarization of complex while with weak Lewis acid no such effect is observed. Temperature is also favour rearrange product.

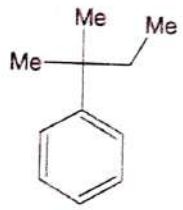
Read the above passage carefully and answer the questions given below it.

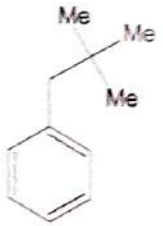
1. What is electrophile in given reaction?
(A) X^+
(B) R^-
(C) (Lewis acid X)$^+$
(D) None of the above

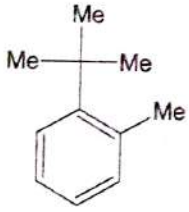
Sol. **(B)**

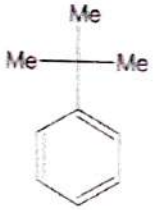
2. 

Product is

(A) 

(B) 

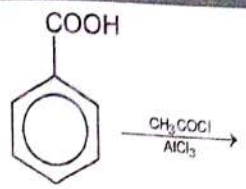
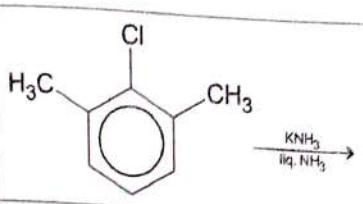
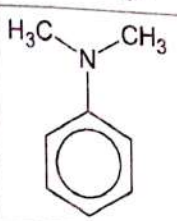
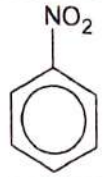
(C) 

(D) 

Sol. **(A)**

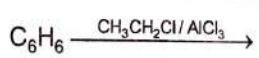
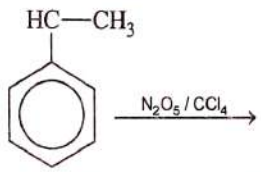
MATRIX-MATCH TYPE QUESTION

1. Match the following question Column-I with Column-II:

Column-I		Column-II	
(A)		(p)	o, p-directing compound
(B)		(q)	Activated compound
(C)		(r)	No reaction
(D)		(s)	Deactivated compound
		(t)	Meta directing

Sol. **A → (s, t), B → (p, q, r), C → (p, q), D → (s, t)**

2. Match the following:

Column – I		Column – II	
(A)	Friedel-Crafts acylation	(p)	Electrophile substitution
(B)	Reimer-Tiemann Reaction	(q)	Cumene
(C)		(r)	Nitronium ion
(D)		(s)	Dichlorocarbene
		(t)	Rearrangement of Carbocation

Sol. **(A → p), (B → p, s), (C → p, q, t), (D → p, r)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Inductive effect	(i)	Temporary effect	(P)	Solvent dependent
(II)	Hyperconjugation	(ii)	Permanent effect	(Q)	Solvent independent
(III)	Mesomeric effect	(iii)	Involves σ -electrons	(R)	Occurs in benzene
(IV)	Electromeric effect	(iv)	Involves π –electrons	(S)	Occurs in toluene

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (II) (iii) (R) [C] (III) (iii) (S) [D] (I) (ii) (Q)

4. Which of the following combination is incorrect ?
[A] (I) (iii) (Q) [B] (III) (ii) (Q) [C] (I) (i) (Q) [D] (III) (iii) (S)

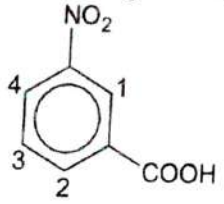
5. Which of the following combination is correct ?
[A] (IV) (i) (Q) [B] (I) (i) (R) [C] (II) (i) (Q) [D] (II) (ii) (S)

Sol.

3. **D**
4. **C**
5. **D**

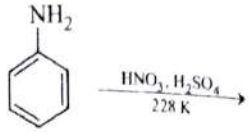
SINGLE INTEGER TYPE QUESTION

1. Nitration of given compound will takes place at which one of the carbons indicated by member.



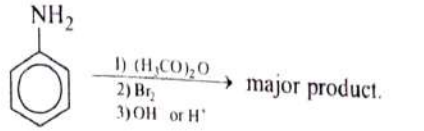
Sol. **(3)**

NUMERICAL VALUE ANSWER TYPE

Prob.1. 

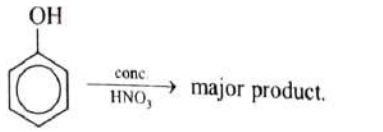
Total number of mononitro derivatives is x. Calculate $\frac{x}{2}$

Sol. **1.5**

Prob.2. 

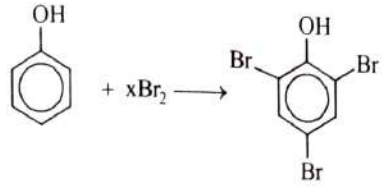
In the final product the number of carbon atom is x and the number of lone pairs is y. Calculate $\frac{x}{y}$

Sol. **1.5**

Prob.3. 

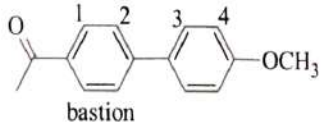
Calculate $\frac{\text{number of O atoms in product}}{\text{number of H atoms in product}}$

Sol. **2.33**

Prob.4. 

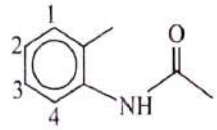
Calculate $\frac{x}{2.5}$

Sol. **1.2**

Prob.5. 

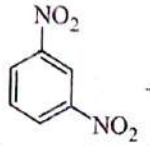
The number which will be attacked most rapidly by $-NO_2^+$ is x. Calculate $\frac{x}{3}$

Sol. **1.33**

Prob.6. 

Sulphonation is most favourable at carbon number x. Calculate $\frac{\text{number of carbon in given compound}}{x}$

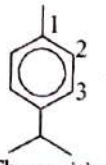
Sol. **4.5**

Prob.7.  $\xrightarrow{HNO_3+H_2SO_4}$ X

Compound X is (p,q,r - Trinitrobenzene). The number of oxygen atom in reactant is Y.

Calculate $\frac{p+q+r}{y}$

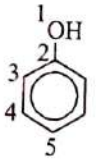
Sol. **2.25**

Prob.8.  $\xrightarrow{Cl_2/AlCl_3}$ Q (major product)

The position at which Cl attaches is x in the product. The number of Hydrogen atoms in reactant is y.

Calculate $\frac{x}{y}$

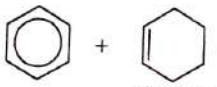
Sol. **6.5**

Prob.9.  $\xrightarrow{HNO_3/H_2SO_4}$ A (more volatile) + B (less volatile)

The position at which NO_2 attaches in product B is x. The number of π bonds in product is y.

Calculate $\frac{x}{y}$

Sol. **1.25**

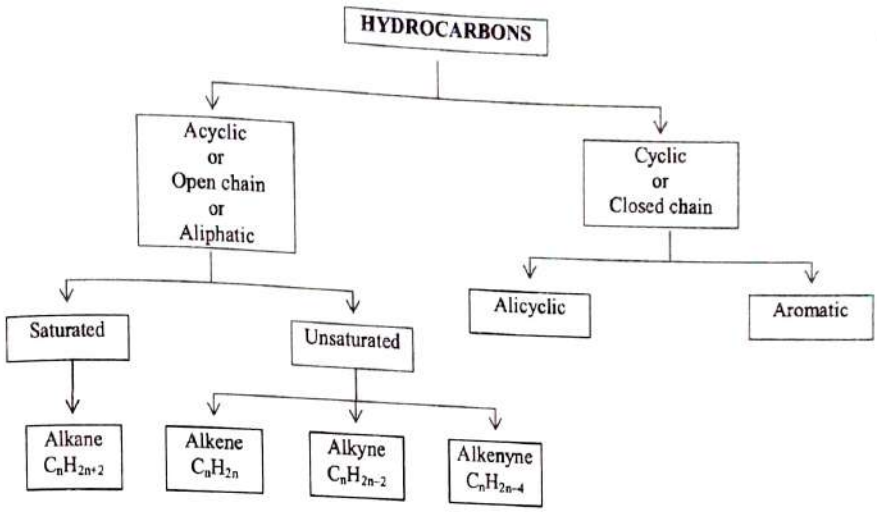
Prob.10.  $\xrightarrow{H_2SO_4}$ P

(2 mole)

The number of carbon-carbon bonds in product P is x and number of π bonds if y. Calculate $\frac{x}{y}$

Sol. **6.67**

HYDROCARBONS 14



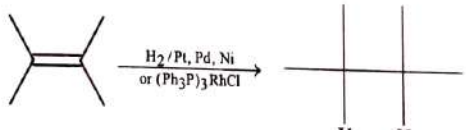
ALKANES

The general formula of alkane: C_nH_{2n+2}

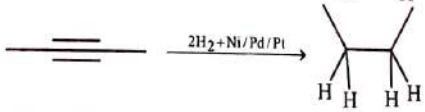
Methods of Preparation

(i) **Reduction:**

(a)



(b)



(ii) **Wurtz reaction:**

$$2R-X \xrightarrow{Na/Dry\ ether} R-R$$

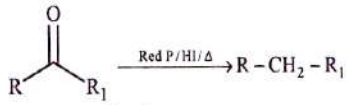
(iii) **Corey–House synthesis:**

$$R-X \xrightarrow{R_2CuLi} R-R$$

(iv) **Reduction:**

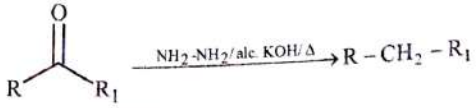
(a) $R-OH \xrightarrow{Red\ P/HI/\Delta} R-H$

(b)

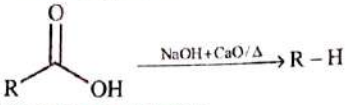


(v) **Clemmensen reduction:**

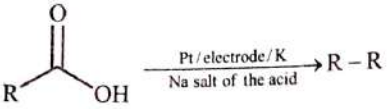
(vi) **Wolff-Kishner reduction:**



(vii) **Decarboxylation:**



(viii) **Kolbe's electrolysis:**



Chemical Properties

(i) **Reaction by X_2:**

$$R-H \xrightarrow{X_2/hv} R-X \text{ (usually a mixture)}$$

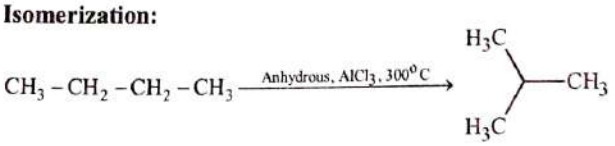
(ii) **Combustion:**

$$C_nH_{2n+2} \xrightarrow{\text{Excess } O_2,\ \text{Flame}} nCO_2 + (n+1)H_2O$$

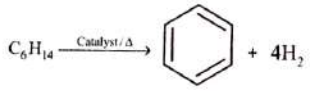
(iii) **Pyrolysis or cracking:**

$$\text{Alkane} \xrightarrow{400-600^\circ C \text{ with or without catalyst}} H_2 + \text{Smaller Alkene} + \text{Alkenes}$$

(iv) **Isomerization:**



(v) **Aromatization:**



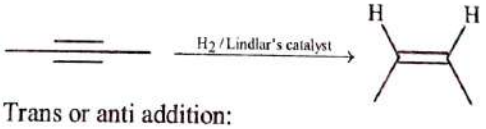
ALKENES

The general formula of alkene: C_nH_{2n}

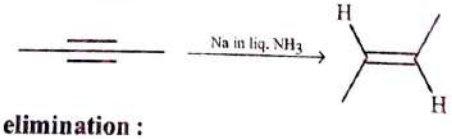
Methods of Preparation

(i) **Addition reaction:**

(a) Syn addition:



(b) Trans or anti addition:



(ii) **β – elimination :**

$$R-CH_2-CH_2-X \xrightarrow{\text{Alc. KOH}/\Delta} R-CH=CH_2$$

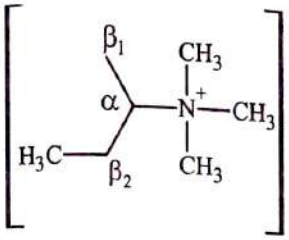
(iii) **Dehydration:**

$$R-CH_2-CH_2-OH \xrightarrow{\text{Conc. } H_2SO_4/\Delta} R-CH=CH_2$$

(iv) **Dehalogenation:**

$$R-CH(X)-CH(X)-R \xrightarrow[\text{or}\ (NaI\ \text{in acetone})]{Zn/C_2H_5OH/\Delta} R-CH=CH-R$$

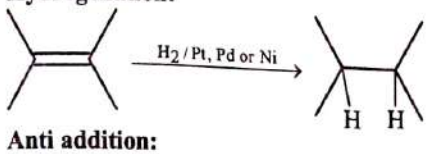
(v) **Hofmann elimination:**



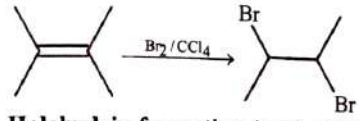
$$\xrightarrow{\Delta} CH_3-CH_2-CH=CH_2\ (\text{Major}) + CH_3-CH=CH-CH_2\ (\text{Minor})$$

Chemical Properties

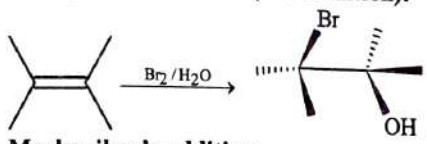
(i) **Hydrogenation:**



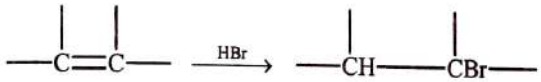
(ii) **Anti addition:**



(iii) **Halohydrin formation (anti addition):**



(iv) **Markonikov's addition:**



(v) **Anti Markonikov's addition:**

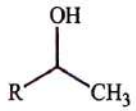
$$CH_3-CH=CH_2 \xrightarrow{HBr/\text{peroxide}} CH_3-CH_2-CH_2-Br$$

(vi) **Syn-hydroxylation:**

$$CH_2=CH_2 \xrightarrow[\text{cold}]{KMnO_4/H_2O} CH_2(OH)-CH_2(OH)$$

(vii) **Hydration reaction:**

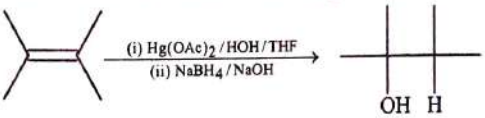
$$R-CH=CH_2 \xrightarrow[(ii)\ HOH]{(i)\ \text{Conc. } H_2SO_4}$$



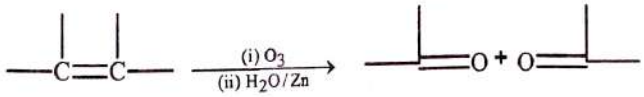
(viii) **Anti Markonikov's**

$$R-CH=CH_2 \xrightarrow[(ii)\ H_2O_2/OH^-]{(i)\ B_2H_6/THF} R-CH_2-CH_2-OH$$

(ix) **Markonikov's without rearrangement:**

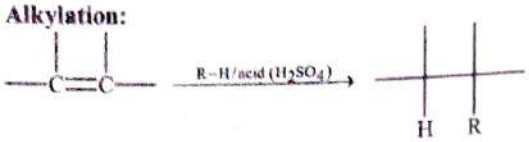


(x)



(xi) **Epoxidation:**

(xii) **Alkylation:**



ALKYNES

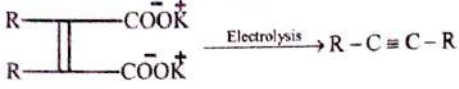
The general formula of alkyne: C_nH_{2n-2}

Methods of Preparation

(i) **Dehydrohalogenation:**

$$R-\underset{X}{CH}-\underset{X}{CH}-R \xrightarrow[(ii)\ NaNH_2/\Delta]{(i)\ KOH\ (Alc.)} R-C\equiv C-R$$

(ii) **Kolbe hydrocarbon synthesis:**



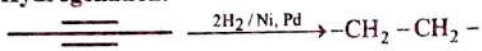
(iii) $-C\equiv C-H \xrightarrow[(ii)\ R-X]{(i)\ LiNH_2\ or\ NaNH_2/NH_3(l)} -C\equiv C-R$

(iv) **Dehalogenation or coupling:**

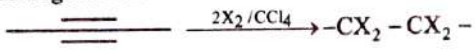
$$2R-CX_3 \xrightarrow{Ag/\Delta} R-C\equiv C-R$$

Chemical Properties

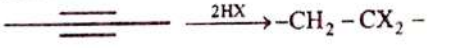
(i) **Hydrogenation:**



(ii) **Halogenation:**



(iii) **Reaction with HX:**



(iv) $-C\equiv C- + H_2O \xrightarrow{H_2SO_4/HgSO_4} -CH=C(OH)- \longleftrightarrow -CH_2-\underset{\underset{O}{\|}}{C}-$

(v) **Markonikov's addition:**

$$R-C\equiv C-H \xrightarrow{HOX} R-\overset{\overset{O}{\|}}{C}-CH_2X$$

(vi) **Oxidation:**

$$R-C\equiv C-R \xrightarrow{KMnO_4/OH^-/HOH\ or\ SeO_2} R-\overset{\overset{O}{\|}}{C}-\overset{\overset{O}{\|}}{C}-R$$

(vii) **Isomerization:**

$$CH_3-C\equiv C-CH_3 \xrightarrow{Alc.\ KOH} CH_3-CH_2-C\equiv CH$$

(viii) **Oxidative ozonolysis:**

$$CH_3-C\equiv \underset{\underset{H}{|}}{C} \xrightarrow[(ii)\ H_2O/H^+]{(i)\ O_3} CH_3COOH + HCOOH$$

ILLUSTRATIVE PROBLEMS

1. Identify (A), (B), (C) and (D)

$$CH_3-C\equiv C-H \xrightarrow[NH_3(l)]{NaNH_2} (A) \xrightarrow{CH_3I} (B) \xrightarrow{Na/NH_3(l)} (C) \xrightarrow{Br_2/H_2O} (D)$$

Sol.

$$CH_3-C\equiv C-H \xrightarrow[NH_3(l)]{NaNH_2} CH_3-C\equiv C^{\ominus}\,{}^{\oplus}Na \xrightarrow{CH_3I} CH_3-C\equiv C-CH_3$$

(A) (B)

$CH_3-C\equiv C-CH_3 \xrightarrow{Na/NH_3(l)}$ (C)

(C) $\xrightarrow{Br_2/H_2O}$ (D)

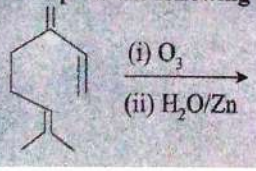
(D)

Erythro form

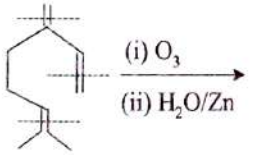
2. For the preparation of CH_3–CH_2–CH_3 which method is suitable ?
(a) Wurtz reaction
(b) Frankland reaction
(c) Kolbe's electrolysis
(d) Corey's house synthesis

Sol. Corey's house synthesis. This Method can be used for the preparation of unsymmetrical Alkanes.

3. Complete the following reaction:



Sol.



$$2CH_2O + CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + OHC-\overset{O}{\overset{\|}{C}}-CH_2\,CH_2\,CHO$$

4. An organic compound (A) of molecular formula C_5H_8 when treated with sodium in liquid ammonia followed by reaction with n–propyl bromide yields (B), C_8H_{14}. (A) gives a ketone (C), $C_5H_{10}O$ when treated with dilute H_2SO_4 and $HgSO_4$. (B) on oxidation with alkaline $KMnO_4$ gives two isomeric acids (D) and (E) $C_4H_8O_2$. Give structures of compound (A) to (E) with proper reasoning.

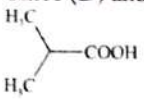
Sol. (i) (A) reacts with sodium in liquid ammonia and thus it is terminal alkyne i.e. $C_3H_7C \equiv C - H$

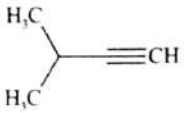
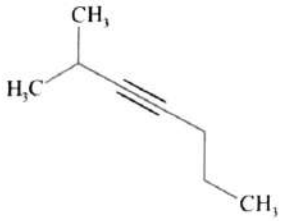
(ii) $C_3H_7C \equiv CH \xrightarrow{Na} C_3H_7C \equiv CNa^+ \xrightarrow{CH_3CH_2CH_2Br} C_3H_7C \equiv C - \underset{(B)}{CH_2} - CH_2 - CH_3$

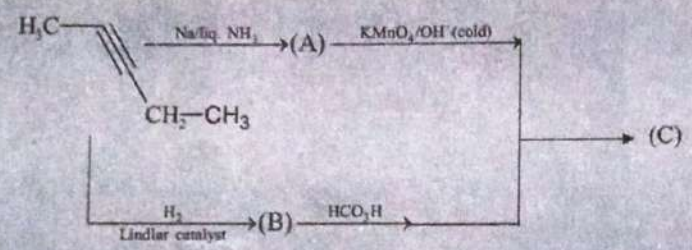
(iii) $C_3H_7C \equiv CH \xrightarrow[H_2SO_4]{HgSO_4} C_3H_7CO - CH_3$

(iv) $C_3H_7C \equiv C - CH_2 - CH_2 - CH_3 \xrightarrow[KMnO_4]{[O]} CH_3 - CH_2 - \underset{(D)}{CH_2} - COOH + \underset{(E)}{C_3H_7COOH}$

Since (D) and (E) are isomers, thus the structure of (E) is

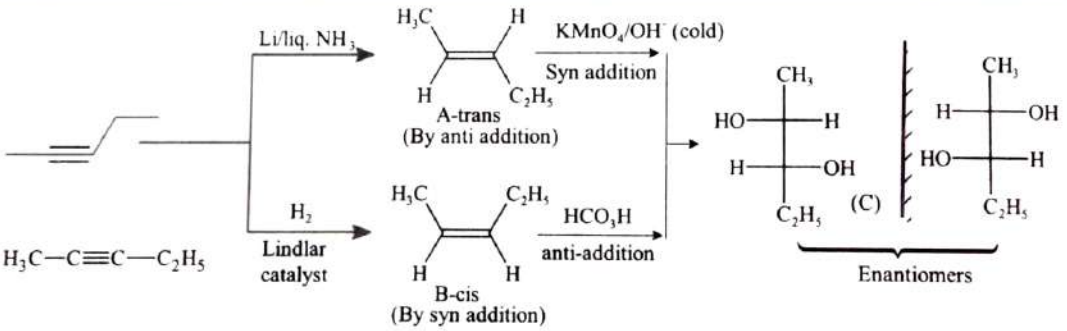


Thus (A) is  and (B) is 

5. 

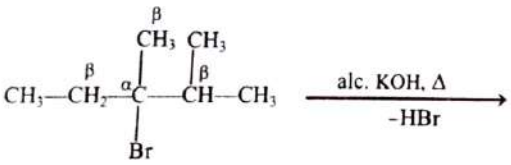
Identify (A), (B), (C) with proper explanation.

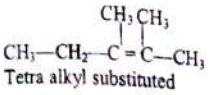
Sol.

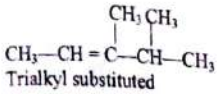


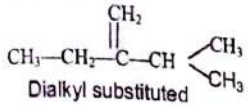
6. List the possible products, in order of decreasing yield, from the reaction of 3-bromo –2,3 – dimethyl pentane with alcoholic KOH.

Sol.




Tetra alkyl substituted (A)

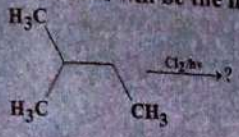

Trialkyl substituted (B)


Dialkyl substituted (C)

The order of decreasing yield is A>B>C.

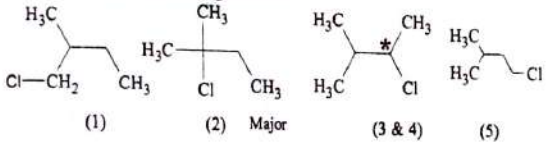
According to Saytzeff rule, more substituted alkene is more stable.

7. (a) How many products can be generated by monochlorination in the following structure and which one will be the major product?

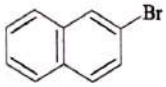


(b)

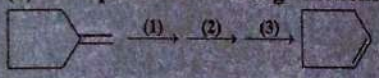
Sol. (a) Four fractions can be generated


(1) (2) Major (3 & 4) (5)

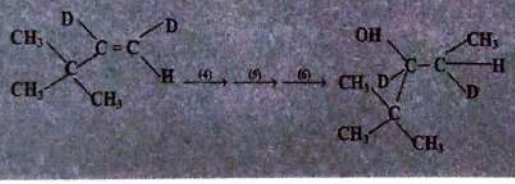
(b)



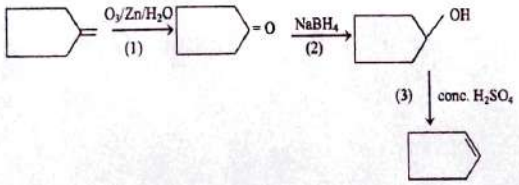
8. (a) Complete the following reactions with appropriate reagents:

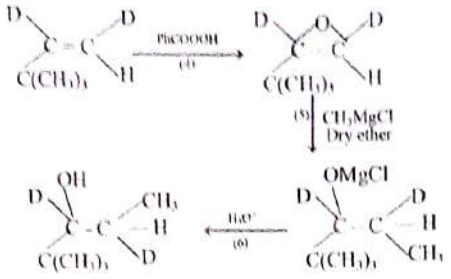


(b)



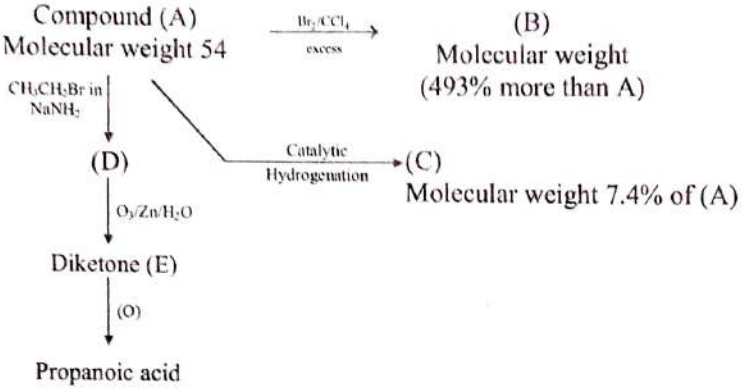
Sol. (a)



(b) 

9. A hydrocarbon (A) of molecular weight 54 reacts with excess of Br_2 in CCl_4 to give a compound (B) whose molecular weight is 493% more than that of (A). However, on catalytic hydrogenation with excess of hydrogen, (A) forms (C) whose molecular weight is only 7.4% more than that of (A). (A) reacts with CH_3CH_2Br in the presence of $NaNH_2$ to give another hydrocarbon (D), which on ozonolysis yields diketone (E). (E) on oxidation gives propanoic acid. Give structures of (A) to (E) with reasons.

Sol.



To 100 gm of (A), add Br_2 and the new weight is 593 gm

$\therefore$ To 54 gm of (A), add Br_2 and the new weight is $\frac{593 \times 54}{100} = 320.2$ gm

$\therefore$ No. of moles of $Br_2 = \frac{\text{Weight}}{\text{Molecular weight}} = \frac{320.2}{160} = 2$ moles

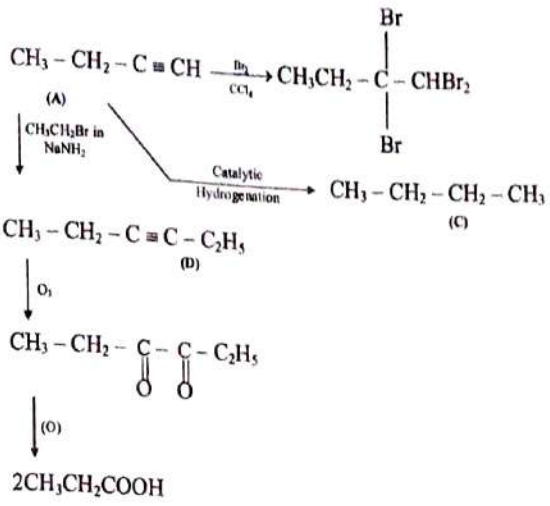
(i) As two moles of Br_2 were added, compound has two π bonds hence general formula should be (C_nH_{2n-2}).

$C_nH_{2n-2} = 54$

$12n + 2n - 2 = 54$

$n = 4$ $\therefore$ molecular formula = C_4H_6

(ii) (A) reacts with CH_3CH_2Br in presence of $NaNH_2 \longrightarrow$ Compound (A) should be terminal alkyne, i.e. 1-butyne

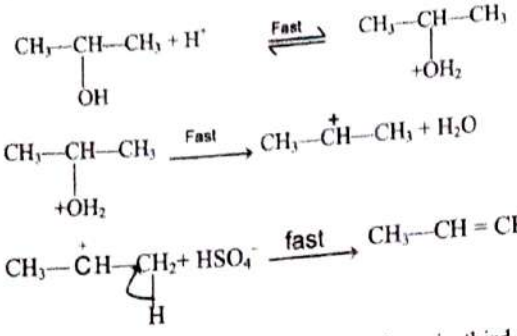

CH3 – CH2 – C ≡ CH
(A)
Br2
CCl4
CH3CH2 – C – CHBr2
Br
Br
CH3CH2Br in NaNH2
Catalytic Hydrogenation
CH3 – CH2 – CH2 – CH3
(C)
CH3 – CH2 – C ≡ C – C2H5
(B)
O3
CH3 – CH2 – C – C – C2H5
O O
(O)
2CH3CH2COOH

SOLVED PROBLEMS

Subjective Question

Prob. 1. Suggest a mechanism for the dehydration of $CH_3CHOHCH_3$ that proceeds through a carbonium ion intermediate. Assign a catalytic role to the acid and keep in mind that the O in ROH is a basic site like O in H_2O.

Sol.



H_2SO_4 act as an acid in first step and base in third step. Since it is regenerated back it is acting as a catalyst.

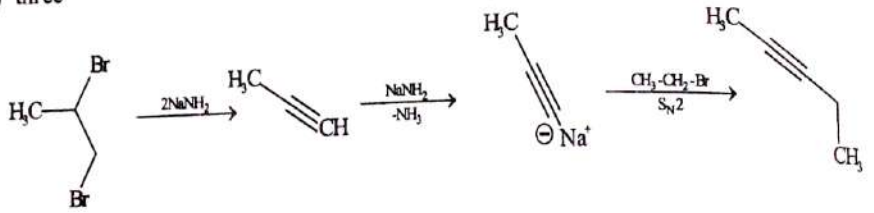
Objective Questions

Prob. 1. 1,2–dibromopropane on treatment with X moles of $NaNH_2$ followed by treatment with ethyl bromide gave a pentyne, the value of X is

(A) one (B) two

(C) three (D) four

Sol.



Thus, X is three.

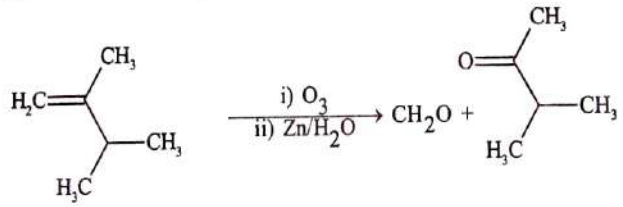
Hence, **(C)** is the correct answer.

Prob. 2. Ozonolysis of 2,3–dimethylbut–1–ene followed by reduction with zinc and water gives

(A) methanoic acid and 3–methyl–2–butanone (B) methanal and 3–methyl–2–butanone

(C) methanal and 2–methyl–3–butanone (D) methanoic acid and 2–methyl–3–butanone

Sol.



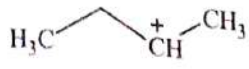
2,3-dimethylbut-1-ene

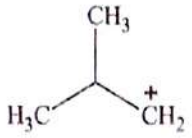
Hence, **(B)** is the correct answer.

Prob. 3. Dehydration of 1-butanol gives 2-butene as a major product, by which of the following intermediate the compound 2-butene is obtained.

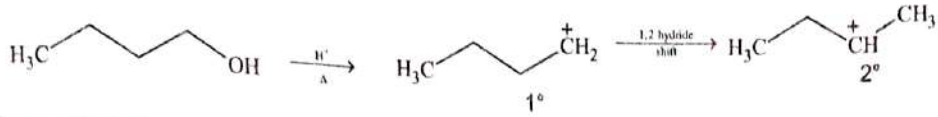
(A)

(B)

(C) 

(D) 

Sol.



Hence, **(C)** is the correct answer.

Prob. 4. Propyne, when treated with B_2H_6 followed by H_2O_2/OH^- gives

(A) Propan-1-ol
(B) Acetone
(C) Propen-1-ol
(D) Propanal

Sol. **(D)**

Prob. 5. But-2-ene reacts with diazomethane to give

(A) Hex-2-ene
(B) Hex-3-ene
(C) 1, 2-dimethyl cyclopropane
(D) 1, 2-dimethyl cyclopropane

Sol. **(C)**

Prob. 6. Hydrocarbons are obtained by electrolysis of

(A) potassium acetate
(B) potassium fumarate
(C) potassium succinate
(D) potassium formate

Sol. **(A,B,C)**

Prob. 7. Which alkenes give CH_3CHO on reductive ozonolysis

(A) Hex–2-ene
(B) But–2-ene
(C) Propene
(D) 1, 2-butadiene

Sol. **(A, B, C, D)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Alkene is more reactive than alkyne for electrophilic addition reactions.
and
STATEMENT-2: Alkyl carbocation is more stable than vinyl carbocation.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

2. **STATEMENT-1:** Melting point of alkyne is more than the corresponding alkane.
and
STATEMENT-2: Alkyne has symmetrical (rode like structure) and hence, compact structure.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

3. **STATEMENT-1:** 2-Methyl-1-Butene is more reactive than 1-butene for electrophilic addition reaction.
and
STATEMENT-2: 2-Methyl-1-butene has more α-hydrogens than 1-butene.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

4. **STATEMENT-1:** Alkynes give two type of reduction: total and partial.
and
STATEMENT-2: H_2/Pd—$BaSO_4$ converts non-terminal alkyne into *cis* alkene.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

5. **STATEMENT-1:** *cis*-2-butene gives (dl) mixture with bromine.
and
STATEMENT-2: Bromine gives *anti* addition reaction with alkene.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

COMPREHENSION

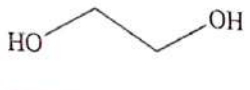
Read the paragraph carefully and answer the following questions:

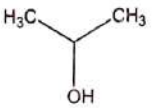
Alkenes can be oxidized in presence of different reagents to give different products. Alkenes on oxidation in presence of silver at high temperature give cyclic ethers, which also can be synthesized by using per acids. Alkenes on reductive ozonolysis give carbonyl compounds in presence of zinc and peroxide. On reaction with acidic $KMnO_4$, it also produces corresponding carbonyl compound. Alkynes can also be oxidized in presence of such reagents but the products are different.

1. Cyclohexene is allowed to react with ozone and subsequently with H_2O_2 and water. The product formed is
(A) oxalic acid (B) malonic acid (C) succinic acid (D) adipic acid

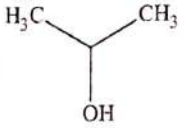
Sol. **(D)**

2. Ethene on reaction with m-CPBA in CH_2Cl_2 forms a compound (A), which on reaction with CH_3–MgI and subsequent hydrolysis gives a compound (B). (A) and (B) are, respectively

(A)  and

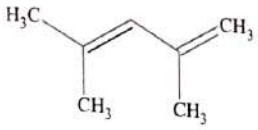
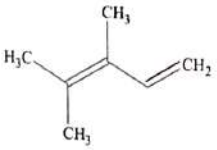
(B) and 

(C) and

(D) and 

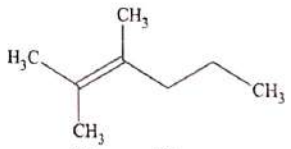
Sol. **(C)**

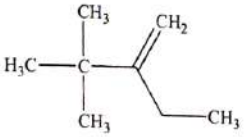
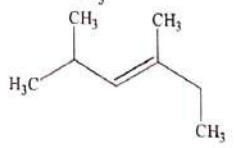
3. An open chain hydrocarbon (C_7H_{12}), on ozonolysis produces propanone, methanal and 2–oxopropanal. The hydrocarbon could be

(A)  (B)  (C) Both (D) None of the above

Sol. **(C)**

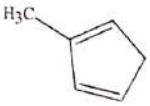
4. A hydrocarbon (C_8H_{16}), on oxidation with a hot acidified solution of $KMnO_4$ forms 2–butanone and isobutyric acid. The hydrocarbon is

(A) $(CH_3)_2C = C(CH_2CH_3)_2$ (B) 

(C)  (D) 

Sol. **(D)**

5. A hydrocarbon (C_6H_8) absorbs two moles of hydrogen in the presence of a platinum catalyst. The reduction product is inert to bromine and $KMnO_4$ solution. The hydrocarbon on ozonolysis produces malonic acid only. The structure of hydrocarbon is

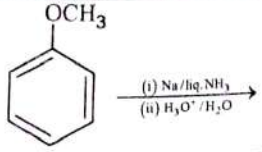
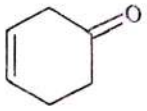
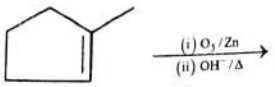
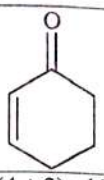
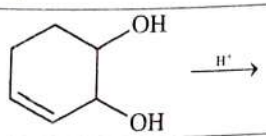
(A) $CH_2 = CH - CH = CH - CH = CH_2$ (B) 

(C) (D)

Sol. **(C)**

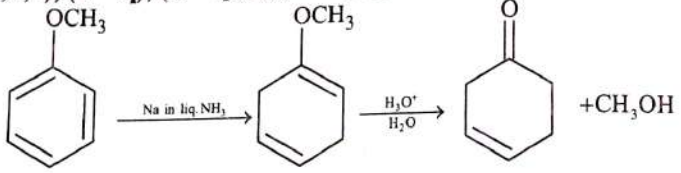
MATRIX-MATCH TYPE QUESTIONS

1. Match the following **Column I** with **Column II**.

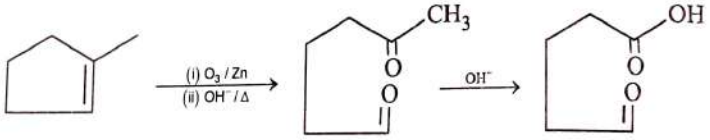
	Column – I (Reaction)		Column – II (Products)
(A)		(p)	
(B)		(q)	
(C)	$H_2C{=}CH{-}CH{=}CH_2 \xrightarrow[(ii)\ aq.\ KOH]{(i)\ CH_2{=}CBr_2/\Delta}$	(r)	(4 + 2) addition reaction
(D)		(s)	Birch reduction
		(t)	CH_3OH

Sol. **(A →p, s, t); (B →q); (C → p, r); (D → p, q)**

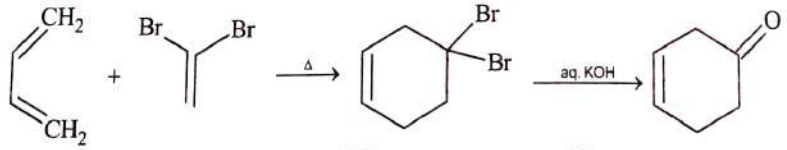
(A)



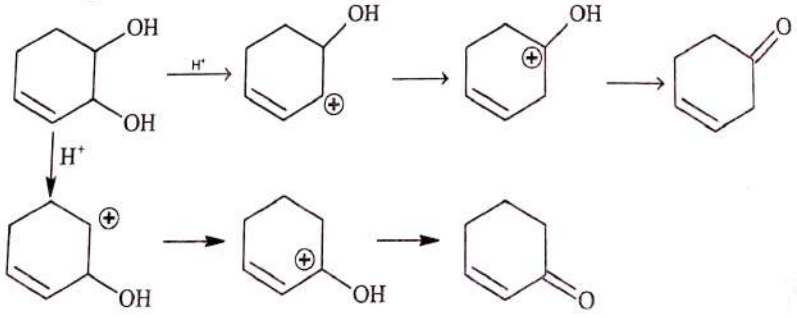
(B)



(C)



(D)

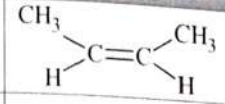
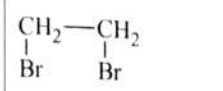
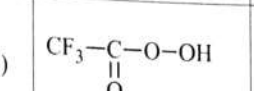
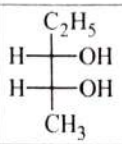
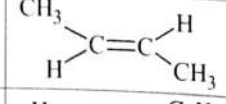
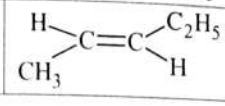
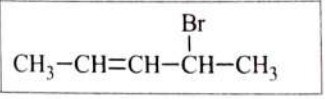


2. Match the following:

Column – I		Column – II	
(A)	$CH_3-CH=CH_2 \xrightarrow{HCl}$	P	Nucleophilic addition
(B)	$CH_3-CH=CH_2 \xrightarrow[h\nu]{NBS}$	Q	Free radical addition
(C)	$CH_3-CH=CH_2 \xrightarrow[Peroxide]{HBr}$	R	Electrophilic addition
(D)	$CH\equiv CH \xrightarrow{HgSO_4/HOH}$	S	Free radical substitution
		t	Carbocation intermediate

Sol. **A→ (r, t), B → (s), C→ (q), D→ (p)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)		(i)	Zinc metal	(P)	Anti-addition
(II)		(ii)		(Q)	
(III)		(iii)	Br_2/CCl_4	(R)	$CH_2=CH_2$
(IV)		(iv)	NBS	(S)	

3. Which of the following combination is correct ?
[A] (I) (iii) (P) [B] (II) (ii) (R) [C] (III) (iii) (Q) [D] (IV) (i) (S)

4. Which of the following combination is incorrect ?
[A] (II) (i) (R) [B] (III) (ii) (P) [C] (IV) (ii) (Q) [D] (IV) (iv) (R)

5. Which of the following combination is correct ?
[A] (IV) (iii) (R) [B] (II) (i) (R) [C] (I) (iv) (Q) [D] (II) (iii) (P)

Sol.

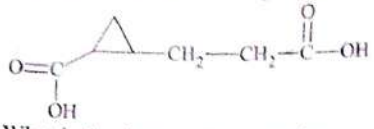
3. **A**

4. **D**

5. **B**

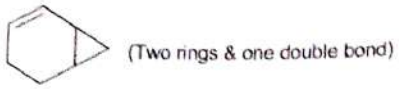
SINGLE INTEGER ANSWER TYPE

1. C_5H_{10} by oxidative ozonolysis form

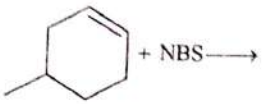


What is the degree of unsaturation.

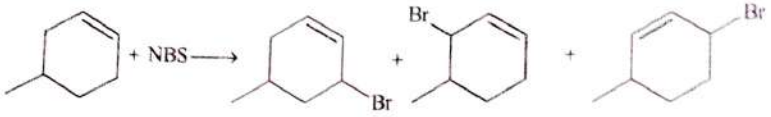
Sol. **(3)**



2. Number of products in the following reaction:

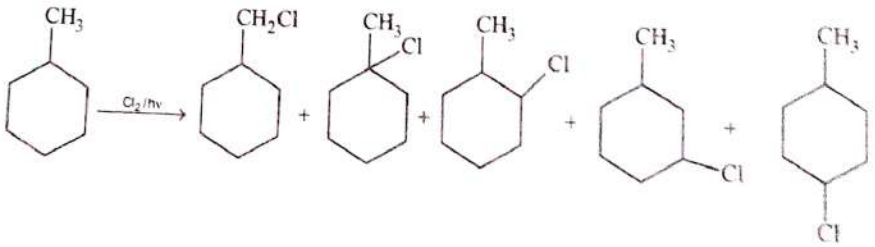


Sol. **(3)**

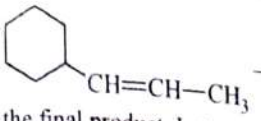


3. How many types of alkyl halides can be produced from monochlorination of A ?

Sol. **(5)**

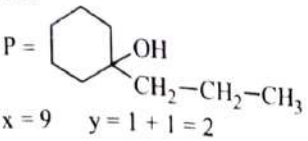


NUMERICAL VALUE ANSWER TYPE

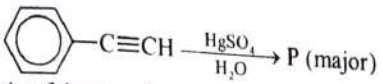
Prob. 1. 

In the final product, Let x = number of carbons and y = degree of unsaturation + number of –OH group then the value of $\frac{x}{y}$ is

Sol. **4.50**



x = 9 y = 1 + 1 = 2

Prob. 2. 

Ratio of degree of unsaturation in reactant/ product is

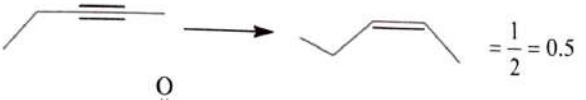
Sol. **1.20**

$\frac{6}{5} = 1.20$

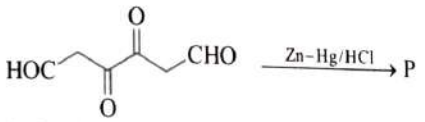
Prob. 3.

Ratio of π bonds in product/ reactants is

Sol. **0.5**

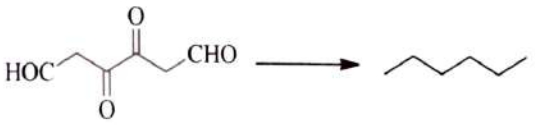


$= \frac{1}{2} = 0.5$

Prob. 4. 

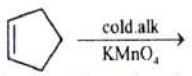
Calculate the ratio of number of carbon atoms in final product by moles of reducing agent utilised for complete reaction ?

Sol. **1.50**



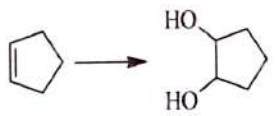
c-atom = 6

moles used = 4

Prob. 5. 

Calculate the ratio of number of hydroxyl groups in the product by number of 'C' atoms ?

Sol. **0.4**



$\frac{2}{5} = 0.4$

Prob. 6. $CH_3-\overset{CH_3}{\underset{|}{C}}=\overset{CH_3}{\underset{|}{C}}-CH_3$ (I) $\quad CH_3-\overset{CH_3}{\underset{|}{C}}=CH-CH_3$ (II) $\quad CH_3-CH=CH_2$ (III) $\quad CH_2=CH_2$ (IV)

Ratio of the most stable compound to most unstable compound out of the following is ?

Sol. **0.25**

Most stable → (I)

Most unstable → (IV)

$\therefore \frac{1}{4} = 0.25$

Prob. 7. $HOOC-(CH_2)_4-COOH \xrightarrow[\substack{CaO \\ \Delta(1eq.)}]{NaOH/} P$

Ratio of number of carboxylic group in product by reactant is :

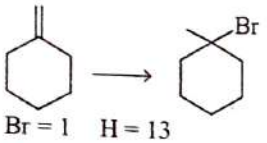
Sol. **0.5**

$COOH-(CH_2)_4-COOH \longrightarrow COOH-(CH_2)_3-CH_3$

Prob. 8. $\xrightarrow{HBr}$ Major product

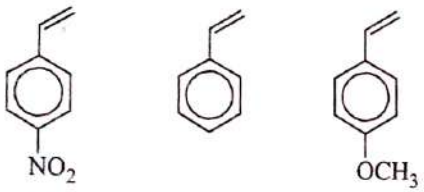
Ratio of bromine by hydrogen atom in product is :

Sol. **0.076**



Br = 1 H = 13

$\frac{1}{13} = 0.076$

Prob. 9. 

(I) (II) (III)

Ratio of least reactive by most reactive compound towards HCl is :

Sol. **0.33**

$\frac{1}{3} = 0.33$

Prob.10. $\underset{3}{CH_3}-\underset{2}{CH_2}=\underset{1}{CH_2} \xrightarrow[\text{Peroxide}]{HBr}$

Ratio of position of Br in product by total number of carbon is ?

Sol. **0.33**

$CH_3-CH_2=CH_2 \longrightarrow \underset{3}{CH_3}-\underset{2}{CH_2}-\underset{1}{CH_2Br}$

$\frac{1}{3} = 0.33$

ALKYL AND ARYL HALIDES 15

ALKYL HALIDE

General formula: $C_nH_{2n+1}X$

Methods of Preparation

(i) **From alcohol:**

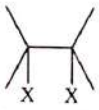
(a) $\text{Alcohol} \xrightarrow{PCl_5/PCl_3/SOCl_2} R-Cl$

(b) $\text{Alcohol} \xrightarrow{P+I_2} R-I$

(c) $\text{Alcohol} \xrightarrow[ZnCl_2\text{ (anhydrous)}]{HX} R-X$

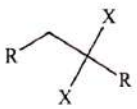
(ii) **From alkene:**

(a) $\text{Alkene} \xrightarrow{HX} R-X$

(b) $\text{Alkene} \xrightarrow{X_2/CCl_4}$ 

vicinal dihalide

(iii) **From alkyne:**

$\text{Alkyne} \xrightarrow{H-X/CCl_4}$ 

gem dihalide

(iv) **Halide exchange:**

$\text{Halide exchange} \xrightarrow[\text{acetone}]{NaI} R-I$

(v) **Silver salt of acid:**

$\text{Silver salt of acid} \xrightarrow[\text{(Hunsdiecker reaction)}]{Br_2/CCl_4} R-Br$

(vi) **From alkane:**

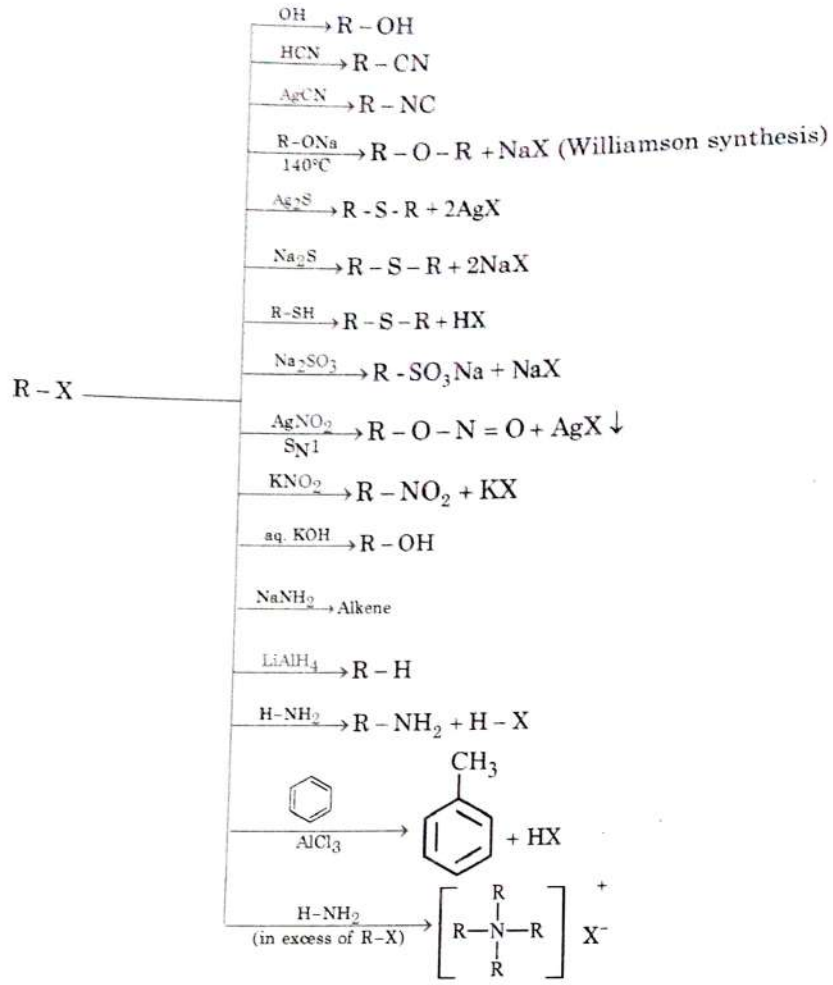
$\text{Alkane} \xrightarrow{X_2/h\nu \text{ or } X_2 \text{ in excess}} R-X$

Chemical Properties

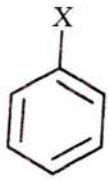
Nucleophilic substitution reaction:

$R-X + Nu^- \longrightarrow R-Nu + X^-$

Reactivity of $R-X$: $R-I > R-Br > R-Cl > R-F$

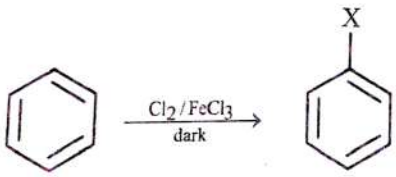


ARYL HALIDE

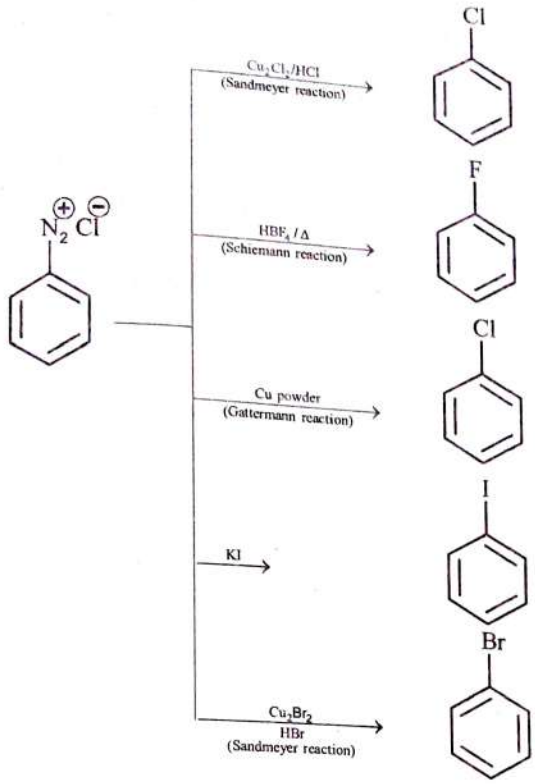


Methods of Preparation

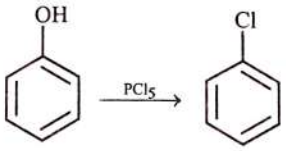
(i) **Halogenation of benzene:**



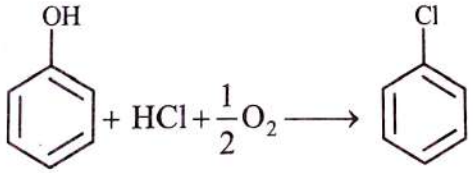
(ii) **From benzene diazonium chloride:**



(iii) **From phenol:**



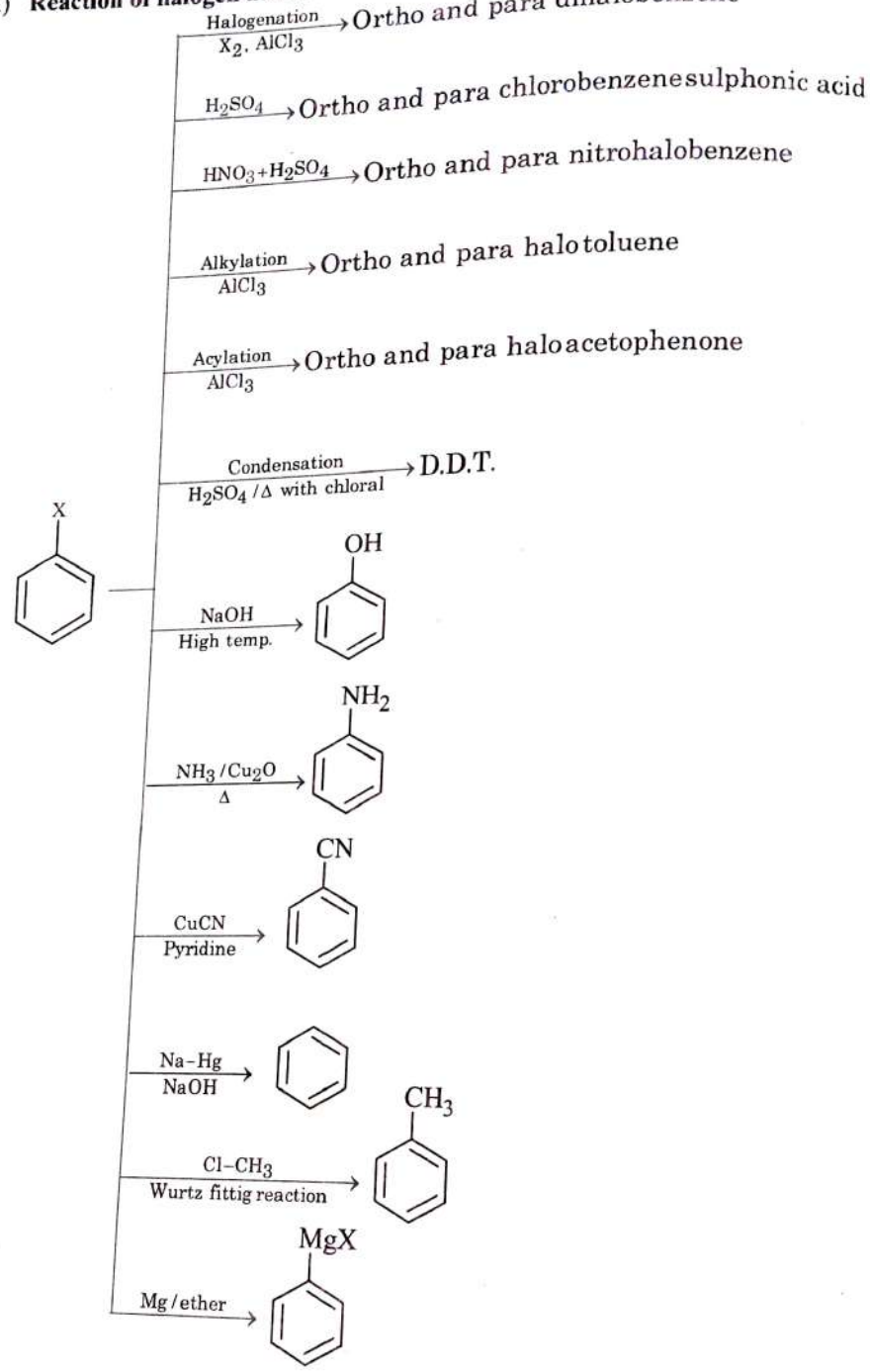
(iv) **Raschig's method:**



Chemical Properties

[–I effect] [+M effect]

(A) **Reaction of halogen atom:**



ILLUSTRATIVE PROBLEMS

1. $CH_3—CH_2—Cl$ gives nucleophilic substitution reaction slowly with water. This reaction is accelerated in the presence of small amount of KI. Explain.

Sol. HOH is weak nucleophile and Cl^- is not a good leaving group. So the nucleophilic substitution of $CH_3—CH_2—Cl$ with water will be slow.

I^- (from KI) is a powerful nucleophile it displaces Cl to form $CH_3—CH_2—I$. Now, I^- being a good leaving group. H_2O can displace it rapidly.

$$CH_3—CH_2—Cl \xrightarrow{I^-} CH_3—CH_2—I \xrightarrow{HOH} CH_3—CH_2—OH$$

2. Explain why aryl halides and vinyl halides do not give insoluble precipitate of silver halide when these are treated with alcoholic silver nitrate solution.

Sol. Organic halides give AgX by SN reaction.

$$R—X + AgNO_3 \rightarrow R—NO_3 + AgX\downarrow$$

The above reaction is ionic reaction. Therefore, if requires the formation of halide. The C—X bond of aryl halides as well as that of vinyl halides are very strong and hence, very difficult to break heterolytically because of the following reasons:

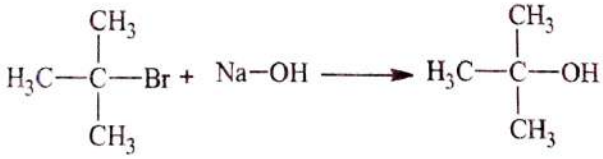
(i) The C—X bond is formed by sp^2-p overlap. This makes the bond short and strong.

(ii) the p-orbital electrons of the halogen atom get delocalized which gives the C—X bond some double bond character as shown below:

Since, halide ions do not form the said compounds, they do not precipitate silver halides.

3. The rate of hydrolysis of tertiary butyl bromide in 50% aqueous methanol is thousand times faster than that in pure methanol. Explain.

Sol.



The above reaction follows S_N1 pathway. We know that rate of an SN1 reaction increases with increasing ionizing power of the solvent. Methanol has a dielectric constant 33 and that of water is 80. Thus, water is a better ionizing solvent than methanol. So, the rate of hydrolysis of tertiary butyl bromide increases largely when pure methanol is replaced by 50% aqueous NaOH.

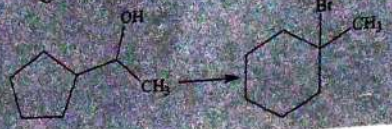
4. Carry out the following conversions:

(a)

(b)

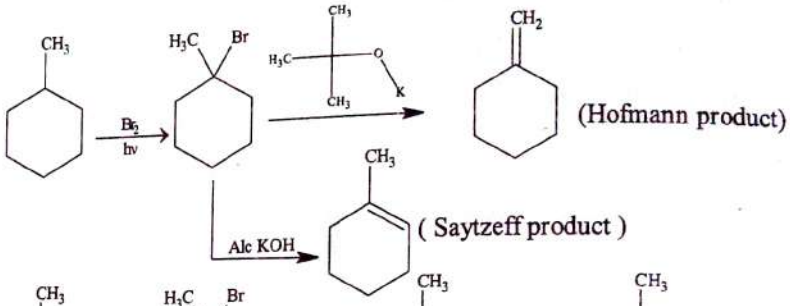
(c)

(d)

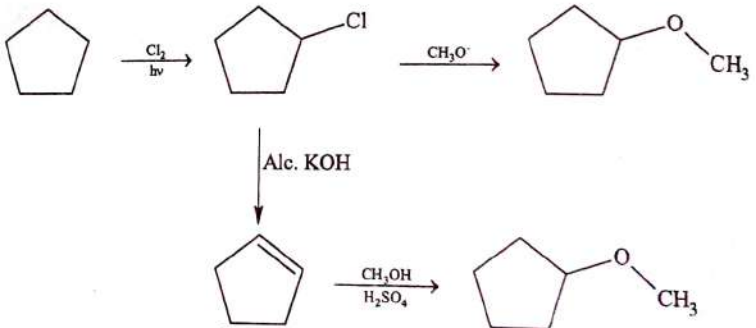


Sol.

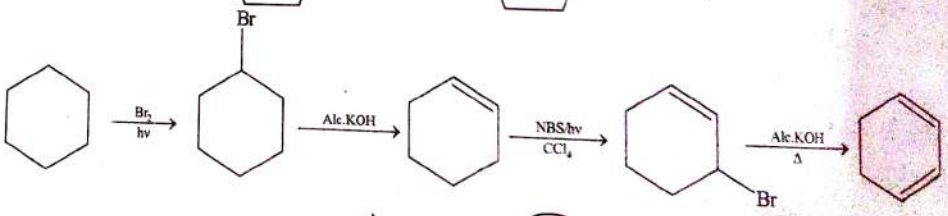
(a)



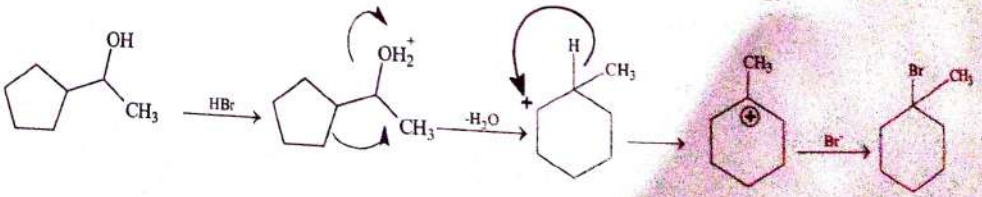
(b)



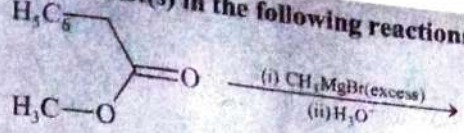
(c)

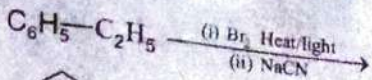


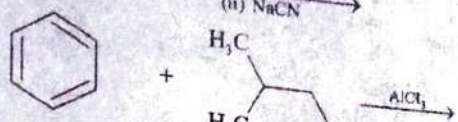
(d)

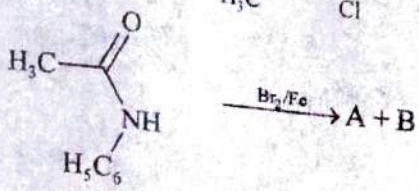


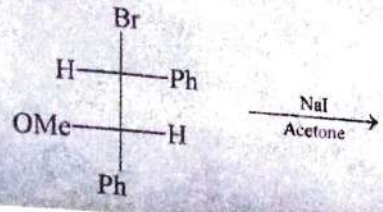
5. **Predict the product(s) in the following reactions:**

(i) 

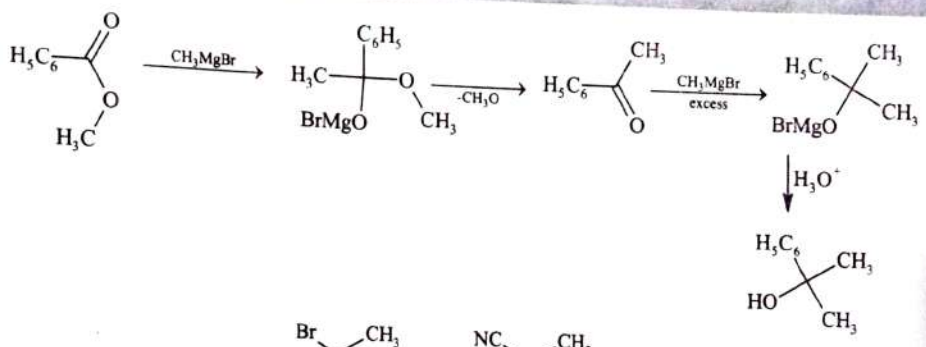
(ii) 

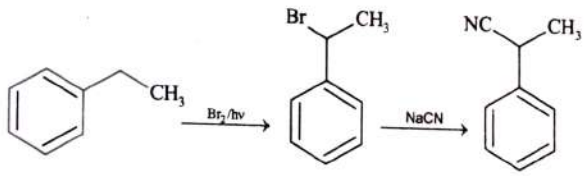
(iii) 

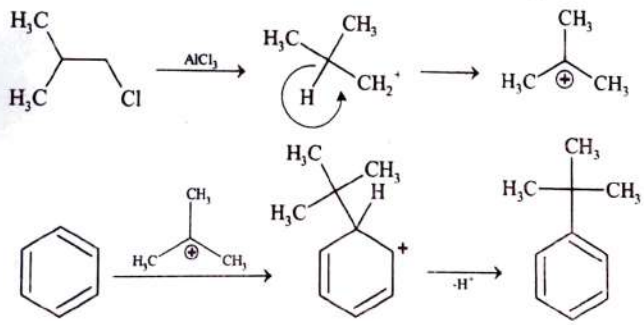
(iv) 

(v) 

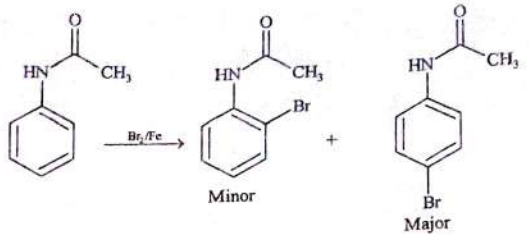
Sol.

(i) 

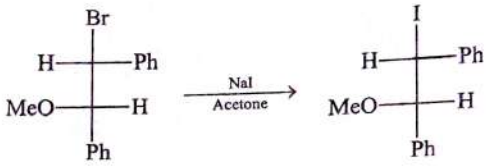
(ii) 

(iii) 

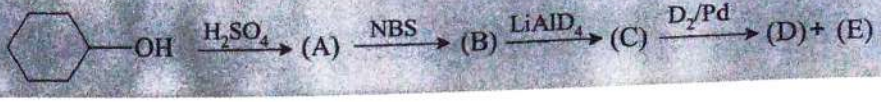
(iv)



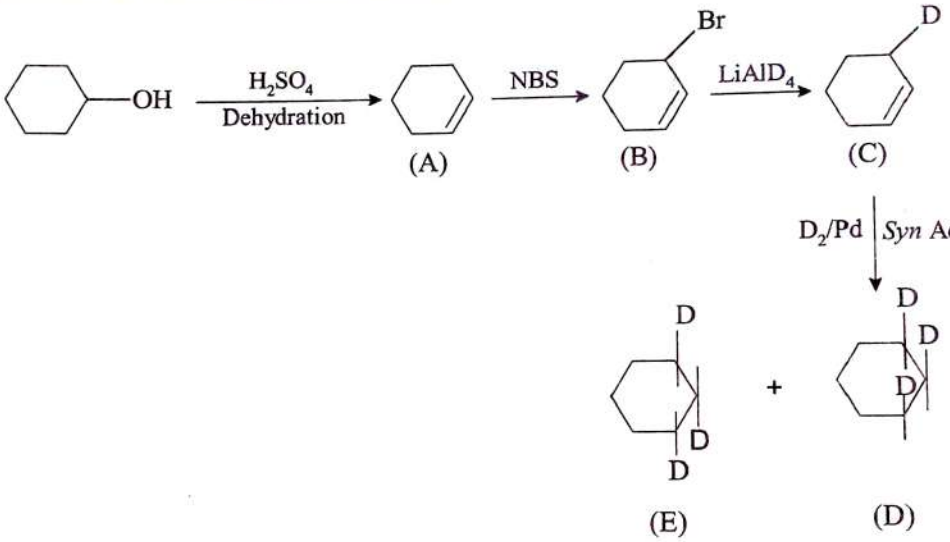
(v)



6. Give structure of (A) to (E) in the sequence of reactions.



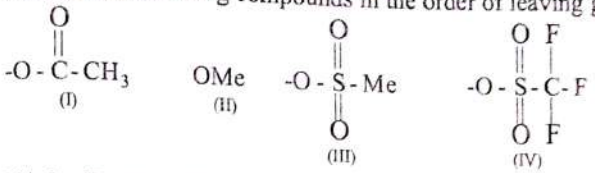
Sol.



SOLVED PROBLEMS

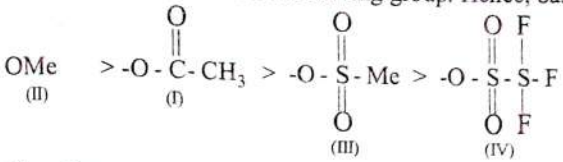
Objective Questions

Prob. 1. Arrange the following compounds in the order of leaving group ability:



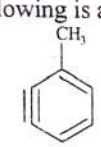
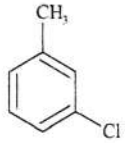
(A) I > II > III > IV
(B) IV > III > I > II
(C) III > II > I > IV
(D) II > III > IV > I

Sol. Weaker the base, better is the leaving group. Hence, basicities decrease in the order



Therefore, the order of leaving group ability decreases in the reverse order (IV) > (III) > (I) > (II).
Hence, **(B)** is the correct answer.

Prob. 2. o–Chlorotoluene reacts with sodamide in liquid NH_3 to give o–toluidine and m–toluidine. Which of the following is an intermediate in the above reaction?

(A) 
(B)
(C) 
(D) None of the above

Sol. The formation of a mixture of o–and m–toluidine suggests that triple bond in benzyne is between o– and m– position.
Hence **(A)** is the correct answer.

Prob. 3. Which of the following will give yellow precipitate with I_2/NaOH?

(A) $CH_3 - CO - O - CO - CH_3$
(B) $I - CH_2 - \overset{\overset{O}{\|}}{C} - CH_2 - CH_3$
(C) $CH_3 - CO - NH_2$
(D) $CH_3 - \underset{\underset{OH}{|}}{CH} - CH_2 - CH_3$

Sol. $ICH_2 - CO - CH_2 - CH_3$ on further treatment with I_2/NaOH first give $I_2CHCOCH_2CH_3$ and then $I_3C - CO - CH_2 - CH_3$ which subsequently undergoes hydrolysis with NaOH to yield CHI_3 and $CH_3 - CH_2 - COONa$.

$CH_3 - \underset{\underset{OH}{|}}{C} - CH_2 - CH_3$ being methyl carbinol on treatment with I_2/NaOH will also give CHI_3.

Hence, **(B)** and **(D)** are the correct answers.

Prob. 4. $C_2H_4Cl_2 \xrightarrow{KCN} (B) \xrightarrow{H_3O^+} (C) \xrightarrow{\Delta} (D)$

(A)

The compound D has the characteristic(s)

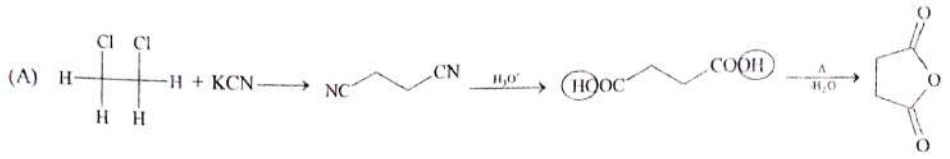
(A) It is succinic anhydride if (A) is vic dihalide

(B) It is propionic acid if (A) is gem dihalide

(C) It is succinic anhydride if (A) is gem dihalide

(D) It is propionic acid if (A) is vic dihalide

Sol.



Succinic anhydride

(B) $H_3C-CHCl_2 \xrightarrow{KCN} H_3C-CH(CN)_2 \xrightarrow{H_3O^+} H_3C-CH(COOH)_2 \xrightarrow[\text{decarboxylation}]{\Delta} CH_3CH_2COOH$

Hence, **(A)** and **(B)** are the correct answers.

Prob. 5. Dipole moment is shown by

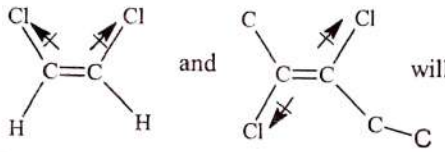
(A) 1, 4-dichlorobenzene

(B) cis-2, 3-dichloroethene

(C) trans-1, 2-dichloroethene

(D) trans-1, 2-dichloro-2-pentene

Sol.

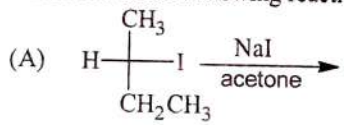


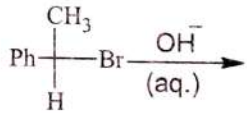
and

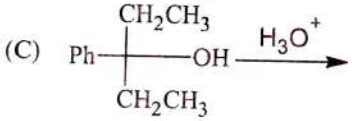
will show dipole moment.

Hence, **(B)** and **(D)** are the correct answers.

Prob.6. Which one of the following reactions leads to formation of optically inactive mixture ?

(A) 

(B) 

(C) 

(D) both (A) & (C)

Sol. **(D)**

Prob.7. The correct order of nucleophilicity in weakly polar aprotic solvents is

(A) $F^- < Cl^- < Br^- < I^-$

(B) $F^- > Cl^- > Br^- > I^-$

(C) $F^- < Br^- < Cl^- < I^-$

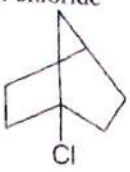
(D) $F^- < I^- < Br^- < Cl^-$

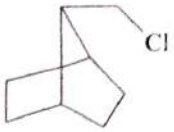
Sol. **(A)**

Prob.8. Which of the following is inert towards S_N1 reaction ?

(A) Ethyl chloride

(B) Isopropyl chloride

(C) 

(D) 

Sol. **(C)**

Prob.9. The reactivity of the following halides towards aqueous solution of KOH is:
CH_3CH_2Cl (I), $CH_2 = CH - CH_2Cl$ (II), $CH_2 = CHCl$ (III), C_6H_5Cl (IV)

(A) I < II < III < IV
(B) I > II > III > IV
(C) II > I > III > IV
(D) I < III < IV < II

Sol. **(C)**

Prob.10. Which statement is correct for an S_N1 reaction at chiral carbon ?

(A) The product will be optically active, but have the opposite configuration.
(B) The reaction will involve racemization and partial inversion.
(C) The rate of reaction is a function of concentration of nucleophile.
(D) The attacking group will be strong electrophile.

Sol. **(B)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Optically active alkyl halide gives enatiomeric pair in SN1 reaction.
and
STATEMENT-2: Alkyl carbocation has planar geometry.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

2. **STATEMENT-1:** $CH_3—CH_2—O—CH_2—Cl$ reacts faster with water than $CH_3—CH_2—O—CH_2—CH_2—Cl$ does.
and
STATEMENT-2: Carbonium ion of $CH_3—CH_2—O—CH_2Cl$ is more stable than the carbonium ion of $CH_3—CH_2—O—CH_2—CH_2—Cl$.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

3. **STATEMENT-1:** Neopentyl chloride always gives SN2 reaction.
and
STATEMENT-2: Neopentyl chloride is primary alkyl halide.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

4. **STATEMENT-1:** SN1 reaction is carried out in the presence of polar protic solvents.
and
STATEMENT-2: Polar protic solvents increases stability of carbocation due to solvation.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

5. **STATEMENT-1:** Treatment of 2-chloro butane with potassium tert butoxide produces 1-butene as the main product.

and

STATEMENT-2: Bulky bases always gives Hofmann product in E2 reactions.

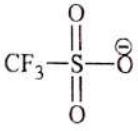
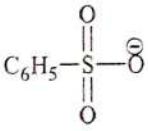
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

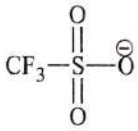
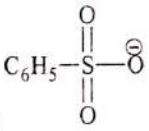
(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

6. **STATEMENT-1:**  is better leaving group than 

and

STATEMENT-2:  is weaker base than 

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

7. **STATEMENT-1:** 2-Pentanone can be distinguished from 3-pentanone by I_2/NaOH.

and

STATEMENT-2: Iodoform test is given by methyl ketones.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

COMPREHENSION

In S_N1 reaction of (S) – 2 bromobutane with water, two substitution products are formed, one has the same relative configuration as the reactant and the other has the inverted configuration. This is because in such reactions, the leaving group departs before the nucleophile attack.
In most S_N1 reactions, the racemic product is non – 50 : 50 mixture, i.e. partial racemization takes place, the inverted product is more than 50%. The formation of different composition of the product is due to following steps involved in S_N reactions.

$R-X$	$\rightarrow R^+X^- \rightarrow R^+$	solvent X^-	$\rightarrow R^+ + X^-$
Undissociated	Intermediate	solvent – separated	dissociated
Molecule	ion pair	ion pair	ions
I	II	III	IV

1. Which type of intermediate is formed in the reaction of (S)-2-bromobutane with water?
   (A) sp^3 hybridized
   (B) sp^2 hybridized
   (C) sp hybridized
   (D) A transition state

Sol. **(B)**

2. 50 : 50 racemic mixture is formed due to the attack of nucleophile on which intermediate species of the reaction?
   (A) I
   (B) II
   (C) III
   (D) IV

Sol. **(D)**

3. The non-50 : 50 racemic mixture during S_N reaction of an alkyl halide is formed due to the attack of the nucleophile on species
   (A) I
   (B) II
   (C) III
   (D) both II and III

Sol. **(D)**

4. In the above reaction, the non-50 : 50 racemic mixture contains more amount of
   (A) original configuration
   (B) inverted configuration
   (C) either of the two
   (D) none of the above

Sol. **(B)**

1. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	E_1	(p)	Carbocations
(B)	E_2	(q)	Strong nucleophile
(C)	S_N1	(r)	Reactivity of a alkyl halides $1^o > 2^o > 3^o$
(D)	S_N2	(s)	Transition state (TS)
		(t)	Reactivity of alkyl halides $3^o > 2^o > 1^o$

Sol. **A → (p, t) B → (s, t) C → (p, t) D → (q, s, r)**

2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	E_1cb	(p)	First order
(B)	S_N2	(q)	No rearrangement
(C)	E_1	(r)	Rearrangement
(D)	S_N1	(s)	Strong nucleophile
		(t)	Inversion of configuration

Sol. **A → (p, q) B → (q, s, t) C → (p, r) D → (p, r)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	CH_3CH_2Br	(i)	alc. KOH/ Δ	(P)	$CH_3CH_2CH_2CH_3$
(II)	$CF_3—CHCl_2$	(ii)	Na, dry ether	(Q)	$CF_2=CCl_2$
(III)	$CH_3-\underset{\displaystyle Br}{\underset{\vert}{CH}}-CH_2-CH_3$	(iii)	$H^{\oplus}/\Delta$	(R)	$CH_3CH_2CH_3$
(IV)	$CH_3-\overset{\displaystyle CH_3}{\overset{\vert}{\underset{\vert}{C}}}-OH$ (with CH_3 below)	(iv)	H_2, Pd, Δ	(S)	$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{C}}=CH_2$

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (I) (ii) (Q) [C] (III) (iii) (P) [D] (I) (ii) (R)

4. Which of the following combination is correct ?
[A] (I) (iii) (Q) [B] (II) (i) (Q) [C] (III) (iv) (P) [D] (IV) (iii) (S)

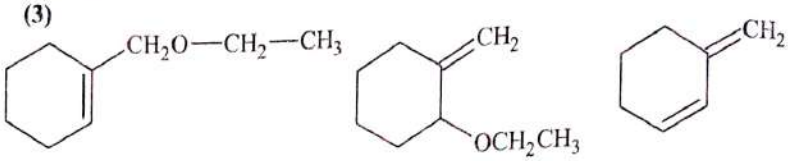
5. Which of the following combination is correct ?
[A] (III) (iii) (P) [B] (II) (i) (Q) [C] (III) (iv) (Q) [D] (II) (i) (P)

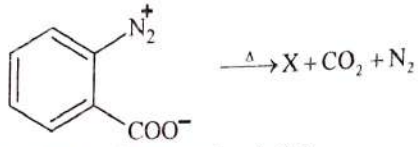
Sol.
3. **D**
4. **A**
5. **B**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. When 1-bromomethylcyclohexene under-goes solvolysis in ethanol, what will be the number of major products.

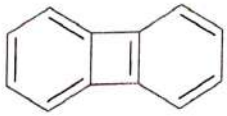
Sol. **(3)**



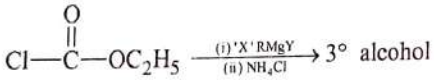
2. 

The degree of unsaturation in X is.

Sol. **(9)**

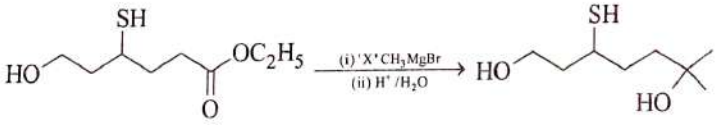
'X' is 

NUMERICAL VALUE ANSWER TYPE

Prob. 1. 

Find out the value of $\frac{x}{2}$ is

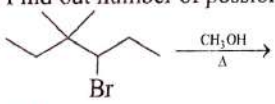
Sol. **1.5**

Prob. 2. 

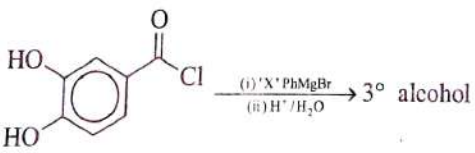
Find out the value of $\frac{x}{6}$ is

Sol. **0.666**

Prob. 3. Find out number of possible E_1 product is x in the following react. Find the value of $0.23\,x$.



Sol. **0.92**

Prob. 4. 

Find the value of $\frac{x}{7}$ is

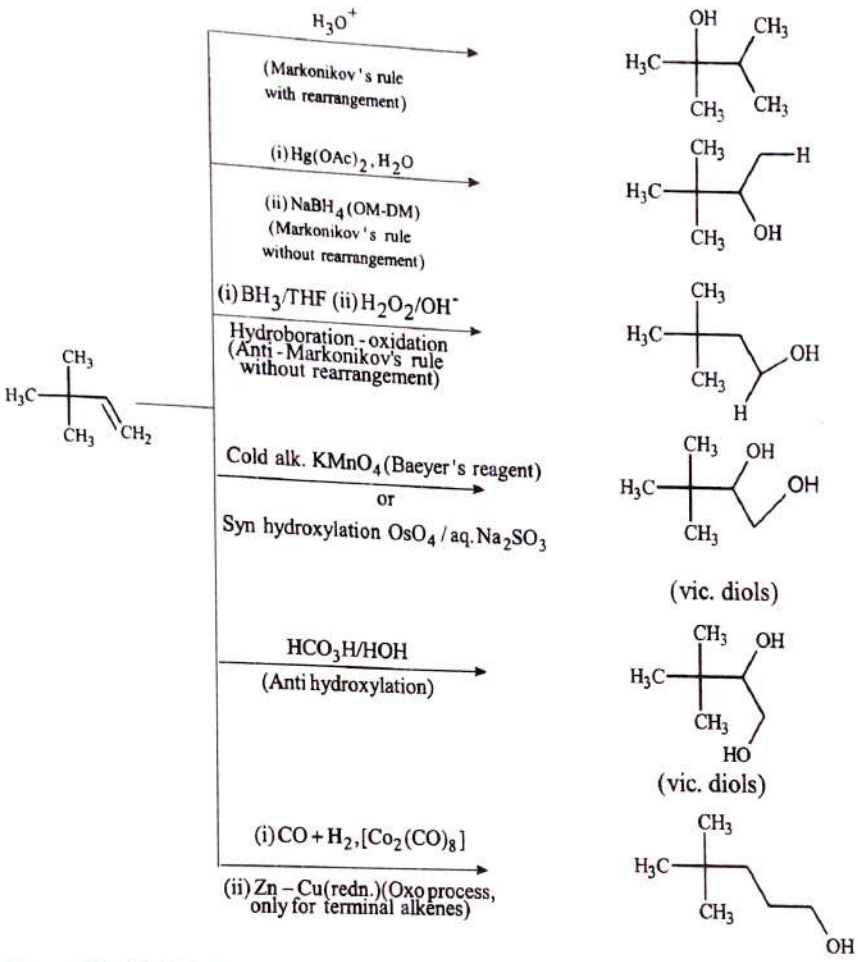
Sol. **0.571**

ALCOHOL, ETHER AND PHENOL 16

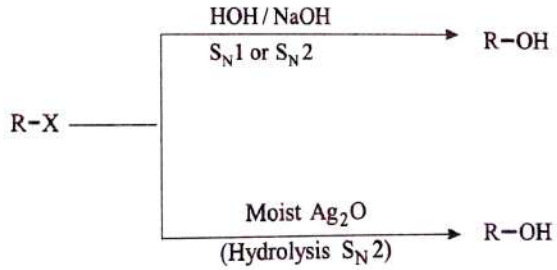
ALCOHOLS

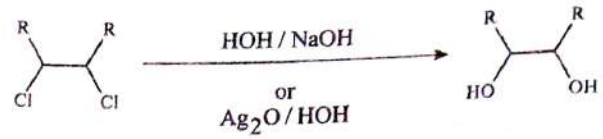
Methods of Preparation

(i) **From alkenes:**

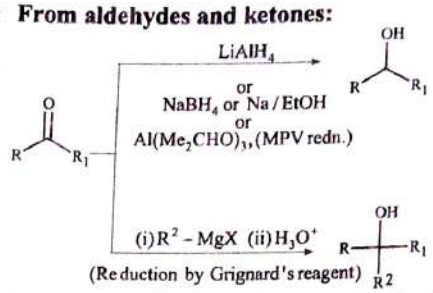


(ii) **From alkyl halides:**

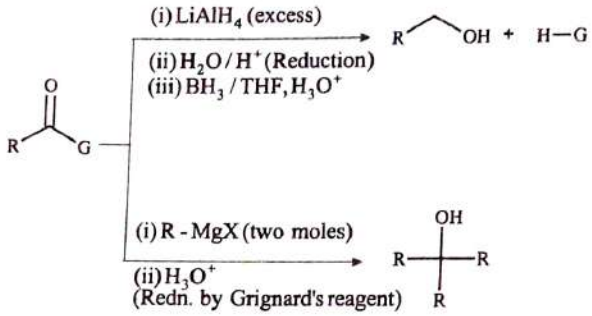


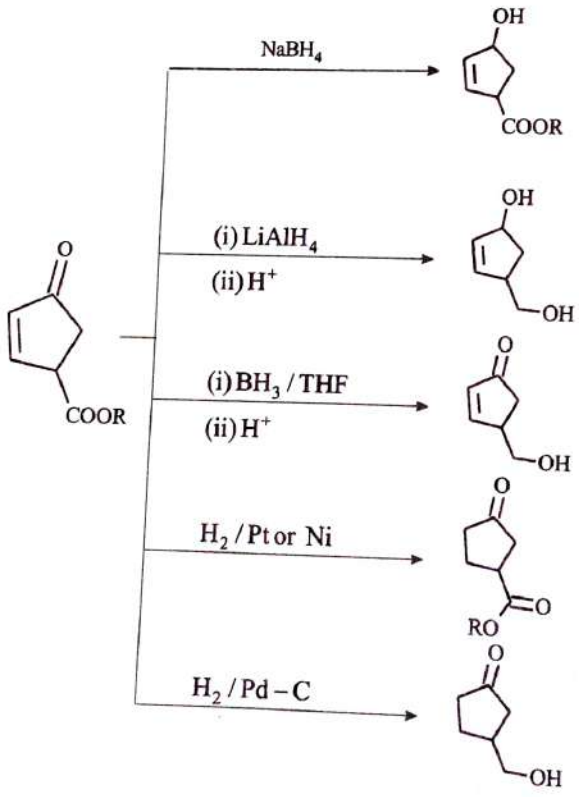


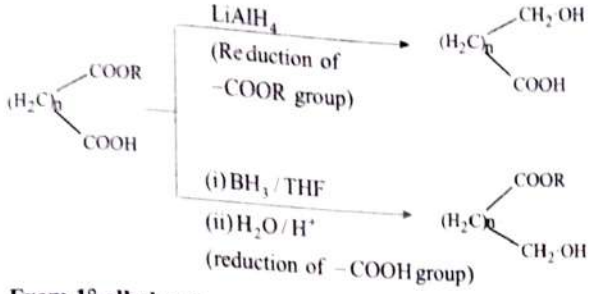
(iii) **From aldehydes and ketones:**



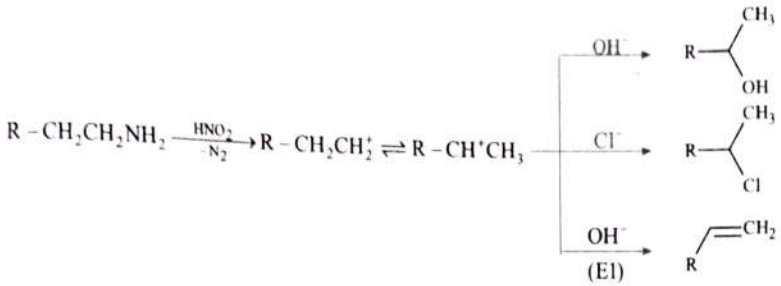
(iv) **From carboxylic acids and their derivatives:**



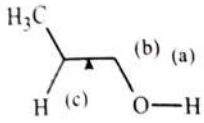




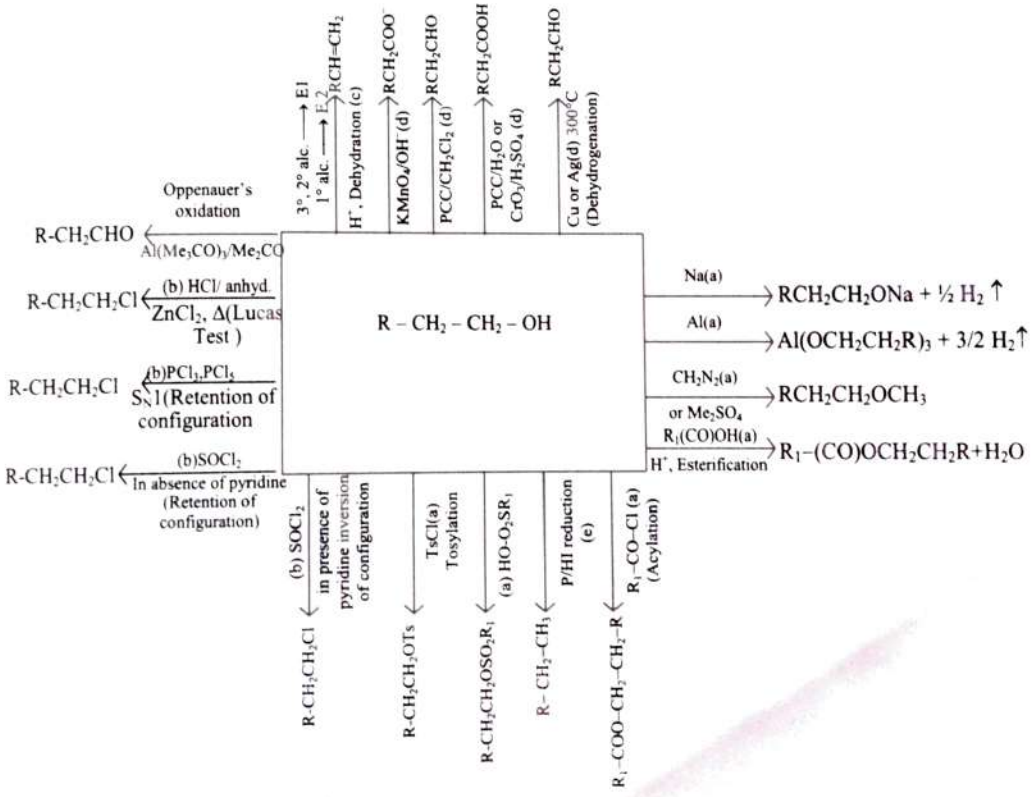
(v) **From 1° alkyl amines:**

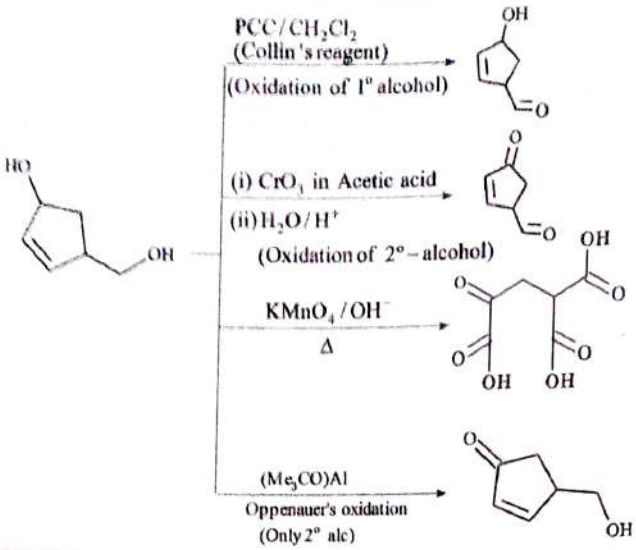


Chemical Properties

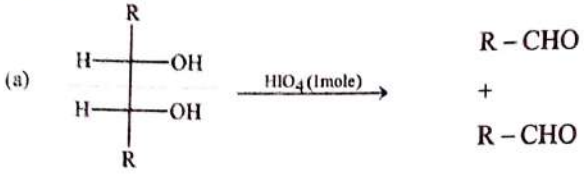


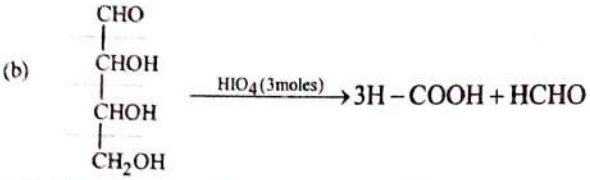
(a) Acidic properties of alcohol
(b) S_N reaction of alcohols
(c) Dehydration of alcohols
(d) Oxidation of alcohols
(e) Reduction of alcohols



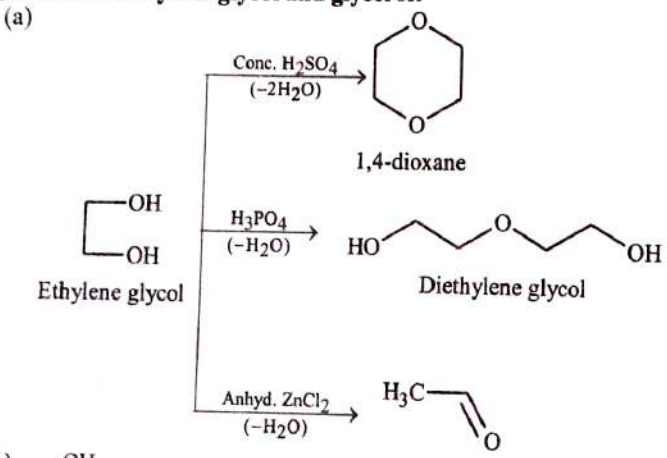


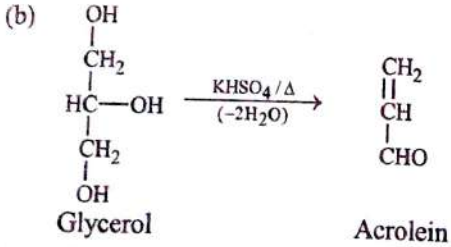
Oxidation:





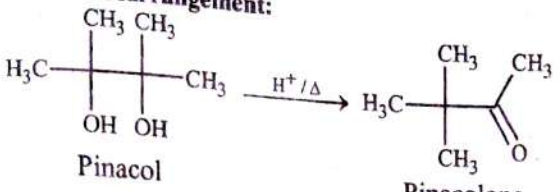
Dehydration of ethylene glycol and glycerol:





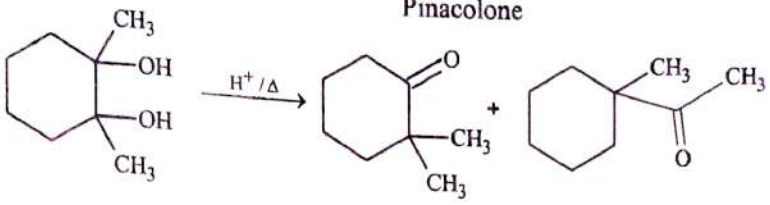
Pinacol Pinacolone rearrangement:

(a)



Pinacol Pinacolone

(b)

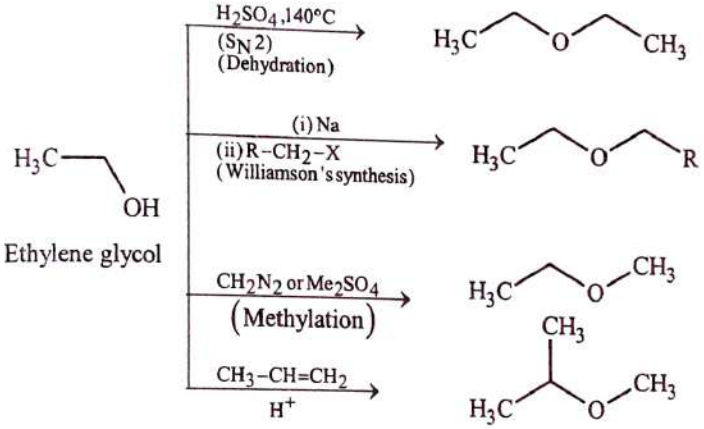


ETHER

It is referred as alkyl derivative of alcohols.

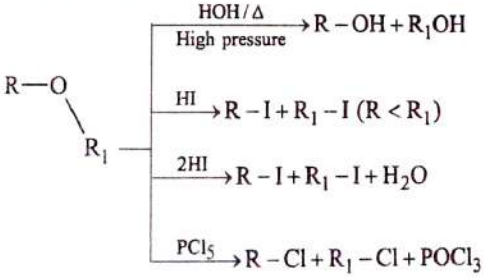
Methods of Preparation

From alcohols:

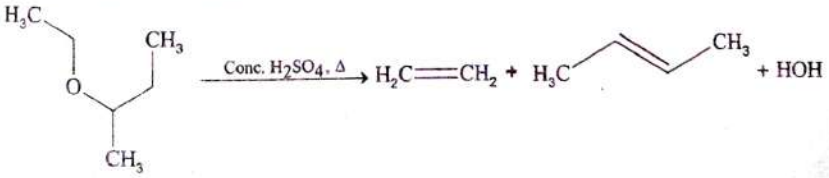


Chemical Properties

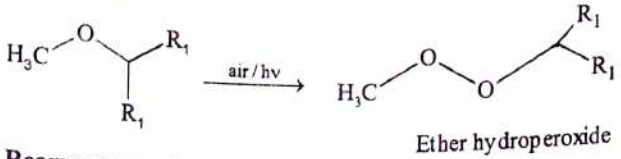
SN reactions:



Dehydration reactions:

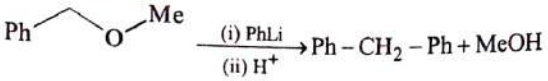


Oxidation:



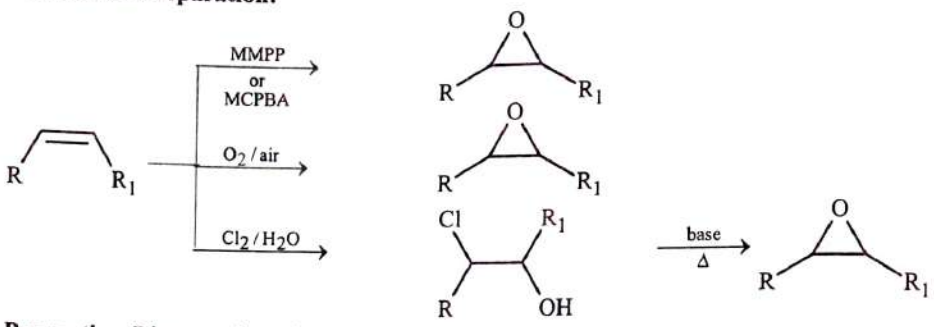
Ether hydroperoxide

Rearrangement:

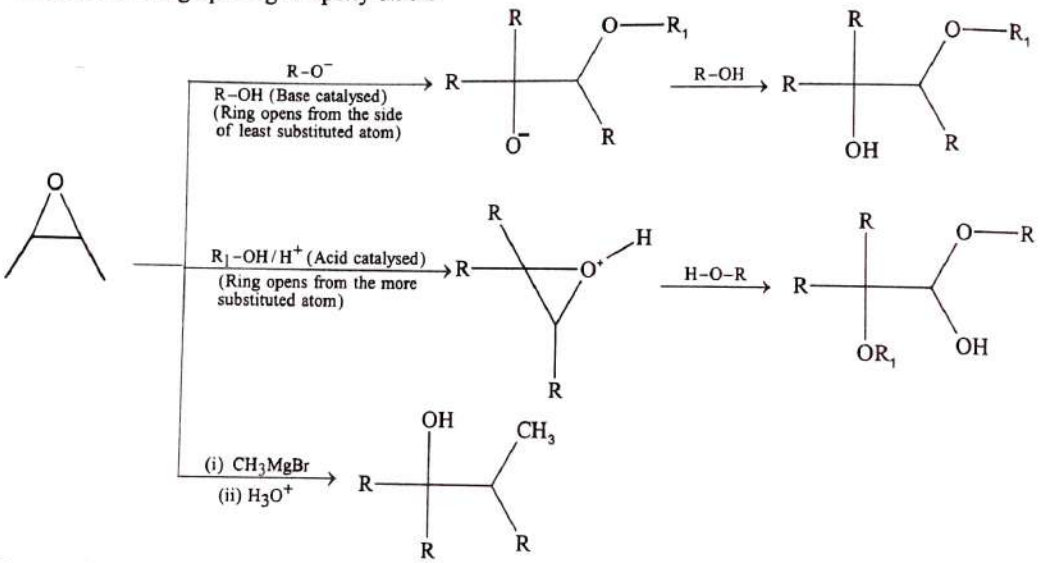


Cyclic Ethers (Oxiranes)

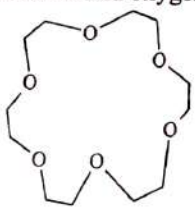
Methods of Preparation:



Properties: Ring opening of epoxy ethers



Crown ethers are cyclic polymers of ethylene glycols, e.g. 18-crown-6 (here 18 denotes the sum of total no. of carbon and oxygen atoms in the ring while 6 denotes no. of oxygen atoms in the ring)

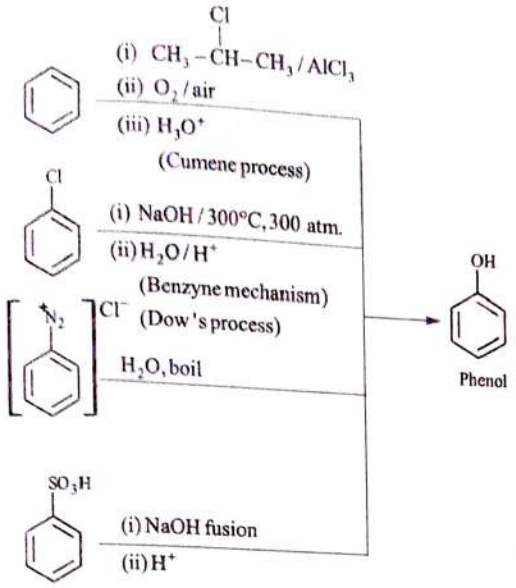


18 – crown – 6

PHENOLS

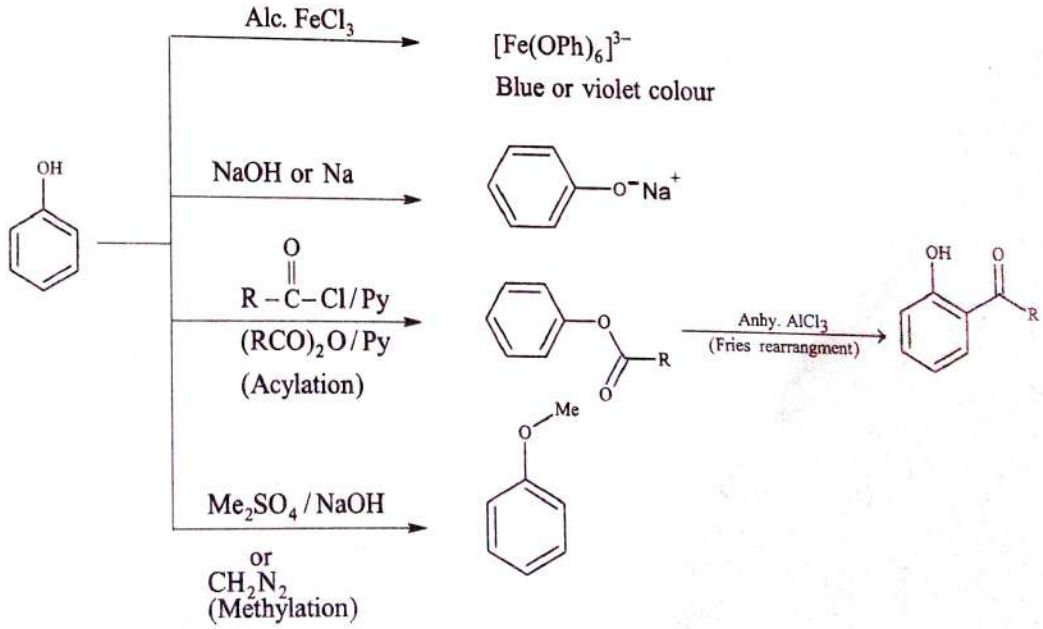
Compounds in which –OH group is attached to sp^2 hybrid carbon atom of aromatic ring.

Methods of Preparation

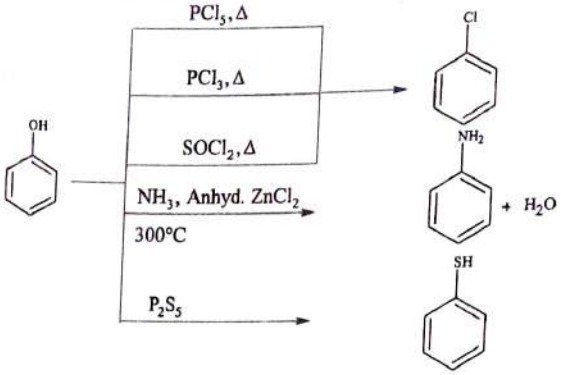


Chemical Properties

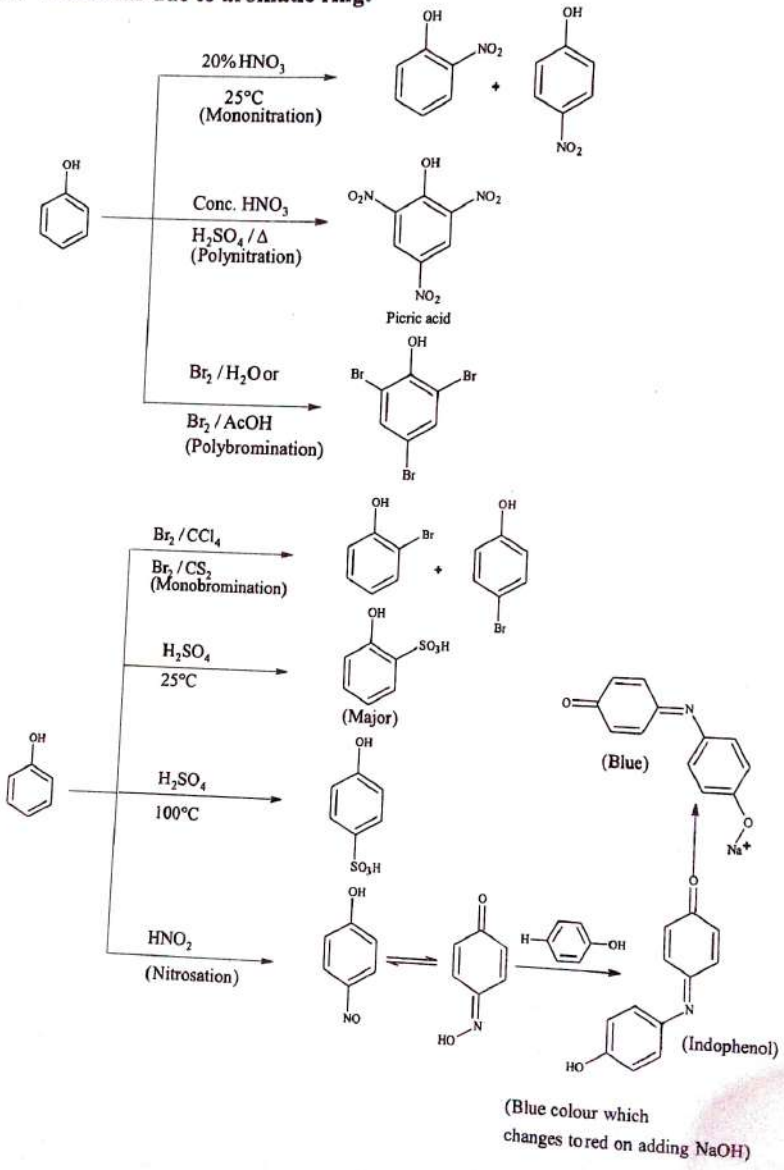
(i) **Reactions due to cleavage of O–H bond:**

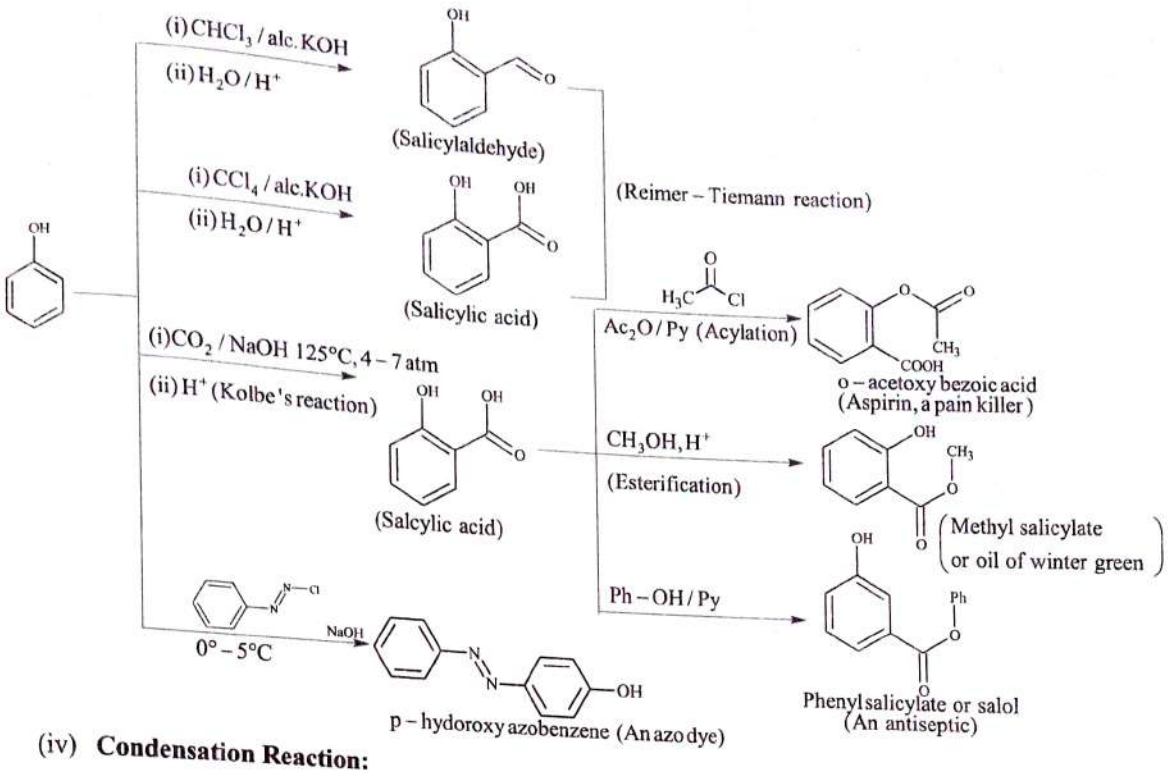


(ii) **Reactions due to cleavage of C–O bond:**

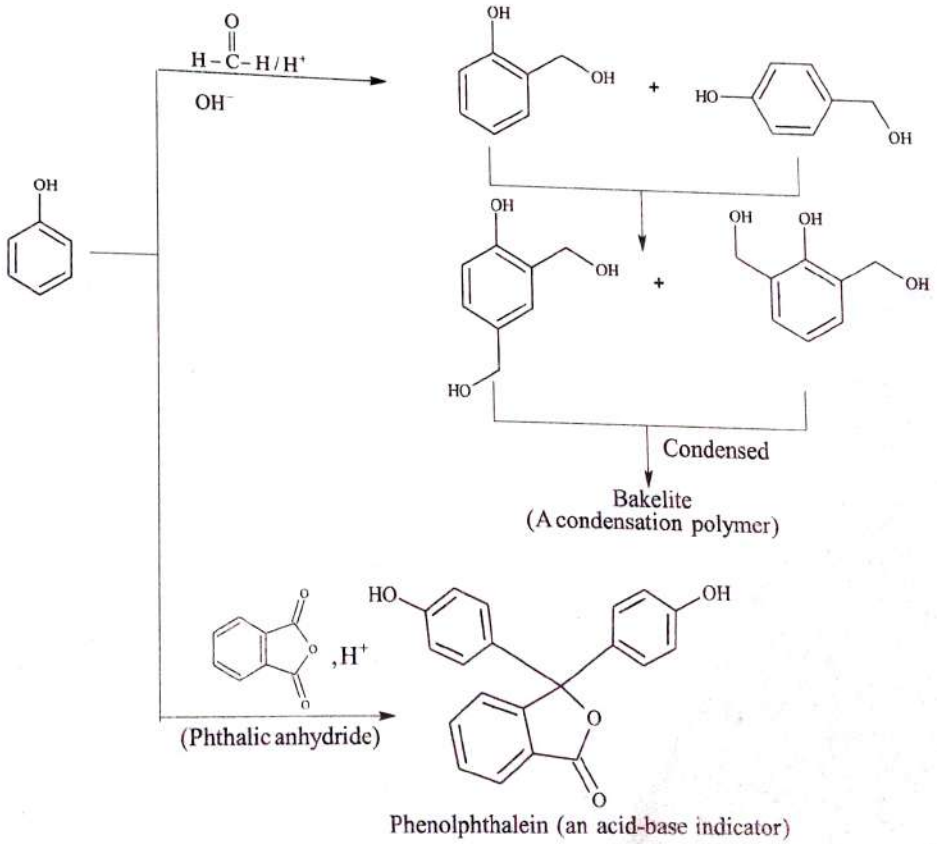


(iii) **Reactions due to aromatic ring:**

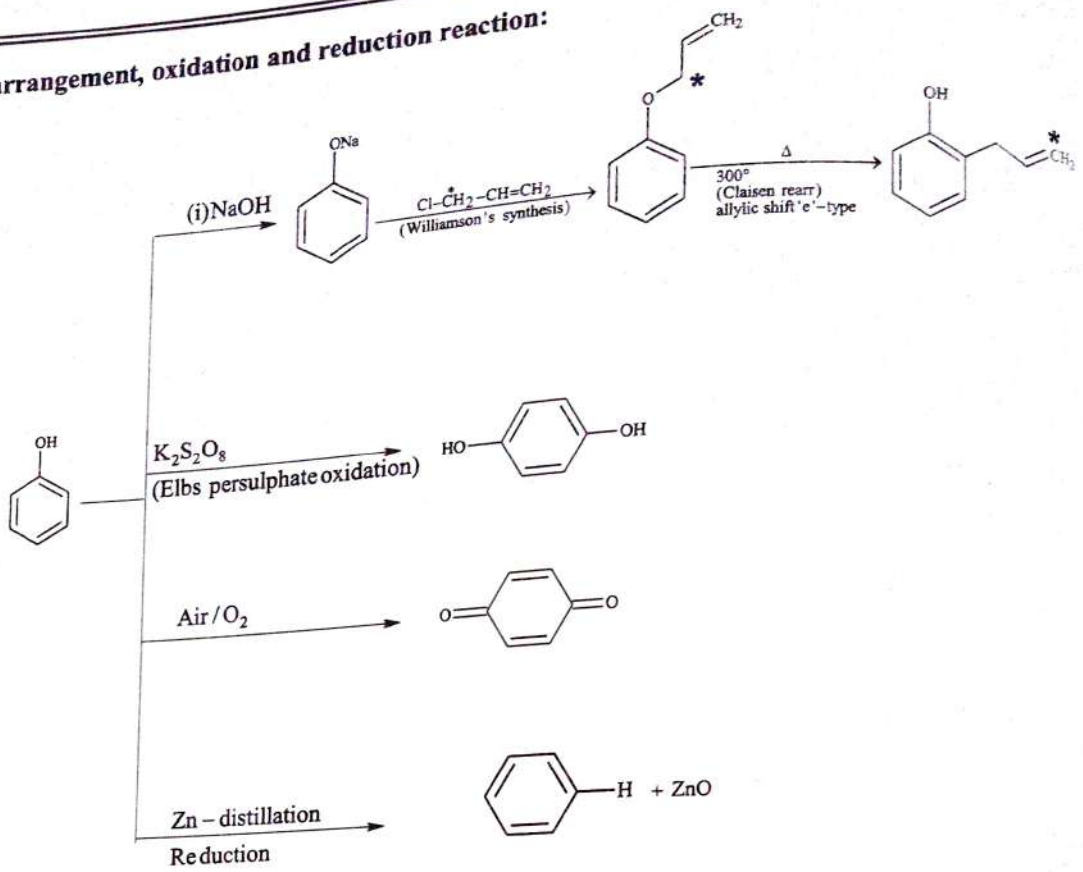




(iv) **Condensation Reaction:**



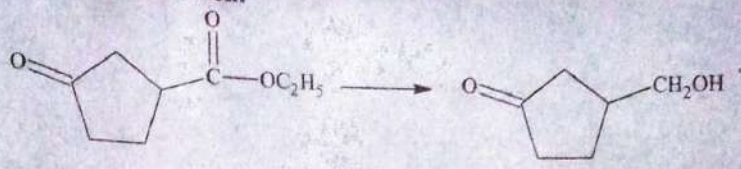
(v) **Rearrangement, oxidation and reduction reaction:**



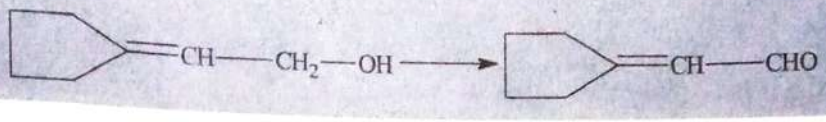
ILLUSTRATIVE PROBLEMS

1. **Effect the following conversion:**

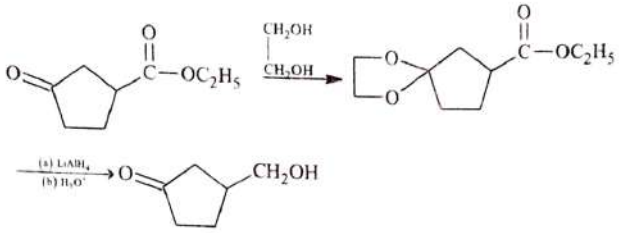
(i)



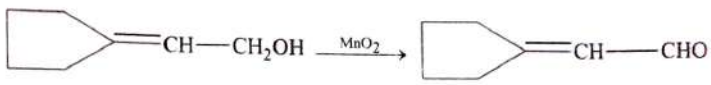
(ii)



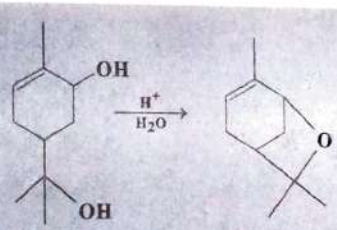
Sol. (i)



(ii)

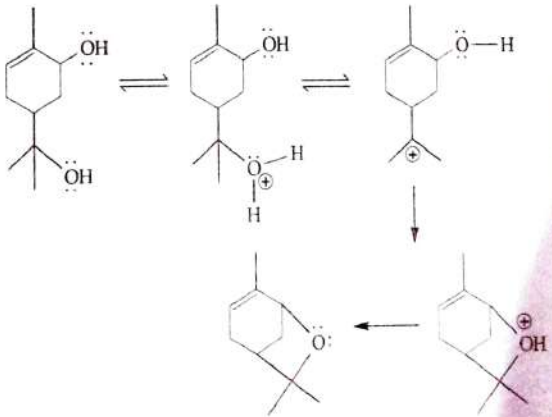


2.

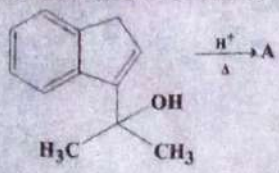


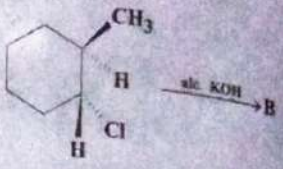
Write the plausible mechanism.

Sol.

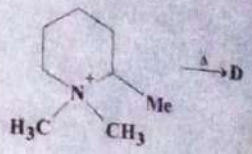


3. **Identify the alkenes in each of the following elimination reactions.**

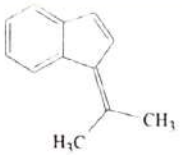
(a) 

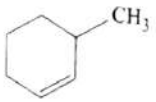
(b) 

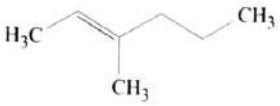
(c)

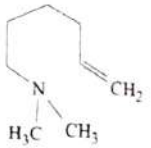
(d) 

(e)

Sol. A = 

B = 

C = 

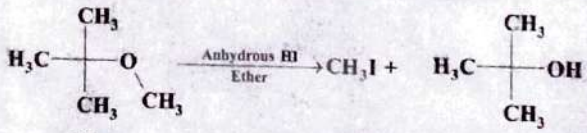
D = 

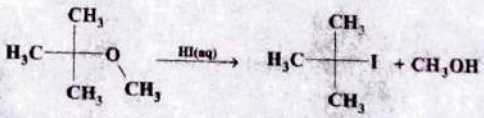
E =

4. **Account for the following:**

(i) Hydration of 3–phenyl–1–butene in dil. H_2SO_4 forms 2–phenyl–2–butanol instead of 3–phenyl–2–butanol.

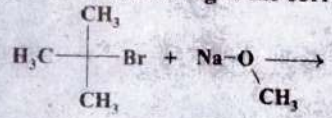
(ii) Acid catalysed dehydration of t–butanol is faster than that of n–butanol.

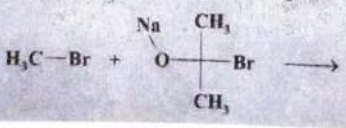
(iii) (a) 

(b) 

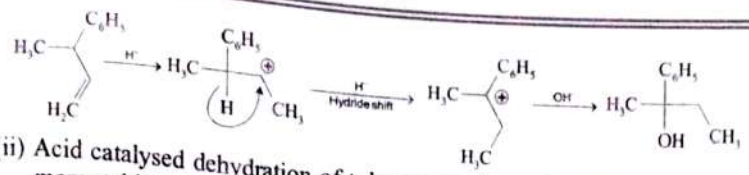
(iv) Although phenoxide ion has more number of resonating structures than benzoate ion, benzoic acid is stronger acid than phenol.

(v) Which of the following is the correct method for synthesizing methyl–t–butyl ether?

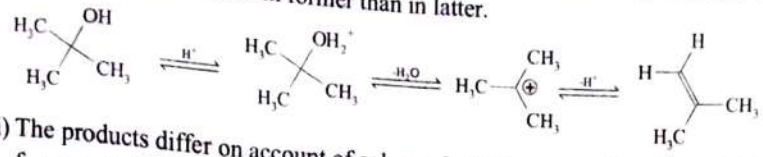
(a) 

(b) 

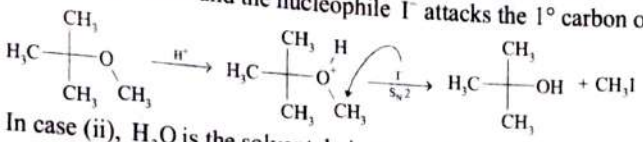
Sol. (i) Hydration of 3–phenyl–1–butene gives 2° carbocation, an intermediate which undergoes a hydride shift.



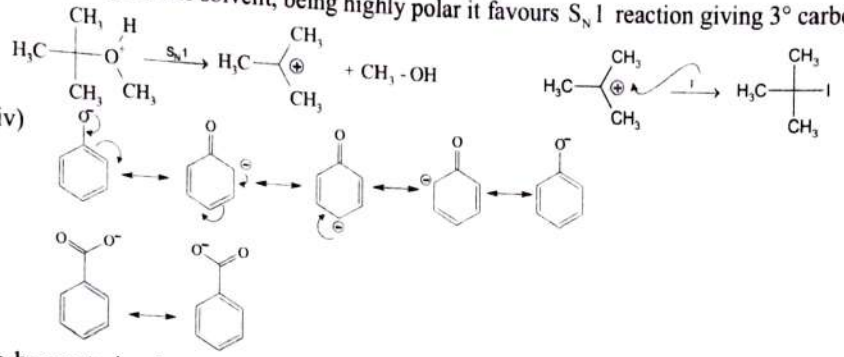
(ii) Acid catalysed dehydration of t–butanol is faster than that of n–butanol. Due to the formation of more stable carbocation in former than in latter.



(iii) The products differ on account of solvent. In the first case, the solvent (ether) is almost inert and it favours S_N2 route and the nucleophile I^- attacks the 1° carbon of CH_3.



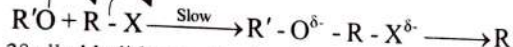
In case (ii), H_2O is the solvent, being highly polar it favours S_N1 reaction giving 3° carbocation.



(iv)

The benzoate ion is more stabilized because the negative charge is on the more electronegative oxygen atom whereas in phenoxide ion, it is on less electronegative carbon atoms. This makes benzoic acid stronger acid than phenol.

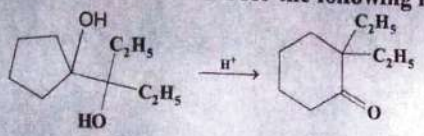
(v) The ether formation involves nucleophilic substitution of alkoxide ion for halide ion.



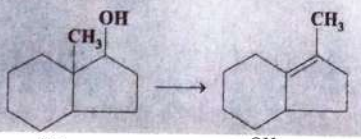
3° alkyl halide can also involve elimination of HX to give alkene in the presence of base. So, it is

5. Provide a suitable mechanism for the following reaction.

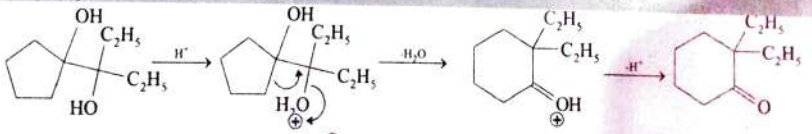
(i)



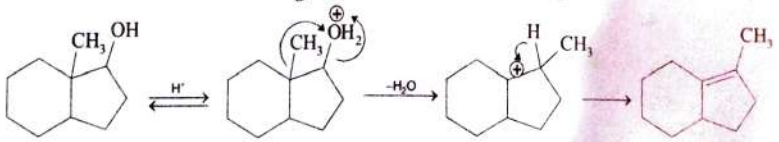
(ii)



Sol. (i)

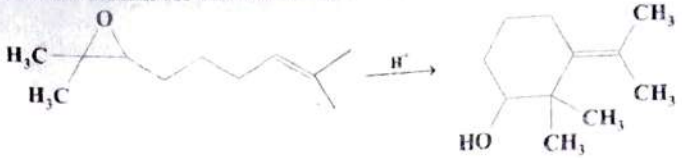


(ii)

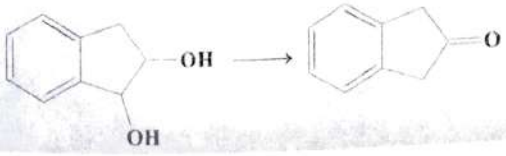


6. **Propose a mechanism for the following reaction.**

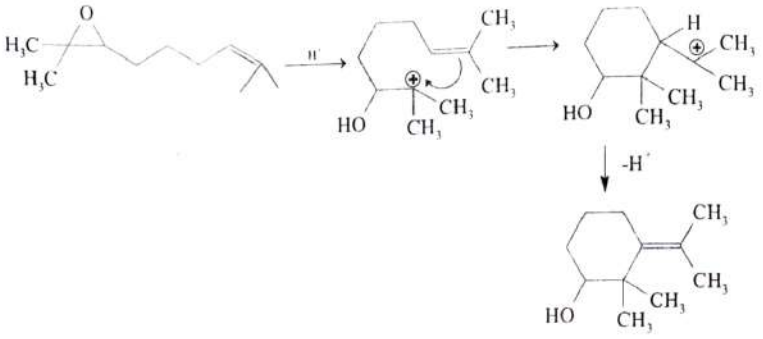
(i)



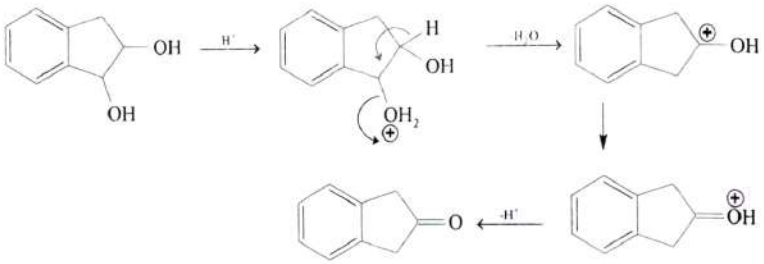
(ii)



Sol. (i)



(ii)



7. **Isopropyl alcohol on treatment with excess of conc. H_2SO_4 gives 'X' which on treatment with HBr gives 'Y'. When 'Y' is dissolved in ether and treated with sodium gives 'Z'. Calculate the molecular weight of 'Z'.**

Sol.

$$CH_3-\underset{OH}{\underset{|}{C}}H-CH_3 \xrightarrow[H_2SO_4]{Conc} CH_3-CH=CH_2 + H_2O$$

Propene

'X'

$$CH_3-CH=CH_2 + HBr \rightarrow CH_3-\underset{Br}{\underset{|}{C}}H-CH_3$$

2 – bromo propane (X)

$$CH_3-\overset{Br}{\overset{|}{C}}H-CH_3 + 2Na + CH_3-\overset{Br}{\overset{|}{C}}H-CH_3 \xrightarrow[ether]{Dry} CH_3-\underset{CH_3}{\underset{|}{C}}H-\underset{CH_3}{\underset{|}{C}}H-CH_3$$

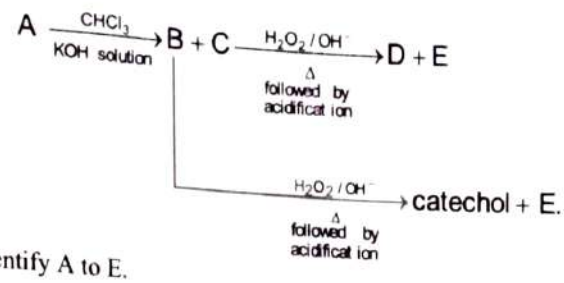
'Z' (2,3 – dimethyl butane)

Molecular weight of Z (C_6H_{14}) = 6 × 12 + 14 = 86 g mol^{-1}

SOLVED PROBLEMS

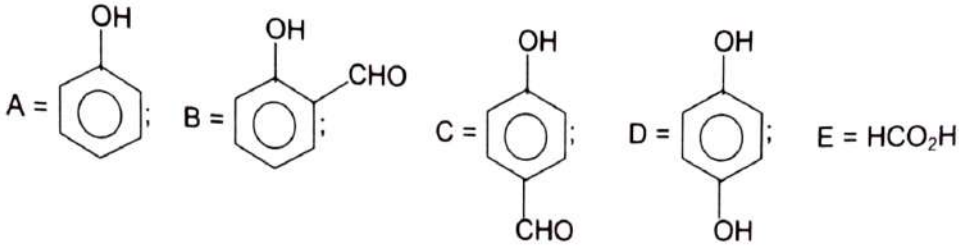
Subjective Questions

Prob. 1.



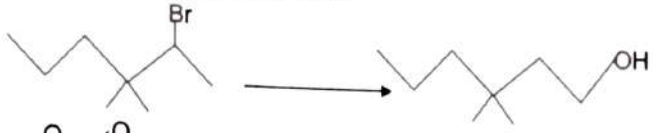
Identify A to E.

Sol.

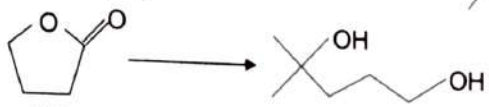


Prob. 2. Carryout the following transformations.

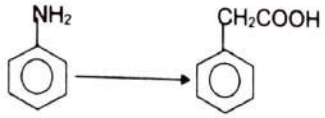
i)



ii)

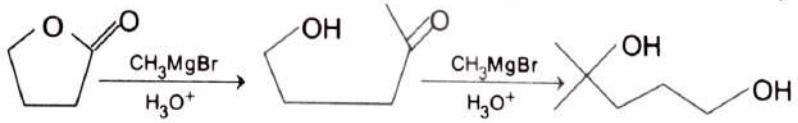


iii)

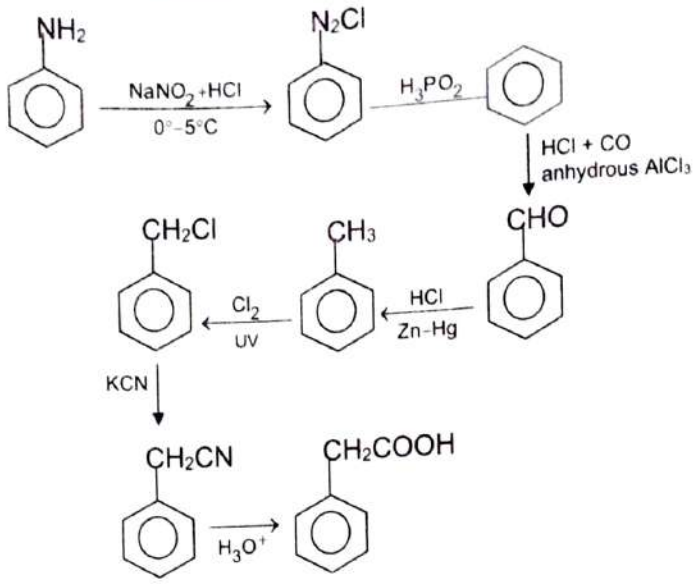


Sol. i)

ii)

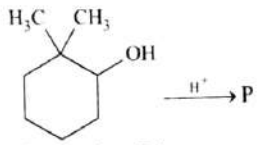


iii)

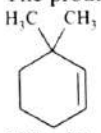


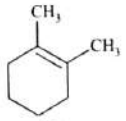
Objective Questions

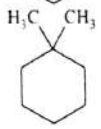
Prob. 1.

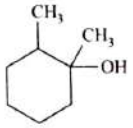


The product P is

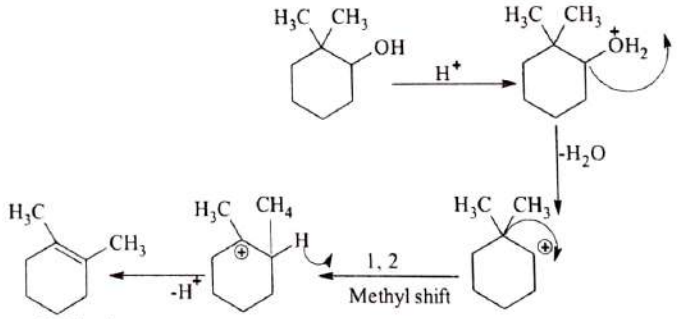
(A) 

(B) 

(C) 

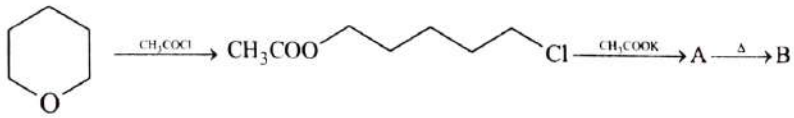
(D) 

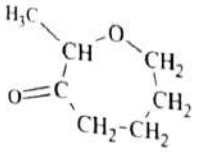
Sol.



Hence, **(B)** is the correct answer.

Prob. 2.

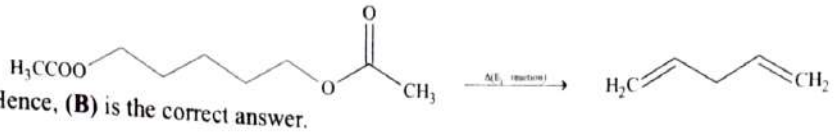


(A) 

(B)

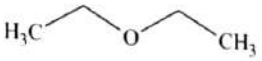
(C)

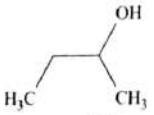
(D) None of the above

Sol. 

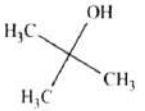
Hence, **(B)** is the correct answer.

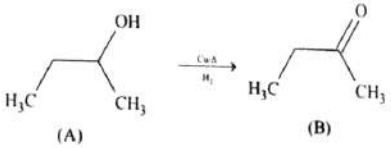
Prob. 3. An organic compound (A) (MF $C_4H_{10}O$) upon dehydrogenation gives a compound (B) which forms phenyl hydrazone with phenylhydrazine and both the compounds (A) and (B) respond to iodoform test. Hence, the compound (A) is

(A) 

(B) 

(C)

(D) 

Sol. 

(A) (B)

Hence, **(B)** is the correct answer.

Prob. 4. Alcohols may acts as

(A) oxidizing agent
(B) reducing agent
(C) Lewis base
(D) Bronsted acid

Sol. In presence of acids, it acts as a Lewis base but in presence of bases it acts a Bronsted acid, i.e.

$$ROH + H^+ \rightarrow R - \overset{+}{O}H_2$$

$$ROH + Na \rightarrow RO^- Na^+ + \frac{1}{2}H_2$$

Further alcohols reduce $KMnO_4$ and $K_2Cl_2O_7$ and themselves get oxidized to aldehydes/ketones and ultimately to carboxylic acids.

Hence, **(B), (C)** and **(D)** are the correct answers.

Prob. 5. Which of the following compounds will give a yellow precipitate with iodine and alkali

(A) 2-Hydroxypropane
(B) Acetophenone
(C) Methyl acetate
(D) Acetamide

Sol. 2-Hydroxypropane ($CH_3CHOHCH_3$) contains the grouping CH_3CHOH – and acetophenone ($C_6H_5COCH_3$) contains the grouping CH_3CO – linked to carbon and hence give yellow ppt. with I_2 and alkali-iodoform test.

In methyl acetate ($CH_3CO - OC_2H_5$) and acetamide ($CH_3CO - NH_2$), CH_3CO is attached to a hetero atom but not to carbon atom and hence both these compounds do not give iodoform test.

Hence, **(A)** and **(B)** are the correct answers.

Prob. 6. How would you get racemic mixture of 1,2 butane diol from cis 2-butene.

$$CH_3CH=CHCH_3 \xrightarrow[(ii)B]{(i)A} CH_3-\underset{\displaystyle OH}{\underset{|}{CH}}-\underset{\displaystyle OH}{\underset{|}{CH}}-CH_3$$, A & B are

(A) $A = KMnO_4/OH^-$, $B = H_2O$ (B) $A = CF_3CO_3H$, $B = H_2O$

(C) $A = OsO_4/OH^-$, $B = H_2O$ (D) $A = O_3/H_2O$, $B = Ph_3P$

Sol. **(B)**

A, C, D will give stereoselective reactions.

Prob. 7. A compound 'X' with molecular formula C_3H_8O can be oxidised to a compound 'Y' with the molecular formula $C_3H_6O_2$, 'X' is most likely to be

(A) Primary alcohol (B) Secondary alcohol

(C) Aldehyde (D) Ketone

Sol. **(A)**

1° alcohols give acid under uncontrolled oxidation.

Prob. 8. Phenol is less acidic than

(A) acetic acid (B) p-methoxyphenol

(C) p-nitrophenol (D) Ethanol

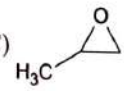
Sol. Phenol is less acidic than acetic acid as well as p-nitrophenol.

Hence, **(A)** and **(C)** are the correct answers.

Prob. 9. Compound which gives alcohol on reduction is/are

(A) $Me-\underset{\displaystyle O}{\underset{\|}{C}}-Cl$

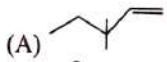
(B) $Me-\underset{\displaystyle O}{\underset{\|}{C}}-NH_2$

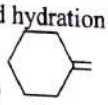
(C) 

(D) $Me-\underset{\displaystyle O}{\underset{\|}{C}}-O-\underset{\displaystyle O}{\underset{\|}{C}}-Me$

Sol. **(A, C, D)**

Prob. 10. HBO, oxymercuration-demercuration and acid catalysed hydration will not give same product in

(A) 

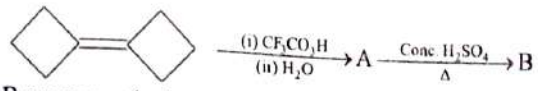
(B) 

(C)

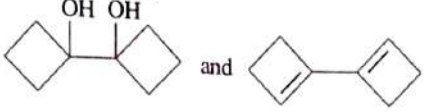
(D)

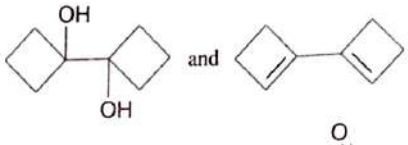
Sol. **(A, B, D)**

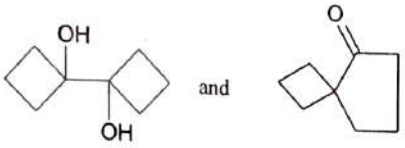
Prob. 11.

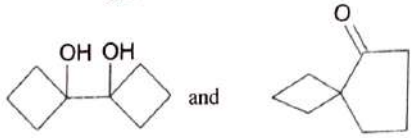


A and B are respectively.

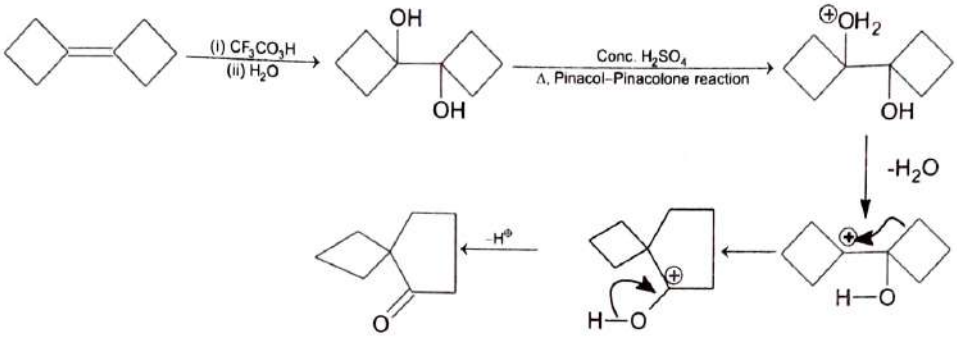
(A) 

(B) 

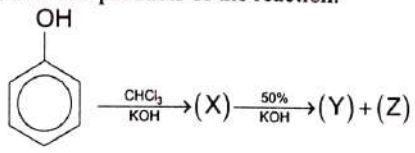
(C) 

(D) 

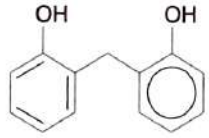
Sol. **(C)**

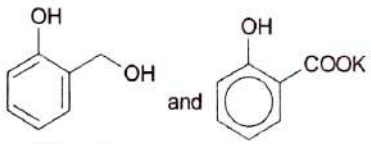


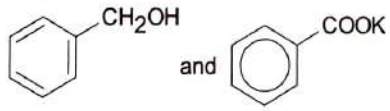
Prob. 12. The end products of the reaction:

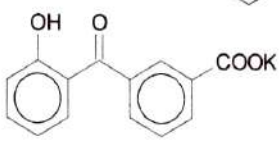


(Y) and (Z) are:

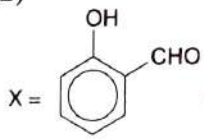
(A) 

(B) 

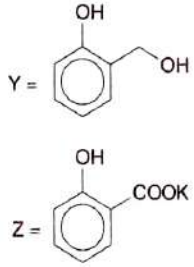
(C) 

(D) 

Sol. **(B)**

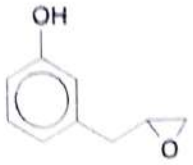


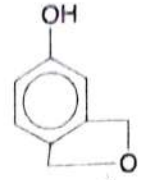
Reimer-Tiemann product

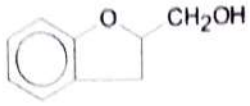


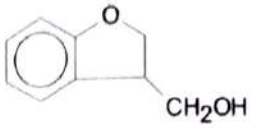
Cannizzaro product

Prob. 13. Consider the following reaction and identify "X"

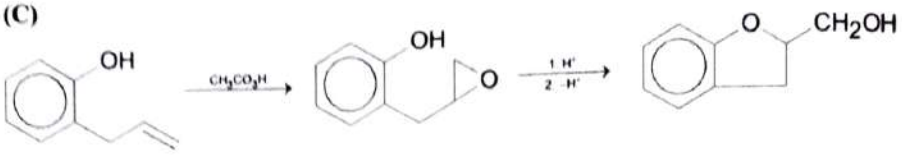
(A) 

(B) 

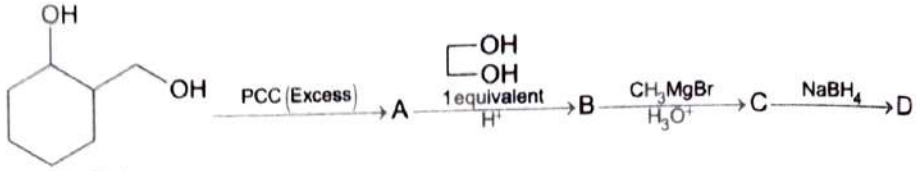
(C) 

(D) 

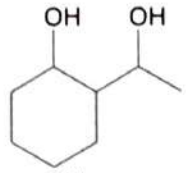
Sol. **(C)**

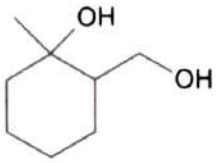


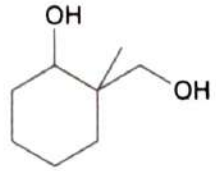
Prob. 14.

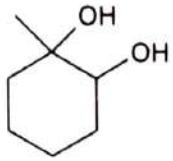


The product D is:

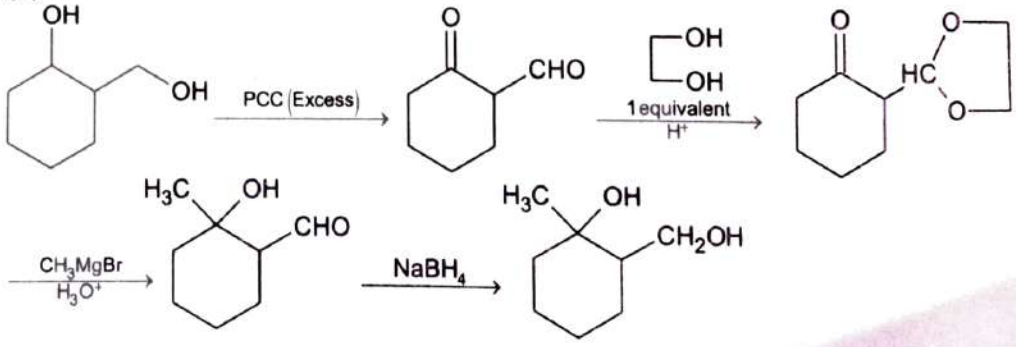
(A) 

(B) 

(C) 

(D) 

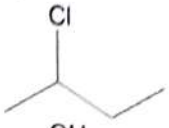
Sol. **(B)**



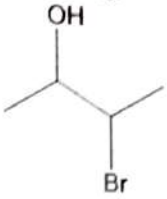
COMPREHENSION

An organic compound (A) of molecular formula C_4H_9Cl on treatment with alcoholic KOH solution gives 3 isomeric alkenes. The stable alkene among the three (B), was treated with aqueous Br_2 solution which resulted in (C). (C) on treatment with NaOH liberated a gas with the formation of (D). (D) on treatment with CH_3MgCl followed by hydrolysis gives (E). (E) on treatment with H'/HBr gave (F) and (F) on treatment with alcoholic solution of NaOH gave (G). If (G) can't show geometrical isomerism identify the structures of (A) to (G).

1. What is 'B':

(A) 

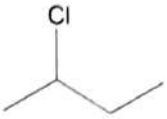
(B)

(C) 

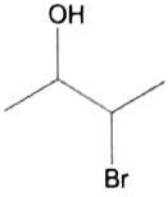
(D)

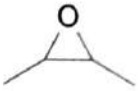
Sol. **(B)**

2. What is 'D' :

(A) 

(B)

(C) 

(D) 

Sol. **(D)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Solubility of n-alcohol in water decreases with increase in molecular weight
and
STATEMENT-2: The hydrophobic nature of alkyl chain increases.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** The solubility of alcohols in water due to H-bonding. As the molecular weight of alcohol increases, the increase in hydrophobic nature of alkyl group results in decrease in solubility.

2. **STATEMENT-1:** Alcohols can not be dried by $CaCl_2$.
and
STATEMENT-2: $CaCl_2$ forms solvated molecules with alcohols.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** Explanation is correct reason for statement.

3. **STATEMENT-1:** Acid catalysed dehydration of t-butanol is faster than n- butanol.
and
STATEMENT-2: The order of stability of carbocation is $3° > 2° > 1°$.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** Explanation is correct reason for statement.

4. **STATEMENT-1:** Alcohols gives only substitution products with HX and not elimination product.
and
STATEMENT-2: X is a very weak base which can not abstract proton from alcohol.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

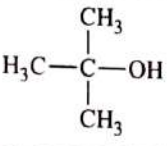
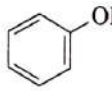
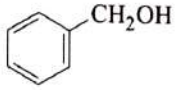
Sol. **(A)** Explanation is correct reason for statement.

5. STATEMENT-1: On oxidation with mild oxidising agents (e.g.-atmospheric oxygen) thiol gives disulfides.
and
STATEMENT-2: The S – H bond in stronger than the O – H bond
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**

MATRIX -MATCH TYPE QUESTIONS

1. Match the following set.

Column-I		Column-II	
(A)	$CH_3—CH_2—OH$	(p)	No response to Victor Meyer,s reagent.
(B)		(q)	Turbidity immediately when treated with Lucas reagent
(C)		(r)	No response to Lucas reagent
(D)		(s)	Red colour in Victor Maeyer's test
		(t)	Gives test with dil. $AgNO_3$

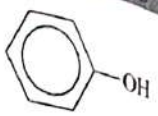
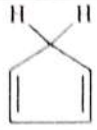
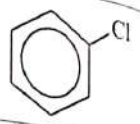
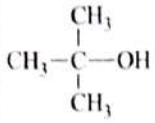
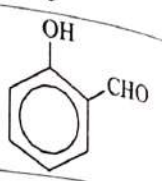
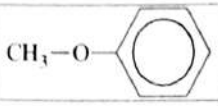
Sol. **(A → r, s; B → p, q; C → p, r; D → q, s)**

2. Match the following set.

Column-I		Column-II	
(A)	Oxidation of 1°alcohol in aldehyde	(p)	$KMnO_4/H^+$
(B)	CrO_3 . HCl	(q)	Collin's reagent
(C)	CrO_3 .2	(r)	Jone's reagent
(D)	Oxidation of alkene into acid	(s)	P.C.C

Sol. **(A) q, s (B) s (C) q (D) p**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)		(i)	Conc.HI	(P)	
(II)		(ii)	$\xrightarrow[\text{(ii) } H^+]{\text{(i) } CHCl_3, KOH, 60^\circ}$	(Q)	
(III)		(iii)	Conc.HBr	(R)	
(IV)		(iv)	$\xrightarrow{\text{Alc. KOH}}$	(S)	C_2H_5I

3. Which of the following combination is correct ?
[A] (I) (ii) (R) [B] (I) (ii) (S)
[C] (I) (i) (S) [D] (I) (ii) (P)

4. Which of the following combination is correct ?
[A] (II) (ii) (Q) [B] (II) (iii) (R)
[C] (II) (iv) (S) [D] (II) (ii) (R)

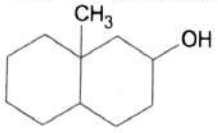
5. Which of the following combination is correct ?
[A] (IV) (i) (P) [B] (IV) (iv) (P)
[C] (IV) (ii) (P) [D] (III) (iii) (R)

Sol.

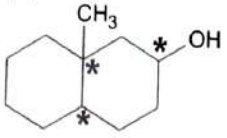
3. **A**
4. **A**
5. **A**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. What will be the number of stereo-isomers of the given compound:



Sol. **(8)**



2. 0.1 mol of a hydroxyl compound reacts with 62.5 g of PCl_5 (mol. wt. 208.5). Determine the number of —OH groups.

Sol. **(3)**

We know that one mol of PCl_5 reacts with one mol of —OH group.

Mole of Alcohol = 0.1, mol of $PCl_5 = \frac{62.5}{208.5} = 0.3$

Hence number of OH group = $\frac{\text{mol of } PCl_5}{\text{mol of Alcohol}} = \frac{0.3}{0.1} = 3$

3. How many of the give red precipitation with Fehling solution:

$C_6H_5 - COC_6H_5$, C_6H_5CHO, $C_6H_5CH_2CHO$, $C_6H_5COCH_3$, CH_3CHO, $CH_3 - \overset{O}{\overset{\|}{C}} - CH_3$

Sol. **(2)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. How many grams of a compound A (MF = $C_4H_{10}O_4$, having 4 alcoholic groups) reacts with excess of CH_3MgBr give 11.2 L of CH_4 at S.T.P.

Sol. **15.25**

1 mole of 'A' compound gives 4 moles of CH_4

moles of $CH_4 = \frac{11.2}{22.4} = \frac{1}{2}$;

hence mole of 'A' $= \frac{1}{2} \times \frac{1}{4} = \frac{1}{8}$

weight of 'A' $= \frac{1}{8} \times 122 = 15.25g$

Prob. 2. A unknown compound (A) (MM = 342) on acetylation gives a product (MM = 678) then find the number of hydroxyl group present in compound (A).

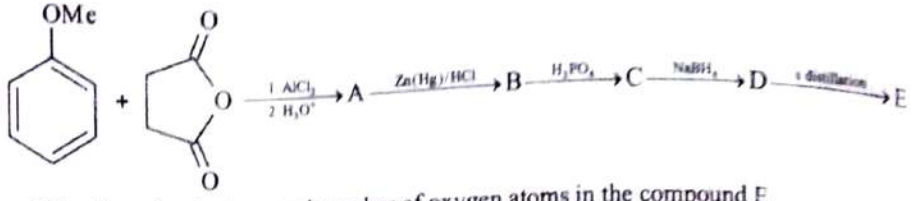
Sol. **8**

Increase in mass due to acetylation = 678 – 342 = 336

Replacement of one hydrogen from hydroxyl group by $-\overset{O}{\overset{\|}{C}} - CH_3$ group increase mass = 42

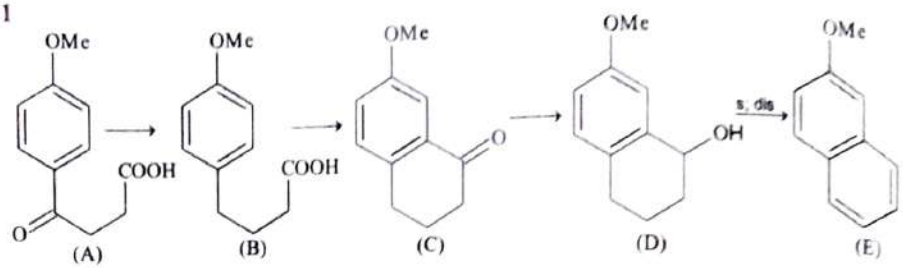
Hence number of OH group $= \frac{336}{42} = 8$

Prob.3.



Sum of Number of π electrons and number of oxygen atoms in the compound E.

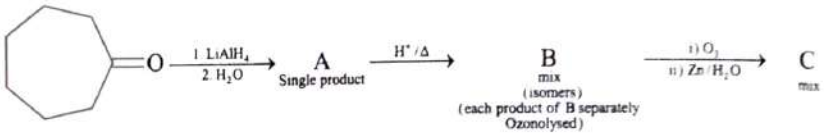
Sol. **11**



Prob.4. If (±) 2-methyl butanoic acid were esterified by reaction with (±) 2-butanol, how many optically active compounds would be present in the final equilibrium reaction mixture.

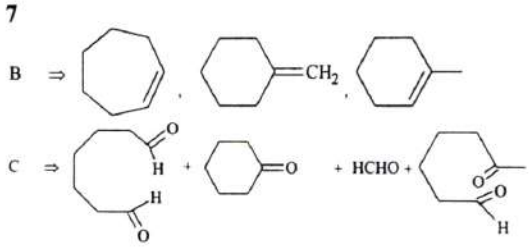
Sol. **4**

Prob.5.



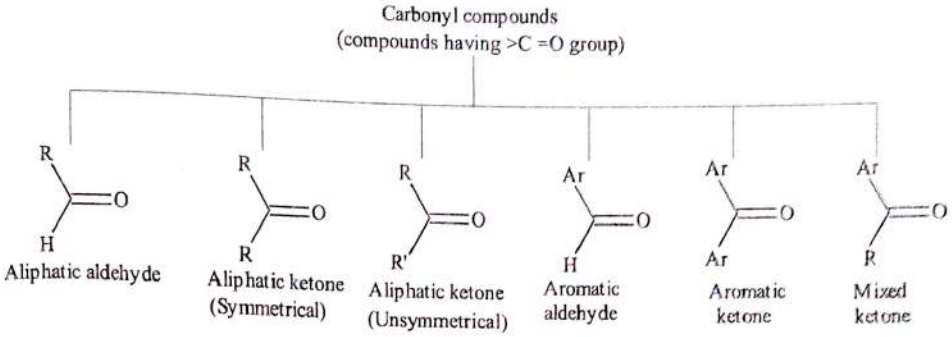
The sum of mix of B and mix of C is

Sol. **7**



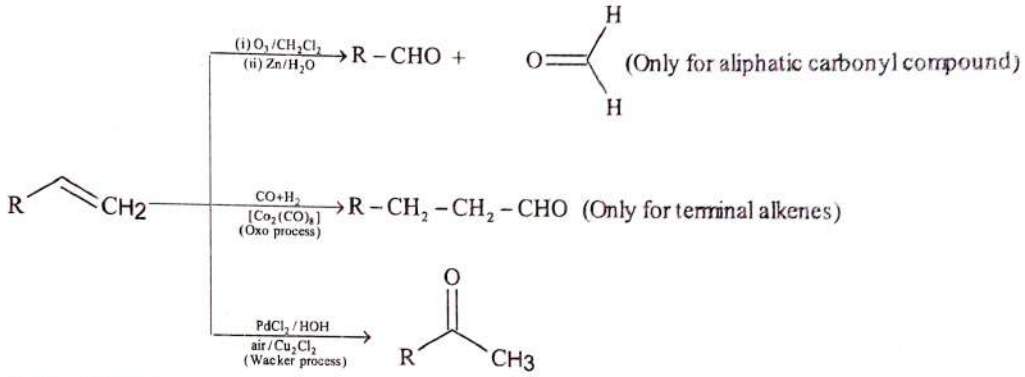
ALDEHYDE & KETONE

17

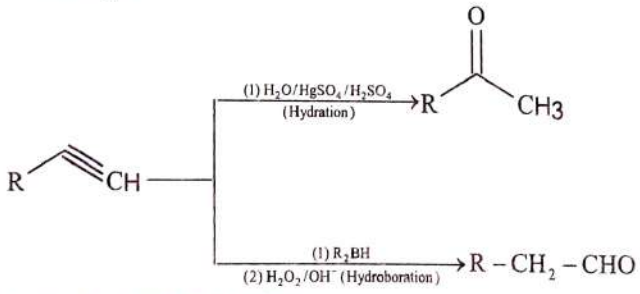


METHODS OF PREPARATION

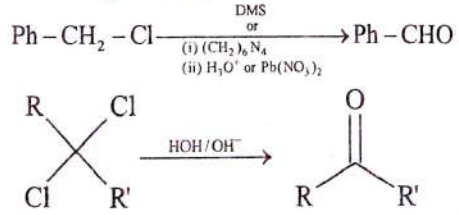
(i) **From alkenes:**



(ii) **From alkynes:**

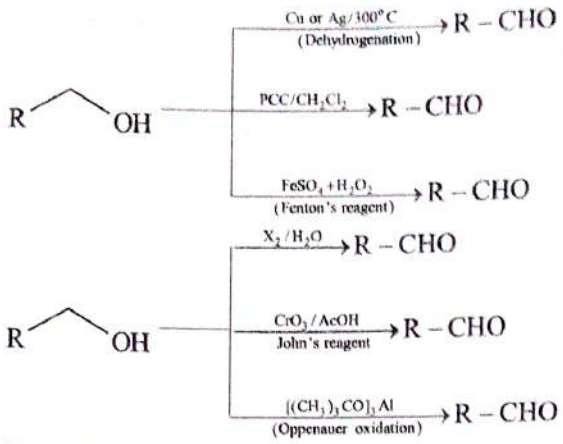


(iii) **From alkyl and aryl halides:**



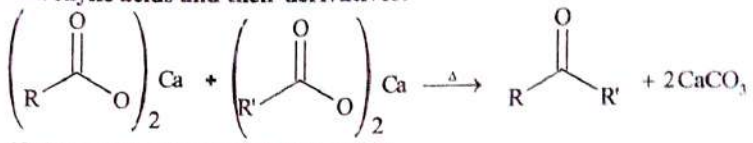
Gem-dihalide

(iv) **From alcohols:**

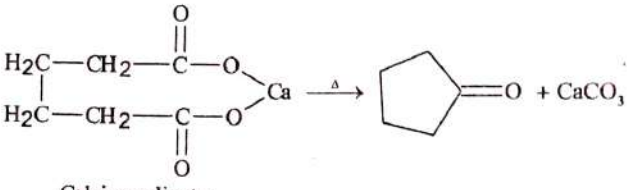


(v) **From carboxylic acids and their derivatives:**

(a)

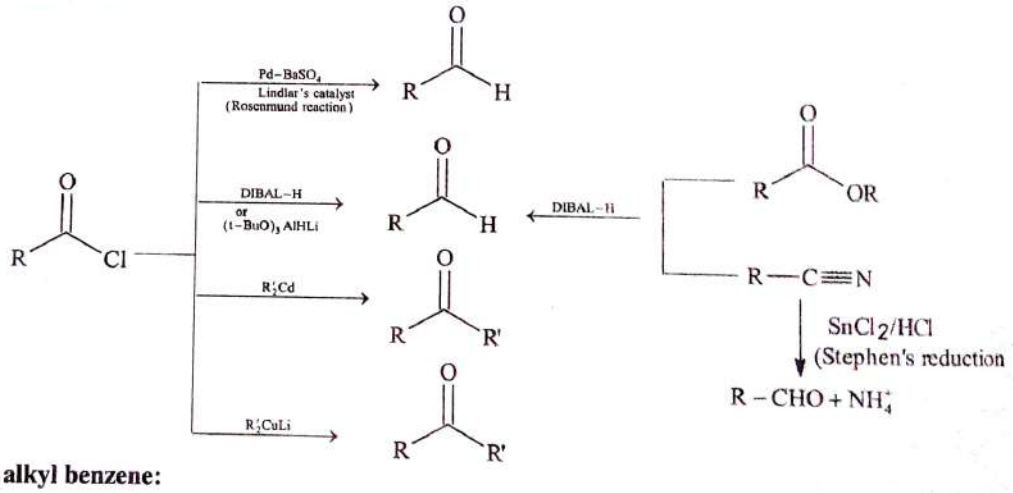


(Calcium salt of monocarboxylic acid)

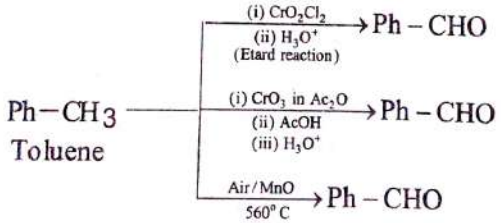


Calcium adipate

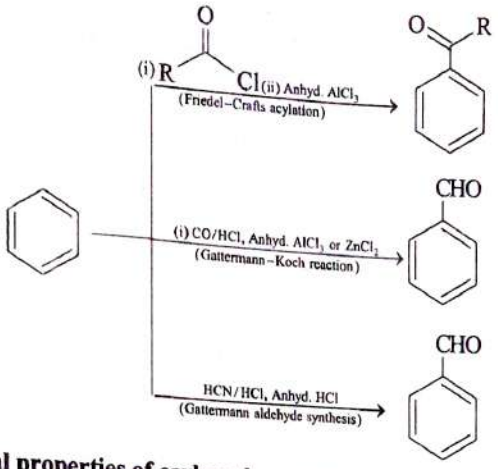
(b) **From acyl halide:**



(vi) **From alkyl benzene:**



(vii) **From benzene:**

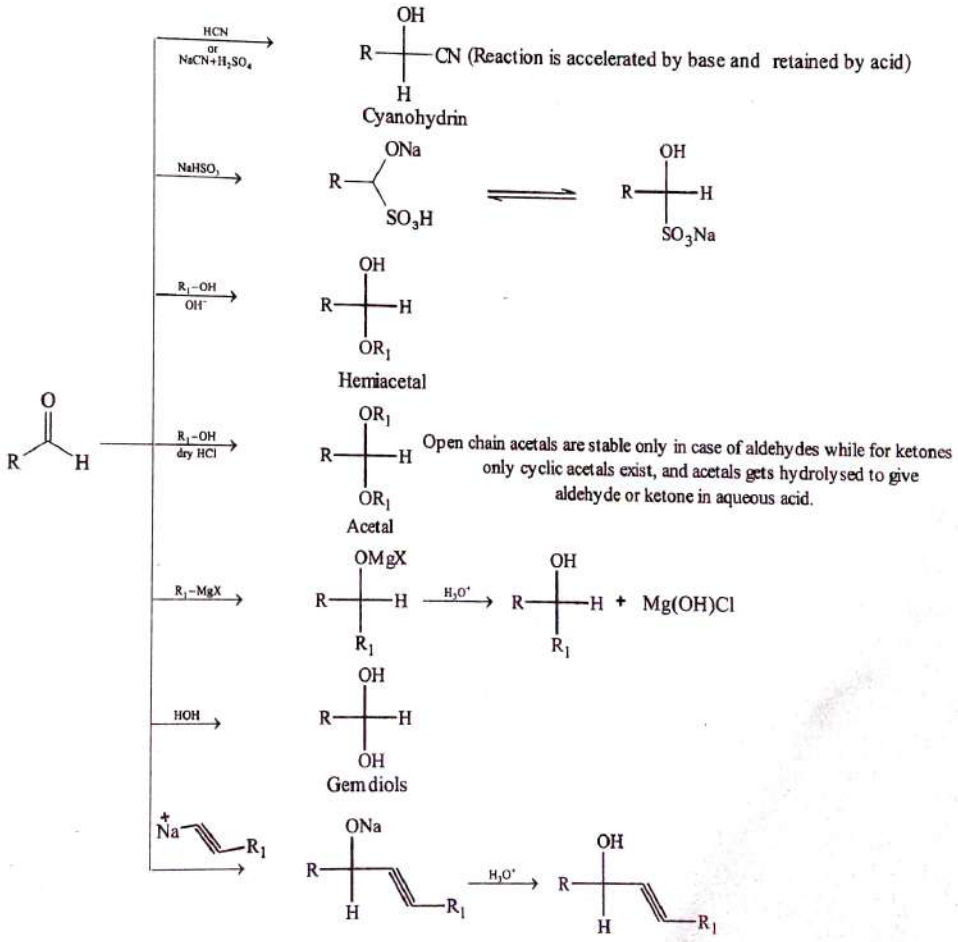


Chemical properties of carbonyl compounds

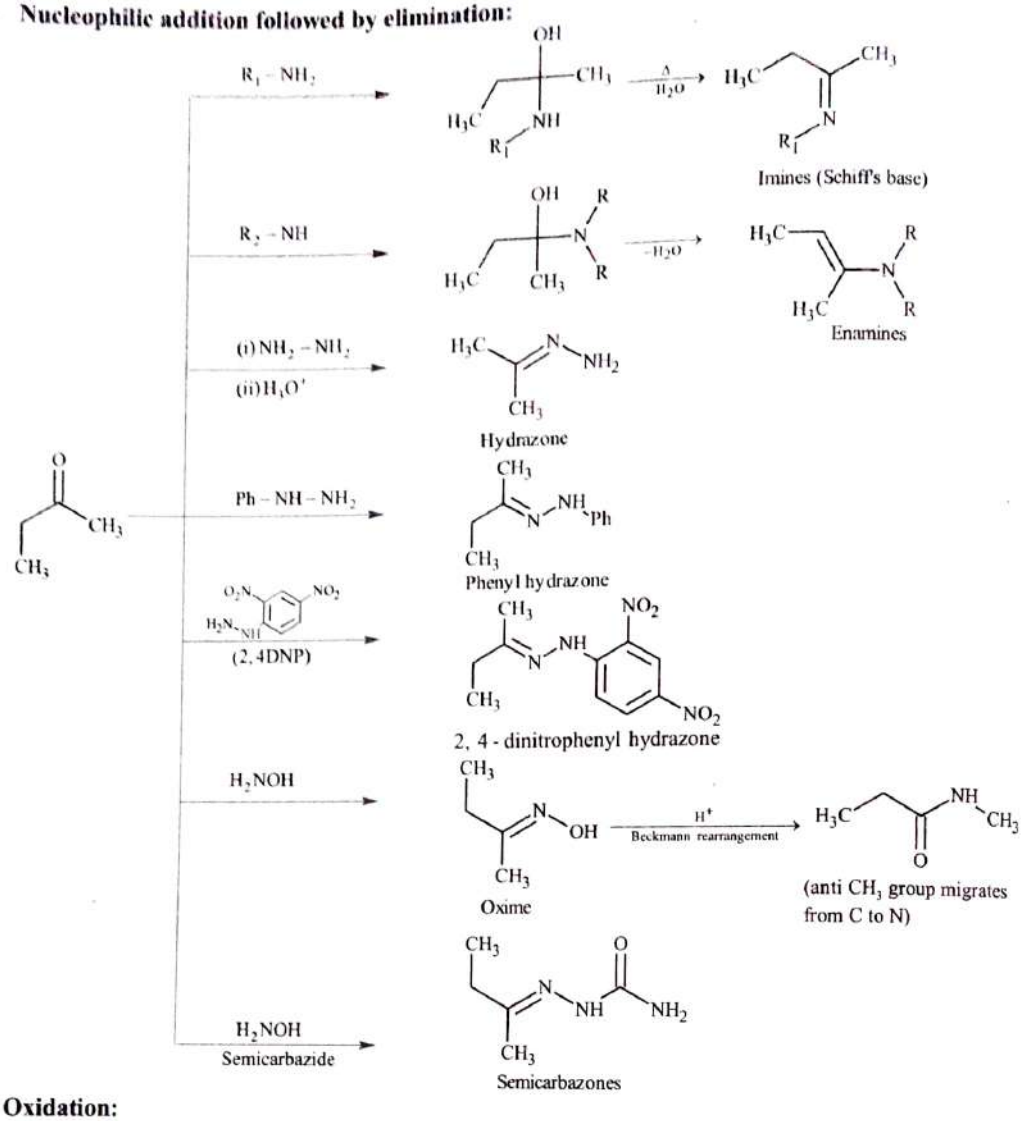
(i) Reactions due to >C = O group
(ii) Reactions due to α–H–atom
(iii) Condensation reaction

Reactions Due to >C = O group

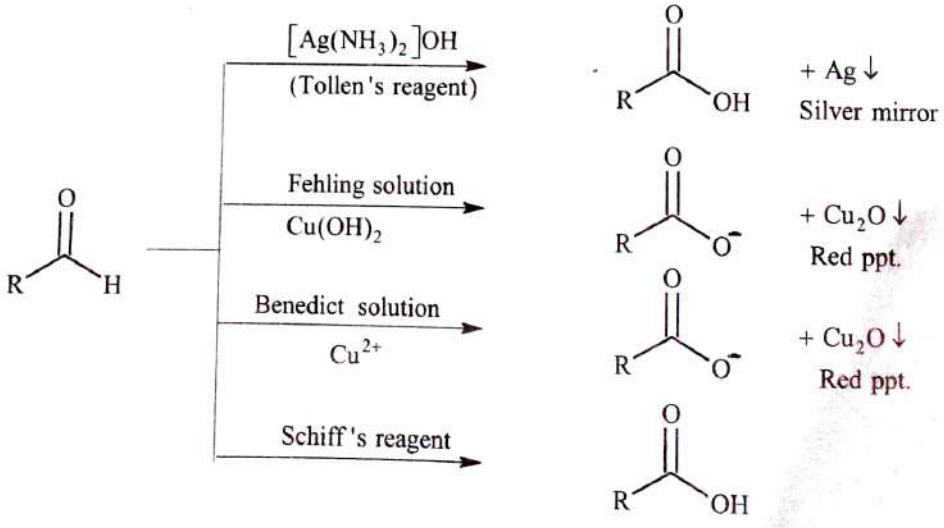
(i) **Nucleophilic addition:**

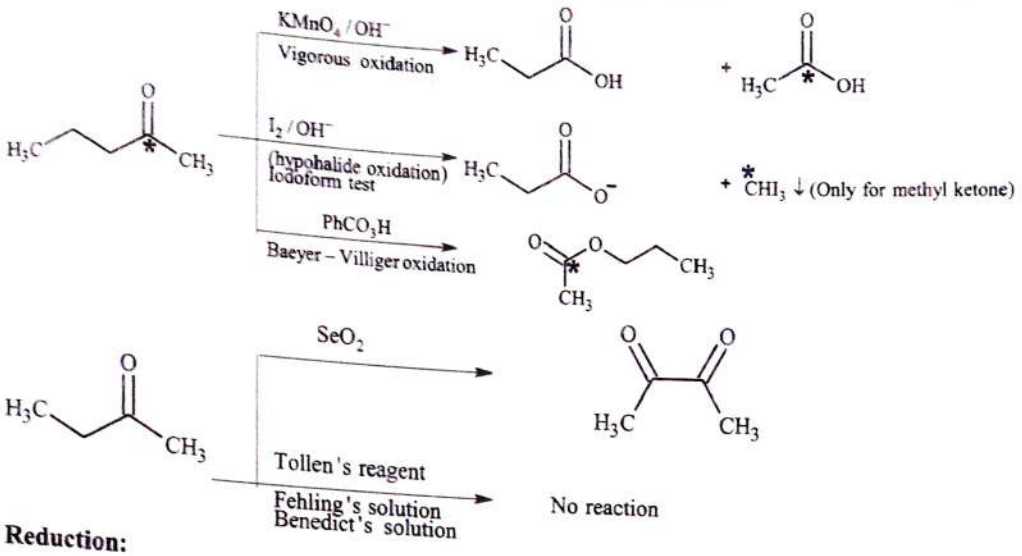


(ii) **Nucleophilic addition followed by elimination:**

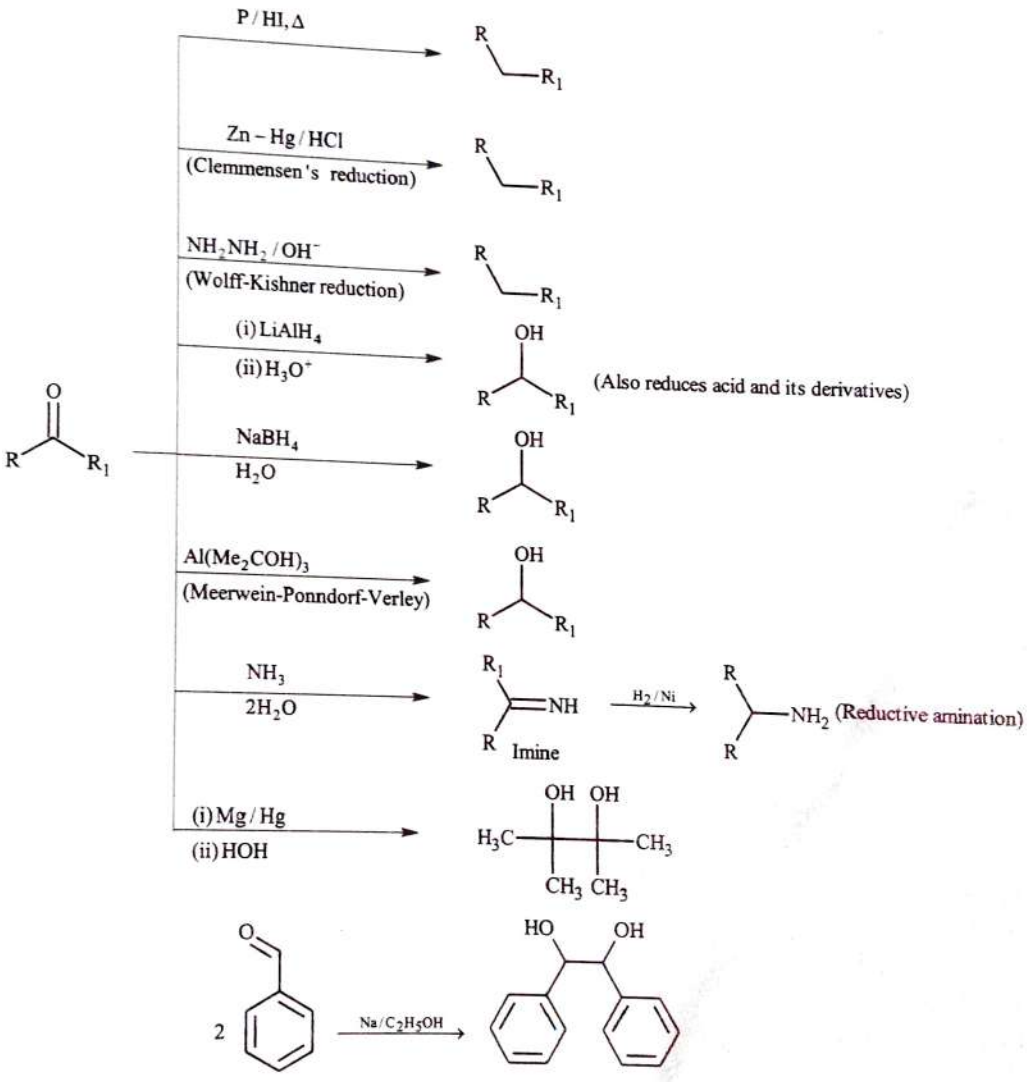


(iii) **Oxidation:**

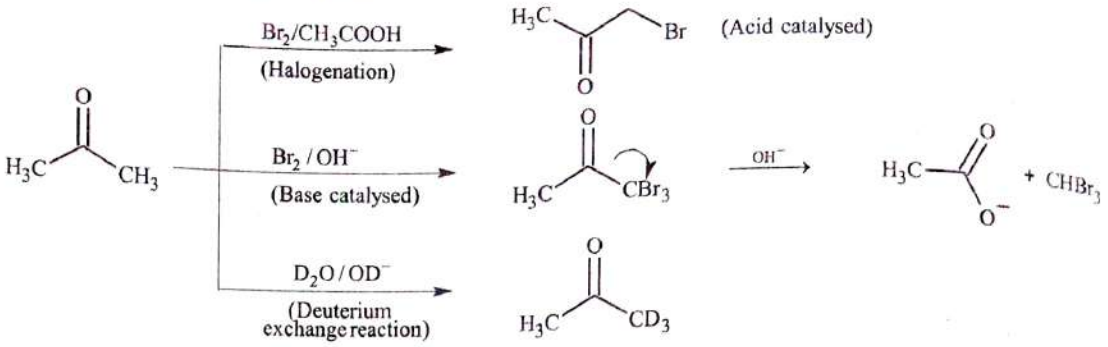


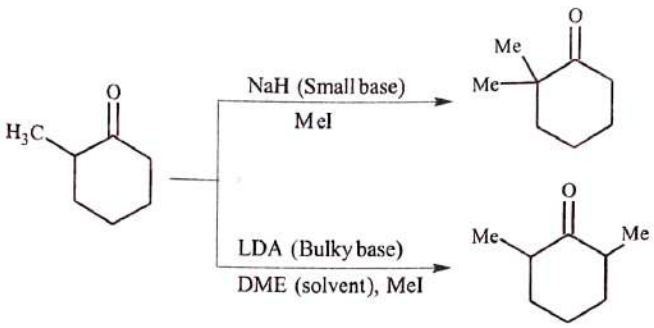


(iv) **Reduction:**



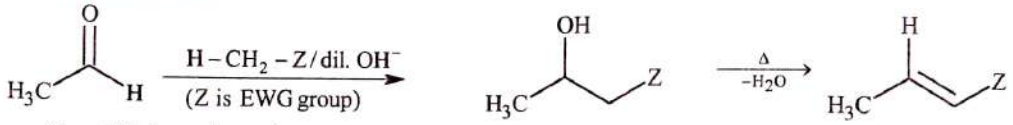
Reactions Due to α-H atom



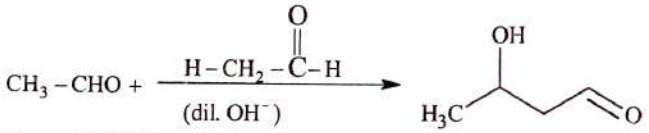


Condensation Reaction

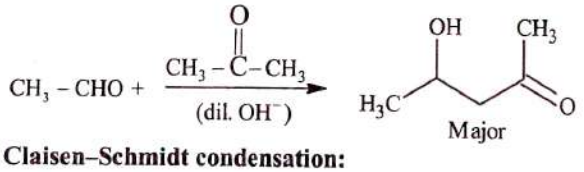
General mechanism:



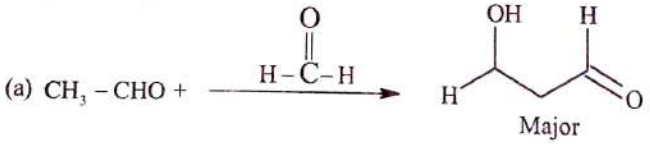
(i) **Aldol condensation:**

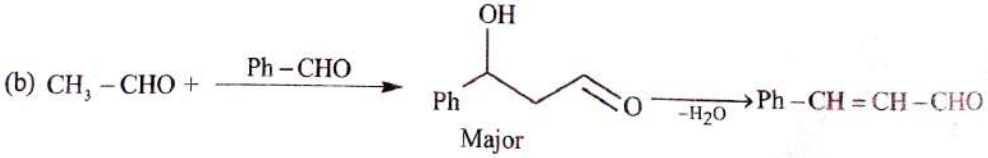


(ii) **Crossed aldol condensation:**

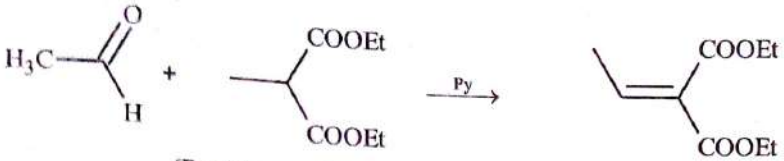


(iii) **Claisen–Schmidt condensation:**

(a) 

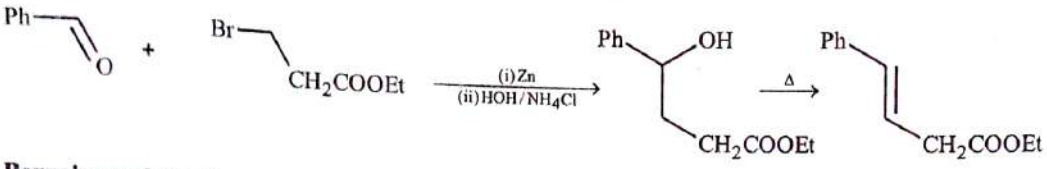
(b) 

(iv) **Knovengel's reaction:**

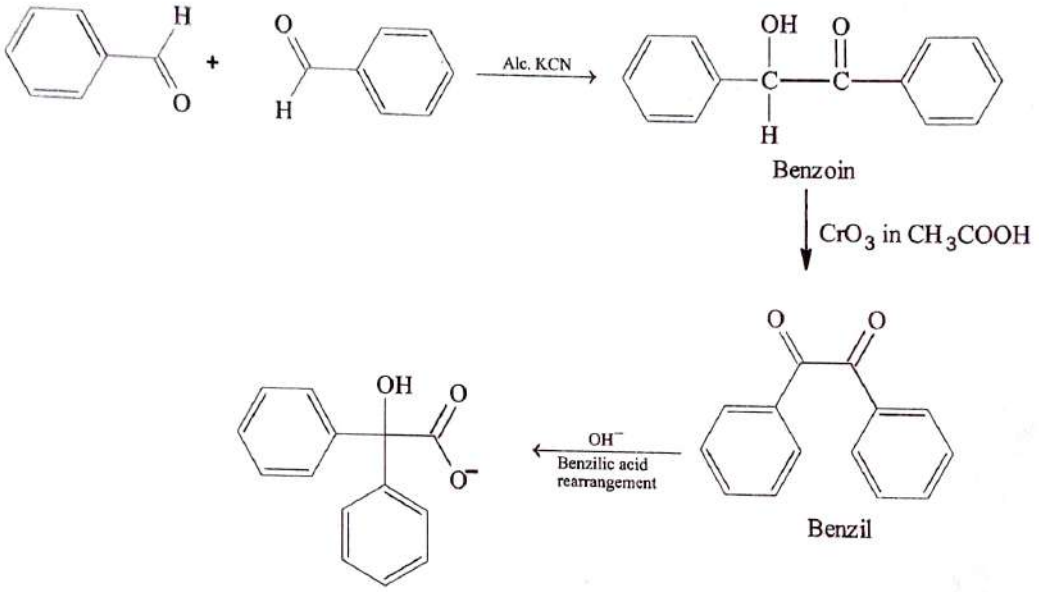


(Reactive methylene comp.)

(v) **Reformatsky reaction:**



(vi) **Benzoin condensation:**



(vii) **Cannizzaro reaction:**

Aldehydes having no α - H atom undergo Cannizzaro reaction.

(a) $H - CHO + H - CHO \xrightarrow{50\%OH^-} H - \overset{O}{\overset{\|}{C}} - O^- + CH_3OH$

(b) $Ph - CHO + H - CHO \xrightarrow{50\%OH^-} Ph - CH_2 - OH + H - \overset{O}{\overset{\|}{C}} - O^-$ (crossed Cannizzaro)

(c) $Ph - CHO + Me_3C - CHO \xrightarrow{50\%OH^-} Ph\text{-}COO^- + Me_3C - CH_2 - OH$ (crossed Cannizzaro)

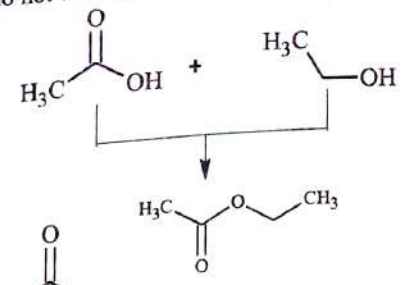
(d) $\begin{matrix} CHO \\ | \\ CHO \end{matrix} \xrightarrow{50\%OH^-} \begin{matrix} COO^- \\ | \\ CH_2OH \end{matrix}$ (intramolecular Cannizzaro reaction)

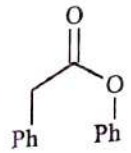
(e) $Ph - \overset{O}{\overset{\|}{C}} - CHO \xrightarrow{50\%OH^-} Ph - \overset{OH}{\overset{|}{C}}H - COO^-$ (intramolecular Cannizzaro)

(f) $2Me_2CH - CHO \xrightarrow[\text{exceptional}]{50\%OH^-} Me_2CH\text{-}CH_2OH + Me_2CH\text{-}COO^-$ (Cannizzaro reaction)

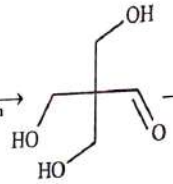
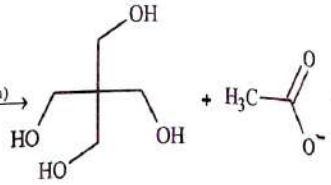
(viii) **Tischenko reaction:**

Only aldehydes show this reaction. Ketones do not undergo this reaction.

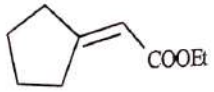
(a) $CH_3-CHO + CH_3-CHO \xrightarrow[Al_2O_3]{Al(OC_2H_5)_3}$ 

(b) $Ph-CHO + Ph\text{-}CHO \xrightarrow{Na_2[Fe(CO)_4]}$ 

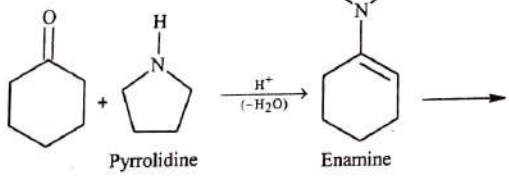
(ix) **Tollen's reaction**

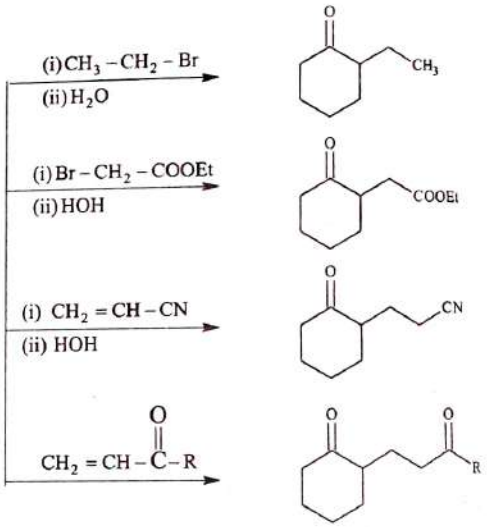
$CH_3-CHO + HCHO\text{(excess)} \xrightarrow[\text{Aldol condensation}]{Cu(OH)_2}$  $\xrightarrow{HCHO\ (Cannizzaro\ reaction)}$ 

(x) **Wittig reaction**

$Cl-CH_2-COOEt \xrightarrow[(ii)RLi]{(i)PPh_3} Ph_3\overset{+}{P}-\overset{-}{C}H-COOEt \xrightarrow{}$ 

(xi) **Reaction with 2° amine (Enamine formation):**



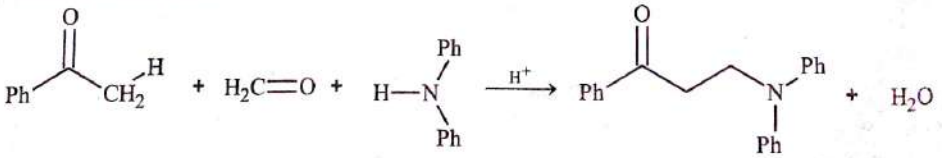


$6H\text{-}CHO + 6NH_3 \xrightarrow[-6H_2O]{OH^-} (CH_2)_6N_4$

Hexamethylene tetramine

(Urotropine)

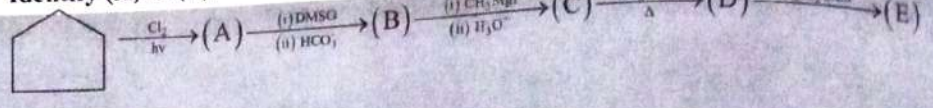
(xii) **Mannich reaction:**



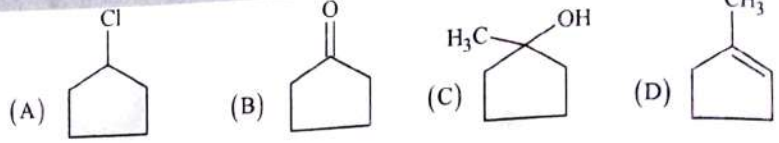
Mannich base

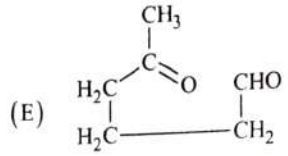
ILLUSTRATIVE PROBLEMS

1. Identify (A) to (F):

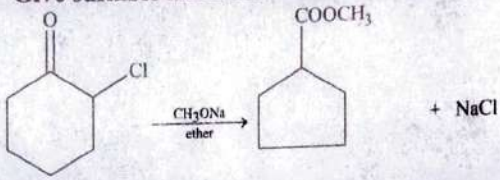


Sol.

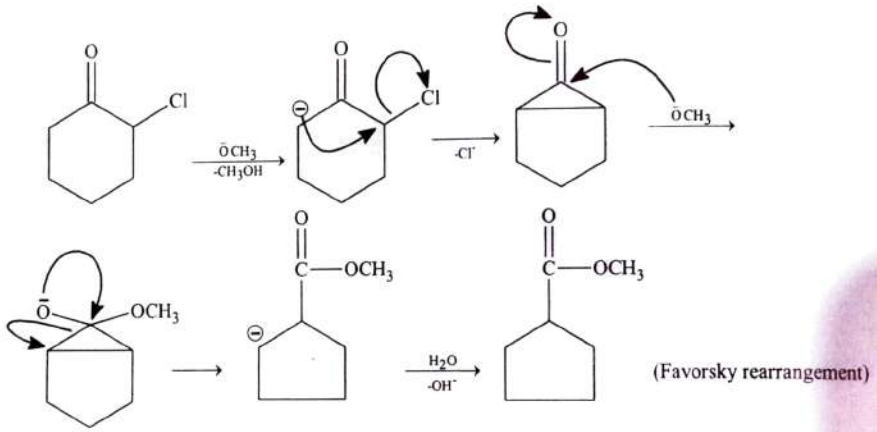




2. Give suitable mechanism for the following transformations:

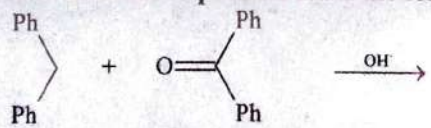


Sol.

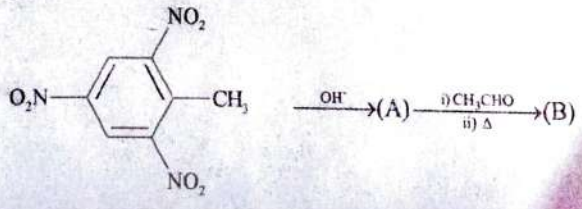


3. Give the structures of the product from the following condensation.

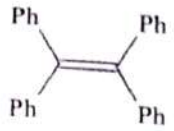
(i)



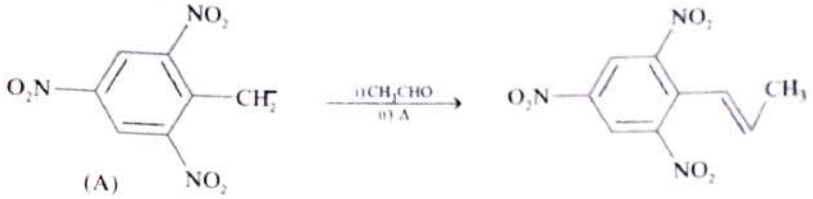
(ii)



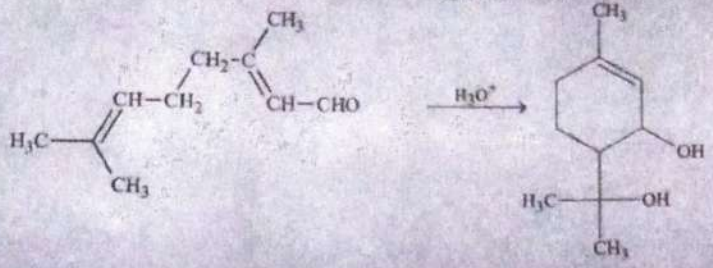
Sol. i)



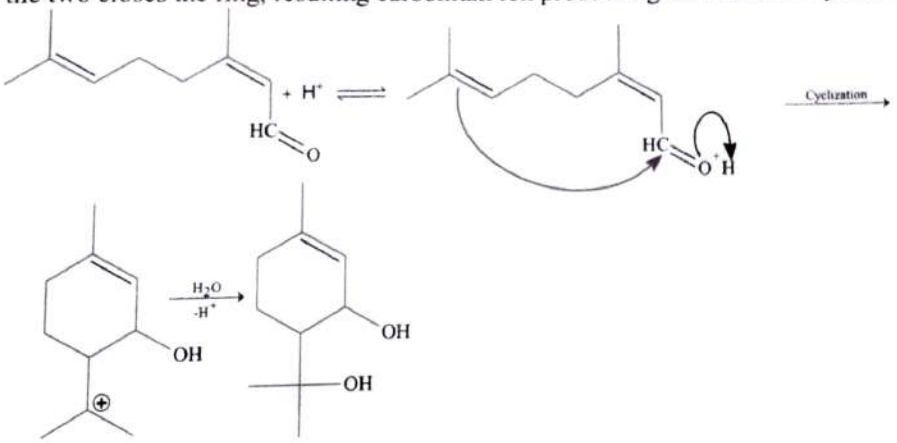
ii)



4. Suggest a mechanism for the following reaction:



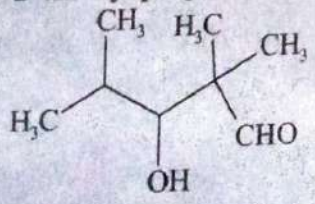
Sol. The protonated aldehyde is an electrophile and the double bond is a nucleophile. The reaction between the two closes the ring, resulting carbonium ion produced gives a diol on hydration.



5. How would you bring about the following conversions?

(i) Ethanal to 2–hydroxy–3–butenoic acid

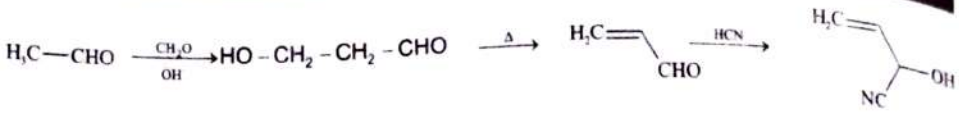
(ii) 2–methyl propanal to



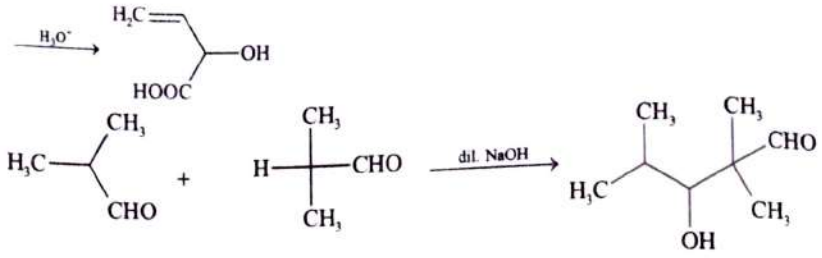
(iii) 2–butanone from ethyl alcohol

(iv) 3–hexanone from n–propyl alcohol

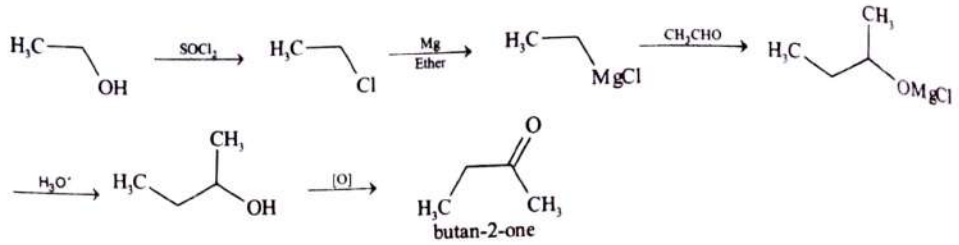
Sol. (i)



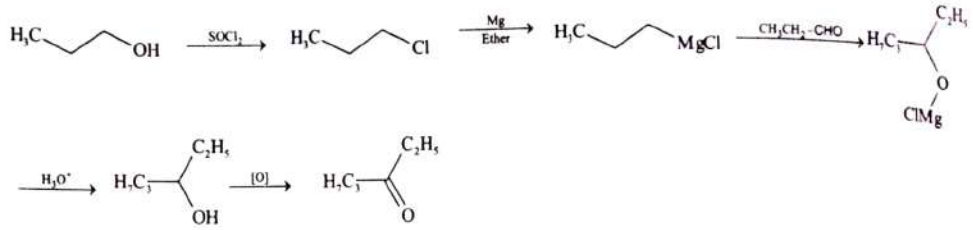
(ii)



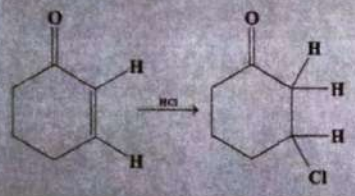
(iii)



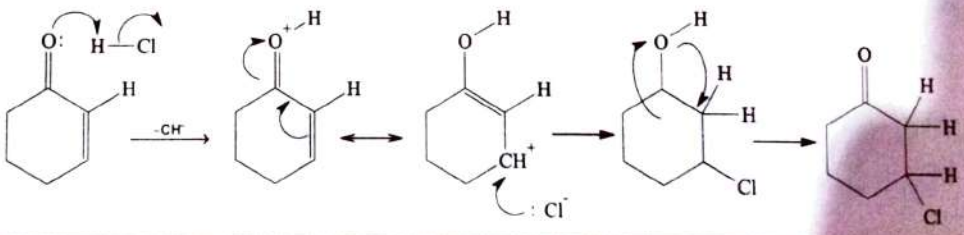
(iv)



6. Show all steps involved in the mechanism of the following reaction.



Sol.



7. Outline a method for synthesis of given compounds from the indicated

(i) Methanal to ethanal

(ii) Tert butyl alcohol from acetone

(iii) Acetylene to acetone

(iv) n–propyl alcohol from ethyl alcohol

(v) Glycerol from acetone

Sol. (i) $HCHO \xrightarrow{[O]} HCOOH \xrightarrow{Ca(OH)_2} (HCOO)_2Ca \xrightarrow[\Delta]{(CH_3COO)_2Ca} 2CH_3CHO$

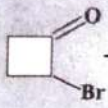
(ii) $(H_3C)_2C{=}O + CH_3MgBr \longrightarrow H_3C-C(CH_3)_2-O-MgBr \xrightarrow{H_3O^+} H_3C-C(CH_3)_2-OH + Mg(OH)Br$

(iii) $CH \equiv CH \xrightarrow{Hg^{+2}/H_2SO_4} CH_3CHO \xrightarrow{[O]} CH_3COOH \xrightarrow{Ca(OH)_2} (CH_3COO)_2Ca \xrightarrow[\text{distillation}]{\text{dry}} CH_3-C(=O)-CH_3$ (Acetone)

(iv) $H_3C-OH \xrightarrow{SOCl_2} H_3C-Cl \xrightarrow[\text{Dry ether}]{Mg} H_3C-MgCl \xrightarrow{HCHO} H_3C-CH_2-O-MgCl \xrightarrow{H_2O/H^+} H_3C-CH_2-OH$

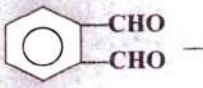
(v) $(H_3C)_2C{=}O \xrightarrow{LiAlH_4} (H_3C)_2CH-OH \xrightarrow[\Delta]{\text{conc } H_2SO_4} H_3C-CH{=}CH_2 \xrightarrow[500^oC]{Cl_2} ClH_2C-CH{=}CH_2 \xrightarrow{NaOH(aq.)} HO-CH_2-CH{=}CH_2 \xrightarrow{HOCl} HO-CH_2-CH(Cl)-OH \xrightarrow{NaOH(aq.)} HO-CH_2-CH(OH)-OH$

8. Identify A to O in the following reactions:

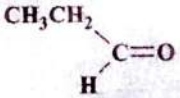
(a)  **A**

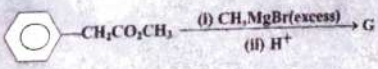
(b) **+ $Ph_3P = CH_2$ → B**

(c) **base → C**

(d)  **NaOH → D**

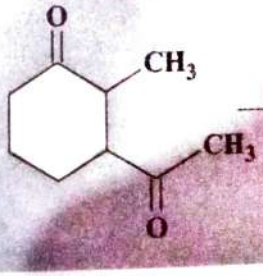
(e) **E**

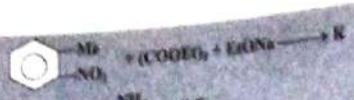
(f)  **1. KCN/H_2SO_4 2. $LiAlH_4$ → F**

(g)  **(i) CH_3MgBr(excess) (ii) H^+ → G**

(h) **KOH → HCOOK + H**

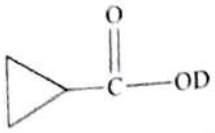
(i) **$Ph-CHO + CH_3-COOC_2H_5$** $\xrightarrow[\Delta]{NaOC_2H_5/EtOH}$ **I**

(j)  **$Br_2/NaOH$ → $CHBr_3$ + J**

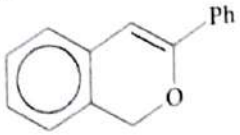
(k)  (l) Propanal $\xrightarrow[\Delta]{NaOH}$ L

(m) $HCHO \xrightarrow{NH_3} M$ (n) $ClCH_2CH_2CH_2COPh + KOH + MeOH \rightarrow N$

(o)

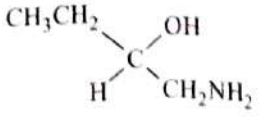
Sol. (a) 

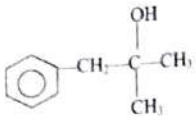
(b) (witting reaction)

(c) 

(d) (Intramolecular Cannizzaro reaction)

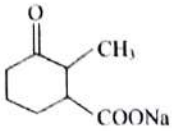
(e) $Al(OEt)_3$ (Tischenko reaction)

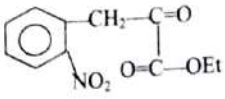
(f) 

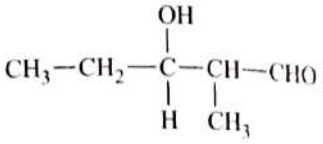
(g) 

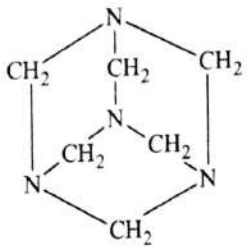
(h)

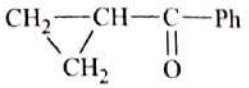
(i) $Ph-CH=CHCOOC_2H_5$

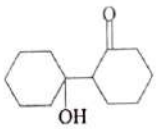
(j) 

(k) 

(l) 

(m)  Urotropine

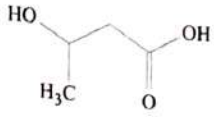
(n) 

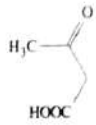
(o) 

9. Complete the following chemical equations:

(i) $C_4H_8O_3\ (A) \xrightarrow{CrO_3} (B) \xrightarrow{Warm} CH_3-\overset{O}{\overset{\|}{C}}-CH_3 + CO_2$

(ii) $C_2H_5Br \xrightarrow{KCN} (A) \xrightarrow{H_3O^+} (B) \xrightarrow{NH_3} (C) \xrightarrow{\Delta} (D) \xrightarrow{Br_2/OH^-} (E)$

Sol. (i) A = 

B = 

(ii) A = C_2H_5CN

C = CH_3COONH_4

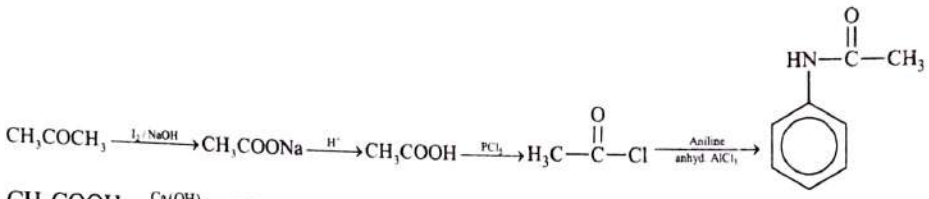
E = $C_2H_5NH_2$

B = $H_5C_2–COOH$

D = $C_2H_5CONH_2$

10. How will you synthesize
(i) acetanilide from acetone?
(ii) iodoform from acetic acid?
(iii) malonic acid from acetic acid?
(iv) crotonic acid from acetaldehyde?

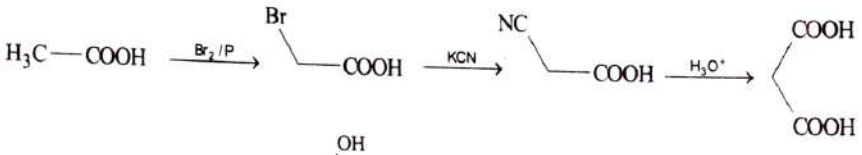
Sol. (i)



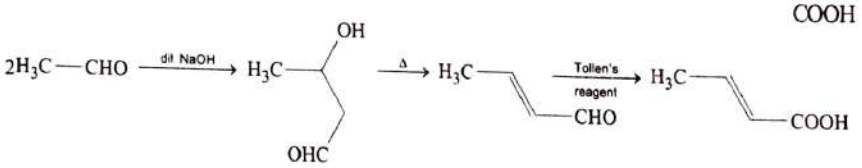
(ii) $CH_3COOH \xrightarrow{Ca(OH)_2} (CH_3COO)_2Ca \xrightarrow{\Delta} CH_3COCH_3$

$\xrightarrow{NaOH/I_2} CHI_3$

(iii)



(iv)



SOLVED PROBLEMS

Subjective Question

Prob. 1. Compound (A) C_8H_9Cl is hydrolysed by aqueous KOH to form (B), $C_8H_{10}O$, (B) on mild oxidation gives (C) which gives positive iodoform test what are (A), (B) and (C)

Sol. (A) $C_6H_5CHClCH_3$
(B) $C_6H_5CHOHCH_3$
(C) $C_6H_5COCH_3$

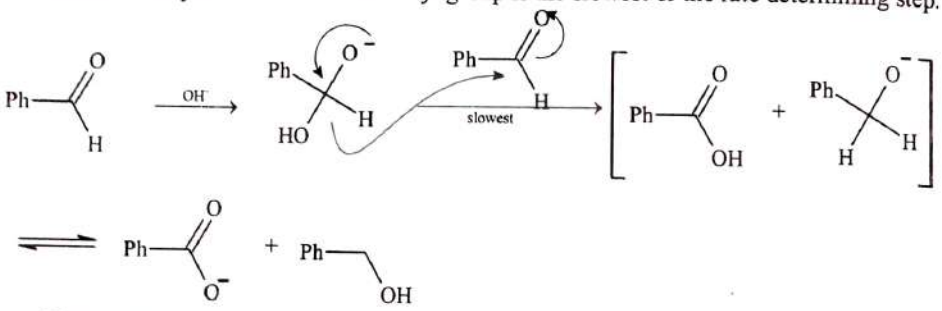
Objective Questions

Prob. 1. In Cannizzaro reaction given below:

$$Ph - CHO \xrightarrow{OH^-} Ph - CH_2 - OH + PhCOO^-$$

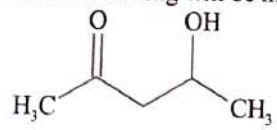
The slowest step is
(A) the attack of OH^- at the carbonyl group.
(B) the transfer of hydride to the carbonyl group
(C) the abstraction of proton from carboxylic acid
(D) the deprotonation of Ph – COOH

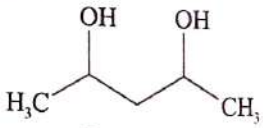
Sol. The transfer of hydride ion to the carbonyl group is the slowest or the rate determining step.

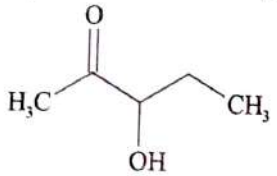


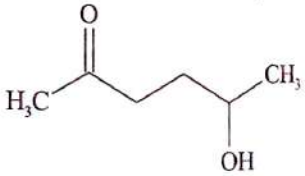
Hence, **(B)** is the correct answer.

Prob.2. Which of the following will be most readily dehydrated in acidic conditions?

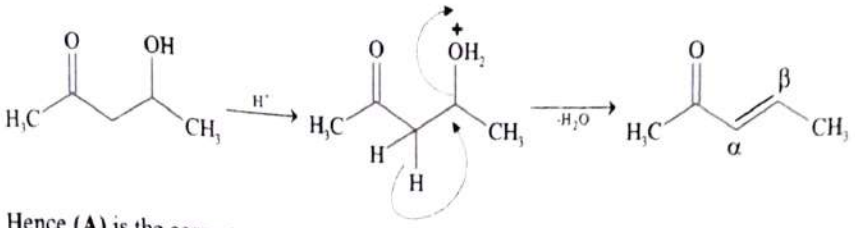
(A) 

(B) 

(C) 

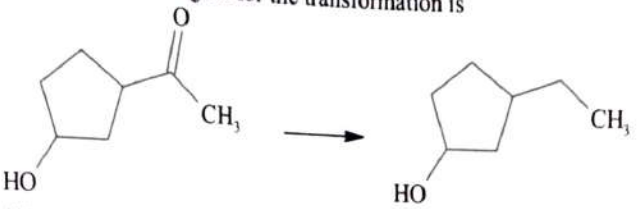
(D) 

Sol. Aldols, i.e. β–hydroxyaldehydes or β–hydroxyketones readily undergo dehydration to form α, β-unsaturated aldehydes or ketones.



Hence **(A)** is the correct answer.

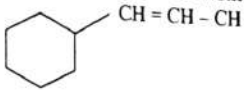
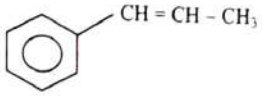
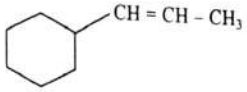
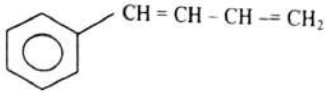
Prob. 3. The appropriate reagent for the transformation is



(A) Zn (Hg)/HCl
(B) $NH_2 - NH_2 / OH^-$
(C) Both (A) and (B)
(D) None of the above

Sol. Both Zn(Hg)/HCl and NH_2–NH_2/OH^- can reduce $COCH_3$ group to CH_2–CH_3 but HCl will also bring about replacement of OH group by Cl. Therefore, the most appropriate reagent is $NH_2 - NH_2/OH^-$.
Hence, **(B)** is the correct answer.

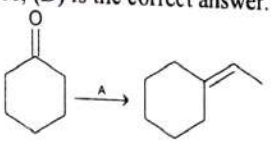
Prob. 4. Benzaldehyde on reaction with $CH_2 = CH - CHPPh_3$ forms

(A) 
(B) 
(C) 
(D) 

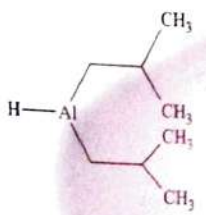
Sol. Wittig reaction

Hence, **(D)** is the correct answer.

Prob. 5.

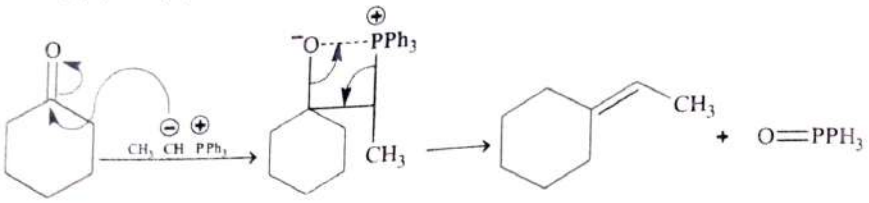


Reagent 'A' can be,

(A) CF_3CO_3H
(B) C_2H_5MgBr followed by conc. H_2SO_4
(C) $CH_3CH = P(Ph)_3$
(D) 

Hence, **(B)** and **(C)** are the correct answers.

Sol.



Prob. 6. Among the following which statement is/are correct statement?

(A) $CH_3-\overset{\overset{O}{\|}}{C}-OH$ will not respond to haloform test.

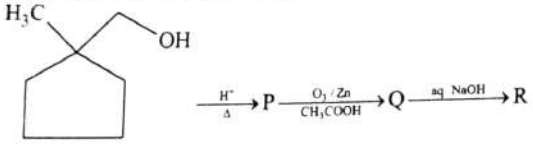
(B) Schiff's reagent and Schiff's base are different compounds.

(C) Fehling solution is a good reagent to detect aromatic aldehydes.

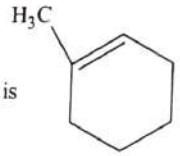
(D) Both aldehydes and ketones can react with 2, 4-dinitrophenylhydrazine reagent.

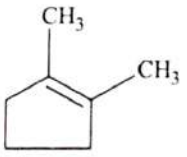
Sol. **(A, B, D)**

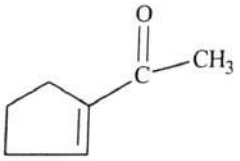
Prob. 7. About the following reaction

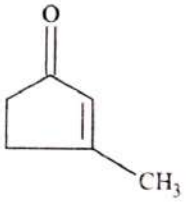


Which of the following option(s) is/are correct?

(A) Compound P is 

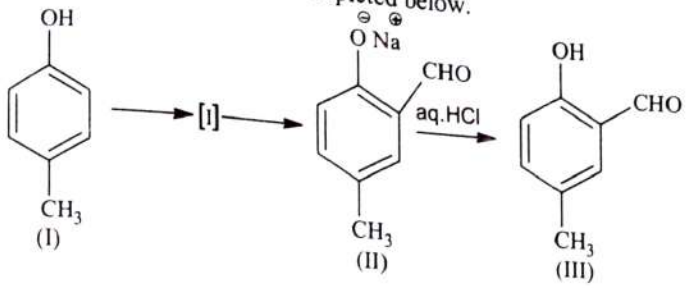
(B) Compound P is 

(C) Compound R is 

(D) Compound R is 

Sol. **(A, C)**

Reimer-Tiemann reaction introduces an aldehyde group, on to the aromatic ring of phenol, *ortho* to the hydroxyl group. This reaction involves electrophilic aromatic substitution. This is a general method for the synthesis of substituted salicyladehydes as depicted below.



1. Which one of the following reagents is used in the above reaction?

(A) aq. $NaOH + CH_3Cl$
(B) aq. $NaOH + CH_2Cl_2$
(C) alc. $NaOH + CHCl_3$
(D) alc. $NaOH + CCl_4$

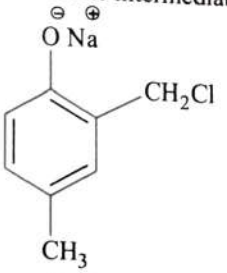
Sol. **(C)**

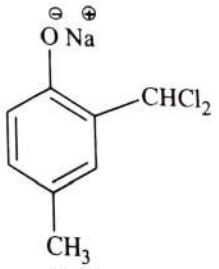
2. The electrophile in this reaction is

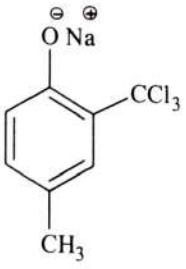
(A) : CHCl
(B) $^{+}CHCl_2$
(C) $:CCl_2$
(D) $.CCl_3$

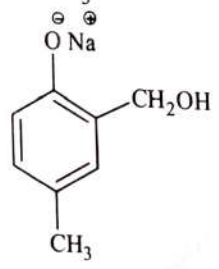
Sol. **(C)**

3. The structure of the intermediate I is

(A) 

(B) 

(C) 

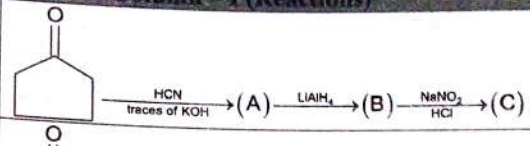
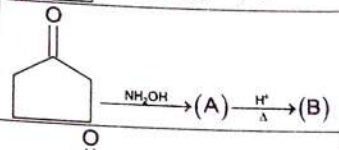
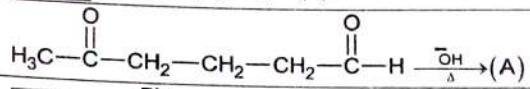
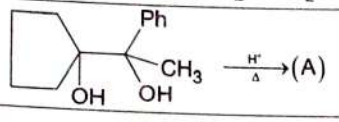
(D) 

Sol. **(B)**

MEGACOSM

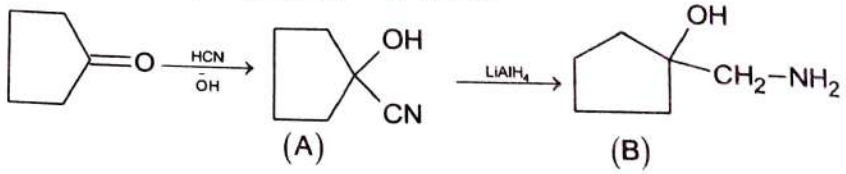
MATRIX – MATCH TYPE QUESTIONS

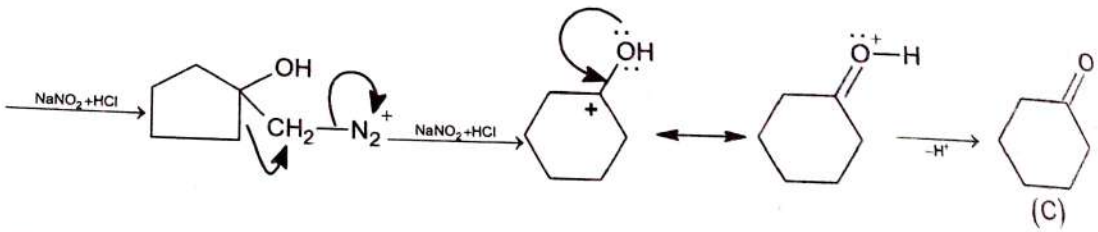
1. Match the following question:

Column – I (Reactions)		Column – II (Products)	
(A)		(p)	Formation of six membered ring takes place
(B)		(q)	Final product is ketone or α,β – unsaturated ketone
(C)		(r)	Final product formed will give positive Tollen's test
(D)		(s)	Final product formed will react with 2, 4-DNP
		(t)	Ring expansion takes place

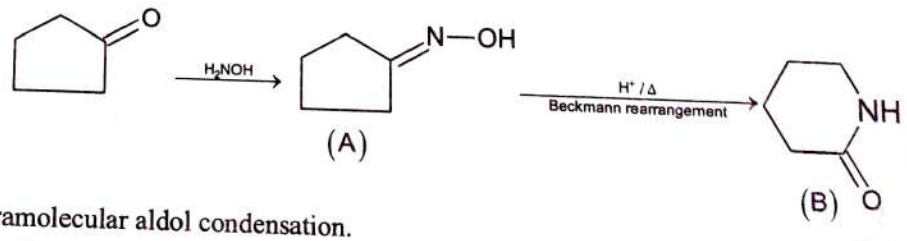
Sol. **((A – p, q, s, t) (B – p, t) (C – p, q, s) (D – p, q, s, t))**

(A)

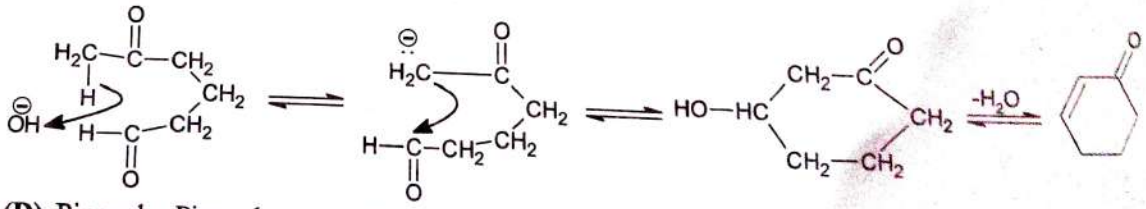




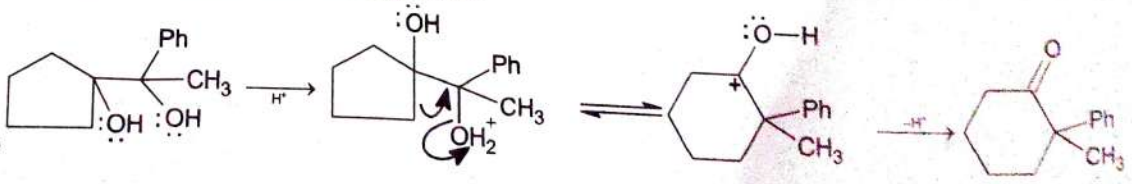
(B)



(C) Intramolecular aldol condensation.



(D) Pinacol – Pinacolone rearrangement.



2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Aldol condensation	(p)	CN^- as catalyst
(B)	Cannizzaro reaction	(q)	Organozinc compound
(C)	Benzoin condensation	(r)	Hydride transfer
(D)	Reformatsky reaction	(s)	β – hydroxyl aldehyde
		(t)	Enolate ion

Sol. **A → (s, t); B → (r); C → (p); D → (q)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Ph—CHO	(i)	Iodoform reaction	(P)	Precipitation
(II)	HCHO	(ii)	Fehling solution	(Q)	CH_3COO^-
(III)	CH_3COCH_3	(iii)	Aldol condensation	(R)	$HCOO^-$
(IV)	CH_3CHO	(iv)	Cannizzaro reaction	(S)	CH_3—CH=CH—CHO

3. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (II) (iii) (R) [C] (III) (iii) (S) [D] (IV) (ii) (P)

4. Which of the following combination is incorrect ?
[A] (IV) (iii) (S) [B] (III) (i) (Q) [C] (I) (i) (Q) [D] (IV) (i) (P)

5. Which of the following combination is correct ?
[A] (IV) (i) (Q) [B] (I) (i) (R) [C] (II) (i) (Q) [D] (II) (i) (P)

Sol.:

3. **D**

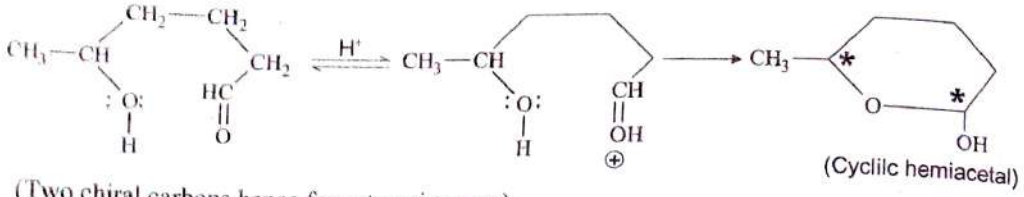
4. **C**

5. **A**

SINGLE INTEGER ANSWER TYPE QUESTIONS

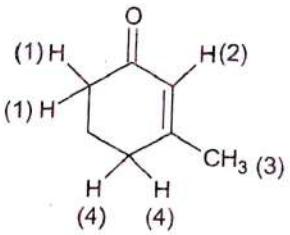
1. 5-Hydroxyhexanal forms a six member hemiacetal, which predominates at equilibrium in aqueous solution. How many stereoisomers are possible for this cyclic hemiacetal?

Sol. **(4)**



(Two chiral carbons hence four stereoisomers)

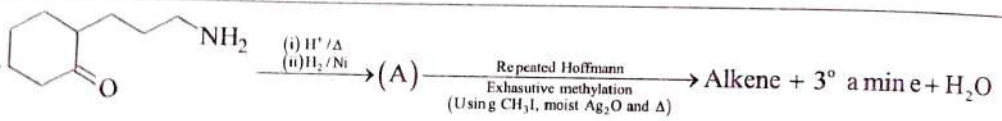
2. Which hydrogen of the given compound is least acidic in nature?



Sol. **(2)**

NUMERICAL VALUE ANSWER TYPE

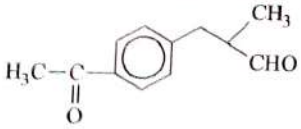
Prob. 1.



The number of moles of CH_3–I consumed per mole of compound (A) is x. The value of $\frac{x}{2}$

Sol. **1.5**

Prob. 2 The number of possible isomeric product obtained from the reduction of



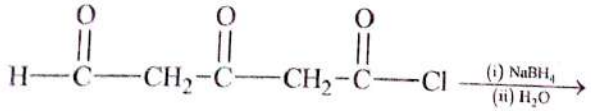
with $NaBH_4$ would be x. Then value of $\frac{2x}{3}$

Sol. **1.33**

Prob. 3. Formaldehyde reacts with ammonia to produce a compound which is used to treat urinary infections. X number of tertiary nitrogen atoms are present in one molecule of the product formed. Then value of $\frac{x}{3}$ is

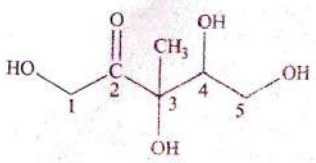
Sol. **1.33**

Prob. 4. The number of –OH groups present in the end product is x. Then value of $\frac{x}{2}$



Sol. **1.5**

Prob. 5. Alcoholic group at position x reacts fastest with Lucas reagent. Then value of $\frac{2x}{5}$



Sol. **1.2**

CARBOXYLIC ACID & ITS DERIVATIVES

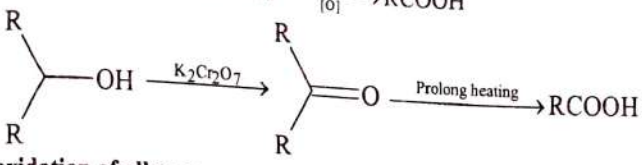
18

The organic compounds containing one or more carboxyl groups are called carboxylic acids.
The general formula of carboxylic acids are
Saturated acid: $C_nH_{2n+1}COOH$ or $C_nH_{2n}O_2$
Unsaturated acid : $C_nH_{2n-1}COOH$

METHODS OF PREPARATION

(i) **By oxidation of alcohol, aldehyde & ketone:**

(a) $RCH_2OH \xrightarrow{K_2Cr_2O_7} RCHO \xrightarrow[[O]]{K_2Cr_2O_7} RCOOH$

(b) 

(ii) **By oxidation of alkene:**

(a) $R-CH=CH-R' \xrightarrow[(ii)\ H_2O,\ \Delta]{(i)\ KMnO_4} RCOOH + R'COOH$

(b) Carbonylation:

$CH_2=CH_2 + CO + H_2O \xrightarrow[300-400^\circ C]{H_3PO_4} CH_3CH_2COOH$

(iii) **From alkyne:**

(a) $CH \equiv CH + 3[O] + H_2O \xrightarrow[KMnO_4]{\text{alkaline}} 2HCOOH$

(b) $R-C \equiv CH + 3[O] + H_2O \xrightarrow[KMnO_4]{\text{alk.}} R-COOH + HCOOH$

(c) $CH \equiv CH + H_2O + [O] \xrightarrow{\text{Chromic acid}} CH_3-COOH$

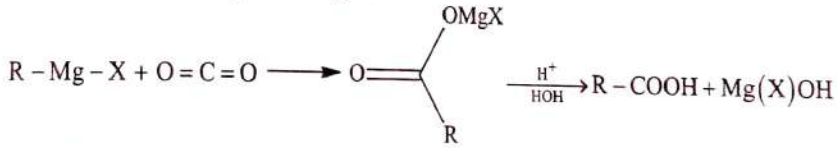
[Note: In presence of chromic acid single carboxylic acid is formed.]

(iv) **By hydrolysis of trihalogen derivatives:**

$R-CX_3 \xrightarrow{H_2O} [R-C(OH)_3] \xrightarrow{-H_2O} R-COOH$

Unstable

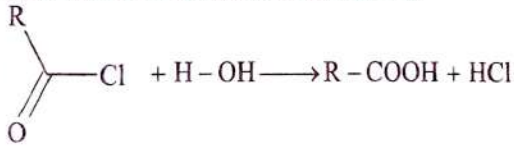
(v) **By carbonation of Grignard reagent:**

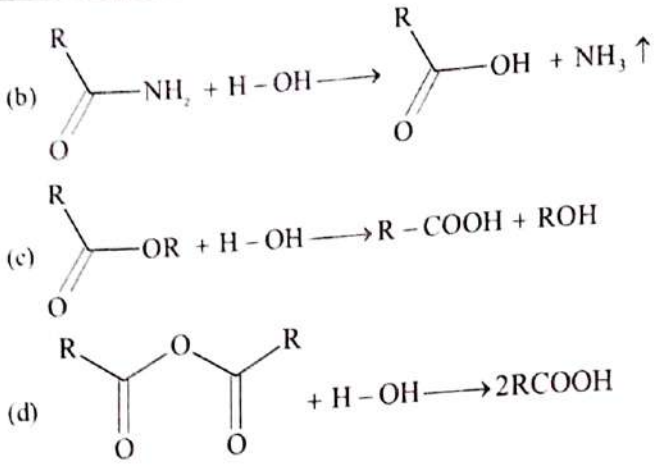


(vi) **Hydrolysis of cyanide:**

$R-C \equiv N + H_2O \xrightarrow{H^+} R-COOH + NH_3$

(vii) **By hydrolysis of carboxylic acid derivatives:**

(a) 

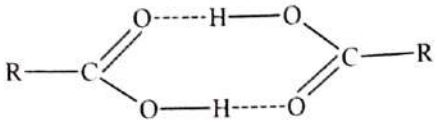


PHYSICAL PROPERTIES

(i) Molecules of carboxylic acid are polar and show intermolecular hydrogen bonding.

(ii) C_1–C_4 are highly soluble in water due to hydrogen bonding.

The boiling point of carboxylic acids is higher than alcohol due to presence of hydrogen bonding. It exist in dimeric form.



CHEMICAL PROPERTIES

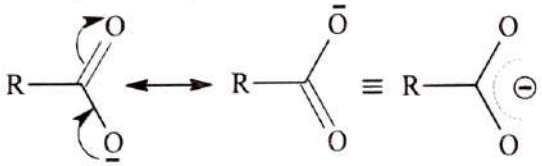
Acidic Strength

(i) Acid strength $\propto$ $-I$ effct

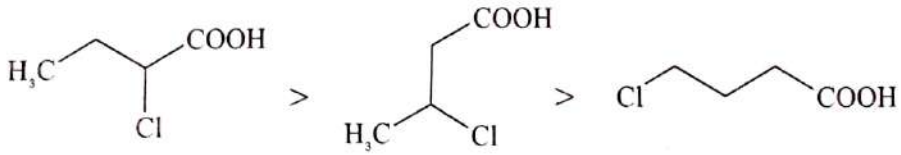
(ii) Acid strength α $\frac{1}{+I \text{ effect}}$

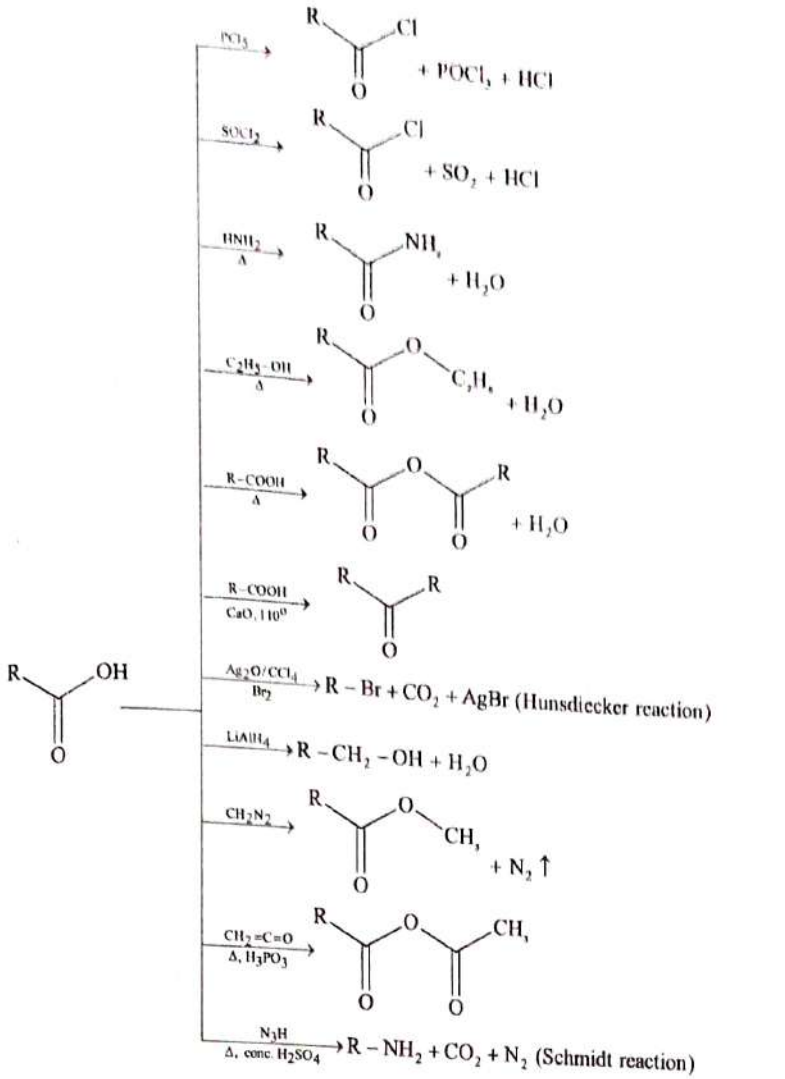
$RCOOH > H\text{–}OH > ROH > CH \equiv CH > NH_3 > RH$

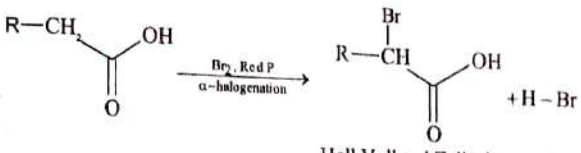
Acidity of carboxylic acid is due to resonance stabilization of its anion.



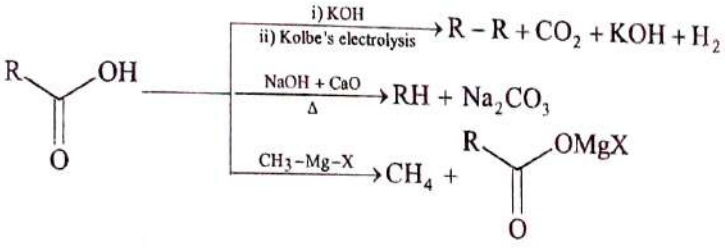
$Cl_3CCOOH > Cl_2CHCOOH > ClCH_2COOH > CH_3COOH$







Hell Volhard Zelinsky reaction



Identification

(i) Aqueous solution of the acid turns blue litmus red.

(ii) The aqueous solution of the acid with $NaHCO_3$ gives effervescence of CO_2 and RCOONa.

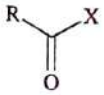
$$R-COOH + NaHCO_3 \longrightarrow R-COONa + H_2O + CO_2 \uparrow$$

(iii) On heating with alcohol fruity smell is formed.

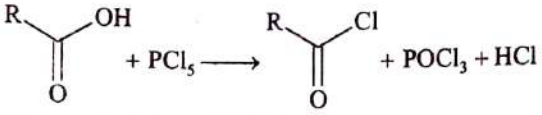
$$RCOOH + ROH \overset{H^+}{\rightleftharpoons} RCOOR + H_2O$$

Acid Derivatives

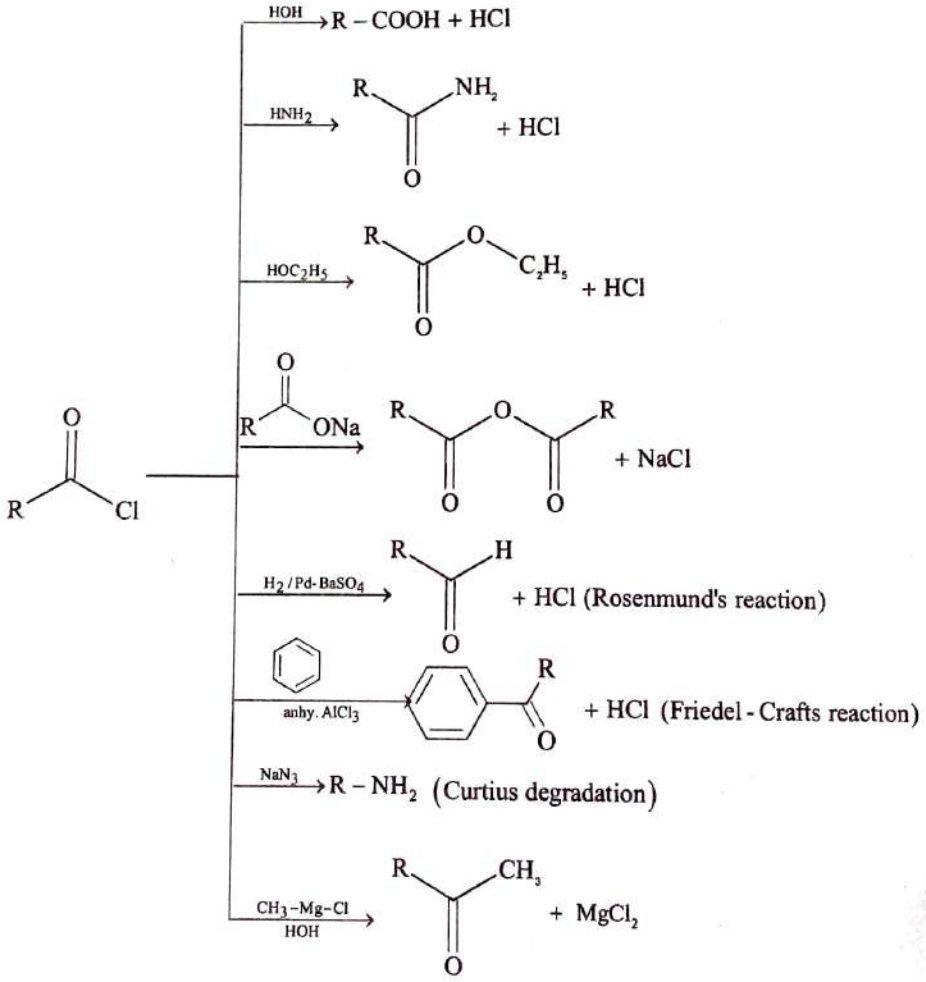
Acid Halide



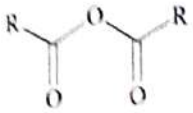
Methods of Preparation



Chemical Properties



ACID ANHYDRIDE



Methods of Preparation

(i) $H_3C-COOH + CH_2=C=O \xrightarrow[\Delta]{AlPO_4} H_3C-CO-O-CO-CH_3$

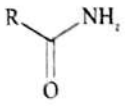
(ii) $2\,CH_3-COOH \xrightarrow[-H_2O]{P_2O_5} H_3C-CO-O-CO-CH_3$

Chemical Properties

$(RCO)_2O$ reacts with:

- H_2O → $2R-COOH$
- HNH_2 → $R-CO-NH_2 + RCOOH$
- C_2H_5OH → $R-CO-O-C_2H_5 + RCOOH$
- PCl_5 → $R-CO-Cl$
- $LiAlH_4$ → RCH_2OH

AMIDE

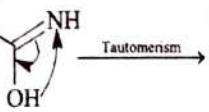
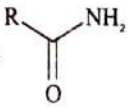


Methods of Preparation

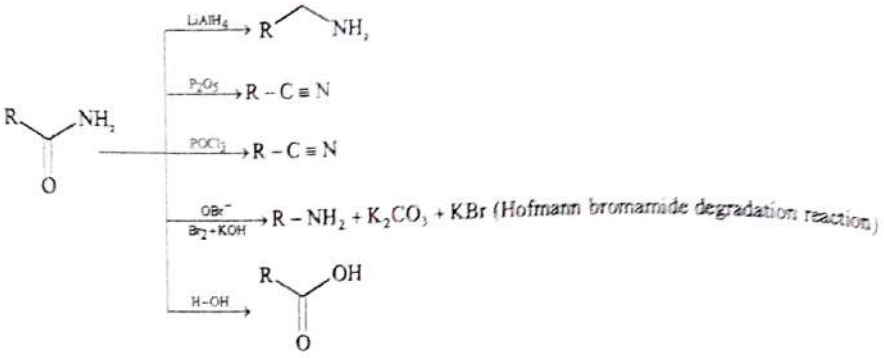
(i) $R-CO-Cl + H-NH_2 \longrightarrow R-CO-NH_2 + HCl$

(ii) $R-CO-OH + H-NH_2 \xrightarrow{\Delta} R-CO-NH_2 + H_2O$

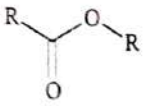
(iii) **By partial hydrolysis of cyanide:**

$R-C\equiv N + H^+ + OH^- \longrightarrow$  $\xrightarrow{\text{Tautomerism}}$ 

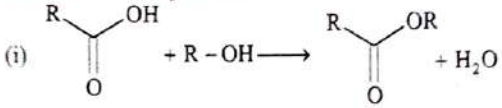
Chemical properties

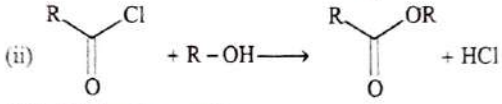


ESTER

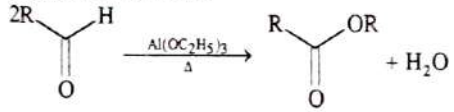


Methods of Preparation

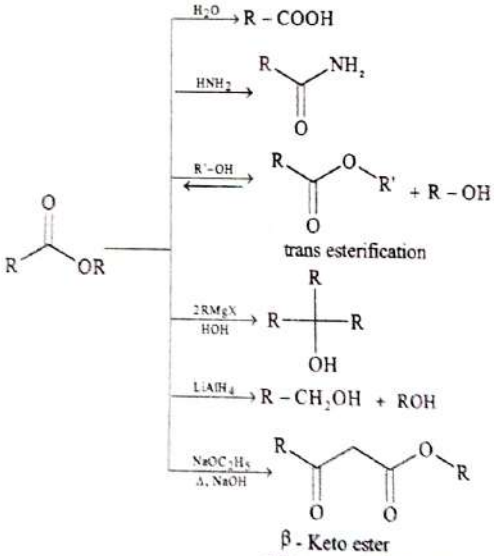
(i) 

(ii) 

(iii) **Tischenko reaction:**



Chemical Properties



ILLUSTRATIVE PROBLEMS

1. The neutralization equivalent of an acid is 116. Fusion of sodium salt of the acid with sodalime gives a hydrocarbon which is a monosubstituted derivative of butane. What structural formula may be assigned to the acid?

Sol. The final product is an alkane derivative; this indicates the acid is mono carboxylic, RCOOH.
As neutral equivalent is 116, therefore, formula weight of alkyl group R is 116 – 45 (Formula weight of –COOH) = 71.
This corresponds to the pentyl group (C_5H_{11}). The formation of butane derivative suggests a branched alkyl radical containing four carbon atoms in the largest chain. Therefore, R should be

$$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2$$

and the acid may be

$$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-COOH$$ (3-methyl pentanoic acid),

$$CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle COOH}{\overset{|}{C}}}}-CH_3$$,

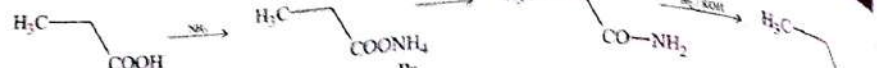
$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle COOH}{\overset{|}{CH}}}}-CH-CH_3$$,

$$CH_2-CH_2-\underset{\displaystyle CH_3}{\underset{|}{\overset{\displaystyle COOH}{\overset{|}{CH}}}}-CH_3$$

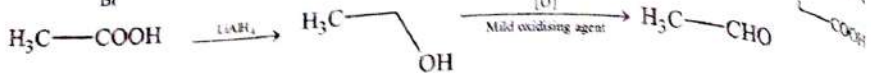
2. Will the following equations proceed easily? Explain.
(a) $CH_3COCl + H_2O \longrightarrow CH_3COOH + HCl$
(b) $CH_3CONH_2 + NaOH \longrightarrow CH_3COO^-Na^+ + NH_3$
(c) $(CH_3O)_2O + NaOH \longrightarrow CH_3COBr + CH_3OH$
(d) $CH_3COOCH_3 + HBr \longrightarrow CH_3COBr + CH_3OH$

Sol. All the above reactions are nucleophilic substitution of acyl compounds.
Such reactions are favoured by the following conditions:
(i) If the incoming group ($Nu^{\cdot\cdot}$) is a stronger nucleophile than the leaving group (Y:).
(ii) If the final product is resonance stabilized.
(a) Yes, ^-OH is a stronger nucleophile than Cl^-
(b) Yes, even though $\ddot{N}H_2$ is a stronger base than $\ddot{O}H$ in basic solution, which shifts the reaction to back side. In basic solution, the resonance stabilized $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-O^-$ formed, which shift the reaction to completion.
(c) Yes, because the leaving group $CH_3-\overset{\displaystyle O}{\overset{\|}{C}}-O^-$ is a weaker base than ^-OH
(d) No, Br^- is a weaker base than OH^-

3. How would you bring about the following conversions?
(i) Propionic acid to ethylamine
(ii) Ethyl bromide to succinic acid
(iii) Acetic acid to acetaldehyde

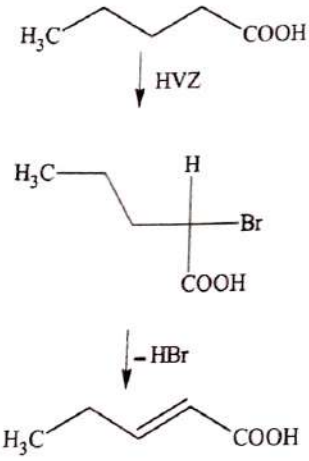
Sol. (i) 

(ii)

(iii) 

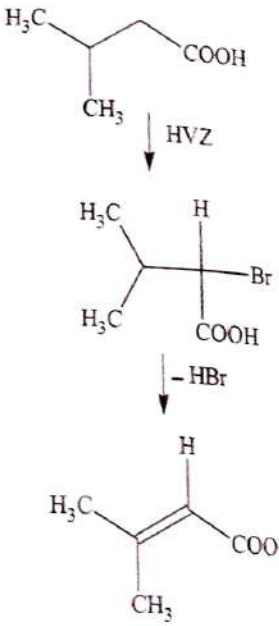
4. **An organic acid (A) ($C_5H_{10}O_2$) reacts with Br_2 in presence of phosphorus to give (B). Compound (B) contains an asymmetric carbon atom and yields (C) on dehydrohalogenation. Compound (C) does not show geometric isomerism and on decarboxylation gives an alkene (D) which on ozonolysis gives (E) and (F). Compound (E) gives a positive Schiff's test but (F) doesn't. Give structures of (A) to (F) with reasons.**

Sol. (E) is an aldehyde and (F) is a ketone. The number of C–atom in (A) , (B) and (C) are 5 and in (D) just four. It's also clear that (A) has atleast one α-hydrogen as it is showing HVZ reaction. Two options are possible for (A)



(Show geometrical isomerism)

or



(Does not show geometrical isomerism)

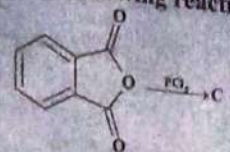
$$\therefore \text{ (A) } CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}H-CH_2-COOH \qquad \text{(B) } CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}H-CH(Br)-COOH$$

$$\text{(C) } (CH_3)_2C=CH-COOH \qquad \text{(D) } CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=CH_2$$

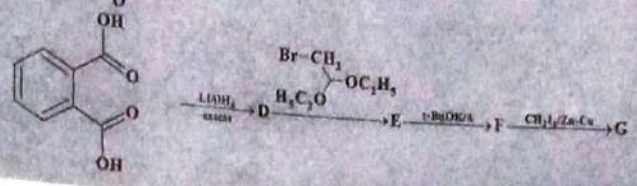
$$\text{(E) } H-\underset{\displaystyle H}{\overset{|}{C}}=O \qquad \text{(F) } CH_3-\underset{\displaystyle CH_3}{\overset{|}{C}}=O$$

5. Complete the following reactions:

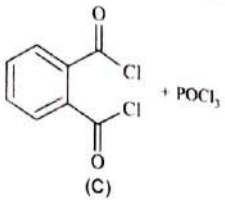
(i)



(ii)

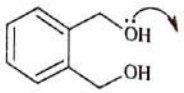


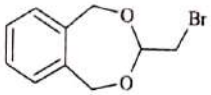
Sol. (i)

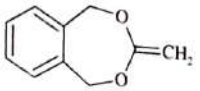


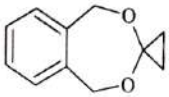
(C)

(ii)

D = 

E = 

F = 

G = 

6. Compound A is acidic in reaction, having the molecular formula $C_4H_8O_3$, is oxidized with mild oxidizing agents to give B which is unstable, syrupy substance. It easily produces, C, C_3H_6O and carbon dioxide. On heating A alone yields D, $C_4H_6O_2$ an acid neutralization equivalent of 104. What are A, B,C and D? Give their structural formulae too.

Sol. Compound A has molecular formula $C_4H_8O_3$ as it is acidic in reaction. So, it should have –COOH group. The remaining group C_3H_7O contains oxygen. The product D, having formula C_3H_5COOH, an unsaturated acid is obtained by heating compound A.

β-hydroxy substituted acids yield substituted acids on heating. Hence, A should be β-hydroxy substituted acid, i.e.

$$(A) = \begin{array}{l} \quad\quad OH \\ CH_3 - \underset{|}{C}H - CH_2 - COOH \end{array}$$

(A) on oxidation with mild oxidizing reagents will give

$$\begin{array}{l} \quad\quad OH \\ CH_3 - \underset{|}{C}H - CH_2 - COOH \end{array} \qquad \begin{array}{l} \quad\quad O \\ CH_3 - \underset{\|}{C} - CH_2 - COOH \end{array}$$

B is keto substituted acid, hence on further decomposition it will eliminate CO_2 to give

$$\begin{array}{l} \quad\quad O \\ CH_3 - \underset{\|}{C} - CH_2 - COOH \end{array} \qquad CH_3COCH_3$$

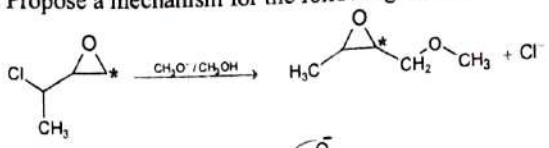
(C) Acetone

$$\begin{array}{l} \quad\quad OH \\ CH_3 - \underset{|}{C} - CH_2 - COOH \\ \quad\quad (A) \end{array} \qquad \begin{array}{l} CH_3 - CH = CH - COOH \\ \quad\quad (D) \end{array}$$

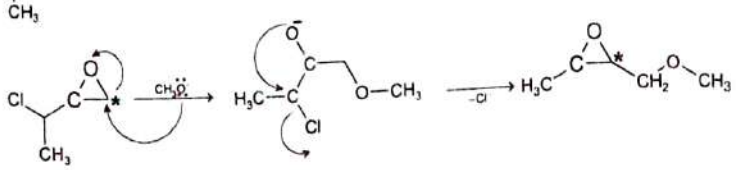
SOLVED PROBLEMS

Subjective Question

Prob. 1. Propose a mechanism for the following reaction.



Sol.

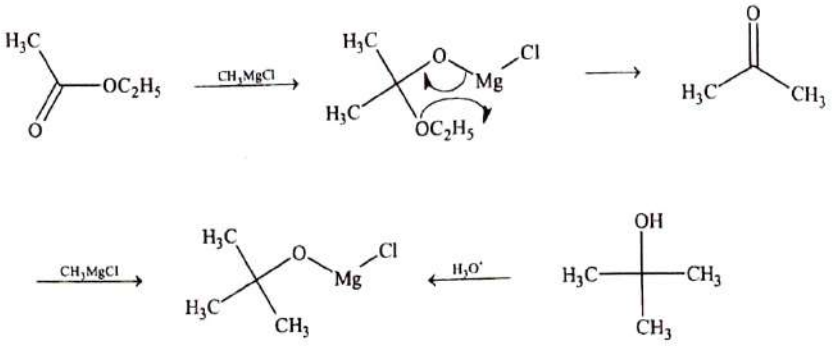


Objective Questions

Prob. 1. Ethyl acetate on reaction with excess of methyl magnesium chloride and dil. H_2SO_4 gives

(A) dimethyl ketone
(B) iso propyl alcohol
(C) ethyl aceto acetate
(D) t- butyl alcohol

Sol.



Hence, **(D)** is the correct answer.

Prob. 2. The correct order of increasing boiling point of the following compound is

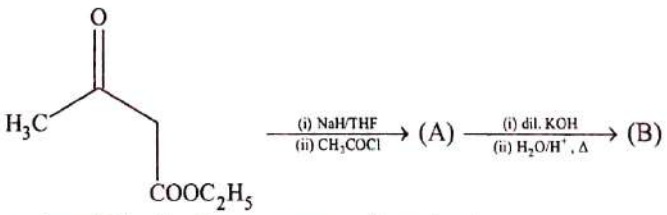
(I) (II) (III) (IV)

(A) I < II < III < IV
(B) II < III < IV < I
(C) II < IV < III < I
(D) II < III < I < IV

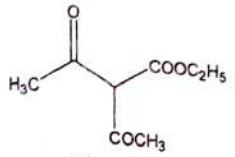
Sol. II < III < IV due to higher mol. wt. and (I) has got highest boiling point because of the presence of H-bonding.

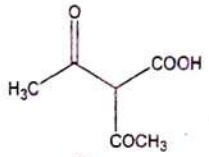
Hence, **(B)** is the correct answer.

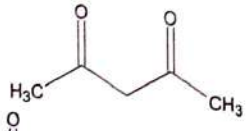
Prob. 3.

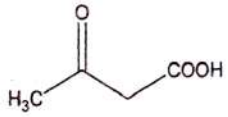


The product (B) in the above sequence of reaction is

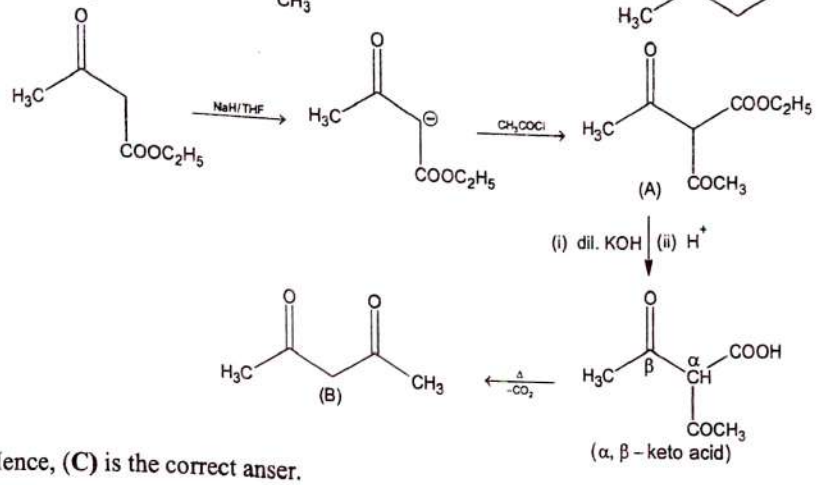
(A) 

(B) 

(C) 

(D) 

Sol.



Hence, **(C)** is the correct anser.

Prob. 4. Chlorination of toluene in presence of light and heat followed by treatment with aqueous NaOH gives

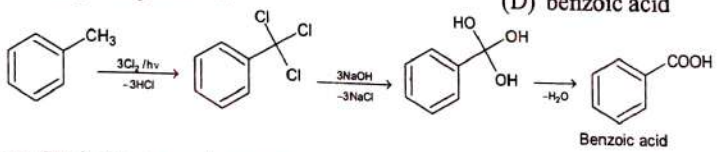
(A) o-cresol

(B) p-cresol

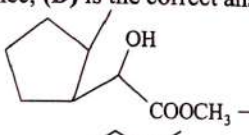
(C) 2,4-dihydroxy toluene

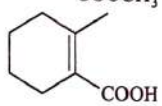
(D) benzoic acid

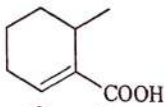
Sol.

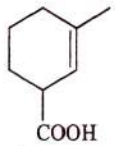


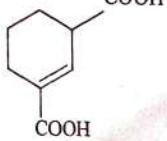
Hence, **(D)** is the correct answer.

Prob. 5.  P; 'P' as a major product must be,

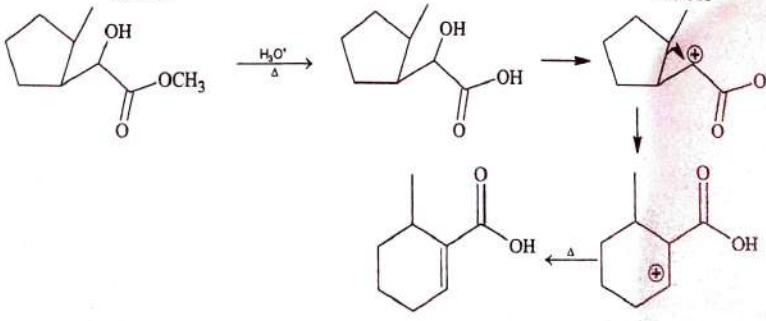
(A) 

(B) 

(C) 

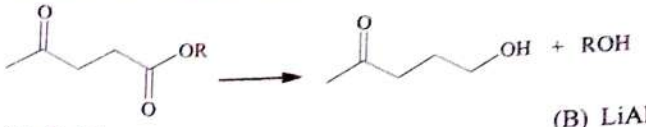
(D) 

Sol.



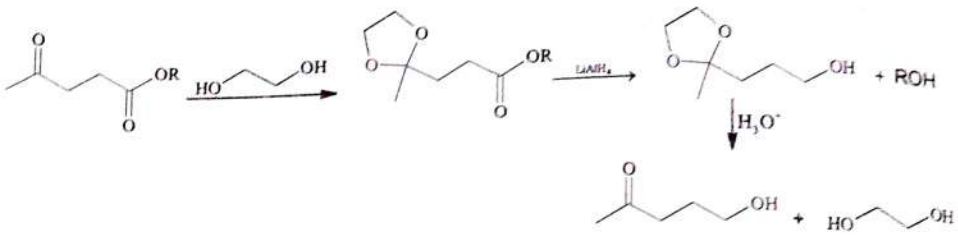
Hence, **(B)** is the correct answer.

Prob. 6. Which of the following reagents are involved in the following transformation?



(A) H_3O^+
(B) $LiAlH_4$
(C) Ethylene glycol
(D) Acetone

Sol.



Hence, **(A), (B)** and **(C)** are the correct answers.

Prob. 7. In the given reaction :

$$RCOOAg + X_2 \xrightarrow{\Delta} R\text{–}X + CO_2 + AgX$$

X_2 can be:

(A) Cl_2
(B) Br_2
(C) I_2
(D) F_2

Sol. **(A, B)**

Prob. 8. Which of the following on reduction with $LiAlH_4$ will give ethyl alcohol?

(A) $(CH_3CO)_2O$
(B) CH_3COCl
(C) CH_3CONH_2
(D) $CH_3COOC_2H_5$

Sol. Only CH_3CONH_2 gives 1° amine, rest all give ethyl alcohol as shown below:

(A) $(CH_3CO)_2O \xrightarrow{LiAlH_4} 2CH_3CH_2OH$
(B) $CH_3COCl \xrightarrow{LiAlH_4} CH_3CH_2OH + HCl$
(C) $CH_3CONH_2 \xrightarrow{LiAlH_4} CH_3CH_2NH_2$
(D) $CH_3COOC_2H_5 \xrightarrow{LiAlH_4} 2CH_3CH_2OH$

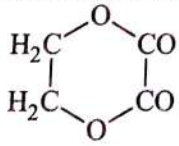
Hence, **(A), (B)** and **(D)** are the correct answer.

Prob. 9. The formula $C_4H_4O_4$ can represent

(A) a cyclic ester of dibasic acid
(B) a trans-dibasic acid
(C) a cis-dibasic acid
(D) an α,β – unsaturated dibasic acid which on heating forms a monobasic acid

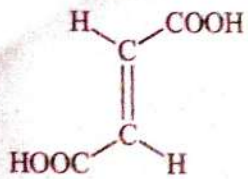
Sol. The $C_4H_4O_4$ can represent

(A)

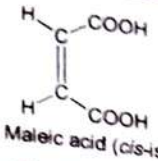


Ethylene oxalate

(B)



Fumaric acid (*trans*-isomer)

(C) 

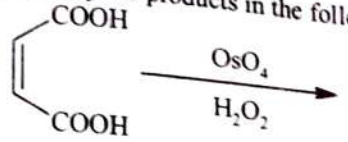
Maleic acid (*cis*-isomer)

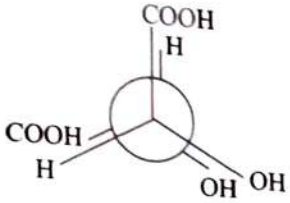
(D)

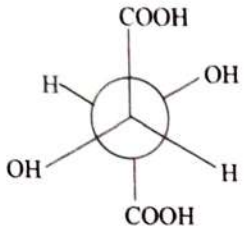
α, β - Unsaturated acid

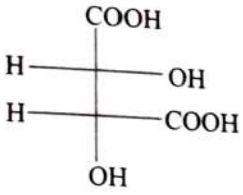
Hence, **(A), (B), (C)** and **(D)** are the correct answers.

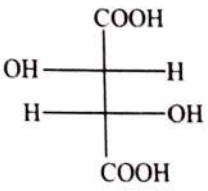
Prob. 10. Identify the products in the following reaction and their configurations ?



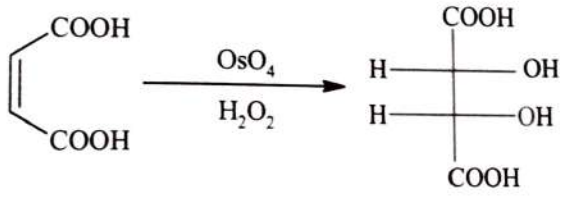
(A) 

(B) 

(C) 

(D) 

Sol. **(B)**



meso

(Formed by syn addition)

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Unlike the >C = O group of aldehydes and ketones, the >C = O of $R-\overset{O}{\overset{\|}{C}}-OH$ does not undergo nucleophilic addition reactions.
and
STATEMENT-2: Carboxylic acids exist as dimers due to intermolecular hydrogen bonding.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

2. **STATEMENT-1:** $CH_3COCH_2COOC_2H_5$ will give iodoform test.
and
STATEMENT-2: It contains CH_3CO group linked to a carbon atom.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

3. **STATEMENT-1:** Acetic acid does not undergo haloform reaction.
and
STATEMENT-2: Acetic acid has no α hydrogen.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**

4. **STATEMENT-1:** Carboxylic acids react with diazomethane.
and
STATEMENT-2: The CH_2 group of diazomethane becomes the o-methyl group of esters.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

5. **STATEMENT-1:** The solubility of maleic acid in water is higher than that of fumaric acid.
and
STATEMENT-2: Maleic acid and fumaric acid are cis-trans isomers.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

COMPREHENSION

The acid strength of saturated aliphatic carboxylic acids depends mainly upon the inductive effect of substituent and its position with respect to the COOH group. The acid strength of aromatic carboxylic acids, on the other hand, depends upon both the inductive and the resonance effect of the substituents. Their trends are, however, similar to those of aliphatic acids except following two differences

(a) The acids weakening effect of electron donating substituents and acid-strengthen effect of the electron – withdrawing substituents is more pronounced at p-position than at m-position.
(b) Due to ortho effect, o-substituted benzoic acids are usually stronger than benzoic acid regardless of the nature of substituent whether electron donating or electron withdrawing.

1. Which of the following acids has the smallest dissociation constant?
(A) $CH_3CHFCOOH$ (B) $FCH_2 - CH_2COOH$
(C) $BrCH_2 - CH_2COOH$ (D) $CH_3 - CHBrCOOH$

Sol. **(C)**

2. Among the following, the strongest acid is:
(A) $ClCH_2COOH$ (B) Cl_2CH_2COOH
(C) Cl_3COOH (D) CH_3COOH

Sol. **(C)**

MATRIX-MATCH TYPE QUESTION

1. Match the following:

Column – I (Reagents reacting with $PhCH_2COOH$)		Column – II (Product formed)	
(A)	CH_3MgBr	(p)	$PhCH_2COCl$
(B)	PCl_5	(q)	$PhCH_2COOCH_3$
(C)	NH_3, followed by heating	(r)	CH_4
(D)	CH_3OH in the presence of conc. H_2SO_4	(s)	$PhCH_2CONH_2$

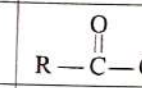
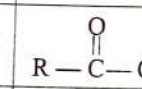
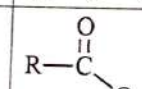
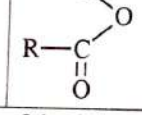
Sol. **A → (r); B → (p); C → (s); D → (q)**

2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	$C_6H_5COCH_3$	(p)	HVZ reaction
(B)	C_6H_5CHO	(q)	Fehling's solution
(C)	$CH_3 - CH_2COOH$	(r)	Sodium bisulphate solution
(D)	HCOOH	(s)	Sodium hypoiodite
		(t)	Reduces Tollen's reagent

Sol. **A → (s) B → (q) C → (p) D → (q, t)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	R—COOH	(i)	$R—CH_2OH$	(P)	H_2/Pd
(II)		(ii)		(Q)	B_2H_6/THF
(III)		(iii)	R'—OH	(R)	$NaBH_4$
(IV)		(iv)	R—CHO	(S)	$LiAlH_4$

3. Which of the following combination is correct ?
[A] (I) (i) (S) [B] (II) (iii) (P) [C] (I) (iv) (Q) [D] (IV) (i) (R)

4. Which of the following combination is incorrect ?
[A] (II) (i) (S) [B] (III) (i) (S) [C] (III) (iii) (P) [D] (II) (ii) (R)

5. Which of the following combination is correct ?
[A] (IV) (ii) (P) [B] (II) (i) (R) [C] (III) (ii) (R) [D] (IV) (iv) (S)

Sol.

3. **A**

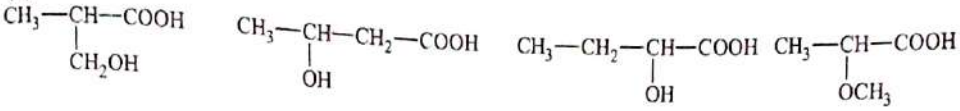
4. **D**

5. **B**

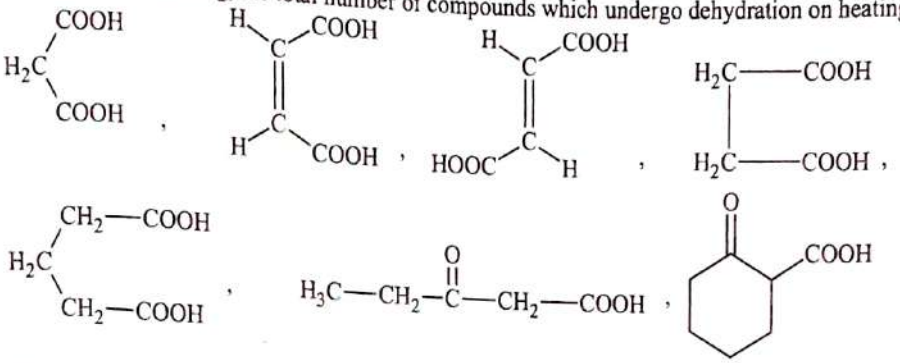
SINGLE INTEGER ANSWER TYPE QUESTIONS

1. What is the number of optically active structural isomers of $C_4H_8O_3$. Which evolve CO_2 with aq. $NaHCO_3$.

Sol. **(4)**



2. Amongst the following, the total number of compounds which undergo dehydration on heating is

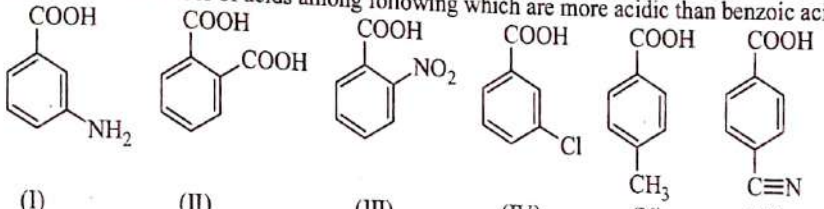


Sol. **(3)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Find the value of 'X' if $X = \frac{Y}{5}$

and Y = Total number of acids among following which are more acidic than benzoic acid ?



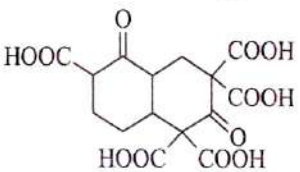
(I) (II) (III) (IV) (V) (VI)

Sol. **0.80**

$Y = 4 \qquad X = \frac{4}{5} = 0.80$

Prob. 2. If 'X' = Number of moles of CO_2 produced on heating compound (A)

Find the value of $Z = \frac{X}{10}$



(Compound – A)

Sol. **0.50**

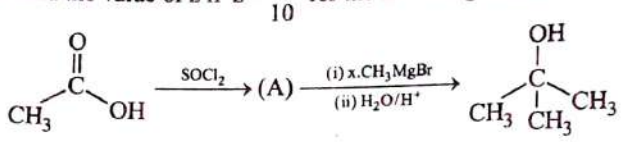
$X = 5 \qquad \therefore Z = \frac{5}{10} = 0.50$

Prob. 3. If 'Y' is total number of chlorine atom present in product obtained on reaction between succinic acid and PCl_5, then what is the value of $\frac{Y}{4}$

Sol. **0.5**

$Y = 2 \quad \frac{2}{4} = 0.5$

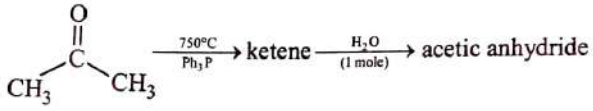
Prob. 4. Find the value of z if $z = \frac{x}{10}$ for the following sequence of reactions.



Sol. **0.2**

$x = 2 \quad \frac{2}{10} = 0.2$

Prob. 5. For the gives sequence of reaction if X is equal to moles of ketene consumed to give acetic anhydride, then what is the value of $\frac{'X'}{10}$.



Sol. **0.2**

$X = 2 \quad \frac{2}{10} = 0.2$

AMINES 19

BASICITY OF AMINES

1. **Medium:** Polar protic
   Order of basicity: $(CH_3)_2NH > CH_3NH_2 > (CH_3)_3N > NH_3$
   Reason: Due to solvation effect.
2. **Medium:** Non-polar
   Order of basicity: $(CH_3)_3N > (CH_3)_2NH > CH_3NH_2 > NH_3$
   Reason: Only inductive effect is considerable.
3. **Medium:** Polar protic
   Order of basicity: $Et_2NH > Et_3N > EtNH_2 > NH_3$
   Reason: Due to solvation effect.
4. **Medium:** Non-polar
   Order of basicity: $Et_3N > Et_2NH > EtNH_2 > NH_3$
   Reason: Only inductive effect is considerable.
5. **Medium:** Any medium
   Order of basicity: $p\text{-}CH_3C_6H_5NH_2 > m\text{-}CH_3C_6H_5NH_2 > C_6H_5NH_2 > o\text{-}CH_3C_6H_5NH_2$
6. **Medium:** Any medium
   Order of basicity: $PhNH_2 > m\text{-}NO_2C_6H_4NH_2 > m\text{-}HOC_6H_4NH_2$
7. **Medium:** Any medium
   Order of basicity: $p\text{-}CH_3OC_6H_4NH_2 > C_6H_5NH_2 > o\text{-}CH_3OC_6H_4NH_2 > m\text{-}CH_3OC_6H_4NH_2$

METHODS OF PREPARATION

(i) $\underset{\text{Alkyl halide}}{RX} \xrightarrow{NH_3} R-NH_2$

If NH_3 is present in excess, then R_2NH, R_3N and $R_4N^+HX^-$ will be formed along with $R\text{–}NH_2$.

(ii) Reduction in acidic medium or catalytic reduction

(a) $R-NO_2 \xrightarrow[Sn/HCl \text{ or } Fe/HCl]{H_2+Ni \text{ or } Pt} R-NH_2$

(b) $Ph-NO_2 \xrightarrow[Sn/HCl \text{ or } Fe/HCl]{H_2+Ni/Pt} Ph-NH_2$

(iii) $Ph-CN \xrightarrow{LiAlH_4} PhCH_2NH_2$

(iv) $R-\overset{\overset{O}{\|}}{C}-NH_2 \xrightarrow[(ii)\ LiAlH_4]{LiAlH_4 \text{ or } (i)\ P_2O_5} RCH_2NH_2$

(v) By reductive amination, 2° amine can be prepared from 1° amine.

$CH_3CHO \xrightarrow[(ii)\ H_2/Ni]{(i)\ NH_3} CH_3CH_2NH_2$

(vi) Curtius reaction

$R-\overset{\overset{O}{\|}}{C}-Cl \xrightarrow[(iii)\ 2OH^-/H_2O]{(i)\ NaN_3,\ (ii)\ \Delta} RNH_2$

(vii) Schmidt reaction

$$RCOOH \xrightarrow{HN_3/\text{conc. } H_2SO_4} RNH_2$$

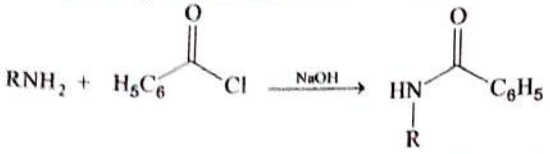
(viii) $RNC \xrightarrow[LiAlH_4]{Ni \text{ or } Pt \text{ or}} RNHCH_3$

(ix) Gabriel synthesis

$$\text{Phthalimide} \xrightarrow[(iii)\ 2KOH/H_2O]{(i)\ OH^- \ (ii)\ R-X} RNH_2$$

Chemical Properties

(i) Benzylation (Schotten–Baumann reaction)



Benzyl alkyl amine

(ii) Carbylamine reaction

Only 1° amines show this reaction (whether aliphatic or aromatic).

$$C_2H_5NH_2 + CHCl_3 + KOH \longrightarrow C_2H_5N \equiv C$$

Ethyl isocyanide

(iii) $C_2H_5NH_2 + CH_3CHO \longrightarrow C_2H_5N = CHCH_3$

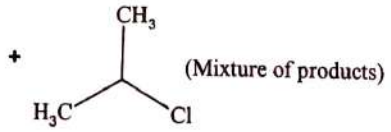
Ethylidine ethylamine

(Schiff's base)

(iv) $C_2H_5NH_2 + COCl_2 \longrightarrow C_2H_5NCO$

Carbonyl chloride — Ethyl isocyanate

(v) $CH_3CH_2CH_2NH_2 + HONO \text{ or } (NaNO_2/HCl) \longrightarrow CH_3CH_2CH_2OH + CH_3CH_2CH_2CH_2Cl + CH_3-CH=CH_2$



(Mixture of products)

(vi) Diazonium salt formation

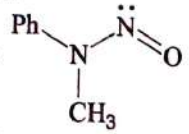
$$PhNH_2 + NaNO_2/HCl \xrightarrow{(0-5^\circ C)} PhN_2^+Cl^-$$

(vii) $(CH_3)_2NH + NaNO_2/HCl \longrightarrow (CH_3)_2\ddot{N}-\ddot{N}=O$

N-nitrosodimethyl amine

The compound formed is yellow oil.

(viii) $PhNHCH_3 + NaNO_2/HCl \longrightarrow$



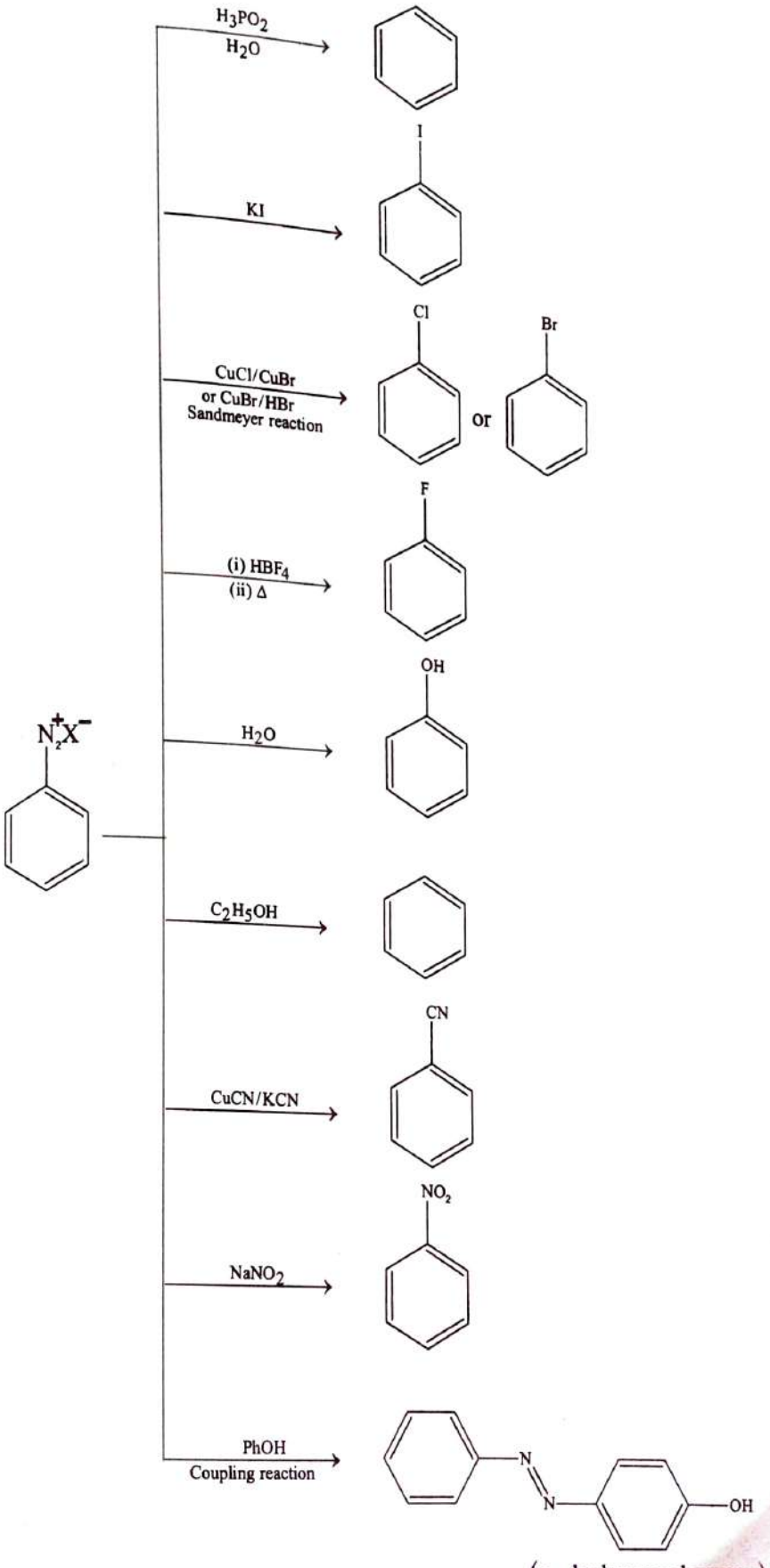
N-nitroso N-methyl aniline

The compound formed is yellow oil.

(ix) $R_3N + NaNO_3/HX \longrightarrow R_3\overset{+}{N}HX^- + R_3N-\overset{+}{N}=OX^-$

N-nitroso ammonium

Reactions of Aryl Diazonium Salts



(p – hydroxyazobenzene)

IMPORTANT REDUCTION REACTIONS

(i) $Ph-NO_2 \xrightarrow[H_2O]{Zn/NH_4Cl} Ph-NHOH$

Phenyl hydroxylamine

This reduction takes place in neutral medium.

(ii) $Ph-NO_2 \xrightarrow{Zn/NaOH \text{ in } CH_3OH[8(H)]} Ph-N=N-Ph$

Azobenzyl

This reduction takes place in basic medium.

(iii) $Ph-NO_2 \xrightarrow[\text{or } Na_3AsO_3+NaOH \text{ or Glucose+NaOH}]{As_2O_3/NaOH \text{ and } 6(H)} Ph-N=\underset{\downarrow}{N}-Ph$
O

Azoxy benzene

This reduction takes place in basic medium.

(iv) $Ph-NO_2 \xrightarrow{Zn/NaOH\ [10(H)]} Ph-NH-NHPh$

Hydroazo benzene

This reduction takes place in basic medium.

(v) $Ph-NO_2 \xrightarrow{LiAlH_4} Ph-N=N-Ph$

(vi) 1, 3-di-nitro benzene $\xrightarrow[Na_2S_x]{(NH_4)_2S}$ m-nitro aniline

This is selective reduction in which only one nitro group is reduced.

(vii) $ArN=NAr \xrightarrow{\text{Mild reduction}} ArNH-NHAr$

Azo compound Hydrazo compound

(viii) $ArN=NAr' \xrightarrow[(ii)\ OH^-]{(i)\ SnCl_2,\ H_3O^+} ArNH_2 + Ar'NH_2$

(ix) $Ph-NO_2 \xrightarrow{Sn/NaOH} PhNHNHPh$

REARRANGEMENT REACTIONS

(i) **Fischer Hepp rearrangement:**

$$Ph-\underset{|}{\overset{NO}{N}}-R \xrightarrow[(ii)\ OH^-]{(i)\ H^+} ONPhNHR$$

(ii) **Phenyl hydroxyl amines rearrangement:**

$$PhNHOH \xrightarrow[(ii)\ OH^-]{(i)\ H^+} p-HOPhNH_2$$

(iii) **Sulphonic acid:**

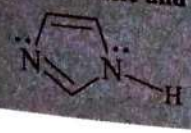
$$PhNHSO_3H \xrightarrow{\Delta(\text{heat})} SO_3^-Ph\overset{+}{N}H_3$$

(iv) **Benzydine rearrangement:**

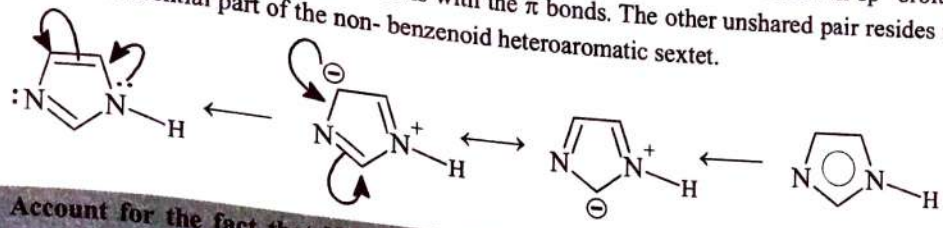
$$Ph-NH-NH-Ph \xrightarrow{H_3O^+} H_2NPh-PhNH_2$$

ILLUSTRATIVE PROBLEMS

1. Which unshared pair in imidazole is more basic and why?

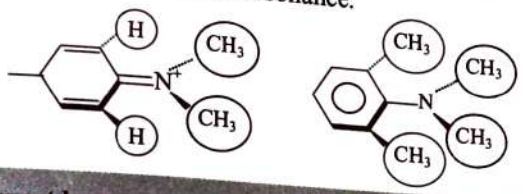


Sol. The unshared pair in unprotonated nitrogen is more basic because it resides in sp^2 orbital and is not involved in any resonance interactions with the π bonds. The other unshared pair resides in a p-orbital and is an essential part of the non- benzenoid heteroaromatic sextet.



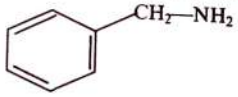
2. Account for the fact that N,N—dimethylaniline is only slightly more basic than aniline, 2, 6-dimethyl-N,N-dimethylaniline is much more basic than 2,6 – dimethylaniline.

Sol. Extended π bonding between the amino N and the ring requires that the σ bonds on N become coplanar with the ring and its ortho bonds. Bulky substituents in the ortho 2,6 positions sterically hinder the attainment of this geometry and interfere with the base–weakening extended π bonding. This effect is called steric inhibition of resonance.

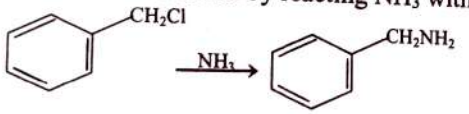


3. Convert benzene to benzylamine.

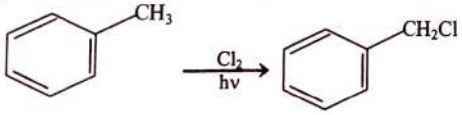
Sol. The structure of benzylamine is



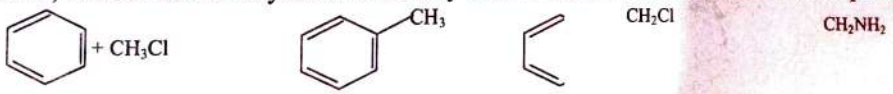
This can be obtained by reacting NH_3 with benzyl chloride.



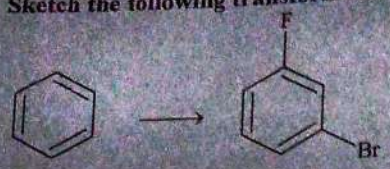
Now, let us find a way to prepare benzyl chloride. It can be easily prepared from toluene by photochemical chlorination.



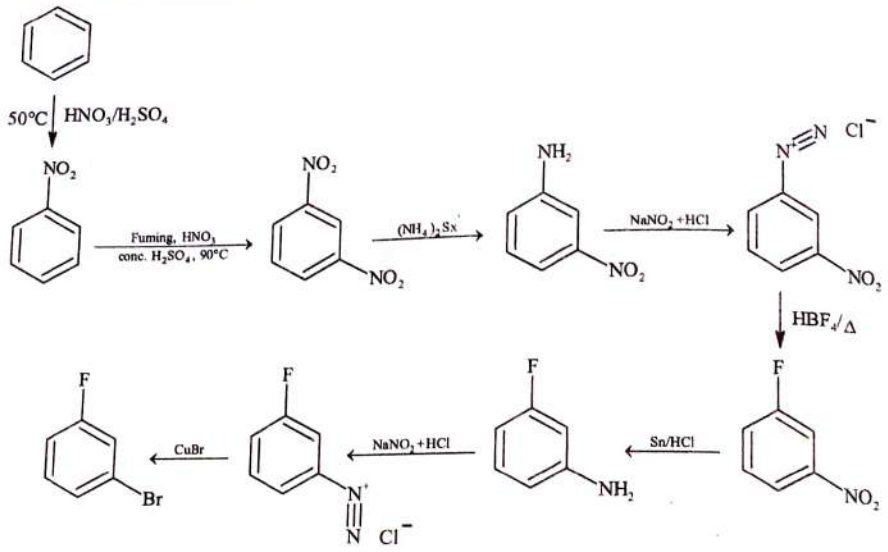
Now, Friedel Crafts alkylation can easily obtain toluene from benzene. The sequence of reaction is



4. Sketch the following transformations:

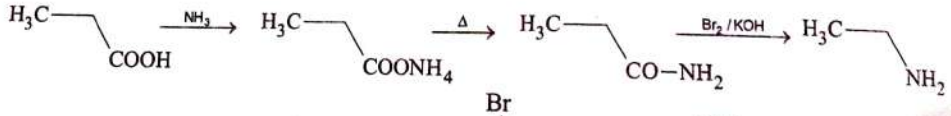


Sol.

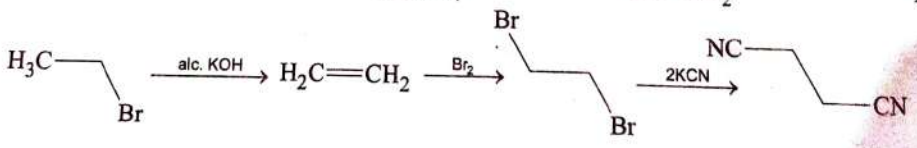


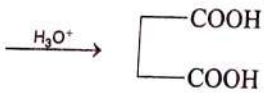
5. How would you bring about the following conversions?
(i) Propionic acid to ethylamine
(ii) Ethyl bromide to succinic acid
(iii) Acetic acid to acetaldehyde

Sol. (i)

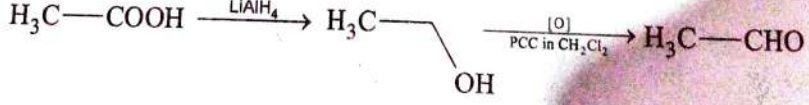


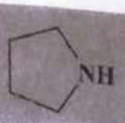
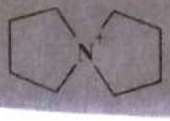
(ii)



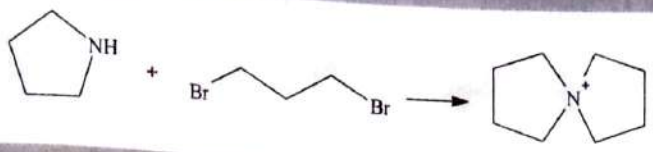


(iii)

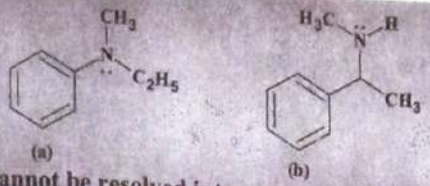


6. Convert  into 

Sol.



7.

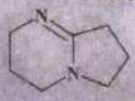
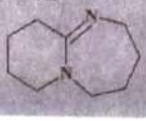


(a) (b)

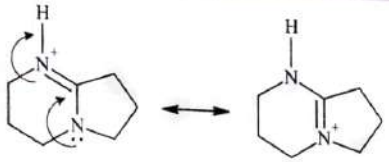
(a) cannot be resolved into enantiomers while (b) can, why?

Sol. In (a) Walden inversion occurs while in (b) carbon is chiral.

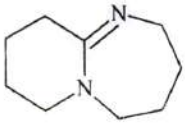
8. **In the following molecules indicate which nitrogen is the one most likely to be protonated and explain the enhanced strength of these relative to simple amine.**

(a)  (b) 

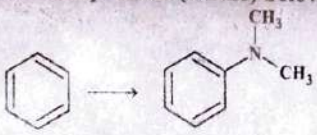
Sol. (a)

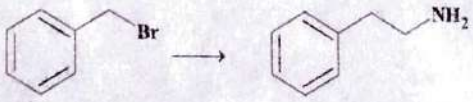


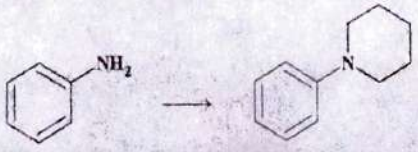
Resonance stabilized

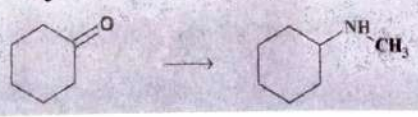
(b) Similarly,  (The conjugate acid is resonance stabilized.)

9. **Suggest a synthesis of each product (amine) below, starting with the indicated material.**

(a) 

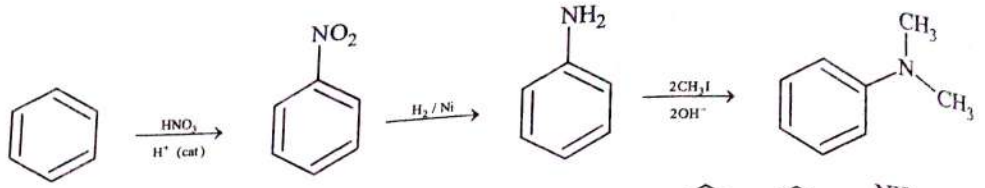
(b) 

(c) 

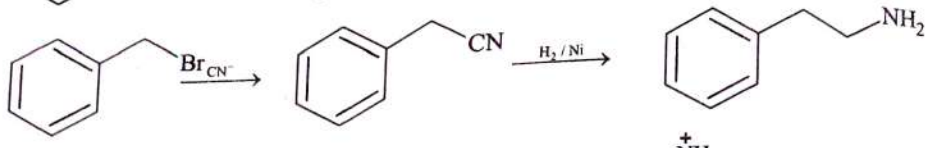
(d) 

MEGACOSM

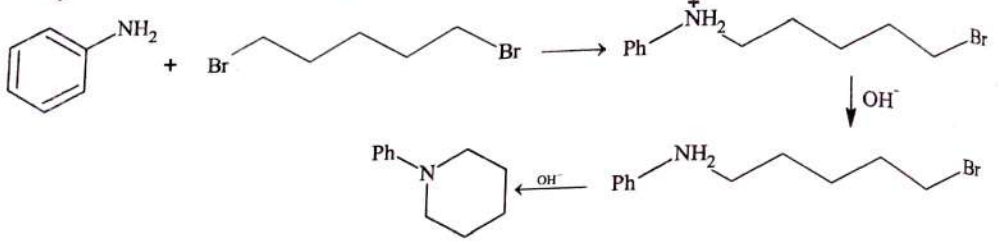
Sol. (a)



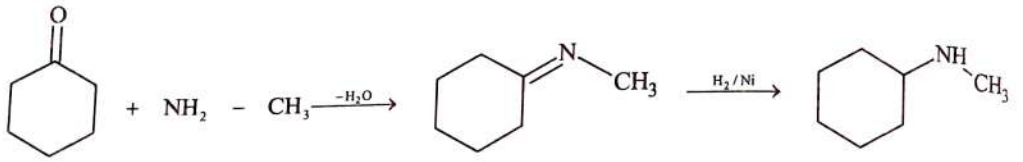
(b)



(c)



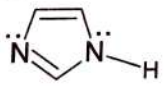
(d)



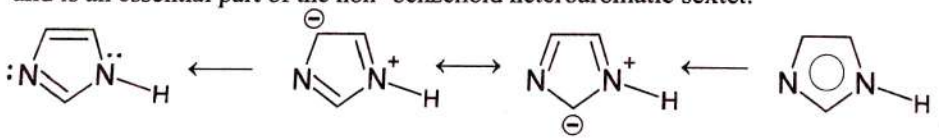
SOLVED PROBLEMS

Subjective Question

Prob. 1. Which unshared pair in imidazole is more basic and why?

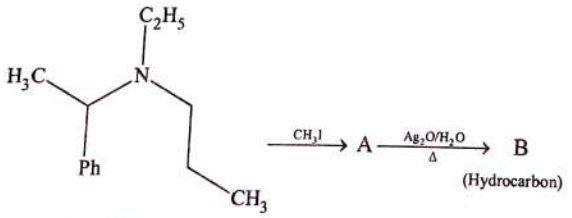


Sol. The unshared pair in unprotonated nitrogen is more basic because it resides in sp^2 orbital and is not involved in any resonance interactions with the π bonds. The other unshared pair resides in a p-orbital and is an essential part of the non- benzenoid heteroaromatic sextet.



Objective Questions

Prob. 1.



Hydrocarbon 'B' is,

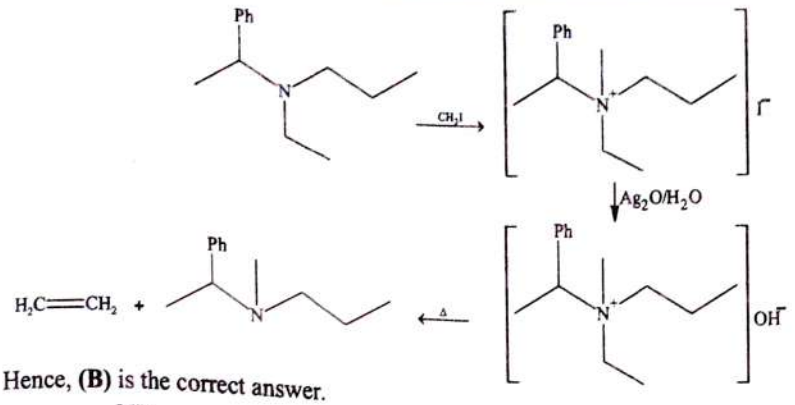
(A) CH_4

(B) $CH_2 = CH_2$

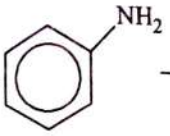
(C) $CH_2 = CH - CH_3$

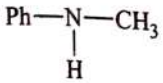
(D) $Ph - CH = CH_2$

Sol.

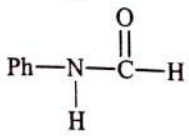


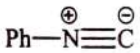
Hence, **(B)** is the correct answer.

Prob. 2.  $\xrightarrow[\Delta]{CHCl_3, KOH} X \xrightarrow[H_2O]{LiAlH_4} Y$; (Y) is :

(A) 

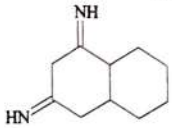
(B) $Ph—CH_2—NH_2$

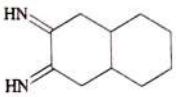
(C) 

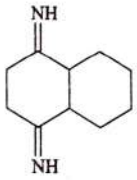
(D) 

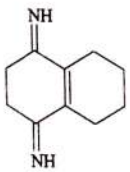
Sol. **(A)**

Prob. 3. Which of the following imine is least stable?

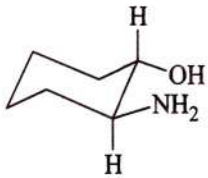
(A) 

(B) 

(C) 

(D) 

Sol. **(D)**

Prob.4.  $\xrightarrow{HNO_2}$ (A); Product (A) is :

(A) cyclopentane carboxyaldehyde
(B) cyclohexane-1, 2-diol
(C) 2-aminocyclohexene
(D) cyclohex-2-enol

Sol. **(A)**

Prob. 5. Which of the following compound(s) will not dissolve in an alkali solution after it has undergone reaction with Hinsberg's reagent?

(A) Diethyl amine
(B) Isopropyl amine
(C) N-methyl ethanamine
(D) N, N-dimethyl aniline

Sol. **(A, C)**

Prob. 6. Which compounds do not give Lassaigne test for nitrogen?

(A) $C_6H_5N_2Cl$ (B) $NH_2 - CH_2 - COOH$

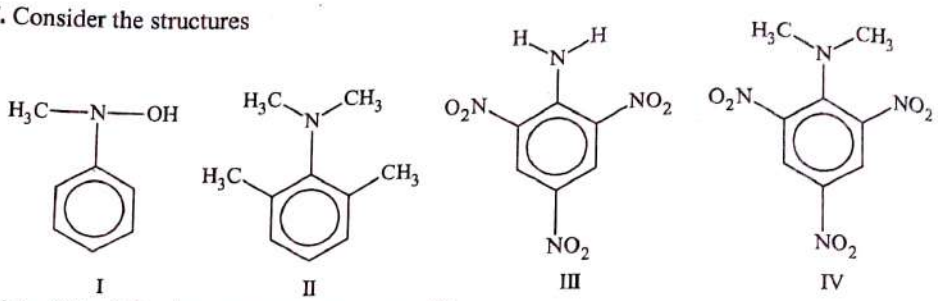
(C) NH_2OH (D) CH_3CONH_2

Sol. **(A)** Because at the required temperature of Lassaigne test, N_2 expels out in the form of a gas in $C_6H_5N_2Cl$.

(C) NH_2OH does not contain carbon.

Hence, **(A, C)** are the correct answers.

Prob. 7. Consider the structures



I II III IV

Which of the following statements are correct?

(A) The base strength of II is more than that of I.

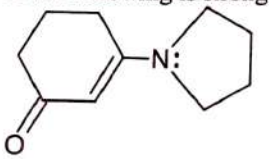
(B) The base strength of II is less than that of I.

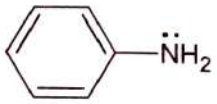
(C) The base strength of IV is more than that of III.

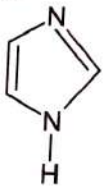
(D) The base strength of IV is less than that of III.

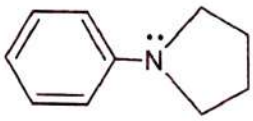
Sol. **(A, C)**

Prob. 8. Which of the following is strongest base ?

(A) 

(B) 

(C) 

(D) 

Sol. **(C)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Aniline does not undergo Friedel – Craft's reaction.
and
STATEMENT-2: $-NH_2$ group of aniline reacts with $AlCl_3$ (Lewis acid) to give acid – base reaction.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

(C) Statement-1 is True, Statement -2 is False.

(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** Aniline is Lewis base and thus consumes $AlCl_3$ which therefore cannot provided electrophile to develop Friedel-Craft's reaction.

2. **STATEMENT-1:** During diazotization, excess of HCl is used with $NaNO_2$.
and
STATEMENT-2: Excess of HCl prevents the hydrolysis of diazonium salt.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** Explanation is correct reason for statement.

3. **STATEMENT-1:** In strongly acidic medium, aniline becomes less reactive towards electrophilic reagent.
and
STATEMENT-2: $-NH_2$ group is o– directing.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

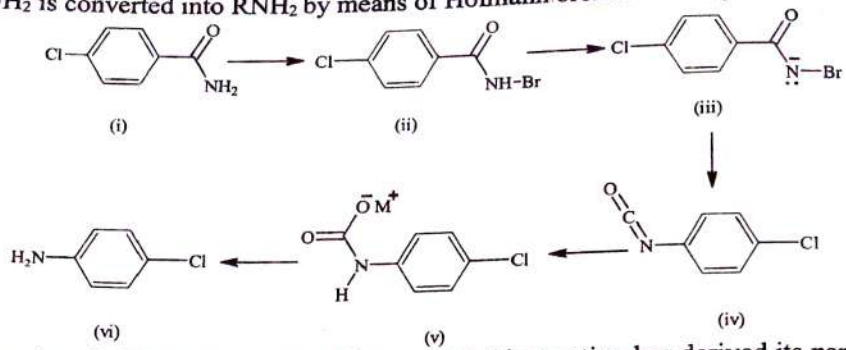
Sol. **(B)** In presence of acid —NH_2 group forms —NH_3^+ i.e. anilium ion and show m – directing nature. Thus, ring is deactivated for S_E reactions.

4. **STATEMENT-1:** Acylation of aniline increases the reactivity of C_6H_6 ring.
and
STATEMENT-2: Acylation of aniline decreases the electron density at C_6H_6 ring.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)** $-NHCOCH_3$ has lesser activating nature than $-NH_2$ group.

COMPREHENSION

$RCONH_2$ is converted into RNH_2 by means of Hofmann bromamide degradation.



In this reaction, RCONHBr is formed from which this reaction has derived its name. Electron donating group at phenyl activates the reaction. Hofmann degradation reaction is an intramolecular reaction.

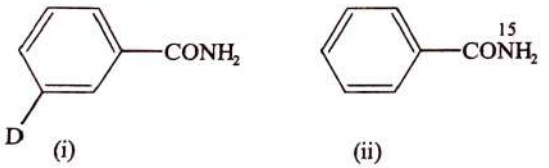
1. How can the conversion of (i) to (ii) be brought about?
   (A) KBr
   (B) KBr + CH_3ONa
   (C) KBr + KOH
   (D) Br_2 + KOH

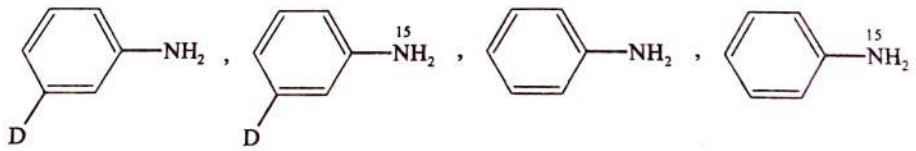
Sol. **(D)**

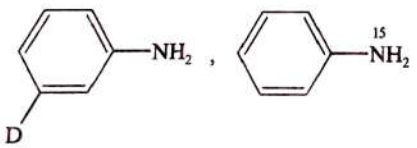
2. Which is the rate determining step in Hofmann bromamide degradation?
   (A) Formation of (i)
   (B) Formation of (ii)
   (C) Formation of (iii)
   (D) Formation of (iv)

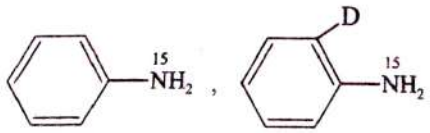
Sol. **(D)**

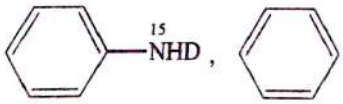
3. What are the constituent amines formed when the mixture of (i) and (ii) undergoes Hofmann bromamide degradation?



(A) 

(B) 

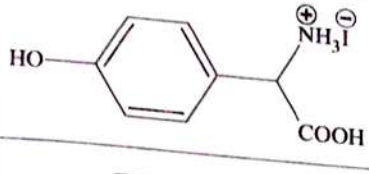
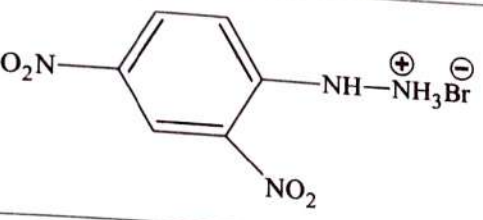
(C) 

(D) 

Sol. **(B)**

MATRIX – MATCH TYPE

1. Match the compounds in **Column I** with their characteristic test(s)/ reaction(s) given in **Column II**. Indicate your answer by darkening the appropriate bubbles of the 4 × 4 matrix gives in the ORS.

	Column I	Column II
(A)		(p) sodium fusion extract of the compound gives Prussian blue colour with $FeSO_4$
(B)		(q) gives positive $FeCl_3$ test
(C)		(r) gives white precipitate with $AgNO_3$
(D)		(s) reacts with aldehydes to form the corresponding hydrazone derivative

Sol. **A →r, s; B →p, q; C →p, q, r; D → p, s**

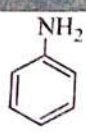
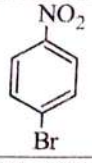
2. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	$CH_3CH_2CH_2CN$	(p)	Reduction with Pd – C/H_2
(B)	$CH_3CH_2OCOCH_3$	(q)	Reduction with $SnCl_2/HCl$
(C)	$CH_3 - CH = CH - CH_2 - OH$	(r)	Development of foul smell on treatment with chloroform and alcoholic KOH
(D)	$CH_3CH_2CH_2CH_2NH_2$	(s)	Reduction with diisobutyl aluminium hydride
		(t)	Alkaline hydrolysis

Sol. **A → (p, q, s, t) B → (p, s, t) C → (p) D → (r)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

DBE is double bond equivalent

Column – 1		Column – 2 DBE		Column – 3	
(I)		(i)	1	(P)	Forms p-nitro phenol on treating with aq. KOH
(II)		(ii)	5	(Q)	Forms Methyl isocyanide on treating with $CHCl_3$ and KOH
(III)	$CH_3—NH_2$	(iii)	4	(R)	Form benzene diazonium salt treating with $NaNO_2 + HCl$
(IV)		(iv)	0	(S)	Forms butanamine on treating it with H_2/Ni

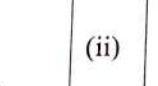
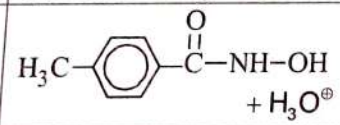
3. Which of the following combination is correct ?
[A] (I) (iii) (R) [B] (II) (iii) (P) [C] (II) (iv) (Q) [D] (IV) (i) (S)

4. Which of the following combination is incorrect ?
[A] (II) (i) (S) [B] (III) (iv) (Q) [C] (IV) (ii) (P) [D] (II) (ii) (R)

5. Which of the following combination is correct ?
[A] (IV) (ii) (P) [B] (II) (iii) (Q) [C] (III) (ii) (R) [D] (IV) (iv) (S)

Sol. 3. **A;** 4. **D;** 5. **A**

Answer 6, 7 and 8 by appropriately matching the information given in the three columns of the following table.

Column – 1 Reactants		Column – 2 Product		Column – 3 Reaction Name	
(I)	$CH_3—\overset{O}{\overset{\|\|}{C}}—NH_2 + Br_2 + NaOH(aq)$	(i)	$Ph—NH_2$	(P)	Curtius rearrangement
(II)	$^*CH_3—\overset{O}{\overset{\|\|}{C}}—Cl + NaN_3 + Heat$	(ii)		(Q)	Schmidt rearrangement
(III)	$PhCOOH + HN_3 + H_2O$	(iii)	$CH_3—NH_2$	(R)	Lossen rearrangement
(IV)		(iv)	$^*CH_3—N=C=O$	(S)	Hoffmann bromamide degradation

6. Which of the following combination is correct ?
[A] (I) (ii) (Q) [B] (II) (iv) (P) [C] (III) (iv) (R) [D] (IV) (iv) (S)

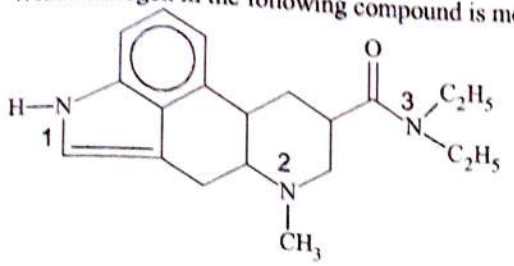
7. Which of the following combination is incorrect ?
[A] (III) (iii) (S) [B] (IV) (ii) (R) [C] (III) (i) (Q) [D] (II) (iv) (P)

8. Which of the following combination is correct ?
[A] (I) (ii) (Q) [B] (II) (i) (P) [C] (III) (iii) (R) [D] (I) (iii) (S)

Sol. 6. **B;** 7. **A;** 8. **D**

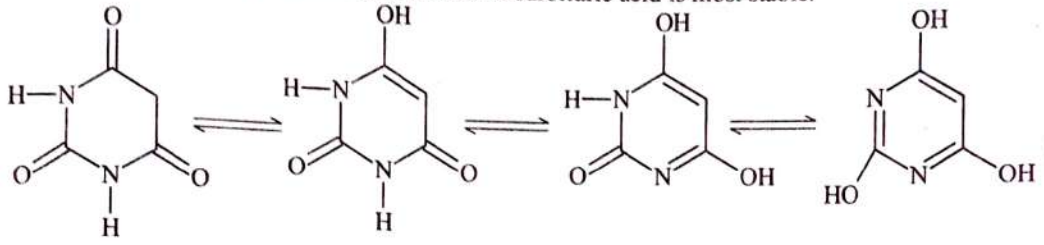
SINGLE INTEGER ANSWER TYPE QUESTIONS

1. Which nitrogen in the following compound is most basic



Sol. **(2)**

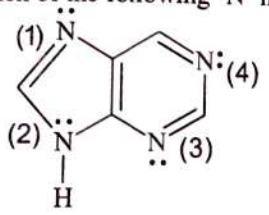
2. Which of the following resonating structures of barbituric acid is most stable.



(1) Barbaturic Acid (2) (3) (4)

Sol. **(4)**

3. Which of the following 'N' in purine is least basic ?



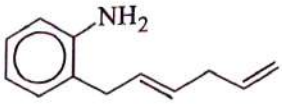
Sol. **(2)**

Loan pair of N number (2) is involved in resonance and is not available for H^+

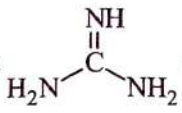
NUMERICAL VALUE ANSWER TYPE

Prob. 1. 10.8 g of CH_3CONH_2 underogoes Hoffmann bromamide degradation to produce _____ g of CH_3NH_2.

Sol. **6.20**

Prob. 2. The degree of unsaturation in  is x and in pyridine is y. The due of $\frac{x}{y}$ is

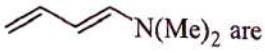
Sol. **1.50**

Prob. 3. Total number of lone pair of electrons in is a, and in  is b. The value of $\frac{b}{a}$ is

Sol. **1.5**

Prob. 4. The number of moles of methyl amine 79.05 g

Sol. **2.55**

Prob. 5. The total number of electrons participating in conjugation in  are

Sol. **6.00**

CARBOHYDRATES, AMINO ACIDS & POLYMERS 20

CARBOHYDRATES

Carbohydrates are a class of naturally occurring organic compounds that include polyhydroxy aldehydes or polyhydroxy ketones or substances which on hydrolysis give polyhydroxy aldehydes or ketones.

CLASSIFICATION OF CARBOHYDRATES

1. On the Basis of Structure

Monosaccharides

(i) These cannot be decomposed by hydrolysis to give simpler carbohydrates, e.g. glucose, fructose etc.
(ii) These are called simple sugars.

Oligosaccharides

(i) These yield definite number (3 to 10) of monosaccharides on hydrolysis.
   (a) Disaccharides: These yield two monosaccharides on hydrolysis, e.g. sucrose, maltose.
   (b) Trisaccharides: These contain three monosaccharide units.

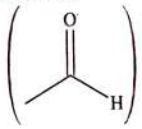
Polysaccharides

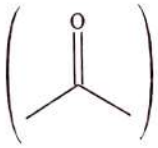
(i) These are high molecular weight compounds which yield many monosaccharide molecules on hydrolysis, e.g. starch, cellulose.
(ii) These are called non-sugars.

2. On the Basis of Reactive Nature of Carbohydrate

(i) Non-reducing sugar: They do not reduce Fehling's solution and Tollen's reagent.
(ii) Reducing sugar: They reduce Fehling's solution and Tollen's reagent, e.g. monosaccharides and disaccharides except sucrose.

Monosaccharides can be further classified into

(i) Aldoses: contains aldehydic  group, e.g. aldo-triose, aldo-tetrose.

(ii) Ketoses: contains ketonic  group, e.g. ketohexoses.

D & L Forms

If the hydroxyl group on the asymmetric carbon atom farthest from aldehydic or ketonic group projects to right, the compound is a member of D-family and if it is towards left it is a member of L-family.

Epimers

A pair of diastereomers that differ only in the configuration about of a C_{20} carbon atom are said to be epimers.

Anomers

A pair of stereoisomers which differ in configuration, only around C_1 are called anomers.

Haworth Projection Formulae

According to this projection, draw a hexagon with oxygen atom at the upper right hand corner. Place all the groups (on C_1, C_2, C_3 and C_4) which are present on left hand side above the plane of the ring and all those groups on the right hand side below the plane of the ring.

Mutarotation

During mutarotation, the ring opens and then closes either in the inverted position or in the original position giving a mixture of α and β -forms.

$$\alpha - D\,Glucose \rightleftharpoons Equilibrium\ mixture \rightleftharpoons \beta - D\,Glucose$$

$$[\alpha]_D = +112^\circ \qquad [\alpha]_D = +52.7^\circ \qquad [\alpha]_D = +19^\circ$$

where $[\alpha]_D$ = specific rotation

In α - form, H is on left hand side and OH is on right hand side of the carbon atom.

REACTIONS OF GLUCOSE

Glucose

- $\xrightarrow{HI/P}$ $CH_3-(CH_2)_4-CH_3$
- $\xrightarrow{Na/Hg}$ $CH_3OH-(CHOH)_4-CH_2OH$ (Sorbital)
- $\xrightarrow{H_2O}$ Forms neutral solution
- $\xrightarrow{NH_2OH}$ $HC{=}N{-}OH$ / $(CHOH)_4$ / CH_2OH
- $\xrightarrow{HCN}$ $CH_2OH{-}(CHOH)_4{-}CH(CN)(OH)$ (Glucose cyanohydrin)
- $\xrightarrow[(ii)\ HNO_3]{(i)\ Br_2/H_2O}$ $HOOC-(CHOH)_4-COOH$ (Glucaric acid) [First gluconic acid is formed which is converted to glucaric acid.]
- $\xrightarrow[reagent]{Tollen's}$ $CH_2OH-(CHOH)_4-COOH + 2Ag$ (gluconic acid; silver mirror)
- $\xrightarrow[Solution]{Fehling's}$ $CH_2OH-(CHOH)_4-COOH + Cu_2O$ (gluconic acid; Red ppt.)
- $\xrightarrow{Acetic\ anhydride}$ $OHC-(CHOOCCH_3)_4-CH_2OOCCH_3$ (Glucose penta-acetate)
- $\xrightarrow{C_6H_5NHNH_2}$ $HC{=}NNHC_6H_5$ / $C{=}NNHC_6H_5$ / $(CHOH)_3$ / CH_2OH (Glucosazone)
- $\xrightarrow{conc.\ HCl}$ Laevulinic acid
- $\xrightarrow{Gaseous\ HCl}$ $\alpha-$ and $\beta-$ Methyl glucopyranoside [Anomeric methyl acetals are formed]

Interconversions of monosaccharides

(i) **Killiani–Fischer synthesis:** This is a method of chain lengthening of carbon chain of an aldose. Addition of HCN to glyceraldehyde produces two epimeric cyanohydrins. The cyanohydrins can be converted to aldose by hydrolysis, acidification and lactonisation and reduction with $Na–Hg/H_2SO_4$.

(ii) **Ruff degradation:** It is used to shorten the chain length. It involves two steps:
(a) Oxidation of aldose to an aldonic acid using Br_2/H_2O.
(b) Oxidative decarboxylation of the aldonic acid to the next lower aldose using H_2O_2 and $Fe_2(SO_4)_3$.

INVERSION OF SUCROSE

$$\underset{[\alpha]_D = +66.5^\circ}{\text{Sucrose}} + H_2O \xrightarrow{H^+} \underbrace{\underset{[\alpha]_D = +53^\circ}{\text{D-glucose}} + \underset{[\alpha]_D = -92^\circ}{\text{D-fructose}}}_{[\alpha]_D = -39^\circ}$$

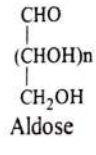
Initially, sucrose is present in dextrorotatory form but on hydrolysis the final mixture formed is laevorotatory. Hence, hydrolysis of sucrose is known as inversion of sucrose.

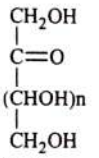
STARCH

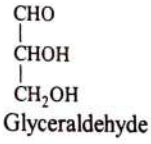
It is a polysaccharide, which consists two fractions. They are
(i) amylase and
(ii) amylopectin

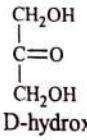
Amylose	Amylopectin
Linear polymer of α – D glucose units	Branched polymer of α – D glucose units
It is soluble in water	It is insoluble in water
Long straight chains containing 200–1000 glucose units linked by α – 1, 4 – glycoside links	Short chains (about 20 glucose units) cross-linked by α – 1,6 – glycoside links
Used for making edible films	Used in textile industry and as a thickener in foods

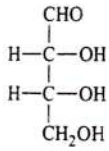
STRUCTURE OF CARBOHYDRATES

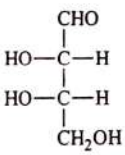

Aldose

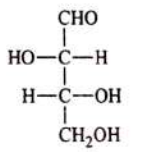

Ketose

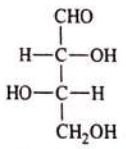

Glyceraldehyde

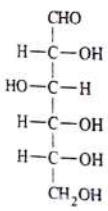

D-hydroxy acetone

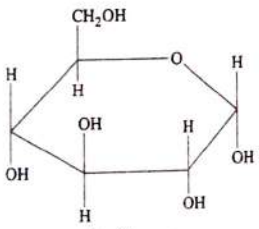

D (–)-erythrose

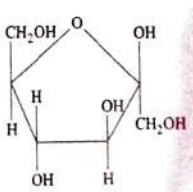

L (+)-erythrose

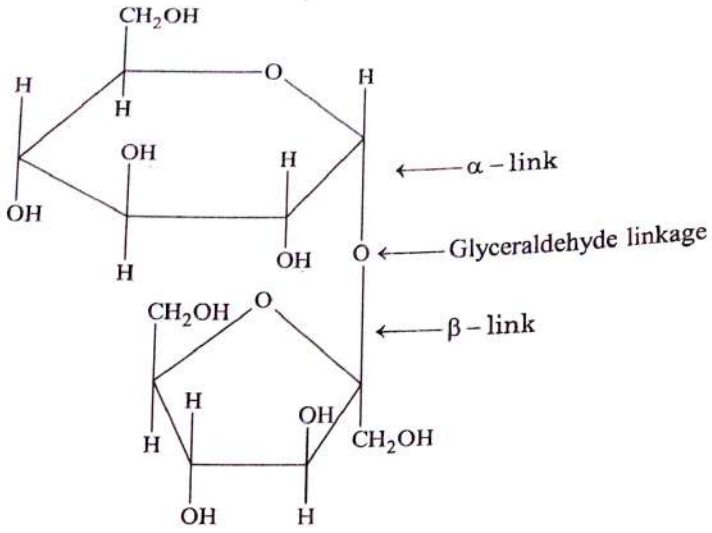

D (–)-threose


L (+)-threose


D(+)-Glucose


α – D – Glucose


β – D – Fructopyranose

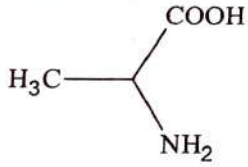


Sucrose

AMINO ACIDS

Amino acids are organic molecules, which contain two functional groups, i.e. carboxylic group and amino group.
For example,

$H_2N - CH_2 - COOH$ Amino acetic acid or glycine



α – Amino propionic acid or alanine

Physical Properties and Structure

(i) Amino acids are non-volatile solids, which melt at fairly high temperature.
(ii) They are insoluble in organic solvents and are highly soluble in polar solvents.
(iii) Their aqueous solution is neutral.
(iv) Their aqueous solution have high dipole moment.
(v) Due to the presence of two oppositely behaving groups, i.e. COOH and NH_2, the acidity and basicity of amino acids is extremely low.

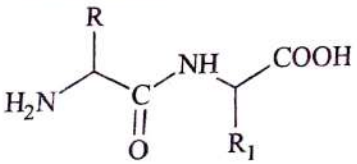
Isoelectric Point

$$H_2N-CHRCOO^- \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} H_2NCHRCOOH \underset{OH^-}{\overset{H^+}{\rightleftharpoons}} H_3N^+ - CHRCOOH$$

The hydrogen ion concentration of the solution in which a particular amino acid does not migrate under the influence of an electric field is called the isoelectric point of amino acid.

PEPTIDES

The amides formed by interaction between amino groups and carboxyl groups of amino acids are called peptides.



$-\overset{\overset{O}{\|}}{C}-NH$ group is referred as peptide bond.

Polymers are high molecular mass compounds which are composed of large number of simple repeating units. The repeating units are called monomers.
Polymers are also known as macromolecules.
Polymers can be natural and man-made (synthetic). The process of forming polymers is known as polymerization.

General Methods of Preparation

(i) **In chain reaction polymerization:** There is a series of reactions each of which consumes a reactive particle and produces another similar particle; each individual reaction thus depends upon the previous one. The reactive particles can be free radicals, cations or anions.

$$Rad. + CH_2 = CH_2 \longrightarrow Rad.CH_2\dot{C}H_2 \xrightarrow{CH_2=CH_2} Rad.CH_3CH_2\dot{C}H_2 \text{ etc.}$$

(ii) **In step reaction polymerization:** There are series of reactions each of which is essentially independent of the preceding one; a polymer is formed simply because the monomer undergoes reaction at more than one functional group.

$$HOCH_2CH_2OH + p-HOOCPhCOOH \longrightarrow HOOCPh-\overset{O}{\overset{\|}{C}}-OCH_2-CH_2-O-\overset{O}{\overset{\|}{C}}-Ph-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-OH$$

(iii) **Free radical vinyl polymerization:** Here, free radicals are generated typically in presence of small amounts of an initiator, such as a peroxide. Reaction occurs at the double bonded carbons, and is called vinyl polymerization.

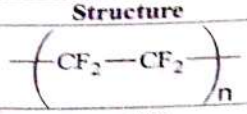
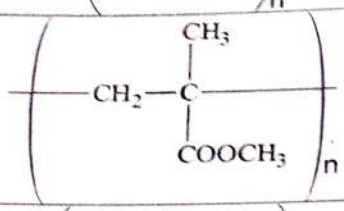
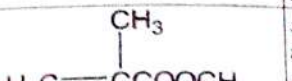
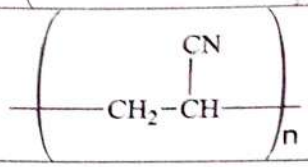
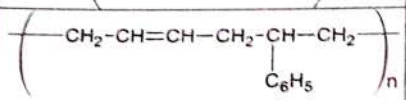
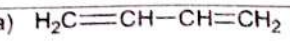
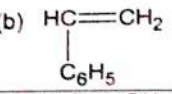
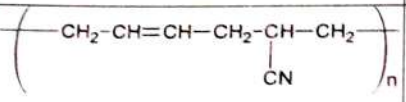
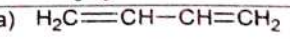
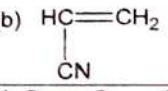
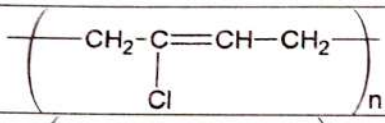
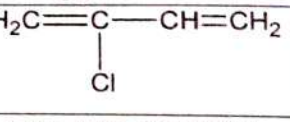
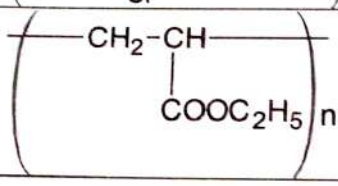
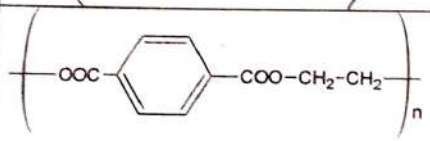
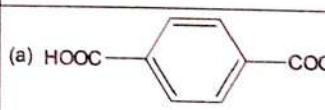
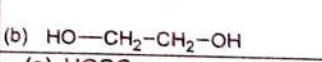
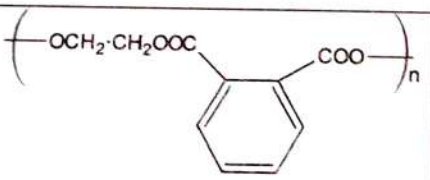
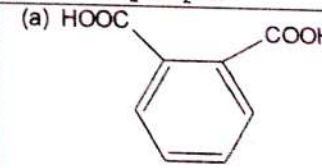
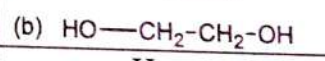
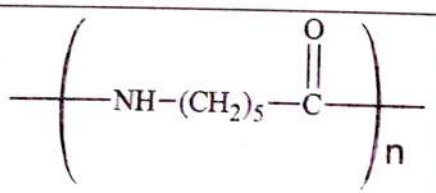
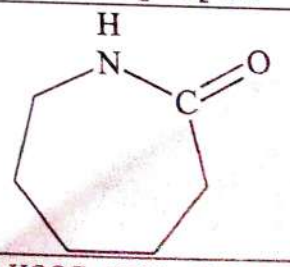
$$\underset{\text{vinyl monomer}}{H_2C=\underset{G}{CH}} \xrightarrow{\text{initiator}} CH_3\underset{G}{CH}CH_2\underset{G}{CH}CH_2\underset{G}{CH}CH_2\underset{G}{CH} \quad \text{or} \quad \left(-CH_2\underset{G}{CH}-\right)_n$$

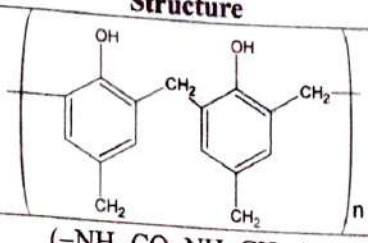
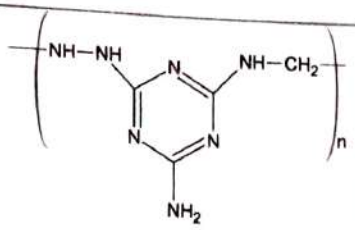
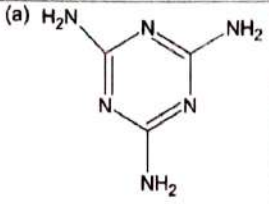
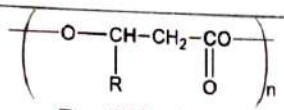
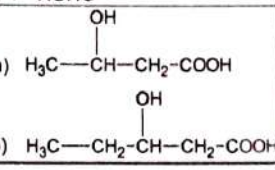
(iv) **Copolymerization:** If mixture of two (or more) monomers is allowed to undergo polymerization, then a copolymer is obtained. It is a polymer that contains two or more bonds of monomeric units in the same molecule.

$$H_2C=\underset{Ph}{CH} + H_2C=\underset{HC-O-CH_3}{\overset{CH_3}{C}} \xrightarrow{\text{initiator}} -CH_2-\underset{Ph}{CH}-CH_2-\overset{CH_3}{\underset{COOCH_2}{C}}-$$

Some Important Polymers

S.N.	Name of Polymer	Structure	Monomer	Uses
1.	Polythene	$\left(-CH_2-CH_2-\right)_n$	$CH_2 = CH_2$	An insulator, anticorrosive, packing material, household and laboratory wares.
2.	Polystyrene	$\left(-CH(C_6H_5)-CH_2-\right)_n$	$HC(C_6H_5)=CH_2$	An insulator, wrapping material, manufacture of toys and household articles.
3.	Polyvinylchloride (PVC)	$\left(-CH(Cl)-CH_2-\right)_n$	$CH_2=CHCl$	In manufacture of raincoats, hand bags, vinyl flooring and leather clothes.

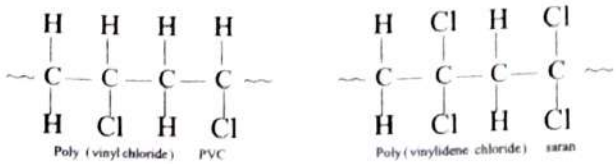
S.N.	Name of Polymer	Structure	Monomer	Uses
4.	Polytetrafluoro ethylene (PTFE) or Teflon		$CF_2=CF_2$	As lubricant, insulator and making cooking wares.
5.	Polymethyl metha acrylate (PMMA) or Plexi glass			As substitute of glass and making decorative materials.
6.	Polyacrylonitrile (Orlon)		$CH_2=CHCN$	In making synthetic fibres and synthetic wool.
7.	Styrene butadiene rubber (SBR or BuNa-S)		(a)  (b) 	In making automobile tyres and footwear.
8.	Nitrile rubber (Buna-N)		(a)  (b) 	In making oil seals, manufacture of hoses and tank linings.
9.	Neoprene or chloroprene			As insulator, making conveyor belts and printing rollers.
10.	Polyethyl acrylate		$CH_2=CH-COOC_2H_5$	In making films, house pipes and finishing fabrics.
11.	Terylene (Dacron)		(a)  (b) 	For making fibres, safety belts, tyre cords, tents etc.
12.	Glyptal		(a)  (b) 	As binding material in preparation of mixed plastics and paints.
13.	Nylon 6			In making fibres, plastics, tyre cords and ropes.
14.	Nylon 66	$(-NH(CH_2)_6NHCO(CH_2)_4CO-)_n$	(a) $HOOC-(CH_2)_4-COOH$ (b) $H_2N-(CH_2)_6-NH_2$	In making brushes, synthetic fibres, parachutes, ropes and carpets.

S.N.	Name of Polymer	Structure	Monomer	Uses
15.	Bakelite		(a) HCHO (b) C_6H_5OH	For making gears, protective coating and electrical fittings.
16.	Urea formaldehyde resin	$(-NH-CO-NH-CH_3-)_n$	(a) HCHO (b) NH_2CONH_2	For making unbreakable cups and laminated sheets.
17.	Melamine formaldehyde resin		(a)  (b) HCHO	In making plastic crockery, unbreakable cups and plates.
18.	Poly-β-hydroxy butyrate-co-β-hydroxy valerate (PHBV)	 $R = CH_3, C_2H_5$		As packaging orthopaedic devices and in controlled drug release.

ILLUSTRATIVE PROBLEMS

1. **Although both polymers are prepared by free radical processes, poly (vinyl chloride) is amorphous and poly (vinylidene chloride) (saran) is highly crystallilne. How do you account for the differenr? (vinylidene chloride is 1,1-dichloroethene).**

Sol.

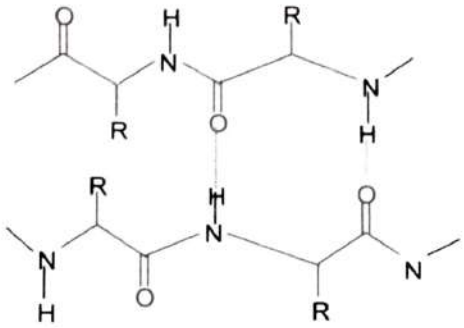


As poly (vinyl chloride) is able to show stereoisomerism and further it is formed by a free radical process, it is atactic (chlorine atoms distributed randomly), the molecules fit together poorly.

Poly (vinylidene chlroide) has two identical substituents on each carbon and the chains fit together well

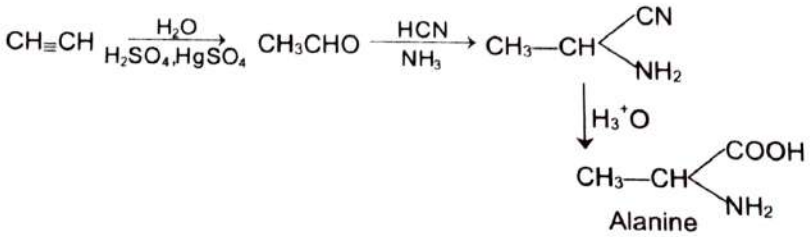
2. **Show the fundamental unit of structure common to all polypeptides and proteins and show how cross linking occurs between two chains by H - bonding.**

Sol.



3. **How will you synthesize Alanine from acetylene.**

Sol.



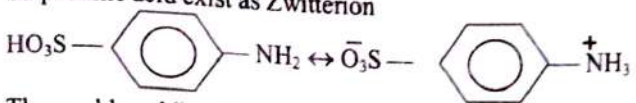
4. **Glycine exists as ($H_3N^+CH_2COO^-$) while anthranilic acid (p^- $-NH_2-C_6H_4-COOH$) does not exist as dipolar ion.**

Sol. –COOH is too weakly acidic to transfer H^+ to the weakly basic $-NH_2$ attached to the electron withdrawing benzene ring. When attached to an aliphatic carbon, the $-NH_2$ is sufficiently basic to accept H^+ form –COOH group.

5. **(i) Sulphanilic acid although has acidic as well as basic group, it is soluble in alkali but insoluble in mineral acid**

(ii) Sulphanilic acid is not soluble in organic solvents.

Sol. (i) Sulphanilic acid exist as Zwitterion

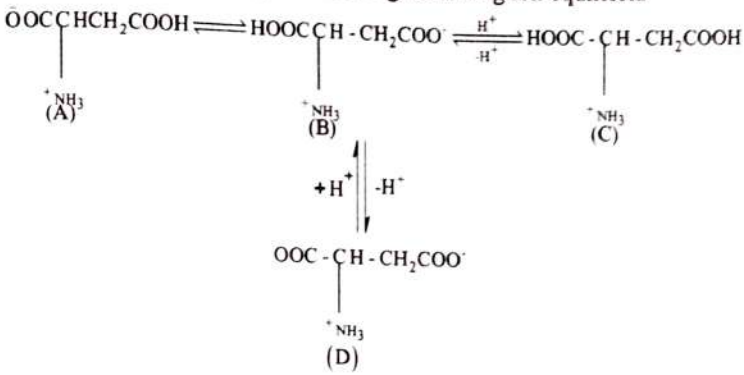


The weakly acidic $-{}^{+}NH_3$ transfers H^+ to OH^- to form a soluble salt, $p^-\text{-}NH_2\text{-}C_6H_4\text{-}SO_3^-\ Na^+$ on the other hand $-SO_3^-$ is too weakly basic to accept H^+ from strong acids.

(ii) Due to its ionic character it is insoluble in organic solvents.

6. Why should isoelectric point for Aspartic acid (2.98) be so much lower than that of leucine?

Sol. This may be explained by considering following ion equilibria



It is apparent that ions (A) and (B) are neutral, while (C) is a cation and (D) is dianion. In species (D), the anion is derived from the second — COOH group present in aspartic acid and is not possible in leucine. At neutral pH a significant concentration of (D), will be present in aqueous solution. It will therefore, be necessary to decrease the pH of such a solution if the formation of (D) is to be suppressed to a stage where anions and cations are present in equal concentration (the isoelectric point).

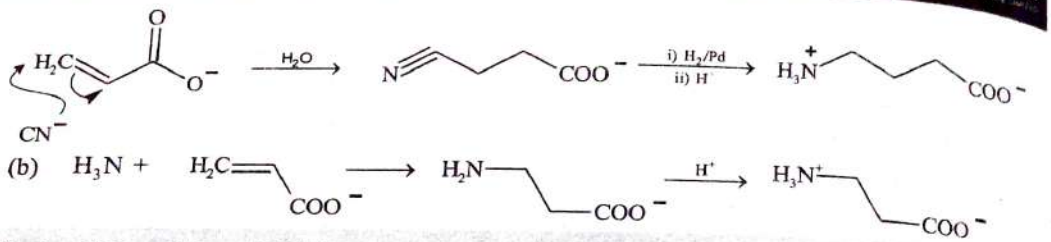
7. What is a peptide bond or linkage? Distinguish between a peptide, oligopeptide polypeptide and protein.

Sol. $-\overset{\overset{O}{\|}}{C}-NH-$, amide linkage

A peptide is an amide formed by intermolecular reaction of the amino group of one amino acid and the carboxyl group of a second amino acid. Dipeptides are made from two amino acids tripeptides from three amino acids etc., which may be same or different. If there are four to ten amino acid residues, the peptide is called an oligopeptide. A poly peptide chain is made up of many amino aicds. The terms peptide and polypeptides are often used interchangeably. A protein consists of one or more polypeptide chains and each chain can contain as much as several hundred amino acids. The total no. of residues may vary from 50 to over 100.

8. From acrylic acid prepare (a) γ–amino butyric acid (b) β–aminopropionic acid (β–alanine).

Sol (a) The $-NH_2$ and the additional C are introduced from CN^- which reacts with the acid by Michael type addition.

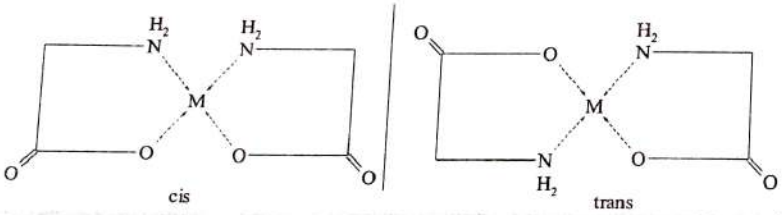


(b) $H_3N + H_2C=CH-COO^- \longrightarrow H_2N-CH_2-CH_2-COO^- \xrightarrow{H^+} H_3N^+-CH_2-CH_2-COO^-$

9. Glycine (NH_2-CH_2-COOH) is a monobasic bidentate chelating agent. A transition metal ion M(II) with d^8 configuration forms bis ligand complex with this chelating agent, which is a non-electrolyte and is diamagnetic. Mention the hybridization of the central metal in the complex and structure of the complex. Will the complex exhibit geometrical isomerism? If yes, draw the structure of these isomers.

Sol. The complex is diamagnetic. This means ligand is strong and it induces spin-pairing in M(II). The complex is 4–coordinate and hence it must be formed by dsp^2 hybridization. The complex is non-electrolyte, i.e. it is a neutral complex. Thus, both the molecules of the ligand chelated to the metal (II) are deprotonated.

So, the complex is $[M(NH_2CH_2COO)_2]$



10. A polymeric substance, tetrafluoroethylene, can be represented by the formula $(C_2F_4)_x$ where x is a large number. The material was prepared by polymerizing C_2F_4 in the presence of a sulphur bearing catalyst that served as a nucleus upon which the polymer grew. The final product was found to contain 0.012% S. What is the value of x if each polymeric molecule contains 2 sulphur atoms? Assume that the catalyst contributes a negligible amount to the total mass of polymer.

Sol. *Final product contains 0.012% S*

It contains two S atoms

$$\therefore \frac{\text{Molecular weight} \times 0.012}{100} = 2 \times 32$$

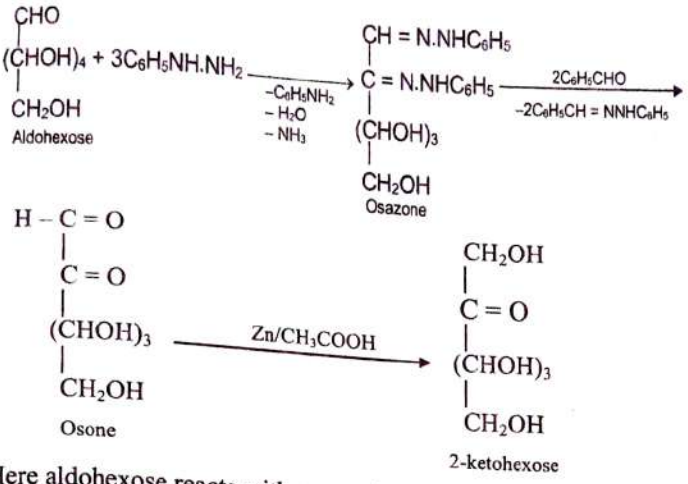
Weight of monomer, i.e. C_2F_4 is 100

$\therefore 100 \times x = 5.33 \times 10^5$

$\therefore x = 5.33 \times 10^3$

11. Show how an aldohexose can be used to synthesize 2-ketohexose. (b) Since glucose is converted to fructose by this method, what can you say about the configurations of C^3, C^4 and C^5 in the sugars.

Sol.



Here aldohexose reacts with one molecule of phenylhyrazine which condenses with the aldehyde group to give phenylhydrazone. When warmed with excess of phenyl hydrazine, the secondary alcoholic group adjacent to the aldehyde group is oxidized by another molecule of phenylhydrazine, to a ketonic group. With this ketonic group, the third molecule of phenylhydrazine condenses to give osazone. The phenylhydrazinyl group is transferred from osazone to C_6H_5CHO giving $C_6H_5CH = N{\cdot}NHC_6H_5$ and a dicarbonyl compound called an osone. The more reactive aldehyde group of the osone is reduced, not the less reactive keto group and it gives the 2-ketohexose.

(b) The configurations of these carbons which are unchanged in the reactions, must be identical in order to get the same osazone.

12. (a) Supply structures for H through K. Given:

An aldohexose $\xrightarrow{NH_2OH/base}$ **H** $\xrightarrow{Ac_2O/NaOAc}$ **I** $\xrightarrow{-HOAC}$ **J** $\xrightarrow{NaOMe/MeOH}$ **K.**

(b) Explain the last step (c) what is net structural change (d) Name this overall method. (e) Discuss the possibility of epimer formation.

Sol. (a) H is an oxime $HOCH_2(CHOH)_4CH = NOH$; I is the completely acetylated oxime, $AcOCH_2(CHOAc)_4CH = NOAc$ that loses 1 mole of HOAc to form J, $AcOCH_2(CHOAc)_4C{\equiv}N$; K is an aldopentose, $HOCH_2(CHOH)_3CHO$.

(b) The acetates undergo transesterification to give methyl acetate freeing all the sugar OH's. This is followed by reversal of HCN addition.

(c) There is loss of one C from the carbon chain.

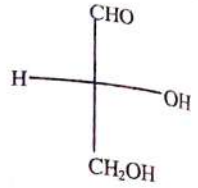
(d) Wohl degradation

(e) The α-CHOH becomes the –CH = O without any configurational changes of the other chiral carbons. Thus no epimers are formed.

SOLVED PROBLEMS

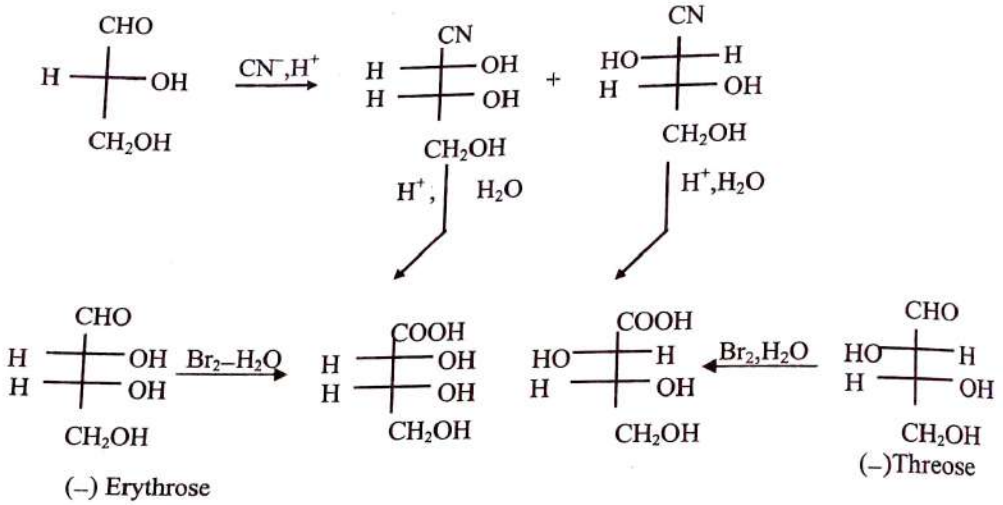
Subjective Questions

Prob. 1. When R glyceraldehyde, $CH_2OHCH(OH)CHO$ is treated with cyanide and the resulting product is hydrolysed, two monocarboxylic acids are formed. These acids are identical with the acids obtained by oxidation with Br_2 - water of (–) threose and (–) erythrose. Assign single structure to (–) erythose and to (–) threose.



R- Glyceraldehyde

Sol.

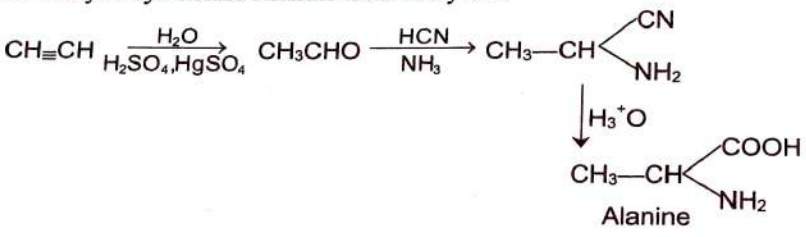


(–) Erythrose

(–)Threose

Prob. 2. How will you synthesize Alanine from acetylene.

Sol.



Alanine

Objective Questions

Prob. 1. What per cent of α anomers of D–glucose present in an equilibrium mixture with a specific rotation is $+52.6^{\circ}$? Given $[\alpha]_{\alpha} = +112.2^{\circ}$ and $[\alpha]_{\beta} = +18.7^{\circ}$.

(A) 50%

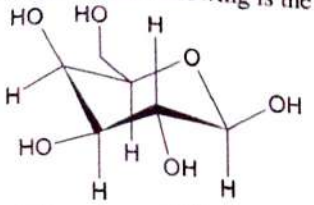
(B) 64%

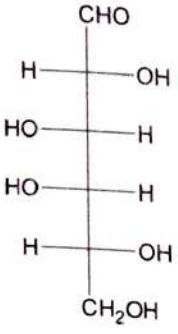
(C) 36%

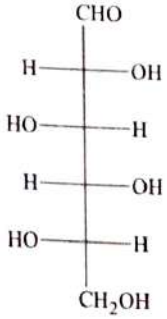
(D) 63%

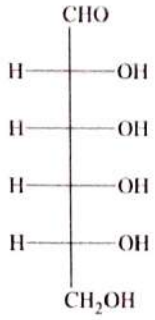
Sol. **(C)**

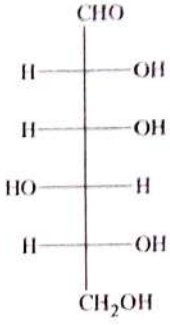
Prob. 2. Which of the following is the open-chain form of the pyranose structure shown below?



(A) 

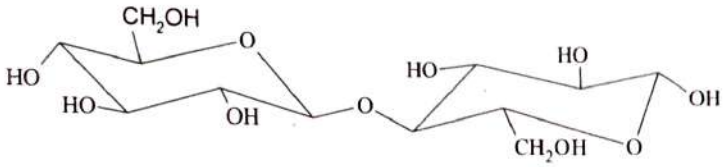
(B) 

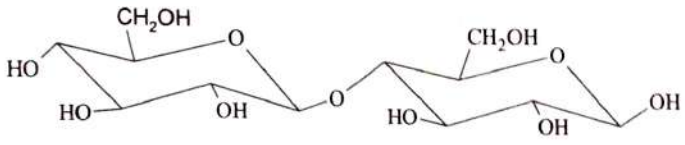
(C) 

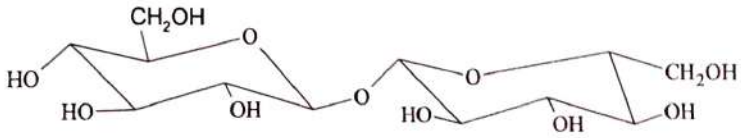
(D) 

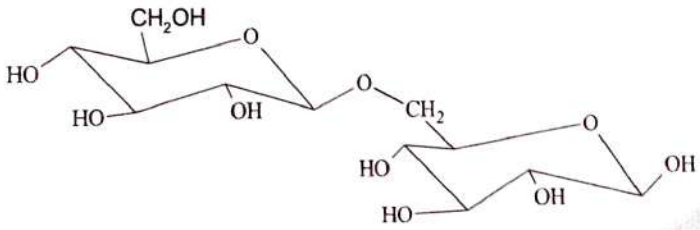
Sol. **(A)**

Prob. 3. Which of the following is a non reducing sugar?

(A) 

(B) 

(C) 

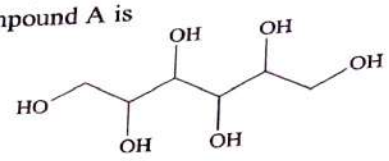
(D) 

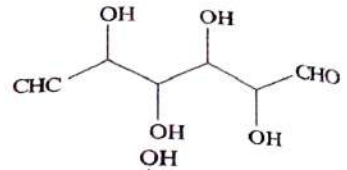
Sol. **(C)**

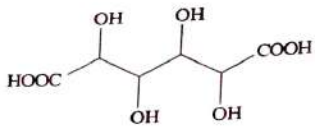
C – 1 of both units are linked through glycosidic linkage.

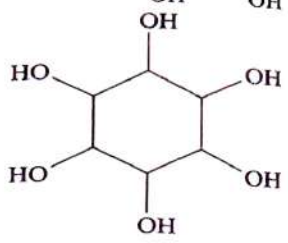
Prob. 4. $\underset{(1\ mole)}{A} + \underset{(6\ mole)}{6HIO_4} \longrightarrow \underset{(6\ moles)}{6HCOOH}$

The compound A is

(A) 

(B) 

(C) 

(D) 

Sol. **(D)**

Prob. 5. Which of the following is not a condensation polymer?

(A) Glyptal
(B) Nylon-66
(C) Dacron
(D) PTFE

Sol. Others are condensed polymer
Hence **(C)** is the correct answer.

Prob. 6. Cellulose is a polymer constituted by

(A) α - D – glucose
(B) β - D – glucose
(C) both α - D and β - D varieties
(D) none of these

Sol. **(B)**

Prob. 7. Which of the following is not a chromophore?

(A) $-NH_2$
(B) $-NO$
(C) $-NO_2$
(D) $-N=N-$

Sol. Chromophore is colour bearing group
Hence **(A)** is the correct answer.

Prob. 8. Polypropylene is not used in

(A) clothes
(B) parachute ropes
(C) ropes
(D) heat-resistant plastics

Sol. **(B)**

Prob. 9. Which of the following alternatives is/are correct regarding the statement(s)?

(A) With bromine water, glucose oxidizes into gluconic acid, but fructose gives no reaction.
(B) With nitric acid, glucose oxidizes into saccharic acid, but fructose oxidizes into mixture of glycolic acid, tartaric acid and trihydroxy glutaric acid.
(C) Glucose on reduction gives sorbitol and fructose on reduction gives mixture of sorbitol and mannitol.
(D) Glucose gives positive test with alcoholic α-naphthol + conc. HCl

Sol. **(A, B, C)**

Prob. 10. Which of the following is not a polyamide fibre
(A) Nylon-6, 6
(B) Nylon-6
(C) Teflon
(D) Terylene

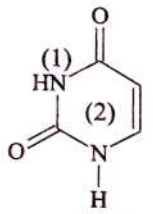
Sol. **(C, D)**
Teflon is polytetrafluoroethene, terylene is a polyester while nylon-6,6 and nylon-6 are polyamides.

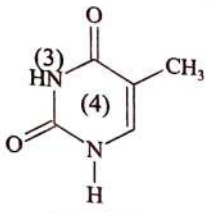
Prob. 11. If sucrose is treated with conc. H_2SO_4, the product formed is
(A) CO_2
(B) C
(C) CO
(D) H_2O

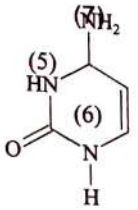
Sol. $C_{12}H_{22}O_{11} \xrightarrow{\text{conc. } H_2SO_4} 12C + 11H_2O$
(absorbed by conc. H_2SO_4)

Hence, **(B)** and **(D)** are the correct answers.

Prob. 12. Consider the following three bases of DNA and RNA.


Uracil
(In RNA)


Thymine
(In DNA)

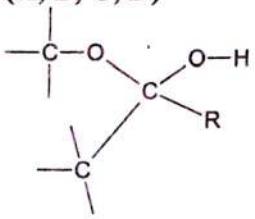

Cytosine
(In both DNA and RNA

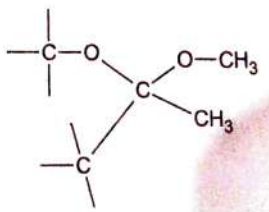
Order of acidic strength on different sites will be
(A) (2) > (1)
(B) (3) > (4)
(C) (7) > (6)
(D) (6) > (5)

Sol. **(A, B, D)**

Prob. 13. Identify the correct statement(s) out of the following sugars:
(A) α - D- Glucopyranose has a hemi acetal structure and is a reducing sugar
(B) Methyl α - D-Glucopyranoside has acetal structure and is a non-reducing sugar
(C) α - D-Glucopyranose and β-D-Glucopyranose are anomers
(D) Lactose is a disaccharide and is a reducing sugar

Sol. **(A, B, C, D)**


α-D-Glucopyranose


Methyl α-D-Glucopyranoside

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: D-glucose and D-fructose can be distinguished by bendict test as well as Br_2/H_2O.
and
STATEMENT-2: D-glucose and D-fructose are reducing sugar.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

2. **STATEMENT-1:** α-D-Glucopyranose is a non-reducing sugar.
and
STATEMENT-2: Six membered pyranose ring is always in equilibrium with its open chain counterpart.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)** is wrong but **(R)** is correct.

3. **STATEMENT-1:** Fructose is a reducing sugar.
and
STATEMENT-2: Fructose does not have aldehyde functional group.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. Both are correct but not correct explanation.

4. **STATEMENT-1:** Sucrose cann't undergo inversion in basic medium.
and
STATEMENT-2: Acetal linkage cannot be broken in basic medium.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. Both are correct and correct explanation.

5. **STATEMENT-1:** The weight average molecular weight of a polymer is necessarily greater than its number average molecular weight.
and
STATEMENT-2: Larger molecules in a sample weigh more than smaller molecules.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

6. **STATEMENT-1:** Sugars having hemiacetal linkages are reducing
and
STATEMENT-2: Hemiacetal can be converted back into aldehyde or ketone group in basic medium.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

7. **STATEMENT-1:** Sugars having acetal linkages are non-reducing
and
STATEMENT-2: Acetal cann't be converted back into aldehyde or ketone group in basic medium.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. A wrong
B correct

8. **STATEMENT-1:** Glucose gives a reddish-brown precipitate with Fehling's solution.
because
STATEMENT-2: Reaction of glucose with Fehling's solution gives CuO and gluconic acid.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1
(C) Statement-1 is True, Statement-2 is False
(D) Statement-1 is False, Statement-2 is True

Sol. $C_6H_{12}O_6 + \text{Fehling solution} \longrightarrow (C_6H_{11}O_7)^- + \underset{\text{(Red ppt.)}}{Cu_2O}$ –

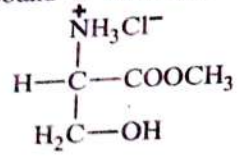
COMPREHENSION – I

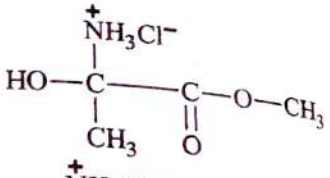
An α-amino acids 'A' has molecular formula $C_3H_7NO_3$. 'A' on treatment with methanol in presence of HCl yields 'B' $[C_4H_{10}NO_3Cl]$. 'B' on further treatment with PCl_5 yields 'C' $[C_4H_9NO_2Cl_2]$ which on acidic hydrolysis yields 'D' $[C_3H_6NO_2Cl]$. 'D' on reduction with Na(Hg) in dilute acidic medium yields alanine.

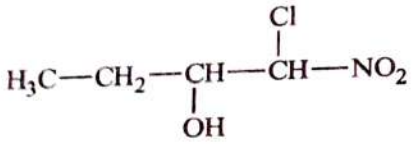
1. Compound 'A' has how many functional groups?
(A) 1 (B) 2
(C) 3 (D) 4

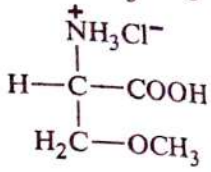
Sol. **(C)**

2. Compound 'B' should be

(A) 

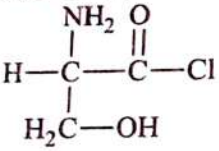
(B) 

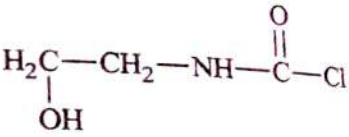
(C) 

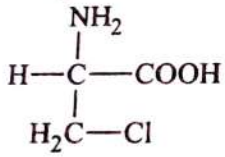
(D) 

Sol. **(A)**

3. Compound 'D' is:

(A) 

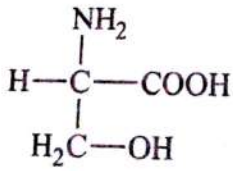
(B) 

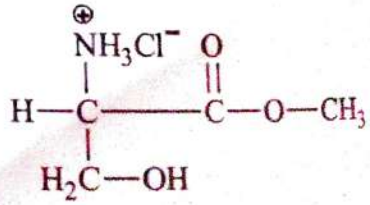
(C) 

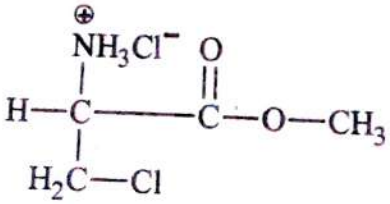
(D) (A) and (B) both are possible

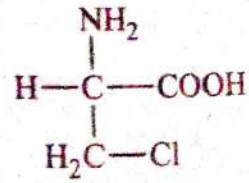
Sol. **(C)**

Solution for the question No. 1 to 3

A 

B 

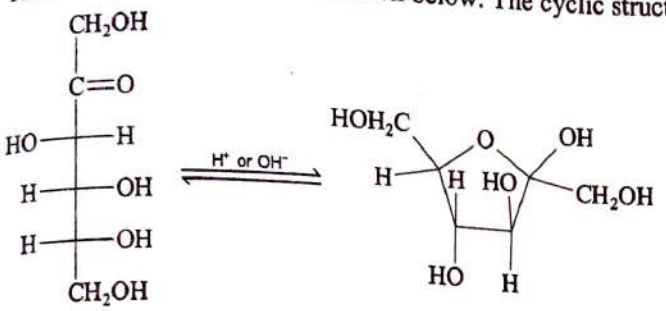
C 

D 

COMPREHENSION – II

Read the following paragraphs and answer the questions given below it:

The structures of fructose are shown below. The cyclic structure predominates in aqueous solution.

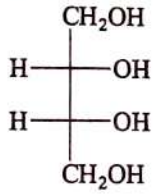


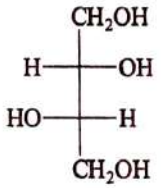
1. What is the functional group at C2 in the cyclic structure?
   (A) Alcohol
   (B) Ether
   (C) Acetal
   (D) Hemiacetal

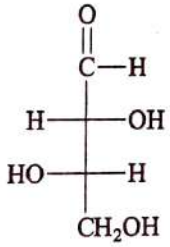
Sol. **(D)**

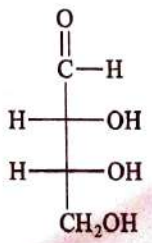
Acid catalysed reaction of D-glucose with benzaldehyde produces 4, 6-oxo-benzylidene derivative. Reduction with $NaBH_4$, followed by excess HIO_4 cleavage and acid hydrolysis yields a tetrose ($C_4H_8O_4$) and benzaldehyde.

2. The Fischer projection of the tetrose is

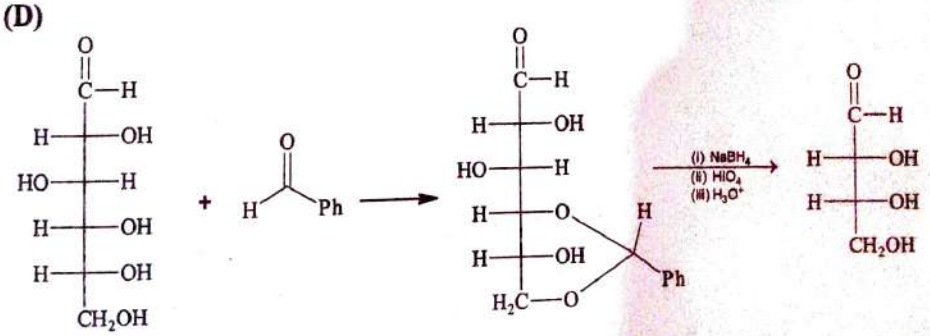
(A) 

(B) 

(C) 

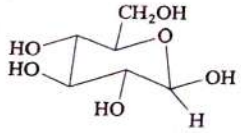
(D) 

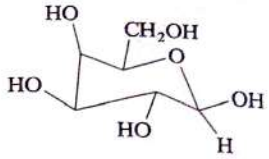
Sol. **(D)**

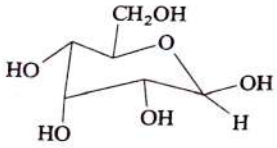


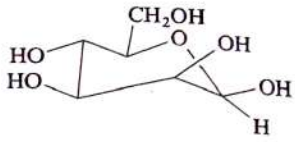
D-aldohexose gives an optically inactive aldonic acid on oxidation with nitric acid. Ruff degradation of this aldohexose produces an aldopentose that is reduced by $NaBH_4$ to an optically active alditol.

3. Which of the following is the above mentioned hexose?

(A) 

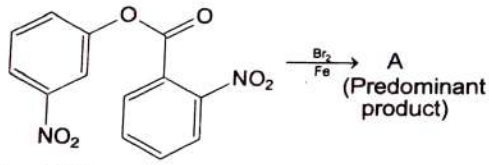
(B) 

(C) 

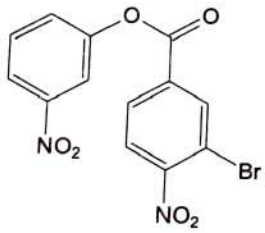
(D) 

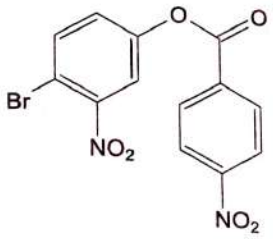
Sol. **(B)**

Alkenes when react with HOCl produce an addition product. This product produces a cyclic ether on reaction with an alkali.

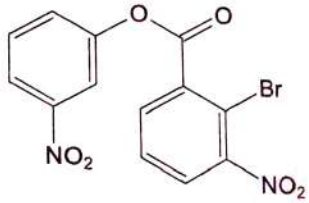
4. 

A will be

(A) 

(B) 

(C)

(D) 

Sol. **(B)**

MATRIX-MATCH TYPE QUESTIONS

1. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Cellulose	(p)	Natural polymer
(B)	Nylon-6,6	(q)	Synthetic polymer
(C)	Protein	(r)	Amide linkage
(D)	Sucrose	(s)	Glycoside linkage
		(t)	Hexamethylenediamine and adipic acid

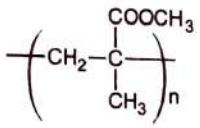
Sol. **A → (p, s); B → (q, r, t); C → (p, r); D → (s)**

2. Match the following:

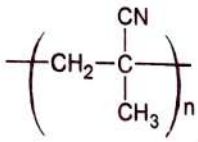
Column-I		Column-II	
(A)	PMMA	(p)	Thermosetting polymer
(B)	Orlon	(q)	Thermoplastic polymer
(C)	Bakelite	(r)	Addition polymer
(D)	Dacron	(s)	Condensation polymer
		(t)	Brittle
		B	

Sol. **A → q, r, t; B → r; C → p, s; D → q, s**

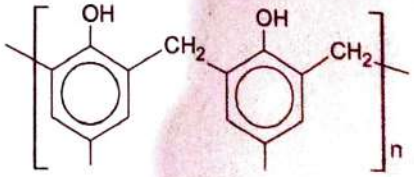
PMMA — Polymethylacrylate



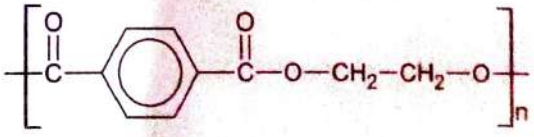
Orlon — Polyacrylonitrile



Bakelite — Phenol formaldehyde



Dacron — Polyethylene tetraphthalate



Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	Sucrose	(i)	β -Glucose	(P)	Non reducing sugar
(II)	Starch	(ii)	β -Fructose	(Q)	Reducing sugar
(III)	Cellulose	(iii)	α -Glucose	(R)	Polysaccharide
(IV)	Glycogen	(iv)	β -Glucose	(S)	Disaccharide

3. Which of the following combination is correct ?
[A] (I) (ii) (Q)
[B] (II) (i) (P)
[C] (II) (iv) (Q)
[D] (I) (iii) (S)

4. Which of the following combination is incorrect ?
[A] (I) (ii) (S)
[B] (I) (iii) (P)
[C] (III) (iv) (R)
[D] (II) (iii) (S)

5. Which of the following combination is correct ?
[A] (I) (iii) (Q)
[B] (III) (iv) (R)
[C] (III) (iii) (R)
[D] (IV) (iv) (S)

Sol.

3. **D**
4. **D**
5. **B**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. A compound of mol wt. 180 is acetylated to give a compound of molecular wt. 390. What is the number of amino groups in the initial compounds?

Sol. **(5)**

Difference in mass of compound = 390 – 180 = 210

Wt. of CH_3CO – group = 43

Therefore, number of $-NH_2$ group $\frac{210}{43}$

$= 4.88 \approx 5$

2. How many tripeptides can be prepared by linking the amino acids glycine, alanine and phenyl alanine?

Sol. **(6)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. If x number of tripeptides possible from three different amino acids if each is used only once. Then value of $\frac{x}{4}$ is

Sol. **1.50**

$$\frac{6}{4}=1.50$$

Prob. 2. If p number of sugar are reducing out of all, glucose, fructose, ribose, deoxyribose, sucrose, maltose, xylose. Find value of $\frac{p}{5}$

Sol. **1.20**

$$\frac{6}{5}=1.20$$

Prob. 3. pK_{a_1} and pK_{a_2} of isoleucine are 2.38 and 9.66. Thus isoelectric point of isoleucine is

Sol. **6.02**

$$\frac{2.38+9.66}{2}=\frac{12.04}{2}=6.02$$

Prob. 4. Amongst the following, the total number of elastomers is x.
Natural rubber, Polypropylene, Polythene, Vulcanized rubber, Nylon-6, Polyvinyl chloride, Buna-N, Chloroprene, Buna-S, Polystyrene

Sol. **2.5**

$$\frac{5}{2}=2.5$$

Prob. 5. Amongst the given polymers: Polyester, bakelite, polyethene, PVC, Teflon, PAN, PMMA, nylon-6, melamine-formaldehyde. If p are number of thermoplastics, find value of $\frac{p}{2}$

Sol. **3.50**

$$\frac{7}{2}=3.50$$

PRACTICAL ORGANIC CHEMISTRY 21

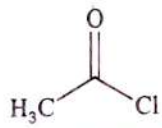
TESTS FOR DIFFERENT ELEMENTS

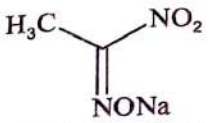
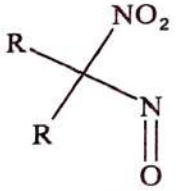
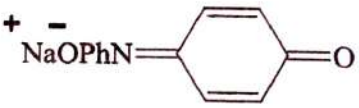
Sodium Fusion Test

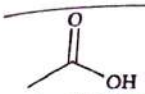
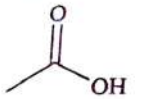
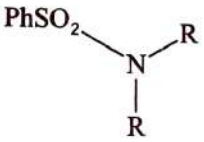
This test is used for the detection of nitrogen, sulphur and halogens in organic compounds. First, the Lassigne's filtrate is prepared where the organic compound is fused with Na metal to form ionizable compounds such as NaCN, Na_2S and NaX, which can be further detected for the presence of N, S and X by performing other tests.

Name of Element	Reagent used	Compound formed	Reactions involved
N	$FeSO_4$	$Fe_4[Fe(CN)_6]_3$	(i) $FeSO_4 + 6NaCN \longrightarrow Na_4[Fe(CN)_6] + Na_2SO_4$ (ii) $3Na_4[Fe(CN)_6] + 2Fe_2(SO_4)_3 \longrightarrow Fe_4[Fe(CN)_6]_3 + 6Na_2SO_4$ Prussian blue
S	Dil. CH_3COOH and $(CH_3COO)_2Pb$	PbS	Black ppt. of PbS is formed.
	$Na_2[Fe(CN)_5NO]$	Purple colour is obtained	The colour slowly fades on standing.
X (halogen, Cl, Br, I but N, S are absent)	(i) dil. HNO_3 (ii) $AgNO_3$ (increases)	ppt. is obtained	If ppt. is white and readily soluble in NH_3 then Cl is present. If ppt. is pale yellow and is not easily soluble then Br is present and if ppt. is yellow and insoluble in aqueous NH_3 then I is indicated.
Br, I (N, S are absent)	Cl_2/H_2O	(Colourless or reddish brown)	If iodine is present then first purple colour is obtained which dissolves and the solution becomes colourless on excess addition of Cl_2/H_2O. If Br is present then colour will be reddish brown.
X (Br, Cl, I) (N and S present)	(i) dil. HNO_3 (ii)evaporation	H_2S, HCN gases are evolved	Since N, S are extracted from mixture now test number 3, i.e. for halogens can be performed.
X (Cl, Br, I) (N and/or S present)	1–2 drops of $Ni(NO_3)_2$	$Ni(CN)_2$ and Ni_2S is formed	$Ni(CN)_2$ and Ni_2S is filtered off. The filtrate is acidified with 2 M HNO_3 and test for halogen is performed.

DETECTION OF FUNCTIONAL GROUPS

Name of functional group	Name of test	Reagent used	Compound formed	Observation
–OH group	Sodium metal test	Active metals like Na	H_2 is formed	The alcohol used should be dry as water also reacts with Na.
–OH group	Acetyl chloride test		HCl is evolved	The reaction is fast for 1° and 2° alcohol. The HCl gas evolved forms white fumes with NH_4OH.

Name of functional group	Name of test	Reagent used	Compound formed	Observation
–OH group	Cerric ammonium test	$(NH_4)_2Cl(NO_3)_6$	$R(OH)_2Cl(NO_3)_4$	Pink or red.
Test for distinguishing 1°, 2° and 3° alcohols	Lucas test	$ZnCl_2$ + HCl		White turbidity may appear. 1° alcohol → no reaction at room temperature. 2° alcohol → white turbidity after 5–10 minutes. 3° alcohol → white turbidity appears instantly.
Test for distinguishing 1°, 2° and 3° alcohols	Victor Meyer test	(i) P/I_2 or HI (ii) $AgNO_2$ (iii) HNO_2 (iv) NaOH	1° alcohol →  (Sodium salt of Nitrolic acid)	Red colour
			2° alcohol →  (Pseudonitrol)	Blue colour obtained after treating pseudonitrol with alkali.
			3° alcohol → colourless	3° alcohol gives colourless solution as $R_3C–NO_2$ formed does not react with HNO_2.
Phenolic –OH group	Litmus test	Blue litmus paper		Red colour is obtained.
Phenolic –OH group	Neutral $FeCl_3$ test	Neutral $FeCl_3$	$[Fe(OC_6H_5)_6]^{3-}$	Violet complex
Phenolic –OH group	Liebermann's test	(i) $NaNO_2/H_2SO_4$ (ii) Excess of H_2O (iii) Excess NaOH	 (Iodophenol anion)	Blue colour First blue colour of p–nitrosophenol is obtained then colour changes to red (iodophenol) then blue colour of iodophenol anion is formed.
Carbonyls (aldehydes and ketones)	2, 4–dinitrophenyl hydrazine	2, 4–dinitro phenyl hydrazine	Hydrazine derivative	Crystalline ppt. The ppt. is oily at first and becomes crystalline on standing
Aldehydes	Schiff's test	Schiff's reagent		Pink, red or manganese colour. Firstly aldehyde is dissolved in alcohol and then treated with Schiff's reagent.
Aldehydes	Tollen's test (silver-mirror test)	Tollen's reagent	Ag is formed	Silver mirror is obtained.
Aldehydes	Fehling's test	Fehling's solution	Cu_2O is formed	Red colour

Name of functional group	Name of test	Reagent used	Compound formed	Observation
 Carboxyl group	Sodium bicarbonate test	$NaHCO_3$	CO_2 is formed	Brisk effervescence
 Carboxyl group	Ester test	(i) CH_3CH_2OH conc. H_2SO_4 (ii) Aqueous Na_2CO_3	Ester is formed	Sweet, fruity smell of an ester.
RNH_2 (1° aliphatic amine)	Nitrous acid test	$NaNO_2/HCl$	N_2 is evolved	
$PhNH_2$	Nitrous acid test	(i) $NaNO_2/HCl$ (ii) β – naphthol	Azo dye	Orange ppt.
R_2NH	Nitrous acid test	$NaNO_2/HCl$	Nitroso amine	Orange-yellow oils
R_3N	Nitrous acid test	$NaNO_2/HCl$	Trialkyl ammonium nitrile	Colourless
RNH_2 (aliphatic or aromatic)	Carbylamine test	$CHCl_3/KOH$	RNC	Unpleasant odour
RNH_2	Hinsberg test	$PhSO_2Cl$	$PhSO_2NHR$	The product formed is soluble in NaOH.
R_2NH	Hinsberg	$PhSO_2Cl$		The product formed is insoluble in NaOH.
Nitro group	Milliken's test	Zn/NH_4Cl	RNHOH	The product obtained is treated with ammoniacal $AgNO_3$ solution and heated in water bath, white to grey and then black ppt. is formed.
Nitro group		(i) Sn/HCl (ii) Carbylamine test	RNC	Unpleasant odour

ILLUSTRATIVE PROBLEMS

1. When primary bromides are treated with a solution of NaI in pure acetone precipitate generates within 3 minutes at 25°C whereas primary chlorides when treated with same reagent precipitate generates within 6 minutes at 50°C. Explain.

Sol. The reaction is probably of the S_N2 type involving a bimolecular attack of the iodide ion upon the carbon atom carrying the chloride or bromine. Now Br^- is better leaving group when Cl^- (as it is a weaker base). So the rate of the reaction for bromides is faster.

$RBr + NaI \longrightarrow RI + NaBr\downarrow$

$RCl + NaI \longrightarrow RI + NaCl\downarrow$

2. A ketone 'A' undergoes the haloform reaction and on reduction gives a compound B. On heating with sulphuric acid B gives C which forms mono ozonide D with ozone. D on hydrolysis in the presence of Zinc dust gives only acetaldehyde. Identify A, B, C and D and write the chemical reaction involved.

Sol. $A = CH_3 - CH_2 - \overset{\overset{O}{\|}}{C} - CH_3$

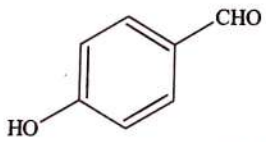
$B = CH_3 - CH_2 - \underset{\underset{OH}{|}}{CH} - CH_3$

$C = CH_3CH = CH - CH_3$

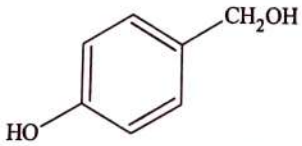
D = Ozonide

3. An organic compound A contains C = 68.85%, H = 4.9% and reacts with NH_2OH. On treatment with NaOH it forms the compound B and C. The compound B is soluble in HCl and can be oxidized to compound C with alk. $KMnO_4$. The compound C on treatment with conc. HCl forms an acid D, which on heating with alkali gives phenol. Identify ABCD.

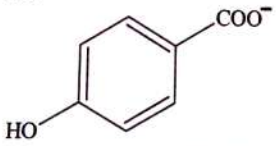
Sol. A =



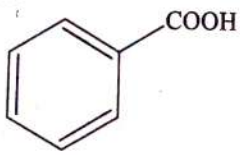
B =



C =

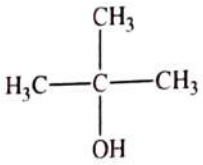


D =



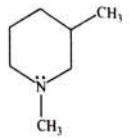
4. **A compound of formula $C_4H_{10}O$ reacts rapidly with metallic sodium but not with Lucas reagent. Treatment of A, with hot conc. sulphuric acid yields a new compound B, C_4H_8. If B is hydrated in aqueous sulphuric acid, a new compound C, $C_4H_{10}O$ is formed. Compound C is almost inert to metallic sodium but reacts rapidly with Lucas reagent. Assign structure to A, B and C.**

Sol. A = $CH_3-\underset{CH_3}{\overset{|}{C}}H-CH_2OH$ B= $CH_3-\underset{CH_3}{\overset{|}{C}}=CH_2OH$ C =

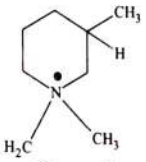


5. **Compound A $C_7H_{15}N$, is insoluble in water but dissolve in dil. HCl. It does not undergo any reaction with nitrous acid. When A is treated with methyl iodide and then with silver oxide, B is formed. On heating B forms C $C_8H_{17}N$. C adds hydrogen in the presence of a catalyst forming D, $C_8H_{19}N$. D on treatment with CH_3I and then with Ag_2O gives $(CH_3)_3N$ and E. E can be hydrogenated to 2-methyl pentane. Assign structures to compound A to E.**

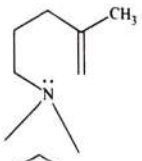
Sol. A =



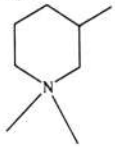
B =



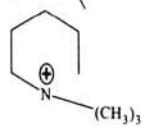
C =



D =



E =



6. **Combustion of .42 g of an organic compound gave .924 g of CO_2 and .243 g of H_2O. Due to distillation of .208 g of the compound with NaOH ammonia evolved required 30 mL of $\frac{N}{20}$ H_2SO_4 for complete neutralization. Calculate the percentage composition of compound.**

Sol. % C = 60% %H = 6.43%
%N = 10.09 % % O = 24.48 %

7. **Four isomeric compounds (A),(B),(C) and (D) have molecular formula C_4H_6**

(i) A gives white ppt with Tollen's reagent
(ii) B consumes one mole of hydrogen and on ozonolysis gives succinaldehyde.
(iii) C consumes two moles of hydrogen and on ozonolysis gives only one product i.e. acetic acid.
(iv) D consumes two moles of hydrogen and on ozonolysis gives 2 moles of formaldehyde and one mole of glyoxal.

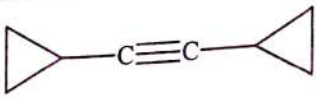
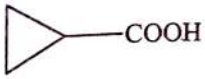
Sol. $A = CH_3 - CH_2 - C = CH$

B=

$C = CH_3C \equiv C - CH_3$

$D = CH_2 = CH - CH = CH_2$

8. Compound (A) C_8H_{10} discharges the colour of Br_2/water and consumes two moles of hydrogen, (A) on oxidation with hot $KMnO_4/H^+$ (A) gives compound (B) $C_4H_6O_2$. B does not discharge colour of Br_2 and gives effervescence with $NaHCO_3$. What will be structure of (A) and (B).

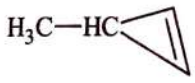
Sol. A =  B = 

9. Which of the following compound would give a positive iodoform test?

(a) 2-butanol
(b) 1- pentanol
(c) Acetone
(d) 3-pentanone
(e) 2-pentanone
(f) 3-pentanol
(g) 1- phenyl ethanol
(h) Aceo phenone

Sol. (a) 2-butanol
(c) Acetone
(e) 2-pentanone
(h) Aceo phenone

10. Six organic compounds have the same molecular formula C_4H_6 A B C D E and F. All decolourise red colour of Br_2/CCl_4, only C gives white ppt. with Amm. $AgNO_3$. A on ozonolysis gives dicarboxylic acid which on heating produce propanoic acid. (B) on ozonolysis gives succinic acid. While D gives two molecules of acetic acid and (E) gives 2-moles of formic acid and one mole of oxalic acid. F gives 1 mole of acetic acid, one mole of formic acid and CO_2. What are A to F.

Sol. A =  B =

C = D = $CH_3 - C \equiv C - CH_3$

E = $CH_2 = CH - CH = CH_2$ F = $H_2C = C = CH - CH_3$

11. Compound X (C_5H_8O) does not react appreciably with Lucas reagent at room temperature but gives a precipitate with ammoniacal silver nitrate. With excess of MeMgBr, 0.42 g of (X) gives 224 ml CH_4 at STP. Treatment of (X) with H_2 in presence of Pt catalyst followed by boiling with excess HI gives n-pentane. Suggest structure for X and write the equations involved.

Sol. (i) (X) C_5H_8O does not react with Lucas reagent appreciably at room temperature but gives precipitate with ammoniacal $AgNO_3$ and thus, (X) has terminal alkyne linkage as well as primary alcoholic group.

(ii) (X) on hydrogenation and then reacting with HI gives pentane and thus, (X) is straight chain compound.

(iii) Keeping in view of the above facts (X) may be

$CH_2.CH_2.CH_2.C \equiv CH$
|
OH Pent-4-yn-1-ol

(iv) Its reaction with MeMgBr gives CH_4 (it has two acidic or active H^* atoms) and thus, 1 mole of (X) will give two moles of CH_4.

$$CH_2.CH_2.CH_2.C \equiv CH + 2CH_3MgBr \longrightarrow 2CH_4$$
|
OH

$\because$ 84 g (X) gives 2×22.4 litre CH_4

$\therefore$ 0.42 g (X) will give $\frac{2 \times 22.4 \times 0.42}{84} = 224$ ml CH_4

Given fact is confirmed.

12. A basic volatile, nitrogen compound gave a foul smelling gas when treated with $CHCl_3$ and alcoholic KOH. A 0.295 g sample of the substance dissolved in aqueous HCl and treated with $NaNO_2$ solution at 0°C liberated a colourless, odourless gas whose volume correspond to 112 ml at STP. After the evolution of the gas was complete, the aqueous solution was distilled to give an organic liquid which did not contain nitrogen and which on warming with alkali and iodine gave yellow precipitate. Identify the original substance. Assume it contains one N atom per molecule.

Sol. (i) Compound gives foul odour with $CHCl_3$ and alcoholic KOH. Thus, it is primary amine, say $R - NH_2$. The reaction is carbylamine reaction.

(ii) $RNH_2 \xrightarrow[0°C]{HCl+NaNO_2} ROH + N_2$

$\because$ 112 ml N_2 is given by 0.295 g

$\therefore$ 22400 ml N_2 is given by $\frac{0.295 \times 22400}{112} = 59$

$\therefore$ Molecular weight of $RNH_2 = 59$

$\therefore$ $R = 59 - (14 + 2) = 43$

i.e., $C_nH_{2n+1} = 43$

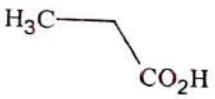
$\therefore$ $n = 3$

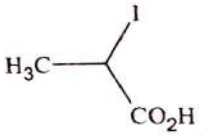
(iii) Given, $C_3H_7NH_2 \xrightarrow{HCl+NaNO_2} C_3H_7OH \xrightarrow{I_2+NaOH}$ Yellow precipitate

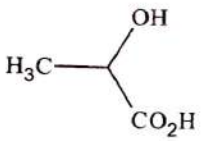
Since alcohol gives iodoform test and thus it is secondary alcohol, i.e. $CH_3CHOHCH_3$ (Propan-2-ol)

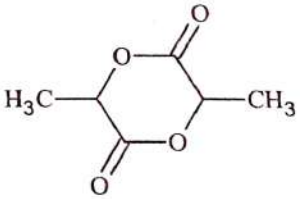
Thus, original compound should be $\underset{\text{Propan-2-amine}}{CH_3-\overset{NH_2}{\overset{|}{C}H}-CH_3}$

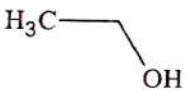
13. An organic compound (A) of molecular formula $C_3H_6O_2$ which gives brisk effervescence on treatment with $NaHCO_3$ and on treatment with red phosphorus in presence of I_2 gives (B). (B) an optically active compound on treatment with AgOH produces (C), which is also optically active. (C) on treatment with H^+ at high temperature produces (D). (C) on decarboxylation produces (E), a compound less acidic than (C). (E) on oxidation gives a compound (F) which is more acidic than (E) and had smell of vinegar. Identify (A) to (F).

Sol. A = 

B = 

C = 

D = 

E = 

F = CH_3COOH

SOLVED PROBLEMS

Subjective question

Prob. 1. There are two mixtures A and B available in the laboratory. 'A' contains benzoic acid and naphthalene, whereas 'B' contains benzoic acid and nitrobenzene. Suggest a method to get benzoic acid and nitrobenzene.

Sol. Steam distillation

Prob. 2. What conclusion would you draw if during Lassaigne's test a blood red colouration is obtained?

Sol. It indicates the presence of both N and S.

Objective question

Prob. 1. Sodium fusion extract of an organic compound gives a blood red colouration with few drops of $FeCl_3$ solution. This indicates the presence of
(A) Nitrogen in the organic compound
(B) Sulphur in the organic compound
(C) Nitrogen and sulphur in the organic compound
(D) Sulphur and chlorine in the organic compound

Sol. **(C)**

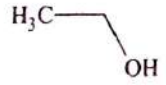
Prob. 2. The method which is not useful in the purification of liquid is
(A) Fractional distillation
(B) Steam distillation
(C) Chromatography
(D) Sublimation

Sol. **(D)**

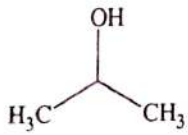
Prob. 3. MIC (which is responsible for Bhopal gas tragedy) can be produced by which of the following test
(A) Mulliken Test
(B) Carbylamine Test
(C) Victor-Meyer Test
(D) Libermann Nitrose Test

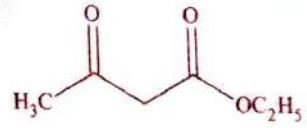
Sol. **(B)**

Prob. 4. Among the following which will not respond to iodoform test

(A) 

(B)

(C) 

(D) 

Sol. **(D)**

Prob. 5. Among the following which statement is not correct

(A) $CH_3-\overset{\overset{O}{\|}}{C}-OH$ will not respond to haloform test

(B) Schiff's reagent and Schiff's base are different compound

(C) Fehling solution is a good reagent to detect aromatic aldehydes

(D) Both aldehyde and ketone can react with 2,4-dinitrophenylhydrazine reagent

Sol. **(C)**

Prob. 6. To separate a mixture of 1°, 2° & 3° amines from each other one should follow

(A) Beckmann's method

(B) Hinsberg's method

(C) Zeisel method

(D) Victor Meyer's Method

Sol. **(B)**

Prob. 7. Phenol can be distinguished from ethanol by

(A) Tollen's reagent

(B) Benedict's reagent

(C) $FeCl_3$

(D) Schiff's base

Sol. **(C)**

Prob. 8. p-Cl – $C_6H_4NH_2$ and $PhNH_3^+Cl^-$ can be distinguished by

(A) NaOH

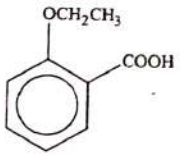
(B) $AgNO_3$

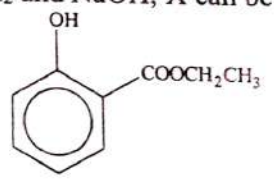
(C) $LiAlH_4$

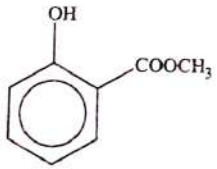
(D) Zn

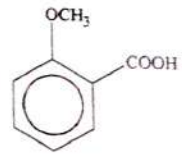
Sol. **(B)**

Prob. 9. A compound (X) on hydrolysis gave an acid and alcohol. Acid gave violet colour with neutral $FeCl_3$ while alcohol gave yellow precipitate on boiling with I_2 and NaOH, X can be

(A) 

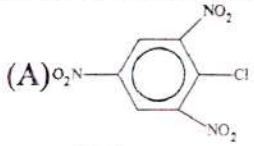
(B) 

(C) 

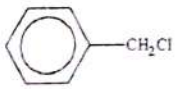
(D) 

Sol. **(B)**

Prob. 10. Detection of the chlorine is not possible without preparing sodium extract in

(A) 

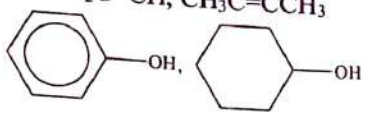
(B) $CHCl_3$

(C) 

(D) $CH_2=CHCH_2Cl$

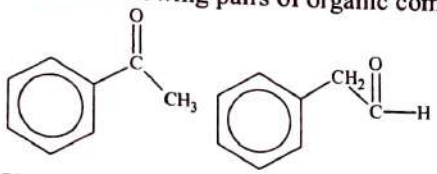
Sol. **(B)**

Prob. 11. Each pair has been matched with the suitable reagent used for making distinction. Which pair has been matched wrongly?

	Pair	**Reagent**
(A)	$CH_2=CH_2$, CH_3-CH_3	MnO_4^-/OH^-
(B)	$CH_3CH_2C{\equiv}CH$, $CH_3C{\equiv}CCH_3$	Br_2 water
(C)	$CH_3CH_2C{\equiv}CH$, $CH_3C{\equiv}CCH_3$	$AgNO_3/NH_4OH$
(D)		$FeCl_3$

Sol. **(B)**

Prob. 12. Consider the following pairs of organic compounds

I: 

II: CH_3OH CH_3CH_2OH

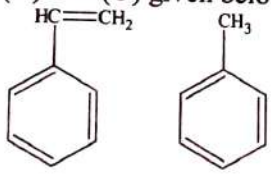
III: $CH_3\overset{\overset{O}{\|}}{C}CH_3$ HCHO

A test that can make distinction between each pair is

(A) Lucas test
(B) Silver-mirror test
(C) Victor-Meyer's test
(D) Iodoform Test

Sol. **(D)**

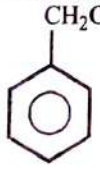
Prob. 13. (X) and (Y) given below can be chemically distinguished using



(A) Br_2 water
(B) alkaline $KMnO_4$
(C) both of these
(D) none of these

Sol. **(A)**

Prob. 14. Which alcohol gives Lucas test immediately ?

(A) 

(B) CH_3CH_2OH
(C) $(CH_3)_3C-OH$
(D) $(CH_3)_2CH-OH$

Sol. **(A, C)**

Prob. 15. Which of the following reagents can be used to distinguish benzaldehyde from acetophenone ?

(A) Ammonical solution of $AgNO_3$
(B) NaOBr
(C) Brady's reagent
(D) Cu^{2+} in sodium citrate

Sol. **(A, B)**

MEGACOSM

Reasoning Type

1. STATEMENT-1: Libermann's nitroso test cannot be used to detect the presence of tertiary amine.
and
STATEMENT-2: Tertiary amines cannot form oily emulsion of nitroso phenol.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

2. STATEMENT-1: Victor Meyers method is used for determination of molecular weight a non-volatile substance.
and
STATEMENT-2: From the volume of vapour produced molecular weight of the organic compound is calculated using Avogadro's hypothesis.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

COMPREHENSION

An aromatic compound contains 69.4% carbon and 5.8% hydrogen. A sample of 0.303g of this compound was analyzed for nitrogen by Kjeldahl's method. The ammonia evolved was absorbed in 50ml of 0.05M H_2SO_4. The excess of the acid required 25ml of 0.1 M sodium hydroxide for neutralization. Molecular mass of the compound is 121.

Prob. 1. Empirical formula of organic compound is
(A) C_7H_7NO (B) C_6H_6NO
(C) C_4H_6NO (D) C_2H_6NO

Sol. **(A)**

Prob. 2. The no. of possible structures of the compound is –
(A) 2 (B) 3
(C) 4 (D) 5

Sol. **(C)**

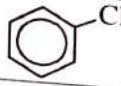
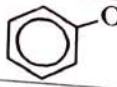
MATRIX – MATCH TYPE

1. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	CH_3COOH	(p)	Dye test
(B)	$C_6H_{12}O_6$ (Fructose)	(q)	Silver mirror test
(C)	C_6H_5CHO	(r)	Haloform test
(D)	C_6H_5OH	(s)	Liebermann's nitroso test
		(t)	Malachite green test

Sol. **A → (r) B → (q) C → (q, t) D → (p, s)**

2. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	Glucose ($C_6H_{12}O_6$)	(p)	Phthalein test
(B)	Benzaldehyde : 	(q)	Schiff's test
(C)	Phenol : 	(r)	Benedicts test
(D)	Acetaldehyde : CH_3CHO	(s)	Molisch's test
		(t)	Tollen's test

Sol. **A → (s, t) B → (q, r) C → (p) D → (q)**

3. Match the Column – I with Column – II:

	Column – I (Estimation/detection of elements)		Column – II (Methods)
(A)	Estimation of halogens	(p)	Kjeldahl's method
(B)	Estimation of carbon and hydrogen	(q)	Dumas method
(C)	Estimation of nitrogen	(r)	Carius method
(D)	Estimation of sulphur	(s)	Leibig's method

Sol. **A → r; B → s; C → p, q; D → r**

Answer 4, 5 and 6 by appropriately matching the information given in the three columns of the following table.

	Column – 1		Column – 2		Column – 3
(I)	Analysis of carbon	(i)	Lassaigne's test	(P)	H_2SO_4 conc. + K_2SO_4 + $CuSO_4$
(II)	Analysis of nitrogen	(ii)	Beilstein test	(Q)	Violet coloured complex on adding sodium nitroprusside
(III)	Analysis of sulphur	(iii)	Kjeldal's method	(R)	Greenish /Bluish flame
(IV)	Analysis of halogens	(iv)	Copper oxide test	(S)	Gas evolved on heating with CuO turns lime water milky

4. Which of the following combination is correct ?
[A] (IV) (iv) (Q) [B] (II) (iii) (R) [C] (III) (iii) (S) [D] (I) (iv) (S)

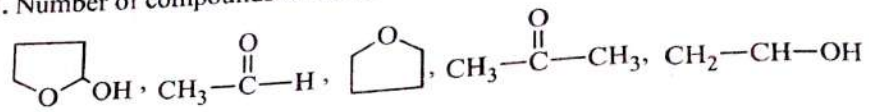
5. Which of the following combination is incorrect ?
[A] (IV) (ii) (R) [B] (III) (i) (Q) [C] (II) (iii) (P) [D] (IV) (i) (P)

6. Which of the following combination is correct ?
[A] (II) (iii) (P) [B] (I) (i) (R) [C] (II) (i) (Q) [D] (II) (i) (P)

Sol. 4. **D;** 5. **D;** 6. **A**

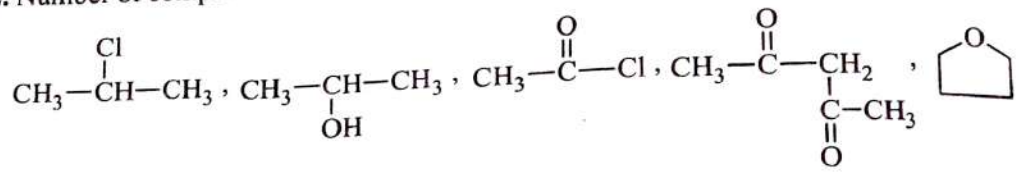
NUMERICAL VALUE ANSWER TYPE

Prob. 1. Number of compounds which gives Tollen's test in the given is x. Then value of $\frac{x}{2}$ is



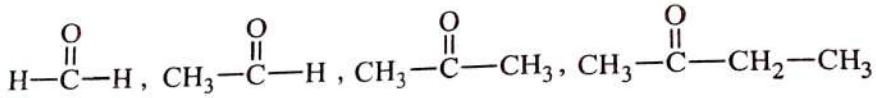
Sol. **1.5**

Prob. 2. Number of compounds which gives haloform test is x. Then value of $\frac{2x}{5}$



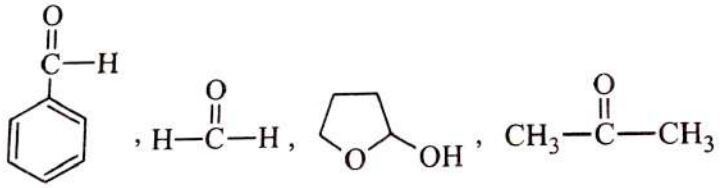
Sol. **1.2**

Prob. 3. Number of compound which gives positive $NaHSO_3$ test is x. Then value of $\frac{x}{3}$



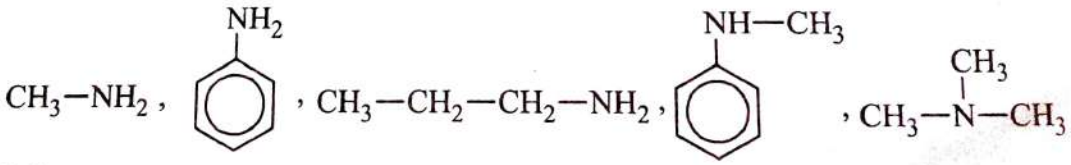
Sol. **1.33**

Prob. 4. Number of organic compound which gives positive Fehling's test, is x then value of $\frac{2x}{3}$



Sol. **1.33**

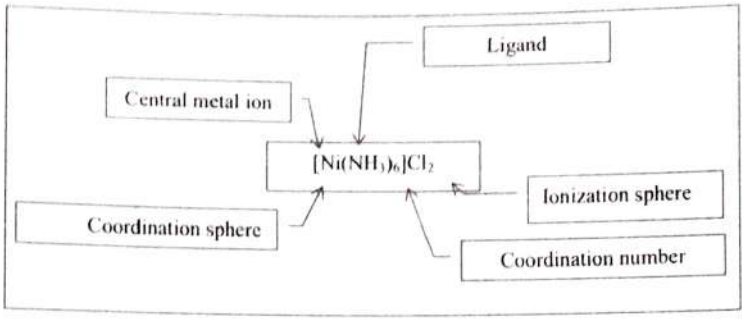
Prob. 5. Number of organic compound which gives positive carbyamine test is x then value of $\frac{x}{2}$



Sol. **1.5**

COORDINATION COMPOUNDS 22

Coordination compounds contain a central metal atom or ion surrounded by a number of oppositely charged ions or neutral molecules.



IMPORTANT TERMS

Coordination Sphere

It is the part of the coordination compound which contains the metal atom and the ligands attached to it. It is enclosed in a square bracket.

Ionization Sphere

The part of the complex compound which is present outside the square bracket.

Coordination Number

The total number of ligands attached to the central metal atom in a coordination sphere.

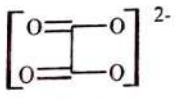
Ligands

Species (neural, negative or positive) which donate the electron pair to the central metal atom.
Depending on the number of donor sites ligands may be classified as unidentate, bidentate and multidentate.
Unidentate ligands: OH^-, Cl^-, Br^-, H_2O, NH_3 (neutral ligands)

(negative ligands)

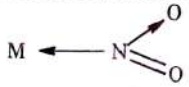
Bidentate ligands:

ethylenediamine
(en)

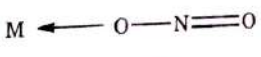


oxalate ions
(ox)

Multidentate: Having more than two donor atoms present in the molecule. They may be tri, tetra, penta or hexadentate depending upon the number of donor atoms present in their molecules, e.g. Ethylenediamine tetra acetic acid (EDTA) is a hexadentate ligand.
Ambident Ligands: Monodentate ligands which can coordinate with the central atom through more than one site are called ambident ligands, e.g. NO_2 can coordinate to the metal atom through N or O.



Nitro



Nitrito

Spectro chemical series: $I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^-, F^- <$ urea, $OH^- <$ ox, $O^{2-}, H_2O < NCS^- <$ py, $NH_3 <$ en < bpy, $< NO_2^- < C_6H_5^- < CN^- < CO$.

NOMENCLATURE OF COORDINATION COMPOUNDS

(i) **Order of naming ions:** Positive ion whether simple or complex is named first followed by negative ion. Non-ionic complexes are given a one word name.
While naming complex part, first ligand should be named and then the metal. Before naming the ligands, the number of ligands should be specified.

(ii) **Naming coordination sphere:** In naming the coordination sphere the ligands are named first and then central metal ion.

(iii) **Naming a ligand:** Name of negative ligand end with o.
Examples: CN^- – cyano, OH^- – hydroxo, SO_4^{2-} –sulphato,
$C_2O_4^{2-}$ – oxalato, NO_2^- – nitrito, NO_3^- – nitrato, NH_2^- – amido,
NH^{2-} – imido, N^{3-} – nitrido, S^{2-} – sulphido or thio,
O_2^- – superoxo, O_2^{2-} – peroxo, HO_2^{2-} – perhydroxo etc.

Neutral Ligands have no specific naming.
Examples: H_2O – aquo or aqua
NH_3 – ammine
O_2 – dioxygen
NO – nitrosyl
CO – carbonyl
$H_2N–CH_2–CH_2–NH_2$–ethylenediamine

Positive ligands end with ium.
Examples: $NH_2 – NH_3^+$ – hydrazinium
NO^+ – nitrosonium
NO_2^+ – nitronium
$C_6H_5N^+H$ – pyridinium

(iv) If several ligands of same type are present, prefixes di (for two), tri (for three), tetra (for four) etc., are used. If the ligand already contains such words, e.g. ethylenediamine, then bis (for two), tris (for three), tetrakis (for four) etc., are used.

(v) **Order of naming the ligands:** All ligands whether negative, neutral or positive are named alphabetically.

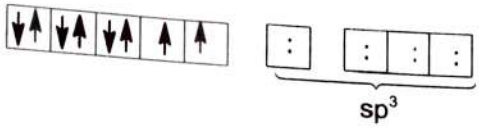
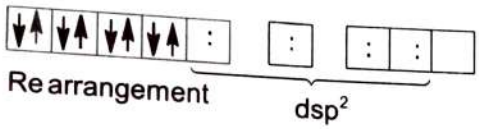
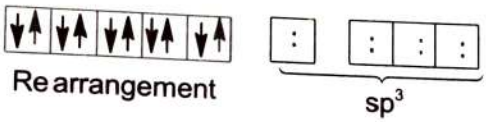
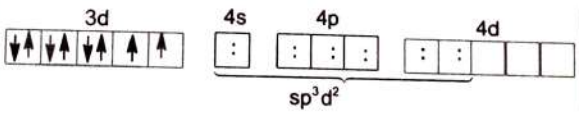
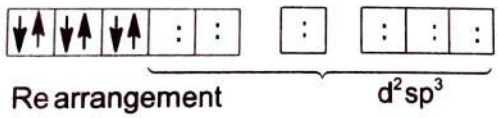
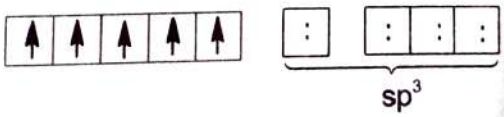
(vi) (a) If the complex ion is a cation or coordination compound is non-ionic, the name of the metal is written as such followed by oxidation state in roman letters inside the bracket.

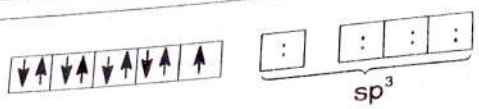
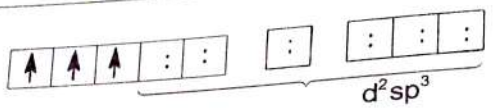
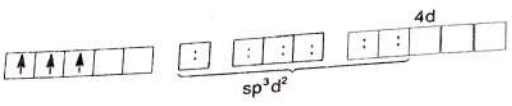
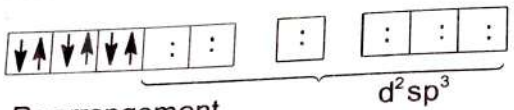
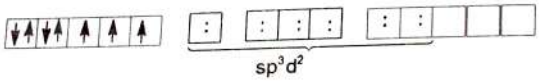
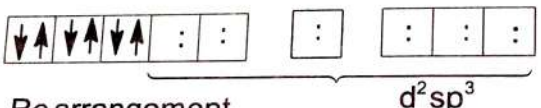
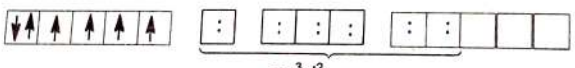
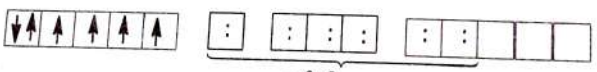
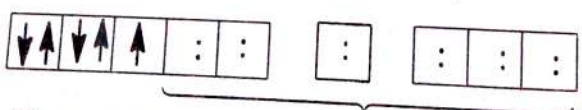
(b) If the complex ion is an anion, name of the metal should end with ate followed by oxidation state of the metal in roman letters in bracket.

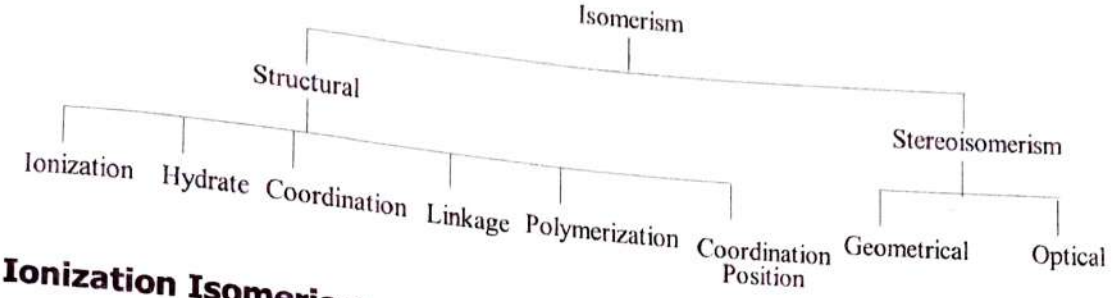
Examples: Cr – chromate
Ag – argentate
Au – aurate
Co – cobaltate
Fe – ferrate
Ni – nickelate

(c) If the complex compound is an acid, i.e. when the cation is H^+, the name of the metal atom is made to end in –ic followed by its oxidation state in bracket and then the word acid is to be added.
Example: $H_4[Pt(CN)_6]$ – hexacyanoplatinic(II)acid.

SOME IMPORTANT COORDINATION COMPOUNDS

Complex	Configuration of Central Ion/ Atom	Oxidation state of metal	Type of hybridization	Geometry shape	No. of unpaired electrons	Magnetic nature
$[NiCl_4]^{2-}$		+2	sp^3	Tetrahedral	2	Paramagnetic
$[Ni(CN)_4]^{2+}$		+2	dsp^2	Square planar	0	Diamagnetic
$Ni(CO)_4$		0	sp^3	Tetrahedral	0	Diamagnetic
$[Ni(NH_3)_6]^{2-}$		+2	sp^3d^2 (Outer)	Octahedral	2	Paramagnetic
$[Mn(CN)_6]^{4-}$		+2	d^2sp^3 (Inner)	Octahedral	1	Paramagnetic
$[MnCl_4]^{2-}$		+2	sp^3	Tetrahedral	5	Paramagnetic

$[CuCl_4]^{2-}$		+2	sp^3	Tetrahedral	1	Paramagnetic
$[Cr(NH_3)_6]^{3+}$		+3	dsp^3 (Inner)	Octahedral	3	Paramagnetic
$[Cr(H_2O)_6]^{3+}$		+3	$sp^3\ d^2$ (Outer)	Octahedral	3	Paramagnetic
$[Co(NH_3)_6]^{3+}$		+3	$d^2\ sp^3$ (Inner)	Octahedral	0	Paramagnetic
$[Co(H_2O)_6]^{2+}$		+2	$sp^3\ d^2$ (outer)	Octahedral	3	Paramagnetic
$[Fe(CN)_6]^{4-}$		+2	$d^2\ sp^3$ (Inner)	Octahedral	0	Diamagnetic
$[Fe(H_2O)_6]^{2+}$		+2	$sp^3\ d^2$ (Outer)	Octahedral	4	Paramagnetic
$[Fe(NH_3)_6]^{2+}$		+2	$sp^3\ d^2$ (Outer)	Octahedral	1	Paramagnetic
$[Fe(CN)_6]^{4-}$		+3	d^2sp^3 (Inner)	Octahedral	1	Paramagnetic



Ionization Isomerism

It arises when coordination compounds give different ions in solution.
Example: $[Co(NH_3)_5Br]SO_4$ and $[Co(NH_3)_5SO_4]Br$

Hydrate Isomerism

It arises when different number of water molecules are present inside and outside the coordination sphere.
Example: $[Cr(H_2O)_6]Cl_3$ and $[Cr(H_2O)_5Cl]Cl_2.H_2O$

Coordination Isomerism

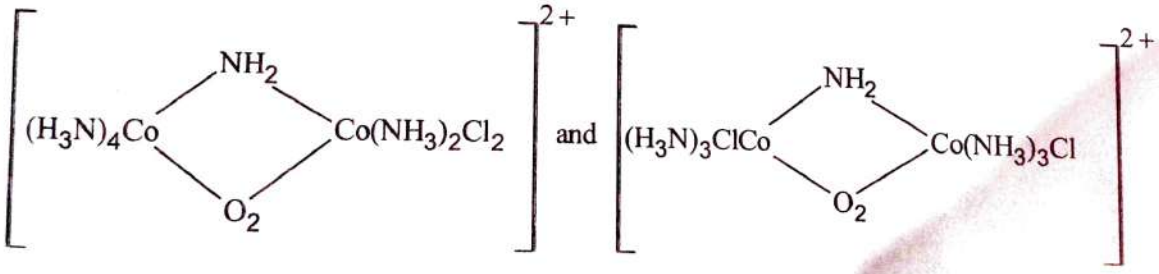
It arises in coordination compounds having both cationic and anionic complex ions. The ligands are interchanged in both the cationic and anionic ions to form isomers.
Example: $[Pt(NH_3)_4[CuCl_4]$ and $[Cu(NH_3)_4][PtCl_4]$

Linkage Isomerism

It arises in complexes having ambident ligands. Linkage isomers differ in the point of attachment of donor atoms of the ambident ligands.
Example: $[Co(NH_3)_5NO_2]Cl_2$ and $[Co(NH_3)_5ONO]Cl_2$

Coordination-Position Isomerism

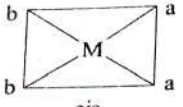
It is exhibited by polynuclear complexes by changing the position of ligands with respect to different metal atoms present in the complex.
Example:



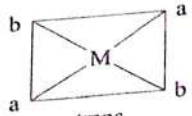
Geometrical Isomerism

In square planar complexes:

(i) $[M(a)_2(b)_2]$



cis



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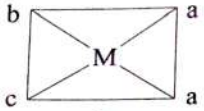
(ii) $[M(a)_2bc]$



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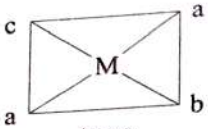
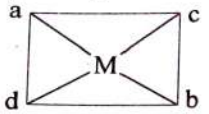
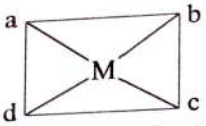
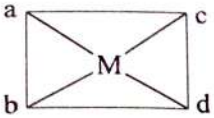


trans

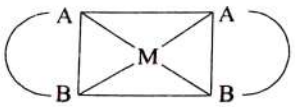
(iii) [Mabcd]







(iv) $[M(AB)_2]$, A and B are different atoms of a chelating ligand



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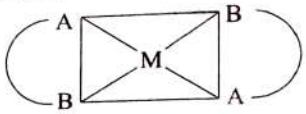
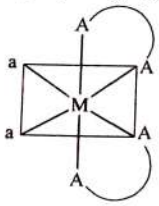


trans

In octahedral complexes:

(i) [M(abcdef)] – There are 15 geometrical isomers

(ii) $[M(AA)_2a_2]$



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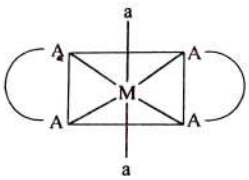
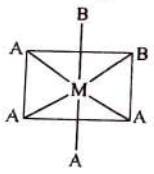
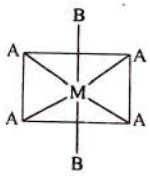


trans

(iii) $[MA_4B_2]$

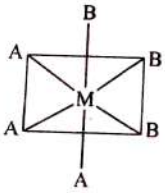


cis - form

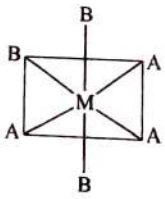


trans - form

(iv) $[MA_3B_3]$



cis - form

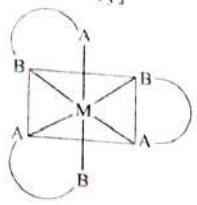


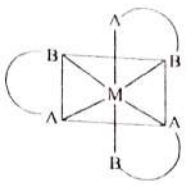
trans - form

Optical Isomerism

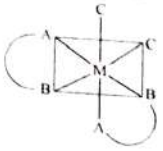
A coordination compound which can rotate the plane of polarized light. Optical isomerism is very common in octahedral complexes.

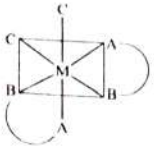
(i) $[M(AB)_3]$



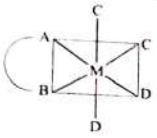


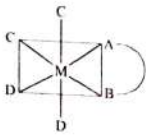
(ii) $[M(AB)_2C_2]$





(iii) $[M(AB)C_2D_2]$





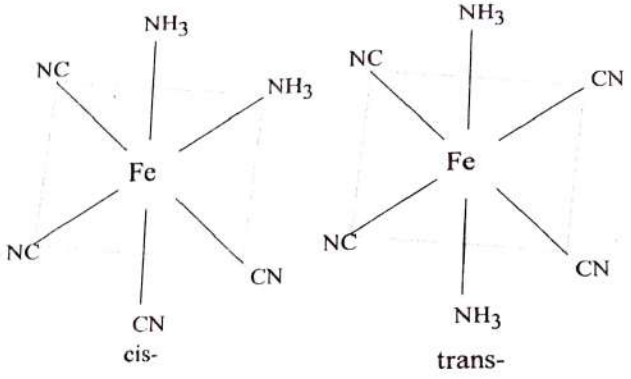
ILLUSTRATIVE PROBLEMS

1. Why is geometrical isomerism not possible in tetrahedral complexes having two different types of unidentate ligands coordinated with the central metal ion?

Sol. Tetrahedral complexes do not show geometrical isomerism because the relative positions of the ligands attached to the central metal atom are same with respect to each other.

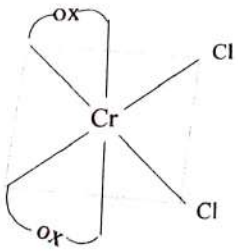
2. Draw structures of geometrical isomers of $[Fe(NH_3)_2(CN)_4]^-$.

Sol.

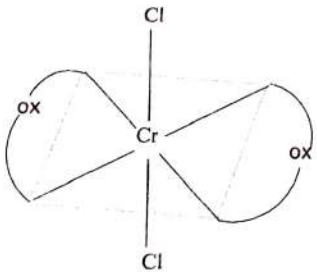


3. Out of the following two coordination entitles, which is chiral (optically active)?
(a) cis – $[CrCl_2(ox)_2]^{3-}$
(b) trans – $[CrCl_2(ox)_2]^{3-}$

Sol.



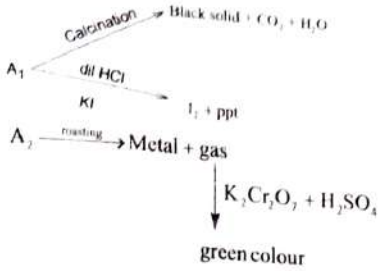
cis-OPTICALLY ACTIVE



trans-OPTICALLY INACTIVE

(because its mirror image is superimposable

4. **A_1 & A_2 are two ores of metal M. A_1 on calcination gives black precipitate, CO_2 & water.**



Sol. **$CuCO_3$, $Cu(OH)_2$, Cu_2S, Cu, CuO, Cu_2I_2, SO_2**

$A_1 = Cu(OH)_2CuCO_3$

$A_2 = Cu_2S$

$$\underset{(A1)}{Cu(OH)_2CuCO_3} \xrightarrow{\text{Calcination}} \underset{\text{(Black Solid)}}{2CuO} + CO_2 + H_2O$$

$$Cu(OH)_2CuCO_3 \xrightarrow{\text{dilHCl}} CuCl_2 + CO_2 + 3H_2O$$

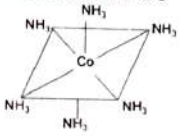
$$2CuCl_2 + 4KI \rightarrow Cu_2I_2 + I_2 + 4KCl$$

$$\underset{(A_2)}{2Cu_2S} + 3O_2 \rightarrow 2Cu_2O + 2SO_2$$

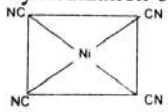
$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$$

5. **Draw the structures of $[Co(NH_3)_6]^{3+}$ $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$ Write the hybridization of atomic orbitals of the transition metal in each case.**

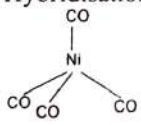
Sol. Hybridization of Co^{+3} is d^2sp^3, shape is octahedral



Hybridization of Ni^{+2} is dsp^2 so shape is square planar.



Hybridisation of Ni is sp^3 so shape is tetrahedral



6. **(a) Calculate magnetic moment of Fe^{3+} in $[Fe(CN)_6]^{3-}$ and in $[Fe(H_2O)_6]^{3+}$.**
(b) Arrange following complexes in decreasing order of magnetic moment:
$[Ni(H_2O)_4]^{2+}$, $[Ni(CN)_4]^{2-}$, $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$

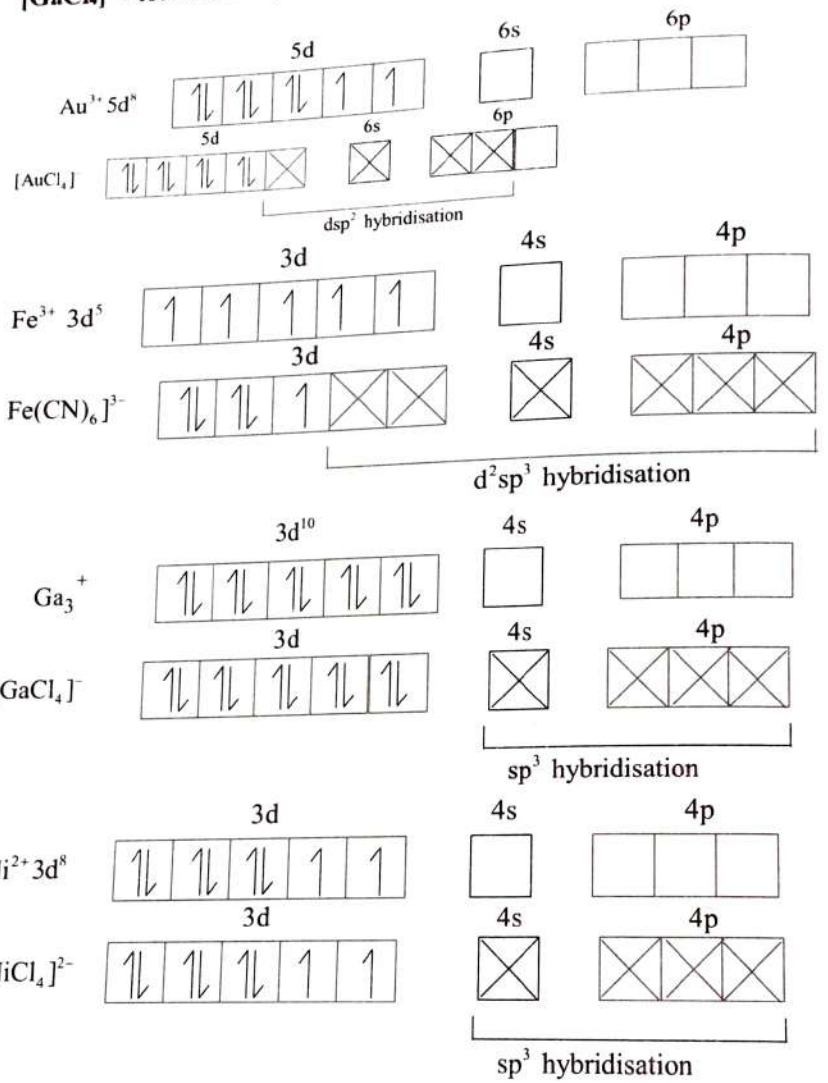
Sol. (a) In $[Fe(CN)_6]^{3-}$, Fe^{3+} ion has only one unpaired electron. Thus, magnetic moment of Fe^{3+} will be $\sqrt{3}$, i.e., 1.732 B. M. $\left(\mu_s = \sqrt{n(n+2)}\right)$ B.M., where, n is number of unpaired electrons). In $[Fe(H_2O)_6]^{3+}$, Fe^{3+} ion has 5 unpaired electrons, hence, its magnetic moment will be $\sqrt{35}$ B.M., i.e., 5.92 B.M.

(b) $[Fe(CN)_6]^{4-} = [Ni(CN)_4]^{2-} < [Fe(CN)_6]^{3-} < [Ni(H_2O)_4]^{2+}$

In $[Fe(CN)_6]^{4-}$ and $[Ni(CN)_4]^{2-}$, central atom has no unpaired electron, hence, their magnetic moment is zero. In $[Fe(CN)_6]^{3-}$, there is only one electron with central Fe^{3+} ion, hence, its magnetic moment will be 1.732 B.M.. In $[Ni(H_2O)_4]^{2+}$, there are two unpaired electrons with Ni^{2+} and hence, its magnetic moment will be $\sqrt{8}$ B.M.

7. How do you account for the indicated molecular geometry for the following compounds in terms of valence bond theory.
$[AuCl_4]^-$ - square planar $[Fe(CN)_6]^{3-}$ - octahedral
$[GaCl_4]^-$ - tetrahedral $[NiCl_4]^{2-}$ - tetrahedral

Sol.



8. Explain the following

(i) Anhydrous $FeCl_3$ cannot be obtained by heating hydrated $FeCl_3$.

(ii) The colour of mercurous chloride changes from white to black when treated with ammonia solution.

(iii) The compounds of Zn, Cd and Hg are usually white.

(iv) A dark blue precipitate is formed when NaOH soluion is added to $CuSO_4$ solution. The precipitate darkens on heating.

Sol. (i) On heating hydrated $FeCl_3$, anhydrous $FeCl_3$ is not formed as water of crystallisation reacts to form Fe_2O_3 and HCl.

(ii) Hg_2Cl_2 reacts with NH_3 to form a mixture of mercury and mercuric aminochloride which is a black substance

$$Hg_2Cl_2 + NH_3 \longrightarrow \underbrace{Hg + Hg(NH_2)Cl}_{\text{black}} + HCl$$

(iii) In the compounds of zinc metals, M^{2+} ions possess the penultimate d-orbitals completely filled i.e., $(n-1)d^{10}$ configuration. There is no d-d transition. Hence the compounds of zinc metals are colourless.

(iv) NaOH reacts with $CuSO_4$ when dark blue precipitate of $Cu(OH)_2$ is formed. This precipitate of heating forms CuO which is black in colour. Hence, the colour darkens on heating.

$$CuSO_4 + 2NaOH \longrightarrow Cu(OH)_2 + Na_2SO_4$$

$$Cu(OH)_2 \longrightarrow \underset{\text{Black}}{CuO} + H_2O$$

9. A, B and C are three complexes of chromium (III) with empirical formula $H_{12}O_6Cl_3Cr$. All the three complexes have water and chloride ions as ligands. Complex A does not react with conc. H_2SO_4, whereas, complexes B and C lose 6.7% and 13.5% of their original weight respectively. On treatment with concentrated H_2SO_4. Identify A, B and C.

Sol. $A = [Cr(H_2O)_6]Cl_3 \xrightarrow{H_2SO_4} \text{No reaction}$

($\therefore$ All H_2O molecules are present in co–ordination sphere)

$B = [CrCl(H_2O)_5]Cl_2 \cdot H_2O \xrightarrow{H_2SO_4} \text{one mole of } H_2O \text{ is lost}$

Molecular weight of complex = 266.5

$$\% \text{ loss} = \frac{18}{266.5} \times 100 = 6.75\%$$

$C = [CrCl_2(H_2O)_4]Cl_2 \cdot 2H_2O \xrightarrow{H_2SO_4} 2 \text{ moles of } H_2O \text{ is removed}$

$$\therefore \% \text{ loss in weight} = \frac{2 \times 18}{266.5} \times 100 = 13.50\%$$

Subjective Question

Prob. 1. $[Fe(H_2O)_6]^{3+}$ is strongly paramagnetic whereas $[Fe(CN)_6]^{3-}$ is weakly paramagnetic. Explain.

Sol. In both the complexes, Fe is in +3 oxidation state with the configuration $3d^5$. CN^- is a strong ligand. In its presence, 3d electrons pair up leaving only one unpaired electron. The hybridization is d^2sp^3 forming inner orbital complex. H_2O is a weak ligand. In its presence, 3d electrons do not pair up. The hybridization is sp^3d^2 forming an outer orbital complex containing five unpaired electrons. Hence, it is strongly paramagnetic.

Objective Questions

Prob. 1. Which equation does not involve the reduction of a transition metal compound?

(A) $Fe_2O_3 + 3CO \longrightarrow 2Fe + 3CO_2$

(B) $TiO_2 + 2C + 2Cl_2 \longrightarrow TiCl_4 + 2CO$

(C) $Cr_2O_3 + 2Al \longrightarrow 2Cr + Al_2O_3$

(D) $TiCl_4 + 4Na \longrightarrow Ti + 4NaCl$

Sol. $\underset{(+4)}{TiO_2} \longrightarrow \underset{(+4)}{TiCl_4}$

No change in oxidation number takes place.

Hence, **(B)** is the correct answer.

Prob. 2. Chemical volcano is produced on heating

(A) $K_2Cr_2O_7$

(B) $(NH_4)_2Cr_2O_7$

(C) $ZnCr_2O_7$

(D) K_2CrO_4

Sol. **(B)**

On heating $(NH_4)_2Cr_2O_7$, N_2 is given out with Cr_2O_3 produced at higher rate giving look of artificial volcano.

Hence, **(B)** is the correct answer.

Prob. 3. Find the statement which is not correct with respect to transition metals.

(A) Most transition metals, should on thermodynamic consideration liberate hydrogen from acids.

(B) The colour of the hydrated ions is due to transition of electrons from different 'd' orbitals of the same principle quantum numbers.

(C) The compounds of the elements in low oxidation states are generally ionic.

(D) The complex formation of transition metal ions is often accompanied by a change of colour and sometimes by a change in the intensity of the colour.

Sol. Transition metals get rendered passive by their coating of the oxide or having high hydrogen over voltage.

Hence, **(A)** is the correct answer.

Prob. 4. The magnetic moment of an iron compound is 5.918 BM, then the oxidation state of iron in this compound will be

(A) 0 (B) 1

(C) 2 (D) 3

Sol. **(D)**

Prob. 5. $CuSO_4$ decolourises on addition of KCN, the product is

(A) $[Cu(CN)_4]^{2-}$ (B) Cu^{2+} get reduced to form $[Cu(CN)_4]^{3-}$

(C) $Cu(CN)_2$ (D) CuCN

Sol. **(A)**

Prob. 6. Which type of isomerism is shown by $[Co(NH_3)_4Br_2]Cl$?

(A) Geometrical and Ionisation (B) Optical and Ionisation

(C) Geometrical and Optical (D) Geometrical only

Sol. **(A)**

Prob. 7. The pair of compounds having metals in their highest oxidation state is

(A) MnO_2, $FeCl_3$ (B) $[MnO_4]^-$, CrO_2Cl_2

(C) $[Fe(CN)_6]^{3-}$, $[Co(CN)_3]$ (D) $[NiCl_4]^{2-}$, $[CoCl_4]^-$

Sol. $[MnO_4]^-$, Mn = +7

CrO_2Cl_2, Cr = +6

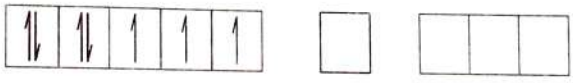
Hence, **(B)** is the correct answer.

Prob. 8. Spin only magnetic moment of the compound $Hg[Co(SCN)_4]$ is

(A) $\sqrt{3}$ (B) $\sqrt{15}$

(C) $\sqrt{24}$ (D) $\sqrt{8}$

Sol. $Hg[Co(SCN)_4]$

Co^{+2} = 

n=3

$\mu_s = \sqrt{n(n+2)} = \sqrt{3\times5} = \sqrt{15}$

Hence, **(B)** is the correct answer.

Prob. 9. Amongst the following, identify the species with an atom in (+6) oxidation state.

(A) MnO_4^- (B) $Cr(CN)_6^{3-}$

(C) NiF_6^{2-} (D) CrO_2Cl_2

Sol. Mn is in +7 oxidation state in MnO_4^-, Cr is in +3 oxidation state in $[Cr(CN)_6]^{3-}$, Ni is in (+4) oxidation state in NiF_6^{2-} and Cr is in (+6) oxidation state in CrO_2Cl_2.

Hence, **(D)** is the correct answer.

Prob. 10. Which of the following statements are correct?

(A) In ferrocyanide ion, the effective atomic number is 36

(B) Ferricyanide ion is more stable than ferrocyanide ion

(C) $[Cr(H_2O)_6]^{3+}$ and $[Cr(NH_3)_6]^{3+}$ both have d^2sp^3 hybridization

(D) In $[Co(H_2O)_6]^{3+}$ the hybridization is sp^3d^2 while in $[Co(NH_3)_6]^{3+}$ the hybridization is d^2sp^3.

Sol. For $[Fe(CN)_6]^{4-}$ effective atomic number = atomic number − oxidation state + 2(co-ordination number)
$= 26 - 2 + 12 = 36$

Hence (A) is correct.

In Ferricyanide ion Fe is +3 oxidation state while in ferricyanide ion Fe is +2 oxidation state. Fe^{3+} complexes are more stable than Fe^{2+} complex due to high charge and small size. Hence choice (B) is correct. Choice (C) is also correct. In Cr^{3+} the electronic configuration is $[M]3d^3$. Already inner d-orbitals are present. Hence, whether with strong field or weak field ligands, inner d-orbitals can be provided for bonding. Hence hybridization is d^2sp^3. The electronic configuration of Co^{3+} is $[M]3d^6$. Hence inner d-orbitals are not available. So with strong field ligands inner complexes are formed but with weak field ligands outer complexes are formed.

Hence, **(A), (B), (C)** and **(D)** are correct options.

Prob. 11 If excess of $AgNO_3$ solution is added to 100mL of a 0.024M solution of dichlorobis(ethylenediamine) cobalt (III) chloride, how many mole of AgCl will be precipitated.

(A) 0.0012 (B) 0.0016

(C) 0.0024 (D) 0.0048

Sol. **(C)**

$$\underset{\text{n moles}}{[CO(en)_2Cl_2]Cl} \longrightarrow [CO(en)_2Cl_2]^+ + \underset{\downarrow AgNO_3}{Cl^-}\ \text{nmoles}$$

$$\underset{\text{n moles}}{AgCl}$$

$$\text{Moles of complex} = \frac{100}{1000} \times 0.024 = 0.024 = \text{Moles of AgCl precipitated.}$$

Prob. 12 Select the incorrect match for $[M(H_2O)_6]^{2+}$ complex

	Metal ions	Electronic configuration	CFSE
(A)	Mn^{2+}	$t_{2g}^3\ eg^2$	$0\Delta_0$
(B)	V^{2+}	$t_{2g}^3\ eg^0$	$-1.2\Delta_0$
(C)	Ni^{2+}	$t_{2g}^6\ eg^2$	$-1.6\Delta_0$
(D)	Ti^{2+}	$t_{2g}^2\ eg^2$	$-0.8\Delta_0$

Sol. **(C)**

$$CFSE = -0.4\Delta_o\ {}^{n}t_{2g} + 0.6\Delta_o\ {}^{n}e_g$$

where n = No. of e^-

$[Ni(H_2O)_6]^{2+}$ has CFSE = $-1.2\Delta_o$

COMPREHENSION-1

Read the following paragraph and answer the questions given below:

A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms (X) and (Y). The form (X) reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous Ammonia, whereas (Y) gives a pale yellow precipitate soluble in concentrate ammonia.

1. Choose the true statement regarding compound X and Y.
(A) Both compound X and Y have different magnetic moment (spin only)
(B) Both complex X and Y have same magnetic moment (spin only)
(C) Magnetic moment of both X and Y is calculated by assuming they are low spin complex.
(D) Both compound X and Y have different primary valency.

Sol. **(B)**

2. The hybridization of compound 'X' and 'Y' is respectively:
(A) sp^3d^2, d^2sp^3
(B) sp^3d^2, sp^3d^2
(C) d^2sp^3, d^2sp^3
(D) d^2sp^3, sp^3d^2

Sol. **(C)**

3. Choose the correct statement regarding the complex X and Y.
(I) Both complexes have same Van't Hoff factor with respect to any % solvent.
(II) Both complexes have exactly same conductivity in H_2O.
(III) In both the complexes NH_3 is involved only to secondary valency.
(IV) In both the complexes electronic configuration of t_{2g} is same.
(A) I, II and III
(B) I, II, III and IV
(C) II, III and IV
(D) I, III and IV

Sol. **(D)**

COMPREHENSION-2

One pink solid has the formula $COCl_3.5NH_3.H_2O$. A solution of this salt is also pink and rapidly gives three mol white ppt. on titration with excess of silver nitrate solution. When pink solid is heated, it loses 1 mole H_2O to give a purple solid with the same ratio of NH_3 : Cl : CO as in pink solid. The purple solid releases two of its chloride ions rapidly.

4. Which of the following products are formed when pink solid reacts with $AgNO_3$?
(A) $[Co(NH_3)_5NO_3](NO_3)_2 + 3AgCl$
(B) $[Co(NH_3)_5(H_2O)](NO_3)_3 + 3AgCl$
(C) $[Co(NH_3)_5Cl](NO_3)_2 + 2AgCl$
(D) $[Co(NH_3)_5Cl](NO_3)_3 + 2AgCl$

Sol. **(B)**

5. Select the correct statement
(A) Pink solid shows geometrical isomerism
(B) Purple solid shows geometrical isomerism
(C) Pink solid and purple solid are hydrated isomer of each other
(D) Pink solid is diamagnetic

Sol. **(D)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: $[Fe(H_2O)_5NO]SO_4$ is paramagnetic.
and
STATEMENT-2: The Fe in $[Fe(H_2O)_5NO]SO_4$ has three unpaired electrons.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

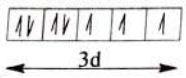
Sol. **(A)**

$\left[Fe(H_2O)_5\,NO\right]SO_4$

Here Fe has +1 oxidation state.

$Fe^+ = 3d^6 4s^1$ in presence of NO^+ $4s^1$ electron are paired in 3d sub shell.

So electronic configuration of Fe^+ is



2. **STATEMENT-1:** Quenched steel is very hard and brittle.
and
STATEMENT-2: During quenching, steel is heated to a temperature well below red hot.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**

3. **STATEMENT-1:** KCN gives $K_3[Cu(CN)_4]$ complex compound with $CuSO_4$ solution
and
STATEMENT-2: KCN reduces Cu^{2+} to Cu^+ and then forms a complex compound.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

4. **STATEMENT-1:** Mercury is liquid at room temperature.
and
STATEMENT-2: In mercury, there is no unpaired d-electron and thus metallic bonding is weakest.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

5. **STATEMENT-1:** During calcination of haematite ore, ferrous oxide is oxidized to ferric oxide.
and
STATEMENT-2: Ferric iron does not form slag during smelting and loss of iron is prevented.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

6. **STATEMENT-1:** Gold is recovered from its solution containing aurocyanide complex by adding zinc dust.
and
STATEMENT-2: Zinc is more electropositive than gold.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

7. **STATEMENT-1:** $KMnO_4$ is purple in colour due to charge transfer.
and
STATEMENT-2: In MnO_4^-, there is no electron present in d-orbitals of manganese.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

8. **STATEMENT-1:** K_2CrO_4 has yellow colour due to charge transfer.
and
STATEMENT-2: CrO_4^{2-} ion is tetrahedral in shape.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

9. **STATEMENT-1:** The highest oxidation state of chromium in its compounds is + 6.
and
STATEMENT-2: Chromium atom has only six electrons in ns and (n – 1)d orbitals.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

10. **STATEMENT-1:** CrO_3 reacts with HCl to form chromyl chloride gas.
and
STATEMENT-2: Chromyl chloride (CrO_2Cl_2) has tetrahedral shape.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

MATRIX – MATCH TYPE QUESTIONS

1. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	$[MnCl_6]^{2-}$	(p)	One unpaired electron
(B)	$[Fe(CN)_6]^{3-}$	(q)	d^2sp^3
(C)	$[CoF_6]^{3-}$	(r)	sp^3d^2
(D)	$[Fe(H_2O)_6]^{2+}$	(s)	Four unpaired electron
		(t)	Inner orbital complex

Sol. **A → (p, q) B → (p, q, t) C → (r, s) D → (r, s)**

2. Match the Column – I with Column – II:

	Column – I Metal Ion		Column – II Magnetic moments (B.M.)
(A)	Cr^{3+}	(p)	$\sqrt{35}$
(B)	Fe^{2+}	(q)	$\sqrt{30}$
(C)	Ni^{2+}	(r)	$\sqrt{24}$
(D)	Mn^{2+}	(s)	$\sqrt{15}$
		(t)	$\sqrt{8}$

Sol. **A → (s) B → (r) C → (t) D → (p)**

3. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	$[M(H_2O)_6]^{\pm n}$	(p)	If M = Fe^{2+} compound is diamagnetic
(B)	$[M(NH_3)_6]^{\pm n}$	(q)	If M = Fe^{3+} compound is paramagnetic
(C)	$[M(Cl_6)]^{\pm n}$	(r)	If M = Cr^{3+} compound is paramagnetic
(D)	$[M(CN)_6]^{\pm n}$	(s)	If M = Co^{3+} compound is diamagnetic
		(t)	If M = Pt^{+4} compound is diamagnetic

Sol. **A → (q, r, s, t); B → (q, r, s, t); C → (q, r, t); D → (p, q, r, s, t)**

4. Match the Column – I with Column – II:

	Column – I		Column – II
(A)	Eg^o	(p)	$[Pt(NH_3)_4(en)]Cl_4$
(B)	Paramagnetic but $\mu \neq \sqrt{15}$ B.M	(q)	$[Cr(en)_3]^{3+}$
(C)	Low spin and diamagnetic complex	(r)	$[Co(H_2O)_3F_3]$
(D)	Exist in more than one stereoisomeric form	(s)	$[Co(C_2O_4)_3]^{3-}$
		(t)	$(NH_4)_3[Fe(CN)_6].3H_2O$

Sol. **A → (p, q, s, t); B → (r, t); C → (p, s); D → (q, r, s)**

5. Match the Column – I with Column – II:

	Column – I Complexes		Column – II Hybridization of central atom
(A)	$Ni(CO)_4$	(p)	sp^3
(B)	$[Ni(CN)_4]^{2-}$	(q)	dsp^2
(C)	$[Fe(CN)_6]^{4-}$	(r)	sp^3d^2
(D)	$[MnF_6]^{4-}$	(s)	d^2sp^3

Sol. **A → (p) B → (q) C → (s) D → (r)**

Answer questions 6, 7 and 8 by appropriately matching the information given in the three columns of the following table.

Column 1		Column 2		Column 3	
(I)	$Ni(CO)_4$	(i)	d^2sp^3	(P)	Zero unpaired electron
(II)	$[Fe(EDTA)]^-$	(ii)	sp^3	(Q)	One unpaired electron
(III)	$Ni(DMG)_2$	(iii)	sp^3d^2	(R)	Polydentate ligand
(IV)	$[Cr(NH_6)]^{3+}$	(iv)	dsp^2	(S)	Three unpaired electron

6. The only correct combination is
[A] (IV) (iv) (Q) [B] (I) (ii) (P)
[C] (III) (iii) (R) [D] (I) (ii) (S)

7. Which one is only incorrect combination is
[A] (III) (iv) (R) [B] (II) (i) (Q)
[C] (III) (iv) (P) [D] (IV) (iii) (S)

8. The only correct combination for $[Cr(NH_6)]^{3+}$ is
[A] (III) (iii) (P) [B] (IV) (i) (S)
[C] (III) (iv) (Q) [D] (III) (i) (P)

Sol.

6. **B**
7. **D**
8. **B**

SINGLE INTEGER ANSWER TYPE

1. What is the oxidation state of the central atom in [Fe(EDTA)] complex ion.

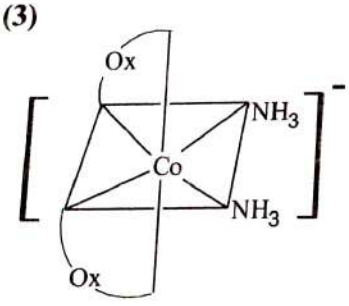
Sol. **(3)**

$x + 1 \times (-4) = -1$

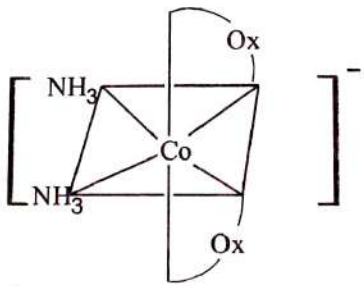
$\therefore\ x = 3$

2. What is the number of possible isomer of an octahedral complex : $[Co(C_2O_4)_2(NH_3)_2]^-$.

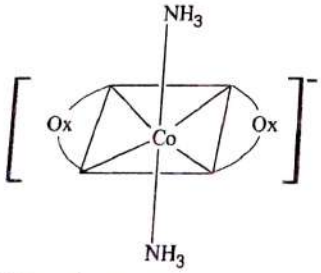
Sol. **(3)**



Cis-d-isomer



cis-l-isomer



Trans isomer

3. Consider the following complex for the given 3 questions dichlorobis (ethylenediamine) chromium (III) chloride

(i) Number of unpaired electrons in complex = X

(ii) One mole complex $\xrightarrow[\text{(excess)}]{AgNO_3}$ Y mole of AgCl

(iii) Total stereoisomeric forms of above complex = Z.

What is the value of X + Y + Z

Sol. **(7)**

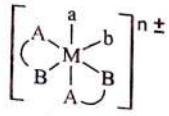
X = 3, Y = 1, Z = 3

$[CrCl_2(en)_2]Cl$

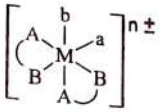
$\therefore$ X + Y + Z = 7

4. How many geometrical isomers are possible for complex $[Mab(AB)_2]^{n\pm}$

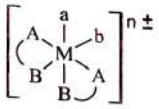
Sol. **(6)**



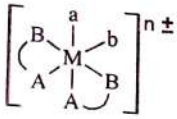
cis-form



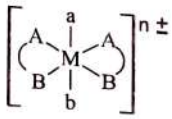
cis-form



trans-form



trans-form



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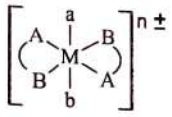


trans-form

NUMERICAL VALUE ANSWER TYPE

Prob. 1. The number of geometrical isomers of $[M(AB)c_2d_2]$ is x, than the value of $\frac{x}{5}$ is

Sol. **0.8**

$x = 4, \frac{4}{5} = .8$

Prob. 2. The number of unpaired electrons are present in t_{2g} orbital of $[Cr(NH_3)_6]^{3+}$ is x, the value of $\frac{x}{2}$ is

Sol. **1.5**

$\frac{3}{2} = 1.5$

Prob. 3. The sum of primary and secondary valency of chromium in the complex $CrCl_3.6NH_3$ is x, the value of $\frac{x}{2}$ is

Sol. **4.5**

$\frac{9}{2} = 4.5$

Prob. 4. The number of 3d electrons occupied in t_{2g} orbitals of hydrated Cr^{+3} ions (octahedral) is x, the value of $\frac{x}{2}$ is

Sol. **1.5**

$\frac{3}{2} = 1.5$

Prob. 5. If Hund's rule violated, then the unpaired electrons present in $[Cr(NH_3)_6]^{3+}$ complex ion are x, the value of $\frac{x}{2}$ is

Sol. **0.5**

Prob. 6. If $AgNO_3$ solution is added in excess to 1 M solution of $COCl_3.xNH_3$, One mole of AgCl is formed. What is the value of $\frac{x}{5}$ is

Sol. **0.8**

$$\frac{4}{5} = 0.8$$

Prob. 7. An alcoholic solution of dimethylglyoxime is added to an aqueous solution of nickel(II) chloride. Slow addition of ammonium hydroxide lead to the precipitation of a rosy-red coloured metal complex. The number of hydrogen bonds present in the structure of the complex is 'x'. Then the value of $\frac{x}{5}$ is

Sol. **.4**

$$\frac{5}{5} = .4$$

Prob. 8. What are the values of $\left(\frac{m+n}{2}\right)$ in the anionic species $[V(CO)_m]^{n-}$ if it is following Sidgwick E.A.N. rule and having octahedral shape.

Sol. **3.5**

Prob. 9. The oxidation state of Co in its oxo-complex species $[Co_2O_4(C_2H_4)_2(H_2O)_2]^{2-}$ is 'x'. The value of $\frac{x}{2}$ is

Sol. **1.5**

Prob.10. The total possible stereoisomer for the molecule $[Ma_2bcde]^{m\pm}$ is x, then $\frac{x}{2}$ is ..... (where a,b,c,d,e are simple monodentate ligand having no chiral centre)

Sol. **7.5**

PREPARATION AND PROPERTIES OF COMPOUNDS

23

COMPOUNDS OF BORON

Diborane, B_2H_6

Preparations:

(i) $4BCl_3 + 3LiAlH_4 \xrightarrow{\text{ether}} 2B_2H_6 + 3LiCl + 3AlCl_3$

(ii) $2BCl_3 + 6H_2 \xrightarrow[\text{discharge}]{\text{Silent electric}} B_2H_6 + 6HCl$

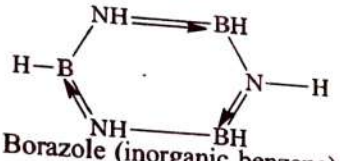
(iii) $8BF_3 + 6LiH \longrightarrow B_2H_6 + 6LiBF_4$

Properties:

(i) $B_2H_6 + 2NH_3 \xrightarrow{\text{Low temp.}} B_2H_6 \cdot 2NH_3$

(ii) $B_6H_6 2NH_3 \xrightarrow{200°C} 2B_3N_3H_6 + 12H_2$

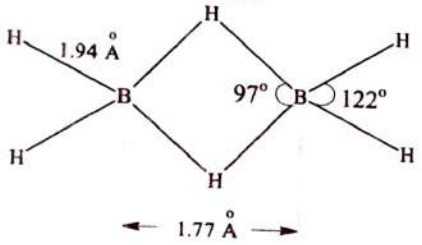
Borazole



Borazole (inorganic benzene)

(iii) $B_2H_6 + HCl \xrightarrow{\text{Anhy. } AlCl_3} B_2H_5Cl + H_2$

Structure of B_2H_6:



Borax (Sodium Tetraborate), $Na_2B_4O_7 \cdot 10H_2O$

Preparations :

(i) $4H_3BO_3 + Na_2CO_3 \longrightarrow Na_2B_4O_7 + 6H_2O + CO_2$

Boric acid

(ii) $Ca_2B_6O_{11} + 2Na_2CO_3 \longrightarrow 2CaCO_3 + Na_2B_4O_7 + 2NaBO_2$

Colemanite

Properties :

(i) Aqueous solution of borax is alkaline in nature due to hydrolysis.

$$Na_2B_4O_7 + 3H_2O \longrightarrow 2NaBO_2 + 2H_3BO_3$$

$$NaBO_2 + 2H_2O \longrightarrow NaOH + H_3BO_3$$

(ii) **Action of heat**

$$Na_2B_4O_7.10H_2O \xrightarrow[-H_2O]{\Delta} Na_2B_4O_7 \xrightarrow{740°C} 2NaBO_2 + B_2O_3$$

Anhy. sodium metaborate — Transparent glassy mass

Boric Acid or Orthoboric Acid, H_3BO_3

Preparations :

(i) $Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$

Borax

(ii) $Ca_2B_6O_{11} + 2SO_2 + 11H_2O \longrightarrow 2Ca(HSO_3)_2 + 6H_3BO_3$

Colemanite

Properties :

Action of heat :

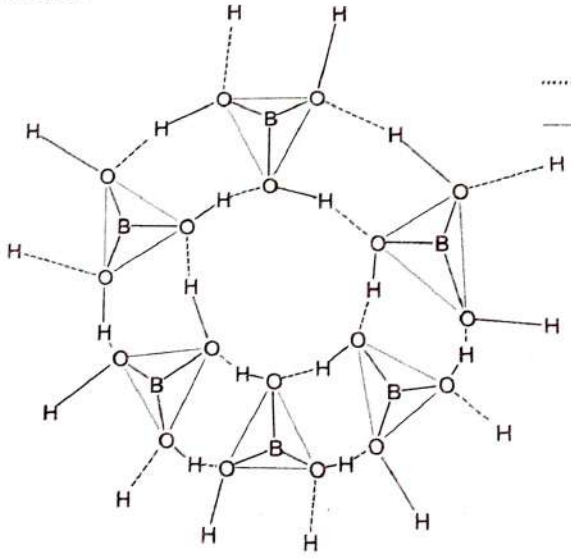
$$4H_3BO_3 \xrightarrow[-4H_2O]{373K} 4HBO_2 \xrightarrow[-H_2O]{413K} H_2B_4O_7 \xrightarrow[-H_2O]{\text{Red heat}} 2B_2O_3$$

Meta boric acid Tetraboric acid Boron trioxide

When hot glassy mass is brought in contact with coloured salt and heated again in the flame, B_2O_3 displaces the volatile oxides and reacts with basic oxides to form metaborates. Metaborates of basic radicals show characteristic colours. This test is known as borax bead test.

Metaborates of	Colour
Cu	Blue
Fe	Green
Co	Blue
Cr	Green
Ni	Brown

Structure of orthoboric acid:



.....Hydrogen bond

_____Covalent bond

COMPOUNDS OF ALUMINIUM

Aluminium Oxide or Alumina, Al_2O_3

Preparations :

(i) $\alpha - Al_2O_3$ (rhombic lattice) occurs in earth's crust as corundum Al_2O_3.

Transparent coloured crystals of corundum, viz., ruby are used as precious stones.

(ii) Prepared by igniting aluminium hydroxide and aluminium sulphate.

$$2Al(OH)_3 \xrightarrow{\Delta} Al_2O_3 + 3H_2O$$

$$Al_2(SO_4)_3 \xrightarrow{\Delta} Al_2O_3 + 3SO_3$$

Aluminium Chloride, $AlCl_3$

Preparations :

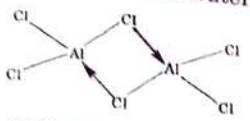
(i) $2Al + 6HCl \xrightarrow[\text{Absence of air}]{\Delta} 2AlCl_3 + 3H_2$

(ii) $Al_2O_3 + 3C + 3Cl_2 \xrightarrow{1000°C} 2AlCl_3 + 3CO \xrightarrow{\text{Cooled}}$ Solid anhydrous aluminium chloride

alumina Vapours

Properties :

(i) On heating it sublimes at 180°C and the vapour density corresponds to the formula Al_2Cl_6. It does not conduct current in fused state. The dimeric formula is retained in non – polar solvent but is broken into $[Al(H_2O)_6]Cl_3$ on dissolution in water on account of high heat of hydration. The dimer is represented as



(ii) $AlCl_3 + 3NH_4OH \longrightarrow Al(OH)_3 + 3NH_4Cl$

Gelatinous ppt. does not dissolve in excess of NH_4OH.

Alums

Aluminium alums have general formula

$[M'(H_2O)_6][Al(H_2O)_6](SO_4)_2$

$(M' \Rightarrow Na^+, K^+ \text{ or } NH_4^+)$

In place of Al^{3+}, some other ions (M^{3+}) also forms alums of similar formula. Some important alums are

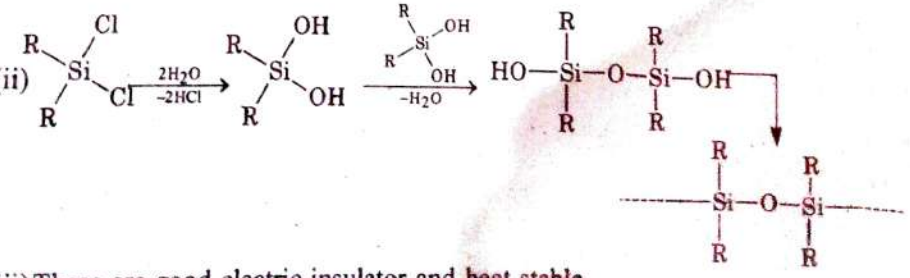
Potash alum	: $K_2SO_4.Al_2(SO_4)_3.24H_2O$
Ammonium alum	: $(NH_4)_2SO_4.\ Al_2(SO_4)_3.24H_2O$
Sodium alum	: $Na_2SO_4.Al_2(SO_4)_3.24H_2O$
Chrome alum	: $K_2SO_4.Cr_2(SO_4)_3.24H_2O$
Ferric alum	: $(NH_4)_2SO_4.Fe_2(SO_4)_3.24H_2O$

COMPOUNDS OF SILICON

Silicones

Silicones

(i) These are organosilicon polymers containing Si – O – Si linkages.

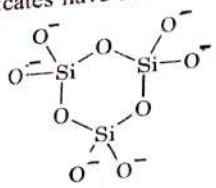
(ii) 

(iii) These are good electric insulator and heat stable.

Silicates

Silicates →
- (i) Sodium silicates, Na_2SiO_3 (water glass). Chemically, it is sodium metasilicates containing an excess of silica. Composition may vary from $Na_2SiO_3.SiO_2$ to $Na_2SiO_3.3SiO_2$
- (ii) Cyclic silicates have formula $(SiO_3)_n^{2n-}$

$Si_3O_9^{6-}$



Carborundum (Silicon Carbide), SiC

- (i) $SiO_2 + 3C \xrightarrow{\text{Heated in electric furnace}} SiC + 3CO$
  Sand
- (ii) It is chemically inert and resists the attack of almost all reagents.

COMPOUNDS OF NITROGEN

Nitrous oxide (Laughing Gas), N_2O

Structure: $:\bar{N}=\overset{+}{N}=\ddot{O}: \longleftrightarrow :N=\overset{+}{N}=\ddot{O}:$ (Linear molecule)

Preparation: (i) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

(ii) $4Zn + 10HNO_3 \longrightarrow 4Zn(NO_3)_2 + N_2O + 5H_2O$

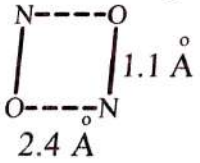
Properties: $N_2O + NaNH_2 \longrightarrow \underset{\text{Azides}}{NaN_3} + H_2O$

Nitric Oxide, NO

Structure:

$:\dot{N}\!=\!\ddot{O}: \longleftrightarrow :\dot{\bar{N}}\!=\!\dot{O}:^{+}$ (Paramagnetic)

It is diamagnetic in liquid and solid state due to formation of dimer.



Preparation:

(i) $3Cu + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

(ii) $4NH_3 + 5O_2 \xrightarrow[750°C,\ 6\ atm]{\text{Pt. gauze}} 4NO + 6H_2O$

Properties:

$2NO + Cl_2 \longrightarrow \underset{\text{Nitrosyl chloride}}{2NOCl}$

Nitrogen Dioxide, NO_2

Structure:

In solid state it exists as dimer (N_2O_4).

Preparation :

$$2Pb(NO_3)_2 \longrightarrow 2PbO + 4NO_2 + O_2$$

Properties :

(i) The gas condenses to a brown liquid which turns pale on cooling and eventually becomes a colourless solid due to dimerization.

(ii) Liquid N_2O_4 is useful as a non-aqueous solvent and itself ionizes.

$$N_2O_4 \rightleftharpoons NO^+ + NO_3^-$$

$$NOCl + NH_4NO_3 \longrightarrow NH_4Cl + N_2O_4$$

Acid Base Salt Solvent

Nitrogen Trioxide or Nitrogen Sesquioxide, N_2O_3

Structure:

Asymmetrical Symmetrical

Properties:

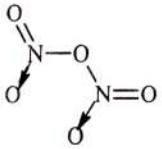
(i) $N_2O_3 + H_2O \longrightarrow 2HNO_2$

(Anhydride of nitrous acid)

(ii) $N_2O_3 + 2HClO_4 \longrightarrow 2NO(ClO_4) + H_2O$

Dinitrogen Peroxide, N_2O_5

Structure:



Properties:

(i) It is anhydride of HNO_3 and it is a strong oxidizing agent.

(ii) Solid N_2O_5 is ionic $NO_2^+NO_3^-$ (nitronium nitrate).

Nitrous Acid, HNO_2

Structure:

$H-O-N=O$

Preparation:

$$2NaNO_2 + H_2SO_4 \longrightarrow Na_2SO_4 + 2HNO_2$$

$$Ba(NO_2)_2 + H_2SO_4 \longrightarrow \underset{\text{Insoluble}}{BaSO_4} + 2HNO_2$$

Properties :

In HNO_2, N has +3 O.S. Therefore, it can acts as an oxidizing as well as reducing agent.

$$SnCl_2 + 2HCl + 2HNO_2 \longrightarrow SnCl_4 + 2NO + 2H_2O \text{ (Oxidation)}$$

$$H_2O_2 + HNO_2 \longrightarrow HNO_3 + H_2O \text{ (Reduction)}$$

Nitric acid, HNO_3

Structure:

Preparation:

(i) $KNO_3 + H_2SO_4 \longrightarrow KHSO_4 + HNO_3$ (Lab method)

(ii) $4NH_3 + 5O_2 \xrightarrow[750-900°C]{Pt} 4NO + 6H_2O + 21,600$ calories

$2NO + O_2 \longrightarrow 2NO_2$

$3NO_2 + H_2O \longrightarrow 2HNO_3 + NO$

Properties :

(i) $Sn + 4HNO_3 \longrightarrow H_2SnO_3 + 4NO_2 + H_2O$

(ii) $H_2S + 2HNO_3 \longrightarrow 2NO_2 + 2H_2O + S$

(iii) $3Pb + 8HNO_3 \longrightarrow 3Pb(NO_3)_2 + 2NO + 4H_2O$

(iv)

Concentration of nitric acid	Metal	Main product
Very dilute HNO_3	Mg, Zn	H_2 + metal nitrates
	Fe, Zn, Sn	NH_4NO_3 + metal nitrates
Dilute HNO_3	Pb, Cu, Ag, Hg	NO + metal nitrates
	Fe, Zn	N_2O + metal nitrates
	Sn	$NH_4NO_3 + Sn(NO_3)_2$
Conc. HNO_3	Zn, Pb, Cu, Ag	NO_2 + metal nitrates
	Sn	$NO_2 + H_2SnO_3$

Nitrates

The salts of nitric acid are called nitrates, e.g. $AgNO_3$, KNO_3, $Ca(NO_3)_2$, $NaNO_3$.

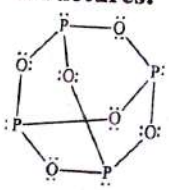
Distinction between nitrites and nitrates

Test	Nitrites	Nitrates
(i) Treat the salt with dil. HCl	(i) Brown fumes which turn $FeSO_4$ solution black	(i) No action
(ii) To the salt solution add $FeSO_4 + H_2SO_4$ (Ring test)	(ii) Dark brown colour even with dilute H_2SO_4	(ii) A dark brown ring is formed with conc. H_2SO_4 only
(iii) To the acidified solution of the salt add KI solution	(iii) I_2 evolved, gives blue colour with starch	(iii) No action

COMPOUNDS OF PHOSPHORUS

Phosphorus Trioxide, P_4O_6

Structures:

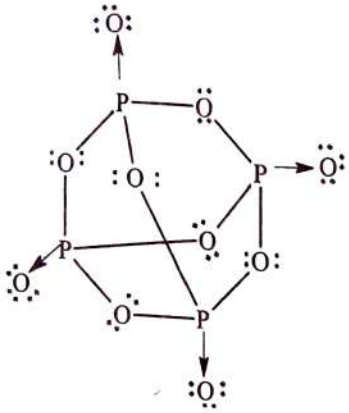


Properties:

$P_4O_6 + 2O_2 \longrightarrow P_4O_{10}$

Phosphorus pentaoxide, P_4O_{10}

Structure:

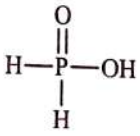
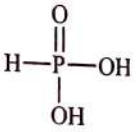
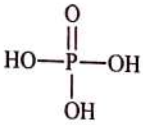
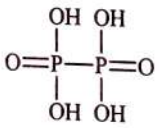
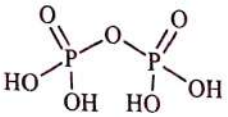


Properties:

$P_4O_{10} \xrightarrow{2H_2O} 4HPO_3 \xrightarrow{2H_2O} 2H_4P_2O_7 \xrightarrow{2H_2O} 4H_3PO_4$

Meta phosphoric acid — Pyrophosphoric acid — Orthophosphoric acid

Oxyacids of Phosphorus

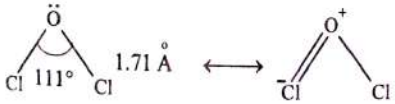
Name of oxyacids	Hypo phosphorus acid	Phosphorus acid	Ortho phosphoric acid	Hypophosphoric Acid	Pyrophosphoric Acid
Formula	H_3PO_2	H_3PO_3	H_3PO_4	$H_4P_2O_6$	$H_4P_2O_7$
Basicity	1	2	3	4	4
Structure					
Oxidation state	+1	+3	+5	+4	+5

Oxides of Chlorine

Dichloro Oxide, Cl_2O

It is a yellowish brown gas.

Structure:



Properties:

$2Cl_2 + 2HgO \xrightarrow{300°C} HgCl_2.HgO + Cl_2O$

ILLUSTRATIVE PROBLEMS

1. Discuss the various reactions that occur in the Solvay process.

Sol. In Solvay ammonia process, CO_2 is passed through brine, (i.e., a concentrated solution of NaCl) saturated with ammonia when sodium bicarbonate being sparingly soluble gets precipitated.

$$NaCl + NH_3 + CO_2 + H_2O \longrightarrow NaHCO_3 \downarrow + NH_4Cl \quad \ldots (i)$$

Sodium bicarbonate on subsequent heating gives sodium carbonate,

$$2\,NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O \quad \ldots (ii)$$

CO_2 needed for the reaction shown in Eq. (i) is prepared by heating calcium carbonate and the quick lime, CaO is dissolved in water to form slaked lime, $Ca(OH)_2$

$$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$$

$$CaO + H_2O \longrightarrow Ca(OH)_2 \quad \ldots (iii)$$

NH_3 needed for the purpose is prepared by heating NH_4Cl obtained in Eq. (i) with $Ca(OH)_2$ obtained in Eq. (iii)

$$2\,NH_4Cl + Ca(OH)_2 \longrightarrow 2\,NH_3 + CaCl_2 + 2\,H_2O$$

Therefore, the only by product of the reaction is calcium chloride, $CaCl_2$.

2. Does the hydrolysis of XeF_6 lead to a redox reaction?

Sol. Complete hydrolysis of XeF_6 gives XeO_3

$$\overset{+6}{Xe}F_6 + 3H_2O \longrightarrow \overset{+6}{Xe}O_3 + 6HF$$

On the other hand, partial hydrolysis of XeF_6 gives $XeOF_4$ and XeO_2F_2

$$\overset{+6}{Xe}\overset{-1}{F_6} + \overset{+1}{H_2}\overset{-2}{O} \longrightarrow \overset{+6}{Xe}\overset{-2}{O}\overset{-1}{F_4} + 2\overset{+1}{H}\overset{-1}{F}$$

$$\overset{+6}{Xe}\overset{-1}{F_6} + 2\overset{+1}{H_2}\overset{-2}{O} \longrightarrow \overset{+6}{Xe}\overset{-2}{O_2}\overset{-1}{F_2} + 4\overset{+1}{H}\overset{-1}{F}$$

Since the oxidation states of all the elements in the products of hydrolysis and reactants remain the same therefore, it is not a redox reaction.

3. Write balanced equations for the reactions of the following compounds with water

(i) Al_4C_3 (ii) CaNCN (iii) BF_3 (iv) NCl_3 (v) XeF_4

Sol. BCl_3 or BBr_3, B_2H_6

$$(i)\ Al_4C_3 + 12H_2O \longrightarrow 4Al(OH)_3 + 3CH_4 \uparrow$$

$$(ii)\ CaNCN + 3H_2O \longrightarrow CaCO_3(\downarrow) + 2NH_3$$

$$(iii)\ 4BF_3 + 3H_2O \longrightarrow \underset{\text{boric acid}}{H_3BO_3} + \underset{\text{fluoboric acid}}{3HBF_4}$$

$$(iv)\ NCl_3 + 3H_2O \longrightarrow NH_3 + \underset{\text{hypochlorous acid}}{3HOCl}$$

$$(v)\ 2XeF_4 + 3H_2O \longrightarrow Xe + \underset{\text{xenon trioxide}}{XeO_3} + F_2 + 6HCl$$

4. In the following equation,

$$A + 2B + H_2O \longrightarrow C + 2D$$

($A = HNO_2$, $B = H_2SO_3$, $C = NH_2OH$). Identify D. Draw the structures of A, B, C and D.

Sol. $$3Ca(OH)_2 + 2Cl_2 \longrightarrow \underbrace{Ca(OCl)_2 + Ca(OH)_2\,CaCl_2.2H_2O}_{\text{bleaching powder is a mixture of } CaOCl_2 \text{ and hydrasted basic calcium chloride}}$$

What happens when?

5. **(a) $K_2Cr_2O_7$ is treated with $SnCl_2$ in acidic media.**
(b) Pb_3O_4 is treated with conc. HNO_3
(c) $Pb(CH_3COO)_2$ is treated with bleaching powder.
(d) PbO_2 is treated with NaOH
(e) Ferrous sulphate is ignited
(f) $FeCl_3$ is treated with sodium carbonate. The resulting precipitate is further heated.

Sol. (a) $K_2Cr_2O_7 + 14HCl + 3SnCl_2 \longrightarrow 3SnCl_4 + 2KCl + \underset{(green)}{2CrCl_3} + 7H_2O$

(b) $Pb_3O_4 + 4HNO_3 \longrightarrow 2Pb(NO_3)_2 + PbO_2\downarrow + 2H_2O$

(c) $Pb(CH_3COO)_2 + Ca(OCl)Cl + H_2O \longrightarrow PbO_2 + CaCl_2 + 2CH_3COOH$

(d) $PbO_2 + 2NaOH \longrightarrow Na_2PbO_3 + H_2O$

(e) $2FeSO_4 \longrightarrow Fe_2O_3 + SO_2 + SO_3$

(f) $2FeCl_3 + 3Na_2CO_3 + 3H_2O \longrightarrow 2Fe(OH)_3\downarrow + 6NaCl + 3CO_2$

$2Fe(OH)_3 \longrightarrow Fe_2O_3 + 3H_2O$

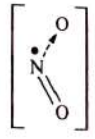
6. **Complete the following reactions:**

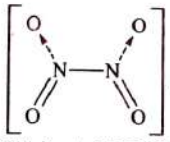
(i) $2NaOH + 2Al + 2H_2O \longrightarrow (A) + 3H_2$
(ii) $2NaOH + Sn + H_2O \longrightarrow (B) + 2H_2$
(iii) $2NaOH + Si + H_2O \longrightarrow (C) + 2H_2$
(iv) $6NaOH + S \longrightarrow (D) + 2Na_2S + 3H_2O$
(v) $3NaOH + P_4 + 3H_2O \longrightarrow 3NaH_2PO_2 + (E)$
(vi) $2KO_2 + (F) \longrightarrow K_2CO_3 + O_2$

Sol. A $= 2NaAlO_2$, B $= Na_2SnO_3$, C $= Na_2SiO_3$, D $= Na_2S_2O_3$, E $= PH_3$, F $= CO$

7. **(a) NO_2 gas is paramagnetic in nature. When it is cooled below 0°C, acquires diamagnetic behaviour. Explain.**
(b) Write the reaction of urea with nitrous acid.
(c) What do you expect if Fe^{3+} is treated first with SCN^- and then with F^- ion?
(d) Write the reactions when excess CN^- is treated with Cu^{2+} ion.
(e) How will you prepare bleaching powder from $CaCO_3$?

Sol. (a)

At room temperature,  there is unpaired electron at nitrogen. On cooling dimer exists,

 and there is no unpaired electron, so diamagnetic.

(b) $Co(NH_2)_2 + 2HNO_2 \rightarrow 2N_2 + CO_2 + 3H_2O$

(c) $Fe^{3+} + 6SCN^- \rightleftharpoons \underset{\text{Red}}{[Fe(SCN)_6]^{3-}}$

$[Fe(SCN)_6]^{3-} + 6F^- \rightleftharpoons \underset{\text{Colourless}}{[FeF_6]^{3-}} + 6SCN^-$

(d) $2Cu^{2+} + 4CN^- \longrightarrow Cu_2(CN)_2 + \underset{\text{Poisonous gas}}{(CN)_2}\uparrow$

$Cu_2(CN)_2 + 6KCN \longrightarrow 2K_3[Cu(CN)_4]$

(e) $CaCO_3 \xrightarrow{\text{Heat}} CaO \xrightarrow{H_2O} Ca(OH)_2 \xrightarrow[40°C]{Cl_2} CaOCl_2$

8. Why hydrated barium peroxide is used in the preparation of H_2O_2 instead of the anhydrous variety?

Sol. If anhydrous barium peroxide is used in the preparation, the barium sulphate, thus formed forms an insoluble protective coating on the surface of solid barium peroxide. This prevents the further reaction of the acid, i.e., causing the reaction to stop. If, however, hydrated barium peroxide (in the form of thin paste) is used, the water causes to dislodge the insoluble $BaSO_4$ from the surface of BaO_2. $BaSO_4$ thus settles at the bottom of the reaction vessel and the reaction continues without any difficulty.

SOLVED PROBLEMS

Subjective Question

Prob. 1. Does the hydrolysis of XeF_6 lead to a redox reaction?

Sol. Complete hydrolysis of XeF_6 gives XeO_3

$$\overset{+6}{Xe}F_6 + 3H_2O \longrightarrow \overset{+6}{Xe}O_3 + 6HF$$

On the other hand, partial hydrolysis of XeF_6 gives $XeOF_4$ and XeO_2F_2

$$\overset{+6}{Xe}\overset{-1}{F}_6 + \overset{+1}{H}_2\overset{-2}{O} \longrightarrow \overset{+6}{Xe}\overset{-2}{O}\overset{-1}{F}_4 + 2\overset{+1}{H}\overset{-1}{F}$$

$$\overset{+6}{Xe}\overset{-1}{F}_6 + 2\overset{+1}{H}_2\overset{-2}{O} \longrightarrow \overset{+6}{Xe}\overset{-2}{O}_2\overset{-1}{F}_2 + 4\overset{+1}{H}\overset{-1}{F}$$

Since the oxidation states of all the elements in the products of hydrolysis and reactants remain the same therefore, it is not a redox reaction.

Objective Questions

Prob.1. The reaction of P_4 with **X** leads selectively to P_4O_6. The **X** is

(A) Dry O_2 (B) A mixture of O_2 and N_2

(C) Moist O_2 (D) O_2 in the presence of aqueous NaOH

Sol. $P_4 + 3O_2 \xrightarrow{N_2} P_4O_6$ (exclusively)

(N_2 is used to retard the further oxidation.)

Hence, **(B)** is the correct answer.

Prob. 2. In the following statements, select the correct statement:

(A) $N(CH_3)_3$ has pyramidal structure. (B) $N(SiH_3)_3$ shows planar arrangement.

(C) both correct. (D) none is correct.

Sol. **(C)**

Prob. 3. In the electrolysis of alumina, cryolite is added to alumina to

(A) lower the m.p. of alumina (B) increase the electrical conductivity

(C) minimise the anode effect (D) remove impurities from alumina

Sol. **(B)**

Prob. 4. Which of the following is/are amphoteric?

(A) BeO (B) Ag_2O

(C) CO_2 (D) SnO_2

Sol. BeO and SnO_2 are amphoteric, CO_2 is acidic whereas Ag_2O is basic in nature.

Hence, **(A)** and **(D)** are the correct answers.

Prob. 5. Which of the following properties decrease from top to bottom down the group in case of hydrides of oxygen family members?

(A) Dipole moment (B) Bond angle

(C) Covalent character (D) Thermal stability

MEGACOSM

Sol. Dipole moment, bond angle and thermal stability decrease from top to bottom down the group in case of hydrides of group 16 elements. Dipole moment and bond angle decrease due to decrease in electronegativity of the central atom and thermal stability decreases due to weakening of the M – H bond as the size of the central atom increases from top to bottom.

Hence, **(A), (B)** and **(D)** are the correct answers.

Prob. 6. White phosphorus (P_4) has

(A) six P – P single bonds
(B) four P – P single bonds
(C) four lone pairs of electrons
(D) P – P – P angle of 60°

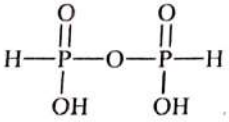
Sol. White phosphorus (P_4) has tetrahedral structure in which each P atom lies at the corners of the regular tetrahedron. There are six single P – P bonds, four lone pairs of electrons and P – P – P bond angle of 60° in the P_4 structure.

Hence, **(A),(C)** and **(D)** are the correct answers.

Prob. 7.Pyrophosphorus acid, $H_4P_2O_5$

(A) contains P in +5 oxidation state
(B) is dibasic acid
(C) is strongly reducing in nature
(D) contains one P – O – P bond

Sol. Pyrophosphorus acid ($H_4P_2O_5$) is dibasic acid as it contains two P – OH bonds, strongly reducing in nature due to the presence of two P – H groups and contains one P – O – P bond.



Hence, **(B), (C)** and **(D)** are the correct answers.

Prob. 8. Which compound does dissolve in hot dilute HNO_3?

(A) HgS
(B) PbS
(C) CuS
(D) CdS

Sol. **(A, B, C, D)**

Prob. 9. Which is/are correct statements about P_4O_6 and P_4O_{10}?

(A) both form oxyacids H_3PO_3 and H_3PO_4 respectively
(B) in P_4O_6 each P is joined to four O and in P_4O_{10} each P is joined to six O atoms
(C) in P_4O_6 each P is joined to three O and in P_4O_{10} each P is joined to four O atoms
(D) in P_4O_6 each P is joined to three O and in P_4O_{10} each P is joined to five O atoms

Sol. **(A, C)**

Prob. 10. Select correct statements.

(A) ClO_2 and Cl_2O are used as bleaching agents for paper pulp and textiles
(B) OCl^- (hypohalites) salts are used as sterilizing agents
(C) OCl^- disproportionates in alkaline medium
(D) BrO_3^- is oxidized to Br_2 by Br^- in acidic medium

Sol. **(A, B, C, D)**

Prob.11. $P_4 \xrightarrow{Ba(OH)_2/\Delta} \underset{(salt)}{X} \xrightarrow{H_2SO_4} Y(oxyacid) \xrightarrow{\Delta} A(oxyacid)$

In the above sequence of reactions Y and A respectively are

(A) H_3PO_2 and H_3PO_4 (B) H_3PO_4 and $H_4P_2O_7$

(C) H_3PO_4 and HPO_3 (D) H_3PO_3 and H_3PO_4

Sol. **(A)**

$$P_4 \xrightarrow{OH^-} PH_3 + H_2PO_2^- (X) \xrightarrow{H^+} \underset{Y}{H_3PO_2} \xrightarrow{\Delta} PH_3 + \underset{A}{H_3PO_4}$$

Prob.12 In which of the following option product gas X and Y (other than water vapour) are same ?

(A) $Mg_2C_3 + H_2O \longrightarrow X$; $Al_4C_3 + H_2O \longrightarrow Y$

(B) $NH_4NO_3 \xrightarrow{\Delta} X$; $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta} Y$

(C) $NH_4Cl \xrightarrow{NaOH(aq.)} X$; $NaNO_3 \xrightarrow{Al/NaOH(aq.)} Y$

(D) $Zn + dil.HNO_3 \longrightarrow X$; $Ag + dil.HNO_3 \longrightarrow Y$

Sol. **(C)**

	X	Y
A)	Propyne	Methane
B)	N_2O	N_2
C)	NH_3	NH_3
D)	N_2O	NO

Prob. 13. Which of the following order is correct for the property mentioned in brackets.

(A) $S^{2-} > Cl^- > K^+ > Ca^{2+}$ (Ionisation energy)

(B) $C < N < F < O$ (2^{nd} Ionisation energy)

(C) $B > Al > Ga > In > Tl$ (Electronegativity)

(D) $Na^+ > Li^+ > Mg^{2+} > Be^{2+} > Al^{3+}$ (Ionic radius)

Sol. **(B)**

COMPREHENSION

COMPREHENSION-1

Read the following paragraph and answer the questions given below:

The noble gases have closed-shell electronic configuration and are monoatomic gases under normal conditions. The low boiling points of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interactions.

The direct reaction of xenon with fluorine leads to a series of compounds with oxidation numbers +2, +4 and +6. XeF_4 reacts violently with water to give XeO_3. The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valence shell.

Prob. 1. Argon is used in arc welding because of its

(A) low reactivity with metal (B) ability to lower the melting point of metal

(C) flammability (D) high calorific value

Sol. Argon is used mainly to provide an inert atmosphere in high temperature metallurgical (arc welding of metals/alloys) extraction.

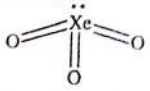
Hence, **(A)** is the correct answer.

Prob. 2. The structure of XeO_3 is

(A) linear (B) planar

(C) pyramidal (D) T-shaped

Sol.



sp^3 hybridized pyramidal structure.

Hence **(C)** is correct.

Prob. 3. XeF_4 and XeF_6 are expected to be

(A) oxidizing (B) reducing

(C) unreactive (D) strongly basic

Sol. $6XeF_4 + 12H_2O \longrightarrow 4Xe + 2XeO_3 + 24HF + 3O_2$

$XeF_6 + 3H_2O \longrightarrow XeO_3 + 6HF$

Hence, **(A)** is the correct answer.

COMPREHENSION-2

An inorganic compound (X) gives brick red flame and also gives the following observation:

(i) Produce smell of chlorine in moist air

(ii) produce brown coloration with CH_3COOH & KI

(iii) Produce an anaesthetic compound with ethanol.

Prob.4 Metal ion which is present in (X) is

(A) Ba^{2+} (B) Sr^{2+}

(C) Ca^{2+} (D) Na^{+}

Sol. C

Prob. 5 What is the oxidation state of chlorine in compound (X) ?

(A) +1 (B) –1

(C) Both + 1 & –1 (D) Zero

Sol. C

Prob. 6 What is the formula of compound which is formed by the reaction of (X) and C_2H_5OH ?

(A) $CHCl_3$ (B) CH_3Cl

(C) CCl_4 (D) C_2H_5Cl

Sol. A

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** Between $SiCl_4$ and CCl_4, only $SiCl_4$ reacts with water.
and
STATEMENT-2: $SiCl_4$ is ionic and CCl_4 is covalent.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**

MATRIX – MATCH TYPE QUESTIONS

1. Match the Column – I with Column – II:

Column I		Column II	
(A)	$Bi^{3+} \longrightarrow (BiO)^+$	(p)	Heat
(B)	$[AlO_2]^- \longrightarrow Al(OH)_3$	(q)	Hydrolysis
(C)	$SiO_4^{4-} \longrightarrow Si_2O_7^{6-}$	(r)	Acidification
(D)	$(B_4O_7^{2-}) \longrightarrow [B(OH)_3]$	(s)	Dilution by water

Sol. **(A → q), (B → r), (C → p), (D → q, r)**

2. Match the Column – I with Column – II:

Column I		Column II	
(A)	Colemanite	(p)	Na_3AlF_6
(B)	Cryolite	(q)	$KAl(SO_4)_2.12H_2O$
(C)	Bauxite	(r)	$Ca_2B_6O_{11}.5H_2O$
(D)	Borax	(s)	$Na_2B_4O_7.10H_2O$
(E)	Potash alum	(t)	$Al_2O_3.2H_2O$

Sol. **(A – r), (B – p), (C – t), (D – s), (E – q)**

3. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	HF	(p)	Reducing agent
(B)	HI	(q)	Oxidising agent
(C)	Cl_2	(r)	Precipitate with $AgNO_3$ solution
(D)	$(CN)_2$	(s)	Attacks glass
		(t)	Pseudo halogen

Sol. **A → (s) B → (p, r) C → (q) D → (q, t)**

4. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Zn + V. dil. HNO_3	(p)	NO
(B)	Sn + dil. HNO_3	(q)	NO_2
(C)	Ag + dil. HNO_3	(r)	NH_4NO_3
(D)	Pb + conc. HNO_3	(s)	$Sn(NO_3)_2$
		(t)	$Zn(NO_3)_2$

Sol. **A → (r, t); B → (r, s); C → (p); D → (q)**

5. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	HNO_3	(p)	Oxidizing Agent
(B)	HNO_2	(q)	Reducing Agent
(C)	H_2SO_4	(r)	Strongest Acid
(D)	H_2SO_3	(s)	Produced when NO_2 dissolve in water
		(t)	Disproportionate on standing

Sol. **A → (p, s); B → (p, q, s, t); C → (p, r); D → (p, q)**

6. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Increasing order of I.E.	(p)	F < O < S < Se
(B)	Increasing order of electron affinity	(q)	O < N < F < Ne
(C)	Increasing order of size	(r)	Na < Mg < Al < Si
		(s)	$O^{2-} < O^{-} < O < O^{+}$

Sol. **A → (q) B → (s) C → (p)**

Answer 7, 8 and 9 by appropriately matching the information given in the three columns of the following table.

Column – 1 Name of oxyacid		Column – 2 Formula		Column – 3 Basicity	
(I)	Metaphosphoric acid	(i)	H_3PO_3	(P)	1
(II)	Phosphorus acid	(ii)	$H_4P_2O_7$	(Q)	2
(III)	Hypophosphoric acid	(iii)	$H_4P_2O_6$	(R)	4
(IV)	Pyrophosphoric acid	(iv)	HPO_3	(S)	4

7. Which of the following combination is correct ?
[A] (I) (iv) (P)
[B] (II) (ii) (R)
[C] (III) (ii) (Q)
[D] (IV) (i) (S)

8. Which of the following combination is incorrect ?
[A] (IV) (ii) (R)
[B] (III) (iii) (S)
[C] (II) (i) (Q)
[D] (II) (iii) (P)

9. Which of the following combination is correct ?
[A] (IV) (i) (R)
[B] (III) (ii) (Q)
[C] (II) (iii) (S)
[D] (IV) (iv) (P)

Sol.

7. **A**

8. **D**

9. **C**

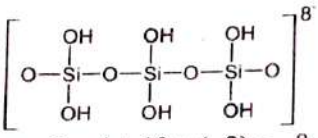
SINGLE INTEGER ANSWER TYPE

1. How many litres of SO_2 taken in standard conditions have to be passed through a solution of $HClO_4$ to reduced to HCl

Sol. **(9) litre**

2. The silicate anion in the mineral kinoitic is a chain of three SiO_4 tetrahedra that share corners with adjacent tetrahedra. What is negative charge on the silicate anion.

Sol. **(- 8),**



Charge = $3 \times 4 + 10 \times (-2) = -8$

3. The number of P – O – H links in orthophosphoric acid molecule is .....

Sol. **(3)**

4. The number of S – O – S links in $H_2S_2O_7$ molecule is .....

Sol. **(1)**

Prob. 1. Al_4C_3 on hydrolysis gives a gas. The number of H atom in gas is x. The value of $\frac{x}{5}$ is

Sol. **0.8**

Prob. 2. The number of P—O—P bonds in cyclic metaphosphoric acid is x. The value of $\frac{x}{2}$ is

Sol. **1.5**

$\frac{3}{2} = 1.5$

Prob. 3. When thiosulphate ion is oxidized by iodine, the new product formed. The number of S—S linkage is x. The value of $\frac{x}{2}$ is

Sol. **1.5**

$\frac{3}{2} = 1.5$

Prob. 4. The sum of number of P = O, P—O—P linkages in $P_5O_{16}^{-7}$ ion is x. The value of $\frac{x}{2}$ is

Sol. **4.5**

$\frac{9}{2} = 4.5$

Prob. 5. The number of nitrate which gives No_2 gas on heating from the following is x. Then find the value of $\frac{x}{5}$ is

Nitrogen dioxide is released on heating; NH_4NO_3, KNO_3, $NaNO_2$, $Pb(NO_3)_2$, $AgNO_3$.

Sol. **0.4**

$\frac{2}{5} = .4$

Prob. 6. The number of ion having S—S bond from following is x. Find the value of $\frac{x}{5}$ is

$S_2O_4^{2-}$, $S_2O_6^{2-}$, $S_2O_2^{2-}$, $S_2O_3^{2-}$, $S_2O_7^{2-}$

Sol. **.2**

Prob. 7. The total number of diprotic acids among the following is x. The value of $\frac{x}{4}$ is

H_3BO_3, H_3PO_2, H_2CrO_4, H_2SO_3.

Sol. **1.5**

$\frac{6}{4} = 1.5$

Prob. 8. The total number of lone pairs of electron in N_2O_3 is x. The value of $\frac{x}{5}$ is

Sol. **1.6**

$$\frac{8}{5} = 1.6$$

Prob. 9. The value of $\frac{n}{2}$ in the molecular formula $Be_nAl_2Si_6O_{18}$ is

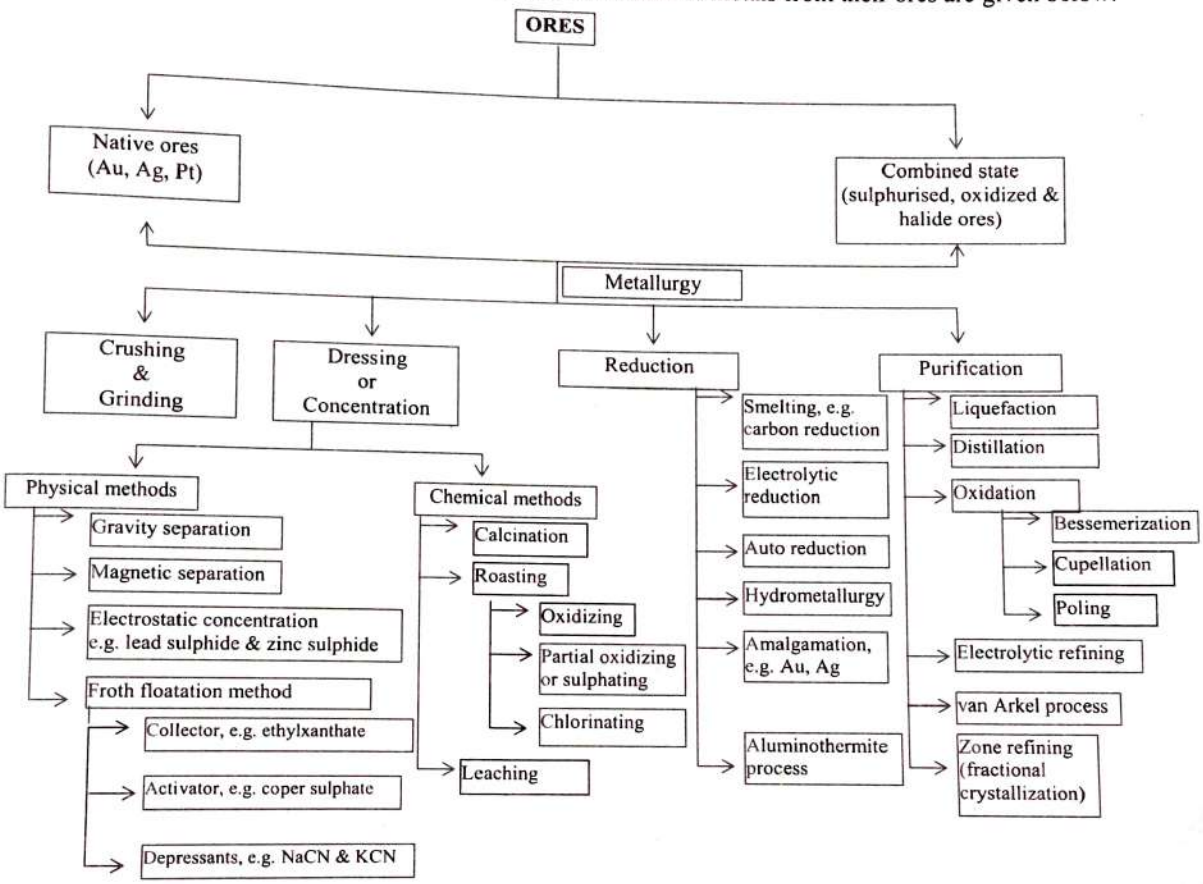
Sol. **1.5**

Prob.10. Three moles of B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is x. The value of $\frac{x}{4}$ is

Sol. **1.5**

ORES AND METALLURGY 24

"All ores are minerals but all minerals are not ores".
The above statement can be explained by the example, copper occurs in nature in the form of several minerals like Cu_2O, Cu_2S and $CuFeS_2$ but copper pyrite (Cu_2S) is considered as most economical mineral for extraction of the metal. Hence, Cu_2S is chief ore of copper.
The various metallurgical processes involved in the extraction of metals from their ores are given below:



IMPORTANT TERMS & FACTS

1. **Collectors:** In froth floatation process, ethyl xanthate and potassium ethyl xanthate are used as collectors.
2. **Activators:** They activate the ore and help in separation of different minerals present in same ore, e.g. copper sulphate is used as an activator.
3. **Depressants:** They depress the floating property of one of the component of the ore and helps it to get separated from others, e.g. NaCN and KCN are used as depressants.
4. **Flux:** Gangue (impurity) + Flux (easily fusible material) $\longrightarrow$ Slag
   (a) **Acidic flux:** These are used when gangue is basic such as lime or other metallic oxides, e.g. MnO, FeO, silica, borax etc.
   $CaO + SiO_2 \longrightarrow CaSiO_3$
   Gangue Flux Slag

(b) **Basic flux:** These are used when gangue is acidic like silica, P_4O_{10} etc. The examples of basic flux are CaO, lime stone ($CaCO_3$), magnesite ($MgCO_3$), haematite (Fe_2O_3) etc.

$SiO_2 + MgCO_3 \longrightarrow MgSiO_3 + CO_2\uparrow$

Gangue Flux Slag

Potassium is highly reactive, hence it is stored under kerosene.

Fluorspar is added to anhydrous $CaCl_2$ in the extraction of metallic calcium for reducing the cost of production.

5. **Cupellation:** When impure metal contains impurities of other metals which forms volatile oxides, then it is heated in a cupel and removed with the blast of air. For example, traces of lead are removed from silver by heating impure silver with a blast of air in a cupel in which lead is oxidized to lead oxide (PbO) which being volatile escapes leaving behind pure silver.
6. **Poling:** Poling is used for refining of metals which contain impurities of its own oxide.
7. Metals which are low in the activity series (e.g. Cu, Hg, Ag, Au) can be obtained by heating their compounds in air, metals which are in the middle of the activity series (e.g. Fe, Zn, Ni, Sn, Pb) are obtained by heating their oxides with carbon while metals which are very high in the activity series (e.g. Na, K, Ca, Mg, Al) are obtained by electrolytic reduction method.
8. $CaCl_2$ and KF are added to NaCl (Down's cell) to lower the melting point of NaCl to $\approx$ 850 K.
9. Potassium is not extracted by electrolysis of fused KCl because
   (i) Potassium boils at the melting point of KCl.
   (ii) Potassium is quite soluble in molten KCl and does not float over the surface of molten KCl from where it can be skimmed off.
10. K_2CO_3 cannot be prepared by Solvay process because $KHCO_3$ is more soluble than $NaHCO_3$ in water and hence does not get precipitated.
11. Lapis lazuli is a rare mineral having blue colour. Ultramarine is an artificial lapis lazuli. It is a complex of silicate of sodium and aluminium containing 12% sulphur.
12. Al is extracted from its ore alumina by electrolysis of fused mixture of alumina, cryolite (Na_3AlF_6) and fluorspar (CaF_2) in 20:60:20 ratio in an iron tank using carbon electrodes. The function of adding cryolite and fluorspar is
   (i) to make alumina a good conductor of electricity and
   (ii) to lower its melting point to $\approx 1140\,K$.
13. $FeSiO_3$ is the slag formed in the extraction of copper.
14. $Cu_2S + FeS$ is called matte.
15. Thomas slag, $Ca_3(PO_4)_2$, is the slag formed from P_2O_5 impurity in the metallurgy of iron.
16. Iron is present as cementite, Fe_3C, in white cast iron.
17. Fool's gold is FeS_2.
18. In cold climate, tin becomes brittle so it cannot be used as structural metal.

ILLUSTRATIVE PROBLEMS

1. A sulphide ore (A) on roasting leaves a residue (B). (B) on heating with chlorine gives (C), soluble in water, addition of excess potassium iodide to a solution of (C) gives a solution (D). A brown precipitate (E) is formed when a solution of ammonium sulphate is added to an alkaline solution of (D). Identify (A) to (E).

Sol.

$$\underset{(A)}{HgS} \xrightarrow{\text{roasting}} \underset{(B)}{HgO} \xrightarrow{Cl_2} \underset{(C)}{HgCl_2} \xrightarrow{\text{excess of KI}} \underset{(D)}{K_2HgI_4}$$

$$\text{alkaline} \Big| \downarrow (NH_4)SO_4$$

$$HgO{\cdot}Hg(NH_2)I$$

Brown ppt.
(E)

2. Dolomite can also be treated to get $MgCl_2$ which in turn is electrolysed to get Mg (Dow's natural brine process). Give reactions of this process.

Sol.

$$MgCO_3.CaCO_3 \xrightarrow[\Delta]{\text{Calcination}} MgO + CaO + 2CO_2 \uparrow$$

$$\downarrow HCl \qquad \downarrow$$

$$MgCl_2 + CaCO_3 \downarrow \xleftarrow{CO_2} MgCl_2 + CaCl_2$$

On passing CO_2 into aq. $MgCl_2$ and $CaCl_2$ solution, $CaCO_3$ is precipitated leaving $MgCl_2$ soluble.

3. Ferrochrome, an iron chromium alloy used in making stainless steel, is produced by reducing chromite ($FeCr_2O_4$) with coke:

$$FeCr_2O_4(s) + 4C(s) \longrightarrow \underbrace{Fe(s) + 2Cr(s)}_{\text{Ferrochrome}} + 4CO(s)$$

(i) How many kilograms of chromium can be obtained by the reaction of 236 kg of chromite with an excess of coke?

(ii) How many litres of CO at 25°C and 740 mm Hg are obtained as a by- product?

Sol. (i) $FeCr_2O_4(s) + 4C(s) \longrightarrow Fe(s) + 2Cr(s) + 4CO(s)$

224 kg of chromite gives = 104 kg of Cr

236 Kg of chromite gives $= \dfrac{104 \times 236}{224}$ kg of Cr

$= 109.57$ Kg Cr

(ii) 224 kg chromite gives = 112 Kg CO

Thus 236 kg chromite gives $= \dfrac{112 \times 236}{224}$ kg CO

$\Rightarrow$ 118 kg CO = 4214.2857 mol.

As $PV = nRT$

$T = 298$ K

$P = \dfrac{740}{760}$ atm

$$V = \frac{nRT}{P} = \frac{4214.28 \times 0.0821 \times 298 \times 760}{740}$$

$$= 105.89 \times 10^3 \text{ litre}$$

4. NaCN is sometimes added in the forth flotation process as depressant when ZnS and PbS minerals are expected because :

(a) $Pb(CN)_2$ is precipitated while no effect on ZnS

(b) ZnS forms soluble complex $Na_2[Zn(CN)_4]$ while PbS forms froth

(c) PbS forms soluble complex $Na_2[Pb(Cn)_4]$ while ZnS forms froth

(d) NaCN is never added in froth floatation process

Sol. (b) Depressant is a substance that is added to suppress the floating characteristic of metal sulphide present as an impurity.

$$PbS + \underset{\text{(Impurity)}}{ZnS} \xrightarrow{4NaCN} \underset{\text{Soluble complex}}{Na_2[Zn(CN)2]} + PbS$$

5. Complete and balance the following equations:

(a) $Zn + Fe_2(SO_4)_3 \longrightarrow ? + ?$

(b) $CuSO_4 \xrightarrow{heat} ? + ?$

(c) $Al + \text{conc. } H_2SO_4 \longrightarrow ? + ? + ?$

(d) $Al_2O_3 + C + N_2 \xrightarrow{heat} ? + ?$

(e) $Pb + O_2 + H_2O \longrightarrow ?$

Sol. (a) $Zn + Fe_2(SO_4)_3 \longrightarrow ZnSO_4 + 2FeSO_4$

(b) $CuSO_4 \xrightarrow{\Delta} CuO + SO_3$

(c) $2Al + \text{conc. } 6H_2SO_4 \longrightarrow Al_2(SO_4)_3 + 3SO_2 + 6H_2O$

(d) $Al_2O_3 + 3C + N_2 \xrightarrow{\Delta} 2AlN + 3CO$

(e) $2Pb + O_2 + 2H_2O \longrightarrow 2Pb(OH)_2$

6. Consider the following metallurgical processes :

(I) Heating impure metal with CO and distilling the resulting volatile carbonyl (b.p. 43°C) and finally decomposition at 150°-200°C to get the pure metal

(II) Heating the sulphide ore in air until a part is converted to oxide and then further heating in the absence of air to let the oxide react with unchanged metal sulphide

(III) Electrolysis of the molten electrolyte containing approximately equal amounts of the metal chloride and NaCl to obtained the metal

(a) I, II and III **(b) II, III and I** **(c) III, I and II** **(d) II, I and III**

Sol. (I) $\underset{\text{(impure)}}{Ni} + 4CO \xrightarrow{50°C} \underset{\text{(volatile)}}{[Ni(CO_4)]} \xrightarrow{230°C} \underset{\text{(pure)}}{Ni} + 4CO\uparrow$

(II) $Cu_2S + \frac{3}{2}O_2 \xrightarrow{\Delta} Cu_2O + SO_2$

$Cu_2S + 2Cu_2O \xrightarrow[\text{temp.}]{\text{high}} 6Cu + SO_2$

(III) $MgCl_2(s) \xrightarrow{\text{electrolysis}} Mg^{2+}(\ell) + 2Cl^-(\ell)$

At cathode : $Mg^{2+} + 2e^- \longrightarrow Mg(s)$

At anode : $2Cl^-(\ell) \longrightarrow Cl_2(g) + 2e^-$

7. **Name the process shown by the following equation.**
$2CuFeS_2 + O_2 \xrightarrow{1400\text{-}1450^\circ C} Cu_2S + Fe_2O_3 + SO_2$
How do we extract blister copper from Cu_2S ? Explain with the help of equations.

Sol. $CuFeS_2 \xrightarrow[1400-1450^\circ C]{O_2} Cu_2S + Fe_2O_3 + 3SO_2$
This is a roasting process of copper pyrites.
In this process, sand is added to remove the iron as iron silicate slag, $Fe_2(SiO_3)_3$, which floats on the surface. Air is blown through the liquid matte of Cu_2S with some FeS and silica, causing partial oxidation.
$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$
$Fe + SiO_2 \longrightarrow Fe_2(SiO_3)_3$
$Cu_2S + O_2 \longrightarrow Cu_2O + SO_2$
After sometime the air is turned off and self-reduction of the oxide and sulphide occurs, giving impure 'blister copper' which is 98–99% pure.
$Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$

8. **Complete the following reactions:**
(a) $Al(OH)_4^- + CO_2 \longrightarrow ? + ? + ?$
(b) $Ag^+ + Na_2S_2O_3 \longrightarrow ? + ?$
(c) $MnO_4^- + H_2S \xrightarrow{\Delta} ? + ?$

Sol. (a) $2Al(OH)_4^- + CO_2 \longrightarrow 2Al(OH)_3 + CO_3^{2-} + H_2O$
(b) $Ag^+ + 2Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2] + Na^+$
(c) $MnO_4^- + H_2S \longrightarrow Mn^{2+} + SO_2 + 2OH^-$

9. **In moist air copper corrodes to produce a green layer on the surface. Explain.**

Sol. In presence of moist air a thin film of green basic copper carbonate is formed on its surface and hence copper corrodes
$2Cu + O_2 + H_2O + CO_2 \longrightarrow \underset{\text{green}}{CuCO_3 \cdot Cu(OH)_2}$

Subjective Question

Prob. 1. Assign suitable terms to the following:

(i) Materials which can withstand very high temperatures without melting and softening.
(ii) Sulphide ores are generally heated in a stream of air.
(iii) The substances used for the removal of gangue in the ores in the form of slags.
(iv) Two metals which are manufactured by the electrolysis of their fused salts.
(v) Two metals used for reduction in metallurgical processes.

Sol. (i) Refractory materials
(ii) Roasting
(iii) Fluxes
(iv) Sodium, aluminium
(v) Aluminium, magnesium or sodium.

Objective Questions

Prob. 1. Of the following acids

I : hypo phosphorus acid II: hydrofluoric acid
III: oxalic acid IV: glycine
(A) I, II are monobasic, III dibasic acid and IV amphoteric
(B) II monobasic, I, III dibasic acid, IV amphoteric
(C) I monobasic, II, III dibasic, IV amphoteric
(D) I, II, III dibasic, IV amphoteric

Sol. **(A)**

Prob. 2. Argentite, a compound of silver, was treated with potassium cyanide solution. The product obtained has the formula

(A) $KAg(CN)_2$ (B) $K_2Ag(CN)_2$
(C) $K_2Ag(CN)_4$ (D) $KAg(CN)_4$

Sol. **(A)**

$$Ag_2S + 4KCN \rightleftharpoons \underset{\text{Potassium argentocyanide}}{2KAg(CN)_2} + K_2S$$

Hence, **(A)** is the correct answer.

Prob. 3. Which ore contains both iron and copper?

(A) Cuprite (B) Chalcocite
(C) Chalcopyrite (D) malachite

Sol. **(A)**

Prob. 4. Butter of tin is

(A) $(NH_4)_2SnCl_6$ (B) $SnCl_2 + Sn(OH)_2$
(C) $SnCl_4, 5H_2O$ (D) H_2SnCl_4

Sol. **(C)**

Prob. 5. On heating a mixture of Cu_2O and Cu_2S, we get :

(A) $Cu + SO_2$ (B) $Cu + SO_3$
(C) $CuO + CuS$ (D) Cu_2SO_3

Sol. The reactions are

$$Cu_2S + 2Cu_2O \longrightarrow 6Cu + SO_2$$

(auto-reduction)

Hence, **(A)** is the correct answer.

Prob. 6. In the process of extraction of gold

$$\text{Roasted gold ore} + CN^- + H_2O \xrightarrow{O_2} [X] + OH^-$$

$$[X] + Zn \longrightarrow [Y] + Au$$

Identify the complexes [X] and [Y]

(A) $X = [Au(CN)_2]^-$, $Y = [Zn(CN)_4]^{2-}$ (B) $X = [Au(CN)_4]^{3-}$, $Y = [Zn(CN)_4]^{2-}$

(C) $X = [Au(CN)_2]^-$, $Y = [Zn(CN)_6]^{4-}$ (D) $X = [Au(CN)_4]^-$, $Y = [Zn(CN)_4]^{2-}$

Sol. **(A)**

Prob. 7. The slag obtained during the extraction of copper pyrites is composed mainly of

(A) Cu_2S (B) $FeSiO_3$

(C) $CuSiO_3$ (D) SiO_2

Sol.

$$2CuFeS_2 + O_2 \longrightarrow Cu_2S + 2FeS + SO_2$$

$$2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2$$

$$FeO + SiO_2 \longrightarrow FeSiO_3 \text{ (slag)}$$

Hence, (B) is the correct answer.

Prob. 8. Roasting of sulphide ore is done before smelting

(A) to remove moisture and volatile impurities

(B) because sulphide cannot be reduced easily as oxide by carbon reduction

(C) to remove earthen impurities

(D) All of the above

Sol. **(A, B)**

Prob. 9. Which of the following process is based on the formation of complex compounds?

(A) Belgian process (B) Mac Arthur Forest process

(C) Poling process (D) Mond process

Sol. **(B)** and **(D)** are the correct answers.

Prob. 10. The extraction of metals from oxide ores involve

(A) reduction with carbon (B) reduction with aluminium

(C) electrolytic reduction (D) reduction with CO

Sol. **(A, B, C, D)**

Prob. 11. Metals which can be extracted by smelting process are

(A) Pb (B) Fe

(C) Zn (D) Al

Sol. **(A, B, C)**

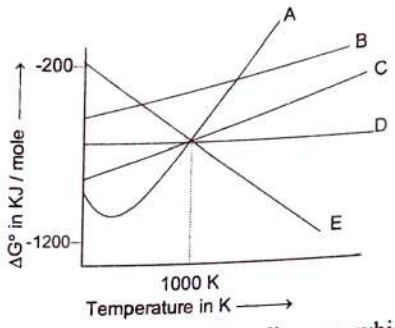
Prob.12. Leaching is a process of :

(A) reduction (B) concentration

(C) refining (D) oxidation

Sol. **(B)**

Prob.13.



Graph A represents $2Zn + O_2 \longrightarrow 2ZnO$
Graph B represents $4Cu + O_2 \longrightarrow 2Cu_2O$
Graph C represents $2CO + O_2 \longrightarrow CO_2$
Graph D represents $C + O_2 \longrightarrow CO_2$
Graph E represents $2C + O_2 \longrightarrow 2CO$

According to above Ellingham diagram, which of the following reaction is/are feasible above 1000K

(A) $Cu_2O + C \longrightarrow 2Cu + CO$ (B) $ZnO + C \longrightarrow Zn + CO$

(C) $Cu_2O + CO \longrightarrow 2Cu + CO_2$(D) $ZnO + CO \longrightarrow Zn + CO_2$

Sol. **(A, B, C, D)**

Less the magnitude of ΔG° for a chemical reaction, less will be the feasibility

Prob.14. In which of the following pair(s), the minerals are converted in to metals by self-reduction process?

(A) Cu_2S, PbS (B) PbS, HgS

(C) PbS, ZnS (D) Ag_2S, Cu_2S

Sol. **A, B**

Prob.15. During the manufacturing of steel from cast iron by open hearth process, a neutral gas was evolved which reacts with a metal X to form a complex compound Y. Cation of metal X gives following tests :

$X^{n+} + SCN^- \longrightarrow$ blood red precipitate.

$X^{n+} + CH_3COO^- \longrightarrow$ deep red precipitate.

$X^{n+} + K_4[Fe(CN)_6] \rightarrow X_4[Fe(CN)_6]_n$

The compound Y was found to have EAN = 36. Which of the following statements is/are true ?

(A) The evolution of neutral gas was accompanied by the formation of MnO, SiO_2 and SO_2.

(B) The magnetic moment in Y is 0.

(C) The bond energy of the bond in the neutral gas increases when it forms Y.

(D) All of the above.

Sol. **(D)**

Reaction taking place in the hearth :

$2Fe_2O_3 + 3Si \longrightarrow 4Fe + 3SiO_2$

$Fe_2O_3 + 3Mn \longrightarrow 2Fe + 3MnO.$

$2Fe_2O_3 + 3S \longrightarrow 4Fe + 3SO_2$

$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO\uparrow$ (neutral gas)

$5Fe_2O_3 + 6P \longrightarrow 10Fe + 3P_2O_5$

$MnO + SiO_2 \longrightarrow MnSiO_3$

$CaO + SiO_2 \longrightarrow CaSiO_3$

$3MgO + P_2O_5 \longrightarrow Mg_3(PO_4)_2$

$2C + O_2 \longrightarrow 2CO$

$4P + 5O_2 \longrightarrow P_4O_{10}$

The qualitative tests indicate the metal to be Fe.

$\therefore$ $Y \equiv Fe(CO)_5$

Which is dsp^3 hybridized having zero magnetic moment (diamagnetic)
In $Fe(CO)_5$, the bond length between C — O decreases due to synergic effect there by increasing the bond energy.

COMPREHENSION

Read the following paragraph and answer the questions given below:

Dow's process of extraction of Mg involves extraction of Mg from sea water. Sea water is concentrated in sun light and is then treated with slaked lime. Magnesium hydroxide is heated in a stream of HCl to give $MgCl_2$ which is electrolyzed to discharge Mg. the mixture is in the ratio 35% $MgCl_2$ + 50% NaCl + 15% $CaCl_2$.NaCl and $CaCl_2$ are added to lower the fusion temperature and to increase the conductance.

$$Mg^{2+} + Ca(OH)_2 \longrightarrow Mg(OH)_2 + Ca^{2+}$$

$$Mg(OH)_2 + 2HCl \longrightarrow MgCl_2 + 2H_2O(l)$$

Electrolysis of fused $MgCl_2 \xrightarrow{\text{Anode}} 2Cl^- \longrightarrow Cl_2 + 2_{-1e^0} \xrightarrow{\text{Cathode}} Mg^{2+} + 2_{-1e^0} \longrightarrow Mg$

Mg electrolysed is protected from atmospheric oxidation by a blanket of inert gases.

Prob. 1. $Mg^{2+} + Ca(OH)_2 \longrightarrow Mg(OH)_2 \downarrow + Ca^{2+}$

This reaction indicates:
(A) $Mg(OH)_2$ is weaker base than $Ca(OH)_2$
(B) Solubility product of $Mg(OH)_2$ is less than that of $Ca(OH)_2$
(C) Polarising power of Mg^{2+} is more than that of Ca^{2+} ion
(D) Both (B) and (C)

Sol. **(D)**

Prob. 2. In the hydrated chloride of Mg the value of x is:
(A) 6
(B) 4
(C) 8
(D) 10

Sol. **(A)**
$MgCl_2.6H_2O$

Prob. 3. Molten mixture contains Mg^{2+}, Na^+ and Ca^{2+} but at cathode only Mg is discharged because:
(A) standard reduction potential of Mg is least among the three
(B) standard oxidation potential of Mg is least among the three
(C) discharge potential of Mg is highest
(D) None of the above

Sol. $E^\circ_{o.p.}$ for Mg^+, Na^+ and Ca^{2+} are 2.37, 2.71 and 2.87 V respectively.

Prob. 4. Molten mixture of NaCl and $CaCl_2$ is added to the heated $MgCl_2.xH_2O$ with dry HCl gas because:
(A) $MgCl_2.xH_2O$ + dry $HCl \xrightarrow{973-1023K}$ Partially dehydrated
$MgCl_2$ and molten (NaCl + $CaCl_2$) makes it fully dehydrated
(B) $CaCl_2$ is dehydrating agent
(C) ($CaCl_2$ + NaCl) lowers the melting point of $MgCl_2$
(D) None of the above

Sol. **(C)**
To lower the melting point of $MgCl_2$

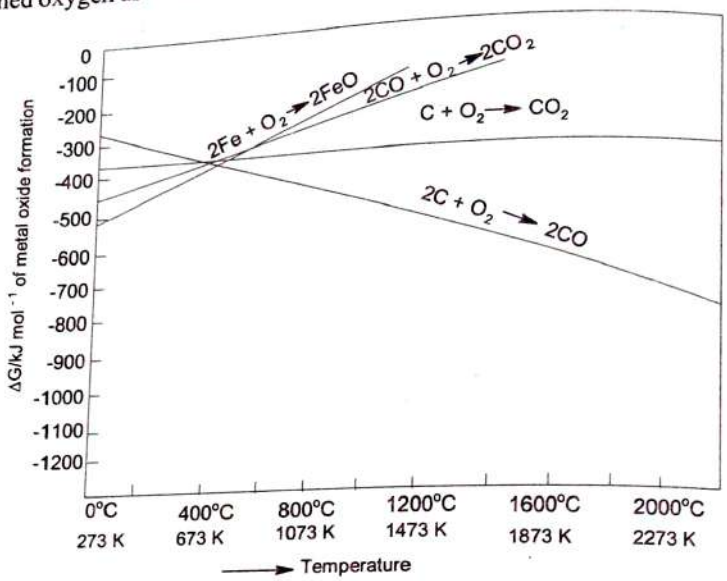
Read the following paragraph and answer the questions given below:

Metals are extracted from their ores by a wide variety of techniques. The most common ores are oxides. Most metals are obtained by direct treatment of theirs ores with chemical agents, but the extraction of certain others require electrolysis. An example of the former type of process is the extraction of iron from its oxide, described by the following equation:

$$Fe_2O_3 + 3C \longrightarrow 2Fe + 3CO$$

The relative ease of extraction of a metal from its oxide can be estimated using the Ellingham diagram, as shown in the figure.

The diagram plots the free energies of formation of various oxides per mole of consumed oxygen as a function of absolute temperature.



Prob.5. All of the following reagents could be used to extract lead from its oxide except
(A) Molecular fluorine
(B) Molecular boron
(C) Molecular Nitrogen
(D) Elemental carbon

Sol. **(A)**

Prob.6. Which of the following best explains why the free energy of formation of Fe_2O_3 becomes less negative as the temperature is increased:
(A) The free energy of formation is independent of the absolute temperature.
(B) Entropy drops as a result of the consumption of oxygen.
(C) At low temperature, the free energy of formation becomes less dependent on the enthalpy of formation.
(D) As entropy increases, the free energy of formation increases.

Sol. **(B)**

Prob.7. When ferric oxide is reduced to obtain iron metal, CO_2 or CO can be produced. The production of CO_2
(A) is directly proportional to the temperature.
(B) is inversely proportional to the temperature.
(C) is independent of the temperature.
(D) requires cooling of the CO.

Sol. **C**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** In aluminothermic process, Al reduction increases the heat of the reaction.
and
STATEMENT-2: During reduction, Al_2O_3 is formed which is an exothermic process.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(B)**

2. **STATEMENT-1:** Sulphide ores are concentrated by froth floatation process.
and
STATEMENT-2: Pine oil acts as a frothing agent in froth floatation process.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(B)**

3. **STATEMENT-1:** Platinum and gold occur in native state in nature.
and
STATEMENT-2: Platinum and gold are noble metals.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

4. **STATEMENT-1:** Levitation is used for the separation of oxide ores from impurities.
and
STATEMENT-2: Ore particles are removed by washing in a current of water.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(C)**

5. **STATEMENT-1:** Pine oil is fused as a foaming agent in froth floatation process.
and
STATEMENT-2: Adsorption principle is involved in froth floatation process.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

6. **STATEMENT-1:** $CuSO_4$ acts as activator in froth flotation process.
and
STATEMENT-2: Activates the frothing property of sulphide ore.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

7. **STATEMENT-1:** Leaching is the process of converting the ore into oxide and then reducing it.
and
STATEMENT-2: Leaching is done in a blast furnace.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(D)**

8. **STATEMENT-1:** In smelting, roasted ore is heated with powdered coke in presence of a flux.
and
STATEMENT-2: Oxides are reduced to metals by C or CO. Impurities are removed as slag.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

MATRIX-MATCH TYPE QUESTIONS

1. Match the extraction processes listed in Column I with metals listed in Column II:

Column I		Column II	
(A)	Ca	(p)	Found as its native state
(B)	Zn	(q)	Found as its sulphide
(C)	Cr	(r)	Found as its carbonate
(D)	Ag	(s)	Found as its oxide

Sol. **(A → r; B → q,r,s; C → r; D → p,q)**

2. Match **Column**-I with **Column**-II:

Column I (Metal)		Column II (Procedure of extraction)	
(A)	Iron	(p)	Carbon reduction method
(B)	Lead	(q)	Self-reduction
(C)	Copper	(r)	Thermite process
(D)	Chromium	(s)	Hydrometallurgical process

Sol. **(A→ p; B→ p, q; C→ q, s; D→ r)**

3. Match the following metals given in **Column I** with the appropriate metal extraction process(es) listed below in **Column II**.

	Column – I		Column – II
(A)	Silver	(p)	Fused salt electrolysis
(B)	Lead	(q)	Cyanide process
(C)	Iron	(r)	Carbon monoxide reduction
(D)	Magnesium	(s)	Self reduction
		(t)	Blast furnace is used

Sol. **(A → q); B → s; C → r, t; D → p)**

Silver→ Ore is dissolved by complex formation using CN^- in presence of air followed by precipitation with zinc.

Lead → Involves roasting of PbS followed by self reduction.

Iron → Involves calcination followed by carbonmonoxide reduction.

Magnesium → Involves calcination (of $MgCl_2.6H_2O$ in presence of dry HCl gas) followed by fused salt electrolysis.

4. Match the following metals given in column – (I) with the appropriate metal extraction process listed below in column – (II)

	Column – I		Column – II
(A)	Fe	(p)	Cyanide process
(B)	Ag	(q)	Leaching
(C)	Pb	(r)	Carbon reduction
(D)	Cu	(s)	Self reduction
		(t)	Electrolysis

Sol. **(A → (r) B → (p, q, t) C → (r, s, t) D→ (r, s, t))**

Answer 5, 6 and 7 by appropriately matching the information given in the three columns of the following table.

	Column – 1 Catalyst		Column – 2 Name		Column – 3 Function
(I)	$TiCl_4$	(i)	Fenton's reagent	(P)	Oxidation of alcohol
(II)	$FeSO_4/H_2O_2$	(ii)	Ziegler-Natta	(Q)	Polythene preparation
(III)	Pd/C	(iii)	Lindlar	(R)	Hydrogenation of alkynes to give cis-alkene
(IV)	Pt/PtO	(iv)	Adam	(S)	Synthesis of CH_3OH

5. Which of the following combination is incorrect ?

[A] (I) (ii) (Q) [B] (II) (i) (P) [C] (III) (iv) (R) [D] (IV) (iv) (S)

6. Which combination will convert ethyene to ethane ?

[A] (III) (iii) (R) [B] (I) (i) (Q) [C] (IV) (iv) (S) [D] (II) (ii) (P)

7. Which combination will oxidize ethanol ?

[A] (I) (ii) (Q) [B] (II) (i) (P) [C] (III) (iii) (R) [D] (IV) (iv) (S)

Sol. 5. **C**

6. **A**

7. **B**

SINGLE INTEGER ANSWER TYPE

1. Of the following reduction processes:
(i) $Fe_2O_3 + C \rightarrow Fe$
(ii) $ZnO + C \rightarrow Zn$
(iii) $Ca_3(PO_4)_2 + C \rightarrow P$
(iv) $PbO + C \rightarrow Pb$
Correct process are:
(1) i, ii, iii & iv
(2) i, ii & iv
(3) i, ii & iii
(4) ii & iv

Sol. **(1)**

2. Sea water contains 1272 g of Mg^{2+} per metric ton (1 mega gram). How much of slaked lime must be added to 1.0 metric ton of sea water to precipitate all the Mg^{2+} ion. (give approx value in kg).

Sol. **4** kg (approx)

$Mg^{2+} + Ca(OH)_2 \downarrow + Ca^{2+}$

74 gm

In sea water

24 gm

24 gm of $Mg^{2+} \equiv 74$ g $Ca(OH)_2$

$\therefore 1272 \text{ g } Mg^{2+} \equiv \frac{74}{24} \times 1272 = 3922 \text{g}.$

$= 3.922 \text{ kg} \approx 4 \text{ kg of } Ca(OH)_2$

3. How many mL of 0.5 M H_2SO_4 are needed to dissolve 0.5 of Copper (II) carbonate?

Sol. **8.09 mL** $\approx$ **8** mL

$$CuCO_3 + H_2SO_4 \longrightarrow CuSO_4 + H_2O + CO_2$$

1 mol 1mol

Let of volume of 0.5 M H_2SO_4 be V

Moles of $CuCO_3 \equiv$ Moles of H_2SO_4

$$\frac{0.5}{123.5} = \frac{0.5 \times V}{1000}$$

$\therefore V = 8.09 \text{ mL} \approx 8 \text{ mL}$

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Balance the following reaction.

$$Al_2O_3 \cdot 2H_2O + xNaOH + H_2O \longrightarrow xNa[Al(OH)_4]$$

What is value of $\frac{x}{5}$?

Sol. **0.40**

Prob. 2. $Ag_2S + \underset{\text{Excess}}{KCN} \longrightarrow [Ag(CN)_x]^{-n}$

$$[Ag(CN)_x]^{-n} + Zn \longrightarrow [Zn(CN)_y]^{-m}$$

Find the value of $\frac{x+y}{8}$

Sol. **0.75**

CONCEPTS OF ACIDS & BASES 25

ARRHENIUS CONCEPT

Acid: Any hydrogen containing compound which gives H^+ ions in aqueous solution, is called acid, e.g. HCl, HNO_3, H_2SO_4, CH_3COOH etc.
Base: A substance which gives OH^- ions in aqueous solution, is called base, e.g. NaOH, KOH, $Ca(OH)_2$ etc.

BRONSTED–LOWRY CONCEPTS

Acid: Proton donor

(a) Molecule : $HCl \longrightarrow H^+ + Cl^-$
(b) Cations : $[Al(H_2O)_6]^{3+} \longrightarrow H^+ + [Al(H_2O_5)(OH)]^{2+}$
(c) Anions : $HCO_3^- \longrightarrow H^+ + CO_3^{2-}$

Base: Proton acceptor

(a) Molecules : $\longrightarrow Py + H^+ \longrightarrow [PyH]^+$
(b) Cations : $[Al(H_2O)_5(OH)]^{2+} + H^+ \longrightarrow [Al(H_2O)_6]^{3+}$
(c) Anions: $HCO_3^- + H^+ \longrightarrow H_2CO_3$

Conjugate acid–base pairs:

$$\underset{\text{Acid 1 – Base 1}}{HCl} + H_2O \rightleftharpoons \underset{\text{Acid 2}}{H_3O^+} + \underset{\text{Base 2}}{Cl^-}$$

Acid 1 – Base 2 : Conjugate acid – base pair
Acid 2 – Base 1 : Conjugate acid – base pair
[**Rule:** More stronger is an acid, the weaker must be its conjugate base.]

LEWIS CONCEPT

According to this concept, an acid is a species which can accept a pair of electrons to form a covalent bond and a base is a species which can donate a pair of electrons to form a covalent bond.
According to Lewis concept, the following species can acts as Lewis acids,

(a) Molecules in which the central atom has incomplete octet. Lewis acids are electron deficient molecules such as BF_3, $AlCl_3$, $GaCl_3$ etc.

$$H_3N: + AlCl_3 \longrightarrow H_3N \rightarrow AlCl_3$$

(b) Molecules in which the central atom has empty d–orbitals can accept an electron pair and act as Lewis acids. The central atom of the halides such as SiX_4, $SnCl_4$, PX_3, PF_5, SF_4, SeF_4, $TeCl_4$ etc., have vacant d–orbitals.

$$\underset{\text{Lewis acid}}{SiF_4} + \underset{\text{Lewis base}}{2\ddot{\underset{\cdot\cdot}{F}}:} \longrightarrow \underset{\text{Complex}}{(SiF_6)^{2-}}$$

(c) Simple cations: All cations are expected to act as Lewis acids since they are electron deficient in nature such as Na^+, K^+, Ca^{2+} etc.

The following species can act as Lewis bases,

(a) Neutral species having at least one lone pair of electrons, e.g. ammonia, amines, alcohols etc., act as Lewis bases as they contain a pair of electrons.

$$NH_3,\ R-\ddot{N}H_2,\ R-\ddot{\underset{\cdot\cdot}{O}}-H,\ H-\ddot{\underset{\cdot\cdot}{O}}-H,\ R-\ddot{\underset{\cdot\cdot}{O}}-R$$

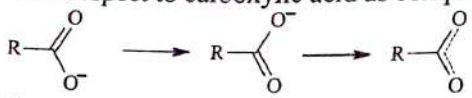
(b) Negatively charged species or simple anions, e.g. chloride, cyanide, hydroxide ions etc., act as Lewis bases CN^-, Cl^-, OH^-, F^-.

ORIGIN OF ACIDITY AND BASICITY IN ORGANIC COMPOUNDS

(a) Among hydrocarbons, acidity increases with the % age of s-character. This is because higher the s-character, closer will be the electrons to the nucleus of carbon and the electrons further moves away from H and it can be easily removed as H^+. Therefore, the increasing order of acidity is, $CH_3 - CH_3 < CH_2 = CH_2 < CH \equiv CH$.

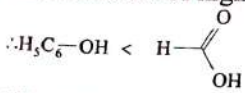
(b) Alcohols are more acidic than alkanes because oxygen is more electronegative than carbon and consequently O – H bond will break easily to give H^+ ions than C – H bond.

(c) Carboxylic acids are more acidic than alcohols because carboxylate anion is more resonance stabilized with respect to carboxylic acid as compared to alkoxide anion with respect to alcohol.



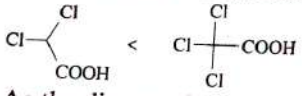
(d) Phenols are more acidic than alcohols because phenoxide ion is more resonance stabilized as compared to alkoxide ion.

(e) Phenols are less acidic than carboxylic acids. This is because in phenoxide ion some of the resonance structures are of higher energy and this makes phenoxide ion not very stable.

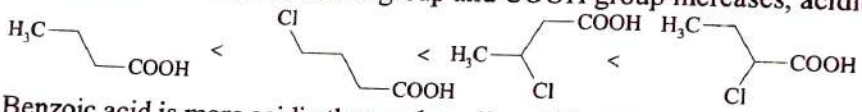


(f) $CH_3COOH > CH_3CH_2COOH$ (greater the number of CH_3 added, less is the acidity due to +I effect of CH_3).

(g) Increase in electron withdrawing substituents in simple aliphatic acids increases acidity.
$CH_3COOH < I–CH_2COOH < Br–CH_2COOH < F–CH_2COH$



(h) As the distance between chloro group and COOH group increases, acidity decreases.



(i) Benzoic acid is more acidic than carboxylic acid because benzoate anion is more resonance stabilized as compared to carboxylate anion.

(j) Aniline is less basic than NH_3. This is because in aniline the nitrogen atom is bonded to an sp^2 hybridized carbon atom but the unshared electron pair on nitrogen interacts with the delocalized orbital.

(k) Introduction of an alkyl group on the nitrogen of aniline results in small increase in pK_a value due to +I effect of alkyl groups. Therefore, $C_6H_5NH_2 < C_6H_5NHMe < C_6H_5NMe_2$.

(l) Introduction of phenyl groups on N lowers basicity because the substituted amine becomes more stable than the ion. Therefore, $Ph_2NH < PhNH_2$.

THE LEVELLING EFFECT

The apparent strength of a protonic acid is dependent on the solvent in which the acid is dissolved. When all the acids which are stronger than H_3O^+, i.e. $HClO_4$, H_2SO_4, HCl and HNO_3 are added to H_2O, they appear to have equal strength, since all these acids are levelled to the strength of H_3O^+.

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

Acid Base Acid Base

↓ (solvent)

Any of the 4 acids mentioned

All the four acids mentioned above react almost completely with water to form hydronium ion, H_3O^+. Therefore, all strong acids in aqueous solution appear almost equally strong and hence their relative strengths in aqueous solution cannot be compared. This phenomenon is called 'levelling effect' and such solvent (water) in this case is called 'levelling solvent'.

The levelling effect is also observed in the case of base, for example

(a) All strong bases like NaH, $NaNH_2$, C_2H_5ONa etc., are of equal strength in water.

(b) Aniline is a very weak base and NaOH is very strong base in water but they appear to be equally strong when acetic acid is used as a solvent (Levelling effect).

Periodic variations of acidic properties and basic properties:

(i) Hydracids of the elements of the same periods, e.g., CH_4, NH_3, H_2O and HF
Order of acidic strength: $CH_4 < NH_3 < H_2O < HF$
Reason: Stability of their conjugate bases, $CH_3^- < NH_2^- < OH^- < F^-$

(ii) Hydracids of the elements of same group, e.g., NH_3, PH_3, AsH_3, SbH_3 and BiH_3: show basic character
Order of basic strength: $NH_3 > PH_3 > AsH_3 > SbH_3 > BiH_3$
Reason: Decrease in the availability of ion pair
HF, HCl, HBr and HI: show acidic character
Order of acidic strength: $HF < HCl < HBr < HI$
Reason: Decrease in bond energy from HF to HI

(iii) **Oxyacids:**

(a) The acidic character of oxyacids of the same element, which is in different oxidation states, increases with the increase of its oxidation state.

$$H\overset{+1}{O}Cl < HClO_2^{+3} < HClO_3^{+5} < HClO_4^{+7}$$

$$H_2\overset{+4}{S}O_3 < H_2\overset{+6}{S}O_4$$

$$H\overset{+3}{N}O_2 < H\overset{+5}{N}O_3$$

(b) The acidic character of oxyacids of different elements which are in the same oxidation state decreases with the increase in the atomic number of the central atom.

$$H\overset{+3}{O}Cl < H\overset{+3}{O}Br > H\overset{+3}{O}I$$

$$H\overset{+7}{Cl}O_4 > H\overset{+7}{I}O_4$$

$$H_2\overset{+4}{S}O_3 > H_2\overset{+4}{Se}O_3$$

Exception to the rule: H_3PO_2, H_3PO_3 and H_3PO_4
Expected order: $H_3PO_2 < H_3PO_3 < H_3PO_4$, Actual order: $H_3PO_2 >> H_3PO_3 > H_3PO_4$

HARD AND SOFT ACIDS AND BASES

A qualitative correlation between the various Lewis acids and bases has been achieved. This is done by classifying the acids or bases into two classes, i.e. hard and soft. Acid-base reactions are then treated as per general principle that hard acids prefer to combine with hard bases and soft acids prefer to combine with soft bases. The criterion of hardness (or softness) is ascribed to the hardness of the electron cloud associated with any species. A firmly held electron cloud with low polarizability makes a species hard while an easily polarizable electron cloud characterizes the species as soft. A third category with intermediate character appears in the border line.

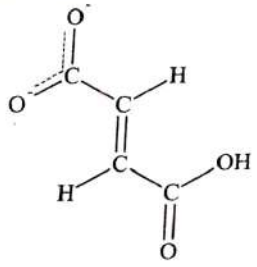
	Hard acids	Soft acids
1.	Small size, high electronegativity, low polarizability.	Large size, low electronegativity, high polarizability.
2.	High positive oxidation state, absence of any outer electron which are easily excited to higher states.	Zero or low positive oxidation state, several easily excitable valence electrons.
3.	H^+, Li^+, Na^+, K^+, Be^{2+}, Mg^{2+}, Ca^{2+}, Sr^{2+}, Al^{3+}, Fe^{3+}, Co^{3+}, Cr^{3+}, CO_2, SO_3. HX (hydrogen bonding molecules)	Cu^+, Ag^+, Hg^+, Cd^{2+}, Pt^{2+}, H^+, Pt^{4+}, I_2, Br_2, Br^+, Cl, Br, I, N M^0 (metal atoms) and bulk atoms.

	Hard Bases	Soft Bases
1.	High electronegativity, low polarizability.	Low electronegativity, high polarizability.
2.	Presence of filled orbitals, empty orbitals may exist at high energy level.	Partly filled orbitals, empty orbitals are low lying.
3.	H_2O, OH^-, F^-, Cl^-, CH_3COO^-, PO_4^{3-}, SO_4^{2-}, NO_3^-, CO_3^{2-}, ClO_4^-, NH_3, N_2H_4, RNH_2	H^-, I^-, SCN^-, CN^-, $S_2O_3^{2-}$, R_2S, RSH, RS^-, R_3P, R_3As

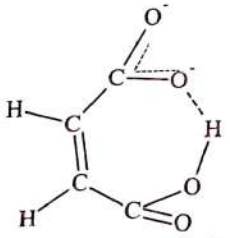
ILLUSTRATIVE PROBLEMS

1. On the basis of H-bonding explain that the second ionization constant K_2 for fumaric acid is greater than for maleic acid.

Sol. We know that H-bonding involving acidic H has an acid weakening effect and H-bonding in conjugate base has an acid strengthening effect.
Both dicarboxylic acids have two ionisable H-atoms considering second ionization step.



Fumarate monoanion (No H-bonding)



(Maleate monoanion, H-bonding)

2. Compare the basic strengths of ortho, meta and para isomers of nitroaniline and predict the decreasing order of basicity.

Sol. The $-NO_2$ group which is a powerful electron withdrawing inductive effect when present at ortho and para positions for the interaction of the unshared pair of the amino nitrogen with the delocalised p-orbital system of the benzene nucleus is then enhanced. The neutral molecule is thus stabilized even further with respect to the cation, resulting in further weakening as a base. Thus the order of basicity of nitroanilines is as

$m\text{-}NO_2C_6H_4NH_2 > p\text{-}NO_2C_6H_4NH_2 > o\text{-}NO_2C_6H_4NH_2$.

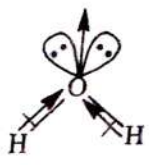
3. Why lithium salts have a greater degree of covalent character than other halides of the group?

Sol. The small size of Li^+ gives it a large polarizing power, hence covalent nature of its compound increases. It can be viewed in terms of Fajan's rule.

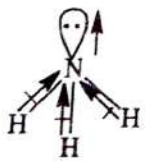
(a) Smaller cation
(b) Larger anion
(c) Greater is charge density as the surface causes greater polarizing power of Li^+.
Hence, greater covalent nature.

4. Which of the following molecules are polar: H_2O, NH_3, SO_2, $CHCl_3$, BF_3?

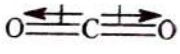
Sol.



(Polar)
$\mu = 1.84$ D

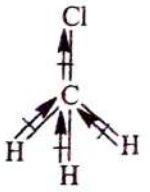


(Polar)
$\mu = 1.86$ D

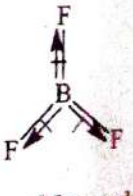


(Non-polar)
$\mu = 0$

(Polar)
$\mu = 1.60$ D



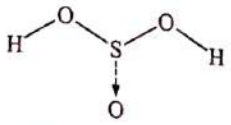
(Polar)
$\mu = 1.86$ D



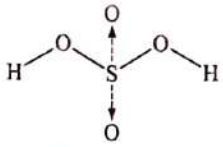
(Non-polar)
$\mu = 0$

5. Why sulphuric acid is stronger than sulphurous acid?

Sol. Sulphuric acid is stronger than sulphurous acid since the sulphur atom in sulphuric acid has more co–ordinated oxygen atoms.



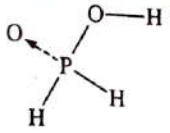
Sulphurous acid

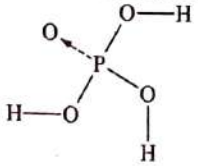


Sulphuric acid

6. In hypophosphorus acid P is in +1 state (H_3PO_2) and in phosphoric acid (H_3PO_4) P is in +5 state but still H_3PO_2 is stronger than H_3PO_4. Explain.

Sol. A consideration of the structures of these acids explains the observed trend.





The hydrogen atoms bonded directly to phosphorus are not acidic as the P – H bond is not polar enough due to equal electronegativity of phosphours and hydrogen. Acidity arises due to O – H groups, one in H_3PO_2 and three in H_3PO_4. The inductive effect of the single oxygen atom acts in the single – O – H bond in H_3PO_2 but it is distributed over three – O – H bonds in H_3PO_4 respectively. So O – H bond in H_3PO_2 much easily broken, explaining its higher acidity over H_3PO_2.

SOLVED PROBLEMS

Subjective Question

Prob. 1. Why $CHCl_3$ is more acidic than CHF_3 ?

Sol. $Cl_3C^{(-)}$ is less basic than $F_3C^{(-)}$; because fluorine can disperse charge only by –I effect while Cl by –I and $p\pi-d\pi$ back bonding

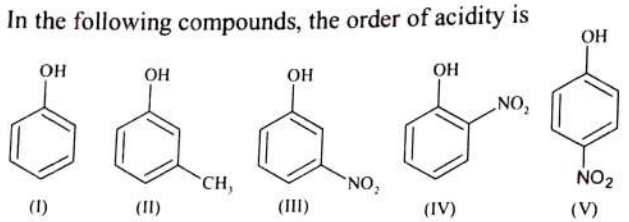
Objective Questions

Prob. 1. Which salt undergoes hydrolysis?

(A) KCl
(B) $NaNO_3$
(C) $CaSO_4$
(D) $HCOONH_4$

Sol. Salts of strong acids and strong bases do not hydrolyse. Thus only $HCOONH_4$ is hydrolyzed.
Hence, **(D)** is the correct answer.

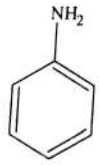
Prob. 2. In the following compounds, the order of acidity is

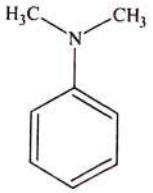


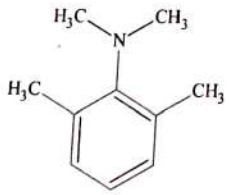
(A) III > IV > I > II > V
(B) I > IV > III > I > V
(C) V > II > I > III > IV
(D) V > IV > III > I > II

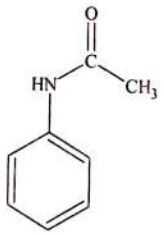
Sol. +I effect decreases acidity, –I and –R effect increases acidity. NO_2 group cannot exert –R effect from the meta position. Ortho-nitro phenol has intramolecular H-bonding.
Hence **(D)** is the correct answer.

Prob. 3. Which is most basic among the following?

(A) 

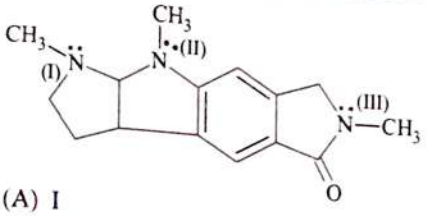
(B) 

(C) 

(D) 

Sol. Due to steric effect of 2 Me–groups present on nitrogen atom Lp on N will become out of plane of π electron. i.e. more available
Hence, **(C)** is the correct answer.

Prob. 4. Which nitrogen in the following compound has least pK_b value?



(A) I
(B) II
(C) III
(D) All have equal pK_b value

Sol. Lone pair on N (II and III) are involved in conjugation and hence less basic than (I) in which lone pair is concentrated on N atom. Hence, (I) 'N' is most basic, i.e. least pK_b value.
Hence, **(A)** is the correct answer.

Prob. 5. From the following reactions:

$$HC \equiv CH + LiNH_2 \rightleftharpoons NH_3 + HC \equiv CLi$$

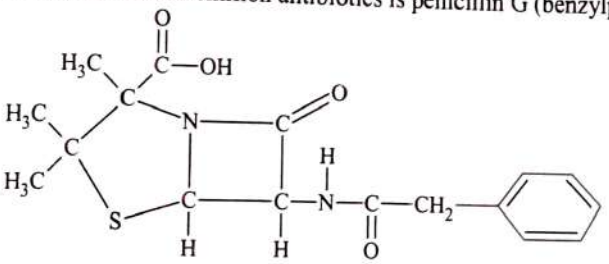
$$NH_3 + R^- \rightleftharpoons NH_2^- + RH$$

Predict which of the following order regarding acid strength is correct?
(A) $RH < NH_3 < HC \equiv CH$
(B) $RH > NH_3 > HC \equiv CH$
(C) $RH > NH_3 < HC \equiv CH$
(D) $RH < NH_3 > HC \equiv CH$

Sol. On the basis of equilibriums, order of basicity,
$R^- > NH_2^- > HC \equiv C^-$
And order of acidity is
$RH < NH_3 < HC \equiv CH$
Hence, **(A)** is the correct answer.

COMPREHENSION

One of the most common antibiotics is penicillin G (benzylpenicillinic acid), which has the following structure:



It is a weak monoprotic acid:

$HP \rightleftharpoons H^+ + P^-$ $\qquad K_a = 1.64 \times 10^{-3}$

where HP denotes the parent acid and P^- the conjugate base. Penicillin G is produced by growing molds in fermentation tanks at 25°C and pH range 4.5 to 5.0. The crude form of this antibiotic is obtained by extracting the fermentation both with an organic solvent in which the acid is soluble. In one stage of purification, the organic extract of the crude penicillin G is treated with buffer solution at pH = 6.50

1. What is the ratio of the conjugate base of penicillin G to the acid at this pH?
(A) 5.2×10^3
(B) 3.2×10^6
(C) 1.8×10^3
(D) 5.6×10^4

Sol. **(A)**

2. Penicillin G is not suitable for oral administration, but the sodium salt (NaP) is because it is soluble. Calculate the pH of a 0.12 M NaP solution formed when a tablet containing the salt is dissolved in a glass of water:
(A) 6.07
(B) 7.93
(C) 0.44
(D) 13.56

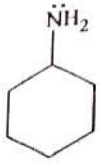
Sol. **(B)**

3. Which hydrogen is most acidic?
(A) H attached to N
(B) H attached to C
(C) H attached to O
(D) All H are acidic

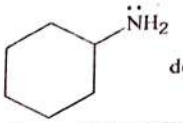
Sol. **(C)**

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** is less basic than 

and

STATEMENT-2: is lonepair in resonance whereas  doesn't

(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

2. **STATEMENT-1:** Conjugate acid of H_2O is H_3O^+.
and
STATEMENT-2: Stronger the conjugate acid weaker the conjugate base.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

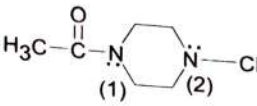
Sol. **(B)**

3. **STATEMENT-1:** Solution of 10^{-4} M CH_3COOH and 0.1 M CH_3COONa is acidic in nature, $pKa = 5$
and
STATEMENT-2: acetic acid is a weak acid
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(B)**

4. **STATEMENT-1:** Phenol is more acidic than water
and
STATEMENT-2: phenoxide ion is resonance stabilized
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

5. **STATEMENT-1:**  nitrogen – 2 is more basic than nitrogen – 1.
and
STATEMENT-2: Conjugate acid of 1-nitrogen is stronger than conjugate acid of nitrogen – 2.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

6. **STATEMENT-1:** Ice floats on water.
and
STATEMENT-2: Density of ice is less than water
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(A)**

MATRIX-MATCH THE FOLLOWING QUESTIONS

1. Match the following:

Column – I		Column – II	
(A)	OH^-	(p)	Conjugate acid of O^{2-}
(B)	$HClO_4$	(q)	Conjugate acid of H_2O
(C)	H_3O^+	(r)	Conjugate base of H_2O
(D)	ClO_4^- is weakest conjugate base of	(s)	Strongest bronsted acid
		(t)	Weakest bronsted acid

Sol. **(A) → (p, r); (B) → (s); (C) → (q); (D) → (s)**

Answer 2, 3 and 4 by appropriately matching the information given in the three columns of the following table.

Column – 1 Acid		Column – 2 pKa		Column – 3 O. S	
(I)	$HClO_4$	(i)	7.46	(P)	+1
(II)	$HClO_3$	(ii)	–10	(Q)	+3
(III)	$HClO_2$	(iii)	1.96	(R)	+5
(IV)	HClO	(iv)	–1	(S)	+7

2. Which of the following combination is correct ?
[A] (I) (ii) (S) [B] (II) (ii) (P) [C] (III) (iv) (P) [D] (IV) (i) (R)

3. Which of the following combination is incorrect ?
[A] (IV) (i) (P) [B] (II) (ii) (R) [C] (III) (iii) (Q) [D] (I) (ii) (S)

4. Order of acidity of compounds of column (I) ?
[A] I > III > II > IV [B] II > I > IV > III [C] I > II > III > IV [D] III > II > I > IV

Sol.
2. **A**
3. **B**
4. **C**

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. 100 mL of 0.2 N HCl is added to 100 mL of 0.18 N NaOH and whole volume is made one litre. What will be the pH of resulting solution.

Sol. **(2)**

2. If the equilibrium constant of the reaction of weak acid HA with strong base is 10^9 then what is the pH?

Sol. **(9)**

3. The % error in $[H^+]$ made by neglecting the ionization of water in 10^{-6} M NaOH is

Sol. **(1)**

4. Total number of diprotic acids among the following is :
H_3PO_4, H_2SO_4, H_3PO_3, H_2CO_3, $H_2S_2O_7$, H_3BO_3, H_3PO_2, H_2CrO_4, H_2SO_4.

Sol. **(6)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. In the titration of a solution of a weak acid HA and NaOH, the pH is 5 after 10 ml of NaOH solution has been added and 5.6 after 20 ml NaOH is added. What is the value of pK_a of HA ?

Sol. **5.30**

Prob. 2. The pH of a solution containing 0.4 M HCO_3^- and 0.2 M CO_3^{2-} is
$[K_{a_1}(H_2CO_3) = 4\times10^{-7},\ K_{a_2}(HCO_3^-) = 4\times10^{-11}]$

Sol. **10.10**

QUALITATIVE ANALYSIS 26

Qualitative inorganic analysis deals with the detection and identification of cations and anions in the salt and mixture of salt.

PROPERTIES OF INORGANIC SALTS

(i) The salts of some transition elements have characteristic colours.

Cations	Colour of salts
Cu^{2+} and Ni^{2+}	Bluish green
Fe^{2+}	Light green
Cr^{3+}	Dark green
Fe^{3+}	Yellowish brown
Co^{2+}	Pink violet
Mn^{2+}	Light pink

(ii) Some salts impart characteristic colours to the flame

Cations	Colour of flame
Ca^{2+}	Brick red
Sr^{2+}	Crimson red
Ba^{2+}	Grassy green
Cu^{2+}	Blue
K^{+}	Pale violet
Na^{+}	Golden yellow
Zn^{2+}, Mn^{2+}	Green flashes

(iii) Salts like $Zn(NO_3)_2$, $MgCl_2$, $ZnCl_2$ absorb moisture and have paste like appearance, are deliquescent.

(iv) Gas evolved by heating the salts

Salt	Gas	Odour/colour	Example
Carbonates & bicarbonates	CO_2	Colourless and odourless	$CaCO_3 \longrightarrow CaO + CO_2\uparrow$
Sulphites	SO_2	Suffocating odour and colourless	$ZnSO_3 \longrightarrow ZnO + SO_2\uparrow$
Hydrated chlorides	HCl	Pungent odour and colourless	$ZnCl_2.6H_2O \longrightarrow Zn(OH)Cl + HCl\uparrow + 5H_2O$
Hydrated sulphides	H_2S	Rotten egg odour and colourless	$BaS.2H_2O \longrightarrow Ba(OH)_2 + H_2S\uparrow$
Nitrates	NO_2	Reddish brown	$2Pb(NO_3)_2 \longrightarrow 2PbO + 2NO_2\uparrow + O_2\uparrow$
Oxalates	$CO + CO_2$	Colourless and odourless	$CaC_2O_4 \longrightarrow CaO + CO\uparrow + CO_2\uparrow$

(v) Residues:

(a) $ZnCO_3 \longrightarrow CO_2 \uparrow + ZnO$ (Residue) → Yellow when hot / White when cold

(b) $2Pb(NO_3)_2 \longrightarrow 2NO_2 \uparrow + O_2 \uparrow + 2PbO \longrightarrow$ Yellow (Residue)

(c) $CuSO_4.5H_2O \longrightarrow 5H_2O + CuSO_4 \longrightarrow$ White

Salts like $Pb(NO_3)_2$, $Ba(NO_3)_2$ decrepitate on heating and the salts such as alums, phosphates, borates swell up.

ACID RADICALS

Radicals	Reagents
CO_3^{2-}, SO_3^{2-}, S^{2-}, NO_2^-, $S_2O_3^{2-}$	dil. HCl/ dil. H_2SO_4
Cl^-, Br^-, I^-, $C_2O_4^{2-}$, NO_3^-	conc. H_2SO_4
SO_4^{2-}	S.E. after neutralization + dil. HCl + $BaCl_2$ solution
PO_4^{3-}	Salt mixture + conc. HNO_3 + ammonium molybdate
BO_3^{3-}	BO_3^{3-} Salt mixture + conc. H_2SO_4 + alcohol

Basic radicals:

Group	Cations	Group Reagent
I	Pb^{2+}, Hg_2^{2+}, Ag^+	dil. HCl
II A / II B	Hg^{2+}, Pb^{2+}, Bi^{3+}, Cu^{2+}, Cd^{2+} / As^{3+}, Sb^{3+}, Sn^{2+}	H_2S in acidic medium
III	Cr^{3+}, Fe^{3+}, Al^{3+}	NH_4Cl/NH_4OH
IV	Co^{2+}, Ni^{2+}, Mn^{2+}, Zn^{2+}	H_2S in alkaline medium
V	Ba^{2+}, Sr^{2+}, Ca^{2+}	$(NH_4)_2CO_3/NH_4OH$ and NH_4Cl
VI	Na^+, K^+, Mg^{2+}	No common group reagent
Zero	NH_4^+	Dil. NaOH

IMPORTANT FACTS

(i) White ppt. of $PbCl_2$ is insoluble in cold water but dissolves in hot water. AgCl and Hg_2Cl_2 remain insoluble in hot water.

(ii) All the sulphides of IIB are soluble in yellow ammonium sulphide but those of IIA do not dissolve.

(iii) Except HgS all the sulphides of group IIA are soluble in 50% HNO_3.

(iv) Copper and cadmium are separated with the help of KCN.

(v) In the filtrate of group II, iron is always present in Fe^{2+} state due to the reduction of Fe^{3+} ion to Fe^{2+} ion by H_2S.

(vi) Ammonium sulphate cannot be used in place of NH_4Cl in third group because the SO_4^{2-} ions will precipitate Ba^{2+}, Sr^{2+} (if present in the mixture) as $BaSO_4$ and $SrSO_4$.

(vii) Excess of ammonium chloride should be avoided in group V, otherwise concentration of CO_3^{2-} ion will not be sufficient enough to precipitate the cations of group V completely.

(viii) A dry test tube is used for performing chromyl chloride test because water hydrolyses CrO_2Cl_2.

$$CrO_2Cl_2 + 2H_2O \longrightarrow 2HCrO_4 + 2HCl$$

(ix) Salts of arsenic, tin, lead, bismuth tend to corrode the platinum wire, hence flame test is avoided in case of such salts.

(x) Chlorides of mercury do not respond to chromyl chloride test because of their ionization.

(xi) Mercury iodide on dry heating vapourizes and forms a yellow sublimate.

(xii) Salts of Hg^{2+}, Pb^{2+}, Ba^{2+} are heavy whereas salts containing carbonates are generally light and fluffy.

(xiii) If the salt mixture is insoluble in water, then anions of the given sample are tested with the help of sodium carbonate extract.

(xiv) All Na^+, K^+ and NH_4^+ compounds are soluble in water.

(xv) All carbonates except first group elements and $(NH_4)_2CO_3$ are insoluble in water.

(xvi) All sulphites, except those of first group and $(NH_4)_2SO_3$ are insoluble in water.

(xvii) The transparent bead in borax bead test is made of $NaBO_2 + B_2O_3$.

(xviii) Microcosmic salt, $Na(NH_4)HPO_4.4H_2O$, can also be used for detecting coloured ions like borax (microcosmic bead test). The salt on heating gives a transparent glassy bead of metaphosphate which forms coloured beads of orthophosphate when heated with coloured salts.

(xix) Sulphite reacts with dil. H_2SO_4 producing SO_2 gas which turns acidified $K_2Cr_2O_7$ solution green.

$$K_2Cr_2O_7 + 3SO_2 + H_2SO_4 \longrightarrow K_2SO_4 + \underset{\text{Green}}{Cr_2(SO_4)_3} + H_2O$$

(xx) Sulphite reacts with $BaCl_2$ solution to give white ppt. of barium sulphite which is soluble in dil. HCl whereas soluble sulphate reacts with $BaCl_2$ solution to give white ppt. of $BaSO_4$ which is insoluble in all the acids.

(xxi) Nitrite reacts with dil. H_2SO_4 to give brown gas (NO_2) which turns $FeSO_4$ black due to $FeSO_4.NO$. On the other hand, nitrate gives a brown ring when reacts with conc. H_2SO_4 in presence of $FeSO_4$ due to the formation of $FeSO_4.NO$.

(xxii) Both CO_2 and SO_2 turn lime water milky.

$$Ca(OH)_2 + CO_2 \longrightarrow \underset{\text{Milkiness}}{CaCO_3\downarrow} + H_2O$$

$$Ca(OH)_2 + SO_2 \longrightarrow \underset{\text{Milkiness}}{CaSO_3\downarrow} + H_2O$$

(xxiii)

Group	Group Reagent	Cations	Precipitated as
I	dilute HCl	Hg_2^{2+}, Pb^{2+}, Ag^+	Chlorides
II	H_2S in the presence of dil. HCl	Pb^{2+}, Hg^{2+}, Cu^{2+}, Bi^{3+}, Cd^{2+}	Sulphides
III	NH_4OH in the presence of excess of NH_4Cl	Fe^{3+}, Al^{3+}, Cr^{3+}	Hydroxides
IV	H_2S in the presence of NH_4OH	Co^{2+}, Ni^{2+}, Zn^{2+}, Mn^{2+}	Sulphides
V	$(NH_4)_2CO_3$ in the presence of NH_4Cl and NH_4OH	Ba^{2+}, Sr^{2+}, Ca^{2+}	Carbonates
VI	Tested individually	Mg^{2+}, Na^+, K^+, NH_4^+	

(xxiv) Yellow ammonium sulphide is used for the separation of group IIA and IIB from each other. Group IIB elements (As, Sb and Sn) are soluble in yellow ammonium sulphide.

(xxv) Lead (Pb^{2+}) is precipitated in group I as well as group II because $PbCl_2$ is slightly soluble in cold water.

(xxvi) Copper salt gives chocolate coloured ppt. with potassium ferrocyanide solution.

$$2CuSO_4 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6] + 2K_2SO_4$$

(xxvii) KCN is used to analyse Cd, directly in presence of Cu. Both Cd and Cu form complexes with KCN.

$$K_3[Cu(CN)_4] \underset{}{\overset{\text{First ionization}}{\rightleftharpoons}} 3K^+ + [Cu(CN)_4]^{3-} \overset{\text{Second ionization}}{\rightleftharpoons} Cu^+ + 4CN^-$$

$$K_2[Cd(CN)_4] \rightleftharpoons 2K^+ + [Cd(CN)_4]^{2-} \rightleftharpoons Cd^{2+} + 4CN^-$$

It has been found that the second dissociation constant in the case of copper complex is negligible. Hence, Cu^{2+} ions concentration in the solution is so low that on passing H_2S gas, solubility product of CuS is never reached. On the other hand, in case of cadmium complex, the second dissociation constant is quite high and is sufficient to precipitate CdS only.

(xxviii) Fe^{3+} salt gives blue ppt. or colouration with potassium ferrocyanide.

$$4FeCl_3 + 3K_4[Fe(CN)_6] \longrightarrow \underset{\text{Prussian blue}}{Fe_4[Fe(CN)_6]_3} + 12KCl$$

(xxix) Acid radicals like $C_2O_4^{2-}$, BO_3^{3-} and PO_4^{3-} interfere in the systematic analysis of basic radicals after group II. Because in group III, the medium is made alkaline by adding NH_4OH, oxalates, borates and phosphates of the cations of group III, IV, V and Mg^{2+} form precipitates in the alkaline medium. However upto group II, the medium is highly acidic and oxalates, borates and phosphates of these cations are soluble.

ILLUSTRATIVE PROBLEMS

1. **(i) An inorganic iodide (A) on heating with a solution of KOH gives a gas (B) and the solution of a compound (C).**
(ii) The gas (B) on ignition in air gives a compound (D) and water.
(iii) Copper sulphate forms a black precipitate on passing (B) through its solution.
(iv) A precipitate of compound (E) is formed on reaction of (C) with copper sulphate solution.
Identify (A) to (E) and give chemical equations for reactions at steps (i) to (iv)

Sol. Gas (B) when passed through copper sulphate solution reduces the latter to copper metal; this indicates that the gas (B) is phosphine (PH_3). Thus, the inorganic iodide must be phosphonium iodide (PH_4I). The various reactions can be written as below.

(i) $PH_4I + KOH \longrightarrow \underset{(B)}{PH_3} + \underset{(C)}{KI} + H_2O$

(ii) $\underset{(B)}{2PH_3} + 4O_2 \longrightarrow \underset{(D)}{P_2O_5} + 3H_2O$

(iii) $3CuSO_4 + 2PH_3 \longrightarrow Cu_3P_2 + 3H_2SO_4$

(iv) $2CuSO_4 + 4KI \longrightarrow Cu_2I_2\downarrow + 2K_2SO_4 + I_2$

2. **A compound (X) of S, Cl and O has a vapour density of 67.5. It reacts with water to form two acids and reacts with KOH to form two salts (Y) and (Z). While (Y) gives white precipitate with $AgNO_3$ solution, (Z) gives white precipitate with $BaCl_2$ solution. Identify X.**

Sol. Reaction of (Y) indicates that it must be a chloride, Cl^-, while reaction of (Z) indicates it to be a sulphate, SO_4^{2-}. Since both of them are derived by the action of KOH on (X), containing S, Cl and O.

$\underset{X}{S,Cl}$ and O compound $\xrightarrow{KOH}$ $\underset{Y}{Cl^-} + \underset{Z}{SO_4^{2-}}$

(X) should be sulphuryl chloride, SO_2Cl_2 (Mol. wt. = 32 + 32 + 71 = 135) which coincides with its vapour density. All the given properties of the compound may be explained as below.

$\underset{X}{SO_2Cl_2} + 2H_2O \longrightarrow 2HCl + H_2SO_4$

$\underset{X}{SO_2Cl_2} + 4KOH \longrightarrow \underset{Y}{2KCl} + \underset{Z}{K_2SO_4}$

$\underset{Y}{KCl} + AgNO_3 \longrightarrow AgCl\downarrow + KNO_3$

$\underset{Z}{K_2SO_4} + BaCl_2 \longrightarrow BaSO_4\downarrow + 2KCl$

3. **The gas liberated on heating a mixture of two salts with NaOH gives a reddish brown precipitate with an alkaline solution of K_2HgI_4. The aqueous solution of the mixture on treatment with $BaCl_2$ gives a white precipitate which is sparingly soluble in conc. HCl. On heating the mixture with $K_2Cr_2O_7$ and conc. H_2SO_4, red vapours of A are produced. The aqueous solution of the mixture gives a deep blue colouration B with potassium ferricyanide solution. Identify the radicals in the given mixture and write the balanced equations for the formation of A and B.**

Sol. Let us summarize the given facts of the question.

Red vapours of (A) $\xleftarrow[\text{conc.}H_2SO_4,\ \text{heat}]{K_2Cr_2O_7}$ Mixture of two salts $\xrightarrow[NaOH]{\text{Heat with}}$ Gas $\xrightarrow{\text{Alk. } K_2HgI_4}$ Reddish brown ppt.

Deep blue colour, (B) $\xleftarrow{K_3[Fe(CN)_6]}$ Aq. solution of the mixture $\xrightarrow{BaCl_2}$ White ppt.

The given reactions lead to following conclusions.

(i) Heating of mixture with NaOH give NH_3 gas (indicated by reddish brown ppt. with alkaline solution of K_2HgI_4) which indicates the presence of NH_4^+ ion in the mixture.

(ii) Heating of mixture with $K_2Cr_2O_7$ and conc. H_2SO_4 give red vapours (of chromyl chloride) indicates the presence of Cl^- ion in the mixture.

(iii) Reaction of aqueous solution of the mixture with barium chloride solution to give white ppt. (of $BaSO_4$) sparingly soluble in conc. HCl indicates the presence of SO_4^{2-} ions in the mixture.

(iv) Reaction of aqueous solution of the mixture with potassium ferricyanide solution to give deep blue colour indicates the presence of Fe^{2+} ions in the mixture.

Hence, the mixture contains the following four ions. NH_4^+, Fe^{2+}, SO_4^{2-} and Cl^-

Equations for the formation of A and B:

$$4NaCl + K_2Cr_2O_7 + 3H_2SO_4 \xrightarrow{\text{heat}} K_2SO_4 + 2Na_2SO_4 + \underset{\substack{\text{Chromyl chloride}\\ \text{(orange)(A)}}}{2CrO_2Cl_2\uparrow} + 3H_2O$$

$$3Fe^{2+} + 2K_3[Fe(CN)_6] \longrightarrow \underset{\text{(Blue ppt.) (B)}}{Fe_3[Fe(CN)_6]_2} + 6K^+$$

4. (i) An inorganic compound (A) is formed on passing a gas (B) through a concentrated liquor containing sodium sulphide and sodium sulphite.

(ii) On adding (A) into a dilute solution of silver nitrate, a white precipitate appears which quickly changes into a black coloured compound (C).

(iii) On adding two or three drops of ferric chloride into an excess of solution of (A), a violet coloured compound (D) is formed. But this colour disappears quickly.

(iv) On adding a solution of (A) into the solution of cupric chloride, a white precipitate is first formed which dissolves on adding excess of (A) forming a compound (E).

Identify (A) to (E) and give chemical equations for the reactions at steps (i) to (iv).

Sol. Let us summarize the given facts.

(i) $Na_2SO_3 + Na_2S + \text{Gas(B)} \longrightarrow A$

(ii) $A + AgNO_3 \longrightarrow \text{White ppt.} \longrightarrow \underset{\text{(black)}}{C}$

(iii) $A + FeCl_3 \longrightarrow \underset{\text{violet}}{D} \longrightarrow \text{Disappears}$

(iv) $A + CuCl_2 \longrightarrow \text{White ppt.} \xrightarrow{\text{excess of A}} \text{ppt. dissolves}$ (E)

The given set of reactions indicate that the compound is sodium thiosulphate which can be formed from equation (i) when the gas (B) is either SO_2 or I_2 and thus the various reactions can be written as below:

(i) $\underset{\text{(B)}}{Na_2SO_3} + 2Na_2S + \underset{\text{(B)}}{3SO_2} \longrightarrow \underset{\text{(A)}}{3Na_2S_2O_3}$

or $\underset{\text{(B)}}{Na_2SO_3} + Na_2S + \underset{\text{(B)}}{I_2} \longrightarrow \underset{\text{(A)}}{Na_2S_2O_3} + 2NaI$

(ii) $Na_2S_2O_3 + 2AgNO_3 \longrightarrow \underset{\text{(white ppt)}}{Ag_2S_2O_3\downarrow} + 2NaNO_3$

$Ag_2S_2O_3 + H_2O \longrightarrow \underset{\text{C(black)}}{Ag_2S\downarrow} + H_2SO_4$

(iii) $\underset{\text{(A)}}{3Na_2S_2O_3} + 2FeCl_3 \longrightarrow \underset{\text{D(violet)}}{Fe_2(S_2O_3)_3} + 6NaCl$

$2Fe^{3+} + 2S_2O_3^{2-} \longrightarrow 2Fe^{2+} + \underset{\text{(colourless)}}{S_4O_6^{2-}}$

(iv) $\underset{\text{(A)}}{2Na_2S_2O_3} + 2CuCl_2 \longrightarrow 2CuCl + Na_2S_4O_6 + 2NaCl$

$2CuCl + Na_2S_2O_3 \longrightarrow \underset{\text{(white)}}{Cu_2S_2O_3\downarrow} + 2NaCl\ ] \times 3$

$3Cu_2S_2O_3 + 2Na_2S_2O_3 \longrightarrow \underset{\text{Soluble complex (E)}}{Na_4[Cu_6(S_2O_3)_5]}$

5. An inorganic halide (A) gives the following reactions:

(i) The cation of (A) on reaction with H_2S in HCl medium, gives a black ppt. of (B). (A) neither gives ppt. with HCl nor blue colour with $K_4Fe(CN)_6$.

(ii) (B) on heating with dil. HCl gives compound (A) and a gas (C) which gives a black ppt. with lead acetate solution.

(iii) The anion of (A) gives chromyl chloride test.

(iv) (B) dissolves in hot dil. HNO_3 to give a solution of (D). (D) gives ring test.

(v) When NH_4OH solution is added to (D), a white precipitate (E) is formed. (E) dissolves in minimum amount of dil. HCl to give a solution of (A). Aqueous solution of (A) on addition of water gives a white turbidity (F).

Sol. Observation of (i) indicates that cation (A) is Bi^{3+} because it does not give ppt. with HCl nor blue colour with $K_4Fe(CN)_6$, hence it is neither Pb^{2+} nor Cu^{2+}. Since anion of (A) gives chromyl chloride test, hence it contains Cl^- ions. Thus, (A) is $BiCl_3$. Its different reactions are given below:

(i) $\underset{\text{(A)}}{2BiCl_3} + 3H_2S \longrightarrow \underset{\text{(B)}}{Bi_2S_3} + 6HCl$

(ii) $\underset{\text{(B)}}{Bi_2S_3} + 6HCl \longrightarrow \underset{\text{(C)}}{3H_2S} + \underset{\text{(A)}}{2BiCl_3}$

(iv) $Bi_2S_3 + 8HNO_3 \xrightarrow{\Delta} \underset{\text{(D)}}{2Bi(NO_3)_3} + 2NO + 3S + 4H_2O$

(v) $Bi(NO_3)_3 + 3NH_4OH \longrightarrow \underset{\substack{\text{(E)}\\\text{White ppt.}}}{Bi(OH)_3\downarrow} + 3NH_4NO_3$

$Bi(OH)_3 + 3HCl \xrightarrow{\Delta} \underset{\text{(A)}}{BiCl_3} + 3H_2O$

$BiCl_3 + H_2O \longrightarrow \underset{\substack{\text{(F)}\\\text{Bismuth oxychloride}\\\text{(White turbidity)}}}{BiOCl} + 2HCl$

(A) = $BiCl_3$,
(B) = Bi_2S_3,
(C) = H_2S,
(D) = $Bi(NO_3)_3$,
(E) = $Bi(OH)_3$,
(F) = BiOCl

6. A metal cation gives black precipitate with H_2S which dissolves in hot conc. HNO_3 to give white ppt. When cation reacts with KI it gives intensely brown solution on further treating with thiosulphate becomes colourless and white ppt. becomes visible. Metal ion also reacts with KCN to form poisonous gas. Identify the cation and give chemical reactions.

Sol.

$$Cu^{++} + H_2S \longrightarrow \underset{\text{black}}{CuS\downarrow} + 2H^+$$

$$3CuS\downarrow + 8HNO_3 \longrightarrow 3\,Cu^{++} + 6NO_3^- + \underset{\text{white ppt}}{3S\downarrow} + 2NO\uparrow + 4H_2O$$

$$2Cu^{++} + 5I^- \longrightarrow \underbrace{\underset{\text{white}}{2CuI\downarrow} + \underset{\text{Violet}}{I_3^-}}_{\text{Brown solution}}$$

$$\underset{\text{Violet}}{I_3^-} + 2S_2O_3^- \longrightarrow 3I^- + \underset{\text{Colurless}}{S_4O_6^-}$$

$$Cu^{++} + 4CN^- \longrightarrow [Cu(CN)_4]^{2-} + (CN)_2$$

7. Two solid laboratory reagents 'A' and 'B' give the following reactions:

Compound A :

(i) On strongly heating it gives two oxides of sulphur.

(ii) On adding aqueous NaOH solution to its aqueous solution a dirty green precipitate is obtained, which starts turning brown on exposure to air.

Compound B :

(i) It imparts green colour to the flame.

(ii)Its solution does not give a precipitate on passing H_2S.

(iii) When it is heated with solid $K_2Cr_2O_7$ and concentrated H_2SO_4, a red gas is evolved. The gas when passed in an aqueous solution of NaOH, turns it yellow.

Sol. **Compound A :**

(i) Compound 'A' is ferrous sulphate which give SO_2 and SO_3 on heating

$$2FeSO_4 \longrightarrow Fe_2O_3 + SO_2\uparrow + SO_3\uparrow$$

Ferrous sulphate

(ii) Ferrous sulphate gives dirty green $Fe(OH)_2$ with aqueous NaOH.

$$FeSO_4 + 2NaOH \longrightarrow \underset{\text{(Dirty green)}}{Fe(OH)_2} + Na_2SO_4$$

$Fe(OH)_2$ turns brown on oxidation, when exposed to atmosphere.

$$Fe(OH)_2 + \frac{1}{2}H_2O + \frac{1}{4}O_2 \longrightarrow \underset{\text{Brown}}{Fe(OH)_3}$$

Compound B: is $BaCl_2$ which imparts green colour to flame and does not give any precipitate with H_2S.

$BaCl_2$ forms chromyl chloride, when treated with $K_2Cr_2O_7$ and conc.H_2SO_4.

$$2BaCl_2 + K_2Cr_2O_7 + 3H_2SO_4 \longrightarrow K_2SO_4 + 2BaSO_4 + \underset{\substack{\text{chromyl chloride}\\ \text{(red gas)}}}{2CrO_2Cl_2} + 3H_2O$$

Chromyl chloride forms, yellow coloured Na_2CrO_4 with aqueous NaOH.

$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\substack{\text{Yellow}\\ \text{(sodium chromate)}}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

SOLVED PROBLEMS

Subjective Question

Prob. 1. On the basis of following reactions, identify (A), (B), (C) and (D) and write down their chemical formulae :

(i) (A)aq. + Zn → (B) gas

(ii) (A)aq. + (C)(g) → PH_3

(iii) (A)aq. + NH_4Cl → (D) gas.

Sol. (i) (A) is an alkali and the equations is,

$$\underset{(A)}{2NaOH} + Zn \text{----------} \underset{(B)}{Na_2ZnO_2} + H_2(g)$$

(ii) (A) is NaOH and (C) is phosphorus. The equation will be,

$$\underset{(A)}{3NaOH} + 3H_2O + \underset{(C)}{P_4} \text{----------} PH_3 + 2NaH_2PO_2$$

(iii) (D) is NH_3 gas and (A) is NaOH. The equation will be,

$$\underset{(A)}{NaOH} + NH_4Cl \text{----------} \underset{(D)}{NH_3} + NaCl + H_2O$$

Prob. 2. An aqueous solution of a gas (X) gives the following reactions.

(i) It decolourises an acidified $K_2Cr_2O_7$ solution.

(ii) On boiling with H_2O_2, cooling it and then adding an aqueous solution of $BaCl_2$, a precipitate insoluble in dil. HCl is obtained.

(iii) On passing H_2S in the solutions white turbidity is obtained. Idenfity X and give equations for steps (i), (ii) & (iii).

Sol.

$$SO_2 + H_2O \longrightarrow H_2SO_3$$

$$H_2SO_3 + K_2Cr_2O_7 \xrightarrow{H^+} Cr^{3+} + H_2SO_4$$

$$H_2SO_3 + H_2O_2 \xrightarrow{\text{boil}} H_2SO_4 \xrightarrow{BaCl_2} BaSO_4$$

$$H_2SO_3 + H_2S \longrightarrow S + H_2O$$

Prob. 3. A white coloured inorganic salt gives following reactions.

(i) When exposed to H_2S becomes black

(ii) It is decomposed by dil HCl evolving a gas with effervescence, the gas turns lime water milky

(iii) When heated strongly to about 450° it decomposes to give three gases and leaves a red coloured residue used by housewives. Identify compound X and the four gases evolved during reactions.

Sol.

$$PbCO_3 \cdot Pb(OH)_2 \xrightarrow{H_2S} PbS$$

$$PbCO_3 \cdot Pb(OH)_2 \xrightarrow{\text{dil HCl}} PbCl_2 + CO_2 \uparrow$$

$$CO_2 + Ca(OH)_2 \longrightarrow \underset{\text{milk shek}}{CaCO_3}$$

$$PbCO_3 \cdot Pb(OH)_2 \xrightarrow{430^\circ C} \underset{\text{Red lead}}{Pb_3O_4} + CO_2 + H_2O + \frac{1}{2}O_2$$

Prob. 4. Identify (A), (B), (C) & (D) and give their chemical formulae:

(i) $(A) + NaOH \xrightarrow{heat} NaCl + NH_3 + H_2O$

(ii) $NH_3 + CO_2 + H_2O \longrightarrow (B)$

(iii) $(B) + NaCl \longrightarrow (C) + NH_4Cl$

(iv) $(C) \xrightarrow{heat} Na_2CO_3 + H_2O + (D)$

Sol. (i) $NH_4Cl + NaOH \xrightarrow{\Delta} NH_3 + NaCl + H_2O$

(A)

(ii) $NH_3 + CO_2 + H_2O \longrightarrow NH_4HCO_3$ (B)

(iii) $NH_4HCO_3 + NaCl \longrightarrow NaHCO_3 + NH_4Cl$

(C)

(iv) $2\,NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$ (D)

Objective Questions

Prob. 1. Na_2CO_3 cannot be used in place of $(NH_4)_2CO_3$ for precipitation of group V cations because

(A) Na^+ interferes in the detection of group V

(B) concentration of CO_3^{2-} is very low

(C) Na^+ will react with acid radicals

(D) Mg^{2+} will precipitate as $MgCO_3$ if present in the mixture

Sol. Na_2CO_3 is highly ionizable electrolyte which produces very high concentration of CO_3^{2-} ions. As a result, ionic product of $MgCO_3$ may increase its K_{sp} and it may get precipitated along with the radicals of group V.

Hence, **(D)** is the correct answer.

Prob. 2. A scarlet compound (A) is treated with conc. HNO_3 to give a chocolate brown precipitate (B) and a colourless solution of C. On adding KI to this solution, a yellow ppt. (D) is obtained. The precipitate (B) on warming with conc. HNO_3 in the presence of $Mn(NO_3)_2$ prdouces a pink coloured solution due to the formation of (E). What is (E)?

(A) $PbCrO_4$ (B) PbI_2

(C) MnO_2 (D) $HMnO_4$

Sol. Scarlet compound (A) is Pb_3O_4.

$$\underset{(A)}{Pb_3O_4} + 4HNO_3 \longrightarrow \underset{\substack{(B)\\(Chocolate\ brown)}}{PbO_2} + \underset{(C)}{2Pb(NO_3)_2} + 2H_2O$$

$$Pb(NO_3)_2 + 2KI \longrightarrow \underset{\substack{(D)\\(Yellow)}}{PbI_2}$$

$$PbO_2 + Mn^{2+} + HNO_3 \xrightarrow{\Delta} \underset{\substack{(E)\\(Pink)}}{HMnO_4} + Pb^{2+}$$

Hence, **(D)** is the correct answer.

Prob. 3. Which gas gives yellow ppt. with chloroplatinic acid?

(A) SO_2 (B) NH_3

(C) CO (D) NO_2

Sol. $2NH_3 + H_2PtCl_6 \longrightarrow \underset{\substack{Ammonium\ chloroplatinate\\(Yellow)}}{(NH_4)_2PtCl_6}$

Hence, **(B)** is the correct answer.

Prob. 4. Aqueous solution of a salt (Y) is alkaline to litmus. On strong heating, it swells-up to give a glassy material. When conc. H_2SO_4 is added to a hot concentrated solution of (Y), white crystals of a weak acid separates out. Hence, the compound (Y) is

(A) $Na_2SO_4.10H_2O$
(B) $Ca_2B_6O_{11}.10H_2O$
(C) $Na_2B_6O_{11}$
(D) $Na_2B_4O_7.10H_2O$

Sol. (Y) is borax $Na_2B_4O_7.10H_2O$

$$Na_2B_4O_7 + 7H_2O \longrightarrow \underbrace{2NaOH}_{\text{strong base}} + \underbrace{4H_3BO_3}_{\text{weak acid}}$$

Alkaline

$$Na_2B_4O_7 \xrightarrow{\Delta} \underbrace{2NaBO_2}_{\text{Glassy-bead}} + B_2O_3$$

$$Na_2B_4O_7 + H_2SO_4 + 5H_2O \longrightarrow Na_2SO_4 + 4H_3BO_3$$

Hence, **(D)** is the correct answer.

Prob. 5. A white powder when strongly heated gives off brown fumes. A solution of this powder gives a yellow precipitate with a solution of KI. When a solution of barium chloride is added to a solution of powder, a white precipitate results. This white powder may be

(A) a soluble sulphate
(B) KBr or NaBr
(C) $Ba(NO_3)_3$
(D) $AgNO_3$

Sol.

$$AgNO_3 \xrightarrow{\Delta} AgO + \underbrace{NO_2}_{\text{Brown fumes}}$$

$$AgNO_3 + KI \longrightarrow \underbrace{AgI}_{\text{Yellow ppt.}} + KNO_3$$

$$AgNO_3 + BaCl_2 \longrightarrow \underbrace{AgCl}_{\text{White ppt.}} + BaNO_3$$

Hence, **(D)** is the correct answer.

Prob. 6. Which of the following react with dil. H_2SO_4?

(A) $CaCO_3$
(B) KNO_2
(C) Na_2S
(D) $ZnSO_3$

Sol. All these salts react with dil. H_2SO_4.
Hence, **(A), (B), (C)** and **(D)** are the correct answers.

Prob. 7. Conc. H_2SO_4 will not give any gas with

(A) $ZnSO_4$
(B) barium phosphate
(C) magnesium borate
(D) sodium oxalate

Sol. SO_4^{2-}, PO_4^{3-} and BO_3^{3-} do not react with conc. H_2SO_4.
Hence, **(A), (B)** and **(C)** are the correct answers.

Prob. 8. Which of the following statement(s) is/are incorrect?

(A) Manganese salts give a violet borax bead test in reducing flame.
(B) From mixed salt of AgCl and AgI, NH_4OH solution dissolves only AgCl.
(C) Fe^{3+} ions given a deep green precipitate, on adding $K_4[Fe(CN)_6]$ solution.
(D) On strong heating $Na_2S_2O_3.5H_2O$, a polysulphide is resulted.

Sol. Mn appears colourless in reducing flame in Borax Bead Test.

$$Fe^{3+} + K_4\left[Fe(CN)_6\right] \longrightarrow \underbrace{Fe_4\left[Fe(CN)_6\right]_3}_{\text{Prussian blue}} \downarrow + K^+$$

$$Na_2S_2O_3.5H_2O \xrightarrow{\Delta} Na_2S_5 + Na_2SO_4 + H_2O$$

Hence, **(A)** and **(C)** are correct answers

Prob. 9. A solution of colourless salt (X) on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt (s) X is (are):

(A) NH_4NO_3 (B) NH_4NO_2

(C) NH_4Cl (D) $(NH_4)_2SO_4$

Sol. (A) $NH_4NO_3 + NaOH \longrightarrow NH_3 + NaNO_3 + H_2O$

$7NaOH + NaNO_3 + 4Zn \longrightarrow 4Na_2ZnO_2 + NH_3 + 2H_2O$

(B) $NH_4NO_2 + NaOH \longrightarrow NaNO_2 + NH_3 + H_2O$

$3Zn + 5NaOH + NaNO_3 \longrightarrow 3Na_2ZnO_2 + NH_3 + H_2O$

Hence, **(A)** and **(B)** are correct answers

Prob. 10. Which of the following will be completely or partially dissolved in NH_4OH?

(A) AgCl (B) AgBr

(C) $PbSO_4$ (D) AgI

Sol. AgCl is completely soluble while AgBr is partially soluble in NH_4OH.
Hence, **(A)** and **(B)** are the correct answers.

Prob. 11. Bon is precipitated as $Fe(OH)_3$ in the third group and not as $Fe(OH)_2$ is because

(A) $Fe(OH)_2$ is light green

(B) $Fe(OH)_3$ is reddish brown

(C) $Fe(OH)_2$ is partially soluble and is not completely precipitated

(D) $Fe(OH)_2$ is oxidant

Sol. **(D)**

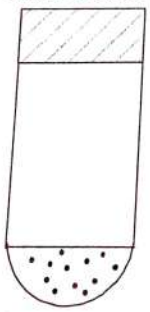
COMPREHENSION

Read the following paragraph and answer the questions given below

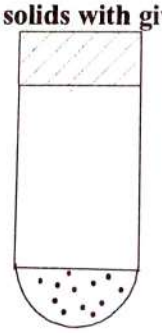
In 1899, a parcel saying 'Handle with care' was received at the following address:
Sherlock Holmes Detective Laboratory
221B, Baker Street
London,
England.

The parcel was opened by Dr. John H. Watson, assistant to Sherlock Holmes. Inside, the following were found:

➢ **Test Tubes containing solids with given colours (4 in number)**



T_1
Black Colour

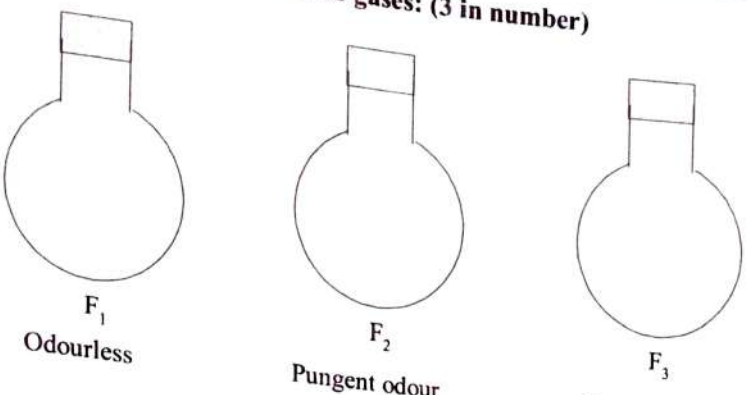


T_2
White Colour

T_3
Vermilion Colour

T_4
Scarlet Colour

➢ **Flasks containing colourless gases: (3 in number)**

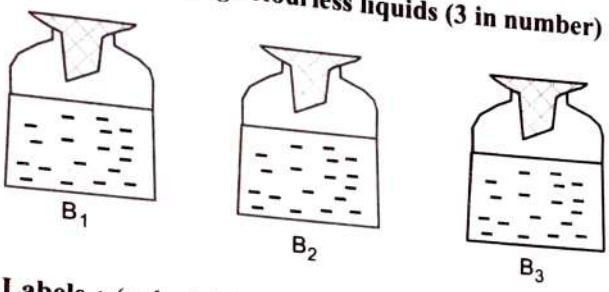


F_1
Odourless

F_2
Pungent odour

F_3
Pungent odour

➢ **Bottles containing colourless liquids (3 in number)**



B_1

B_2

B_3

➢ **Labels :** (only 4 labels were found, that too detached from their containers and lying on the floor of parcel. Rest 6 labels were missing)

$K_2Cr_2O_7(s)$	$HgI_2(s)$	$H_2S(g)$	$NH_4OH(l)$

1. Watson set about experimenting with the contents of the parcel in the laboratory. Here are the excerpts: T_1 and T_4 were found not to dissolve in water. T_2 and T_3 were much soluble in water. Also, T_2 and T_3 imparted lilac colour to the Bunsen flame.
2. T_4 dissolves in an aqueous solution of T_2 due to complexation. The complex formed reacts with B_3 to form a dark-brown precipitate.
3. Both B_1 and T_3 on strong heating evolve gas F_1.
4. $T_1 + B_1 \longrightarrow PbSO_4 \downarrow$

   $T_1 + B_2 \longrightarrow PbSO_4 \downarrow + F_3$

   $B_1 + T_2(aq) \longrightarrow I_2 \downarrow$

   $B_1 + F_3 \longrightarrow S \downarrow (\text{colloidal})$

   $T_3 + B_1 + B_2 \longrightarrow CrO_5(\text{blue, stable in ether})$

   $T_3(aq) + F_2 \longrightarrow Cr_2(SO_4)_3(\text{green})$
5. Mixing equal moles of B_2 and B_3 results in neutralization due to which a salt is formed having pH < 7.

While Watson was trying to make some sense of all these observations, Sherlock Holmes entered the lab. He smiled at Watson who looked bewildered with 'papers' in his hands. Sherlock Holmes took the piece of paper wherein the observations were scribbled, took a deep look at them for a minute, and there it was- Holmes spelled out each of the ten unknowns in quick succession while Watson was dazed at his spontaneity.

'How did you manage to think so fast' asked Watson; to which Sherlock Holmes replied (taking a deep puff off his cigar and blowing out the smoke)- "Elementary, dear Watson".

Answer the following questions:

1. Of the 10 compounds, the number of sulphur containing compounds are:
(A) 2 (B) 3
(C) 4 (D) 5

Sol. $T_1 = PbS$
$F_2 = SO_2$
$F_3 = H_2S$
$B_2 = H_2SO_4$
Hence, **(C)** is the correct answer.

2. In which of the following, the end precipitate has a colour different from others:
(A) $T_1 \xrightarrow{\text{dil.HCl}} \text{ppt.} \xrightarrow{\Delta} \text{ppt. dissolves} \xrightarrow{T_2} \text{ppt.}\downarrow$
(B) $F_2 + F_3 \longrightarrow H_2O + \text{ppt.}\downarrow$
(C) $T_3 + B_2 + NaCl \xrightarrow{\Delta} \text{fumes} \xrightarrow{NaOH} \text{solution} \xrightarrow{Pb^{+2}} \text{ppt.}\downarrow$
(D) $T_4 \xrightarrow{\text{dil.HCl}} \text{solution} \xrightarrow{F_2} \text{ppt.}\downarrow$

Sol. $T_4 = HgS \xrightarrow{HCl} HgCl_2 \xrightarrow{H_2S} HgS\downarrow$ (other ppt's are yellow)
(Black)
Hence, **(D)** is the correct answer.

3. Which reagent is wrongly matched with one of its components:
(A) Tollen's reagent – B_3 (B) Fenton's reagent - B_1
(C) Schiff's reagent – F_2 (D) Lucas reagent – T_1

Sol. Lucas reagent is anhydrous $ZnCl_2$ and has no Pb present in it.
Hence, **(D)** is the correct answer.

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. **STATEMENT-1:** K_2CrO_4 has yellow colour due to charge transfer.
and
STATEMENT-2: CrO_4^{2-} ion is tetrahedral in shape
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(B)** CrO_4^{2-} does not posses d–electrons thus d—d transition is not possible.

2. **STATEMENT-1:** A very dilute acidic solution of Cd^{2+} and Ni^{2+} gives yellow precipitate of CdS on passing hydrogen sulphide.
and
STATEMENT-2: Solubility product of CdS is more than that of NiS.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

Sol. **(B)**

3. **STATEMENT-1:** In acidic medium, Zn^{2+} is not precipitated by S^{2-} ions.
and
STATEMENT-2: Common ion effect reduces the concentration of S^{2-} ions to the minimum level.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

4. **STATEMENT-1:** PbS can be separated from ZnS by electrostatic separation method.
and
STATEMENT-2: PbS is a good conductor and ZnS is a poor conductor of electricity.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

5. **STATEMENT-1:** Sulphate is estimated as $BaSO_4$ and not as $MgSO_4$
and
STATEMENT-2: Ionic radius of Mg^{2+} is smaller than that of Ba^{2+}
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

6. **STATEMENT-1:** Iodine is liberated when KI is added to Cu^{2+} ions but Cl_2 is not liberated when KCl is added to Cu^{2+} ions.
and
STATEMENT-2: The reducing power of I^- is more than Cl^-.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

Sol. **(A)**

MATRIX – MATCH TYPE QUESTION

1. Match the following set.

Column – I		Column – II	
(A)	CuS	(p)	Soluble in yellow Ammonium Sulphide.
(B)	Bi_2S_3	(q)	Soluble in aqua-regia
(C)	HgS	(r)	Insoluble in yellow ammonium sulphide.
(D)	PbS	(s)	Soluble in 50% HNO_3.
		(t)	Soluble in pure water.

Sol. **A →q, r, s; B → p, q, s; C → q, r; D → q, r, s**

2. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	HgS	(p)	2^{nd} group of basic radicals
(B)	NiS	(q)	Buff colour
(C)	MnS	(r)	Dirty colour
(D)	ZnS	(s)	Black
		(t)	Group IV basic radicals

Sol. **(A → (p, s); B → (s, t); C → (q, t); D → (r, t)**

Answer 3, 4 and 5 by appropriately matching the information given in the three columns of the following table.

Column – 1		Column – 2		Column – 3	
(I)	$Cr^{3+}(aq)$ + excess Na_2O_2	(i)	$[Cr(OH)_4]^-$	(P)	Green solution
(II)	$Cr^{3+}(aq)$ + excess Na_2O	(ii)	$CaCrO_4$	(Q)	Yellow solution
(III)	CrO_4^{2-} + $BaCl_2/CH_3COOH$	(iii)	CrO_4^{2-}	(R)	White ppt
(IV)	CrO_4^{2-} + $CaCl_2/CH_3COOH$	(iv)	$BaCrO_4$	(S)	Yellow ppt

3. Which of the following combination is incorrect ?
[A] (I) (iii) (Q) [B] (II) (i) (P)
[C] (III) (iv) (S) [D] (III) (iv) (R)

4. Which of the following combination is correct ?
[A] (I) (iii) (R) [B] (I) (ii) (Q)
[C] (IV) (ii) (Q) [D] (I) (iv) (Q)

5. Which of the following combination is correct ?
[A] (I) (ii) (Q) [B] (II) (i) (P)
[C] (III) (i) (R) [D] (IV) (i) (S)

Sol. 3. **D**
4. **C**
5. **B**

SINGLE INTEGER ANSWER TYPE

1. Which one of the following pairs of ions cannot be separated by H_2S in dilute HCl.
(1) Bi^{3+}, Sn^{4+} (2) Al^{3+}, Hg^{2+}
(3) Cu^{2+}, Zn^{2+} (4) Ni^{2+}, Cu^{2+}

Sol. **(1)**

2. Concentrated aqueous sodium hydroxide can separate a mixture of
(1) Al^{3+} and Sn^{2+} (2) Al^{3+} and Fe^{3+}
(3) Al^{3+} and Zn^{2+} (4) Zn^{2+} and Pb^{2+}

Sol. **(2)**

3. Out of Cu_2S, HgS, Ag_2S and ZnS roasting will convert the minerals into metal in case of
(1) Cu_2S, ZnS (2) HgS, ZnS
(3) Cu_2S, Ag_2S (4) HgS, Ag_2S

Sol. **(4)**

4. Following reaction is not involved in Thermite process:
(1) $3M_3O_4 + 8Al \rightarrow 9Mn + 4Al_2O_3$
(2) $Cr_2O_3 + 2Al \rightarrow Al_2O_3 + 2Cr$
(3) $2Fe + Al_2O_3 \rightarrow 2Al + Fe_2O_3$
(4) $B_2O_3 + 2Al \rightarrow 2B + Al_2O_3$

Sol. **(3)**

5. Number of oxygen atoms in molybdate ion is

Sol. **(4)**

NUMERICAL VALUE ANSWER TYPE

Prob. 1. Consider the following reaction

$$Na_2SO_4 + C \xrightarrow{\Delta} (A) \xrightarrow{Na_2[Fe(CN)_5NO]} \underset{\text{Purple complex}}{(B)}$$

$$(A) \xrightarrow{Cd(NO_3)_2} \underset{\text{Yellow ppt}}{(C)}$$

Then calculate value of $\frac{|x^2 - y^2|}{x+2}$, where x and y are total number of electron present in 't_{2g}' and 'e_g' orbitals respectively in d-block metal ion of compound B.

Sol. **4.5**

Prob. 2. Number of basic radicals among the following cations, which can form soluble complex on adding excess of NH_3 solution out of total B cations.
Cd^{2+}(aq.), Pb^{2+}(aq), Ni^{2+}(aq), Mn^{2+}(aq), Zn^{2+}(aq), Ag^{+}(aq), Hg^{2+}(aq), Fe^{3+}(aq), Mg^{2+}(aq)

Find value of $\frac{B}{A}$

Sol. **2.25**

$$\frac{9}{4} = 2.25$$

Prob. 3. Out of total P anions, if Q anions give colourless acid vapour/gas with conc. H_2SO_4 on reaction with following given anion. CH_3COO^-, Cl^-, Br^-, S^{2-}, SO_3^{2-}, BO_3^{2-}, NO_2^-, $C_2O_4^{2-}$, I^-.

Find value of $\frac{P}{Q}$

Sol. **1.50**

$$\frac{9}{6} = \frac{3}{2} = 1.50$$

Prob. 4. $X(s) \xrightarrow{\text{dil. HCl}} Y\uparrow \xrightarrow[NaOH]{Na_2[Fe(CN)_5(NO)]}$ Purple solution. Gas Y has been allowed to react with the following species in neutral/ acidic medium:
(a) $FeCl_3$, (b) $CuSO_4$, (c) $BaCl_2$, (d) SO_2 (e) $Cr_2O_7^{2-}$ (f) CH_3COONa (g) Hg^{2+}
If P: number of species which undergo redox reaction with gas Y.
Q: number of species with which gas Y undergo precipitation.

Find value of $\frac{P}{Q}$

Sol. **1.50**

$$\frac{3}{2} = 1.50$$

Prob. 5.

$$\underset{\substack{\text{1 mpart violet}\\ \text{colour to flames}}}{(T)} \xrightarrow{\text{Compound (U) + Conc.}H_2SO_4} \underset{\text{Red gas}}{(V)} \xrightarrow{NaOH + AgNO_3} \underset{\text{Red ppt}}{(W)} \xrightarrow[\text{solution}]{NH_3} (X)$$

$$W \xrightarrow{\text{dil. HCl}} (Y) \text{ white ppt}$$

$$(U) \xrightarrow[\Delta]{NaOH} (Z) \text{ gas (gives white fumes with HCl)}$$

Find molar mass of (T) ÷ molar mass of (Z)

Sol. **17.29**

$$\frac{294}{17} = 17.29$$

TEST YOURSELF

NUMERICAL BASED PROBLEMS (SUBJECTIVE)

1. Determine the frequency of revolution of the electron in 2nd Bohr's orbit in hydrogen atom.

2. The energy of the electron in the second and third Bohr orbit of hydrogen atom is -542×10^{-12} erg and -2.41×10^{-12} erg respectively. Calculate the wavelength of the emitted radiation when the electron drops from third to second orbit.

3. A 100 ml flask containing O_2 at 1.2 atm and 300 K is connected to a 250 ml flask containing NO(g) at 0.6 atm and 300 K, by means of a narrow tube of negligible volume, when they combine quantitatively to form NO_2. Finally NO_2 (g) dimerized, partially into N_2O_4(g) and pressure inside the flask was found to be 0.41 atm at the same temperature. Find % dimerization of NO_2 as

$$2NO_2(g) \rightleftharpoons N_2O_4(g)$$

4. A sample of lithium was burnt in to give mixture of Li_2O and Li_3N. The ash was dissolved in 50 meq of HCl and resulting solution back titrated with NaOH. 20 meg of NaOH were required to reach the end point. An excess of NaOH was then added and solution distilled. The ammonia released was then trapped in 10 meq of second acid solution required 6 meq of the base. Calculate percentage Lithium burnt to the oxide.

5. Find the work done when 1 mole of the gas is expanded reversibly and isothermally from 5 atm to 1 atm at 25°C.

6. A vessel at 100 K contains CO_2 with a pressure of 0.5 atm. Some of the CO_2 is converted into CO on addition of graphite. Calculate the value of K_P if the total pressure at equilibrium is 0.8 atm.

7. For the reaction:

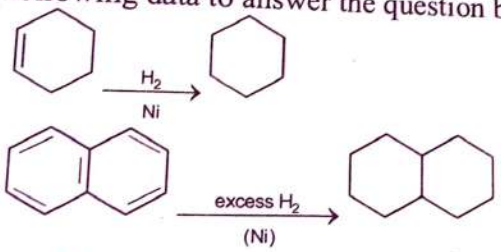
$$SnO_2(s) + 2H_2(g) \rightleftharpoons 2H_2O(g) + Sn(s)$$

Calculate K_P at 900 K, where the equilibrium steam-hydrogen mixture was 45% H_2 by volume.

8. A Galvanic cell is constructed with $Ag^+|Ag$ and $Fe^{3+} | Fe^{2+}$ electrodes. Find the concentration of Ag^+ at which the emf of the cell is zero at equimolar concentrations of Fe^{2+} and Fe^{3+}.

$\left(E^\circ_{Ag^+|Ag} = 0.80\ V \text{ and } E^\circ_{Fe^{3+}|Fe^{2+}} = 0.77\ V\right)$

9. Use the following data to answer the question below:



$\Delta H = -28.6$ kCal mol^{-1}

$\Delta H = -89$ kCal mol^{-1}

naphthalene

Calculate the resonance energy of naphthalene in kCal/mol.

10. Find the value of K_f (freezing point depression constant) for the solvent, p-chlorobenzene, from the following data.

Molar mass = 147.01, Melting point (K) = 326.28, $\Delta H^\circ_{fusion}\left(KJ/mol\right) = 17.88$

Answer must be in the units of 10^{-2} K kg/mol.

11. Calculate the value of molal elevation constant for water if $\Delta S_{vapourisation}$ is 26.33 cal K^{-1} mol^{-1}.

12. Consider the reaction, $2Ag^+ + Cd \longrightarrow 2Ag + Cd^{2+}$
The standard electrode potentials for $Ag^+ \longrightarrow Ag$ and $Cd^{2+} \longrightarrow Cd$ couples are 0.80 volt and 0.40 volt respectively.
(i) What is the standard potential E^o for this reaction?
(ii) For the electrochemical cell, in which this reaction takes place which electrode is negative electrode?

13. At 353 K, the vapor pressure of pure ethylene bromide and propylene bromide are 22.93 and 16.93 K N m^{-2} respectively and these compounds form nearly ideal solution. 3 mole of ethylene bromide and 2 mole of propylene bromide are equilibrated at 353 K and at a total pressure of 20.4 K N m^{-2}.
(a) What is the composition of the liquid phase?
(b) How many moles of each compound are present in the vapour phase?

14. The unit cell length of NaCl is observed to be 0.5627 nm by X-ray diffraction studies; the measured density of NaCl is $2.164 gcm^{-3}$. Calculate the difference of observed and calculated density and calculate % of missing Na^+ and Cl^- ions.

15. $[Fe(CN)_6]^{4-}$ and $[Fe(H_2O)_6]^{2+}$ are of different colours in dilute solutions. Why?

16. Explain from the VB method the possibility of the formation of the C_2N_2 molecule.

17. Water can react with hydrogen ions to make the oxonium ion, H_3O^+. What is present in a water molecule that allows it react with a hydrogen ion? Describe the bonding in the oxonium ion.

18. Why the bond angle of H – C – H in methane (CH_4) is 109° 28' while H – N – H bond angle in NH_3 is 107° though both carbon and nitrogen are sp^3 hybridized.

19. The melting point of KCl is higher than that of AgCl though the crystal radii of Ag^+ and K^+ ions are almost same.

20. An element of group V on reaction with concentrated H_2SO_4 gives an acid (B) and an oxide (C). (C) on reaction with acidified dichromate produces the acid (D). (D) is having very strong affinity towards water. (A) with metal (E) produces as hydrolysis a gas (F) of odour of decaying fish. Find (A) to (F).

21. From the following metallurgical flow chart, identify steps A, B and C.

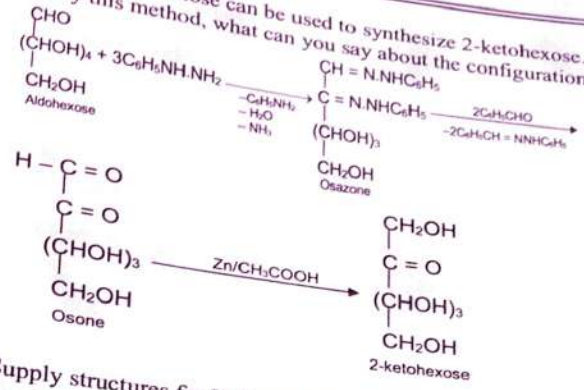
$$\text{Sulphide} \xrightarrow{(A)} \text{Oxide} \xrightarrow{(B)} \text{Impure metal} \xrightarrow{(C)} \text{Pure metal}$$

22. Silver nitrate gives a white precipitate with aqueous sodium thiosulphate; the precipitate dissolves in excess of thiosulphate. If the precipitate is heated with water it turns black, and the supernatant liquid then gives a white precipitate with acidified barium nitrate solution. Give the reactions for the above observations.

23. A yellow solid (A) is unaffected by acids and bases. It is not soluble in water. It dissolves slowly in hot conc. HNO_3 and brown gas (B) is released. The solid (A) dissolves only in a boiling solution of sodium sulphite giving a clear solution (C). Acidification of solution (C) causes a colourless gas (D) to be liberated, accompanied by the appearance of a milky precipiate (E) in the solution. Identify (A) to (E).

24. Dehydrohalogenation of 1, 2-dihalides (with the elimination of two molar equivalents of HX) normally leads to an alkyne rather than to a conjugated diene. However, when 1-2-dibromocycloxehane is dehydrohalogenated, 1,3-cyclohexadiene is produced in good yield. What is the reason?

25. Show how an aldohexose can be used to synthesize 2-ketohexose. (b) Since glucose is converted to fructose by this method, what can you say about the configurations of C^3, C^4 and C^5 in the sugars.



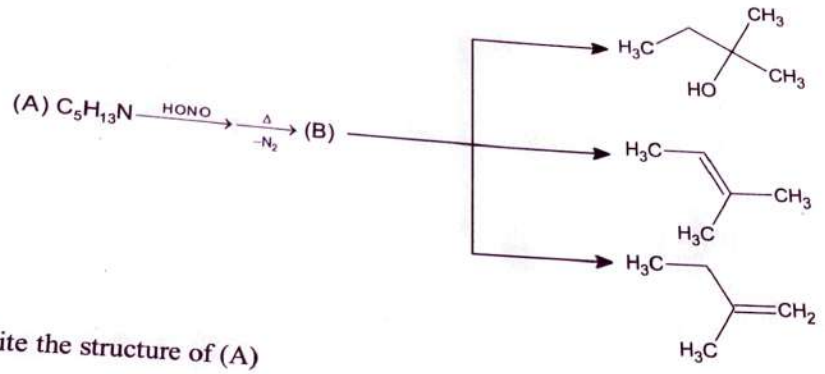
26. (a) Supply structures for H through K. Given:

An aldohexose $\xrightarrow{NH_2OH\ /\ base}$ H $\xrightarrow{Ac_2O\ /\ NaOAc}$ I $\xrightarrow{-HOAC}$ J $\xrightarrow{NaOMe\ /\ MeOH}$ K.

(b) Explain the last step (c). What is net structural change (d) Name this overall method. (e) Discuss the possibility of epimer formation.

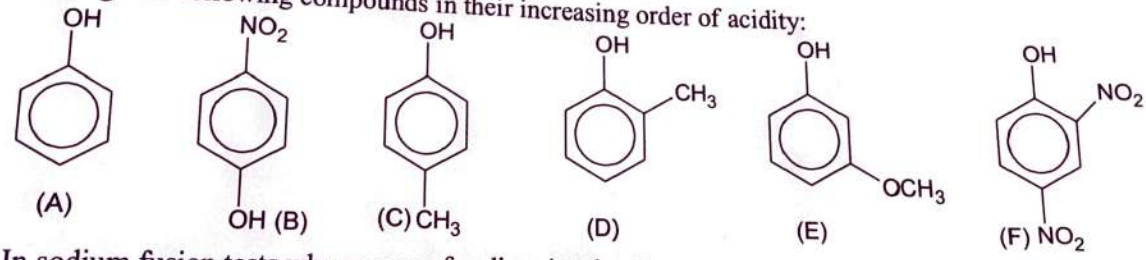
27. A polyhydroxy alcohol having molecular weight = 168. On acetylation the molecular weight increases to 294. Determine the number of –OH groups present in the alcohol.

28.



Write the structure of (A)

29. Arrange the following compounds in their increasing order of acidity:



30. In sodium fusion tests why excess of sodium is taken?

31. Chlorobenzene when treated with ethanolic $AgNO_3$ it does not give white precipitate. Explain

32. Nitrobenzene does not undergo Friedel Crafts Alkylation reaction why?

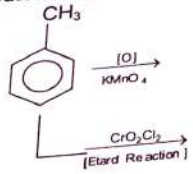
33. Starting from benzene how will you prepare m-nitrobenzoic acid.

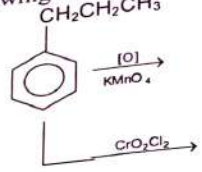
34. A variety of substituent group are listed. Indicate the effect of these group on electrophile aromatic substitution. Classify them as activating or deactivating and also reaction their directing influence.
$-C_6H_5$, $-NH_2$, $-Cl$, $-N(CH_3)_2$, $-OH$, $-CN$, $-NHCOCH_3$, $-NHCH_3$, $-OC_2H_5$, $-CH_3$, $-SO_3H$, $-F$, $-I$,
$-OCH_3$, $-C_2H_5$, $-\overset{\oplus}{N}(CH_3)_3$, $-COOH$, $-COOR$, $-CHO$, $-\overset{O}{\overset{\|}{C}}-$, $-Br$.

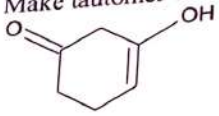
35. Glycine exists as ($H_3N^+CH_2COO^-$) while anthranilic acid ($p-NH_2-C_6H_4-COOH$) does not exist as dipolar ion.

36. Compound A (C_8H_9Cl) is hydrolysed by dilute acids to give compound B ($C_8H_{10}O$) compound B can be oxidised under mild conditions to compound C (C_8H_8O). Compound (C) upon reduction with Zn/Hg and concentrated HCl gives Ethyl benzene. Identify structures from (A) to (C).

37. Predict the product of the following reactions

(i) 

(ii) 

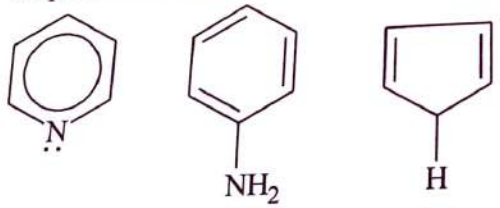
38. Make tautomer of 

39. Why $CH_3-\overset{O}{\overset{\|}{C}}-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$ is known as active methylene compound?

40. Explain the order of hydrolysis of following

CH_3COCl	$(CH_3CO)_2O$	$CH_3COOC_2H_5$	CH_3CONH_2
(a)	(b)	(c)	(d)

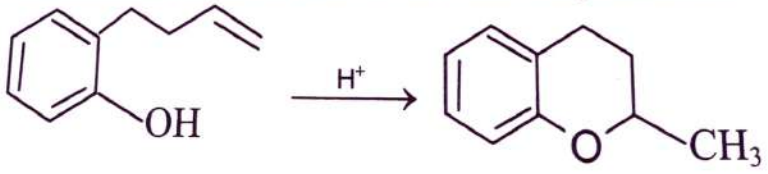
41. Explain basic strength of following



(a) (b) (c)

42. An unsaturated hydrocarbon A (C_6H_{10}) readily gives B on treatment with $NaNH_2$ in liquid NH_3. When B is allowed to react with 1-chloropropane a compound C is obtained. On partial hydrogenation in the presence of Lindlar catalyst, compound C gives D (C_9H_{18}). On ozonolysis, D gives 2, 2-dimethyl propanal and 1-butanal. With proper reasoning give the structures of A, B, C and D.

43. Write the intermediate steps for each of the following reactions.

(i) 

(ii) $C_6H_5CH(OH)C \equiv CH \xrightarrow{H_3O^+} C_6H_5 - CH = CHCHO$

(iii) $CH_3 - \underset{CH_3}{\underset{|}{CH}} - CH_2OH \xrightarrow{HCl} CH_3 - \underset{CH_3}{\overset{Cl}{\overset{|}{\underset{|}{C}}}} - CH_3$

44. Suggest a mechanism for the dehydration of $CH_3CHOHCH_3$ that proceeds through a carbonium ion intermediate. Assign a catalytic role to the acid and keep in mind that the O in ROH is a basic site like O in H_2O.

45. List the possible products, in order of decreasing yield, from the reaction of 3-bromo –2,3 – dimethyl pentane with alcoholic KOH.

46. Account for the formation of both 3–bromo–2,2– dimethyl butane and 2–bromo–2,3 dimethyl butane from the reaction of HBr with 2,3–dimethyl –1- butene.

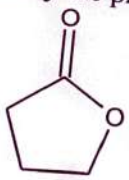
47. Optically active 2-iodooctane when treated with radioactive NaI^{128} in dry acetone rate of loss of optical activity is twice the rate of isotopic exchange. Explain

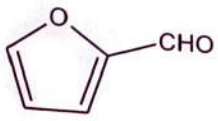
48. A solution of 0.2 g of a compound containing Cu^{+2} and $C_2O_4^{-2}$ ions on titration with 0.02 M $KMnO_4$ in presence of H_2SO_4 consumes 22.6 mL of the oxidant. The resultant solution is neutralized with Na_2CO_3, acidified with dilute acetic acid and treated with excess KI. The liberated iodine requires 11.3 mL of 0.05 N $Na_2S_2O_3$ solution for complete reduction. Find out the molar ratio of Cu^{+2} to $C_2O_4^{-2}$ in the compound. Write down the balanced redox reactions involved in the above titration.

49. An equal volume of a reducing agent is treated separately with 1 M $KMnO_4$ in acid, neutral and alkaline medium. The volumes of $KMnO_4$ required are 20 ml in acid, 33.4 ml in neutral and 100 ml in alkaline medium. Find out the oxidation state of manganese in each reaction product. Find out the volume of 1 M $K_2Cr_2O_7$ consumed, if the same volume of the reducing agent is treated in acid medium.

50. H_2O_2 is reduced rapidly by Sn^{2+}, the products being Sn^{4+} and water. H_2O_2 decomposes slowly at room temperature to yield O_2 and water. Calculate the volume of O_2 produced at 20°C and 1 atm. when 200 gm. of 10% by mass H_2O_2 in water is treated with 100 ml. of 2 M Sn^{2+} and then the mixture is allowed to stand until no further reaction occurs.

51. Identify the product of following reactions

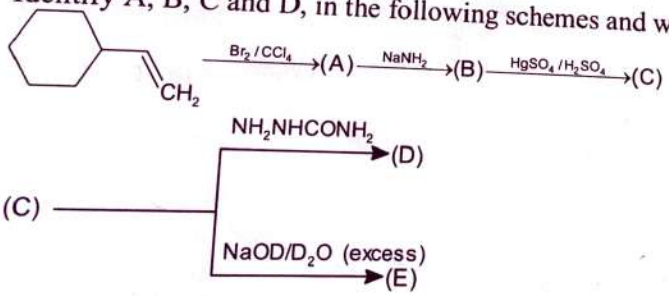
(a)  $\xrightarrow{(Ph_3)_3\bar{C}Na^+}$ C

(b)  $\xrightarrow{50\%NaOH}$ D + E

Furfural

Also write the mechanism involved.

52. Identify A, B, C and D, in the following schemes and write their structures.

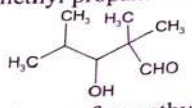


53. Cyclobutylbromide on treatment with magnesium in dry ether forms an organometallic compound (A). The organometallic compound reacts with ethanol to give an alcohol (B) after mild acidification. Prolonged treatment of alcohol (B) with an equivalent amount of HBr gives 1-bromo-1-methylcyclopentane (C). Write the structures of (A), (B) and explain how C is obtained by (B).

54. An alkene (A) $C_{16}H_{16}$ on ozonolysis gives only one product (B) C_8H_8O. Compound (B) on reaction with $NaOH/I_2$ yields sodium benzoate. Compound (B) reacts with KOH/NH_2-NH_2 yielding a hydrocarbon (C) C_8H_{10}. Write the structure of (B) and (C). Based on this information give two isomeric structures of (A).

55. An organic compound (A) $C_8H_4O_3$ in dry benzene in the presence of anhydrous $AlCl_3$ gives compound (B), the compound (B) on treatment with PCl_5 followed by reaction with H_2/Pd ($BaSO_4$) gives compound (C) which on reaction with hdyrazine gives a cyclized compound (D) $C_{14}H_{10}N_2$. Identify (A), (B), (C), and (D). Explain the formation of (D) from (C).

56. How would you bring about the following conversions?
(i) Ethanal to 2–hydroxy–3–butenoic acid
(ii) 2–methyl propanal to

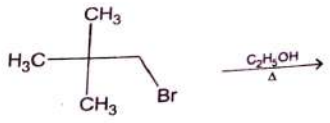


(iii) 2–butanone from ethyl alcohol
(iv) 3–hexanon from n–propyl alcohol

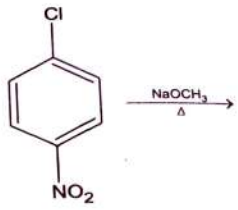
57. Cyclopentyl bromide on treatment with magnesium in dry ether from an organometallic compound (A). The organometallic compound reacts with ethanol to give an alcohol (B) after mild oxidation. Prolonged treatment of alcohol (B) with an equivalent amount of HBr gives 1-bromo – 1- methyl cyclopentane (C). Write the structure of (A), (B) and explain how (C) is obtained from (B)

58. What would be the major products in following reactions?

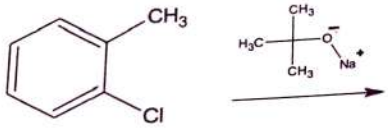
a)



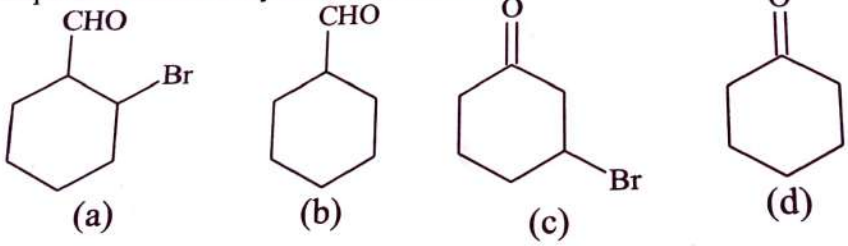
b)



c)



59. Explain the order of hydration of following:

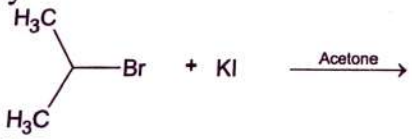


(a) (b) (c) (d)

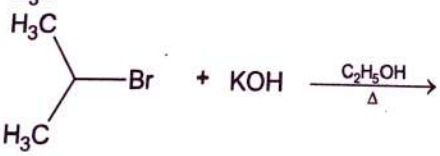
60. Hydrolysis of 2- bromo – 3 – methyl butane (2°) yield only 2-methyl -2- butanol (3°). Explain

61. Complete the following equations showing major product, which reaction mechanism (SN^1, SN^2, E_1, E_2) is likely

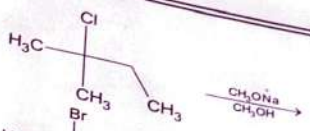
a)



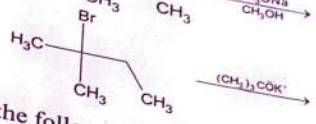
b)



c)

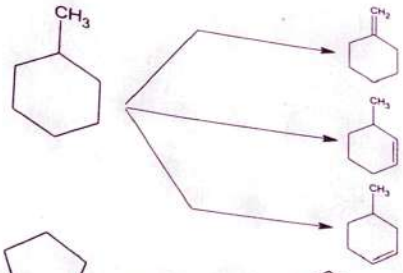


d)



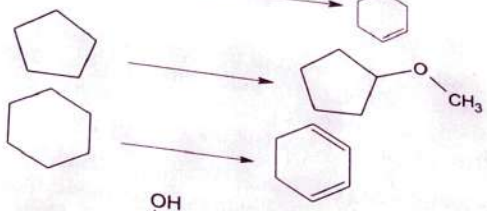
62. Carry out the following conversion

a)

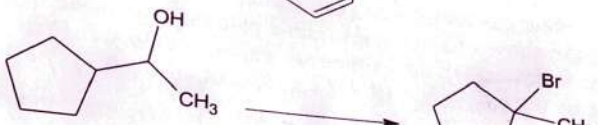


b)

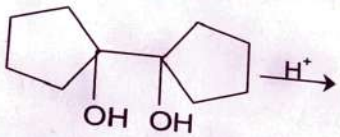
c)



d)

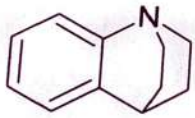


63.

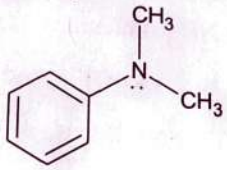


64. An optically active amine (A) is subjected to exhaustive methylation and Hofmann elimination to yield an alkene (B). (B) on ozonolysis gives an equimolar mixture of formaldehyde and butanal. Deduce the structures of (A) & (B). Is there any structural isomer to (A), if yes draw its structure.

65. Which one is more basic and why?

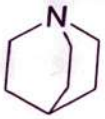


(a)



(b)

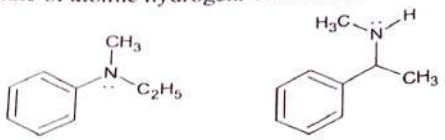
66. Which one is more nucleophilic and why?



(a)

(b)

67. Calculate the energy required to excite one litre of hydrogen gas at 1 atm and 298 K to the first excited state of atomic hydrogen. The energy for the dissociation of H-H bond is 436 kJ mol^{-1}.

68. 

(a) (b)

(a) cannot be resolved into enantiomers while (b) can, why?

69. Pyrrolidine can be readily extracted from an ether solution into an aqueous solution of acetic acid (pK_a = 4.75) but pyrrole cannot. Explain.

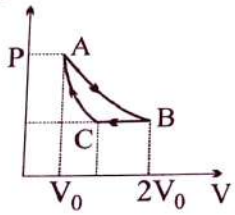
70. A pungent smelling liquid (A) was analysed and found to contain carbon, hydrogen, chlorine and possibly oxygen and upon reaction with aqueous ammonia a neutral compound (B) was formed. When this was treated with bromine and potassium hydroxide a base (C) resulted. In acidic solution, substance (C) reacted with sodium nitrite, giving nitrogen and an alcohol (D). Upon mild oxidation, the alcohol (D) was converted into compound (E) of M.F. C_2H_4O which gave a positive test with ammoniacal silver nitrate solution. Describe the chemistry involved in these reactions and identify the compounds (A), (B), (C), (D) and (E).

71. (a) Calculate the velocity (cm/sec) of an electron placed in the third orbit of the hydrogen atom. Also calculate the number of revolutions per second that this electron makes around the nucleus.
(b) An electron in a - H atom in its ground state absorbs 1.50 times as much as energy as the minimum required for its escape (13.6eV) from the atom. Calculate the wavelength of emitted electrons.

72. One mole of an ideal monoatomic gas initially at 1200 K and 64 atm is expanded to a final state at 300 K and 1.00 atm. To achieve the above change, a reversible path is constructed that involve an adiabatic expansion in the beginning followed by an isothermal expansion to the final state. Determine the magnitude of net work done by the gas, in cal. (R = 2 Cal/K-mol, $ln\ 2 = 0.7$)

73. In a cycle ABCA consisting of isothermal expansion AB, isobaric compression BC & adiabatic compression CA. Find the % (approximate) efficiency of cycle.

[Given : $T_A = T_B = 400$ K ; γ = 1.5, $ln\ 2 = 0.693$, $2^{\frac{1}{3}} = 1.26$]



74. In solid NH_3, each NH_3 molecule has six other NH_3 molecules as nearest neighbours. The enthalpy of sublimation of NH_3 (s) at its melting point is 30.4 kJ mol^{-1} and the estimated value of enthalpy of sublimation of NH_3 (s), when there were no hydrogen bonds, is 15.4 kJ mol^{-1}. What is the average strength of hydrogen bonds in solid NH_3 in kJ mol^{-1} ?

75. Draw a graph between degree of dissociation (α) Vs time for zero order reaction.

76. A 50 litre vessel is equally divided into three parts with the help of two stationary semi permeable membrane (SPM). The vessel contains 60 g H_2 gas in the left chamber, 160 g O_2 in the middle & 140 g N_2 in the right one. The left SPM allows transfer of only H_2 gas while the right one allows the transfer of both H_2 & N_2. Calculate the final ratio of pressure in the three chambers.

77. 5.22×10^{-4} moles of a gas containing H_2, O_2 and N_2 exerted a pressure of 67.4 mm in a certain volume. The gas was passed over a hot Pt-filament which combined H_2 and O_2 into H_2O which was frozen out. When the gas was returned to the same volume, the pressure was 14.3 mm. Extra O_2 was added to increase the pressure to 44.3 mm. The combustion was repeated after which the pressure read 32.9 mm. What was the mole fraction of H_2, O_2 and N_2 in the gas sample.

78. A flask containing same He gas at 1.5 atm & 300 K is connected to another flask, four times larger in volume containing N_2 at same temperature and pressure by means of narrow negligible volume tube. The bigger flask was then kept in a thermostat both maintained at 500 K, while other still at 300 K. Find final pressure.

79. Iron crystallizes in several modifications. At about 910°C, the body centred cubic α-form undergoes a transition to the face centred cubic γ-form. Assuming that the distance between nearest neighbours is same in the two forms at the transition temperature. Calculate the ratio of density of γ-iron to that of α-iron at the transition temperature.

80. Titanium crystallizes in a face centred cubic lattice. It reacts with C or H interstitially, by allowing atoms of these elements to occupy holes in the host lattice. Hydrogen occupies tetrahedral holes, but carbon occupies octahedral holes.
(a) Predict the formula of titanium hydride and titanium carbide formed by saturating the titanium lattice with "foreign" element.
(b) What is the maximum ratio of "foreign" atom radius to host atom radius that can be tolerated in a tetrahedral hole without causing a strain in the host lattice?

81. 5.35 g of a salt ACl (of weak base AOH) is dissolved in 250 ml of solution. The pH of the resultant solution was found to be 4.827. Find the ionic radius of A^+ and Cl^- if ACl forms CsCl type crystal having 2.2 g/cm^3. Given $K_{b(AOH)} = 1.8 \times 10^{-5}$, $\frac{r^+}{r^-} = 0.732$ for this cell unit

82. In a face centred lattice with all the positions occupied by A atoms the body centred octahedral hole in it is occupied by an atom B of an appropriate size for such a crystal, calculate the void space per unit volume of unit cell. Also predict the formula of the compound.

83. 0.10 mol sample of $AgNO_3$ is dissolved in 1.00 L of 1.00 M NH_3. If 0.010 mol of NaCl is added to this solution will AgCl(s) precipitate?

84. In hypophosphorus acid P is in +1 state (H_3PO_2) and in phosphoric acid (H_3PO_4) P is in +5 state but still H_3PO_2 is stronger than H_3PO_4. Explain.

85. On the basis of H-bonding explain that the second ionization constant K_2 for fumaric acid is greater than for maleic acid.

86. The fixing of photographic plate film involves dissolution of unexposed silver bromide in hypo solution according to the equation

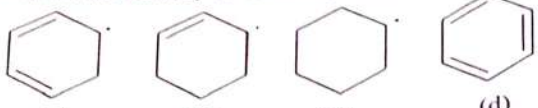
$$AgBr(s) + 2S_2O_3^{2-}(eq) \rightleftharpoons Ag(S_2O_3)_2^{3-}(aq) + Br^-(aq)$$

Calculate molar solubility of AgBr in 0.1 M hypo solution assuming its complete dissociation

K_{sp} of $AgBr = 4.5 \times 10^{-13}$

K_f of $Ag(S_2O_3)_2^{3-} = 2.0 \times 10^{13}$

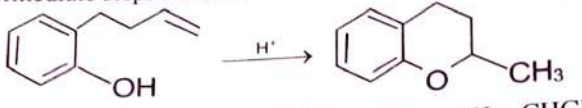
87. Explain the stability of following free radicals .



(a) (b) (c) (d)

88. An unsaturated hydrocarbon A (C_6H_{10}) readily gives B on treatment with $NaNH_2$ in liquid NH_3. When B is allowed to react with 1-chloropropane a compound C is obtained. On partial hydrogenation in the presence of Lindlar catalyst, compound C gives D (C_9H_{18}). On ozonolysis, D gives 2, 2-dimethyl propanal and 1-butanal. With proper reasoning give the structures of A, B, C and D.

89. Write the intermediate steps for each of the following reactions.

(i) 

(ii) $C_6H_5CH(OH)C \equiv CH \xrightarrow{H_3O^+} C_6H_5 - CH = CHCHO$

(iii) $CH_3 - \underset{\displaystyle CH_3}{\underset{|}{CH}} - CH_2OH \xrightarrow{HCl} CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle Cl}{\overset{|}{\underset{|}{C}}}} - CH_3$

90. A buffer solution contains 0.2 mole of acetic acid and 0.25 mole of potassium acetate per litre. Calculate the change of pH of the solution if 0.5 ml of 1 M HCl is added to it. The pK_a of CH_3COOH is 4.76. The volume change on the addition of HCl may be neglected.

91. A solution of mono basic acid (3×10^{-2} M) has a freezing point of depression of 0.06°C. Calculate pKa for the acid. (log 1.6992 = 0.2302) [Molal depression constant 1.86°C/m for water]

92. The e.m.f of cell $Zn_{(s)}|ZnSO_4||\ CuSO_4|Cu_{(s)}$ at 25°C is 0.03 V and temperature coefficient of e.m.f is (-1.4×10^{-4} V) per degree. Calculate the heat of reaction for the change taking place inside the cell.

93. A test for complete removal of Cu^{2+} ions from a solution of $Cu^{2+}_{(aq)}$ is to add NH_3(aq). A blue colour signifies the formation of complex $[Cu(NH_3)_4]^{2+}$ having $K_f = 1.1 \times 10^{13}$ and thus confirms the presence of Cu^{2+} in solution. 250 ml of 0.1M $CuSO_{4(aq)}$ is electrolyzed by passing a current of 3.5 ampere for 1350 seconds. Then sufficient quantity of $NH_{3(aq)}$ is added to the electrolyzed solution maintaining $[NH_3]$ = 0.1 M. If $[Cu(NH_3)_4]^{2+}$ is detectable upto its concentration as low as 1×10^{-5}, would a blue colour be shown by the electrolyzed solution on addition of NH_3?

94. Show that the reaction $CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$ at 300 K is spontaneous and exothermic. When standard entropy change is -0.094 KJ/mol^{-1}. The standard Gibbs free energy of formation for CO_2 and CO are -394.4 and -137.2 KJ mol^{-1}, respectively.

95. Give the valence bond description of $[Ni(CN)_4]^{-2}$ & $[NiCl_4]^{2-}$ i.e. hybridization, shape structure & magnetic moment of above complexes.

96. A, B & C are three complexes of chromium (III) with the empirical formula $Cr(H_2O)_6Cl_3$. These complexes have either water or water and chloride ions as ligands. Complexes A does not react with concentrated H_2SO_4, where as complexes B & C lose 6.75% & 13.5% of their original mass, respectively on treatment with conc. H_2SO_4. Identify A, B & C.

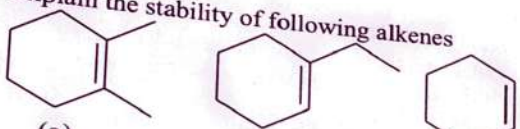
97. Compare and calculate the magnetic moment of Fe^{3+} in $[Fe(CN)_6]^{3-}$ and $[Fe(H_2O)_6]^{3+}$

98. Two metals A and B belongs to the same group of periodic table. Metal (A) forms an insoluble oxide (AO) but soluble metal sulphate (ASO_4) while B forms a soluble oxide and insoluble sulphate. The hydroxide of metal (A) is soluble in NaOH while hydroxide of (B) is insoluble in NaOH. Identify (A) and (B)

99. A white substance A reacts with dilute H_2SO_4 to produce a colourless gas B and a colourless solution C. The reaction between B and acidified $K_2Cr_2O_7$ solution produces a green solution and a slightly coloured precipitate D. The substance D burns in air to produce a gas E which reacts with B to yield D and a colourless liquid. Anhydrous copper sulphate is turned blue on addition of this colourless liquid. Addition of aqueous NH_3 or NaOH to C produces first a precipitate which dissolves in the excess of the respective reagent to produce a clear solution in each case. Identify A, B, C, D and E. Write the equation of the reactions involved.

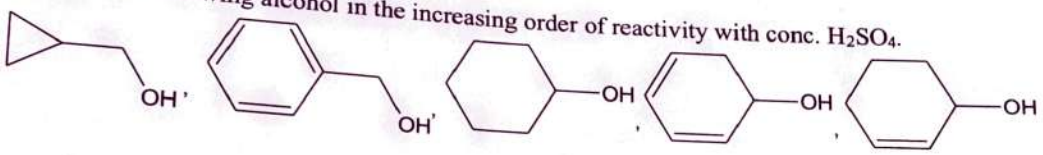
100. Identify the solid laboratory reagent (A) which imparts green colour to the flame and its solution does not forms precipitates on passing H_2S. When it is heated with $K_2Cr_2O_7$ and concentrated H_2SO_4, a red gas is evolved. The gas when passed in aqueous NaOH solution turns it yellow. Give chemical reactions involved.

101. Explain the stability of following alkenes



(a) (b) (c)

102. Arrange the following alcohol in the increasing order of reactivity with conc. H_2SO_4.



MCQ (SINGLE OPTION CORRECT)

1. The potential energy of an electron in the hydrogen atom is –6.8 eV. Indicate in which excited state, the electron is present?
(A) first (B) second
(C) third (D) fourth

2. What atomic number of an element "X" would have to become so that the 4th orbit around X would fit inside the 1st Bohr orbit of hydrogen atom ?
(A) 3 (B) 4
(C) 16 (D) 25

3. Two successive reactions, A $\longrightarrow$ B and B $\longrightarrow$ C, have yields of 82% and 65% respectively. What is the overall percent yield for conversion of A to C?
(A) 53.3 (B) 65
(C) 26.65 (D) Data insufficient

4. Reduction of V_2O_5, followed by addition of a strong base yields $Na_{12}V_{18}O_{42}.24H_2O$ on crystallization if the addition of strong base causes no change in oxidation state of vanadium (obtained after reduction) the equivalent weight for reduction of V_2O_5 in this case will be (if M denotes molecular weight of V_2O_5)
(A) M/2 (B) M/5
(C) M (D) Cannot determine

5. What is the C—C bond length in diamond unit cell if edge length of unit cell is 'a'
(A) $\frac{\sqrt{3}a}{4}$ (B) $\frac{\sqrt{3}a}{2}$
(C) $\sqrt{3}a$ (D) none

6. If two immiscible liquid 'A' and 'B' with vapour pressure P^0_A and P^0_B are taken together in a container, and 'A' forming upper layer then the vapour in the container will be
(A) P^0_B (B) P^0_A
(C) $P^0_A + P^0_B$ (D) can't predict

7. A tribasic weak acid dissociates as followes

$$H_3A \underset{}{\overset{Ka_1}{\rightleftharpoons}} H_2A^{-1} + H^+$$

$$H_2A^{-1} \overset{Ka_2}{\rightleftharpoons} HA^{-2} + H^+$$

$$HA^{-2} \overset{Ka_3}{\rightleftharpoons} H^+ + A^{3-}$$

What will be pH of 0.01M Na_2HA solution.
(A) $\frac{pKa_1 + pKa_2}{2}$ (B) $\frac{pKa_1 + pKa_2 + pKa_3}{3}$
(C) $\frac{pKa_2 + pKa_3}{2}$ (D) can't predict

8. Choose the correct statement of the following:
(A) The specific heat capacity at constant pressure for $H_2(g)$ is 7 cal/unit amount.
(B) The critical volume of a gas is three times of its excluded volume.
(C) Dalton's Law of partial pressure is applicable to the mixture of H_2 and He at all temperatures.
(D) In the diffusion process of a mixture of He and Ne the remaining gas in the container is enriched in Ne.

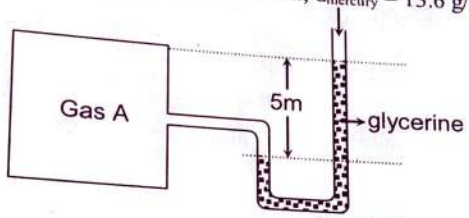
9. Out of following which one will have highest B.pt. ?
(A) 0.1 M aq. NaCl
(B) 0.1 M aq. CH_3COOH with $\alpha = 0.05$
(C) 0.1 M urea
(D) all have same B.pt.

10. The vapour pressure of a solution of 5 g of non-electrolyte in 100 g of water at particular temperature is 2985 Nm^{-2}. The vapour pressure of pure water at that temperature is 3000 Nm^{-2}. The molecular weight of the solute is
(A) 180
(B) 90
(C) 270
(D) 200

11. If vapour pressure of the two volatile liquids are given below. The mole fractions of the components in the idial solutions are also provided. Then which one will be most easily separated by fractional distillation.
(A) $p_A^0 = 100$ mm, $X_A = 0.5$; $p_B^0 = 90$ mm, $X_B = 0.5$
(B) $p_A^0 = 1000$ mm, $X_A = 0.4$; $p_B^0 = 100$ mm, $X_B = 0.6$
(C) $p_A^0 = 50$ mm, $X_A = 0.9$; $p_B^0 = 55$ mm, $X_B = 0.1$
(D) $p_A^0 = 200$ mm, $X_A = 0.7$; $p_B^0 = 201$ mm, $X_B = 0.3$

12. Nitrogen terms several gaseous oxides. One of them has a density of 1.33 gm/lit at 764 mm Hg and 150°C. Formula of the compounds is:
(A) NO
(B) N_2O
(C) NO_2
(D) N_2O_5

13. Find number of moles of gas A, present in the container of volume 10 L and temperature 300 K. Use the diagram shown below. [$d_{glycerine}$ = 2. 72 g/ml, P_{atm} = 1 atm, $d_{mercury}$ = 13.6 g/ml]



(A) 0.94
(B) 0.47
(C) 0.64
(D) 0.74

14. Which of the following is correct?
(A) De-Broglie wavelength associated to a gaseous particles is directly proportional to square-root of its molar mass.
(B) Slope of a plot for 'u_{avg} v/s T' for an ideal gas is –ve.
(C) The molar ratio of O_2 and SO_2 in a mixture is 32 : 1 in a container. It require 6-successive diffusion operations to get 2 : 1 molar ratio of O_2 and SO_2.
(D) R (molar gas constant) does not represents the work-done per degree per mole.

15. When the energy of activation is zero, K (rate constant) is equal to
(A) 0
(B) infinity
(C) A
(D) A^{-1}

16. In terms of mole fraction, equilibrium constant is written as K_x. The relationship between K_c and K_x is
(A) $K_c = K_x\left(\frac{RT}{P}\right)^{\Delta n}$
(B) $K_C = K_X \Big/ \left(\frac{P}{RT}\right)^{\Delta n}$
(C) $\frac{K_C}{K_X} = \left(\frac{RT}{P}\right)^{-\Delta n}$
(D) $K_X = K_C\left(\frac{P}{RT}\right)^{-\Delta n}$

17. For $HCl(aq) + NaOH(aq) \rightarrow NaCl(aq) + H_2O(l)$, $\Delta H = -13.6$ Kcal
for the reaction $\frac{1}{2}H_2SO_4(aq) + NaOH(aq) \rightarrow \frac{1}{2}Na_2SO_4(aq) + H_2O(\ell)$, ΔH is
(A) -6.8 Kcal (B) -13.6 Kcal
(C) +13.6 Kcal (D) +6.8 Kcal

18. The wave number of a given spectral like in the atomic spectrum of a hydrogen like species increases 9/4 times when deuterium nucleus is introduced into its nucleus, then which of the following will be the hydrogen like species?
(A) Li^{+2} (B) Li^{+}
(C) He^{+} (D) Be^{+3}

19. Among the following metal carbonyls, the C – O bond order is lowest in
(A) $[Mn(CO)_6]^+$ (B) $[Fe(CO)_5]$
(C) $[Cr(CO)_6]$ (D) $[V(CO)_6]^-$

20. Which of the following cell can produce more electrical work? (Consider same electrode pressure of H_2 gas in all cases)
(A) $Pt,H_2(g) \mid 0.2M\ NH_2\overset{\oplus}{N}H_3Cl^- \parallel 0.2\ M\ HCOOH \mid H_2(g),Pt$
(B) $Pt,H_2(g) \mid 0.05M\ H_2SO_4 \parallel 0.05\ M\ Ca(OH)_2 \mid H_2(g),Pt$
(C) $Pt,H_2(g) \mid 0.05M\ H_2SO_4 \parallel 0.05\ M\ Ca(CH_3COO)_2 \mid H_2(g),Pt$
(D) $Pt,H_2(g) \mid 0.2M\ CH_3COONa \parallel 0.2\ M\ HCl \mid H_2(g),Pt$

21. The standard oxidation potentials of the electrodes Ag/Ag^+, Sn/Sn^{2+}, Ca/Ca^{2+}, Pb/Pb^{2+} are –0.8, –1.36, 2.86 and 10.12V respectively. The most powerful oxidizing agent among these metals is
(A) Pb (B) Ca
(C) Sn (D) Ag

22. How many minutes will it take to plate out 5.0 g of Cr from a $Cr_2(SO_4)_3$ solution using a current of 1.50 amperes?
(Atomic wt. of Cr = 52)
(A) 254 (B) 30
(C) 152 (D) 103

23. For the reaction between CO_2 and graphite
$CO_2(g) + C(s, \text{graphite}) \rightleftharpoons 2CO(g)$ $\Delta H = 170$ kJ $\Delta S = 170$ J/K
Reaction is spontaneous at
(A) 1200K (B) 900 K
(C) 500 K (D) 298 K

24. The melting point of a solid is 27°C and its latent heat of fusion is 600 calories/mole the entropy change for the fusion of one mole of the solid in cal/K is
(A) 2.22 (B) 2
(C) 22.2 (D) 0.5

25. The correct order of increasing bond angle is :
(A) $H_2O > Cl_2O > F_2O$ (B) $Cl_2O > H_2O > F_2O$
(C) $F_2O > Cl_2O > H_2O$ (D) $F_2O > H_2O > Cl_2O$

26. Among the following species, identify the isostructural pairs (CO_2, N_3^-, N_2O, SO_2) :
(A) CO_2, N_3^-, N_2O (B) CO_2, N_2O, SO_2
(C) CO_2, N_3^-, SO_2 (D) CO_2, N_3^-, N_2O, SO_2

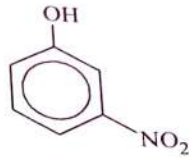
27. Which one of the following have fractional bond order ?
(A) CO^+ (B) CN^-
(C) N_2 (D) CO

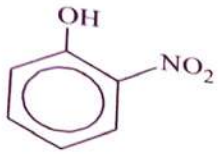
28. The number of lone pairs are same in which pair ?
(A) N_3^-, CO_2 (B) N_2O, CO_2
(C) N_3^-, N_2O (D) All

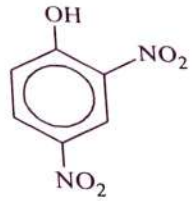
29. The ONO bond angle is maximum in :
(A) NO_3^- (B) NO_2^+
(C) N_2O_4 (D) NO_2

30. Which of the following don't react with $NaHCO_3$?

(A)

(B) 

(C) 

(D) 

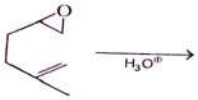
31. Which of the following can stabilize to O_2^+ ion more strongly?
(A) H^- (B) F^-
(C) O_2^- (D) $[PtCl_6]^-$

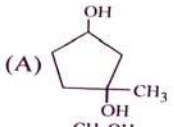
32. Select incorrect reduction process:
(A) $2[Ag(CN)_2]^- + Zn \rightarrow [Zn(CN)_4]^{2-} + 2Ag$ (B) $2FeO + C \xrightarrow{\Delta} Fe + CO$
(C) $8Al + 3Mn_3O_4 \xrightarrow{\Delta} 4Al_2O_3 + 9Mn$ (D) $2ZnS + C \xrightarrow{\Delta} 2ZnS + CS_2$

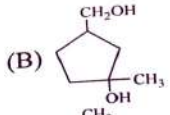
33. The pair of compounds, which cannot exist together in a solution is
(A) $NaHCO_3$ and NaOH (B) Na_2CO_3 and NaOH
(C) $NaHCO_3$ and Na_2CO_3 (D) $NaHCO_3$ and H_2O

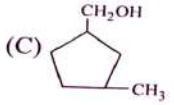
34. The compound $(SiH_3)_3N$ is expected to be
(A) pyramidal and more basic than $(CH_3)_3N$ (B) planar and less basic than $(CH_3)_3N$
(C) pyramidal and less basic than $(CH_3)_3N$ (D) planar and more basic than $(CH_3)_3N$

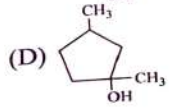
35. p-Cl – $C_6H_4NH_2$ and $PhNH_3^+Cl^-$ can be distinguished by
(A) NaOH (B) $AgNO_3$
(C) $LiAlH_4$ (D) Zn

36. 

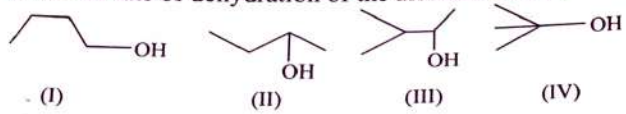
(A) 

(B) 

(C) 

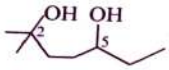
(D) 

37. Order of rate of dehydration of the alcohols will be



(I) (II) (III) (IV)

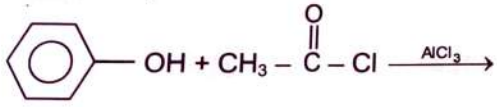
(A) IV > III > II > I
(B) I > II > III > IV
(C) IV > II > III > I
(D) II > IV > I > III

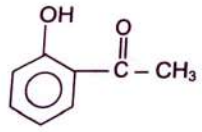
38.  In this diol

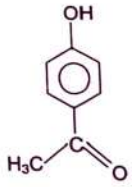
(A) OH at C_2 is more basic than that of at C_5
(B) OH at C_2 is more acidic than at C_5.
(C) both have same basicity
(D) both have same acidic strength

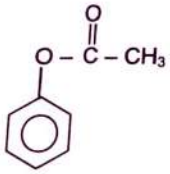
39. Phenol does not react with $NaHCO_3$ because
(A) phenol is a weaker acid than carbonic acid
(B) phenol is a stronger acid than carbonic acid
(C) phenol is as strong as carbonic acid
(D) phenol is insoluble in water.

40. Predict the product of the reaction below



(A) 

(B) 

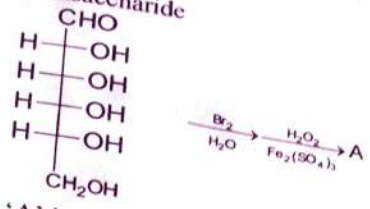
(C) 

(D) Both (A) and (B)

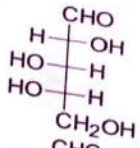
41. Acidic nature is more for
(A) o-amino phenol
(B) m-amino phenol
(C) p-amino phenol
(D) All have equal Ka's

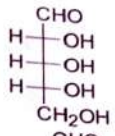
42. Anti Markovnikoffs addition of HBr is not observed in
(A) Propene
(B) 1-Butene
(C) 2-Butene
(D) Pentene-2

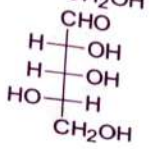
43. A carbohydrate which cannot be hydrolyzed to simpler compounds is called:
(A) Monosaccharide
(B) Polysaccharide
(C) Disaccharide
(D) Trisaccharide

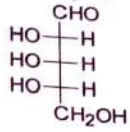
44. 

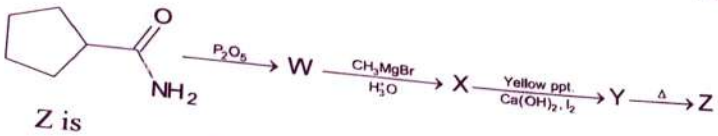
'A' is

(A) 

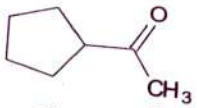
(B) 

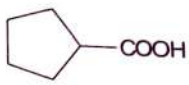
(C) 

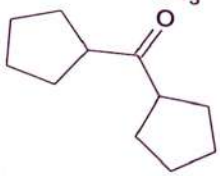
(D) 

45. 

Z is

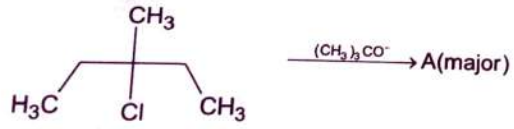
(A) 

(B) 

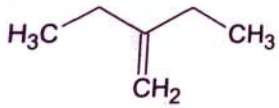
(C) 

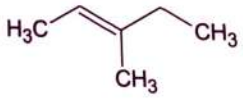
(D)

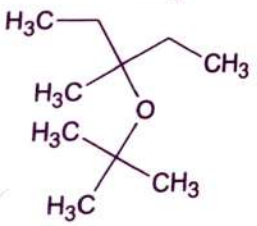
46. A nitrogeneous substance X is treated with HNO_2 and the product so formed is further treated with NaOH solution, which produces blue colouration. X can be
(A) $CH_3CH_2NH_2$
(B) $CH_3CH_2NO_2$
(C) CH_3CH_2ONO
(D) $(CH_3)_2CHNO_2$

47. 

What is A?

(A) 

(B) 

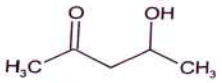
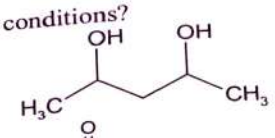
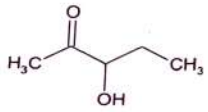
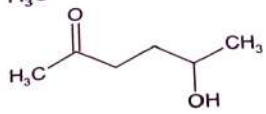
(C) 

(D) None

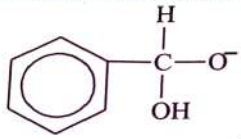
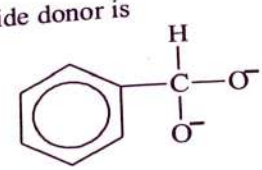
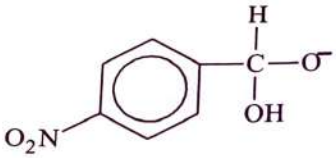
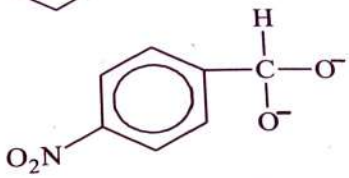
48. How many alcohols are possible with different boiling points with molecular formula $C_6H_{14}O$?
(A) More than 20
(B) Between 10 – 15
(C) Between 16 – 20
(D) < 10

49. Which of the following compound will have highest dipole moment?
(A) o-nitrophenol
(B) p-nitrophenol
(C) m-nitrophenol
(D) nitrobenzene

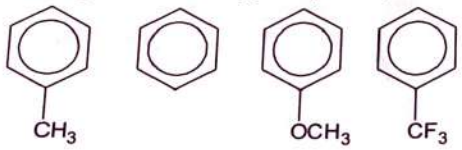
50. Which of the following will be most ready dehydrated in acidic conditions?

(A) 
(B) 
(C) 
(D) 

51. In a Cannizzaro reaction, the intermediate that will be best hydride donor is

(A) 
(B) 
(C) 
(D) 

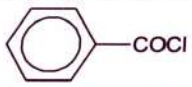
52. Among the following compound, the decreasing order of reactivity towards electrophilic substitution is



I II III IV

(A) III > I > II > IV
(B) IV > I > II > III
(C) I > II > III > IV
(D) II > I > III > IV

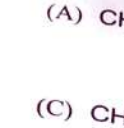
53. Consider the following compounds:

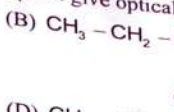
(I) 
(II)
(III)
(IV)

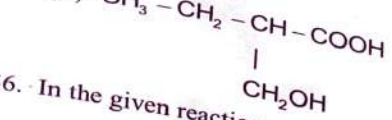
The correct decreasing order of their reactivity towards hydrolysis is
(A) (IV) > (III) > (I) > (III)
(B) (II) > (IV) > (I) > (III)
(C) (II) > (IV) > (III) > (I)
(D) (I) > (II) > (III) > (IV)

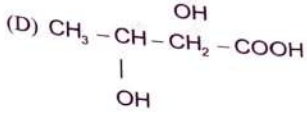
54. Which one of the following on heating gives unsaturated acid:
(A) α-Hydroxy acid
(B) β-Hydroxy acid
(C) γ-Hydroxy acid
(D) δ-Hydroxy acid

55. Which optically active compound on reduction with $LiAlH_4$ will give optically inactive compound?

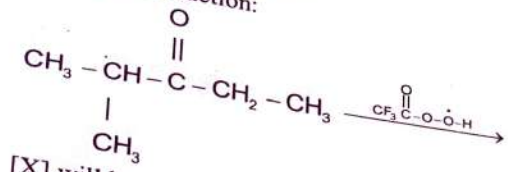
(A) 

(B) 

(C) 

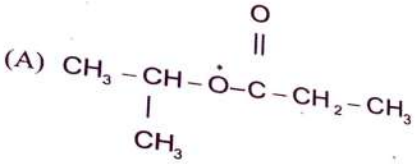
(D) 

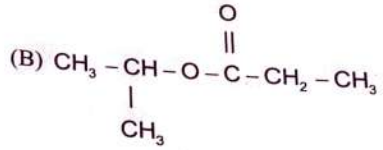
56. In the given reaction:

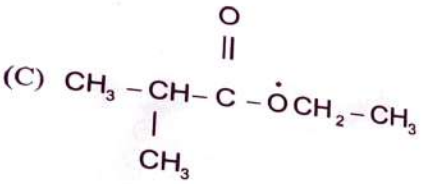


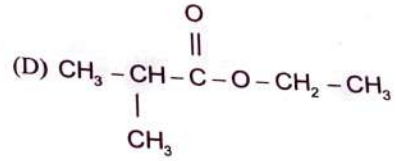
[X] will be:

[X] as main product

(A) 

(B) 

(C) 

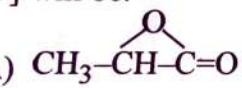
(D) 

57. Which acid can be oxidised by Fehling solution:
(A) Malonic acid
(B) Acetic acid
(C) Oxalic acid
(D) formic acid

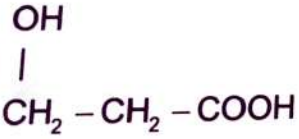
58. Which of the following compounds gives carbondioxide with $NaHCO_3$?
(A) Acetic acid
(B) Hexanol
(C) Phenol
(D) Acetylene

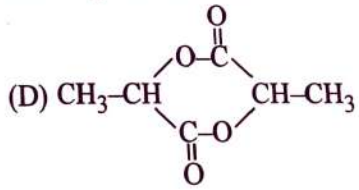
59. In the reaction sequence:

[Y] will be:

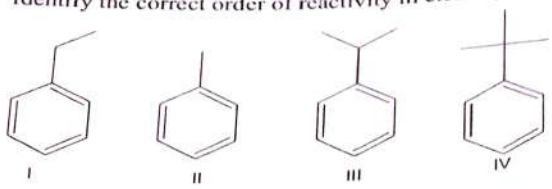
(A) 

(B) CH_2=CH–COOH

(C) 

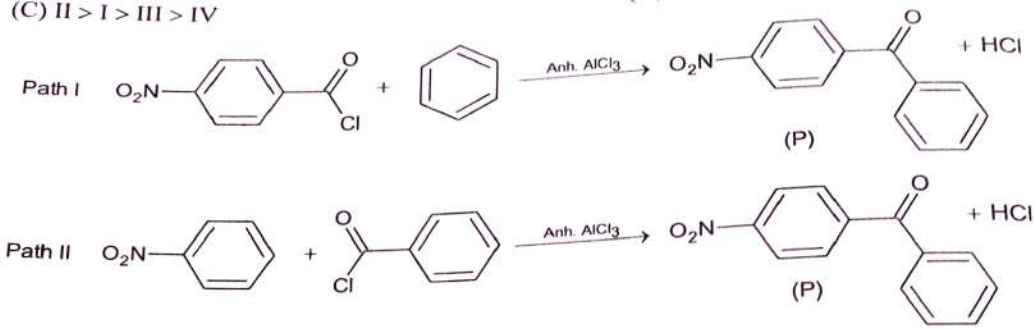
(D) 

60. Identify the correct order of reactivity in electrophilic substitution reactions of the following compounds:



(A) I > II > III > IV
(B) IV > III > II > I
(C) II > I > III > IV
(D) II > III > I > IV

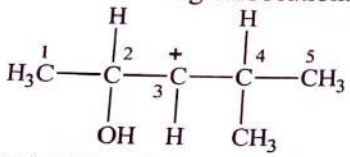
61.



The correct statement among the following about the above two paths of synthesizing the organic product P is:

(A) Path I is better than Path II
(B) Path II is better than Path I
(C) Both the paths are equally good
(D) Path I will give less yield of the product as compared to that obtained by part II

62. In the following carbocation. H/CH_3 that is most likely to migrate to the positively charged carbon is



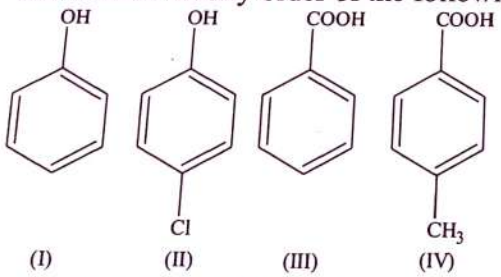
(A) CH_3 at C-4
(B) H at C-4
(C) CH_3 at C-2
(D) H at C-2

63. The correct stability order of the following resonance structures is

$H_2\overset{+}{C}-\overset{-}{N}=N$ (I) $H_2\overset{+}{C}-N=\overset{-}{N}$ (II) $H_2\overset{-}{C}-\overset{+}{N}\equiv N$ (III) $H_2\overset{-}{C}-N=\overset{+}{N}$ (IV)

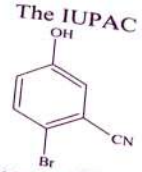
(A) (I) > (II) > (IV) > (III)
(B) (I) > (III) > (II) > (IV)
(C) (II) > (I) > (III) > (IV)
(D) (III) > (I) > (IV) > (II)

64. The correct acidity order of the following is



(A) (III) > (IV) > (II) > (I)
(B) (IV) > (III) > (I) > (II)
(C) (III) > (II) > (I) > (IV)
(D) (II) > (III) > (IV) > (I)

65. The IUPAC name of the following compound is



(A) 4-Bromo-3-cyanophenol
(B) 2-Bromo-5-hydroxybenzonitrile
(C) 2-Cyano-4-hydroxybromobenzene
(D) 6-Bromo-3-hdyroxybenzonitrile

66. Which of the following reagent is used for the separation of acetaldehyde from acetophenone –
(A) NH_2OH
(B) NaOI
(C) Tollens reagent
(D) $C_6H_5NHNH_2$

67. If N and S both are present in an organic compound, during Lassaigne's test both changes to
(A) Na_2S and NaCN
(B) NaSCN
(C) Na_2SO_3 and NaCN
(D) Na_2S and NaCNO

68. A blue colouration is not obtained when:
(A) Ammonium hydroxide dissolves in copper sulphate
(B) Copper sulphate solution reacts with $K_4[Fe(CN)_6]$
(C) Ferric chloride reacts with sodium ferrocyanide
(D) Anhydrous white $CuSO_4$ is dissolved in water

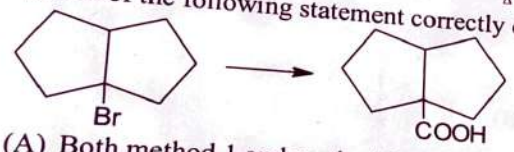
69. In a FCC lattice, atom 'A' occupy all the corners and face centres while atom 'B' is present at alternate tetrahedral voids present around 'A'. What is the co-ordination number of 'A'.
(A) 4
(B) 6
(C) 8
(D) none

70. Compare the two methods shown for the preparation of carboxylic acids:

Method # 1: $R-Br \xrightarrow[Et_2O]{Mg} R-MgBr \xrightarrow[(ii)\ H_3O^+]{(i)\ CO_2} RCO_2H$

Method # 2: $R-Br \xrightarrow{NaCN} R-CN \xrightarrow[\Delta]{H^+/H_2O} RCO_2H$

Which of the following statement correctly describes this conversion

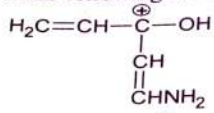


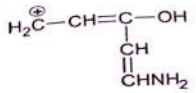
(A) Both method 1 and method 2 are appropriate for carrying out this conversion.
(B) Neither method 1 nor method 2 is appropriate for carrying out this conversion.
(C) Method 1 will work well, but method 2 is not appropriate.
(D) Method 2 will work well, but method 1 is not appropriate.

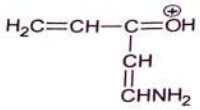
71. Which of the following sequence of reagents can be best utilized for the given transformation

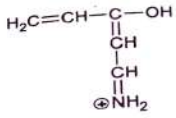
(A) Cl_2, AcOH; D_2O; $LiAlH_4/H_2O$
(B) D_2O/D_3O^+, Cl_2, ACOH; $NaBH_4/H_2O$
(C) $H_2C(SH)-H_2C(SH)$, base; D_2O and then $HgCl_2$ /$CdCO_3$ - H_2O
(D) Base; $H_2C(SH)-H_2C(SH)$ $HgCl_2/CdCO_3$ - H_2O and then D_2O

72. Which of the following is the most stable resonance structure

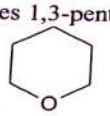
(A) 

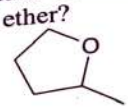
(B) 

(C) 

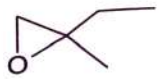
(D) 

73. A chiral $C_5H_{10}O$ ether reacts with hot HI to give a $C_5H_{10}I_2$ product. Treatment of this with hot KOH in ethanol produces 1,3-pentadiene. What is the structure of the original ether?

(A) 

(B) 

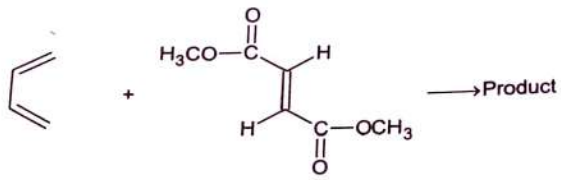
(C)

(D) 

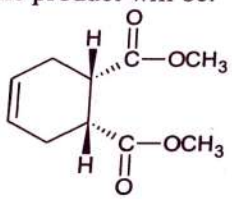
74. Which of the following on treatment with $CH \equiv CH$ produce gas(es):

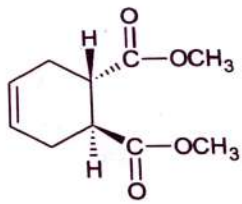
(i) $\bar{N}H_2$ (ii) $C\bar{H}_3$ (iii) $O\bar{H}$ (iv) Na in liq. NH_3

(A) (ii), (iii) and (iv)
(B) (i), (ii) and (iii)
(C) (i), (ii) and (iv)
(D) (i), (iii) and (iv)

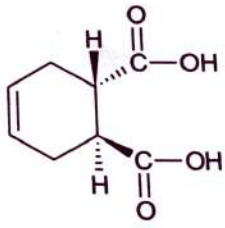
75. 

Here, the product will be:

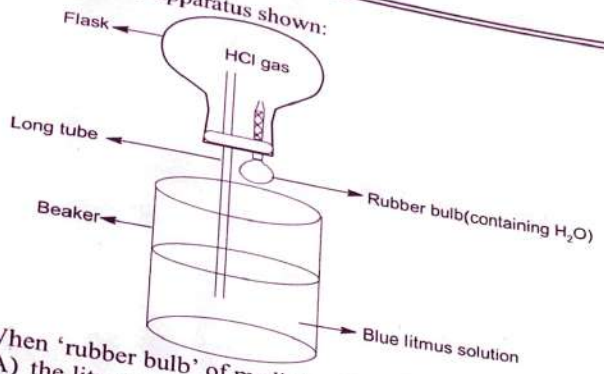
(A) 

(B) 

(C) Both (A) and (B) in equimolar ratio

(D)  + CH_3OH

76. Consider the apparatus shown:



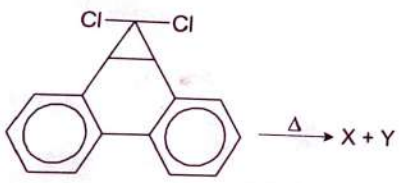
When 'rubber bulb' of medicine dropper will be squeezed:
(A) the litmus solution in beaker will turn red with a slight increase in volume
(B) the litmus solution in the beaker will turn red with slight decrease in volume
(C) the litmus solution will rush up through the long tube to create a blue fountain in the flask
(D) the litmus solution will rush up through the long tube to create a red fountain in the flask

77. ***Experiment # 1:*** A mixture of water and AgCl(s) is shaken until a saturated solution is obtained. Now, the solution is filtered and to 100 ml of the clear filterate, 100 ml of 0.003 M NaBr is added.
Experiment # 2: A mixture of water and AgBr (s) is shaken until a saturated solution is obtained. Now, the solution is filtered and to 100 ml of the clear filterate, 100 ml of 0.003 M NaCl is added.
Given K_{sp} (AgCl) = 10^{-10} and K_{sp}(AgBr) = 10^{-14} at 25°C; precipitation would happen in:
(A) Experiment 1 (B) Experiment 2
(C) Both experiments (D) Neither experiments

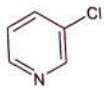
78. Consider the following equilibria:
I. $A(s) \rightleftharpoons B(s)$ II. $2A(\ell) \rightleftharpoons B(\ell)$
III. $A(s) \rightleftharpoons B(g)$ IV. $A(g) \rightleftharpoons B(g)$
Which of the above will not be disturbed by increase in pressure:
(A) I and IV (B) I, II and IV
(C) II, III and IV (D) IV only

79. A Buffer solution having pH 4.72 is prepared by mixing of 1M NaOH and 1M HCN. What is the ratio of volume of acid to base? (pk_b of CN^- = 9.28)
(A) 1 : 1 (B) 1 : 2
(C) 2 : 1 (D) 3 : 1

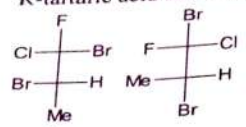
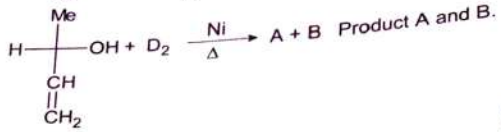
80.



Regarding X and Y which of the following statement is/are correct.
I. Both X and Y are aromatic.
II. One of them is anti aromatic.
III. All the carbon in X and Y are sp^2 hybridized.
IV. One of the above species can give addition reaction with Pyrrole and can produce

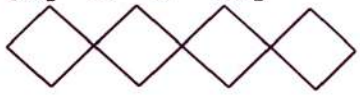


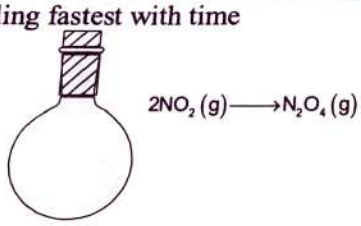
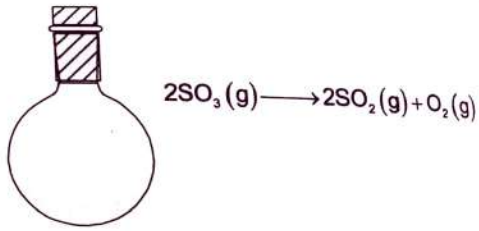
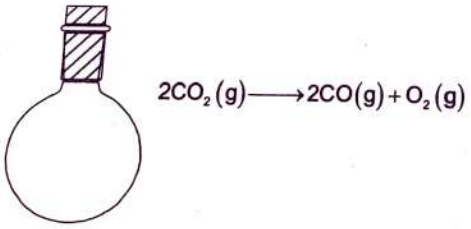
as a major product.
(A) I, III (B) I, III, IV
(C) III, IV (D) II, IV

81. Which of the following pairs are Enantiomers.
(i) D-Glucose and L-Glucose
(ii) R-tartaric acid and S-tartaric acid
(iii) 
(iv) 
(A) I, II
(B) I, II, III
(C) I, II, III, IV
(D) II, IV

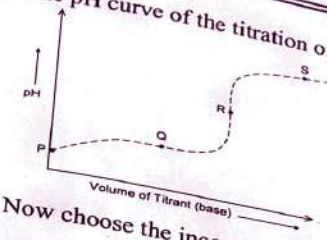
82. In which of the following reaction change of oxidation state of Br is more than unity.
(I) $H_2SO_4 + 2KBr \longrightarrow$
(II) $KBr + H_3PO_4 \longrightarrow$
(III) $Cl_2 + KBr + OH^- \longrightarrow$
(IV) $Br_2 + NaClO_3 \longrightarrow$
(A) I, II, III
(B) III, IV
(C) II, III, IV
(D) I, IV

83. In the following set of Aromatic compound (e.g. P, Q and R). The sum of the total no of position isomers. In all the given compound.
C_6H_4XY, $B_3N_3H_4XY$ C_6H_3XYZ
(P) (Q) (R)
(A) 17
(B) 16
(C) 18
(D) 15

84. Which of the following is planer molecule
(A) $CH_2 = C = CH_2$
(B) $CH_2 = C = C = CH_2$
(C)
(D) 

85. The flasks shown below are having gaseous reactions as indicated. Identify the flask in which density may be falling fastest with time
(A) 
(B) 
(C) 
(D) Density will remain constant in every case

86. The pH curve of the titration of weak acid with a strong base, is given below.



Now choose the incorrect option among the following:

(A) pH at point 'P' $= \frac{1}{2}pK_a - \frac{1}{2}\log[A_0]$, where A_0 is the initial concentration of weak acid.

(B) pH at point 'Q' $= pK_a - \frac{1}{2}\log\frac{[\text{weak acid}]}{[\text{Salt}]}$

(C) pH at point 'R' $= \frac{1}{2}pK_a + \frac{1}{2}pK_w + \frac{1}{2}\log[\text{salt}]$

(D) pH at point 'S' $= \frac{1}{2}pK_a + \frac{1}{2}\log[\text{Base}]$

87. Maximum yield of n-pentane can be obtained by which reaction?

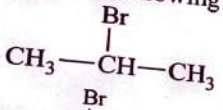
(A) $CH_3—CH_2—CH_2—Br + CH_3—CH_2—Br \xrightarrow[\text{dry ether}]{Na}$

(B) $CH_3—CH_2—CH_2—Br + CH_3—CH_2—Br \xrightarrow[\Delta]{Zn/C_2H_5OH}$

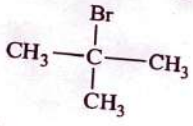
(C) $CH_3—CH_2—CH_2—Br + (CH_3—CH_2)_2CuLi \xrightarrow{\text{Ether}}$

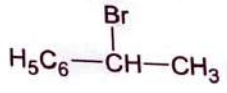
(D) $CH_3—CH_2—CH_2—COONa + CH_3—CH_2—COONa \xrightarrow{\text{Electrolysis}}$

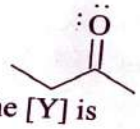
88. Which one of the following compounds is most reactive towards DMSO?

(A) 

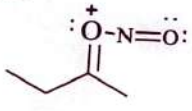
(B) $CH_3 - CH_2 - CH_2 - Br$

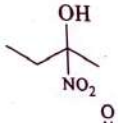
(C) 

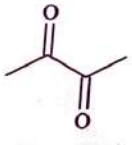
(D) 

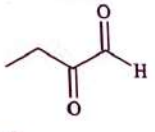
89. 

The [Y] is

(A) 

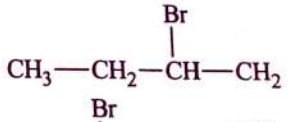
(B) 

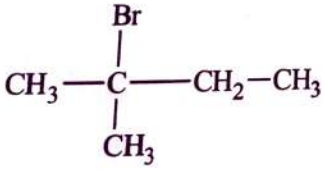
(C) 

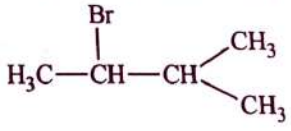
(D) 

90. 2-Butene is obtained by oxidation of which compound with DMSO?

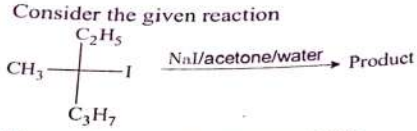
(A) $CH_3 - CH_2 - CH_2 - CH_2 - Br$

(B) 

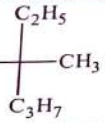
(C) 

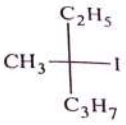
(D) 

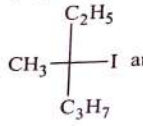
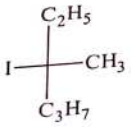
91. Consider the given reaction



The reaction mixture will contain(s):

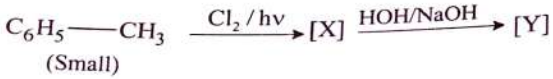
(A) Only 

(B) Only 

(C) Mixture of  and 

(D) No reaction will take place.

92. In the reaction sequence:



[X] and [Y] will respectively be:

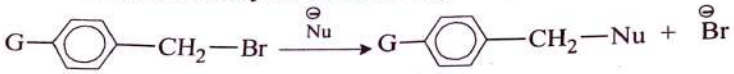
(A) C_6H_5—CH_2Cl and C_6H_5—CH_2OH

(B) C_6H_5—$CHCl_2$ and C_6H_5—CHO

(C) C_6H_5—CCl_3 and C_6H_5COOH

(D) $C_6H_5CCl_3$ and C_6H_5COONa

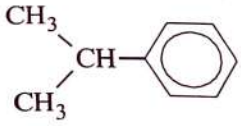
93. *Para*-substituted benzyl bromide undergoes S_N1 reaction with nucleophiles:

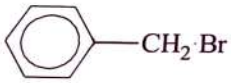


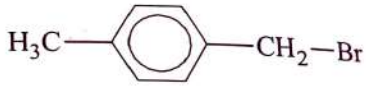
Arrange given four compounds in their decreasing order of reactivity for the above reaction:

(1)

(A)

(2) 

(3) 

(4) 

(A) 2 > 1 > 3 > 4

(B) 1 > 4 > 3 > 2

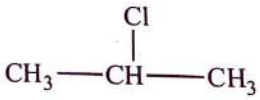
(C) 1 > 4 > 2 > 3

(D) 2 > 1 > 4 > 3

94. Predict the compound which will undergo S_N2 reaction in fastest rate :

(A)

(A) $CH_3 - CH_2 - CH_2 - Cl$

(B) 

(C) $CH_2 - CH_2 - \ddot{O} - CH_2 - CH_2 - Cl$

(D) $CH_3 - CH_2 - CH_2 - F$

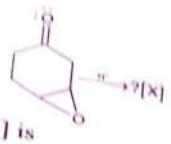
95. Two isomeric halo alkenes (A) and (B) have molecular formula C_5H_9Cl. (A) gives optically inactive compound while (B) gives optically active compound in hydrogenation. The two isomers respectively are:

(A) 3-chloro-1-pentene and 4-chloro-2-pentene

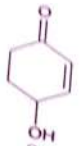
(B) 4-chloro-1-pentene and 2-chloro-2-pentene

(C) 3-chloro-2-pentene and 1-chloro-2-pentene

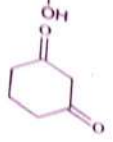
(D) 1-chloro-1-pentene and 5-chloro-1-pentene

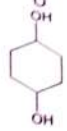
96. 

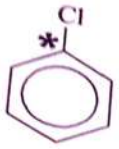
[X] is

(A) 

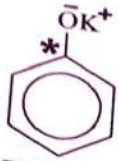
(B)

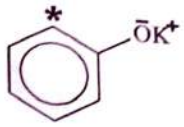
(C) 

(D) 

97. 

Product 'P' is

(A) 

(B) 

(C) Both (A) and (B)

(D) No reaction

98. Arrange the following halides in the decreasing order of S_N1 reactivity:

(I) $CH_3—CH_2—CH_2Cl$

(II) $CH_2=CH—CH(Cl)CH_3$

(III) $CH_3—CH_2—CH(Cl)CH_3$

(A) I > II > III

(B) II > I > III

(C) II > III > I

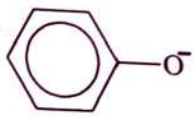
(D) III > II > I

99. The most reactive nucleophile among the following is:

(A) CH_3O^-

(B) $(CH_3)_3CO^-$

(C) $CH_3-\overset{\overset{O}{\|}}{C}-O^-$

(D) 

100. In the given reaction (X) is:

$$CH_3-CH_2-\underset{F}{CH}-CH_3 \xrightarrow{\overset{\ominus}{C}H_3O/CH_3OH} (X)$$

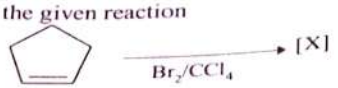
(A) $CH_3-CH_2-\underset{OCH_3}{CH}-CH_3$

(B) $CH_3 - CH = CH - CH_3$

(C) $CH_2 - CH_2 - CH = CH_2$

(D) $CH_3 - CH_2 - CH_2 - OCH_3$

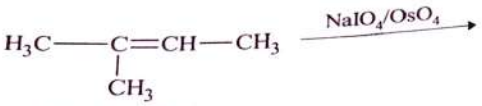
101. In the given reaction



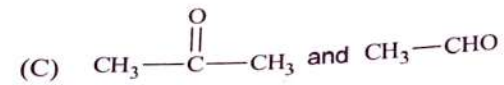
[X] will be:
(A) A *meso* form
(B) pair of diastereomes
(C) pair of enantiomers in which 'Br' atoms are trans to each other
(D) pair of enantiomers in which 'Br' atoms are cis to each other

102. When phenol is treated with excess Br_2 water, it gives
(A) m-bromophenol
(B) o and p- bromo phenol
(C) 2,4- dibromo phenol
(D) 2,4,6-tri bromo phenol

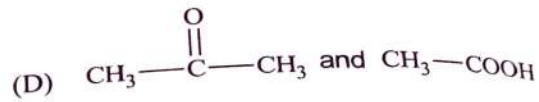
103. Products of the reaction which is given below will be



(A) *Vic* diol
(B) *Vic* dicarbonyl compound



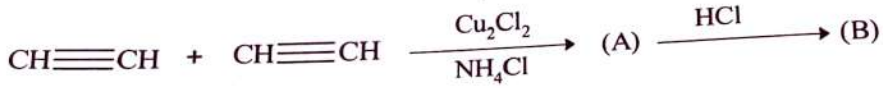
(C) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ and CH_3-CHO



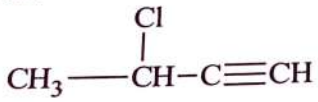
(D) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ and CH_3-COOH

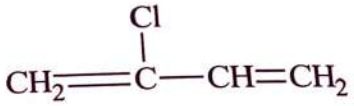
104. The freezing point of one lit of an aqueous solution of non electrolyte having an osmotic pressure of 2 atm at 300 K, should be
(A) 272.84 K
(B) - 0.151 K
(C) 235 K
(D) 298 K

105. In the reaction sequence:



(B) will be:

(A) 

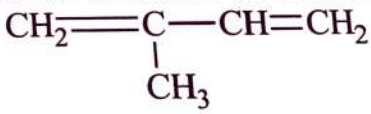
(B) 

(C) $CH_2=CH-C\equiv C-Cl$

(D) $CH_3-CH_2-C\equiv C-Cl$

106. Reductive ozonolysis of allene, $CH_3 - CH = C = CH_2$, will give:
(A) Only CH_3—CHO
(B) Only CH_2O
(C) Only CO_2
(D) Mixture of CH_3CHO, HCHO and CO_2

107. Which of the following will yield a mixture of 2-chlorobutene on treatment with HCl?
(A) $CH_2 = C = CH - CH_3$

(B) 

(C) $CH_2 = CH - CH = CH_2$

(D) $CH\equiv C-CH=CH_2$

108. O—Xylene $\xrightarrow[\text{ozonolysis}]{\text{Reductive}}$ Products possible products are :

(A) $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-CH_3$, $OCH-CHO$

(B) $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H$, $OCH-CHO$

(C) $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-CH_3$, $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H$

(D) $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H$, $CH_3-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H$, $H-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H$

109. An alkene gives two moles of HCHO, one mole of CO_2 and one mole of CH_3COCHO on ozonolysis. What is its structure?

(A) $CH_2=CH-\underset{\underset{CH_3}{|}}{CH}-CH=CH_2$

(B) $CH_2 = C = CH - CH_2 - CH_3$

(C) $CH_2=CH-\underset{\underset{CH_3}{|}}{CH}-CH_3$

(D) $CH_2=C=\underset{\underset{CH_3}{|}}{C}-CH=CH_2$

110. In the given reaction

(A) $\xleftarrow{H_2/Pd\text{-}BaSO_4}$ $CH_3-C\equiv C-CH_3$ $\xrightarrow{Na/NH_3(l)}$ (B)

(A) and (B) are respectively

(A) *cis*, *trans*-2-butene

(B) both *trans*-2-butene

(C) *trans*, *cis*-2-butene

(D) both *cis*-2-butene

111. When 4.0 g of a metal oxide M_xO_y were reduced in hydrogen. 2.8 g of the metal remained. What was the formula of the metal oxide if the relative atomic mass of the metal M was 56?

(A) MO

(B) M_2O

(C) M_2O_3

(D) M_3O_4

112. The oxidation state of Cr in Na_3CrO_8 is

(A) 5

(B) $+\frac{8}{3}$

(C) 6

(D) 3

113. The wave number of electromagnetic radiation emitted during the transition of electron in between two levels of Li^{2+} ion whose principal quantum numbers sum is 4 and difference is 2 is

(A) 3.5 R_H

(B) 4 R_H

(C) 8 R_H

(D) $\frac{8}{9} R_H$

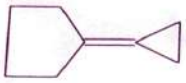
114. 1 mole of a gaseous aliphatic compound $C_nH_{3n}O_m$ is completely burnt in an excess of oxygen. The contraction in volume at room temperature is (consider water is in liquid state)

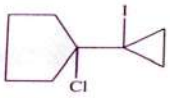
(A) $\left(1+\frac{1}{2}n-\frac{3}{4}m\right)$

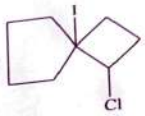
(B) $\left(1+\frac{3}{4}n-\frac{1}{4}m\right)$

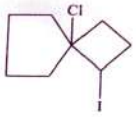
(C) $\left(1-\frac{1}{2}n-\frac{3}{4}m\right)$

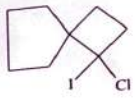
(D) $\left(1+\frac{3}{4}n-\frac{1}{2}m\right)$

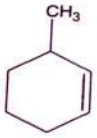
115.  + ICl → Thermodynamic product is

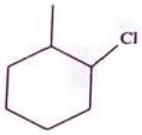
(A) 

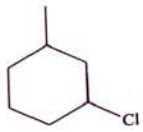
(B) 

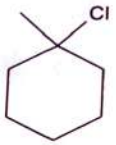
(C) 

(D) 

116.  $\xrightarrow{HCl}$ Major Product is

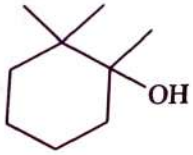
(A) 

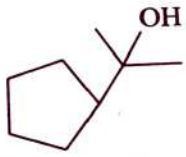
(B) 

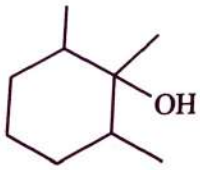
(C) 

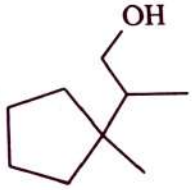
(D)

117. $\xrightarrow{H^+/H_2O}$ Product is

(A) 

(B) 

(C) 

(D) 

118. Between p-nitrophenol and saliylaldehyde, the solubility in base is

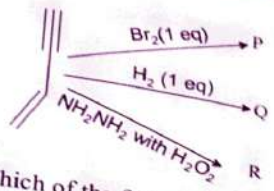
(A) equal in both case

(B) higher in p-nitrophenol

(C) higher in salicyl aldehyde

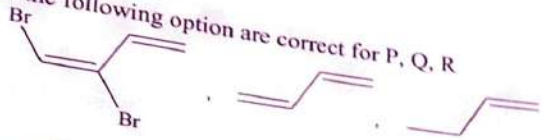
(D) Both are insoluble

119.



Which of the following option are correct for P, Q, R

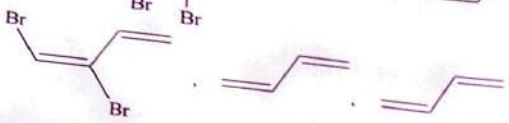
(A)



(B)

(C)

(D)



120. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with the $AgNO_3$ gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y'

(A) $X = CO_2$, $Y = Cl_2$
(B) $X = Cl_2$, $Y = CO_2$
(C) $X = Cl_2$, $Y = H_2$
(D) $X = H_2$, $Y = Cl_2$

121. A solution contains Na_2CO_3 and $NaHCO_3$. 10 ml of this solution required 2.5 ml of 0.1 M H_2SO_4 for neutralization using phenolphthalein as indicator. Methyl orange is then added when a further 2.5 ml of 0.2 M H_2SO_4 was required. The amount of Na_2CO_3 and $NaHCO_3$ respectively, in 1litre of solution is

(A) 5.3 g and 4.2 g
(B) 3.3 g and 6.2 g
(C) 4.2 g and 5.3 g
(D) 6.2 g and 3.3 g

122. For the reaction

$$2NO_{2(g)} \rightleftharpoons 2NO_{(g)} + O_{2(g)},$$

$(K_c = 1.8 \times 10^{-6}$ at $184°C)$

When k_p and k_c are compared at $184°C$, it is found that

(A) K_p is greater than K_c
(B) K_p is less than K_c
(C) $K_p = K_c$
(D) whether K_p is greater than, less than or equal to K_c depends upon the total gas pressure

123. The ionization enthalpy of hydrogen atom is 1.312×10^6 $Jmol^{-1}$. The energy required to excite the electron in the atom from n = 1 to n = 2 is

(A) 8.51×10^5 $Jmol^{-1}$
(B) 6.56×10^5 $Jmol^{-1}$
(C) 7.56×10^5 $Jmol^{-1}$
(D) 9.84×10^5 $Jmol^{-1}$

124. The solubility of calomel($Hg_2Cl_2(s)$), i.e. $Hg_2Cl_2(s) \rightleftharpoons Hg_2^{2+}(aq) + 2Cl^{-1}(aq)$ in water at 25°C is x moles/litre. Its solubility product is
(A) $4x^3$ (B) $12x^3$
(C) $108x^3$ (D) x^2

125. For the reaction $FeC_2O_4 + KMnO_4 \xrightarrow{H^+} Fe^{3+} + CO_2 + Mn^{2+}$. Equivalent weight of FeC_2O_4 will be, if M is molecular weight of FeC_2O_4
(A) M/3 (B) M/2
(C) M/1 (D) M/5

126. 10 ml of H_2O_2 solutions (volume strength = x) required 10 ml of $\frac{M}{0.56}MnO_4^-$ solution in acidic medium. Hence x is
(A)0.56 (B) 5.6
(C)0.1 (D) 50

127. A substance $A_xB_yC_zD_a$ crystallizes in close packed structure. A atoms are present at the corners, B at the centre of cube, C at alternate face & D at alternate edge. What is the formula of compound?
(A)$A_2B_2C_3D_3$ (B) $A_2B_2C_3D_2$
(C)ABCD (D) none of the above in correct

128. If 0.965 ampere current is passed through the aqueous solution of CH_3COONa at 25°C and 1 atm pressure, for one hour, total volume of gases evolved is
(A) 1.762 litre (B) 2.762 litre
(C) 0.762 litre (D) 0.075 litre

129. The entropy change in the fusion of 1 mole of solid melting at 27°C (latent heat of fusion, 2930 J mol^{-1}) is
(A) 9.77 J k^{-1} mol^{-1} (B) 10.73 J k^{-1} mol^{-1}
(C) 2930 J k^{-1} mol^{-1} (D) 108.5 J k^{-1} mol^{-1}

130. Hydrogen sample is in first excited state. A photon of energy 4.9 eV in used to excite the hydrogen sample. Then the de-Broglie wavelength of the electron is
(A)5.5 Å (B) 1 Å
(C)10 Å (D) 100 Å

131. The difference between heat of reaction at constant pressure and constant volume for the reaction given below at 25°C in kJ is

$$3C(s) + 4H_2(g) \longrightarrow C_3H_8(g)$$

(A) −7.43 (B) +3.72
(C) −3.72 (D) 7.43

132. The equilibrium constant for the reaction

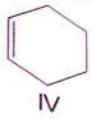
$$2SO_{2(g)} + O_{2(g)} \rightleftharpoons 2SO_{3(g)}$$

is 5. If the equilibrium mixture contains equal moles of SO_3 and SO_2, the equilibrium partial pressure of O_2 gas is
(A)0.2 atm (B) 2 atm
(C)0.02 atm (D) 0.04 atm

133. One litre of a buffer solution containing 0.01 M NH_4Cl and 0.1 M NH_4OH having pK_b of 5 has pH of
(A)9 (B) 10
(C)4 (D) 6

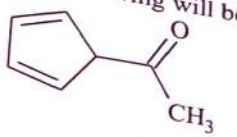
134. Which of the following will have the lowest heat of hydrogenation?

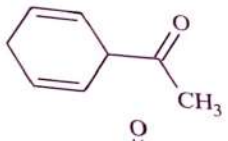
1 – hexene, (I) 2 – hexene, (II) 3 – hexene, (III)

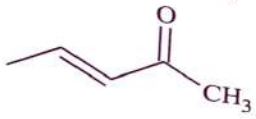

IV

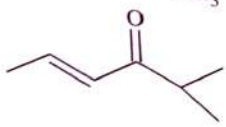
(A) I
(B) II
(C) III
(D) IV

135. Which of the following will be most acidic ?

(A) 

(B) 

(C) 

(D) 

136. At the equilibrium of the reaction $N_2O_4(g) \rightleftharpoons 2NO_2(g)$ the observed molar mass of N_2O_4 is 77.7 gm. The percentage dissociation of N_2O_4 is

(A) 28.4
(B) 46.7
(C) 22.4
(D) 18.4

137. Copper crystallises in fcc lattice with a unit cell edge of 361 pm. The radius of copper atom is:

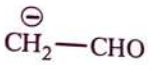
(A) 157 pm
(B) 181 pm
(C) 108 pm
(D) 128 pm

138. In a reaction, $OI^- + Cl^- \xrightarrow{OH^-} OCl^- + I^-$, rate $= \dfrac{K[OI^-][Cl^-]}{[OH^-]}$ the role of OH^- is

(A) catalyst
(B) reagent
(C) inhibitor
(D) none

139. Which of the following is the most stable carbanion?

(I)


(II)

(III)

(IV)

(A) I
(B) II
(C) III
(D) IV

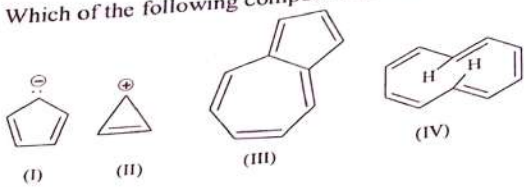
140. For the following equilibria in a close container

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

At a constant temperature, the volume of the container is halved suddenly. Which of the following is correct for K_p and α ?

(A) Both K_p and α changes
(B) None of the K_p and α change
(C) K_p changes but α does not
(D) α - changes but K_p does not

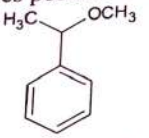
141. Which of the following compound is not aromatic?

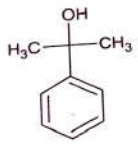


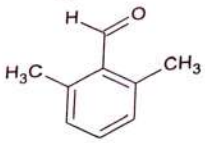
(A) I
(B) II
(C) III
(D) IV

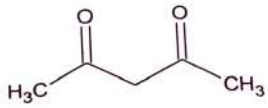
142. $FeCl_3$ gives positive test with
(A) Phenol
(B) Actylacetone
(C) Acetoacetic ester
(D) All of the above

143. IO^- gives positive test with

(A) 

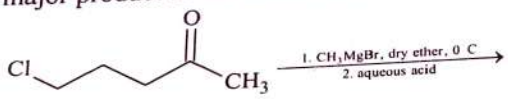
(B) 

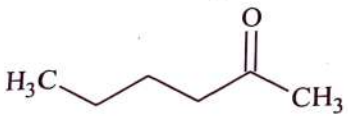
(C) 

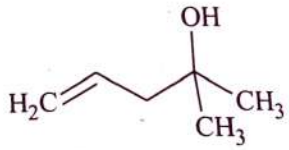
(D) 

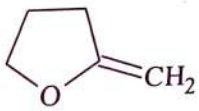
144. Methyl cyclohexene cannot be converted to 1-methyl cyclohexanol by
(A) acid catalysed hydration
(B) oxymercuration followed by reduction
(C) hydroboration followed by oxidation
(D) epoxidation followed by $LiAlH_4$ reduction

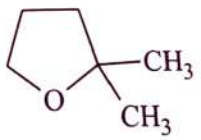
145. The major product in the following reaction is



(A) 

(B) 

(C) 

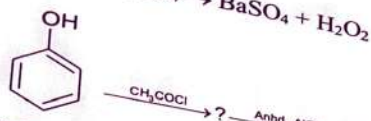
(D) 

146. On conversion into Grignard reagent followed by treatment with ethanol, how many alkyl chlorides will give 2- methyl butane?
(A) 2
(B) 3
(C) 5
(D) 4

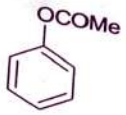
147. For the preparation of t-butyl methyl ether the correct choice of reagent is
(A) methoxide and t-butyl bromide
(B) methanol and 2-bromobutane
(C) 2 butanol and methyl bromide
(D) t-butoxide and methyl bromide

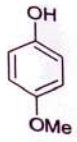
148. Which of the following is a redox reaction?
(A) $N_2O_5+H_2O \rightarrow 2HNO_3$
(B) $AgNO_3 + KI \rightarrow AgI + KNO_3$
(C) $BaO_2 + H_2SO_4 \rightarrow BaSO_4 + H_2O_2$
(D) $SnCl_2 + HgCl_2 \rightarrow SnCl_4 + Hg$

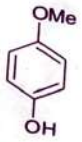
149.

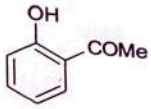


P is

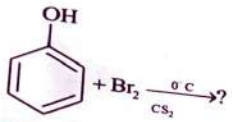
(A) 

(B) 

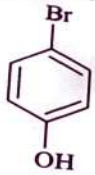
(C) 

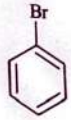
(D) 

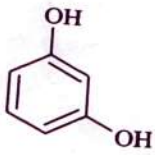
150.



The product is

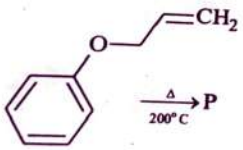
(A) 

(B) 

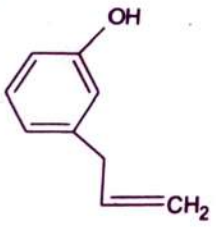
(C) 

(D)

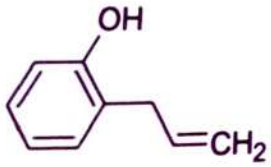
151.



P is

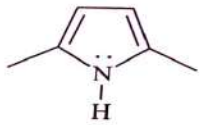
(A) 

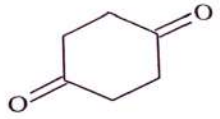
(B)

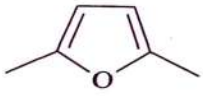
(C) 

(D) None of the above

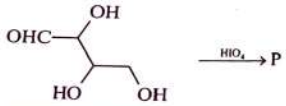
152. $\xrightarrow[100-115°C]{(NH_4)_2CO_3}$ P ; Product P is

(A) 

(B) 

(C) 

(D) None of the above

153. 

The products are

(A) 3HCOOH + HCHO

(B) 2HCHO + 2HCOOH

(C) 3HCHO

(D) CO_2 + 2HCHO + HCOOH

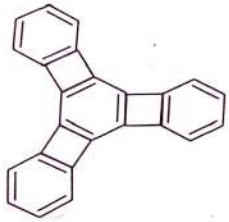
154. In which of the following ionization processes, the bond order has increased and the magnetic behaviour has changed?

(A) $N_2 \longrightarrow N_2^+$

(B) $C_2 \longrightarrow C_2^+$

(C) $NO \longrightarrow NO^+$

(D) $O_2 \longrightarrow O_2^+$

155. 

For 1 mole of the above hydrocarbon, how many moles of H_2 can be added to it at low pressure conditions in presence of Pd catalyst:

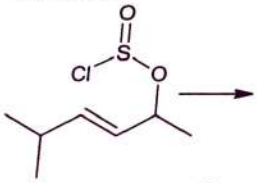
(A) 0

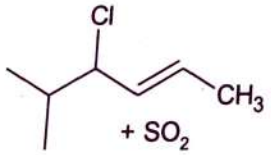
(B) 1

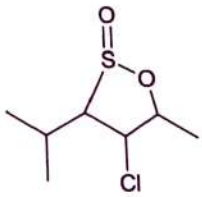
(C) 2

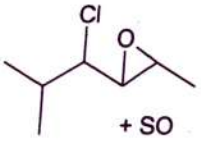
(D) 3

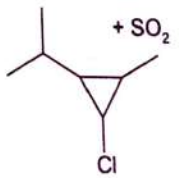
156. Which of the following is the most likely product from the reaction illustrated by the curved arrows in the formula?

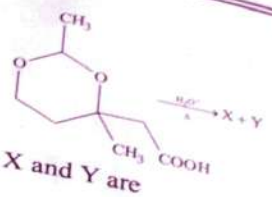


(A) 

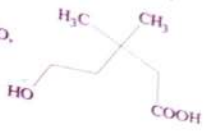
(B) 

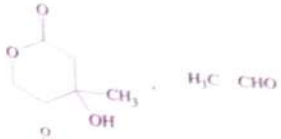
(C) 

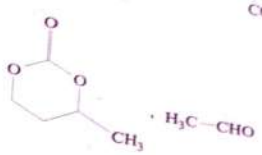
(D) 

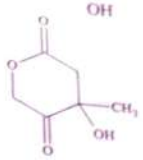
157. 

X and Y are

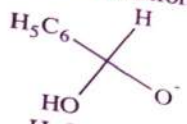
(A) H_3C-CHO, 

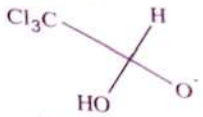
(B) 

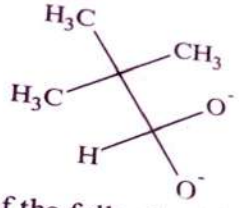
(C) 

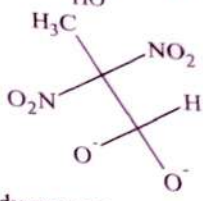
(D) 

158. In Cannizzaro reaction, the intermediate which is best hydride donor is

(A) 

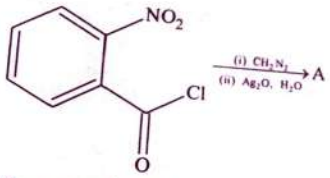
(B) 

(C) 

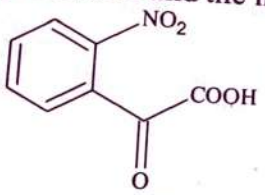
(D) 

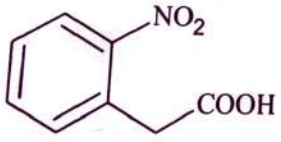
159. Which of the following on oxidative ozonolysis will NOT produce a gaseous product at room temperature:

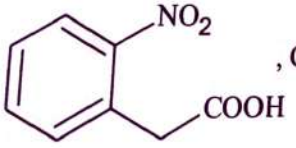
(A) Propadiene
(B) Propene
(C) Cyclohexene
(D) 1, 3-butadiene

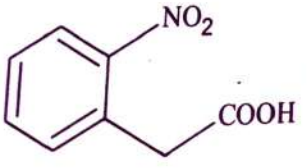
160. 

The product A and the intermediate of this reaction is

(A)  , Carbene

(B)  , Carbanion

(C)  , Carbanion and carbene both

(D)  , Carbene only

161. The rate of diffusion of a gas X is 2 time of gas Y and 0.5 times of gas W then which of the following is true for these three gases
(A) Rate of diffusion of W is 8 times of rate of diffusion of Y
(B) Mass of gas W is 8 times the mass of gas Y
(C) Mass of gas Y is 2 times the mass of gas X
(D) Rate of diffusion of Y is $\frac{1}{4}^{th}$ of rate of diffusion of gas W.

162. ClO_3^- shows
(A) diamagnetism (B) paramagnetism
(C) even number electrons in the structure (D) none of the above

163. In which of the following compound S–S bond is present
(A) $H_2S_2O_5$ (B) $H_2S_2O_7$
(C) $H_2S_2O_8$ (D) both (A) and (C)

164. In P_4O_{10}, the number of oxygen atoms attached to each phosphorus atom is
(A) 3 (B 2
(C) 4 (D) 2.5

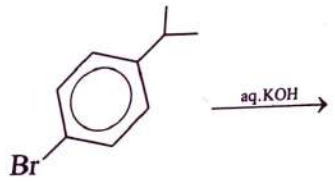
165. Passing H_2S gas through nitric acid produces
(A) rhombic sulphur (B) monoclinic sulphur
(C) amorphous sulphur (D) plastic sulphur

166. HI cannot be prepared by the reaction of NaI with conc. H_2SO_4 because
(A) NaI is chemically inert towards conc. H_2SO_4.
(B) NaI reacts with conc. H_2SO_4 to form HI which being a strong reducing agent, is oxidized by conc. H_2SO_4 (OA) to give iodine.
(C) NaI being considerably covalent does not pass into aqueous solution.
(D) NaI is oxidized by conc. H_2SO_4 to give $NaIO_3$.

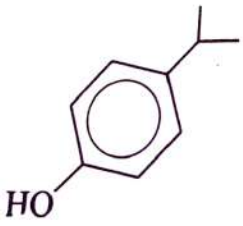
167. In the following reaction,
$B(OH)_3 + H_2O \longrightarrow [B(OH)_4]^- + H^+$
(A) $B(OH)_3$ is a Lewis acid (B) $B(OH)_3$ is a Lewis base
(C) $B(OH)_3$ is amphoteric (D) None of the above

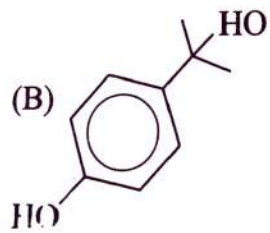
168. Borax is converted into B by the following steps
$\text{Borax} \xrightarrow{I} H_3BO_3 \xrightarrow{\Delta} B_2O_3 \xrightarrow{II} B$
I and II reagents are
(A) acid, Al (B) acid, C
(C) acid, Fe (D) acid, Mg

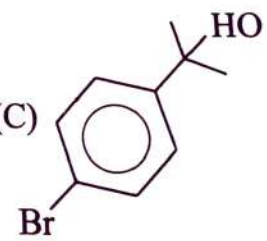
169.



The end product in the above reaction is

(A)


(B)


(C)


(D) None of the above

170. The molecular shapes of SF_4, SiF_4 and XeF_4 are:
(A) the same, with 2, 0 and 1 lone pair of electrons respectively
(B) the same, with 1, 1 and 1 lone pair of electrons respectively
(C) different with 0, 1 and 2 lone pair of electrons respectively
(D) different with 1, 0 and 2 lone pair of electrons respectively

171. Total number of geometrical isomers formed by Mabcd is
(A) three
(B) four
(C) six
(D) zero

172. The general formula of cyclic or ring silicates is
(A) $(Si_2O_5)_n^{2n-}$
(B) $(SiO_3)_n^{2n-}$
(C) $(Si_2O_3^{2-})_n$
(D) both (B) and (C)

173. Oxidation of hydrogen halide, HX, affords a method for industrial and laboratory preparation of the halogen, X_2, in the free state with respect to all of the following except
(A) fluorine
(B) chlorine
(C) bromine
(D) iodine

174. For which one of the following ions the colour is not due to a d-d transition?
(A) CrO_4^{2-}
(B) $Cu(NH_3)_4^{2+}$
(C) $Ti(H_2O)_6^{3+}$
(D) CoF_6^{3-}

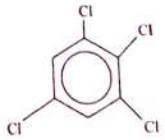
175. Sodium nitroprusside when added to an alkaline solution of sulphide ions produces purple colouration due to the formation of
(A) $Na_4[Fe(CN)_5NOS]$
(B) $Na_3[Fe(CN)_5NOS]$
(C) $Na_2[Fe(H_2O)_5NOS]$
(D) $Na[Fe(H_2O)_5NOS]$

176. What will be the volume strength of 100 ml of H_2O_2 required to react completely with 61 ml of $KMnO_4$ in acidic medium? (Given that 61 ml of $KMnO_4$ reacts completely with 5 ml of 1 M of $K_4[Fe(CN)_6]$ where it converts into K^+, Fe^{3+}, CO_3^{2-} and NO_3^-)
(A) 17.08
(B) 34.2
(C) 18.8
(D) 19.8

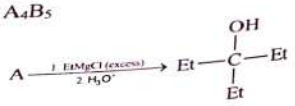
177. 10 g of oxalate was dissolved in 300 ml of solution. This solution required 250 ml of $\left(\frac{M}{10}\right)$ $KMnO_4$ in acidic medium for complete oxidation. The percentage purity of oxalate ion in the salt is
(A) 55%
(B) 57.8%
(C) 45%
(D) None of the above

178. One mole of water vapourizes at 373 K under atmospheric pressure. The work done is
(A) +3140.3 J
(B) −3140.3 J
(C) 101.3 J
(D) −101.3 J

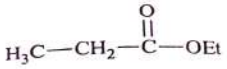
179. What is the spin magnetic moment of cobalt in $Hg[Co(SCN)_4]$?
(A) $\sqrt{3}$ (B) $\sqrt{15}$
(C) $\sqrt{8}$
(D) $\sqrt{24}$

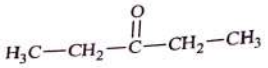
180. The dipole moment of Chlorobenzine is 1.5 D. The dipole moment of  is
(A) 0 D
(B) 1.5 D
(C) 2.86 D
(D) 2.25 D

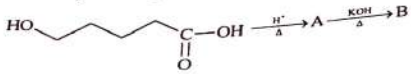
181. What will be the new formula for a compound AB having ZnS type (sphaterite) lattice, if all ions lying on a body plane are removed?
(A) AB_2 (B) A_3B_2
(C) A_4B_5 (D) A_4B_3

182. 

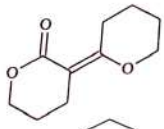
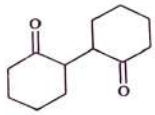
The structure of A is

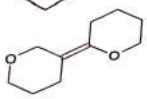
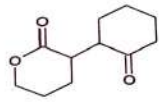
(A) $COCl_2$ (B) 

(C)  (D) All

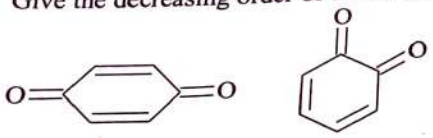
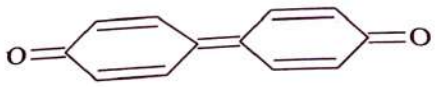
183. 

B is

(A)  (B) 

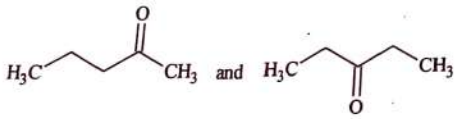
(C)  (D) 

184. Give the decreasing order of stability of following compounds.

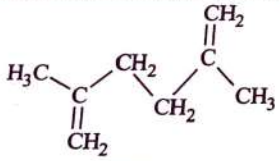
(I) (II) (III)

(A) I > II > III (B) II > I > III
(C) III > I > III (D) III > II > I

185. 

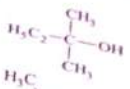
Can be distinguished by
(A) $NaHSO_3$ (B) Fehling solution
(C) Tollen's reagent (D) DMSO

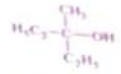
186. An organic compound A, C_8H_{12} on reaction with O_3 followed by Zn gave one mole each of $(CHO)_2$ and $CH_3COCH_2CH_2COCH_3$. The structure of A is

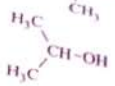
(A)  (B)

(C) (D)

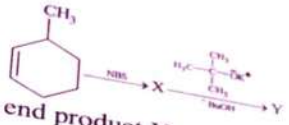
187. $(C_2H_5O)_2C=O \xrightarrow{CH_3MgBr(excess)} A$
A can be

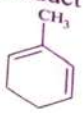
(A) 

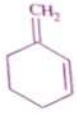
(B) 

(C) 

(D) $(CH_3)_3C-OH$

188. 
The end product Y would be

(A) 

(B) 

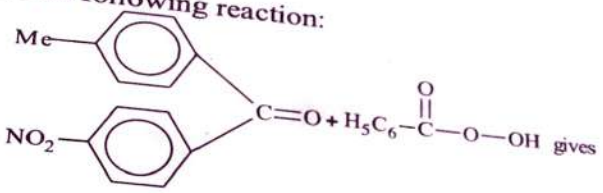
(C)

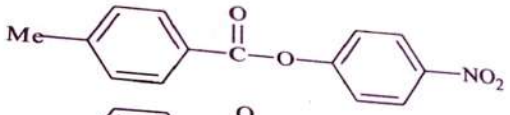
(D) None

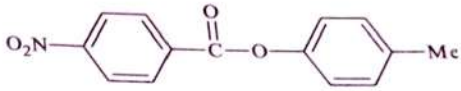
189. The product formed in the following reaction $C_6H_5-O-CH_3 \xrightarrow[\text{heat}]{HI}$ are:

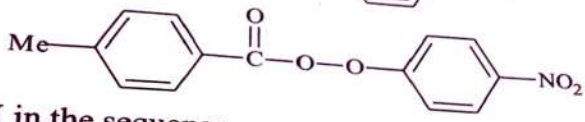
(A) C_6H_5OH and CH_3I
(B) C_6H_5I and CH_3OH
(C) C_6H_5I and CH_3I
(D) C_6H_6 and CH_3OI

190. The following reaction:



(A) 

(B) 

(C) 

(D) None

191. Identify X in the sequence

$$\underset{(X)}{C_3H_9N} \xrightarrow{HNO_2} C_3H_8O \xrightarrow[H_2SO_4]{K_2Cr_2O_7} C_3H_8O_2$$

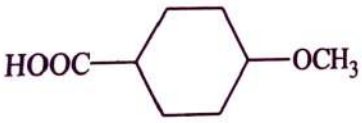
(A) $CH_3CH_2NHCH_3$
(B) $CH_3CH_2CH_2NH_2$
(C) $(CH_3)_3N$
(D) $(CH_3)_2-CH-NH_2$

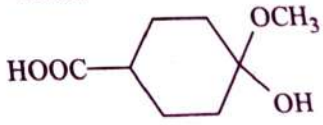
192.

$\xrightarrow[\Delta]{H_3O^+}$ A A is

(A)

(B)

(C) 

(D) 

193. In a metal oxide, the oxide ions are arranged in hexagonal close packing and metal ions occupy two third of the octahedral voids. The formula of the oxide is
(A) MO (B) M_2O_3
(C) MO_2 (D) M_4O_6

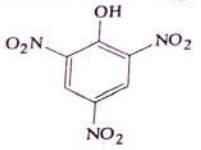
194. Which of the following system has minimum number of unpaired electrons?
(A) d^4 (octahedral, low spin) (B) d^6 (tetrahedral)
(C) d^6 (octahedral, low spin) (D) d^9 (octahedral)

195. In the complex $[Co(NH_3)_5CO_3]ClO_4$, the coordination number, oxidation number, number of d electrons and number of unpaired electrons on the metal are, respectively,
(A) 6, 3, 6, 0 (B) 7, 2, 7, 1
(C) 7, 1, 6, 4 (D) 6, 2, 7, 3

196. A white crystalline solid (A) on boiling with caustic soda solution gave a gas (B), which when passed through an alkaline solution of potassium mercuric iodide gave brown ppt. The substance (A) on heating gave a gas (X), which rekindled a glowing splinter but did not give brown fumes with nitric oxide. The gas (B) is
(A) H_2S (B) NH_3
(C) HCl (D) CO_2

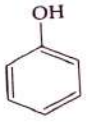
197. For the preparation of t-butylmethyl ether by Willianson's method the correct choice of reagents is
(A) Methoxide and ter-butylbromide (B) Methanol and 2-bromobutane
(C) 2-Butanol and methyl bromide (D) t-butoxide and methyl bromide

198. Which of the following compound cannot liberate CO_2 when treated with $NaHCO_3$?

(A) 

(B) CH_3CO_2H

(C) HCO_2H

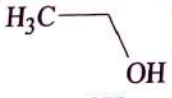
(D) 

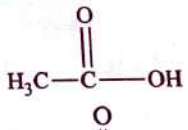
199. Phenol + Phthalic anhydride $\xrightarrow[\text{fusion}]{\text{Conc. } H_2SO_4}$ B

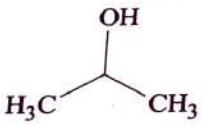
B gives which of the following colour in alkaline medium?
(A) Yellow (B) Colourless
(C) Pink (D) Violet

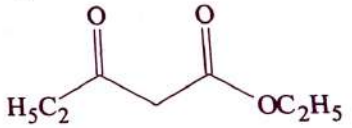
200. MIC (which is responsible for Bhopal gas tragedy) can be produced by which of the following test?
(A) Mulliken test (B) Carbylamine test
(C) Victor Meyer test (D) Liebermann nitroso test

201. Among the following which will not respond to iodoform test?

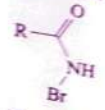
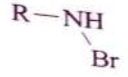
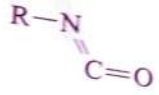
(A) 

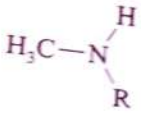
(B) 

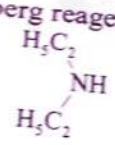
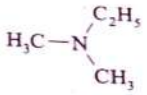
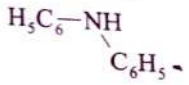
(C) 

(D) 

202. The compound which on reaction with aqueous nitrous acid at low temperature produces an oily nitrosamine is
(A) methylamine
(B) ethylamine
(C) diethylamine
(D) triethylamine

203. Reaction of $RCONH_2$ with a mixture of Br_2 and KOH gives RNH_2 as the main product. The intermediates not involved in the reaction are
(A) 
(B) 
(C) 
(D)

204. Which of the following amines form m-Nitroso derivative when treated with $NaNO_2$ and HCl?
(A) H_3C-NH_2
(B) $Ph-NH_2$
(C) 
(D) $R-NH_2$

205. Which of the following compounds will dissolve in an alkali solution after it has undergone reaction with Hinsberg reagent?
(A) 
(B) 
(C) H_3C-NH_2
(D) 

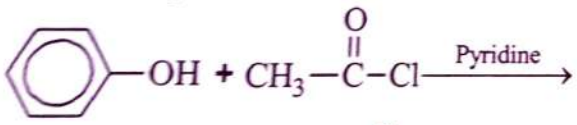
206. Glucose reacts with acetic anhydride to form
(A) monoacetate
(B) tetraacetate
(C) pentaacetate
(D) hexaacetate

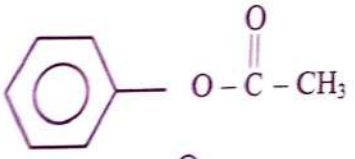
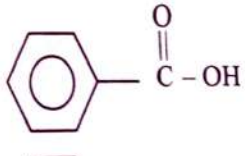
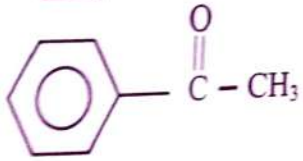
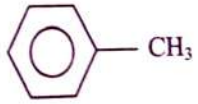
207. Identify the product 'Z' in the following series of reactions.

$$C_6H_{12}O_6(\text{Glucose}) \xrightarrow{HCN} X \xrightarrow{H_2O} Y \xrightarrow[P]{HI} Z$$

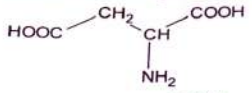
(A) Hexanoic acid
(B) α–methyl caproic acid
(C) Heptanoic acid
(D) None of the above

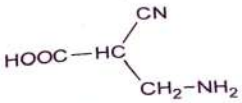
208. Predict the product of reaction below

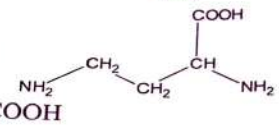


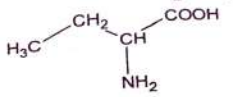
(A) 
(B) 
(C) 
(D) 

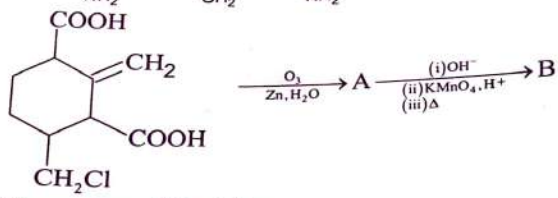
209. Which of the following is a basic amino acid?

(A) 

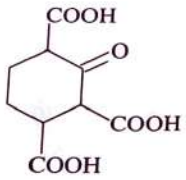
(B) 

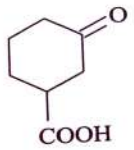
(C) 

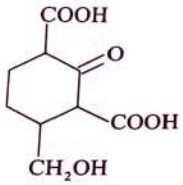
(D) 

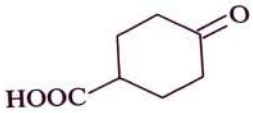
210. 

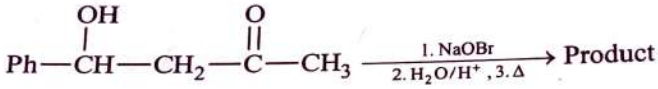
The compound B will be

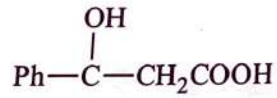
(A) 

(B) 

(C) 

(D) 

211. 

(A) 

(B) $PhCOCH_3$, CO_2, $CHBr_3$

(C) PhCOOH, $H_2C_2O_4$, $CHBr_3$

(D) Ph—COOH, $CHBr_3$, CO

212. For the reaction, $A \longrightarrow B + C$, difference in threshold energy and normal energy of the reactant is 20 kJ / mole. Find the percentage of reactant molecules which are able to cross-over the P.E. barrier at $T = 27^\circ C$.

(A) 3.29×10^{-4}

(B) 3.29×10^{-2}

(C) 1.6×10^{-2}

(D) 2.1×10^{-3}

213. What volume of 0.15 N KI solution is required to react in basic solution with 34.1 ML of 0.216 N solute of $KMnO_4$? The products in the reaction include MnO_2 and IO_3^-.

(A) 25.4 ML

(B) 37.9 ML

(C) 49.1 ML

(D) 98.2 ML

214. The standard enthalpy of formation of gaseous H_2O at 298 K is $-241.82\ kJmol^{-1}$. Calculate ΔH° at 373 K given the following values of the molar heat capacities at constant pressure.

Molar heat capacity of $H_2O(g) = 33.58\ JK^{-1}mol^{-1}$

Molar heat capacity of $H_2(g) = 28.84\ JK^{-1}mol^{-1}$

Molar heat capacity of $O_2(g) = 29.37\ JK^{-1}mol^{-1}$

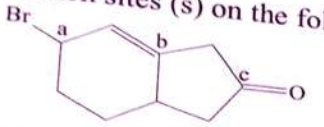
Assume that the heat capacities are independent of temperature.

(A) $-242.6\ kJ\ mol^{-1}$
(B) $-485.2\ kJ\ mol^{-1}$
(C) $-121.3\ kJ\ mol^{-1}$
(D) $-286.4\ kJ\ mol^{-1}$

215. Calculate the enthalpy change when 50 ml of 0.01 M $Ca(OH)_2$ reacts with 25 ml of 0.01 M HCl. Given that ΔH°_{neut} of a strong acid and strong base is 140 kcal per equivalent.

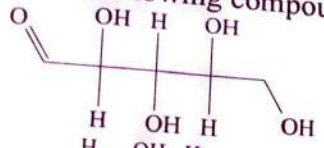
(A) 14.0 cal
(B) 35.0 cal
(C) 10.0 cal
(D) 7.5 cal

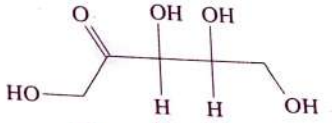
216. Which sites (s) on the following molecule are susceptible to nucleophilic attack?

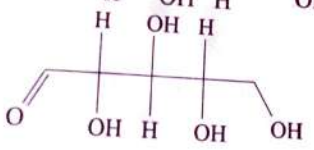


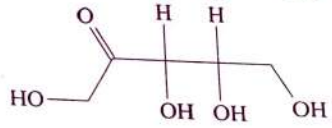
(A) a and b
(B) a and c
(C) b and c
(D) a, b and c

217. Which of the following compound is D-aldopentose?

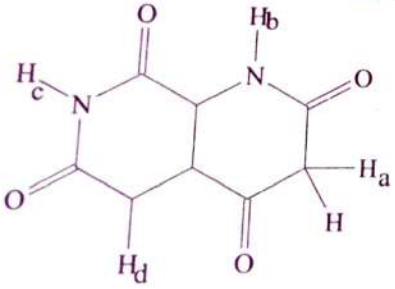
(A) 

(B) 

(C) 

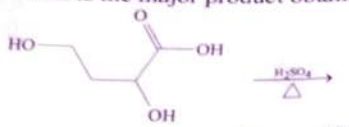
(D) 

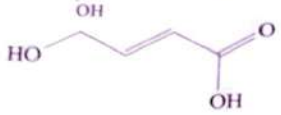
218. The most acidic H in the following compounds is

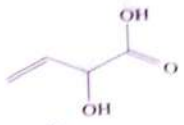


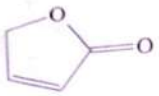
(A) a
(B) b
(C) c
(D) d

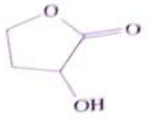
219. What is the major product obtained from the following reaction?



(A) 

(B) 

(C) 

(D) 

220. The end product Y for the following reaction is

$(C_6H_5)_2CHCH_2OH \xrightarrow{Na} X \xrightarrow{CH_3I} Y$

(A) $(C_6H_5)_2C{=}CH_2$

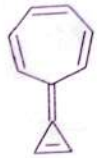
(B) $(C_6H_5)_2CHCH_2ONa^+$

(C) $(C_6H_5)_2CHCH_2O-\overset{\overset{O}{\|}}{C}-S^-Na^+$

(D) $(C_6H_5)_2CHCH_2OCH_3$

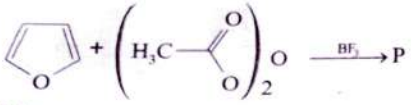
221. Which of the following compound have maximum ionic character?

(A)

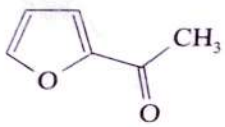
(B) 

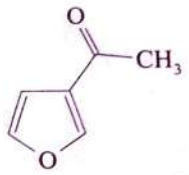
(C)

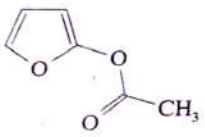
(D) All are having equal ionic character

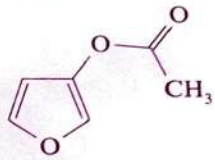
222. 

P is

(A) 

(B) 

(C) 

(D) 

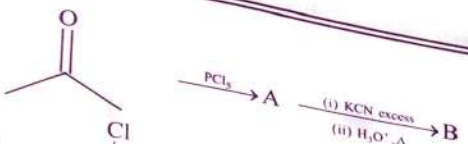
223. An organic compound X (C_4H_9Br) on treatment with alc. KOH gave isomeric product with the formula C_4H_8. On ozonolysis, one of these gave only product CH_3CHO while the other gave two different products. The compound X is

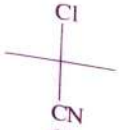
(A) 2-bromobutane

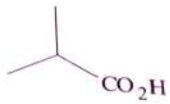
(B) 2-bromopentane

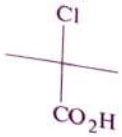
(C) 1, 2-dibromo butane

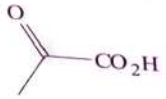
(D) 3-bromopentene

224. 

(A) 

(B) 

(C) 

(D) 

225. The correct decreasing order of acidic strength is
(A) $C_6H_5OH > C_6H_5CH_2OH > C_6H_5COOH > C_6H_5SO_3H$
(B) $C_6H_5CH_2OH > C_6H_5OH > C_6H_5SO_3H > C_6H_5COOH$
(C) $C_6H_5COOH > C_6H_5CH_2OH > C_6H_5OH > C_6H_5SO_3H$
(D) $C_6H_5SO_3H > C_6H_5COOH > C_6H_5OH > C_6H_5CH_2OH$

REASONING TYPE QUESTIONS

This section contains reasoning type questions. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. STATEMENT-1: C_3O_2 is non-planar molecule.
and
STATEMENT-2: Terminal π-bonds of the molecules are lying in different planes.
(A) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is correct explanation for STATEMENT-1
(B) STATEMENT-1 is True, STATEMENT-2 is True; STATEMENT-2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT-1 is True, STATEMENT-2 is False
(D) STATEMENT-1 is False, STATEMENT-2 is True

2. STATEMENT-1: Most probable velocity is the velocity possessed by maximum fraction of molecules at the same temperature.
and
STATEMENT-2: On collision more & more molecules acquire higher velocity at the same temperature.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

3. STATEMENT-1: In any ionic solid [MX] with Schottky defects, the number of positive and negative ions are same.
and
STATEMENT-2: Equal number of cation and anion vacancies are present.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

4. STATEMENT-1: When a cell placed in hypertonic solution, it shrinks.
and
STATEMENT-2: Reverse osmosis is used for desalination of water.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

5. STATEMENT-1: Basic radical of V group are precipitated as their carbonates in presence of NH_4Cl.
and
STATEMENT-2: NH_4OH maintains the pH of the solution basic.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

6. STATEMENT-1: Hydrolysis of ethylacetate in acid medium is pseudo first order reaction.
and
STATEMENT-2: In hydrolysis of ethylacetate water does not take part in the reaction.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

7. STATEMENT-1: Due to adiabatic free expansion, temperature of a real gas always increases.
and
STATEMENT-2: If a real gas is at inversion temperature then no change in temperature is observed in adiabatic free expansion.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

8. STATEMENT-1: Equivalent mass of a substance cannot be greater than the molar mass of the substance.
and
STATEMENT-2: The number of moles of electrons released or accepted by a given substance cannot be less than one.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

9. STATEMENT-1: Reimer-Tiemann reaction of phenol with CCl_4 in NaOH at 340 K gives salicyclic acid as the major product.
and
STATEMENT-2: The reaction occurs through intermediate formation of dichlorocarbene.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

10. STATEMENT-1: $[CoCl_3(NH_3)_3]$ does not give ppt. with $AgNO_3$ solution.
and
STATEMENT-2: $[CoCl_3(NH_3)_3]$ complex is optically inactive.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

11. STATEMENT-1: $[FeCl_2(SCN)_2]^{2-}$, (high spin complex) can exist as diastereoisomers.
and
STATEMENT-2: In tetrahedral geometry all positions are adjacent to each others.
(A) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is correct explanation for STATEMENT-1
(B) STATEMENT – 1 is True, STATEMENT-2 is True; STATEMENT -2 is NOT a correct explanation for STATEMENT-1
(C) STATEMENT – 1 is True, STATEMENT-2 is False
(D) STATEMENT – 1 is False, STATEMENT-2 is True

12. STATEMENT-1: Nucleophilic substitution reaction of an optically active alkyl halide gives a mixture of enantiomers.
and
STATEMENT-2: The reaction occurs by S_N1 mechanism.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

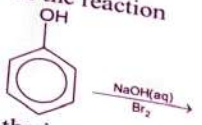
13. STATEMENT-1: Formaldehyde is more reactive than benzaldehyde for nucleophilic addition reaction.
and
STATEMENT-2: Formaldehyde is more stable than benzaldehyde.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

14. STATEMENT-1: Treatment of fumeric acid with OsO_4 followed by hydrolysis with an aqueous solution of $NaHSO_3$ gives a mixture of (+) and (–) tartaric acids.
and
STATEMENT-2: OsO_4 brings about trans-addition of hydroxyl groups to the double bond.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

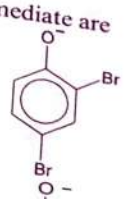
15. STATEMENT-1: When freshly precipitated nickel sulphide (NiS) is shaken with a solution of hydrogen sulphide, a colloidal solution is obtained.
and
STATEMENT-2: A colloidal solution is stable because all the colloidal particles carry same charge.
(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement -2 is False.
(D) Statement-1 is False, Statement-2 is True.

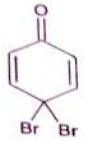
1. Select correct order between given cmpounds as specified
(A) $COCl_2 > COF_2$: $X\hat{C}X$ bond angle (X = F or Cl)
(B) $NO_2F > NO_2Cl$: $O\hat{N}O$ bond angle
(C) $SO_2F_2 < SOF_2$: $F\hat{S}F$ bond angle
(D) $N_2F_2 < N_2(CH_3)_2$: N – N bond length

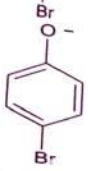
2. In the reaction

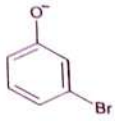


the intermediate are

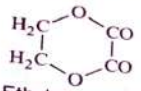
(A) 

(B) 

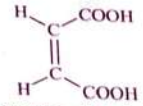
(C) 

(D) 

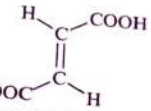
3. The $C_4H_4O_4$ can represent

(A) 

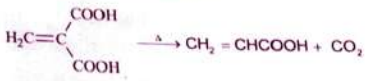
Ethylene oxalate

(B) 

Maleic acid (*cis*-isomer)

(C) 

Fumaric acid (*trans*-isomer)

(D) 

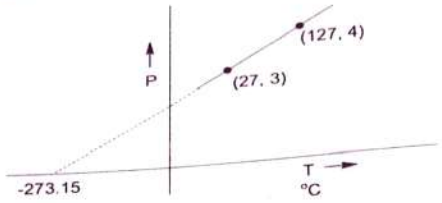
α, β – Unsaturated acid

4. The vapour pressure of a dilute solution of a solute is influenced by
(A) temperature of solution
(B) mole fraction of solute
(C) melting point of solute
(D) degree of dissociation of solute

5. A gas obeys $(P + K)V = RT$. Which of the following is/are not correct about this gas?
(A) slope of its isotherm is negative and curve is exponentially decreasing.
(B) slope of its isotherm is $\left(\frac{K-P}{V}\right)$
(C) Its compressibility factor is independent of P and V.
(D) Slope of its isochore is +ve and have low value at high volume.

6. If a gas expands at a constant pressure by providing heat, which of the followings are incorrect:
(A) The temperature increases.
(B) The kinetic energy remains same.
(C) The kinetic energy of gaseous molecule decreases.
(D) Expansion at constant pressure is not possible.

7. Two moles of ideal gas undergoes the following process $\left(\frac{\partial P}{\partial T}\right)_v = x \times 10^{-y}$ then



(A) $x = 1$ (B) $y = 2$
(C) $x = 2$ (D) $y = 5$

8. Rate of effusion of an ideal gas through a hole depends upon
(A) pressure of the gas (B) molecular wt. of the gas
(C) surface area of the hole (D) shape of the hole

9. Among 0.1 M solution of urea and Na_3PO_4 and aluminium sulphate
(A) the vapour pressure and freezing point are the highest for urea
(B) the vapour pressure and freezing point are the lowest for urea
(C) the elevation in boiling point is the highest for $Al_2(SO_4)_3$
(D) the depression in freezing point is the highest for $Al_2(SO_4)_3$

10. Which of the following statements is/are correct?
(A) 0.1 M NaCl solution will have the same osmotic pressure as 0.1 M glucose solution.
(B) 0.1 M NaCl solution will have the same boiling point as 0.1 M KBr solution.
(C) 0.1 M glucose and 0.1 M urea are isotonic.
(D) 0.1 M $CaCl_2$ solution will have less relative lowering of vapour pressure than 0.1 M NaCl.

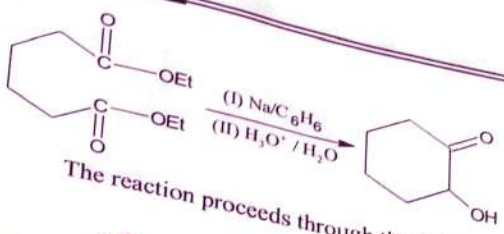
11. Which of the following are true for the first order reaction?
(A) $t_{3/4} = 2t_{1/2}$ (B) $t_{15/16} = 4t_{1/2}$
(C) $t_{15/16} = 3t_{3/4}$ (D) $t_{7/8} = 2t_{3/4}$

12. Gas A (mol. Mass = 16) and gas B (mol. Mass = 4) are confined in to a vessel at partial pressure of 2 atm each at 27°C. The gases are then allowed to effuse out at 27°C through a valve till the pressure inside the vessel becomes 2 atm after which the valve is closed and the gases are heated to 127°C and allowed to react as $A + B \rightleftharpoons C$ for which $k_p = 0.1\ atm^{-1}$. By assuming that C is not formed till the effusion process is completed, Identify the correct statements in the following
(A) The no. of moles of A and B diffused out are in the ratio 1:2.
(B) The no. of moles of A and B left after diffusion at 27°C are in the ratio 2 : 1
(C) Partial pressure of B at 127°C is $\frac{8}{9}$ atm.
(D) Partial pressure of C at 127°C is $\frac{1}{8}$ atm

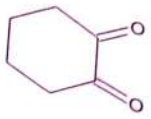
13. Which of the following statements regarding the compound A_xB_y is/are correct?
(A) 1 mole of A_xB_y contains y mole of A and x while of B.
(B) 1 mole of A_xB_y contains x mole of A and y mole of B
(C) 1 equivalent of A_xB_y contains 1 equivalent of A and 1 equivalent of B
(D) equivalent weight of A_xB_y = equivalent weight of A + equivalent weight of B

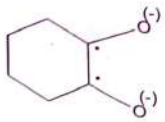
14. Which complex species does/do not exhibit geometrical isomerism and only have two stereo isomers
(A) $[Co(EDTA)]^-$ (B) $[PtBrCl(gly)]^-$
(C) $[Co(acac)_2(en)]^+$ (D) $[Pd(NO_2)(ox)(gly)]$

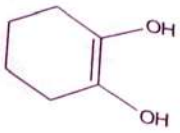
15.

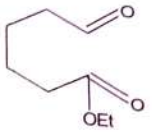


The reaction proceeds through the following species?

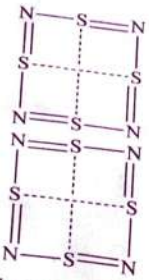
(A) 

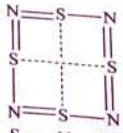
(B) 

(C) 

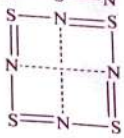
(D) 

16. Which of the following are structure of S_4N_4:

(A) 

(B) 

(C)

(D) 

17. Be and Al resembles in
(A) both become passive on reaction with HNO_3 due to formation of oxide layer
(B) their chlorides are Lewis acids
(C) chlorides exist in polymeric form
(D) hydroxides are soluble in alkali as well as in acid

18. Which of the following statement(s) is (are) correct when a mixture of NaCl and $K_2Cr_2O_7$ is gently warmed with conc. H_2SO_4?
(A) A deep red vapour is evolved
(B) The vapours when passed into NaOH solution gives a yellow solution of Na_2CrO_4
(C) Chlorine gas is evolved
(D) Chromyl chloride is formed

19. $X + H_2SO_4 \longrightarrow Y$ (a colourless gas with pungent smell)

$Y + K_2Cr_2O_7 + H_2SO_4 \longrightarrow$ Green solution

(A) X is SO_3^{2-}
(B) X is CO_3^{2-}
(C) Y is H_2S
(D) Y is SO_2

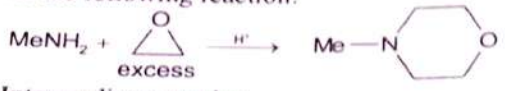
20. O_2 required for complete oxidation of 1 litre of ethane at NTP is
(A) 3.5 lit.
(B) 0.156 mole
(C) 5.00 g
(D) 7.0 litre

21. Correct statement(s) regardring the following reaction is/are

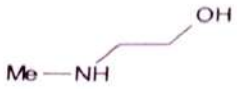
$$ROH + \underset{\text{conc}}{HCl} \xrightarrow{\text{anhydrous } ZnCl_2} \underset{\text{White turbidity}}{RCl\downarrow} + H_2O$$

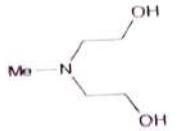
(A) ROH behaves as a base
(B) greater the value of pK_a (alcohol), greater the reactivity with conc. HCl and thus sooner the formation of white turbidity.
(C) alcohol which reacts fastest with Na metal, will give turbidity at fastest rate
(D) alcohol which gives red colour during Victor Meyer test, will give turbidity at slower rate then those giving blue or white colour during Victor Meyer test.

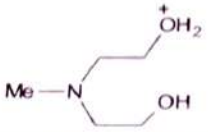
22. In the following reaction:

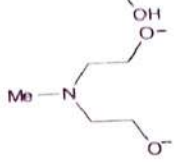


Intermediates can be:

(A) 

(B) 

(C) 

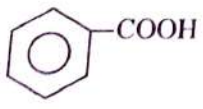
(D) 

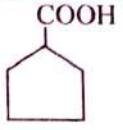
23. Which of the following reactions can be used to prepare acetophenone?
(A) Reaction of benzene with acetyl chloride in presence of anhydrous $AlCl_3$
(B) Dry distillation of a mixture of calcium benzoate and calcium acetate
(C) Oxidation of ethyl benzene by vanadium peroxide.
(D) By the action of $(CH_3)_2Cd$ on benzoyl chloride followed by acid hydrolysis

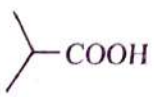
24. Which of the following statements are correct?
(A) The crystal lattice of ice is mostly formed by covalent as well as hydrogen bonds
(B) The density of water increases when heated from 0° C to 4° C due to the change in the structure of the cluster of water molecules
(C) Above 4° C the thermal agitation of water molecules increases. Therefore, intermolecular distance increases and water starts expanding
(D) The density of water increases from 0° C to a maximum at 4° C because the entropy of the system increases

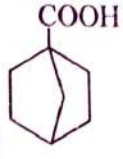
25. Which of the following oxyacids of sulphur contain s – s bonds?
(A) $H_2S_2O_8$ (B) $H_2S_2O_6$
(C) $H_2S_2O_4$ (D) $H_2S_2O_5$

26. Which one of the following compounds will give HVZ reaction?

(A) 

(B) 

(C) 

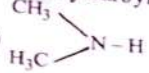
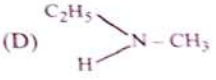
(D) 

27. Which of the following compounds will give acetic acid with $KMnO_4/H^+$:
(A) CH_3–CHO (B) CH_3–CH=CH–CH_3 (C) CH_3–C ≡ C–CH_3 (D) CH_3CH_2OH

28. Which of the following pairs can be distinguished by Fehling's solution?
(A) Glucose and fructose (B) Glucose and sucrose
(C) Benzaldehyde and ethanal (D) Hydroxy propanone and benzaldehyde

29. Which of the following compound has/have effective atomic number equal to the atomic number of a noble gas ?
(A) $K[Co(CO)_4]$ (B) $K_2[Fe(CO)_4]$ (C) $[Co(NH_3)_6]Cl_2$ (D) $[CoCl_3(H_2O)_3]$

30. Compound $C_4H_{11}N$ exhibits
(A) position isomerism (B) metamerism
(C) functional isomerism (D) none of the above

31. Which of the following can be detected by carbylamine test ?
(A) CH_3NH_2 (B)  (C) $C_6H_5NH_2$ (D) 

32. Identify the complexes which are expected to be coloured. Give explanation.
(A) $\left[Ti(NO_3)_4\right]$ (B) $\left[Cu(NCCH_3)\right]^+ BF_4$ (C) $\left[Cr(NH_3)_6\right]^{3+} 3Cl^-$ (D) $K_3[VF_6]$

33. For a solution of non – electrolyte, non – volatile solute, which of the expression is/are correct where π is osmotic pressure, and temperature is constant?
(A) $\frac{d\pi}{dv} = -\frac{\pi}{v}$
(B) $\frac{d^2\pi}{dv^2} = \pi + 1\ v^{-2}$
(C) $\frac{d^2\pi}{dv^2} + v^{-1}\frac{d\pi}{dv} - \pi v^{-2} = 0$
(D) $\frac{d^2\pi}{dv^2} = \frac{2\pi}{v^2}$

34. The pH of which solutions do not change with dilution?
(A) 1 M CH_3COONH_4
(B) $CH_3COONa + CH_3COOH$ (Both 1 M each)
(C) $CH_3COOH + CH_3COONH_4$ (Both 1 M each)
(D) 1 M CH_3COONa

35. Which of the following salt solutions are acidic?
(A) NaOCl (B) NH_4NO_3 (C) Na_2SO_4 (D) $FeCl_3$

36. Consider the ionization of HCl in aqueous solution of CH_3COOH

$$CH_3COOH + HCl \rightleftharpoons CH_3COOH_2^+ + Cl^-$$

The correct statement among the following are
(A) Cl^- is the conjugate base of HCl
(B) CH_3COOH is the conjugate base of $CH_3COOH_2^+$
(C) $CH_3COOH_2^+$ is the conjugate base of CH_3COOH
(D) Cl^- is the conjugate base of CH_3COOH

37. Which of the following statement (s) is /are correct?
(A) The phenomena of gaseous effusion and diffusion are identical
(B) In case of effusion the particles of gas pass through the small hole, independently, not collectively
(C) effusion rate is the rate at which molecules strike a unit area of the wall of the poured container, in which the gas is taken
(D) gaseous diffusion involves the effect of collisions between molecules /particles of gas.

38. Which of the following is correct for critical temperature?
(A) It is the highest temperature at which liquid and vapour can coexist.
(B) Beyond this temperature, there is no distinction between the two phases and a gas cannot be liquefied by compression.
(C) At this temperature, the gas and the liquid phases have different critical densities.
(D) All are correct.

39. Which of the following factors will favour the backward reaction:
$Cl_2(g) + 3F_2(g) \rightleftharpoons 2ClF_3(g) \quad \Delta H = -ve$
(A) Addition of inert gas at constant pressure
(B) Addition of Cl_2 gas
(C) Increase in temperature
(D) Increasing the volume of the container.

40. The equilibrium of which of the following reactions will not be disturbed by the addition of inert gas at constant volume.
(A) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(B) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
(C) $N_2O_4(g) \rightleftharpoons 2NO_2(g)$
(D) $N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$

41. A 2 mole Ne gas sample in a 30 L container fitted with frictionless piston is expanding adiabatically reversibly to 60 L at initial temperature 400 K. Which of the following is/are correct statement(s)?
(A) Work done in the above process is around – 1867.5 kJ
(B) Change in temperature is of 150 K (approx.)
(C) The initial pressure is 2.2 atm and final pressure is 0.68 atm
(D) The final temperature is 250 K (approx.)

42. The normal boiling point of a liquid "A" is 350K ΔH_{vap} at normal boiling point is 35 kJ/mol. Pick out the correct statement(s). (Assume ΔH_{vap} to be independent of pressure)
(A) $\Delta S_{vap} > 100$ J / k – mol at 350 K and 0.5 atm
(B) $\Delta S_{vap} < 100$ J / k – mol at 350 K and 0.5 atm
(C) $\Delta S_{vap} < 100$ J / k – mol at 350 K and 2 atm
(D) $\Delta S_{vap} > 100$ J / k – mol at 350 K and 2 atm

43. Which of the following statement(s) is/are false:
(A) $\Delta_r S$ for $\frac{1}{2}N_2(g) \longrightarrow N(g)$ is positive
(B) ΔG_{system} is always zero for a reversible process in a closed system
(C) ΔG° for an ideal gas is a function of temperature and pressure
(D) Entropy of a closed system is always maximized at equilibrium

44. The extraction of metals from oxide ores involve
(A) reduction with carbon
(B) reduction with aluminium
(C) electrolytic reduction
(D) reduction with CO

45. Which of the following metal(s) is/are commercially extracted by self reduction method from their corresponding ore.
(A) Cu (B) Fe (C) Pb (D) Hg

46. The process/processes by which lighter earthy particles are freed from the heavier particles using water is/are
(A) gravity separation (B) levigation (C) hydraulic washing (D) leaching

47. Acidic fluxes are
(A) SiO_2 (B) B_2O_3 (C) Fe_2O_3 (D) $CaCO_3$

48. A ligand having an unshared pair of electrons can be a
(A) neutral molecule
(B) positively charged ion
(C) negatively charged ion
(D) group containing a lone pair of electrons

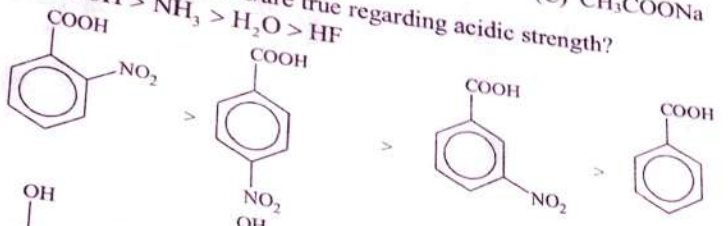
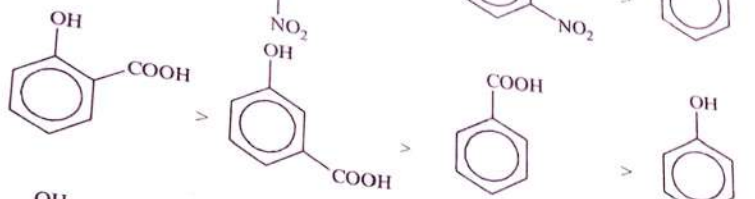
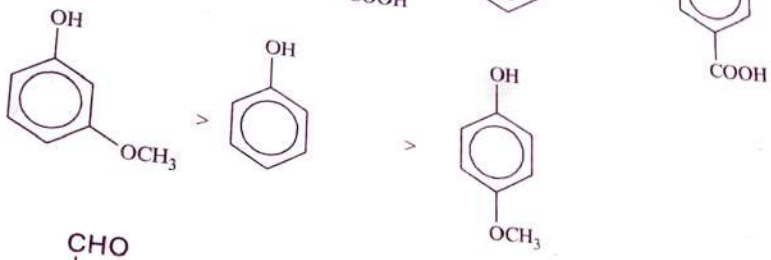
49. $p\pi - d\pi$ bonding occurs between oxygen and
(A) phosphorus in P_4O_{10} (B) Xenon in XeO_2F_2 (C) Nitrogen in N_2O_5 (D) Sulphur in SO_2

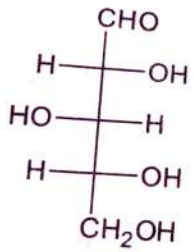
50. On strong heating $Pb(NO_3)_2$ gives
(A) PbO (B) NO_2 (C) O_2 (D) NO

51. White phosphorous can be separated from red phosphorous by
(A) Sublimation (B) dissolving in CS_2
(C) distillation (D) heating with alkali solution

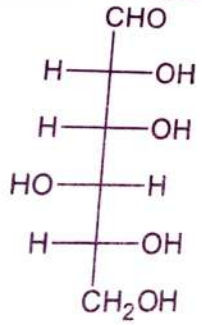
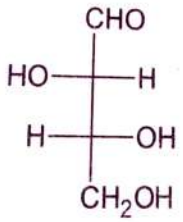
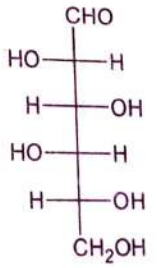
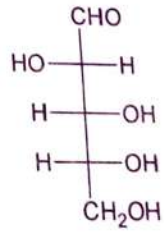
52. The compounds which are coloured only due to charge transfer are
(A) $KMnO_4$ (B) $K_2Cr_2O_7$ (C) $Fe(OH)_3$ (D) $Cr(OH)_3$

53. Both methane and ethane may be obtained by one step reaction from
(A) methyl iodide (B) ethylidene iodide (C) CH_3COONa (D) ethylene iodide

54. Which of the following order is/are true regarding acidic strength?
(A) $CH \equiv CH > NH_3 > H_2O > HF$
(B) 
(C) 
(D) 

55. 
is subjected to Kiliani-Fischer synthesis.

The product formed is

(A) 
(B) 
(C) 
(C) 

56. D-erythrose → (HNO_3) A; D-erythrose → (Br_2, H_2O) B

Which of the following is/are true about A and B
(A) A is optically active but B is not
(B) A is a dicarboxylic acid
(C) B can form a lactone
(D) A compound having m.f. of A can have 4 optical isomers

57. $H_3\overset{+}{N}CHRCOOH \rightleftharpoons H_3\overset{+}{N}-CHR-COO^- \rightleftharpoons H_2N-CHR-COO^-$
(A) (B) (C)

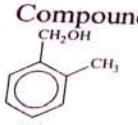
$pK_{a_1} = 2.3$

$pK_{a_2} = 9.7$

Which of the following statements is/are correct
(A) at pH = 2.3, half of A is converted into B
(B) at pH = 6, [B] is at its maximum
(C) at pH = 6, the solution having the above equilibrium becomes non-conducting
(D) pH = 9.7 is called isoelectric point

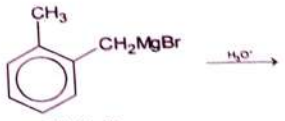
58. Which compounds do not give Lassaigne test for nitrogen?
(A) $C_6H_5N_2Cl$ (B) NH_2-CH_2-COOH (C) NH_2OH (D) CH_3CONH_2

59. Compound (A) given below can be synthesized from which of the following route(s)

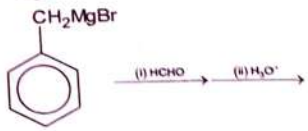


(A)

(A)

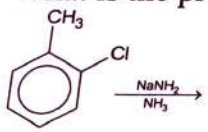
(B) 

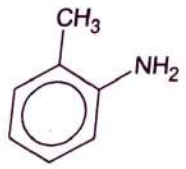
(C)

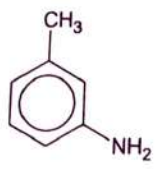
(D) 

60. Aldehydes and ketones give addition reactions with
(A) HCN (B) $NaHSO_3$ (C) RMgX (D) ROH

61. What is the product expected in the following reaction



(A) 

(B) 

(C)

(C) None of these

Comprehension – I

Molar conductivity (Λ_m) is defined as conducting power of the ions produced by 1 mole of an electrolyte in a solution.

$\Lambda_m = \frac{\kappa}{C}$ where κ is the conductivity (S cm^2 mol^{-1}) and 'C' is molar concentration (in mol/cm^3).

The molar conductivity of 0.04 M solution of $MgCl_2$ is 200 S cm^2 mol^{-1} at 298 K. A cell with electrodes that are 2.0 cm^2 in surface area and 0.50 cm apart is filled with $MgCl_2$ solution.

1. Conductance of $MgCl_2$ solution is:
(A) 8×10^{-3} S
(B) 32 S
(C) 0.032 S
(D) none of these

2. How much current will flow when the potential difference between the two electrodes is 5.0 V?
(A) 156.25 A
(B) 0.16 A
(C) 160 A
(D) None of these

3. Equivalent conductinity of $Fe_2(SO_4)_3$ is related to molar conductinity by expression
(A) $\lambda_{eq} = \lambda_m$
(B) $\Lambda_{eq} = \frac{\Lambda_m}{3}$
(C) $\Lambda_{eq} = 3\Lambda_m$
(D) $\Lambda_{eq} = \frac{\Lambda_m}{6}$

Comprehension – II

Read the paragraph carefully and answer the following questions:

In HCP or CCP, constituent particles occupy 74% of the available space. The remaining space (26%) in between the spheres remains unoccupied and is called interstitial voids or holes.

Considering the close packing arrangement, each sphere in the second layer rests on the hollow space of the first layer, touching each other. The void created is called tetrahedral void.

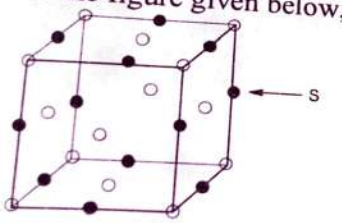
If r is the radius of the spheres in the close packed arrangement then,

r (radius of tetrahedral void) = 0.225 R

In a close packing arrangement, the interstitial void formed by the combination of two triangular voids of the first and second layer is called octahedral void. Thus, double triangular void is surrounded by six spheres. The centre of these spheres on joining, forms octahedron. If R is the radius of the sphere in a close packed arrangement then,

r (radius of octahedral void = 0.414 R).

4. In the figure given below, the site marked as S is a/an



(A) tetrahedral void
(B) cubic void
(C) octahedral void
(D) None of the above

5. In the spinel structure, oxide ions are cubic close packed whereas 1/8th of tetrahedral voids are occupied by A^{2+} cations and 1/2 of octahedral voids are occupied by B^{3+} cations. The general formula of the compound having spinel structure is
(A) $A_2B_2O_4$
(B) AB_2O_4
(C) $A_2B_4O_2$
(D) $A_4B_2O_2$

6. Number of octahedral voids in bcc structure
(A) 0 (B) 1
(C) 2 (D) 4

Comprehension –III

Concrete is produced from a mixture of cement, water, sand and small stones. It consists primarily of calcium silicates and calcium aluminates formed by heating and grinding of clay and limestone. In later steps of cement production a small amount of gypsum, $CaSO_4.2H_2O$ is added to improve subsequent hardening of concrete. The use of elevated temperatures during the final production may lead to formation of unwanted hemihydrate, $CaSO_4.1/2H_2O$. Consider the following reaction:

$$CaSO_4.2H_2O(s) \longrightarrow CaSO_4.\frac{1}{2}H_2O(s) + \frac{3}{2}H_2O(g)$$

The following thermodynamic data apply at 25°C, standard pressure: 1 bar

Compound	$\Delta H^\circ_{f(kJ/mol)}$	$S^\circ(JK^{-1}\ mol^{-1})$
$CaSO_4.2H_2O(s)$	- 2021.0	194.0
$CaSO_4.\frac{1}{2}H_2O(s)$	- 1575.0	130.5
$H_2O(g)$	- 241.8	188.6

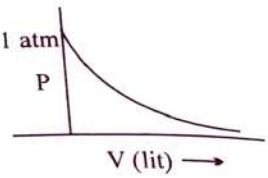
$R = 8.314\ JK^{-1}\ mol^{-1}$

7. ΔH° for the formation of 1.00 kg of $CaSO_4.\frac{1}{2}H_2O(s)$ from $CaSO_4.2H_2O(s)$ is
(A) + 446 kJ (B) + 484 kJ
(C) - 446 kJ (D) - 484 kJ

8. Equilibrium pressure (in bar) of water vapour in closed vessel containing $CaSO_4.2H_2O(s)$, $CaSO_4.\frac{1}{2}H_2O(s)$, $H_2O(g)$ at 25°C
(A) 17.35×10^{-4} bar (B) 2.15×10^{-4} bar
(C) 8.10×10^{-3} bar (D) 7.00×10^{-4} bar

Comprehension – IV

On the recently discovered 10th planet, it has been found that the gases follow the relationship $\mathbf{Pe^{V/2} = nCT}$ where C is constant and other notations are as usual (V in lit., P in atm and T in Kelvin). A curve is plotted between P and V at 500 K & 2 moles of gas as shown in figure



9. The value of constant C is
(A) 0.01 (B) 0.001
(C) 0.005 (D) 0.002

10. Find the slope of the curve plotted between P Vs T for closed container of volume 2 lit. having same moles of gas
(A) $\frac{e}{2000}$ (B) 2000 e
(C) 500 e (D) $\frac{2}{1000e}$

11. If a closed container of volume 200 lit. of O_2 gas (ideal gas) at 1 atm & 200 K is taken to planet, find the pressure (atm) of oxygen gas at the planet at 821 K in same container
(A) $\frac{10}{e^{100}}$ (B) $\frac{20}{e^{50}}$
(C) 1 atm (D) 2 atm

Comprehension – V

The freezing point of 0.2 molal solution of acetic acid in benzene is 277.65 K. Freezing point of pure benzene is 278.4 K and heat of fusion of benzene is 10.042 kJ/mol. If molarity of solution is equal to molality, and acetic acid is found in equilibrium with its dimmer, then answer the following based upon above data

12. K_f for benzene is
(A) $3\ K \times molality^{-1}$
(B) $2\ K \times molality^{-1}$
(C) $5\ K \times molality^{-1}$
(D) $10\ K \times molality^{-1}$

13. The degree of association of acetic acid should be
(A) 0.60
(B) 0.48
(C) 0.92
(D) 0.50

Comprehension – VI

It is a well known fact that the elements above hydrogen in the electrochemical series displace hydrogen from acids and form a salt. One such unknown metal on treatment with excess HCl gives some volume of hydrogen gas and the metal chloride salt. The reaction mixture is then heated to evaporate water and the remaining HCl from the solution to give a white crystalline salt. This salt is then dissolved in a certain mass of H_2O and the freezing point of the resulting solution was measured.

14. 2.0 g of metal on reaction with excess of HCl gave 328.4 ml of hydrogen at 27°C and 750 mm Hg pressure. The mass of metal chloride salt formed would be
(A) 1.93 g
(B) 2.50 g
(C) 2.93 g
(D) 3.60 g

15. If the salt was dissolved in 25 g of H_2O and the freezing point of the solution was –0.97°C, what would be the temperature at which solution will boil? Given: K_f of $H_2O = 1.86°C\ kg\ mol^{-1}$ and K_b of $H_2O = 0.5°C\ kg\ mol^{-1}$.
(A) 100°C
(B) 100.26°C
(C) 101°C
(D) 100.52°C

16. What is the formula of metal chloride?
(A) MCl
(B) MCl_2
(C) MCl_3
(D) M_2Cl_3

Comprehension – VII

K_{sp} of $CdS = 8 \times 10^{-27}$, K_{sp} of $ZnS = 1 \times 10^{-21}$, K_a of $H_2S = 1 \times 10^{-21}$.

17. Which metal sulphide is precipitated first when H_2S is added?
(A) ZnS
(B) CdS
(C) Both at the same time
(D) Both do not form ppt

18. What $[H^+]$ must be maintained in a saturated HS (0.1 M) to precipitate CdS but not ZnS, if $[Cd^{+2}] = [Zn^{+2}] = 0.1$ M initially ?
(A) > 0.1 M
(B) > 0.01
(C) > 0.2 M
(D) > 0.02 M

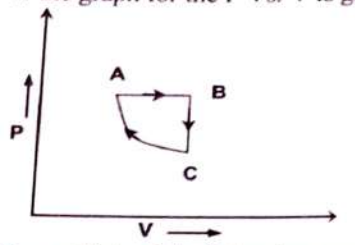
19. When ZnS starts precipitating. What is the concentration of Cd^{+2} is left?
(A) 8×10^{-7}
(B) 0.1
(C) 4×10^{-10}
(D) 2×10^{-9}

Comprehension – VIII

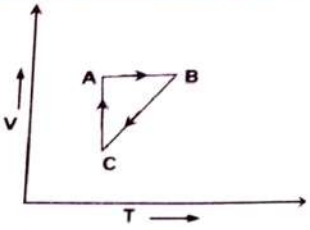
The behaviour of gases have been expressed in terms of various gas laws as Charles Law, Boyles law, Dalton law, etc. In accordance to these laws the ideal gas equation in derived as PV = nRT. Maxwell derived kinetic equation $PV = \frac{1}{3}mnu_{rms}^2$, where u_{rms} represents root mean square speed of molecules.

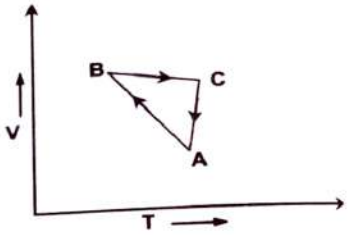
20. Which of the following statement is false?
(A) The product of pressure and volume of a fixed amount of a gas is independent of temperature.
(B) Molecules of different gases have the same K.E. at a given temperature.
(C) The ideal gas equation is not valid at high pressure and low temperature.
(D) The gas constant per molecule is known as Boltzmann constant.

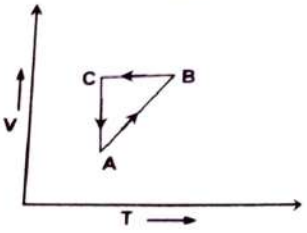
21. If the graph for the P Vs. V is given as in a process:

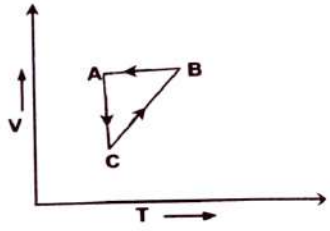


Then which of the following graph is / are correct.

(A)


(B)


(C)


(D)


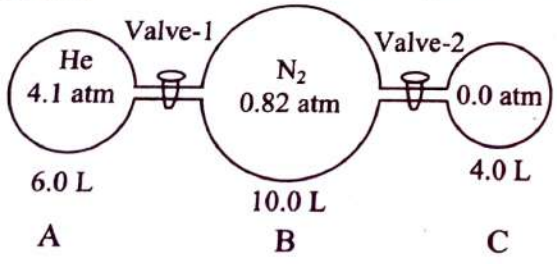
22. Calculate the rms velocity of ozone kept in a closed vessel at 20°C and 82 cm of mercury pressure
(A) 390.3 ms^{-1}
(B) 290 ms^{-1}
(C) 342 ms^{-1}
(D) 412 ms^{-1}

Comprehension – IX

The figure given below shows three glass chambers that are connected by valves of negligible volume. At the outset of an experiment, the valves are closed and the chambers contain the gases as detailed in the diagram. All the chambers are at the temperature of 300K and external pressure of 1.0 atm.

P_{ext} = 1.0 atm

Throughout T = 300K



23. What will be the work done by N_2 gas when the valve–2 is opened and valve–1 remaining closed?
(A) 8.2 L atm
(B) –8.2 atm
(C) 0
(D) –3.28 L atm

24. Which of the following represents the total kinetic energy of all the gas molecules after both valves are opened? ($R = 0.082$ atm L $K^{-1}mol^{-1} = 8.314$ L $K^{-1}mol^{-1}$)
(A) 2836.2 J
(B) 3280.0 J
(C) 4520.6 J
(D) 4988.4 J

Comprehension – X

Read the paragraph carefully and answer the following questions:

The metal-insoluble salt electrode consists of a metal M coated with a porous insoluble salt MX in a solution of X^-. A good example is the silver, silver-chloride electrode for which the half-cell reaction is $AgCl(s)+e \rightleftharpoons Ag(s)+Cl^-(aq)$, where the reduction of solid silver chloride produces solid silver and releases chloride ion into solution for the cell $M(s)\,|\,M^{2+}(aq)\,||\,Ag^+(aq)\,|\,Ag(s)$; $E^\circ_{M/M^{2+}} = +2.37$ V and $E^\circ_{Ag^+/Ag} = +0.80$ V

25. Emf of given cell, when concentration of M^{2+} ion and concentration of Ag^+ ion both are 0.01 M
(A) 3.17
(B) 3.11
(C) 3.01
(D) 3.21

26. If cell reaction is in equilibrium stage, value of logarithms of equilibrium constant is
(A) 1.0745
(B) 107.45
(C) 1074.5
(D) 10.745

Comprehension – XI

The reaction between A and B is of first order with respect to A and of zero order with respect to B. The following data was obtained for four experiments

Experiment	[A] (M)	[B] (M)	Initial Rate (M min^{-1})
I	0.1	0.1	2.0×10^{-2}
II	x	0.2	4.0×10^{-2}
III	0.4	0.4	y
IV	z	0.2	2.0×10^{-2}

27. The value of x is
(A) 0.2
(B) 0.8
(C) 0.1
(D) 0.5

28. The value of y is
(A) 0.2
(B) 0.08
(C) 0.1
(D) 0.5

29. The value of z is
(A) 0.2
(B) 0.8
(C) 0.1
(D) 0.5

Comprehension – XII

10 moles of NH_3 is heated at 15 atm from 27°C to 347°C assuming volume constant. The pressure at equilibrium is found to be 50 atm. The equilibrium constant for the dissociation of NH_3

$$2NH_3(g) \rightleftharpoons N_2(g)+3H_2(g), \quad \Delta H = 91.94 \text{ kJ}$$

Can be written as

$$K_P = \frac{P_{N_2} \times (P_{H_2})^3}{(P_{NH_3})^2} (atm)^2$$

Answer, the following questions:

30. The percentage of dissociation of NH_3 is
(A) 61.2% (B) 20%
(C) 48% (D) 30%

31. The volume of the container in which the gas is heated is
(A) 16.42 L (B) 8.21 L
(C) 20 L (D) 15 L

32. The equilibrium constant K_P for the reaction is:
(A) 7.08×10^2 (B) 3.06×10^2
(C) 7.6×10^2 (D) 1.53×10^3

Comprehension – XIII

Read the paragraph carefully and answer the questions given below it:

According to Bohr's theory, when electron jumps from higher orbit to lower orbit, then it radiates energy in the form of electromagnetic radiations and provides emission spectrum. If

$\Delta E = E_{n_2} - E_{n_1}$

$$\Delta E = 13.6 \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{eV / atom}$$

$$\Rightarrow \frac{1}{\lambda} = \bar{\nu} = R_H \times Z^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \text{eV / atom} \quad \ldots (1)$$

Where R_H is Rydberg's constant and $R_H = 10^7\ m^{-1}$ in Bohr theory, we have assumed that nucleus is at rest and the electron is revolving around it. The assumption of nucleus to be at rest is entirely valid if and only if the mass of nucleus is infinite so far, for the finite mass of nucleus, both the nucleus and the electron will revolve around the common centre of mass, which lies on the line connecting two particles.

Thereby Bohr's theory needs to be modified. So Rydberg's constant of Ritz equation (i) needs to be modified. Simply the mass of electron, i.e. m_e has to be replaced by reduced mass (μ) which is related as

$\frac{1}{\mu} = \frac{1}{M_N} + \frac{1}{m_e}$, Where M_N = mass of nucleus, m_e = mass of electrons.

Answer the following questions:

33. What will be the value of modified Rydberg's constant, if the ratio of mass of nucleus to the mass of electron is 3 : 2.
(A) $\frac{3}{5} \times 10^7\ m^{-1}$ (B) $\frac{5}{3} \times 10^7\ m^{-1}$
(C) $\frac{7}{5} \times 10^7\ m^{-1}$ (A) $\frac{8}{5} \times 10^7\ m^{-1}$

34. If the ratio of mass of electron to mass of nucleus be 1 : 1 and both the nucleus and electron revolve around the common centre of mass which is on the same line connecting the two; then what will be the value of ionization energy for Li^{2+} ions? Given that I.P. of H-atom is 13.6 eV/atom.
(A) 61.2 eV (B) 13.6 eV
(C) 54.4 eV (D) 6.8 eV

Comprehension – XIV

The coordination number of Ni^{2+} is 4.

$NiCl_2 + KCN$ (excess) $\rightarrow$ A (cyano complex)

$NiCl_2 +$ Conc. KCl (excess) $\rightarrow$ B (chloro complex)

35. The IUPAC name of A and B are
(A) Potassium tetracyanonickelate (II), potassium tetrachloronickelate (II)
(B) Tetracyanopotassiumnickelate (II), tetrachlorpotassiumnickelate (II)
(C) Nickel (II), tetrachloronickel (II)
(D) Potassium tetracyanonickel (II), potassium tetrachloronickel (II)

36. Predict the magnetic nature of A and B.
(A) Both are diamagnetic.
(B) A is diamagnetic and B is paramagnetic with one unpaired electron.
(C) A is diamagnetic and B is paramagnetic with two unpaired electrons.
(D) Both are paramagnetic.

Comprehension –XV

Electron pairs tend to minimize repulsion
(a) When lone pairs are present, the bond angles are smaller than predicted.
(b) Lone pairs choose the largest site, e.g. equatorial in trigonal bipyramid.
(c) If all sites are equal, lone pairs will be trans to each other.
Repulsion are of the order of:
LP – LP > LP – BP > BP – BP

37. Which of the following statements is incorrect?
(A) In ClF_3, the axial Cl – F bond length is larger than equatorial Cl – F bond length
(B) In SF_4, F – S – F equatorial bond angle is not 120° but 104° due to lp – bp repulsions
(C) In $[ICl_4]^-$, Cl – I – Cl bond angle is 90°
(D) In OBr_2, the bond angle is less than OCl_2

38. The structure of F_2SeO is analogous to:
(A) SO_3 (B) ClO_3^-
(C) SO_3^{2-} (D) (B) and (C) both

Comprehension – XVI

A black mineral (A) on heating in the presence of air gives a gas (B). The mineral (A) on reaction with dilute H_2SO_4 gives a gas (C) and the solution of a compound (D). On passing (C) into an aqueous solution of (B), white turbidity is obtained. The aqueous solution of (D) on reaction with $K_3[Fe(CN)_6]$ gives a blue compound (E).

39. The aqueous solution of (D) contains
(A) $FeSO_4$ (B) $FeSO_4.(NH_4)_2SO_4$
(C) $FeCl_2$ (D) $FeCl_3$

40. The blue compound (E) must be
(A) $K_4Fe^{II}[Fe^{III}(CN)_6]$ (B) $K_2Fe^{II}[Fe^{II}(CN)_6]$
(C) $K_3Fe^{III}[Fe^{II}(CN)_6]$ (D) $KFe^{II}[Fe^{III}(CN)_6]$

Comprehension – XVII

Test (Q). A compound 'X' was fused with Na metal and the extract gave a white precipitate with $AgNO_3$. The Lassaigne's extract gave red colouration with neutral $FeCl_3$. **Test R** While compound 'Y' when fused with Na metal and subsequent analysis on its Lassaigne's extract did not give any characteristic test. **Test S** While compound Y on fusion with fusion mixture (sodium carbonate + potassium nitrate) or Na_2O_2, followed by extraction, followed by addition of conc. HNO_3 and ammonium molybdate gave a yellow precipitate.

41. Compound X contains
(A) N (B) S
(C) N + S (D) P

42. Compound X
(A) will contain halogens (B) may contain halogens
(C) may contain only sulphur (D) will contain nitrogen

43. Compound Y contains
(A) N (B) S
(C) X (D) P

44. The chemical reaction taking place in Y, when it is fused with fusion mixture is
(A) $3Na + P + 4O \rightarrow Na_3PO_4$
(B) $3Na_2CO_3 + 2P + [O] \rightarrow 2Na_3PO_4 + 3CO_2$
(C) $3KNO_3 + P + 3O \rightarrow K_3PO_4 + 3NO_2 + O_2$
(D) None of these

45. The formula of yellow precipitate is
(A) $(NH_4)_3PO_4$
(B) $Mo(PO_4)$
(C) $(NH_4)_3PO_4.12Mo_3O_3$
(D) $Mo(PO_4)_2$

Comprehension – XVIII

Adolf von Baeyer suggested that, since carbon prefers to have tetrahedral geometry with bond angles of approximately 109°, ring sizes other than five and six may be too strained to exist. Baeyer based his hypothesis on the simple geometrical notion that a three-membered ring (cyclopropane) should be an equilateral triangle with bond angles of 60°, a four membered ring (cyclobutane) should be a square with bond angles of 90° and so on. According to Baeyer's analysis, cyclopropane, with a bond angle compression of $109^0 - 60^0 = 49^0$, should have a large amount of angle strain and must therefore be highly reactive. Cyclohexane becomes puckered to relieve its strain. The angular deviation of cycloalkane is (−11°). Greater is the angular deviation more is the torsional strain.

46. Which of the following is most reactive cycloalkane?
(A) Cyclopropane
(B) Cyclobutane
(C) Cyclopentane
(D) Cyclohexane

47. Which among the following is most strained cycloalkane?
(A) Cyclopropane
(B) Cyclobutane
(C) Cyclopentane
(D) Cyclohexane

48. The tendency of cyclopropane (I), cyclobutane (II) and cyclopentane (III) to form addition compounds is in the order
(A) I > II > III
(B) I = II > III
(C) I > II = III
(D) I = III > II

Comprehension – XIX

Carbohydrates which differ in configuration of glycosidic carbon (C – 1 in aldoses and C – 2 in ketoses) are called anomers. The carbohydrates which differ in configuration at any asymmetric carbon atom other than glycosidic carbon are called epimers. The anomers whose configuration differ at C – 1 are diastereomers. Epimers are also configurational diester

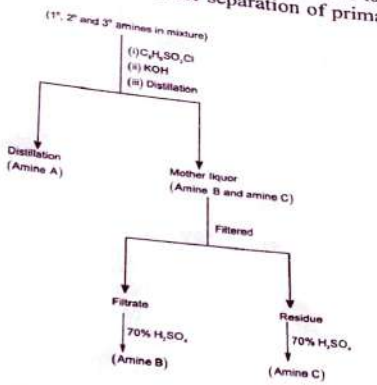
49. In starch and maltose, the glucose units are linked between
(A) C_1 and C_2
(B) C_1 and C_2
(C) C_1 and C_4
(D) C_4 and C_4

50. Methyl-α-D-glucoside and methyl-β-D-glucoside are
(A) anomers
(B) epimers
(C) enantiomers
(D) functional isomers

51. Which of the following statement is correct with respect to two epimeric hexoses?
(A) C – 1 and C – 2 of both have the same configuration
(B) C – 3, C – 4 and C – 5 of both have the same configuration
(C) C – 2, C – 3 and C – 4 of both have the same configuration
(D) One of them must be an aldose and other must be a ketose

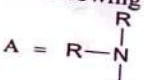
Comprehension – XX

Benzene sulphonyl chloride ($C_6H_5SO_2Cl$) is called Hinsberg's reagent. It is used for the distinction between primary, secondary and tertiary amines. It is also used for separation of primary, secondary and tertiary amines from their mixture.

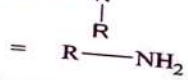


Answer the following questions:

52. Which of the following is correctly matched?

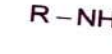
(A) A = 

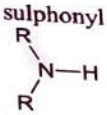
(B) B =

(C) C = 

(D) A =

53. Which of the following amines does not react with benzene sulphonyl chloride?

(A) 

(B) 

(C)

(D) Both (A) and (B)

54. Which of the following is primary amine $R - NH_2$?

(A) A

(B) B

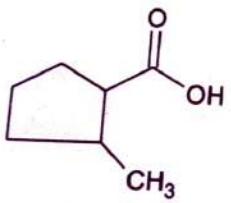
(C) C

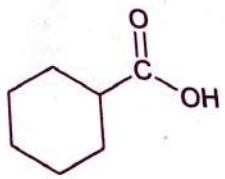
(D) Cannot be predicted

Comprehension – XXI

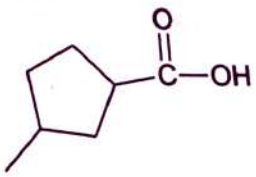
A compound 'A' with molecular formula $C_8H_{12}O$ undergoes hydrogenation and requires 2 mole of H_2 for complete hydrogenation to give compound (B). Compound (B) gives positive iodoform test to give compound (C) $C_7H_{12}O_2$ after hydrolysis. Compound (C) on treatement with soda lime gives cyclohexane.

55. Which of the following can be compound (C)?

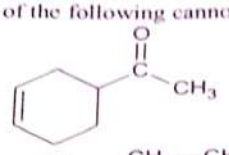
(A) 

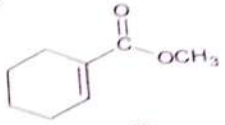
(B) 

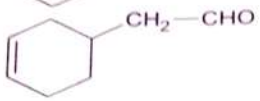
(C)

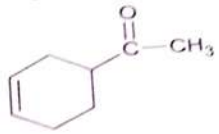
(D) 

56. Which of the following cannot be the structure of compound (A)?

(A) 

(B) 

(C) 

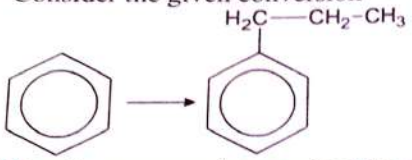
(D) 

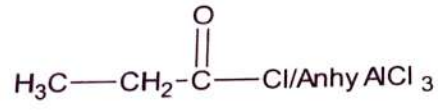
Comprehension – XXII

The introduction of alkyl or acyl group into an aromatic ring is called Friedel-Crafts reaction. The introduction of an alkly group into an aromatic ring is called alkylation. The reaction is usually brought about by alkyl halides, olefins, alcohols and carbonyl compounds in the presence of Lewis acid catalyst.

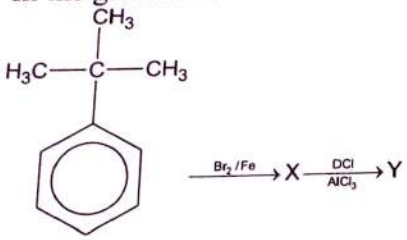
57. Consider the following statements for Friedel-Crafts reaction
(1) Rearrangement in alkylating group takes place.
(2) The entering alkyl group is an activating group, hence polyalkylation occurs frequently.
(3) Aromatic compounds containing m-directing groups do not give this reaction.
(4) Aryl halides can also be used in place of alkyl halides.
Of these statements:
(A) 1 and 2 are correct
(B) 1, 2 and 3 are correct
(C) 2, 3 and 4 are correct
(D) 1, 2, 3 and 4 are correct

58. Consider the given conversion

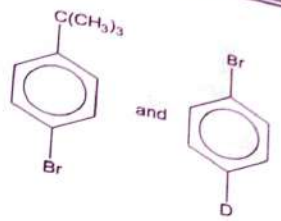


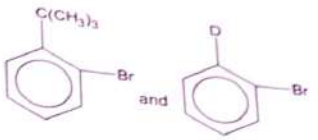
The given conversion can best be performed by
(A) $CH_3–CH_2–CH_2–Cl$/Anhy $AlCl_3$
(B) $CH_3–CH=CH_2/H^+$
(C) $CH_3–CH_2–CH_2–OH/BF_3$
(D)  followed by reduction with NH_2NH_2/alc KOH

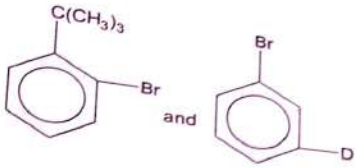
59. In the given reaction

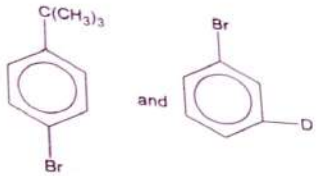


[X] and [Y] respectively will be

(A) 

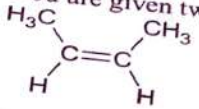
(B) 

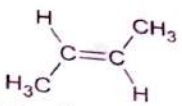
(C) 

(D) 

Comprehension – XXIII

You are given two compounds





First, you name both the compounds. Yes, both of them have the same name. Yet, are they really the same structure, or are they somehow different? Careful inspection should convince you that they are indeed different. Now, how their physical and chemical properties differ?

Generally,

(i) cis isomer has more dipole moment than trans

(ii) cis is less stable than trans

(iii) boiling points of cis isomers are more than those of trans isomers (remember boiling points are directly related to polarity which in turn is determined by dipole moment).

Chemical reactivities of cis and trans also differ from each other.

60. Arrange the following compounds in the increasing order of their boiling points:

$CH_3CH = CHCH_3$ (cis) (i) $CH_3CH = CHCH_3$ (trans) (ii) $CH_3CH = CHCl$ (cis) (iii) $CH_3CH = CHCl$ (trans) (iv)

(A) (ii)<(i)<(iii)<(iv)

(B) (iv)<(ii)<(i)<(iii)

(C) (ii)<(i)<(iv)<(iii)

(D) (iii)<(i)<(ii)<(iv)

61. Which of the following compounds exhibits geometrical isomerism?

$CH_2 = CHCH_3$ (i) $CH_3CH = CHCH_3$ (ii) $CH_3CH = C = CHCH_3$ (iii)

$CH_3CH = C = C = CHCH_3$ (iv)

(A) (ii), (iv)

(B) (ii), (iii) and (iv)

(C) (ii), (iii)

(D) (ii), (iii)

Comprehension – XXIV

A research scholar synthesised alcohol by hydration of 1-butene. He was unaware of the fact that the vessel he used had some coating of a metal and in addition to alcohol (b.p. 100°C) he also got compound (X). X (b.p. 80°C) forms bisulphite compound as well as 2,4-dinitrophenylhydrazone. He however, was successful in the separation based on physical and chemical properties. Read the above technical report and answer the following questions.

62. Which alcohol does research scholar want to sysnthesise and what is the other product?

(A) $CH_3-CH_2CH_2CH_2OH$, $CH_3CH_2-CH_2-CHO$

(B) $H_3C-\underset{\underset{OH}{|}}{CH_2}-CH_2-CH_3$, $H_3C-CH_2-\underset{\underset{O}{||}}{C}-CH_3$

(C) $H_3C-CH_2-CH_2-OH$, $H_3C-CH_2-\underset{\underset{OH}{|}}{CH}-CH_3$

(D) $CH_3-CH_2-CH_2-CH_2-OH$, $CH_3-CH_2-CH=CH_2$

63. He also got yellow precipitate on heating with $NaOH/I_2$. This yellow ppt. is due to
(A) alcohol and side product both
(B) side product only
(C) alcohol only
(D) none of the above

64. Carbonyl compound can be converted into required alcohol by
(A) Clemmensen reduction
(B) Wolf-Kishner reduction
(D) $LiAlH_4$
(D) All of the above

Comprehension – XXV

Volumetric titrations involving $KMnO_4$ are carried out only in presence of dilute H_2SO_4 but not in the presence of HCl or HNO_3. This is because oxygen produced from $KMnO_4$ + dil. H_2SO_4 is used only for oxidizing the reducing agent. Moreover, H_2SO_4 does not give any oxygen of its own to oxidize the reducing agent. In case HCl is used, the oxygen produced from $KMnO_4$ + HCl is partly used up to oxidize HCl and in case HNO_3 is used, it itself acts as oxidizing agent and partly oxidizes the reducing agent. $KMnO_4$ in various mediums gives following products–

$Mn^{7+} \xrightarrow{H^+} Mn^{2+}$

$Mn^{7+} \xrightarrow{H_2O} Mn^{4+}$

$Mn^{7+} \xrightarrow{H_2O} Mn^{6+}$

Read the above paragraph carefully and answer the questions given below it:

65. The equivalent weight of $KMnO_4$ in the given reaction is–

$KMnO_4 \longrightarrow K_2MnO_4 + MnO_2 + O_2$

(A) $\frac{M}{6}$ (B) $\frac{M}{2}$

(C) M (D) $\frac{M}{3}$

66. MnO_4^{2-} (1 mole) in neutral medium disproportionates to,

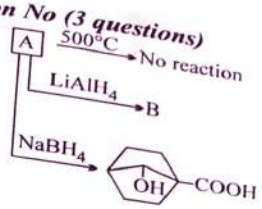
(A) $\frac{2}{3}$ mole of MnO_4^- and $\frac{1}{3}$ mole of MnO_2 (B) $\frac{1}{3}$ mole of MnO_4^- and $\frac{2}{3}$ mole of MnO_2

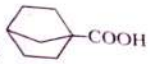
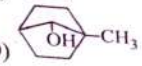
(C) $\frac{1}{3}$ mole of Mn_2O_7 and $\frac{1}{3}$ mole of MnO_2 (D) $\frac{2}{3}$ mole of Mn_2O_7 and $\frac{1}{3}$ mole of MnO_2

67. Number of electrons transferred in each case when $KMnO_4$ acts as an oxidizing agent to give MnO_2, Mn^{++}, $Mn(OH)_3$ and MnO_4^{2-} are respectively–
(A) 3, 5, 4, 1
(B) 4, 3, 1, 5
(C) 1, 3, 4, 5
(D) 5, 4, 3, 1

Question No (3 questions)

Comprehension – XXVII



68. Compound 'B' is?
(A)
(B) 
(C)
(D) 

69. Organic compound A does not undergo decarboxylation reaction because?
(A) Intermediate does not follow Saytzeff'n rule
(B) Intermediate does not follow Hofmann's rule
(C) Intermediate does not follow Bredt's rule
(D) Intermediate does not follow MarkwoniKoffs rule

70. Total number of stereoisomers of B are?
(A) 2
(B) 4
(C) 3
(D) 6

Comprehension – XXVIII

A mixture containing two organic compounds is dissolved in ether and then treated with $NaHCO_3$ solution. The aqueous layer is separated and neutralized with conc. hydrochloric acid. A white precipitate is formed.

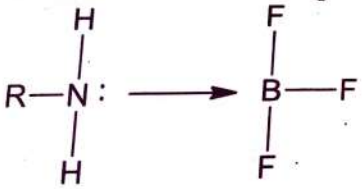
71. The ether insoluble compound is soluble in water. This solution is mixed with Tollen's reagent and warmed. A silver mirror is formed. The compound is –
(A) urea
(B) phthalic acid
(C) glucose
(D) sucrose

72. The other compound obtained as a white precipitate in the above procedure is –
(A) benzoic acid
(B) phthalic acid
(C) phenol
(D) nitrobenzene

73. The ether insoluble component forms a osasone which is also formed by –
(A) fructose
(B) Arabinose
(C) erythrose
(D) galactose

Comprehension – XXIX

Amines are basic compounds. They act as Lewis base due to the presence of lone pair of electrons at nitrogens.



Lewis base Lewis acid

Amines also behave as Arrhenius base as well as Bronsted base

$CH_3NH_2 + HOH \rightleftharpoons CH_3NH_3^+ + OH^-$

$CH_3NH_2 + HCl \rightleftharpoons CH_3NH_3^+ + Cl^-$

Basic character of amines depends on the factors like inductive effect, steric hindrance and resonance. Alkyl groups are electron releasing groups hence these groups increase the electron density at nitrogen as well as the basic character of amines. Basic character of tertiary amines is reduced due to the steric hindrance of three alkyl groups. Experimentally it is observed that stronger bases have smaller pK_b values.

74. Which of the following is most basic in gas phase
(A) NH_3 (B) CH_3NH_2
(C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

75. Which of the following factors influence the basicity of amines?
(i) The inductive effect of alkyl group.
(ii) The polar effect
(iii) The resonance
(A) I, II (B) I, III
(C) II, III (D) I, II, III

76. Which of the following is the most basic in aqueous medium?
(A) NH_3 (B) CH_3NH_2
(C) $(CH_3)_2NH$ (D) $(CH_3)_3N$

Comprehension – XXX

Ester gives nucleophilic addition reaction followed by elimination reaction with carbon nucleophile. When carbon nucleophile is of an ester then the reaction is known as Claisen condensation reaction. This reaction is also carried out between ester and a ketone. A successful Claisen condensation requires an ester with two α – hydrogens and an equivalent amount of base rather than a catalytic amount of base.

77. Consider the given reaction

$$CH_3-COOC_2H_5 \xrightarrow[C_2H_5OH]{C_2H_5ONa} \text{enolate ion} \xrightarrow[\text{Claisen condensation}]{\text{ester(X)}} \text{Product}$$

For the above reaction the most reactive ester is:

(A) $C_6H_5COOC_2H_5$

(B) $C_2H_5O-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-OC_2H_5$

(C) $HCOOC_2H_5$

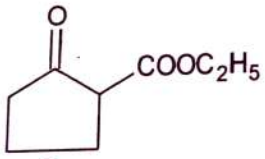
(D) $C_2H_5O-\overset{O}{\overset{\|}{C}}-OC_2H_5$

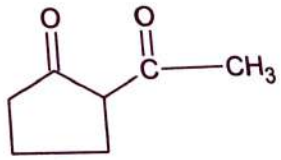
78. Intramolecular Claisen condensation given by diester is known as:
(A) Stobbe condensation (B) Dieckmann condensation
(C) Mannich reaction (D) Reformatsky reaction

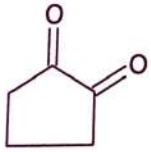
79. In the given reaction:

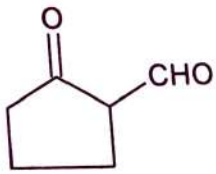
$$C_2H_5O-\overset{O}{\overset{\|}{C}}-(CH_2)_3-CH_2-COOC_2H_5 \xrightarrow[(ii)\ H_2O/HCl]{(i)\ C_2H_5ONa/C_2H_5OH} [X]$$

[X] is:

(A) 

(B) 

(C) 

(D) 

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. An unknown volume and unknown concentration of weak acid HX is titrated with NaOH of unknown concentration. After addition of 10.0 cm^3 of NaOH solution, pH of solution is 5.7 and after the addition of 20.0 cm^3 of NaOH solution, the pH is 6.3. Calculate the pK_a for the weak acid, HX. (Given: antilog of $0.6 \approx 4$)

2. One mole of monoatomic ideal gas ..... in 5 L, piston fitted cylinder at 300 K is heated to 800 K as well as followed to expand, to a volume of 8 L simultaneously. The change in enthalpy of the piston nearly 1.0×10^x J, value 'x' is:

3. How many faradays of electricity are required for the reduction of 1 mole nitrobenzene into N-phenyl hydroxylamine?

4. A large irregularly–shaped closed tank is first evacuated and then connected to a 5.5 litre cylinder containing compressed nitrogen gas. The gas pressure in the cylinder, originally at 2.13 atm, falls to 1.562 atm after it is connected to the evacuated tank. Calculate the volume of the tank in litres.

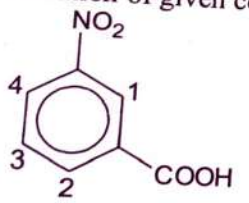
5. Freezing point of 0.2 M KCN solution is –0.7°C. On adding 0.1 mole of $Hg(CN)_2$ to one litre of the 0.2 M KCN solution, the freezing point of the solution becomes –0.53°C due to the reaction $Hg(CN)_2 + mCN^- \longrightarrow Hg(CN)^m_{m+2}$.
What is the value of m assuming molality = molarity?

6. 22.21 ml solution of HIO_3 is reacted with excess of an aqueous solution of SO_2. The excess of SO_2 and I_2 formed are removed by heating the solution. The remaining solution in neutralized by 35.5 ml of 0.16 (N) NaOHsolution. The strength of HIO_3 in gm/lt. (Round off the answer to the nearest whole number) is...?

7. How many lone pairs are present on the central atom in $[ICl_4]^-$.

8. Fructose is subjected to bond cleavages by HIO_4. The number of HCHO unit(s) per unit of fructose is ...

9. Nitration of given compound will takes place at which one of the carbons indicated by number.

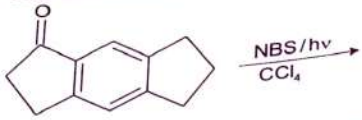


10. When 1-bromomethylcyclohexene under-goes solvolysis in ethanol, what will be the number of products.

11. What are the number isomers for the compound with molecular formula $C_2BrClFI$.

12. For how many 2-substituted ethanols, GCH_2CH_2OH, the gauche conformation is more stable than the anti:
G = - OH, - NH_2, - F, - Cl, - OCH_3, - $NHCH_3$, - $N(CH_3)_3$, - NO_2.

13. Find the number of monobromo derivatives in the given reaction.

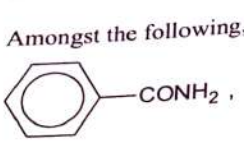


14. When meta-disubstituted benzene $A(C_6H_4X_2)$ is further converted into $B(C_6H_4X_3)$, number of isomers of B formed is.....

15. How many pairs of diastereoisomers are possible in the following molecule?
Ph–CH(Cl)CH(Ph)CH(Cl)Ph

16. How many of the following compounds undergo HVZ reaction?
HCOOH, C_6H_5COOH, CH_3COOH, $CH_3 - CH_2 - COOH$

17. Amongst the following, the total number of compounds which give Hoffmann Bromamide reaction is:



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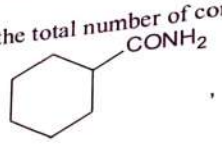

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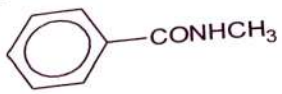

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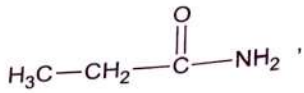

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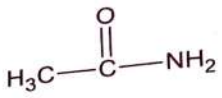


18. In the electrolysis of KI, I_2 is formed at the anode by the reaction;

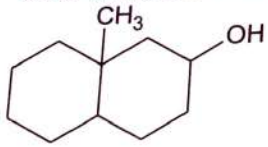
$$2I^- \longrightarrow I_2 + 2e^-$$

After the passage of current of 0.5 ampere for 9650 seconds, I_2 formed required 40 ml of 0.1 M $Na_2S_2O_3 \cdot 5H_2O$ solution in the reaction;

$$I_2 + 2S_2O_3^{2-} \longrightarrow S_4O_6^{2-} + 2I^-$$

What is the current efficiency?

19. What will be the number of stereo-isomers of the given compound:



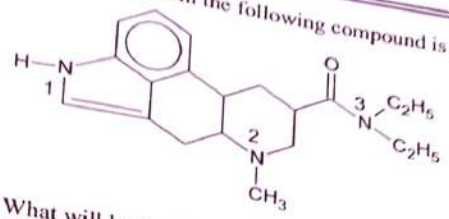
20. How many moles of Br_2 is required to prepare for the complete electrophilic aromatic substitution on phenol.

21. 500 ml of 0.150 M $AgNO_3$ solution is mixed with 500 ml of 1.09 M Fe^{2+} solution and the reaction is allowed to reach equilibrium at 25°C.

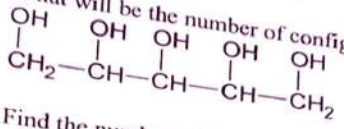
$$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Fe^{3+}(aq) + Ag(s)$$

For 25 ml of the equilibrium solution, 30 ml of 0.0833 M $KMnO_4$ were required for oxidation. Calculate the approximate equilibrium constant for the reaction at 25°C.

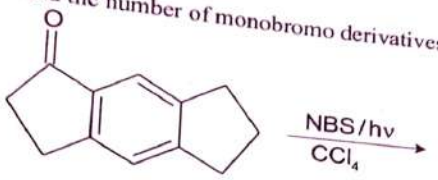
22. Which nitrogen in the following compound is most basic



23. What will be the number of configurational isomers of the compound ?



24. Find the number of monobromo derivatives in the given reaction.



25. A compound of iron and chlorine ($FeCl_x$) is soluble in water. An excess of silver nitrates was added to precipitate the chloride ion as silver chloride. If a 134.8 mg of the compound gave 304.8 mg of AgCl. What is the value of x?

26. A solution containing 2.665 g of $CrCl_3.6H_2O$ is passed through cation exchanger. The chloride ions obtained in solution react with $AgNO_3$ and give 2.87 g of AgCl. What is the number of chloride ion in co-ordination sphere.

27. In alkaline H_2O_2, medium $Cr_2O_7^{2-}$ changes to CrO_8^{3-}. What is the oxidation number of Cr in CrO_8^{3-}

28. How many unpaired electrons are present in $[NiCl_4]^{2-}$complexes?

29. How many isomers are possible for square planar complex $K_2[PdClBr_2SCN]$?

30. A weak field complex of Ni^{2+} has magnetic moment value of 2.82 BM. The number of electrons in the t_{2g} level of Ni^{2+} will be:

31. What is the number of possible isomer of an octahedral complex : $[Co(C_2O_4)_2 (NH_3)_2]^-$.

32. What is the number of unpaired electron in the complex ion $[CoF_6]^{3-}$.

33. Pyrolusite on heating with KOH in presence of air gives a dark green compound (A). The solution of (A) on treatment with sulphuric acid gives a purple colour compound (B). Compound (B) with conc. H_2SO_4 gives (C) which can decompose to (D) and O_2. What is the number of unpaired electrons in the cation of (D)?

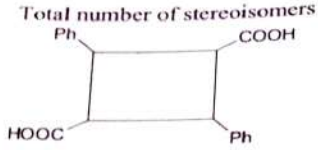
34. What are the number of geometrical isomers of $[Co(NH_3)_3(NO_2)_3]$.

35. When 66.83 g of VCl_4 is dissolved in 1Kg of CCl_4 the freezing point of CCl_4 was lowered by 5.97°. If the equilibrium constant for the reaction is $Y \times 10^{-2}$. Then find the value of Y.

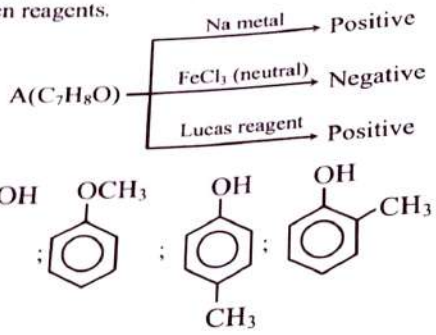
$V_2Cl_8 \rightleftharpoons 2VCl_4$

Assuming that only these two molecular species remain in the solution. The melting point of pure CCl_4 is $-22.9°C$ and its heat of fusion is 4.2 cal gm^{-1}. [Mol. wt. of $VCl_4 = 193$]

36. Total number of stereoisomers of the compound will be:



37. How many of the following isomers of an aromatic compound $A(C_7H_8O)$ give(s) following tests with the given reagents.



38. For the reaction

$A(s) + 2B(g) \rightleftharpoons 2C(g) + D(\ell)$ at TK

find the value of K_p when the equilibrium contains 25% of B(g) by volume of the gaseous species.

39. Find the pH of mixture obtained by mixing equal volume each of 0.1 M Na_3PO_4 and 0.1 M NaH_2PO_4 at TK.

Given: K_{a_1}, K_{a_2} and K_{a_3} of H_3PO_4 at TK are respectively 10^{-3} M, 10^{-6} M and 10^{-12} M.

40. Ammonia gas evolved on treatment of 0.30 g of an organic compound for the estimation of nitrogen was passed in 100 mL of 0.1 M sulphuric acid. The excess of acid required 20 mL of 0.5 M sodium hydroxide solution for complete neutralization. The molar mass of the unknown organic compound is divisible by 10 and results into....

41. 80 ml of $\frac{M}{24}$ $K_2Cr_2O_7$ solution oxidised 22.4 ml H_2O_2 solution. The volume strength of H_2O_2 solution is .....

42. A solid crystal is composed of X, Y and Z atoms. Y atoms are occupying 50% of octahedral voids, whereas X atoms are occupying the 100% tetrahedral void with Z atoms in ccp array arrangement, then the number of Y atoms in rational formula the given cell

43. K_{sp} of $Cu(OH)_2$ is 5×10^{-16} at 25°C, then the pH of a saturated solution of $Cu(OH)_2$ is:

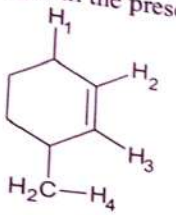
44. x cc volume is occupied by molecules in motion for one mole of gas at NTP if molecules are spherical having the radius 2.155×10^{-9} cm. The value of x is.....

45. An alloy of Pb-Ag weighing 1.08 g was dissolved in dilute HNO_3 and the volume made to 100 mL. A silver electrode was dipped in the solution and the emf of the cell set-up
$Pt(s), H_2(g) \mid H^+(1\ M) \mid\mid Ag^+(aq) \mid Ag(s)$ was 0.62 V. If E°_{cell} is 0.80 V, what is the percentage of Ag in the alloy? (At 25°C, RT/F = 0.06)

46. The rate constant of the reaction,
$2H_2O(aq) \longrightarrow 2H_2O(\ell) + O_2(g)$ is $3\times10^{-3}\ min^{-1}$. The rate of the reaction is $2 \times 10^{-4}\ Ms^{-1}$ when concentration of H_2O_2 is x M. What is the value of x?

47. Potassium crystallizes in a body-centered cubic lattice, with a unit cell length a = 5.20 Å: What is the ratio of no of third nearest neighbours to the number of second nearest neighbours?

48. Which hydrogen in the given compound has the least tendency to get chlorinated when it is exposed to sunlight in the presence of chlorine?



49. An ideal engine operates in a carnot cycle between 327°C and 127°C. If absorbs 9 kcal of heat at a high temperature. The amount of heat in kcal rejected to sink is:

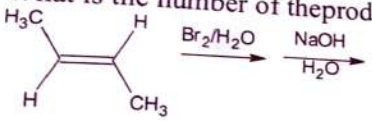
50. Which electronic transition in Balmer series of hydrogen atom has same frequency as that of n = 6 to n = 4 transition in He^+.

51. Sulphide ions react with $Na_2[Fe(NO)(CN)_5]$ to form a purple coloured compound $Na_4[Fe(CN)_5NOS]$. In the reactions, the change in oxidation state of iron is

52. If the freezing point of 0.1 molal $HA_{(aq)}$ is – 0.2046°C, then pH of the solution is:

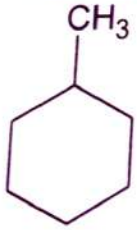
53. How many mL of 0.5 M H_2SO_4 are needed to dissolve 0.5 g of Copper (II) carbonate?

54. What is the number of theproducts of the following sequence of reactions?



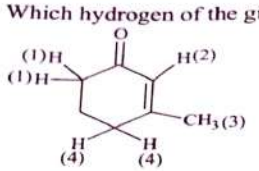
55. An unknown inorganic compound (X) loses its water of crystallization on heating and its aqueous solution gives the following reactions.
(i) It gives a white turbidity with dilute hydrochloric acid solution.
(ii) It decolourizes a solution of iodine in potassium iodide.
(iii) It gives a white precipitate with silver nitrate solution which turns black on standing.
(iv) It imparts a golden yellow colour to the flame in the flame test.
How many molecules of water of crystallisation are there in 1 molecule of (X).

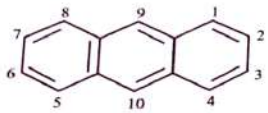
56. What is the total number of alkyl halides can be produced from monochlorination of A.



A

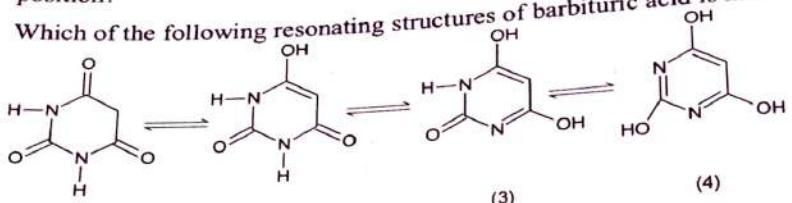
57. Which hydrogen of the given compound is least acidic



58.  $+ CH_3Cl \xrightarrow{AlCl_3}$ major product

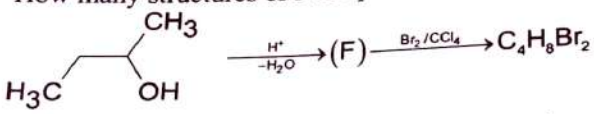
In this reaction the electrophile will attack at the two positions one of them is '10' ; which is the other position?

59. Which of the following resonating structures of barbituric acid is most stable.

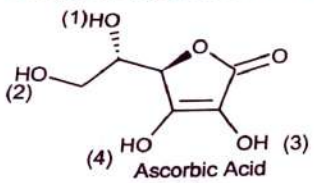


Barbituric Acid

60. How many structures of F are possible.....



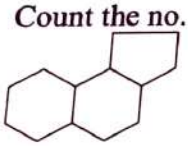
61. Ascorbic acid (Vitamin C) does not have a carboxylic group (—COOH) but it is sufficiently acidic to liberate CO_2 when aqueous $NaHCO_3$ is added.
Which of the labeled H atoms will be most acidic.



Ascorbic Acid

62. Three faradays of electricity are passed through molten Al_2O_3, aqueous solution of $CuSO_4$ and molten NaCl in three different electrolytic cells. The amount of Al, Cu, Na deposited at the cathodes is in the ratio X:Y:Z; The sum X+Y+Z =?

63. Count the no. of chiral atoms in the following compounds



64. Calculate the ratio of sodium formalc and formic acid ($K_a = 2 \times 10^{-4}$) in a buffer solution of Ph = 4.3

65. Calculate k_p for the reaction $k_c = 0.2$ at 305K

66. Number of resonating st. having 2° carbocation

MATRIX MATCH TYPE QUESTIONS

1. Match the Column – I with Column – II:

Column – I	Column – II
(A) Density of gas	(p) depends on absolute temperature.
(B) Avg. KE of gas	(q) depends on molecular mass
(C) Root mean square speed	(r) depends on total molecules of gas
(D) effusion rate of gas	(s) independent of molecular mass of gas

2. Match the symbols in Column – I with their corresponding cell type Column – II:

Column – I	Column – II
(A) $Pt \mid Fe^{3+}, Fe^{2+}$	(p) Metal-metal ion half cell
(B) $Pt \mid H_2 \mid H^+$	(q) Gas-Gas ion half cell
(C) $Pt \mid Hg \mid Hg_2^{2+}$	(r) Oxidation-reduction half cell
(D) $Pb \mid PbSO_4(s) \mid SO_4^{2-}$	(s) Metal-sparingly soluble salt half cell
	(t) Standard reduction potential of the half cell is zero

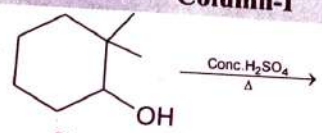
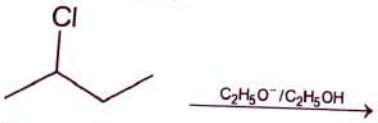
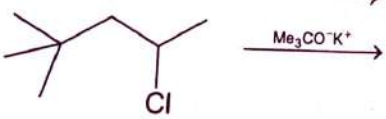
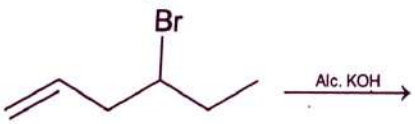
3. Match the Column – I with Column – II:

Column – I	Column – II
(A) Zn + very dil. HNO_3	(p) NO
(B) Sn + dil. HNO_3	(q) NO_2
(C) Ag + dil. HNO_3	(r) NH_4NO_3
(D) Pb + conc. HNO_3	(s) $Sn(NO_3)_2$
	(t) $Zn(NO_3)_2$

4. Match the Column – I with Column – II:

Column – I	Column – II
(A) $[Fe(CN)_6]^{4-}$	(p) Inner orbital complex
(B) $[Fe(CN)_6]^{3-}$	(q) sp^3d^2
(C) $[Co(H_2O)_6]^{2+}$	(r) Paramagnetic
(D) $[Fe(H_2O)_6]^{2+}$	(s) Diamagnetic
	(t) d^2sp^3

5. Match the following question Column-I with Column-II:

Column-I	Column-II
(A) 	(p) Hoffmann's alkene
(B) 	(q) Saytzeff
(C) 	(r) Transition state
(D) 	(s) Carbocation
	(t) 1, 2-alkyl shift

6. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	$C_6H_5COCH_3$	(p)	HVZ reaction
(B)	C_6H_5CHO	(q)	Does not reduce Fehling's solution
(C)	$CH_3 - CH_2COOH$	(r)	Sodium bisulphate solution
(D)	HCOOH	(s)	Sodium hypoiodite
		(t)	Reduces Tollen's reagent

7. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Aldol condensation	(p)	CN^- as catalyst
(B)	Cannizzaro reaction	(q)	Organozinc compound
(C)	Benzoin condensation	(r)	Hydride transfer
(D)	Reformatsky reaction	(s)	β – hydroxyl aldehyde
		(t)	Enolate ion

8. Match the following:

Column-I(organic compounds oxidised by HIO_4)		Column-II	
(A)	CH_3COCHO	(p)	$PhCH = O + HCOOH$
(B)	1,2-cyclohexane dione	(q)	$CH_3CH_2CHO + HOOCCH_3$
(C)	PhCH (OH) CHO	(r)	$HOOC(CH_2)_4COOH$
(D)	$CH_3CH_2CH(OH)COCH_3$	(s)	$CH_3COOH + HCOOH$

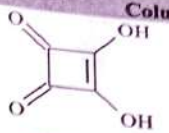
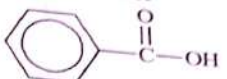
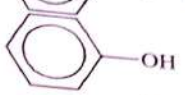
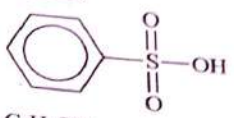
9. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	$CH_3CH_2CH_2CN$	(p)	Reduction with $Pd - C/H_2$
(B)	$CH_3CH_2OCOCH_3$	(q)	Reduction with $SnCl_2/HCl$
(C)	$CH_3 - CH = CH - CH_2 - OH$	(r)	Development of foul smell on treatment with chloroform and alcoholic KOH
(D)	$CH_3CH_2CH_2CH_2NH_2$	(s)	Reduction with diisobutyl aluminium hydride
		(t)	Alkaline hydrolysis

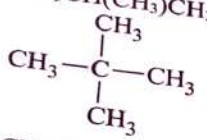
10. Match the Column – I with Column – II:

Column – I		Column – II	
(A)	Identification of 1°, 2°& 3° Alcohols	(p)	Oxymercuration demercuration
(B)	Identification of 1°, 2°& 3° Nitro alkanes	(q)	Cu/300° heat
(C)	Formation of alcohol by anti Markovnikov's addition of H_2O	(r)	Victor Meyer's test
(D)	Formation of alcohol by Markovnikov's addition	(s)	Hydroboration oxidation
		(t)	Lucas test

11. Match the Column – I with Column – II:

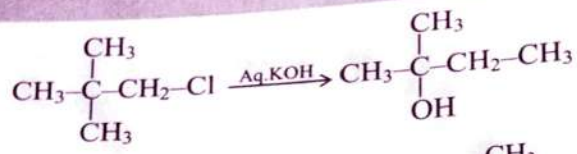
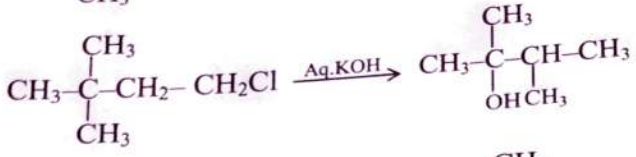
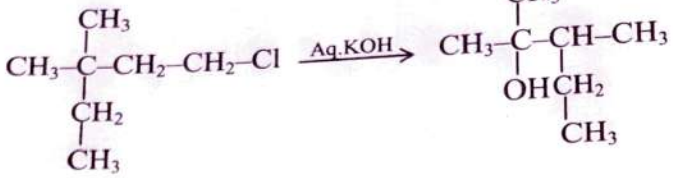
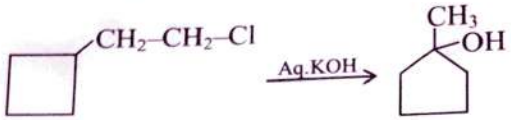
Column – I	Column – II
(A) $NaHCO_3$	(p) 
(B) Na will react with	(q) 
(C) NaOH will react with	(r) 
(D) $NaNH_2$ will react with	(s) 
	(t) C_2H_5OH

12. Match the Column – I with Column – II:

Column – I	Column – II (Boiling point)
(A) $CH_2CH_2CH_2CH_2CH_3$	(p) 28°C
(B) $CH_3CH(CH_3)CH_2CH_3$	(q) 36°C
(C) 	(r) 10°C
(D) $CH_3CH_2CH_2CH_3$	(s) –0.4°C
	(t) 100°C

MATCHING LIST TYPE

1. Match the Column – I with Column – II:

List-I		List-II
(P) 		1. Bond shift
(Q) 		2. Hydride shift
(R) 		3. Methanide shift
(S) 		4. Ethanide shift

Codes:

	P	Q	R	S
(A)	3	2, 3	2, 4	1, 2
(B)	2	3	1	4
(C)	2,3	3,1	4,2	1,3
(D)	3	2	1	4

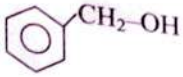
2. Match the Column – I with Column – II:

List-I (Reaction)	List-II (Reagent)
(A) $CH_3-CH=CH_2 \longrightarrow CH_3-\underset{\displaystyle Br}{\underset{\vert}{CH}}-CH_3$	1. HBr
(B) $CH_3-CH=CH_2 \longrightarrow CH_3-CH_2-CH_2Br$	2. NBS
(C) $CH_3-CH=CH_2 \longrightarrow BrCH_2-CH=CH_2$	3. Br_2/CCl_4
(D) $CH_3-CH=CH_2 \longrightarrow CH_3-CHBr-CH_2Br$	4. HBr (Peroxide)
	5. $[(CH_3)_3C]_2O_2$

Codes:

	A	B	C	D
(A)	3	2	1	4
(B)	2	3	1	4
(C)	1	4	2	3
(D)	3	2	1	4

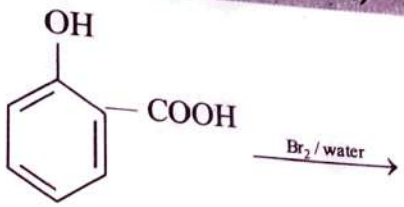
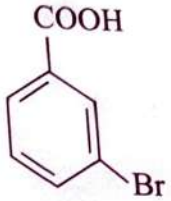
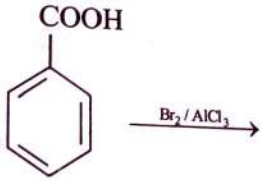
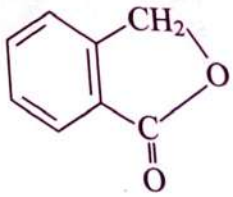
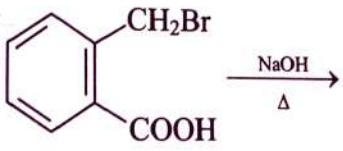
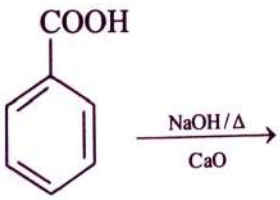
3. Match the List – I with List – II:

List-I	List-II
(P) CH_3-CH_2-OH & $CH_3-\underset{OH}{\underset{\mid}{CH}}-CH_3$	1. Victor Meyer Test
(Q) $CH_2=CH-\underset{OH}{\underset{\mid}{CH}}-CH_3$ & $CH_3-CH_2-\underset{OH}{\underset{\mid}{CH_2}}$	2. $FeCl_3$ Test
(R) $CH_3-\underset{CH_3}{\overset{CH_3}{\underset{\mid}{\overset{\mid}{C}}}}-OH$ &	3. Lucas Test
(S) & 	4. Br_2 water Test

Codes:

	P	Q	R	S
(A)	3	2	1	4
(B)	1, 3	1, 3, 4	2, 3, 4	1, 2, 3, 4
(C)	2	3	4	1
(D)	3,2	2,4,1	1,4,3,2	4

4. Match the following:

List-I (Reactions)	List-II (Products)
(P) 	1. 
(Q) 	2. 
(R) 	3. 2, 4, 6-tribromophenol
(S) 	4. Benzene

Codes:

	P	Q	R	S
(A)	3	2	1	4
(B)	2	3	1	4
(C)	2	3	4	1
(D)	3	1	2	4

5. Match the column

List-I		List-II (Rate)	
(P)	$C_6H_6 + \text{conc. } H_2SO_4 \longrightarrow C_6H_5SO_3H + H_2O$	1.	$C_6H_6 > C_6D_6 > C_6T_6$
(Q)	$C_6H_6 + \text{conc. } HNO_3 \longrightarrow C_6H_5NO_2 + H_2O$	2.	$C_6H_3D_3 > C_6H_3T_3 > C_6T_6$
(R)	$C_6H_6 + I_2 \xrightarrow{HNO_3} C_6H_5I + HI$	3.	$C_6H_6 \simeq C_6H_2D_2T_2 \simeq C_6HT_5$
(S)	$C_6H_6 + Cl_2 \xrightarrow{Fe} C_6H_5Cl + HCl$	4.	$C_6H_2D_2T_2 > C_6HDT_4 > C_6T_6$

Codes:

	P	Q	R	S
(A)	3	2	1	4
(B)	1, 2, 4	3	1, 2, 4	3
(C)	2	3	4	1
(D)	3,1,4	2,4,1	1,3,2	4

6. Match List-I with List-II for the given reaction and select the correct answer from the codes given below the lists:

$$Z - CH_2 - Br + CH_3O^{\ominus} \longrightarrow Z - CH_2 - OCH_3 + \overset{\ominus}{B}r$$

List-I (Z)	List-II (relative reactivity)
(P) H –	(1) 0.1
(Q) CH_3 –	(2) 3
(R) C_2H_5 –	(3) 1
(S) $(CH_3)_2CH$–	(4) 100

Codes:

	P	Q	R	S
(A)	4	2	3	1
(B)	4	2	1	3
(C)	2	4	3	1
(D)	1	2	3	4

7. Match List-I with List-II and select the correct answer from the codes given below:

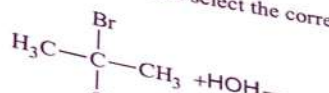
List-I	List-II
(P) $CH_3 - O - SO_2CH_3 + C_2H_5\overset{\ominus}{O}$	1. $CH_3 - CH_2 - PH_2$
(Q) $CH_3 - CH_2 - I + PH_3$	2. $CH_3 - O - C_2H_5$
(R) $HC \equiv \overset{\ominus}{C}\overset{\oplus}{Na} + CH_3 - CH_2 - Br$	3. $CH_3 - O - CH_3$
(S) $CH_3 - Cl + CH_3 - \overset{\ominus}{O}$	4. $CH \equiv C - CH_2 - CH_3$

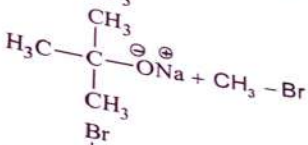
Codes:

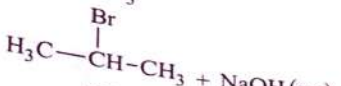
	P	Q	R	S
(A)	4	3	2	1
(B)	1	2	3	4
(C)	2	1	4	3
(D)	2	1	3	4

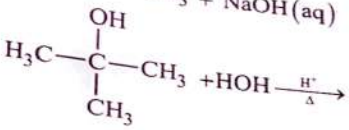
8. Match List-I with List-II and select the correct answer from the codes given below the lists:

List-I

(P) 

(Q) 

(R) 

(S) 

List-II

1. Ether
2. Sec. alcohol
3. Tert alcohol
4. Alkene

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	3	1	2	4
(C)	3	1	4	2
(D)	1	3	2	4

9. Match the column

List-I (Metallurgical Process)		List-II (Elements)
(P)	Thermal decomposition of its compound.	1. Lead
(Q)	Liquation methord	2. Nickel
(R)	Electrolytic reduction method.	3. Bismuth
(S)	Carbon reduction method.	4. Aluminium

Codes:

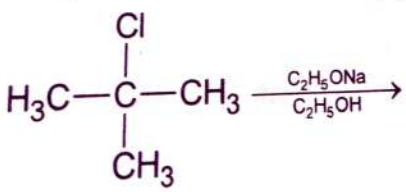
	P	Q	R	S
(A)	3	2,1	1	4
(B)	2	3	1	4
(C)	2	1,3	4	1
(D)	3	2	1	4

10. Match List-I with List-II and select the correct answer from the codes given below the lists:

List-I

(Substrate + reagent)

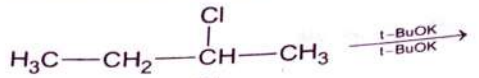
(P) $CH_3 - CH_2 - CH_2 - Cl \xrightarrow[C_2H_5OH]{C_2H_5ONa}$

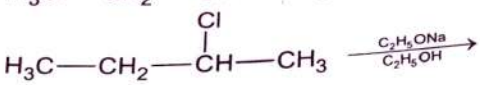
(Q) 

List-II

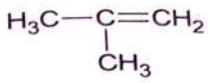
(Product)

1. $CH_3 - CH_2 - CH = CH_2$
2. $CH_3 - CH = CH - CH_3$

(R) 

(S) 

3. $CH_3 - CH = CH_2$

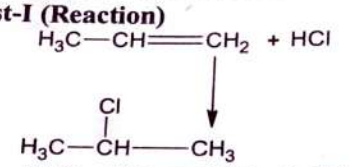
4. 

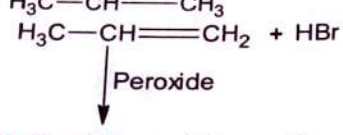
Codes:

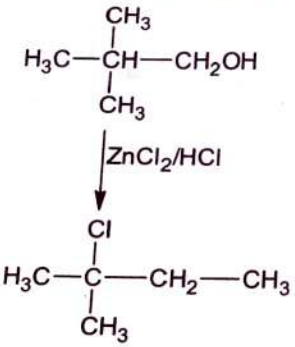
	P	Q	R	S
(A)	3	4	1	2
(B)	3	4	2	1
(C)	4	3	1	2
(D)	4	3	2	1

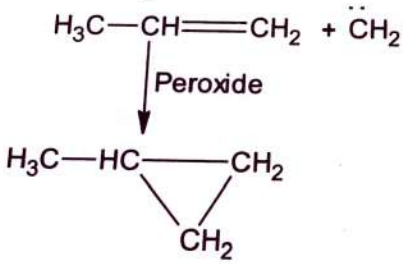
11. Match List-I with List-II and select the correct answer using the codes given below the lists:

List-I (Reaction)

(P) 

(Q) 

(R) 

(S) 

List-II(Mechamism)

1. S_N1
2. Electrophilic addition
3. Free radical addition
4. Concerted addition

Codes:

	P	Q	R	S
(A)	3	1	2	4
(B)	2	3	1	4
(C)	2	3	4	1
(D)	3	2	1	4

12. Match List-I with List-II and select the correct answer from the codes given below the lists:

List-I (Reaction)	List-II (Product)
(P) $CH_3-CH=CH_2 \xrightarrow[(ii)\ HCOOH]{(i)\ BH_3/THF}$	1. n-propyl bromide
(Q) $CH_3-CH=CH_2 \xrightarrow[(ii)\ AgNO_3]{(i)\ BH_3/THF}$	2. n-propylamine
(R) $CH_3-CH=CH_2 \xrightarrow[(iii)\ HOH]{(i)\ BH_3/THF,\ (ii)\ NH_2Cl}$	3. n-hexane
(S) $CH_3-CH=CH_2 \xrightarrow[(ii)\ Br_2]{(i)\ BH_3/THF}$	4. n-propane

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	4	3	2	1
(C)	2	4	1	3
(D)	2	4	3	1

13. Match List-I with List-II and select the correct answer from the codes given below the lists

List-I (Alkali metal)	List-II (Flame colour)
(P) Li	1. Pale violet
(Q) K	2. Crimson red
(R) Rb	3. Red violet
(S) Cs	4. Bluish

Codes:

	P	Q	R	S
(A)	1	2	3	4
(B)	2	1	3	4
(C)	3	1	2	4
(D)	4	3	2	1

14. Match List-I with List-II and select the correct answer from the codes given below the lists

List-I (Metal)	List-II (Ore)
(P) Zinc	1. Azurite
(Q) Tin	2. Carnallite
(R) Copper	3. Calamine
(S) Magnesium	4. Cassiterite

Codes:

	P	Q	R	S
(A)	3	4	1	2
(B)	1	2	3	4
(C)	2	1	3	4
(D)	3	1	2	4

ANSWERS

SUBJECTIVE PROBLEMS

1. $\nu = 8.13 \times 10^{14}$ Hz.
2. 6.6×10^{-5} cm.
3. 68.63%
4. mili moles of $Li_2O = x/2$
   mili moles of $Li_3N = y/3$
5. W = - 3.988 kJ
6. 1.8 atm
7. 1.5
8. 0.3121 M
9. 54 K.cal/mol
10. 7.26
11. 0.51 K molality^{-1}
12. (i) 1.20 Volt
13. (a) 1 – 0.578 = 0.422
    (b) a = 0.9967 mole
    b = 0.537 mole
14. 0.0165; 0.775 %
20. A = P
    B = H_3PO_4
    C = SO_2
    D = H_2SO_4
    E = Al
    F = PH_3
21. (A) Roasting
    (B) Reduction by carbon or more electropositive metal
    (C) Electrolysis
27. 3
29. D < C < A < E < B < F
32. Nitro group is a deactivating group.
40. (i) Solvent extraction (ii) distillation
41. Carius method
48. 0.5
49. 16.66 ml
50. 4.66 L
67. 98.19 kJ
71. (a) 2.44×10^{14} (b) 4.7A°
72. 3120
73. 0011
74. 0005
75. Slope equal to Ea
76. 4 : 7 : 5
77. 0.638, 0.262, 0.1
78. 2.2 atm
79. 1.0887
80. (b) 0.225
81. 1.715 Å
82. 0.75
83. $9.8 \times 10^{-11} < 1.8 \times 10^{-10}$ $K_{sp}AgCl$
86. 0.043 M
87. 7
90. 0.002
91. +3.769
92. –13.842 KJ/mol
93. 2.07×10^{-3}
94. $\Delta H° = -0.285.4$ KJ/mol
96. % loss= 13.5%
    Hence compounds
    A= $[Cr(H_2O)_6]Cl_{3fc}$
    B= $[Cr(H_2O)_5Cl]Cl_2H_2O$
    C=$[Cr(H_2O)_4Cl_2]Cl.2H_2O$
97. Magnetic moment = $\sqrt{1(1+2)} = \sqrt{3}$ B.M
    Magnetic moment = $\sqrt{5(5+2)} = \sqrt{35}$ B.M
98. a) PBr_2Cl_3
    b) $XeOF_2$

MCQ (SINGLE OPTION CORRECT)

1.	A	2.	C	3.	A	4.	A
5.	A	6.	B	7.	A	8.	B
9.	A	10.	A	11.	C	12.	C
13.	A	14.	A	15.	C	16.	B
17.	B	18.	C	19.	D	20.	D
21.	D	22.	B	23.	B	24.	C
25.	B	26.	A	27.	A	28.	D
29.	B	30.	B	31.	D	32.	D
33.	A	34.	B	35.	B	36.	B
37.	A	38.	A	39.	A	40.	D
41.	B	42.	C	43.	A	44.	B
45.	C	46.	D	47.	A	48.	C
49.	A	50.	A	51.	B	52.	A
53.	B	54.	B	55.	C	56.	A
57.	D	58.	A	59.	D	60.	C
61.	A	62.	D	63.	B	64.	A
65.	B	66.	C	67.	B	68.	B
69.	A	70.	C	71.	C	72.	D
73.	B	74.	C	75.	B	76.	B
77.	D	78.	C	79.	B	80.	C
81.	C	82.	B	83.	C	84.	B
85.	D	86.	D	87.	C	88.	B
89.	C	90.	B	91.	C	92.	D
93.	C	94.	C	95.	A	96.	A
97.	C	98.	C	99.	A	100.	D
101.	C	102.	D	103.	C	104.	A
105.	B	106.	D	107.	A	108.	D
109.	D	110.	A	111.	C	112.	A
113.	C	114.	D	115.	D	116.	C
117.	A	118.	B	119.	B	120.	C
121.	A	122.	A	123.	D	124.	A
125.	A	126.	D	127.	A	128.	A
129.	A	130.	C	131.	A	132.	A
133.	B	134.	B	135.	A	136.	D
137.	D	138.	C	139.	A	140.	D
141.	D	142.	D	143.	D	144.	B
145.	D	146.	D	147.	D	148.	D
149.	D	150.	A	151.	C	152.	A
153.	A	154.	C	155.	D	156.	A
157.	B	158.	D	159.	C	160.	C
161.	C	162.	A	163.	A	164.	C
165.	C	166.	B	167.	A	168.	A
169.	D	170.	D	171.	A	172.	D
173.	A	174.	A	175.	A	176.	A
177.	A	178.	A	179.	B	180.	B
181.	B	182.	D	183.	A	184.	A
185.	A	186.	D	187.	D	188.	A
189.	A	190.	B	191.	B	192.	A
193.	B	194.	C	195.	A	196.	A
197.	D	198.	D	199.	C	200.	B
201.	B	202.	D	203.	B	204.	B

205.	C	206.	C	207.	C	208.	A
209.	C	210.	C	211.	B	212.	B
213.	B	214.	A	215.	A	216.	B
217.	B	218.	A	219.	C	220.	D
221.	C	222.	A	223.	A	224.	B
225.	D						

REASONING TYPE QUESTIONS

1.	D	2.	C	3.	B	4.	B
5.	B	6.	C	7.	D	8.	D
9.	C	10.	B	11.	D	12.	A
13.	C	14.	C	15.	B		

MCQ (MULTIPLE OPTIONS CORRECT)

1.	A, B, D	2.	A, C	3.	A, B, C, D	4.	A, B, D
5.	A, B, C	6.	A, C, D	7.	A, B	8.	A, B, C
9.	C, D	10.	B, C	11.	A, B	12.	A, B, C, D
13.	B, C, D	14.	A, C	15.	A, B, C	16.	A, C
17.	A, B, C, D	18.	A, B, D	19.	A, D	20.	A, B, C
21.	A, B, D	22.	A, B, C	23.	A, B, C, D	24.	A, B, C
25.	B, C, D	26.	B, C	27.	A,B,C,D	28.	B, C, D
29.	A, B, D	30.	A, B, C	31.	A, C	32.	C, D
33.	A, C, D	34.	A, B, C	35.	B, D	36.	A, B
37.	B, C, D	38.	A, B	39.	A, C, D	40.	A, B, C, D
41.	B, C, D	42.	B, D	43.	B, C, D	44.	A, B, C, D
45.	A, C, D	46.	A, B, C	47.	A, B	48.	A, B, C, D
49.	A, B, D	50.	A, B, C	51.	B	52.	A, B
53.	A,C	54.	B, C, D	55.	A, C	56.	B, C
57.	A, B, C	58.	A, C	59.	A, D	60.	A, B, C, D
61.	A, B						

COMPREHENSIONS

1.	C	2.	B	3.	D	4.	C
5.	B	6.	A	7.	B	8.	C
9.	B	10.	D	11.	A	12.	C
13.	D	14.	C	15.	B	16.	B
17.	B	18.	A	19.	A	20.	A
21.	C	22.	A	23.	C	24.	D
25.	B	26.	B	27.	A	28.	B
29.	C	30.	A	31.	A	32.	D
33.	A	34.	A	35.	A	36.	C
37.	B	38.	D	39.	A	40.	D
41.	C	42.	B	43.	D	44.	B
45.	C	46.	A	47.	A	48.	A
49.	C	50.	A	51.	B	52.	A
53.	C	54.	B	55.	B	56.	C
57.	B	58.	D	59.	A	60.	A
61.	A	62.	B	63.	A	64.	C
65.	B	66.	A	67.	A	68.	C
69.	C	70.	B	71.	C	72.	A
73.	A	74.	D	75.	D	76.	C
77.	B	78.	B	79.	A		

SINGLE INTEGER ANSWER TYPE QUESTIONS

1.	6	2.	4	3.	4	4.	2
5.	2	6.	9	7.	2	8.	2
9.	3	10.	3	11.	6	12.	7
13.	3	14.	3	15.	8	16.	2
17.	4	18.	8	19.	8	20.	3
21.	3	22.	2	23.	4	24.	3
25.	2	26.	1	27.	5	28.	2
29.	4	30.	6	31.	3	32.	4
33.	3	34.	2	35.	2	36.	5
37.	1	38.	9	39.	9	40.	6
41.	5	42.	1	43.	9	44.	1
45.	1	46.	4	47.	2	48.	2
49.	6	50.	3	51.	0	52.	2
53.	8	54.	2	55.	5	56.	7
57.	2	58.	9	59.	4	60.	3
61.	4	62.	2	63.	4	64.	4
65.	2	66.	4				

MATRIX-MATCH THE FOLLOWING

1. A → p, q; B → p, s; C → p, q; D → p, q, r
2. A → (r) B → (q, t) C → (p) D → (s)
3. A → (t); B → (r, s) C → (p) D → (q)
4. A → (p, s, t) B → (p, r, t) C → (q, r) D → (q, r)
5. A → (q, r, s, t), B → (q, r), C → (p, r), D → (p, r)
6. A → (s) B → (q) C → (p) D → (q, t)
7. A → (s, t) B → (r) C → (p) D → (q)
8. A → (s); B → (r); C → (p); D → (q)
9. A → (p, q, s, t) B → (p, s, t) C → (p) D → (r)
10. A → (q,r,t); B → (r); C → (s); D → (p)
11. A → (p,q,s); B → (p,q,r,s,t); C → (p,q,r,s); D → (p,q,r,s,t)
12. A → q; B → p; C → r; D → s

MATCHING LIST TYPE

1.	A	2.	C	3.	B	4.	D
5.	B	6.	A	7.	C	8.	B
9.	C	10.	A	11.	B	12.	B
13.	B	14.	A				

CHEMISTRY

HINTS & SOLUTIONS

NUMERICAL BASED PROBLEMS (SUBJECTIVE)

1. The frequency of revolution of electron is given by:

$$\text{Frequency} = \frac{1}{\text{time period}}$$

$$\text{Time period} = \frac{\text{total distance covered in 1 revolution}}{\text{velocity}} = \frac{2\pi r}{v}$$

$$\text{Hence frequency } \nu = \frac{v}{2\pi r}$$

First we will have to calculate velocity (v) and radius (r) of electron in second Bohr's orbit by using the standard relations for ν and v. ($Z = 1$ for H-atom)

$$r_n = \frac{5.3 \times 10^{-11} \times n^2}{Z} \text{m} = (0.53 \times 10^{-10}\, n^2)\ \text{m}$$

Now $n = 2$, $Z = 1$

$$r_2 = 0.53 \times 10^{-10}\,(2)^2\ \text{m} = 2.12 \times 10^{-10}\ \text{m}$$

$$v_n = \frac{2.18 \times 10^6}{n}\ \text{m/s}$$

$$v_2 = \frac{2.165 \times 10^6}{2} = 1.082 \times 10^6\ \text{m s}^{-1}$$

$$\text{Hence frequency} = \frac{v_2}{2\pi r_2} = \frac{1.082 \times 10^6}{2\pi(2.12 \times 10^{-10})}$$

$\nu = \mathbf{8.13 \times 10^{14}\ Hz}$.

2. We have,

$\Delta E = E_3 - E_2$

$= -2.41 \times 10^{-12} - (-5.42 \times 10^{-12})$

$= 3.01 \times 10^{-12}$ erg

We know, $\Delta E = h\nu$ and $c = \nu\lambda$

Thus $\Delta E = \frac{hc}{\lambda}$

$$\therefore \frac{hc}{\lambda} = 3.01 \times 10^{-12}$$

$$\therefore \lambda = \frac{hc}{3.01 \times 10^{-12}}$$

$$= \frac{6.62 \times 10^{-27} \times 3 \times 10^{10}}{3.01 \times 10^{-12}}$$

$= \mathbf{6.6 \times 10^{-5}\ cm}$.

3.

	$2NO$	+	$O_2 \longrightarrow 2NO_2 \rightleftharpoons N_2O_4$	
Initial	0.4285		0.3428	
partial press.	0		0.12855	0.4285
			$0.4285 - P$	$\frac{P}{2}$

Total pressure x at the end

$$P = 0.55704 - \frac{P}{2} = 0.41$$

$$\Rightarrow P = 0.2941$$

$$\% \text{ dimerization} = \frac{0.2941}{0.4285} \times 100$$

$$= 68.63\%$$

4. Let no. of mill moless of Li used for Li_2O and Li_3N are x and y respectively.

$$4Li + O_2 \longrightarrow 2Li_2O$$

$$6Li + N_2 \longrightarrow 2Li_3N$$

mili moles of $Li_2O = x/2$
mili moles of $Li_3N = y/3$

5. $W = -3.988$ kJ

6. $$\underset{(0.5-P)}{\overset{0.5}{CO_2(g)}} + \underset{0}{\overset{0}{C(s)}} \rightleftharpoons \underset{2P}{\overset{0}{2CO(g)}}$$

Pressure at equilibrium $= 0.8 = 0.5 - P + 2P$
$P = 0.3$ atm.

$$K_P = \frac{(P_{CO})^2}{P_{CO_2}} = \frac{(0.6)^2}{(0.2)} = 1.8 \text{ atm}$$

7. $$K_P = \frac{\left(P_{H_2O}\right)^2}{\left(P_{H_2}\right)^2} = \frac{(0.55)^2}{(0.45)^2} = 1.5$$

8. As E° for silver electrode is more positive, the cell may be represented as
$Fe^{2+}, Fe^{3+} \,||\, Ag^+, Ag$, $\quad E^\circ_{cell} = 0.80 - 0.77 = 0.03$ V

$$E_{cell} = 0.03 - \frac{0.0591}{1} \log \frac{\left[Fe^{3+}\right]}{\left[Fe^{2+}\right]\left[Ag^+\right]}$$

$$\therefore E_{cell} = 0.03 + \frac{0.0591}{1} \log\left[Ag^+\right] \quad \because \left[Fe^{3+}\right] = \left[Fe^{2+}\right]$$

$$\therefore 0 = 0.03 + \frac{0.0591}{1} \log\left[Ag^+\right] \quad \therefore \left[Ag^+\right] = 0.3121\,M$$

9. Naphthalene is 10 π e's system
i.e. there are 5 π bonds
Expected (theoretical) heat of hydrogenation $= -28.6 \times 5 = -143$ kCal/mol
Observed (experimental) heat of hydrogenation $= -89$ KCal/mol
R.E. $= -89 - (-143)$

10. $$K_f = \frac{M_1 R T_f^{\circ 2}}{\Delta H_f \times 1000} = \frac{(147.01)(8.314)(326.28)}{(17880)(1000)}$$

$K_f = 7.26$ K kg/mol

11. $K_b = \frac{RT_b^2}{1000\ell_v} = \frac{RT_b^2 M}{1000\,\Delta H_v}$

$\Rightarrow K_b = \frac{RK_b.M}{1000 \times \frac{\Delta H_v}{T_b}} = \frac{RT_bM}{1000\Delta S_v}$

For water, $K_b = \frac{2\times373\times18}{1000\times26.33} = 0.51\,K\ molality^{-1}$

12. (i) The half reactions are:

at cathode

$2Ag^+(aq) + 2e^- \longrightarrow 2Ag(s)$

$E^\circ_{Ag^+/Ag} = 0.80$ volt (Reduction potential)

at anode

$Cd(s) \longrightarrow Cd^{2+}(aq) + 2e^-$

$E^\circ_{Cd^+/Cd} = -0.40$ volt (Reduction potential)

or $E^\circ_{Cd/Cd^{2+}} = +0.40$ volt

$E^\circ = E^\circ_{Cd/Cd^{2+}} + E^\circ_{Ag^+/Ag} = 0.40 + 0.80 = 1.20$ volt

(ii) The negative electrode is always the electrode whose reduction potential has smaller value or the electrode where oxidation occurs. Thus, Cd electrode is the negative electrode.

13. (a) Let ethylene bromide $\rightarrow$ A

And propylene bromide $\rightarrow$ B

Then from question,

$P^0_A = 22.93$ K Nm^{-2}

$P^0_B = 18.93$ K Nm^{-2}

$n_A = 3$ mole

$n_B = 2$ mole

Total pressure $= P_T = 20.4$ K Nm^{-2}

$\because P_T = P^0_A X_A + P^0_B X_B$

$= P^0_A X_A + P^0_B(1 - X_A) = (P^0_A - P^0_B)X_A + P^0_B$

$\Rightarrow X_A = \frac{P_T - P^0_B}{P^0_A - P^0_B} = \frac{20.4 - 16.93}{22.93 - 16.93} = 0.578$

$\therefore X_B = 1 - 0.578 = 0.422$

(b) Let mole fraction in vapour phase $= X'_A$

$X'_A = \frac{P^0_A}{P_T}$

$X'_A = \frac{22.93\times0.598}{20.4} = 0.649$

Assuming that the no. of moles of A and B that are vaporized are a and b then

$X'_A = \frac{a}{a+b} = 0.64$ ... (1)

But composition of A in liquid phase,

$$X_A = \frac{3-a}{(3-a)+(2-b)} = 0.578$$

$$= \frac{3-a}{5-(a+b)} = 0.578 \quad \ldots (2)$$

Solving equation (1) and (2)

a = 0.9967 mole

b = 0.537 mole

14. $$\text{Density } (\rho) = \frac{z \times M}{N_o \times a^3}$$

$$= \frac{4 \times 58.5}{6.023 \times 10^{23} \times \left(0.5627 \times 10^{-7}\right)^3}$$

$$= 2.1805 \text{ g/cm}^3$$

Observed density = 2.164 g/cm^3 which is less than calculated density because some ions are missing.

Actual constant units per unit cell can be calculated as:

$$z = \frac{\rho \times N_o \times a^3}{M}$$

$$= \frac{2.164 \times 6.023 \times 10^{23} \times \left(0.5627 \times 10^{-7}\right)^3}{58.5}$$

$$= 3.969$$

Missing units = 4 - 3.969 = 0.031

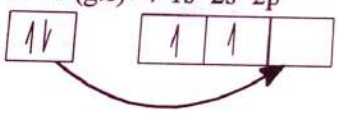
$$\therefore \% \text{ missing of } Na^+ \text{and } Cl^- = \frac{0.031}{4} \times 100 = \mathbf{0.775\%}$$

15. In both the complexes, Fe is in +2 state with the configuration $3d^6$, i.e., it has four unpaired electrons. A the ligands H_2O and CN^- posses different crystal field splitting energy (Δ_o), they absorb differen components of the visible light (VIBGYOR) for d-d transition. Hence, the transmitted colours are different

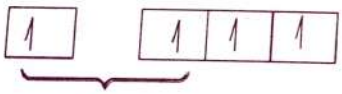
16. $N \equiv C - C \equiv N$

(sp) (sp)

$C \rightarrow (g.s) \Rightarrow 1s^2\ 2s^2\ 2p^2$



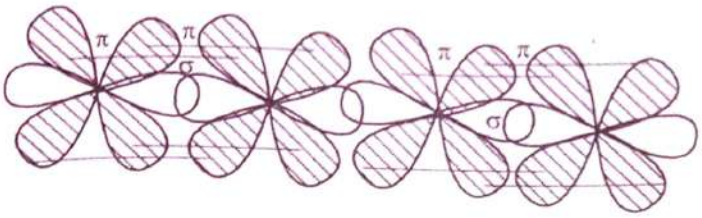
1^{st} e.s. $\Rightarrow$

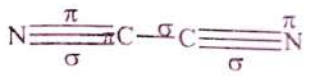


sp

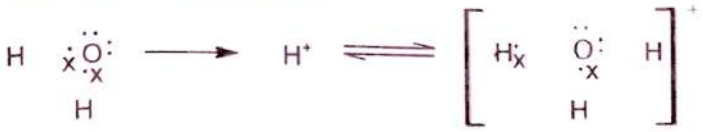
sp sp

unhybridised p-orbital





17. There are two lone pairs, but only one of them makes a co–ordinate bond with an empty 1s orbital of a H atom. There are three bonds.



18. In CH_4 there are 4 bond pair of electrons while in NH_3 are 3 bond pair of electrons and 1 lone pair of electrons. Since bond pair bond pair repulsion is less than lone pair bond pair repulsion, in NH_3 bond angle is reduced from 109°28' to 107°.

19. Now whenever any comparison is asked about the melting point of the compounds which are fully ionic from the electron transfer concept it means that the compound having lower melting point has got lesser amount of ionic character than the other one. To analyse such a question first find out the difference between the 2 given compounds. Here in both the compounds the anion is the same. So the deciding factor would be the cation. Now if the cation is different, then the answer should be from the variation of the cation. Now in the above example, the difference of the cation is their electronic configuration. $K^+ = [Ar]$; $Ag^+ = [Kr]\ 4d^{10}$. This is now a comparison between a noble gas core and pseudo noble gas core, the analysis of which we have already done. So try to finish off this answer.

20. A = P B = H_3PO_4
C = SO_2 D = H_2SO_4
E = Al F = PH_3

21. (A) Roasting
(B) Reduction by carbon or more electropositive metal
(C) Electrolysis

22. When sodium thiosulphate ($Na_2S_2O_3$) solution is added to silver nitrate ($AgNO_3$) solution, there is a white precipitate due to the formation of silver thiosulphate ($Ag_2S_2O_3$); this precipitate dissolve in excess of sodium thiosulphate due to the formation of the complex $[Ag(S_2O_3)_2]^{3-}$

$$2AgNO_3 + Na_2S_2O_3 \longrightarrow \underset{\substack{\text{silver thiosulphate}\\ \text{(white precipitate)}}}{Ag_2S_2O_3\downarrow} + 2NaNO_3$$

$$Ag_2S_2O_3 + 3Na_2S_2O_3 \longrightarrow \underset{\text{sodium argentothiosulphate}}{2Na_3[Ag(S_2O_3)_2]}$$

If the precipitate of $Ag_2S_2O_3$ is heated, there is disproportionation giving black precipitate of Ag_2S and soluble H_2SO_4 which remains in the upper (supernatant) layer which gives white precipitate (of $BaSO_4$) with $Ba(NO_3)_2$ solution.

$$\underset{\text{white precipitate}}{Ag_2S_2O_3} + H_2O \longrightarrow \underset{\text{black precipitate}}{Ag_2S}\downarrow + H_2SO_4$$

$$H_2SO_4 + Ba(NO_3)_2 \longrightarrow \underset{\text{white precipitate}}{BaSO_4}\downarrow + 2HNO_3$$

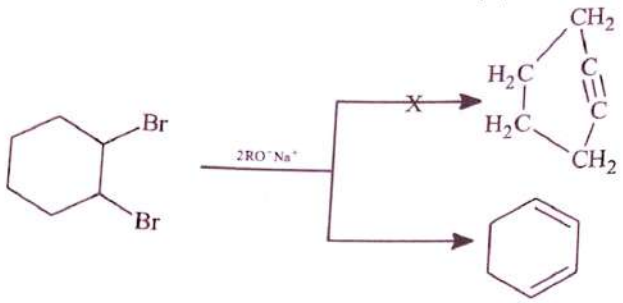
23. Properties of the given compound, especially its solubility in boiling solution of Na_2SO_3 indicates that (A) is sulphur which explains all the given reactions.

$$\underset{(A)}{S} + 6HNO_3 \longrightarrow H_2SO_4 + \underset{(B)}{6NO_2}\uparrow + 2H_2O$$

$$\underset{(A)}{S} + Na_2SO_3 \longrightarrow \underset{(C)}{Na_2S_2O_3}$$

$$\underset{(C)}{Na_2S_2O_3} + H_2SO_4 \longrightarrow Na_2SO_4 + \underset{(D)}{SO_2}\uparrow + H_2O + S\downarrow$$

24.



Too highly strained

A six membered ring cannot accommodate a triple bond because of the strain that would be introduced.

25. Here aldohexose reacts with one molecule of phenylhyrazine which condenses with the aldehyde group to give phenylhydrazone. When warmed with excess of phenyl hydrazine, the secondary alcoholic group adjacent to the aldehyde group is oxidized by another molecule of phenylhydrazine, to a ketonic group. With this ketonic group, the third molecule of phenylhydrazine condenses to give osazone. The phenylhydrazinyl group is transferred from osazone to C_6H_5CHO giving $C_6H_5CH = N{\cdot}NHC_6H_5$ and a dicarbonyl compound called an osone. The more reactive aldehyde group of the osone is reduced, not the less reactive keto group and it gives the 2-ketohexose.

(b) The configurations of these carbons which are unchanged in the reactions, must be identical in order to get the same osazone.

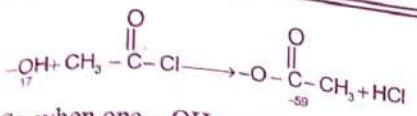
26. (a) H is an oxime $HOCH_2(CHOH)_4CH = NOH$; I is the completely acetylated oxime, $AcOCH_2(CHOAc)_4CH = NOAc$ that loses 1 mole of HOAc to form J, $AcOCH_2(CHOAc)_4\ C{\equiv}N$; K is an aldopentose, $HOCH_2(CHOH)_3CHO$.

(b) The acetates undergo transesterification to give methyl acetate freeing all the sugar OH's. This is followed by reversal of HCN addition.

(c) There is loss of one C from the carbon chain.

(d) Wohl degradation

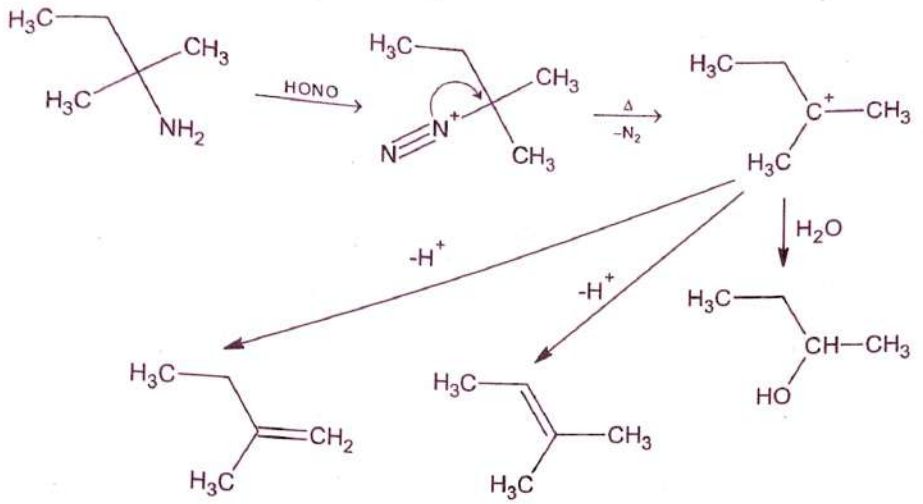
(e) The α-CHOH becomes the –CH = O without any configurational changes of the other chiral carbons. Thus no epimers are formed.

27. 

So when one – OH group is replaced by acetyl group molecular weight increases by 59 – 17 = 42.

Here total increase in molecular weight 294 – 168 = 126

So number of –OH groups present = $\frac{126}{42} = 3$

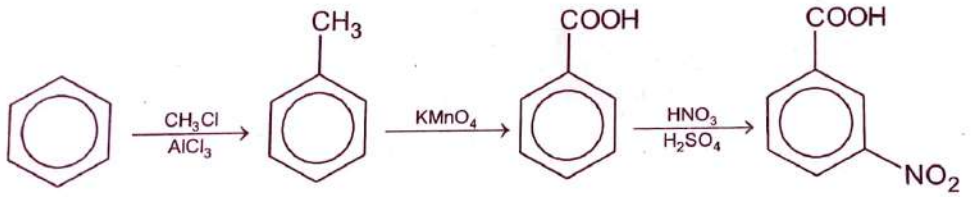
28. 

29. D < C < A < E < B < F

30. If sulphur and nitrogen both are present sodium thiocyanate (NaCNS) may be produced this may give a red coloration with Fe^{3+} but will not respond to tests for cyanide or sulphide ions. With excess of sodium the thiocyanate if formed will be decomposed like this $NaCNS + 2Na \longrightarrow NaCN + Na_2S$

31. Aromatic compounds in which halogen is attached directly to the aromatic nucleus (e.g. chlorobenzene) do not react with ethanolic $AgNO_3$ even on heating probably due to high bond energy of C – Cl bond which has partial double bond character.

32. Nitro group is a deactivating group.

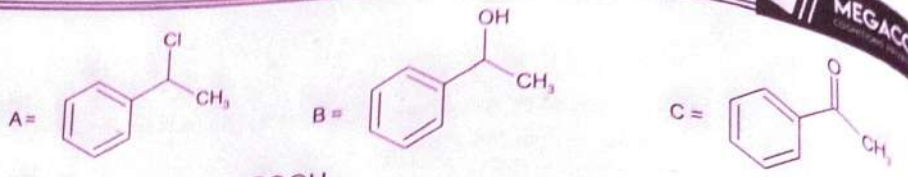
33. 

34. Meta directing group:

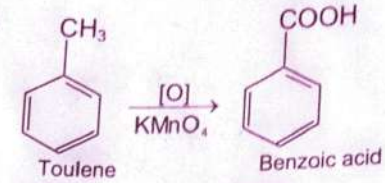
$-NO_2$, $-\overset{\oplus}{N}(CH_3)_3$, –CN, –COOH, –COOR, $-SO_3H$, –CHO, $-\overset{O}{\overset{\|}{C}}-$.

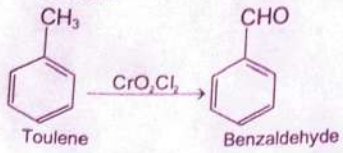
35. –COOH is too weakly acidic to transfer H^+ to the weakly basic $-NH_2$ attached to the electron withdrawing benzene ring. When attached to an aliphatic carbon, the $-NH_2$ is sufficiently basic to accept H^+ form – COOH group.

36

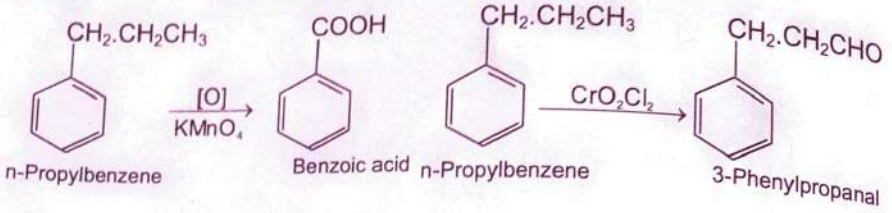


37 a)

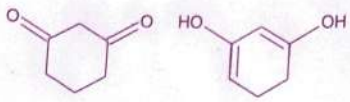




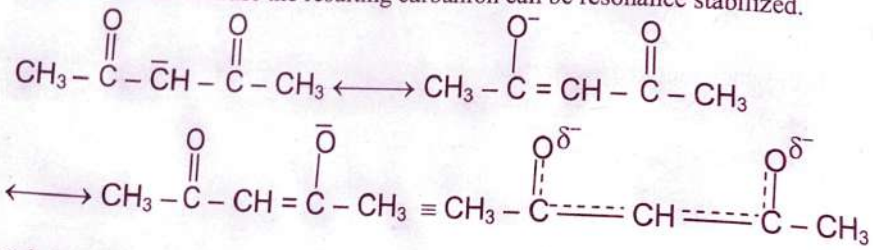
b)



38.



39. The $-CH_2$ group (methylene group) is acidic in nature, i.e. these hydrogens can be easily lost as H^+, on action of a base because the resulting carbanion can be resonance stabilized.



40. a > b > c > d

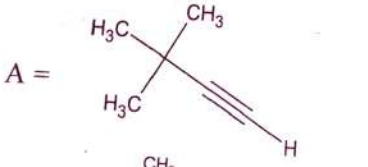
More will be hydrolysis if good leaving group

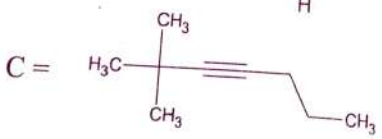
Leaving group $Cl^- > CH_3COO^- > \bar{O}C_2H_5 > NH_2^-$

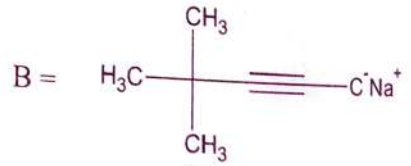
41. a > b > c

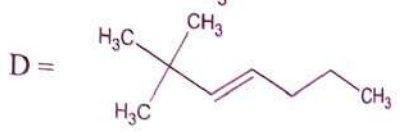
Lone pair involve in delocalination in (c) or (b)

42.

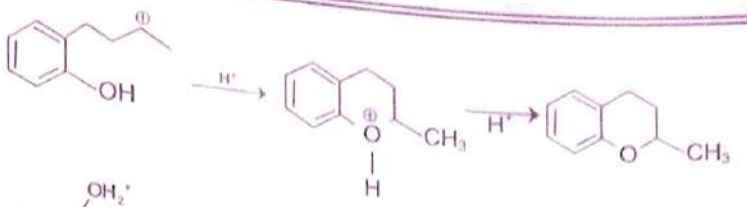




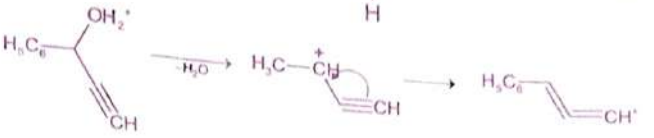


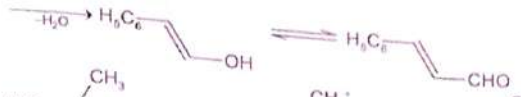


43. (i)

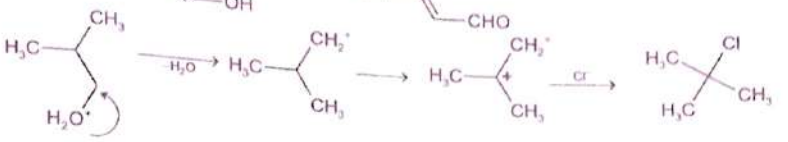


(ii)

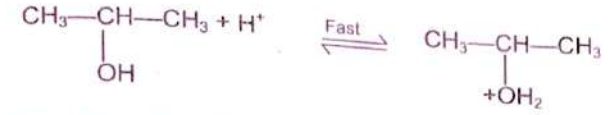


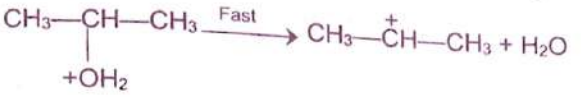


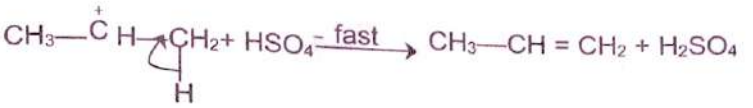
(iii)



44.

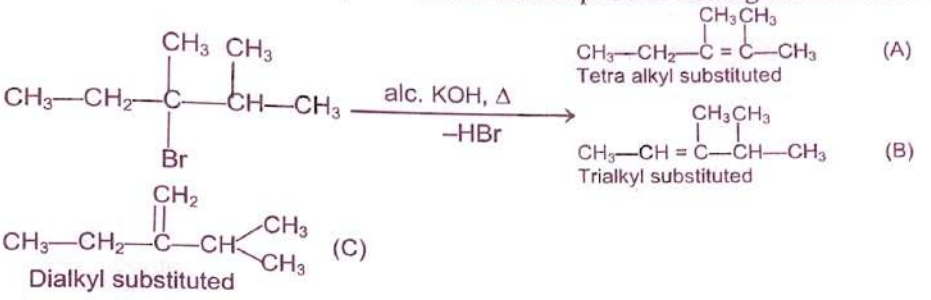






H_2SO_4 act as an acid in first step and base in third step. Since it is regenerated back it is acting as a catalyst.

45.



(A) Tetra alkyl substituted

(B) Trialkyl substituted

(C) Dialkyl substituted

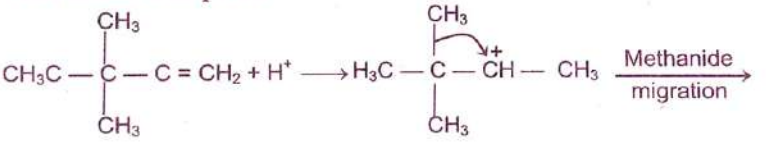
The order of decreasing yield is A>B>C.

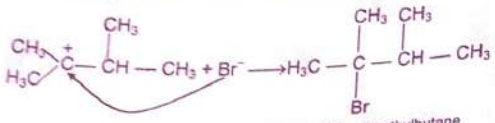
According to Saytzeff rule, more substituted alkene is more stable.

46.

3–bromo–2,2–dimethyl butane

The reaction steps are

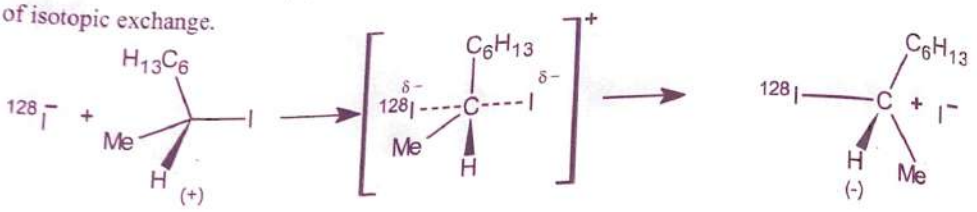




2- Bromo –2,3 - dimethylbutane

The addition of HBr to some alkene gives a mixture of expected alkyl bromide and an isomer formed by rearrangement.

47. This reaction follows S_N2 mechanism. In S_N2 inversion takes place. These will happen inversion of the configuration of the molecules of (+) results in formation of a molecule of its mirror image (-), which pairs of with second molecule of (+) to form a racemate (±). Thus the rate of racemisation will be twice of rate of isotopic exchange.



48. The mixture of Cu^{2+} and $C_2O_4^{2-}$ are reacting separately first with $KMnO_4$ solution and then with solid KI to liberate iodine. It can be seen that Cu^{+2} cannot be oxidised and $C_2O_4^{-2}$ cannot be reduced. This is because Cu is already in its highest oxidation state +2.

$\therefore$ Equivalents of $KMnO_4$ solution $= \dfrac{0.02 \times 5 \times 22.6}{1000} = 2.26 \times 10^{-3}$

$\therefore$ moles of $C_2O_4^{2-} = \dfrac{2.26 \times 10^{-3}}{2} = 1.13 \times 10^{-3}$

This is because only $C_2O_4^{-2}$ is oxidised by $KMnO_4$ to CO_2 ('n' factor 2)

Equivalents of $Na_2S_2O_3$ solution $= \dfrac{0.05 \times 11.3}{1000} = 5.65 \times 10^{-4}$

$\therefore$ moles of $Cu^{+2} = \dfrac{5.65 \times 10^{-4}}{1} = 5.65 \times 10^{-4}$

This is because only Cu^{+2} is reduced by KI to Cu^{+}

(Note: Whenever a metal ion is reduced it always goes to lower oxidation state but generally never goes to oxidation state zero).

$\therefore$ molar ratio of Cu^{+2} to $C_2O_4^{2-} = \dfrac{5.65 \times 10^{-4}}{1.13 \times 10^{-3}} = \mathbf{0.5}$

Reactions: $2Mn\,O_4^- + 5C_2\,O_4^{-2} + 16H^+ \rightarrow 2Mn^{+2}\,10CO_2 + 8H_2O$

$(2Cu^{+2} + 4I^- \rightarrow Cu_2I_2 + I_2)I_2 + 2S_2\,O_3^{-2} \rightarrow 2I^- + S_4\,O_6^{2-}$.

49. Let N_1, N_2 and N_3 be the normalities of 1 M $KMnO_4$ solution in acidic, neutral and alkaline medium respectively.

20 ml $N_1 \equiv$ 33.4 ml $N_2 \equiv$ 100 ml N_3

In acidic medium

1 M $KMnO_4$ = 5 N $KMnO_4$

Thus $N_2 = \dfrac{20}{33.4} N_1 = \dfrac{20}{33.4} \times 5N = 3\ N$

$$N_3 = \frac{20}{100} N_1 = \frac{20 \times 5N}{100} = 1N$$

The volume required for titration of the same volume of reducing agent with acidified $K_2Cr_2O_7$ solution

20 ml 5N $KMnO_4$ = V6N$K_2Cr_2O_7$

$$V = \frac{20 \times 5}{6} = 16.66 \text{ ml}$$

50. Equivalents of H_2O_2 intially $= \frac{200 \times 10}{100} \times \frac{1}{34} \times 2 = 1.176$

Equivalents of Sn^{2+} $= \frac{2 \times 2 \times 100}{1000} = 0.4$

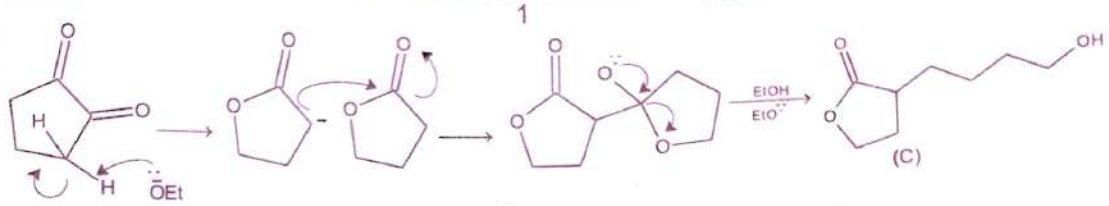
Equivalents of H_2O_2 left $= 1.176 - 0.4 = 0.776$

Moles of H_2O_2 left $= \frac{0.776}{2} = 0.388$

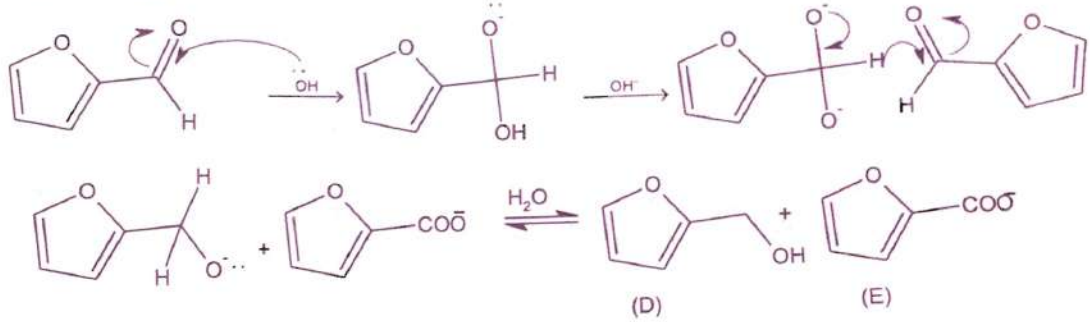
Moles of O_2 produced $= \frac{0.388}{2}\left[\because H_2O_2 \rightarrow H_2O + \frac{1}{2}O_2\right] = 0.194$

Volume of O_2 $= \frac{0.194 \times 0.082 \times 293}{1} =$ **4.66 L**

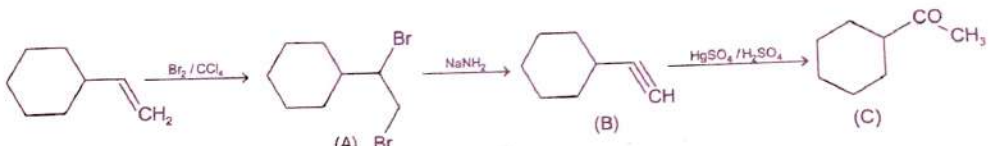
51. a)

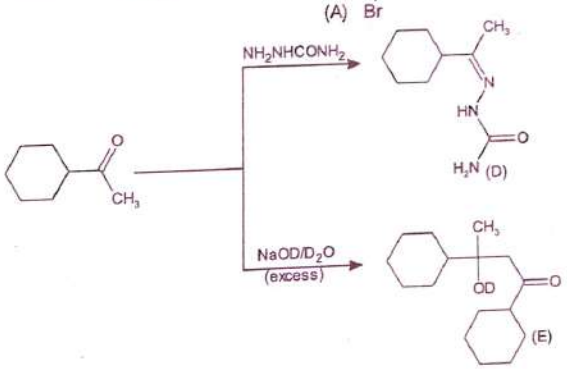


b)



52.





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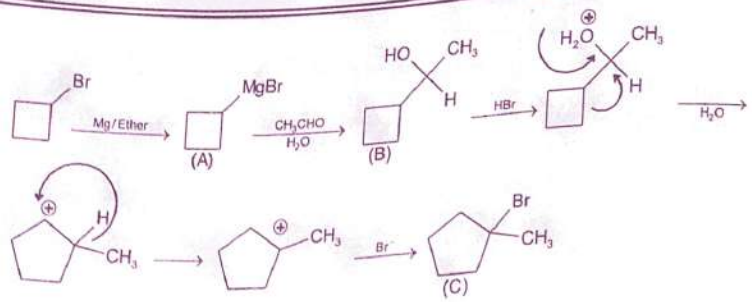

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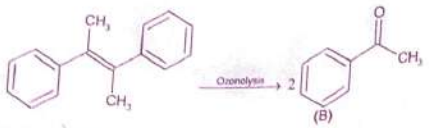
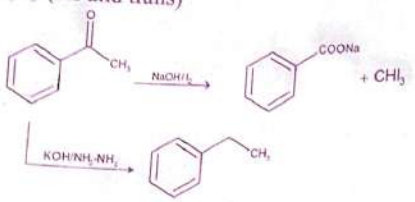
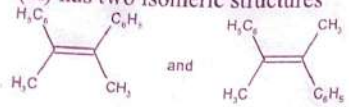


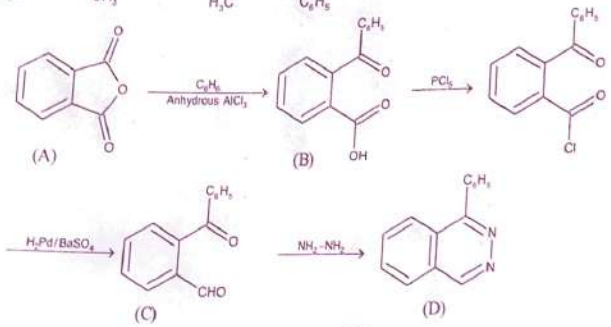
(A) (cis and trans)



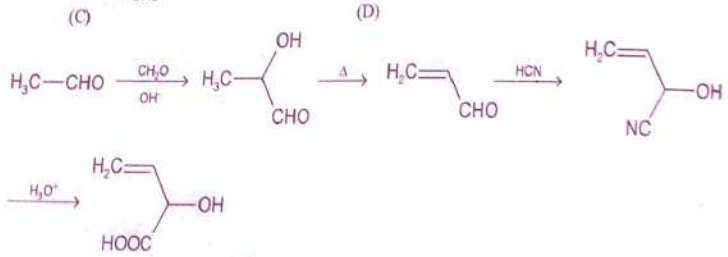
(A) has two isomeric structures



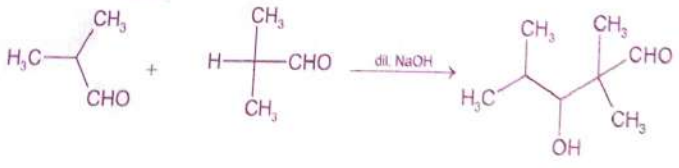
55.



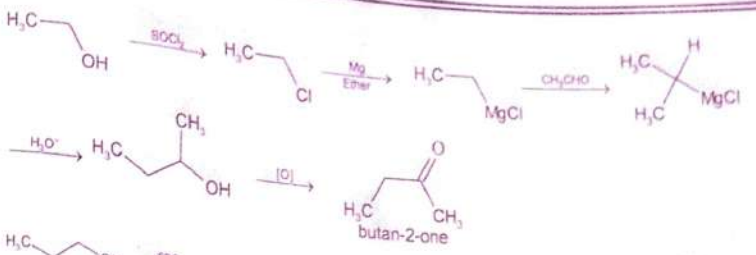
56. (i)



(ii)

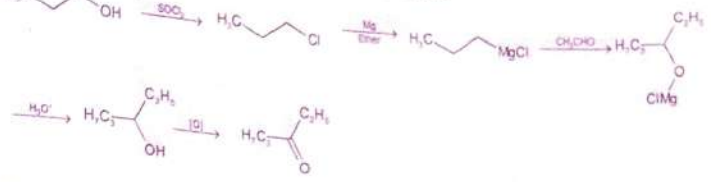


(iii)

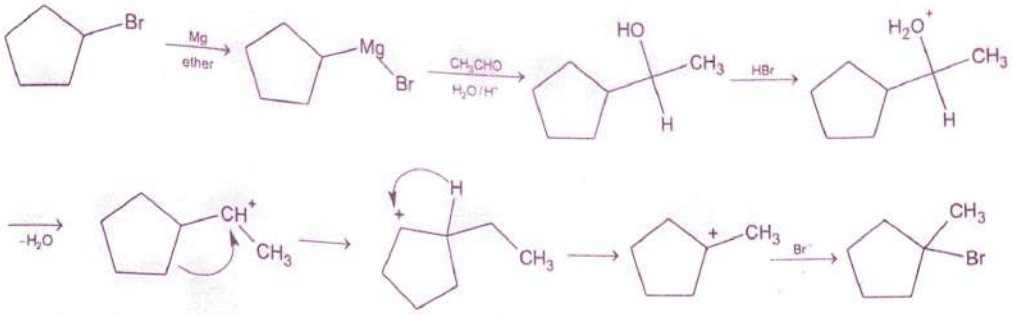


butan-2-one

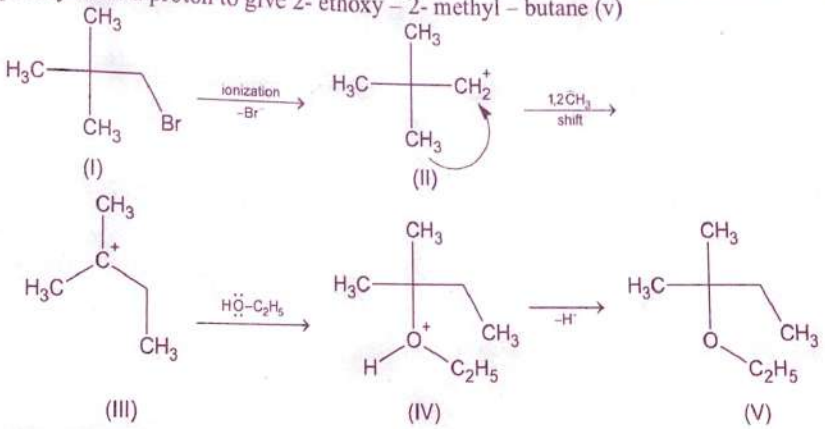
(iv)



57.

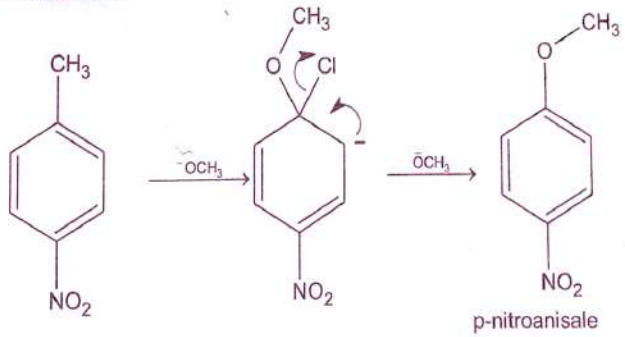


58. Neopentyl bromide (i) First undergoes ionization to give 1° carbocation (II) which rearranges to give a more stable 3° carbocation (III), which, undergoes nucleophilic attack by C_2H_5OH to give (IV) which subsequently loses a proton to give 2- ethoxy – 2- methyl – butane (v)



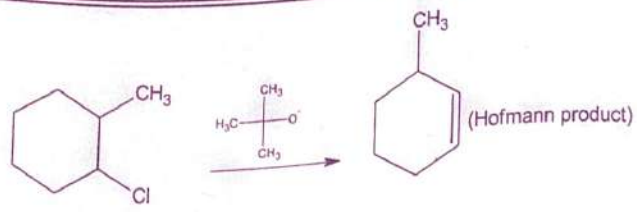
(I) (II) (III) (IV) (V)

b) Here, Cl is activated towards nucleophilic substitution by electron - withdrawing effect of NO_2 group. Nucleophilic attack by CH_3O^- first gives carbanion (i) which loses Cl^- resulting in formation of p – nitro anisole



p-nitroanisale

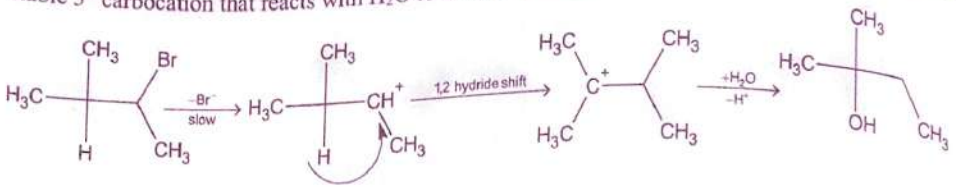
c)



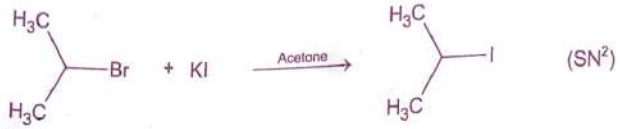
59. a > b > c > d

Aldehyde are more hydrated than ketones. The –I effect of Br increase the case of hydration.

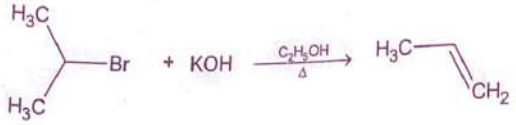
60. First step involves SN solvolysis producing 2° carbocation. Which rearranges by hydride shift to more stable 3° carbocation that reacts with H_2O to form 3° alcohol?



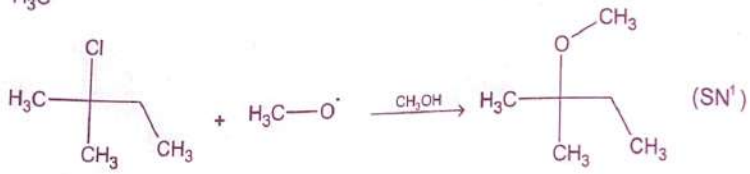
61. a)



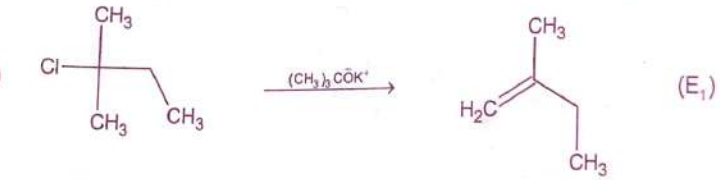
b)



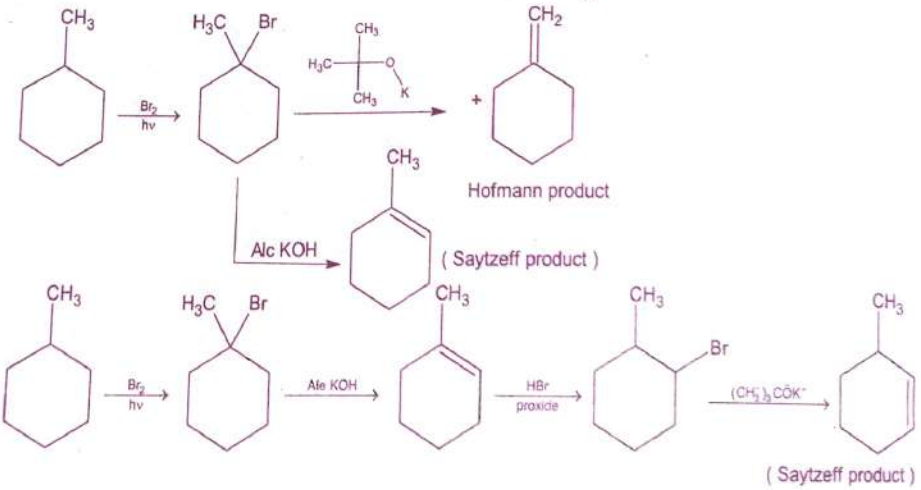
c)



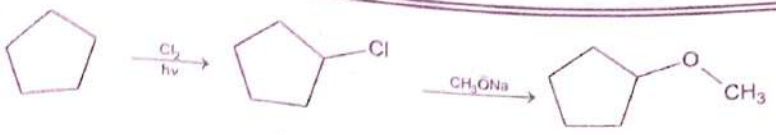
d)

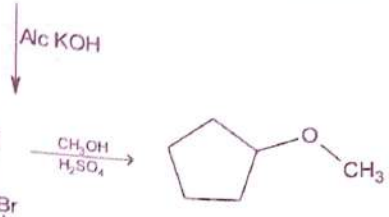


62. a)

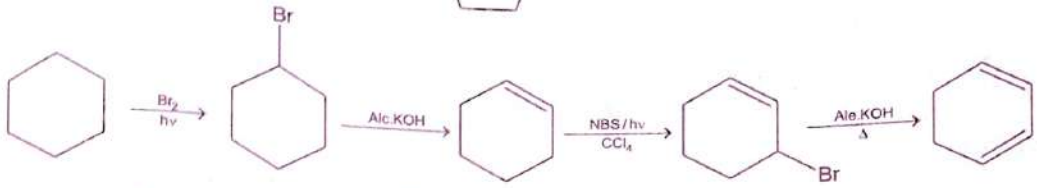


b)

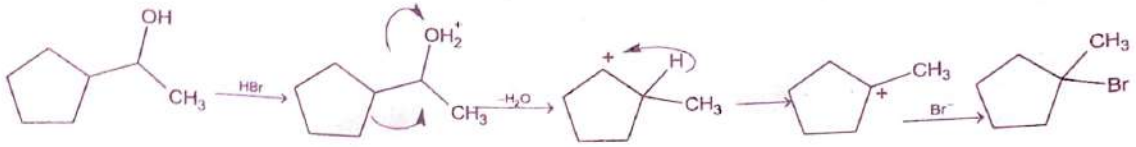




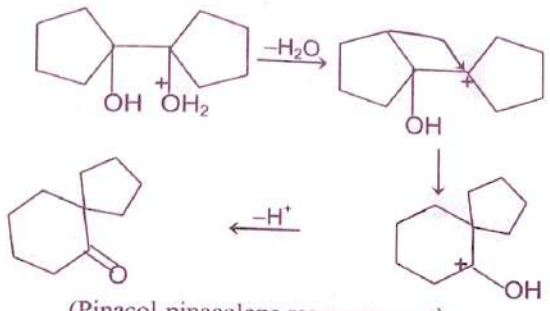
c)



d)



63.



(Pinacol-pinacolone rearrangement)

64. NH_2

(A) $CH_3–CH–CH_2–CH_2–CH_3$ (B) $CH_2=CH–CH_2–CH_3$

65. The unshared electron pair of (a) is prevented by the bicyclic system from becoming parallel to the aromatic system. As a result there is no base weakening resonance into the phenyl ring as there is in (b).

66. Nucleophilicity is very sensitive to *steric* effects. In (a) all three alkyl groups are 'tied back' and prevented from interfering with electrophiles approaching the nitrogen. In (b) all the three groups are free to adopt an infinite number of conformations, some of which block access to nitrogen.

67. For n moles of an ideal gas $PV = nRT$, $n = \frac{PV}{RT} = \frac{1\times1}{0.0821\times298} = 0.0409$

Energy required to effect dissociation of 0.0409 mole of hydrogen = $436 \times 0.0409 = 17.83$kJ.

Energy required to excite one hydrogen atom to the first excited state = $13.6\left(1-\frac{1}{4}\right)$

$= 10.2$ eV $= 10.2 \times 1.6 \times 10^{-19}$ kJ $= 1.632 \times 10^{-21}$ kJ

No. of H-atoms in the sample $= 0.0409 \times 2 \times 6.02 \times 10^{23}$

So energy required for the excitation $= 1.632 \times 10^{-21} \times 0.0409 \times 2 \times 6.02 \times 10^{23} = 80.36$ kJ

The energy required = 17.83 + 80.36= 98.19 kJ

68. In (a) chirality is due to asymmetric substitution at nitrogen and because of nitrogen rapid inversion, the enantiomers cannot be separated. In (b) the chirality is both at asymmetric carbon and nitrogen and the chirality at carbon is not affected by inversion at nitrogen.

69. Pyrrolidine ($pK_b = 2.73$) is 10^{12} times more basic than pyrrole ($pK_b = 14.3$). Therefore, pyrolidine can be protonated by weak acid acetic acid and dissolved in the aqueous phase because pyrrolidine is stronger base than acetate ion. Pyrrole being a weaker base cannot be protonated by such a weak base.

70. A = CH_3CH_2COCl, B = $CH_3CH_2CONH_2$,
C = $CH_3CH_2NH_2$, D = CH_3CH_2OH,
E = CH_3CHO

71. a) $v_n = \sqrt{\frac{Ze^2}{mr_n}}$

where $e = 4.8 \times 10^{-10}$ esu
$m = 9.1 \times 10^{-28}$ g
radius of IIIrd orbital $= 0.52 \times 10^{-8} \times 9$ cm

$$\therefore v_n = \sqrt{\frac{1 \times (4.8 \times 10^{-10})^2}{(9.1 \times 10^{-28})(0.529 \times 10^{-8} \times 9)}}$$

$v_n = 7.29 \times 10^7$ cm sec^{-1}
Now perimeter of IIIrd orbit
$= 2\pi \times 0.529 \times 10^{-8} \times 9$
$= 29.93 \times 10^{-8}$ cm

$\therefore$ no of revolutions per sec = frequency = $\frac{v_n}{2\pi r_n} = \frac{7.29 \times 10^7}{29.93 \times 10^{-8}}$

$= 2.44 \times 10^{14}$

b) $\lambda = \frac{h}{\sqrt{2m(KE)}}$, where $h = 6.625 \times 10^{-34}$ Js

m = mass of $e^- = 9.1 \times 10^{-31}$ kg
$KE = E_2 - E_1$
Where, $E_1 = 13.6$ eV, $E_2 = 13.6 \times 1.5$eV
K.E. $= 6.8$eV $= 6.8 \times 1.6 \times 10^{-19}$ J

Now, $\lambda = \frac{6.625 \times 10^{-34}}{2 \times 9.1 \times 10^{-31} \times 6.8 \times 1.6 \times 10^{-19}} = 4.70$ A°

72. 3120

73. 0011

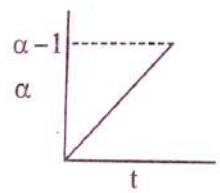
74. 0005

75.

	A ⟶ P	
t = 0	a	–
t = t	a – x	x

$\alpha = \frac{x}{a} \Rightarrow \alpha = \frac{kt}{a}$

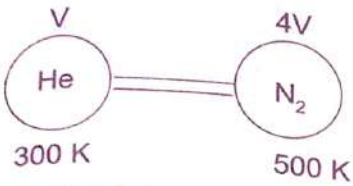
$\alpha = 1$ (maximum)



76. 4 : 7 : 5

77. 0.638, 0.262, 0.1

78. 2.2 atm



$n_1 = x$ moles $\quad n_2 = 4x$ moles

$$\frac{V_1}{n_1T_1} = \frac{V_2}{n_2T_2}$$

$$\Rightarrow \frac{V}{n_1 \times 300} = \frac{4V}{n_2 \times 500} \quad \text{... (i)}$$

And $n_2 + n_1 = x + 4x = 5x$ ... (ii)

From (i) & (ii) $n_1 = \frac{25x}{17}$

Then $\frac{P_1}{P_2} = \frac{n_1}{n_2}$ in flask 1.

$$\frac{1.5}{P} = \frac{x}{n_1} \Rightarrow P = 2.2 \text{ atm.}$$

79. In bcc structure (α form) of iron

$4r = \sqrt{3}a$

$n = 2$ atoms per unti cell from (one atom from 8 corners, one atom from the body centre)

So, ρ (γ–form) $= \frac{n \times M_m}{a^3 \times N_o} = \frac{2 \times M_m}{\left(\frac{4}{\sqrt{3}}r\right)^3 \times N_o}$ ... (i)

In fcc structue (r form) of iron

$4r = \sqrt{2}\,a$

$\therefore a = 2\sqrt{2}\,r$

$n = 4$ atoms per unit cell

$\therefore \rho$ (γ form) $= \frac{4 \times M_m}{(2\sqrt{2}r)^3 \times N_o}$ ... (ii)

Dividing Eq. (ii) by Eq. (i)

$$\frac{\rho(\gamma \text{ form})}{\rho(\alpha \text{ form})} = \frac{2}{\left(\frac{4}{\sqrt{3}}r\right)^3} \times \frac{\left(2\sqrt{2}r\right)^3}{4} = \mathbf{1.0887}$$

80. (a) There are 4 atoms per unit cell and 8 tetrahedral sites per unit cell.

So, the ratio of atoms to tetrahedral sites = 1 : 2

So, the formula would be **TiH_2**

For carbide,

Since carbon occupies octahedral holes

So, ratio of octahedral hole to atom = 1 : 1

The formula of carbide is **TiC**

(b) Since for tetrahedral hole, the limiting ratios, radius ratio $\frac{r^+}{r^-} = 0.225$ to 0.414

Without causing a strain in the host lattice, i.e. $\frac{r^+}{r^-} = 0.225$

81. $ACl(s) \longrightarrow A^+(aq) + Cl^-(aq)$

$A^+(aq) + H_2O \rightleftharpoons AOH(aq) + H^+(aq)$

At equilibrium $C(1-\alpha)$ $C\alpha$ $C\alpha$

For salts of weak base and strong acid, $\alpha = \sqrt{\frac{K_w}{K_b}}$

$$[H^+] = C\alpha = \sqrt{\frac{K_w \times C}{K_b}}$$

$$10^{-4.827} = \sqrt{\frac{1\times10^{-4}}{1.8\times10^{-5}} \times \frac{5.35\times1000}{M\times250}}$$

$M = 53.5$

For CsCl type structure

$$\rho = \frac{zM}{a^3 \times 6.023\times10^{23}}$$

$$a = \left[\frac{1\times53.5}{2.2\times6.023\times10^{23}}\right]^{1/3} = 3.43\ \text{Å}$$

$$r_{A^+} + r_{Cl^-} = \frac{\sqrt{3}}{2} \times a = 0.866\, a = 0.866 \times 3.43\ \text{Å} = 2.97\ \text{Å}$$

$$\because \frac{r_{A^+}}{r_{Cl^-}} = 0.732$$

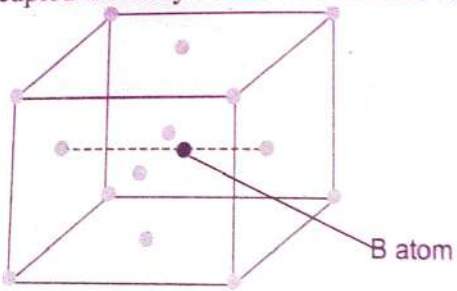
On solving $r_{A^+} = 1.255$ Å $\Rightarrow r_{Cl^-} = \mathbf{1.715\ Å}$

82. Let a be the edge of the cube so that

$4r_A = \sqrt{2}\, a$

$a = 2\sqrt{2}a_A$

Now since the atoms B has occupied the body centre octaheedral void



$2r_A + 2r_B = a$

$2r_A + 2r_B = 2\sqrt{2}\, r_A$

$2rB = \left(2\sqrt{2} - 2\right)r_A\ ;\ \frac{r_B}{r_A} = \sqrt{2} - 1 = 1.414 - 1 = 0.414$

Volume of the cube $a^3 = 16\sqrt{2}r_A^3$

Volume occupied by A and B=

$$\left[4\times\frac{4}{3}\pi r_A^3\right] + \left[\frac{4}{3}\pi r_B^3\right] = \frac{4}{3}\pi\left(4r_A^3 + r_B^3\right)$$

Volume occupied per unit volume of unit cell

$$\phi = \frac{4/3\pi\left(4r_A^3 + r_B^3\right)}{16\sqrt{2}r_A^3} = \frac{\pi}{12\sqrt{2}}\left[4 + \left(\frac{r_B}{r_A}\right)^3\right]$$

$$= \frac{3.14}{1.2\times1.414}\left(4 + (0.414)^3\right) = \frac{3.14\times4.071}{12\times1.414} = \mathbf{0.75}$$

83.

	Ag^+ +	$2NH_3$	$\rightleftharpoons$	$Ag(NH_3)_2$
Initial	0.1 M	1.00 M		0
	x	(1-0.2) M		0.10 M

It has been assumed that almost all Ag^+ ion has been complexed and only its part x is left

$$K_f = \frac{\left[Ag(NH_3)_2^+\right]}{\left[Ag^+\right]\left[NH_3\right]^2} = \frac{0.10}{x\times(0.80)^2} = 1.6\times10^{-9}$$

$x = 9.8\times10^{-9}$ M = Ag^+ (undissolved)

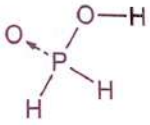
$[Cl^-] = 1.0\times10^{-2}$ M

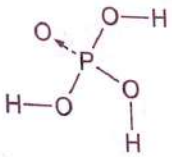
$\therefore [Ag^+][Cl^-] = 9.8\times10^{-9}\times1.0\times10^{-2}$

$= 9.8\times10^{-11} < 1.8\times10^{-10}$ $K_{Sp}AgCl$

Hence AgCl(s) will not precipitate

84. A consideration of the structures of these acids explains the observed trend.

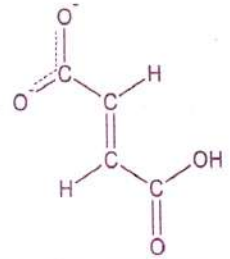




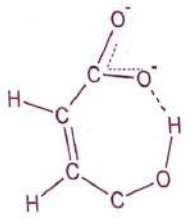
The hydrogen atoms bonded directly to phosphorus are not acidic as the P – H bond is not polar enough due to equal electronegativity of phosphorus and hydrogen. Acidity arises due to O – H groups, one in H_3PO_2 and three in H_3PO_4. The inductive effect of the single oxygen atom acts in the single – O – H bond in H_3PO_2 but it is distributed over three – O – H bonds in H_3PO_4 respectively. So, O – H bond in H_3PO_2 much easily broken, explaining its higher acidity over H_3PO_2.

85. We know that H-bonding involving acidic H has an acid weakening effect and H-bonding in conjugate base has an acid strengthening effect.

Both dicarboxylic acids have two ionisable H-atoms considering second ionization step.



Fumarate monoanion (No H-bonding)



(Maleate monoanion, H-bonding)

86. $AgBr(s) \rightleftharpoons Ag^+ + Br^-$, $K_1 = K_{sp}$

$Ag^+ + 2S_2O_3^{3-} \rightleftharpoons Ag(S_2O_3)_2^{3-}$, $K_2 = K_f$

$AgBr(s) + 2S_2O_3^{3-}(aq) \rightleftharpoons Ag(S_2O_3)_2^{3-}(aq) + Br^-(aq)$ $K_c = K_{sp} \cdot K_f = 9$

	0.1 M	0	0 ..... initially
	(0.1-2x) M	xM	xM ..... at equilibrium

$$\frac{x^2}{(0.1-2x)^2} = 9, \qquad \frac{x}{0.1-2x} = 3$$

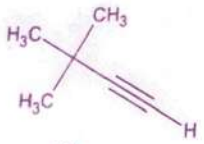
$x = 0.3 - 6x$, $4x = 0.3$, $x = \frac{0.3}{7} = 0.043\,M\,M$

Thus, molar solubility of AgBr 0.1M hypo = 0.043 M

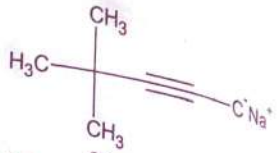
87. a > b > c > d

Conjugative > allylic > 2° > vinylic

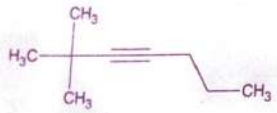
88. A =



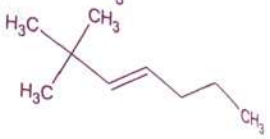
B =



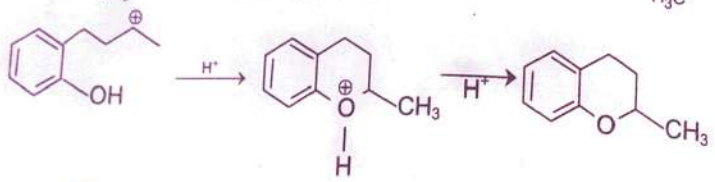
C =



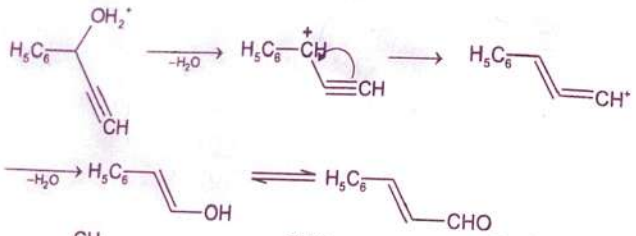
D =



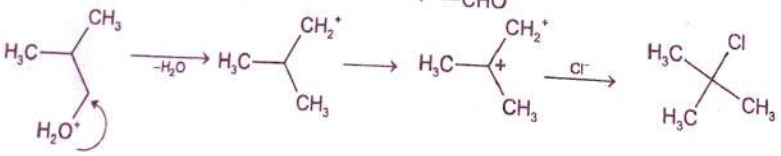
89. (i)



(ii)



(iii)



90. pH of the solution before adding HCl

$$pH = pK_a + \log\frac{0.25}{0.20} = 4.8569$$

HCl added = 0.0005 mole

After adding HCl

$[CH_3COOH] = 0.20 + 0.0005 = 0.2005$

$[CH_3COOK] = 0.25 - 0.0005 = 0.2495$

$$pH = 4.76 + \log\frac{0.2495}{0.2005} = 4.8549$$

Change in pH = 4.8569 – 4.8549 = 0.002

91. We know that,
$\Delta T_f = i \times k_f \times m$ (i)
Where i = Vant Hoff factor
$\Delta T = 0.06$
$K_f = 1.86$
m = molality
Molality may be taken as molarity because solution is dilute and solvent is water.
$=3 \times 10^{-2}$
Substituting the values in equation (i), we get,
$0.06 = i \times 1.86 \times 3 \times 10^{-2}$

$$i = \frac{0.06}{1.86 \times 3 \times 10^{-2}}$$

$i = 1.07526$
Let acid is NH

$$NH \rightleftharpoons N^- + H^+$$

$T_{equilibrium}$ $\quad C(1-\alpha) \quad C\alpha \quad C\alpha$

$$K_a = \frac{C\alpha \cdot C\alpha}{C(1-\alpha)} = C\alpha^2$$

$$i = \frac{\text{Total mole after dissociation}}{\text{initial mole}} = \frac{C(1-\alpha)+C\alpha+C\alpha}{C} = 1+\alpha$$

$\alpha = i - 1$
$\alpha = 1.07526 - 1$
$\alpha = 0.07526$
$K_a = C\alpha^2$
$= 3 \times 10^{-2} \times (0.07526)^2$
$K_a = 1.6992 \times 10^{-4}$
$PK_a = -\log Ka$
$= -\log (1.6992 \times 10^{-4})$
$= +3.769$

92. $F = 96500$ C

$E = 0.03$ V

$T = 273+25 = 298$ K

$n = 2$

$$\left(\frac{dE}{dT}\right)_P = -1.4\times10^{-4} \text{ v. per degree}$$

$$\text{As } \Delta H = nF\left[T\left(\frac{dE}{dT}\right)_P - E\right]$$

$= 2\times96500\ [298\times(-1.4\times10^{-4})-0.03]$

$= -13842$ J/mol $= -13.842$ KJ/mol

93. $Cu^{2+} + 4NH_3 \rightleftharpoons [Cu(NH_3)_4]^{2+}$

$$K_f = \frac{[Cu(NH_3)_4]^{2+}}{[Cu^{2+}][NH_3]^4}$$

Blue colour will be noticed upto $[Cu(NH_3)_4]^{2+} = 1 \times 10^{-5}$
Thus at this stage

$$[Cu^{2+}] = \frac{1.0\times10^{-5}}{1.1\times10^{13}\times(0.1)^4} = 9.1 \times 10^{-15}\text{M}$$

$$\text{Cu deposited (w)} = \frac{\text{E.i.t.}}{96500} = \frac{63.5\times3.5\times1350}{2\times96500} = 1.5546 \text{ gms}$$

Millimoles of Cu^{2+} present (initial) $= 250 \times 0.1$

Weight of $Cu^{2+} = \frac{250 \times 0.1 \times 63.5}{1000} = 1.5875$ gms

Weight of Cu^{2+} left in solution $= 1.5875 - 1.5546 = 0.0329$ gms

$[Cu^{2+}]$ left $= \frac{0.0329 \times 1000}{63.5 \times 250} = 2.07 \times 10^{-3}$

Thus solution will show blue colour, as it will provide appreciable Cu^{2+} to form complex.

94. $\Delta G^\circ = G^\circ_{f(products)} - G^\circ_{f(reactants)}$

$= G^0_{fCO_2} - \left[G^0_{f(CO)} + \frac{1}{2}G^0_f(O_2)\right]$

$= -394.4 - [-137-2+0], \qquad \left[G^0_{f(O_2)} = 0\right]$

$= -257.2$ KJ/mol

Since, ΔG° for reaction is negative, so the reaction will be spontaneous

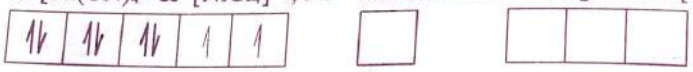
Also, $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$\Rightarrow -257.2 = \Delta H^\circ - [300 \times (-0.294)]$

$\therefore \Delta H^\circ = -0.285.4$ KJ/mol

As ΔH° is negative and thus reaction is exothermic

95. In $[Ni(CN)_4]^{-2}$ & $[NiCl_4]^{-2}$, Ni^{+2} has electronic configuration $[Ar]3d^8$



$\bar{C}N$ is a strong ligand so it cause pairing where as $\bar{C}l$ does not cause paining hence hybridisation of Ni in $[Ni(CN)_4]^{-2}$ is dsp^2 where as in $[NiCl_4]^{-2}$ is sp^3

$[Ni(CN)_4]^{-2}$		$[NiCl_4]^{2-}$	
Hybridisation	- dsp^2	Hybridisation	-sp^3
Structure	- square planar	structure -tetrahedral	
shape	-square planar	shape	-tetrahedral
Magnetic moment	-μ =0	magnetic moment	-$\mu = \sqrt{n(n+2)}$BM = 2.828 BM

96. $A = [Cr(H_2O)_6]Cl_3$

$[Cr(H_2O_6]Cl_3 + H_2SO_4 \longrightarrow$ No reaction

Because in it all H_2O molecules present in co-ordination sphere

$B = [Cr(H_2O)_5Cl]Cl_2.H_2O$

$[Cr(H_2O)_5Cl]Cl_2.H_2O + H_2SO_4 \longrightarrow$ one mole of H_2O is removed because it is present out side the coordination sphere

Molar wt = 266.5

$$\% \text{ loss} = \frac{18}{266.5} \times 100 = 6.75\%$$

$C = [Cr(H_2O)_4Cl_2]Cl\ 2H_2O$

$[Cr(H_2O)_4Cl_2]Cl2H_2O + H_2SO_4 \longrightarrow$

two moles of H_2O are removed

(Because these are present out side the coordination sphere)

$$\% \text{ loss} = 2 \times \frac{18}{266.5} \times 100 = 13.5\%$$

Hence compounds

$A = [Cr(H_2O)_6]Cl_{3fc}$

$B = [Cr(H_2O)_5Cl]Cl_2H_2O$

$C = [Cr(H_2O)_4Cl_2]Cl.2H_2O$

97. In case of $[Fe(CN)_6]^{3-}$ central Fe – metal has +3 oxidation. State and pairing of e$^-$ takes place due to strong ligand CN^-, this resultant it has finally one unpaired e.

$\therefore$ Magnetic moment $= \sqrt{1(1+2)} = \sqrt{3}$ B.M

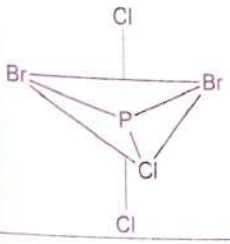
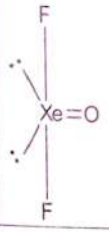
but in case of $[Fe(H_2O)_6]^{3+}$. Central metal ion Fe^{+3} has five unpaired e$^-$ [no rearrangement takes place]

$\therefore$ Magnetic moment $= \sqrt{5(5+2)} = \sqrt{35}$ B.M

98. Predict the hybridisation and draw the shape of the following molecules

a) PBr_2Cl_3

b) $XeOF_2$

Hybridisation	Shape
sp^3d	
Sp^3d	

99. A = ZnS, B = H_2S, C = $ZnSO_4$, D = S, E = SO_2

$$\underset{(A)}{ZnS} + H_2SO_4 \longrightarrow \underset{(C)}{ZnSO_4} + \underset{(B)}{H_2S} \xrightarrow[H_2SO_4]{K_2Cr_2O_7} \underset{(D)}{Cr_2(SO_4)_3} + 3S + H_2O + K_2SO_4$$

$$\underset{(D)}{S} + O_2 \longrightarrow \underset{(E)}{SO_2} \xrightarrow{H_2S} \underset{(D)}{S} + H_2O \text{ (colourless)}$$

$$\underset{(C)}{ZnSO_4} + 2NaOH \longrightarrow Na_2SO_4 + Zn(OH)_2 \downarrow \xrightarrow{2NaOH} Na_2ZnO_2 + 2H_2O$$

100. $A \longrightarrow BaCl_2$

Reactions : $2BaCl_2 + K_2Cr_2O_7 + 3H_2SO_4 \longrightarrow K_2SO_4 + 3H_2O + 2CrO_2Cl_2 + 2BaSO_4$

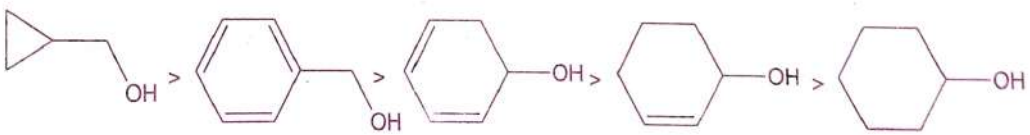
red gas

$$CrO_2Cl_2 + 4NaOH \longrightarrow \underset{\text{yellow solution}}{Na_2CrO_4} + 2NaCl + 2H_2O$$

101. More substituted alkene are more stable stability order

= a > b > c

102.



3. Over percent yield $= \frac{82}{100} \times \frac{65}{100} \times 100 = 53.3\%$

10. $W_2 = 5\ g, W_1 = 100\ g$
$P_s = 2985\ Nm^{-2}$
$P^0 = 3000\ Nm^{-2}$
$M_2 = ?$

$$\frac{P^0 - P_s}{P^0} = \frac{W_2}{M_2} \times \frac{M_1}{W_1}$$

$$M_2 = \frac{W_2 M_1}{\frac{P^0 - P_s}{P^0} W_1} = \frac{5 \times 18}{\left(\frac{3000 - 2985}{3000}\right) \times 100}$$

$M_2 = 180$

12. Using Ideal gas equation: $PM = dRT$
$\Rightarrow M = 46$

18. $Z = 2, \therefore He^+$

20. $$E_{cell} = -\frac{0.0591}{1} \log \frac{[H^+]_a}{[H^+]_c}$$

22. $$W = \frac{Mit}{nF} \qquad \therefore 5 = \frac{52 \times 1.50 \times t}{3 \times 96500}$$

37. Reactivity order
$3^o > 2^o > 1^o$ alcohols

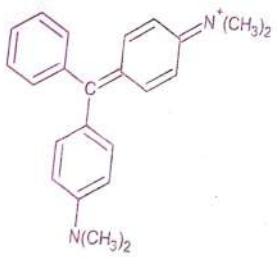
38. C_2 is 3^oC and C_5 is 2^oC.

39. Carbonic acid is stronger than phenol
Hence (A)

40. – OH group is ortho para directing.
Hence (D)

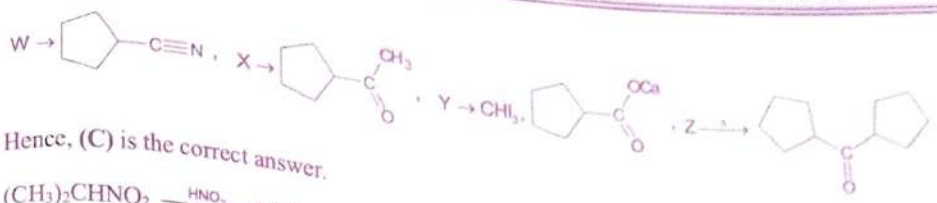
41. $-NH_2$ is electron donating group, at meta position the donation of electron will be least.
Hence (B)

42.



Hence, (C) is the correct answer.

44. It is Ruff degradation.

45. 

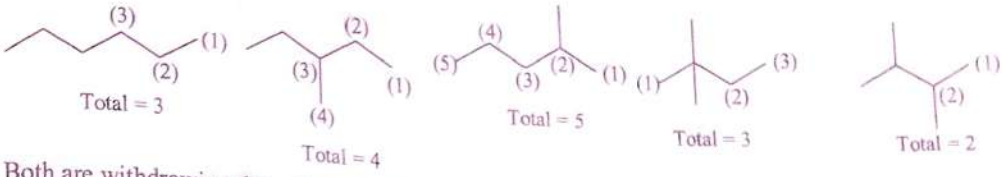
Hence, **(C)** is the correct answer.

46. $(CH_3)_2CHNO_2 \xrightarrow{HNO_2} (CH_3)_2 C(NO_2)(N=O)$ (Pseudo nitrol blue)

Hence, **(D)** is the correct answer.

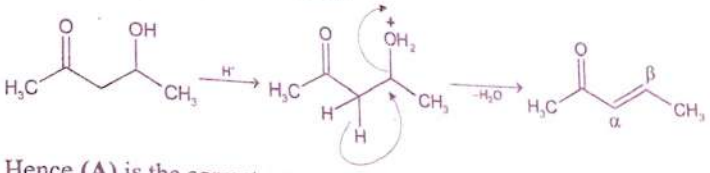
47. Base is bulky so if prefers terminal β - carbon
Hence, **(A)** is the correct answer.

48. 17

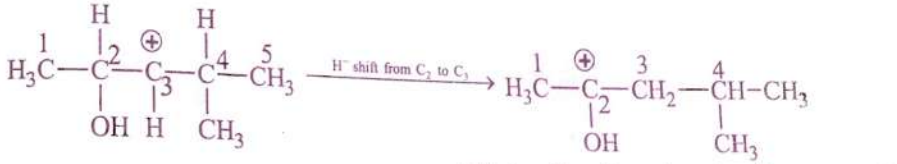


49. Both are withdrawing through σ-bond and have least bond angle.

50. Aldols, i.e. β–hydroxyaldehydes or β–hydroxyketones readily undergo dehydration to form α, β-unsaturated aldehydes or ketones.



Hence **(A)** is the correct answer.

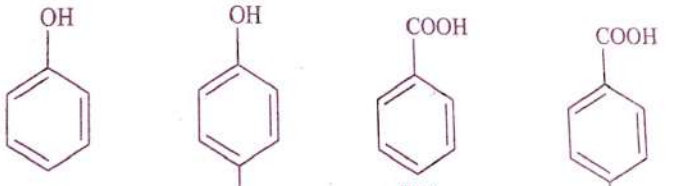
62. 

(driving force is conjugation from oxygen)

Hence, (D) is the correct answer.

63. On the basis of stability of resonating structures.
Hence, **(B)** is the correct answer.

64. A



(I) pKa = 9.98

(II) pKa = 9.38

(III) pKa = 4.17

(IV) pKa = 4.37

Decreasing order of acidic strength: III > IV > II > I
Hence, **(A)** is the correct answer.

65. Priority of CN is highest.
Hence, **(B)** is the correct answer.

68. $2Cu^{2+} + [Fe(CN)_6]^{4-} \longrightarrow Cu_2[Fe(CN)_6]^{-}$
(red brown)

70. Since alkyl halide is 3°, so in presence of NaCN it will follow E_2 path rather S_N2, so method 2 is not appropriate.

71. Aldehyde is first converted into cyclic thioacetal and then st. base like Ph – Li is used to generate carbanion and then treated with D_2O. Finally, the aldehyde is regenerated.

72. As nitrogen is less electronegative than oxygen.

74. $CH \equiv C - H + \bar{N}H_2 \longrightarrow CH \equiv C^- + NH_3$
$CH \equiv C - H + \bar{C}H_3 \longrightarrow CH \equiv C^- + CH_4$
$CH \equiv CH + Na \text{ in liq. } NH_3 \longrightarrow CH_2 = CH_2$

75. Diels-Alder reaction is highly stereo specific, and it is *syn* addition where the configuration of dienophile is retained.

76. HCl(g) is highly soluble in water and is also acidic in nature.

77. $(\text{Ionic product})_1 = \frac{10^{-5}}{2} \times \frac{0.003}{2} = 7.5 \times 10^{-9} > K_{sp}(AgBr)$

$(\text{Ionic product})_2 = \frac{10^{-7}}{2} \times \frac{0.003}{2} = 7.5 \times 10^{-11} < K_{sp}(AgCl)$

Hence, Experiment 1 will produce a precipitate.

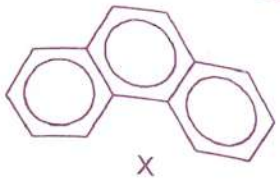
79. Let the volume of NaOH is V_1, Volume of HCN is V_2.
M eq of HCN = $V_2 - V_1$
M eq of NaCN = V_1
As the pH of buffer solution is 4.72 which is equal to pk_a of HCN. Hence this is a condition of Half of Neutralization.

$$4.72 = 4.72 + \log \frac{V_1}{(V_2 - V_1)}$$

but $\frac{V_1}{V_2 - V_1} = 1$

$V_1 = V_2 - V_1$
$2V_1 = V_2$
$V_1/V_2 = ½$
Hence volume of HCN : NaCN = 2 : 1.

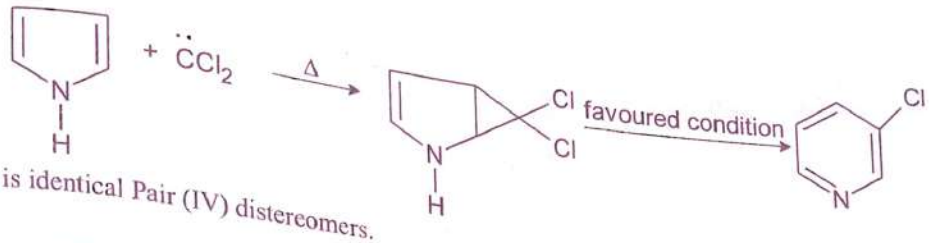
80.



X

$\ddot{C}Cl_2$ (Singlet carbon)

Y



81. (III) is identical Pair (IV) distereomers.

82. (I) $2H_2SO_4 + 2KBr^{-1} \longrightarrow \overset{0}{Br_2} + SO_2 \uparrow + K_2SO_4 + 2H_2O$ (Change is +1)

(II) $K\overset{-1}{Br} + H_3PO_4 \longrightarrow H\overset{-1}{Br} + KH_2PO_4$ (Change is zero)

(III) $Cl_2 + K\overset{-1}{Br} \xrightarrow{OH^-} K\overset{+5}{Br}O_3 + KCl + H_2O$ (Change is +6)

(IV) $\overset{0}{Br_2} + NaClO_3 \longrightarrow Na\overset{+5}{Br}O_3 + Cl_2$ (Change is +5)

85. Mass of the system remains conserved.

86. Point 'S' represents the situation beyond end point at which excess base is present in the titration flask. So its $[H_3O^+]$ can be written as $(H_3O^+) \approx \frac{K_w}{(OH^-)}$

$\therefore pH = p^{k_w} + \log(\text{base})$

104. $\pi V = nST$
$2 \times 1 = n \times 0.0821 \times 300$
$n = 0.0821$
now
$\Delta T_f = K_f \times m$
$= 1.8 \times 0.082 = 0.151$
$f.p = 273 - 0.15$
$= 272.849$

118. B

121. Let, moles of Na_2CO_3 and $NaHCO_3$ in 10 ml mixture = x and y respectively.
Now, $x \times 1 = 2.5 \times 10^{-3} \times 0.1 \times 2$
And $x + y = 2.5 \times 10^{-3} \times 0.2 \times 2$

So, mass of Na_2CO_3 in 1000 ml solution $= \frac{2.5 \times 10^{-3} \times 0.1 \times 2 \times 106 \times 1000}{10} = 5.3$ g

$\therefore$ mass of $NaHCO_3$ in 1000 ml solution $= \frac{2.5 \times 10^{-3} \times 0.2 \times 2 \times 84 \times 1000}{10} = 4.2$ g

128. At anode:

$2CH_3COO^- \longrightarrow C_2H_6(g) + 2CO_2(g) + 2e^-$

At cathode:

$2H^+ + 2e^- \longrightarrow H_2(g)$

Equivalent $\left(\frac{W}{E}\right) = \frac{i \times t}{96500} = \frac{0.965 \times 1 \times 60 \times 60}{96500} = 0.036$

Thus, total equivalent of $C_2H_6 + CO_2 + H_2 = 0.036 + 0.036 + 0.036$

Total moles of gases (n) $= \frac{0.036}{2} + \frac{0.036}{1} + \frac{0.036}{2} = 0.072$

$\therefore V = \frac{nRT}{P} = \frac{0.072 \times 0.0821 \times 298}{1} = 1.762$ litre

Hence, (A).

129. A

131. For reaction $\Delta n^{gases} = -3$

$\Delta H = \Delta E + \Delta nRT$

$\Delta H - \Delta E = -3 \times 8.314 \times 298$

$= -7.43$ kJ

134. Five membered ring becomes aromatic

135. The resultant carbocation is most stable.

139. The charge density of (I) is delocalized to the maximum extent.

140.

$$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$$

Let initially no. of moles	x	–	–
at equilibrium (volume V)	$x(1-\alpha)$	$x\alpha$	$x\alpha$

$K_c = \frac{x^2\alpha^2}{x(1-\alpha)V}$, also $K_p = K_c(RT)^{\Delta n_g}$

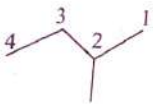
on reducing volume to $\frac{V}{2}$, initial concentration of PCl_5 doubles and its degree of dissociation is increased but K_p does not change, as it is characteristic constant for a reaction at constant temperature.

141. Plane of both rings will not remain same.

142. All enols will give positive test with $FeCl_3$.

143. $-COCH_3$ group is available after oxidation.

144. In OM–DM, H_2O attacks at that carbon which can accommodate more +ve charge.

146. 

In these positions we can have the chlorine atom. Hence, the answer.

147. S_N2 will dominate.

148. D

149. Fries rearrangement.

150. Non-polar solvent will reduce the activity of Br^+, single derivative is obtained.

151. Claisen rearrangement.

154. $N_2\,(B.O. \to 3) \longrightarrow N^{2+}\,(B.O. \to 2.5)$
$C_2\,(B.O. \to 2) \longrightarrow C^{2+}\,(B.O. \to 1.5)$
$NO(B.O. \to 2.5) \longrightarrow NO^{+}\,(B.O. \to 3)$
Paramagnetic diamagnetic
Hence, (C) is the correct answer.

156. Reaction takes place through a 6 membered cyclic T.S.

161. After attack of H^+, carbonium ion formed is stabilized by phenyl shift and the deprotonation from adjacent carbon.

162. Paramagnetic due to one unpaired electron.

163. Structure is $HO-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-\overset{\overset{O}{\|}}{S}-OH$.

164. According to structure, each phosphorus atom is attached to four oxygen atoms.

167. As $B(OH)_3$ accepts a lone pair of electrons from water as OH^-.

168. Acid converts $Na_2B_4O_7$ to H_3BO_3 and B_2O_3 can only be reduced by Mg out of these.

171. Only three geometrical isomers are possible for the complex Mabcd.

173. Fluorine is most reactive halogen.

174. Colour of (A) is due to d-d transition.

175. It is a test for sulphide.

176. The n-factor of $K_4Fe(CN)_6$ is 61.

177. Apply, $N_1V_1 = N_2V_2$.

178. $\Delta V = V_f - V_i$
$= \left(\frac{nRT}{P} - \frac{18}{1000}\right) L = 30.6053$
$W = -P\Delta V = -1 \times 30.6053\ L\ atm = -30.6053 \times 101.3$
$= -3140.3\ J$

181. On a body plane 4 corners B^- and 2 face centre B^- will be lying. On a body plane total 4 tetrahedral voids will be lying but out of 4 only 2 are occupied by A^+ so, effective $A^+ = 2$

effective $B^- = \frac{1}{8}\times 4 + \frac{1}{2}\times 4 = 2.5$

So, A_4B_5

Hence, (C) is the correct answer.

184. They regain aromatics

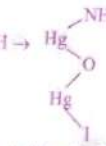
189. A

190. Baeyer villager oxidation: More electron rich ring will migrate.

193. Empirical formula is M_2O_3 as only one octahedral void is associated with one oxide ion but unit cell formula is M_4O_6.

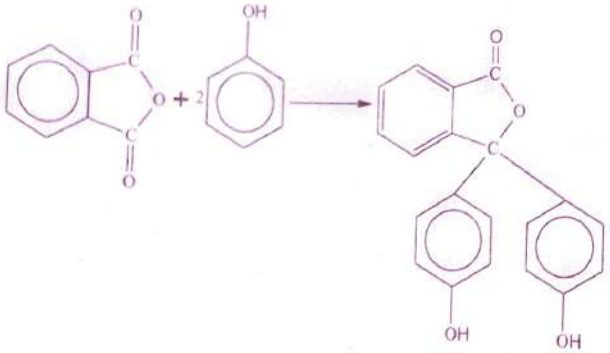
194. Low spin prefer pairing of electrons.

196. $\underset{(A)}{NH_4Cl} + NaOH \rightarrow NaCl + \underset{(B)}{NH_3} + H_2O$

$NH_4Cl + 2K_2HgI_4 + 4KOH \rightarrow$  $+ 7KI + KCl + 3H_2O$

Iodide of Millon's base
(Brown ppt.)

197. D

199. 

Phenolphthalein turns alkaline solution pink.

200. Carbylamine test gives isocyanide.

203. The intermediate should contain oxygen atom.

204. Removal of –OH from HNO_2 and –H from CH_3-NH-R.

205. In Hinsberg's method, primary amines form N-alkyl aromatic sulphonamides.

206. –OH groups of glucose react with anhydride to form esters.

207. X is glucose cyano hydrin, Y is glucose mono carboxylic acid and Z is heptane.

208. A

209. It contains two $-NH_2$ groups against one –COOH group.

214. By using Kirchoff's equation.

216. Allylic C atom having negative atom/group and carbonyl C atom are susceptible for nucleophilic attack.

217. Tilt the molecule in clockwise direction and go for relative comparison with D-glyceraldehyde.

218. It is linked to more electronegative group, which is in between two negative groups.

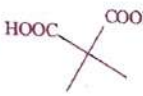
219. 1. Dehydration on 5th carbon,
2. Cyclization and
3. Dehydration followed by deprotonation

220. D

221. Due to aromaticity.

222. 2nd and 5th positions will have higher electron densities than 3rd and 4th.

223. Justify the answer based on ozonized products.

224. Upon heating,  undergoes decarboxylation.

225. D

REASONING TYPE QUESTIONS

1. D

6. Rate $\propto$ [ethylacetate]

9. Dichlorocarbene is electrophile.

12. Both statement I and statement II are true and statement 2 is the correct expnation of statement I.

MCQ (MULTIPLE CHOICE QUESTIONS)

1. A, B, D

7. $\text{Slope} = \frac{1}{100} = 1\times10^{-2}$

11. (A) $\frac{t_{3/4}}{t_{1/2}} = \frac{\log 4}{\log 2}$

(B) $\frac{t_{15/16}}{t_{1/2}} = \frac{\log 16}{\log 2}$

12. At 27°C, total pressure $P_A + P_B = 2 + 2 = 4$ atm

$\frac{r_A}{r_B} = \frac{n_A}{n_B} = \frac{P_A}{P_B}\sqrt{\frac{M_B}{M_A}} \Rightarrow \frac{n_A}{n_B} = \frac{1}{2}$

$\therefore$ moles diffused out at 27°C are in the ratio 1 : 2

	A	B
Initial moles	a	a
moles left	a – x	a – 2x

Total moles after diffusion $= a - x + a - 2x = a$

($\because$ P drops from 4 atm to 2 atm $\Rightarrow$ moles should decrease from 2a to a)

$\therefore a = 3x$

$\therefore$ After diffusion moles of A and B are $3x - x = 2x$ and $3x - 2x = x$

$\therefore$ moles of A and B left are in the ratio 2 : 1

$\therefore$ Partial pressures after diffusion at 27°C

For A it is $\frac{2}{3} \times P = \frac{2}{3} \times 2 = \frac{4}{3}$

For B it is $\frac{1}{3} \times P = \frac{1}{3} \times 2 = \frac{2}{3}$

When the vessel is heated to 127°C,

Partial pressure of A $= \frac{4}{3} \times \frac{4}{3} = \frac{16}{9}$ atm $\left(\because \frac{P_1}{T_1} = \frac{P_2}{T_2}\right)$

Partial pressure of B $= \frac{2}{3} \times \frac{4}{3} = \frac{8}{9}$ atm

Now

A	+	B $\rightleftharpoons$ C
$\frac{16}{9}$		$\frac{8}{9}$ 0
$\frac{16}{9} - P$		$\frac{8}{9} - P$ P

$k_P = 0.1 = \frac{P}{\left(\frac{16}{9} - P\right)\left(\frac{8}{9} - P\right)} \Rightarrow$ partial pressure of C, P = 0.126 atm

22. One mole of $MeNH_2$ can attack on 2 moles of . So will result above species as an intermediate.

28. Glucose, fructose and ethanal give Fehling's test. But C_6H_5CHO undergo Cannizzaro reaction in alkaline medium and does not give Fehling's test.

32. (A) In $\left[Ti(NO_3)_4\right]$, O. N. of Ti is +4.

Hence, in Ti (IV) $\left(1s^2, 2s^2, 2p^6, 3s^2, 3p^6\right)$ incomplete d-orbital is not present. So, it is colourless.

(B) In $\left[Cu(NCCH_3)\right]^+ BF_4$, O. N. of Cu is +1. Hence, in Cu(I) $\left[1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 3d^{10}\right]$ incomplete d-orbital is not present. So it is also colourless.

(C) In $\left[Cr(NH_3)_6\right]^{3+} 3Cl^-$, O.N. of Cr is +3. Hence, in Cr (III) $\left(1s^2, 2s^2 2p^6, 3s^2, 3p^6, 3d^3\right)$ incomplete d-orbitals are present. So, it is coloured.

(D) In $K_3\left[VF_6\right]$, O.N. of V is +3. Hence, in V(III) $\left(1s^2, 2s^2 2p^6, 3s^2, 3p^6, 3d^2\right)$ incomplete d-orbital is present, so it is coloured.

Hence, **(C)** and **(D)** are the correct answers.

34. pH of CH_3COONa solution changes on dilution because it is a salt solution where as other are buffer solutions.

35. NH_4NO_3 and $FeCl_3$ salts undergo cationic hydrolysis.

36. $H^+ \longrightarrow H^+ + Cl^-$

$CH_3COOH_2^+ \longrightarrow CH_3COOH + H^+$

39. Addition of Cl_2 favours forward reaction, remaining backward reaction.

40. Equilibrium is not disturbed by the addition of inert gas at constant volume.

41. $\frac{T_2}{T_1} = \left(\frac{V_1}{V_2}\right)^{\gamma-1}$ on solving

$T_2 = 250\ K$

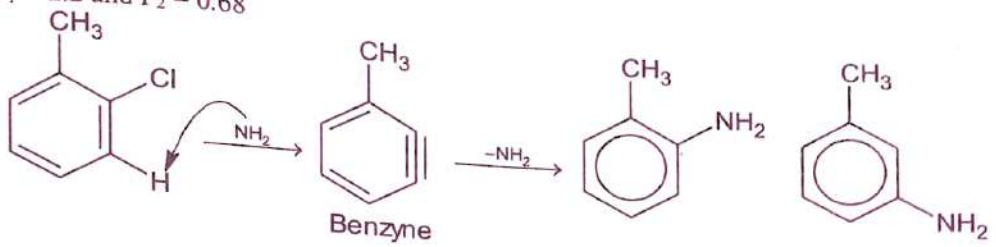
$W = nC_v(T_2 - T_1)$

$= 2 \times 12.5\ kJ\ mol^{-1}\ K^{-1} \times -150\ K = -3735\ kJ$

$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$

$P_1 = 2.2$ and $P_2 = 0.68$

61.



COMPREHENSIONS

7. $\Delta H^\circ = H\left[CaSO_4.\frac{1}{2}H_2O(s)\right] + H\left[\frac{3}{2}H_2O(g)\right]$

$-H\left[CaSO_4.2H_2O(s)\right] = 833\ kJ\ mol^{-1}$

$= +484\ kJ$ for 1 kg

8. $\Delta H^\circ = \Delta H^\circ - \Delta S^\circ$

$= 17920\ J\ mol^{-1}$

$\Delta G^\circ = -2.303RT \log K_p$

$\log K_p = \frac{\Delta G^\circ}{2.303RT} = 7.22 \times 10^{-4}\left(p_{H_2O}\right)^{\frac{3}{2}}$

$p_{H_2O} = 8.1 \times 10^{-3}$ atm

14. Let the metal be M and the metal chloride is represented as MCl_x (where $+x$ is the charge on metal in metal chloride)

$M + xHCl \longrightarrow MCl_x + \frac{x}{2}H_2$

$n = x \quad n = 1 \qquad n = 2$

Equivalents of metal (M) = $\frac{2}{E}$.

Equivalents of H_2 = moles of $H_2 \times 2 = \frac{750}{760} \times \frac{328.4}{1000} \times \frac{1}{0.0821 \times 300} \times 2$

$= \frac{75}{76} \times \frac{328.4}{82.1 \times 300} \times 2 = \frac{75 \times 4}{76 \times 300} \times 2$

$= \frac{1}{76} \times 2 = \frac{1}{38}$

Equivalents of metal = Equivalents of H_2

$\frac{2}{E} = \frac{1}{38}$

$\therefore E = 38 \times 2 = 76$

Moles of M = Moles of $MCl_x = \frac{2}{76x}$

Mass of $MCl_x = \frac{2}{76x}(76x + 35.5x) = \mathbf{2.93\ g}$.

15. $\Delta T_f = K_f \times \text{molality} = K_f \times \frac{n_{solute} \times 1000}{w(\text{in g})}$

$0.97 = 1.86 \times m$

$m = 0.52$

$\therefore \Delta T_b = K_b \times m = 0.5 \times 0.52 = 0.26°C$

$\therefore T_b(\text{solution}) = 100 + 0.26 = \mathbf{100.26°C}$.

16. $0.97 = 1.86 \times \frac{2}{76x} \times \frac{1000}{25}$

$\therefore x = 2.$

$\therefore$ Formula of metal chloride = $\mathbf{MCl_2}$.

17. B

18. For ZnS

$10^{-21} = [S^{2-}] \times 0.1$

$[S^2] = 10^{-20}$

$H_2S \rightleftharpoons 2H^+ + S+$

$10^{-21} = \frac{[H^+]^2 \times 10^{-20}}{0.1}$ $\qquad K_a = 10^{-21}$

$[H^+] = .1$

19. $K_{sp} = [cd^{2+}][s^{2-}]$

$8 \times 10^{-27} = [cd^{2+}] \times 10^{-20}$

$8 \times 10^{-7} = [cd^{2+}]$

22. $\mu_{rms} = \sqrt{\frac{3RT}{M}} = 390.2\ ms^{-1}$

30. $\underset{10-2x}{2NH_3} \rightleftharpoons \underset{x}{N_2} + \underset{3x}{3H_2}$

Pressure increases due to increase in temperature as well as due to increase in moles.

$15\ \alpha\ 300$

$P\ \alpha\ 620$

$P = \frac{620}{300} \times 15 = 31$ atm at 10 moles of NH_3 at 620 K.

Now, NH_3 is dissociated to attain 50 atm at 620 K

$P \alpha n$ or $10 \alpha 31$

$(10 + 2x) \alpha 50$

$\frac{10+2x}{10} = \frac{50}{31} = 1.61$, $10 + 2x = 16.12$

$2x = 6.12$

$\alpha = \frac{2x}{10} \times 100$

$\alpha = \frac{6.12 \times 100}{10} = 6.12$

31. For NH_3

$\frac{P_1}{T_1} = \frac{P_2}{T_2}$ $\quad P_2 \frac{T_2 P_1}{T_1} = \frac{15 \times 620}{300} = 31$

$2NH_2 \rightleftharpoons N_2 + 3H_2$

31

$31 - 2P \quad P \quad 3P$

$31 - 2P + P + 3P = 50$

$P = 9.5$

32. $K_P = \frac{n_{N_2} \times (n_{H_2})^3}{(n_{NH_3})^2} \times \left(\frac{P}{\sum n}\right)^2$

$= \frac{\frac{6.12}{2} \times \left(\frac{6.12 \times 3}{2}\right)^2}{(10 - 6.12)^2} \times \left(\frac{50}{10 + 6.13}\right)^2$

$= 1.53 \times 10^3$

33. $\Delta E = \frac{hc}{\lambda} = \frac{2\pi^2 m_e e^4 Z^2}{h^2}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\Delta E = \frac{2\pi^2 m_e e^4 Z^2}{h^3 C}\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$R_H \propto m_e$

$\frac{1}{\mu} = \frac{1}{M_N} + \frac{1}{m_e} = \frac{m_e + M_N}{M_N m_e} = \frac{\frac{M_N}{m_e} + 1}{\left(\frac{M_N}{m_e}\right) m_e}$

$\frac{1}{\mu} = \frac{\frac{3}{2} + 1}{\frac{3}{2} m_e} = \frac{\frac{5}{2}}{\frac{3}{2} m_e} = \frac{5}{3} m_e$

$\mu = \frac{3}{5} m_e$

34. $E_{\text{Ionization energy}} = \dfrac{2\pi m_e e^4 z^2}{n^2 h^2}$

me mass of electron in Bohr's theory reduced mass of electrons

$m_e = m_N$

$\dfrac{1}{\mu} = \dfrac{1}{m_e} + \dfrac{1}{m_e} \Rightarrow \mu = \dfrac{m_e}{2}$

$E = \dfrac{+13.6}{2} \times 9 = 61.2\text{eV / atoms}$

41. Choice (C) from test Q, given in passage.
Lassaigne's extract (X) $+FeCl_3 \rightarrow$ Red colour $\rightarrow$ Hence CNS confirmed.
Hence, choice (A), (B) and (D) incorrect.

42. Choice (B) is correct because we cannot be definite about halogens, i.e., choice (A) is incorrect because we have not eliminated the presence of CN, for even AgCN is white precipitate hence choice (C) and (D) are incorrect.

43. Compound Y contains phosphorous as test 's' is confirming this, hence choice (D) is correct while (A), (B) and (C) are incorrect.

44. The fusion mixture ($KNO_3 + Na_2CO_3$) on heating undergoes the following reaction.

$KNO_3 \xrightarrow{\Delta} KNO_2 + [O]$

$3Na_2CO_3 + 2P + [O] \rightarrow 2Na_3PO_4 + 3CO_2$

Hence, choice (B) is correct, while (A), (C) and (D) are incorrect.

45. The formula of the yellow precipitate is $(NH_4)_3PO_4.12MoO_3$.

$H_3PO_4 + 12(NH_4)_2MoO_4 + 21HNO_3 \rightarrow (NH_4)_3PO_4.12MoO_3 + 21NH_4NO_3 + 12H_2O$

Hence, choice (B) is correct while (A), (B) and (D) are incorrect.

55. Only (B) an decarboxylation gives cyclohexane.

SINGLE INTEGER ANSWER TYPE QUESTIONS

1. **6;**

Let the molarity and volume of HX be M_1 and V_1 ml respectively while the molarity of NaOH be M_2.

	HX	+ $NaOH \longrightarrow NaX$	+ H_2O	
Initial millimoles	M_1V_1	$10M_2$	–	–
After the addition of 10 c.c of NaOH	$M_1V_1 - 10M_2$	0	$10\,M_2$	–

Since weak acid HX and NaX are left after the reaction they will constitute an acidic buffer.

$\therefore pH = pK_a + \log \dfrac{[Salt]}{[Acid]}$

$5.7 = pK_a + \log \dfrac{10\,M_2}{(M_1V_1 - 10M_2)}$ ...(1)

Let V_2 ml be the volume of NaOH required to neutralize given HX completely.

$\therefore$ At equivalence point, $M_1V_1 = M_2V_2$

$$V_2 = \frac{M_1V_1}{M_2}$$

Dividing the numerator and denominator of log term by M_2, we get

$$\therefore\ 5.7 = pK_a + \log \frac{\frac{10M_2}{M_2}}{\frac{M_1V_1 - 10M_2}{M_2}} = pK_a + \log \frac{10}{V_2 - 10} \qquad \ldots(2)$$

Similarly, after the addition of 20 c.c of NaOH, we have

$$6.3 = pK_a + \log \frac{20}{V_2 - 20} \qquad \ldots(3)$$

Subtracting equation (2) from equation (3)

$$6.3 - 5.7 = \log \frac{20}{V_2 - 20} - \log \frac{10}{V_2 - 10}$$

$$0.6 = \log\left(\frac{20}{V_2 - 20} \times \frac{V_2 - 10}{10}\right)$$

Taking antilog, $4 = \frac{2(V_2 - 10)}{(V_2 - 20)}$

$\therefore\ V_2 = 30$ ml.

Putting V_2 in equation (2),

$$5.7 = pK_a + \log \frac{10}{30 - 10} = pK_a + \log \frac{1}{2}$$

$$pK_a = 5.7 + \log 2 = 5.7 + 0.3 = \mathbf{6.0}.$$

2. $\Delta H = nC_p dT$

$$= 1 \times \frac{5}{2} \times 8.314 \times 500 = 10^4 \text{ J}$$

3. n-Factor of the half cell $= 3 - (-1) = 4$

4. **2;**

Let the volume of the tank be V litres.

As the number of moles of N_2 before and after connecting it to the tank will be same,

$$\frac{2.13 \times 5.5}{RT} = \frac{1.562 \times (5.5 + V)}{RT} \quad \left[n = \frac{PV}{RT}\right]$$ (moles of N_2 before connection) = (moles of N_2 after connection)

$\therefore\ V = 2$ **litres.**

5. $0.7 = 2 \times K_f \times 0.2 \Rightarrow K_f = \frac{0.7}{0.4} = \frac{7}{4}$

$$Hg(CN)_2 + mCN^- \longrightarrow Hg(CN)_{m+2}^{m^-}$$

0.1	0.2	0
0	$(0.2 - 0.1m)$	0.1

Now, Molarity of $K^+ = 0.2$ M

Molarity of $CN^- = (0.2 - 0.1m)$ M

Molarity of complex = 0.1 M

$0.53 = K_f(0.2 + 0.2 - 0.1m + 0.1) = K_f(0.5 - 0.1m)$

$\Rightarrow 0.53 = \frac{7}{4}(0.5 - 0.1m)$ $\Rightarrow 2.1 = 3.5 - 0.7m$

$\Rightarrow 0.7m = 1.4$ $\Rightarrow m = 2.$

6. **9**

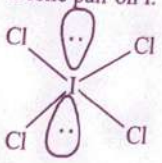
$22.21 \times N_{HIO_3} = 35.5 \times 0.16$

$\Rightarrow N_{HIO3} = \frac{35.5 \times 0.16}{22.21} = 0.256\,(N)$

N factor of $HIO_3 = 5$

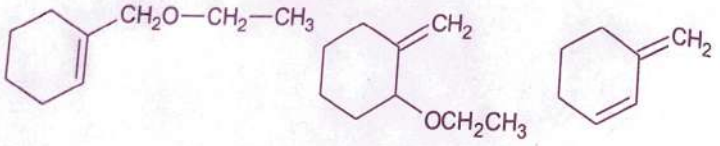
Mass of $HIO_3 = 0.256 \times \frac{176}{5} \approx$ **9 gm/lt.**

7. 2 lone pair on I.

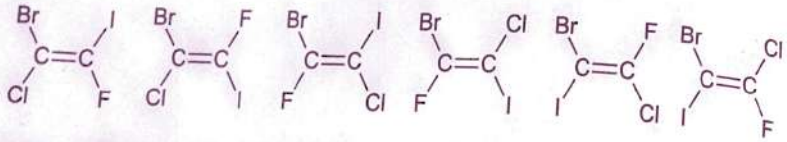


I is sp^3d^2 hybridised with square planar shape in $\left[ICl_4^-\right]$

10.



11.



12. Except – Cl all can undergo intramolecular H-bonding.

18. **8;**

Number of moles of hypo $= \frac{MV}{1000}$

$= \frac{0.1 \times 40}{1000} = 4 \times 10^{-3}$

$\therefore$ Number of moles of $I_2 = 2 \times 10^{-3}$

Mass of $I_2 = 2 \times 10^{-3} \times 254$ g

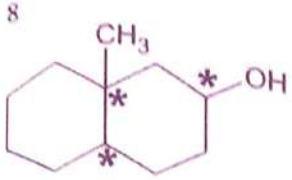
$w = \frac{i \times t \times E}{96500}$

$2 \times 10^{-3} \times 254 = \frac{i \times 9650 \times 127}{96500}$

$i = 0.04$ ampere

Current efficiency $= \frac{0.04}{0.50} \times 100 =$ **8%.**

19. 8



21. 3

$Ag^+(aq) + Fe^{2+}(aq) \rightleftharpoons Ag(s) + Fe^{3+}(aq)$

No. of mmoles initially	$500 \times 0.15 = 75$	$500 \times 1.09 = 545$	0	0
mmoles at equilib.	$75 - x$	$545 - x$	x	x

On titration of reaction mixture with $KMnO_4$, only Fe^{2+} reacts with it.

$\therefore$ Equivalents of Fe^{2+} in 25 ml = Equivalents of $KMnO_4$

$$\frac{(545-x)}{1000} \times 25 \times 10^{-3} \times 1 = 30 \times 10^{-3} \times 0.0833 \times 5$$

$$\frac{(545-x) \times 25}{1000} \approx 12.5 \; ; \; 545 - x = 500 \; ; \; x = 45$$

$$\therefore K_c = \frac{[Fe^{3+}]}{[Ag^+][Fe^{2+}]} = \frac{\frac{45}{1000}}{\frac{(75-45)}{1000} \times \frac{(545-45)}{1000}} = \frac{45 \times 1000}{30 \times 500} = \mathbf{3.0}.$$

26. AgCl is obtained only with free (ionisable) ions.

$CrCl_3.6H_2O + AgNO_3 \longrightarrow AgCl$

mol of AgCl obtained $= \frac{2.87}{143.5} = 0.02$

mol of $CrCl_3.6H_2O \rightarrow \frac{2.665}{266.5} = 0.01$

0.01 mol of complex gives = 0.02 mol of AgCl

$\therefore$ 1 mol of complex gives = 2 mol of AgCl

Number of free Cl^- = 2

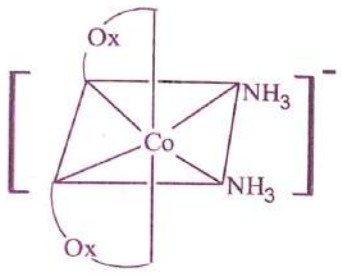
$\therefore$ Number of Cl^- ion co-ordination sphere = 1

30. $M = 2.82 = \sqrt{n(n+2)}$

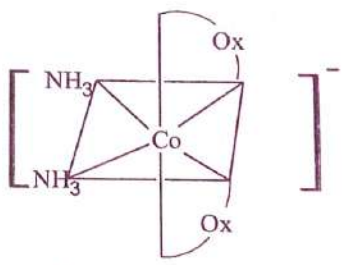
$\therefore n = 2$

2 unpaired electrons are present, i.e. Ni^{2+} configuration is $3d^8$. The number of electrons on t_{2g} are six.

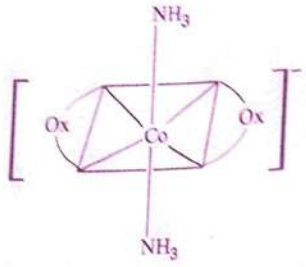
31.



cis-d-isomer



cis-l-isomer



Trans (meso) isomer

33. $MnO_2 + 2KOH + O \longrightarrow \underset{(A)}{K_2MnO_4} + H_2O$

Dark green

$2KMnO_4 + H_2SO_4 \xrightarrow[\text{conc.}]{} \underset{(C)}{Mn_2O_7} + H_2O + K_2SO_4$

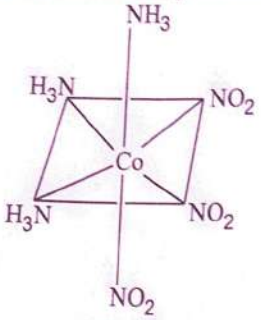
$2Mn_2O_7 \longrightarrow 4MnO_2 + 3O_2$

The cation in (D) is Mn^{4+}

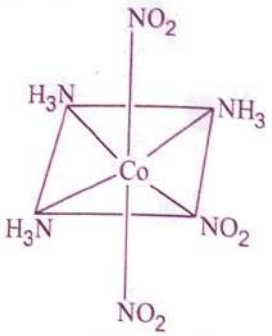
Electronic configuration of $Mn^{4+} = [Ar]3d^3$

Thus number of unpaired electrons in $M^{4+} = 3$

34.



(Facial)



(Meridonial)

35. $K_f = \dfrac{RT^{0^2}_{fusion}M_2}{\Delta H_{molar}} = \dfrac{RT^{0^2}_{fusion}}{\Delta H_{molar}/M_2} = \dfrac{1.987\times(250.1)^2}{4.2\times1000}\text{kgmol}^{-1} = 29.6$

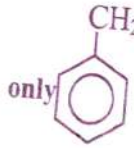
	$2VCl_4 \rightleftharpoons$	V_2Cl_8
$t=0$	0.346	0
	$-2x$	$+x$
$t=t$,	$(0.346-2x)$	x

$\Delta T_f = K_f \times M_{actual}$ $\qquad n_T = 0.346 - x$

on solving $x = 0.146$

$K_c = \dfrac{(0.346-2x)^2}{x} = 0.02 = 2\times10^{-2}$

37. **1;**



38. $Kp = \left(\frac{0.75}{0.25}\right)^2 = 9$

39. $\log[H^+] = -(\log K_{a_2} K_{a_3})^{1/2}$

$= -\frac{1}{2}\log K_{a_2} - \frac{1}{2}\log K_{a_3}$

$= \frac{pK_a + pKa_3}{2}$

$[H^+] = \sqrt{K_{a_2} K_{a_3}} = \sqrt{(10^{-6})(10^{-12})} = 10^{-9}$

$\therefore pH = 9$

40. 6

43. For $Cu(OH)_2$, $K_{sp} = 4s^3$

$S = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{5\times10^{-16}}{4}}$

$\sqrt[3]{125\times10^{-18}} = 5\times10^{-6}$

$[OH^-] = 2S = 2\times5\times10^{-6} = 10^{-5}$

pOH = 5, pH = 9

46. Rate = $K[H_2O_2]$

$2\times10^{-4} = \frac{3\times10^{-3}}{60}\times[H_2O_2]$

$[H_2O_2] = 4M$, $x = 4$

47. 2

distance between two second nearest neighbours = a; ie these are

Distance between two third nearest neighbours = 1.414 a;(two opposite corners of a face)ie these are 12.

48. 2

50. $\bar{\nu}_{He} = RZ^2\left[\frac{1}{4^2} - \frac{1}{6^2}\right]$ ... (i)

$4R\left[\frac{36-16}{36\times16}\right] = \frac{5R}{36}$ ... (ii)

$\because \bar{\nu}_{He} = R\times1^2\left[\frac{1}{2^2} - \frac{1}{n^2}\right]$

$\because \bar{\nu}_{He} = \bar{\nu}_H$

On solving above equation

$n^2 = 9$

$\therefore n = 3$

or corresponding transition from $3 \rightarrow 2$ in Balmer series of hydrogen atom has some frequency that of $6 \rightarrow 4$ transition in He^+.

53. 8.09 mL $\approx$ **8** mL

$CuCO_3 + H_2SO_4 \longrightarrow CuSO_4 + H_2O + CO_2$

1 mol 1mol

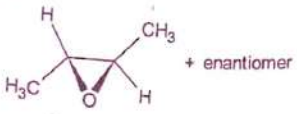
Let of volume of 0.5 M H_2SO_4 be V

Moles of $CuCO_3 \equiv$ Moles of H_2SO_4

$$\frac{0.5}{123.5} = \frac{0.5 \times V}{1000}$$

$\therefore V = 8.09 \text{ mL} \approx 8 \text{ mL}$

54. 2



+ enantiomer

55. Since the compound X decolourizes a solution of iodine in potassium iodide, it should contain thiosulphate ion, $S_2O_3^{2-}$ which also coincides with the two other given facts, i.e. (i) and (iii). Hence, the compound X is sodium thiosulphate, $Na_2S_2O_3.5H_2O$ which explains the given reactions as below.

$$Na_2S_2O_3.5H_2O \xrightarrow{\text{heat}} Na_2S_2O_3 + 5H_2O\uparrow$$

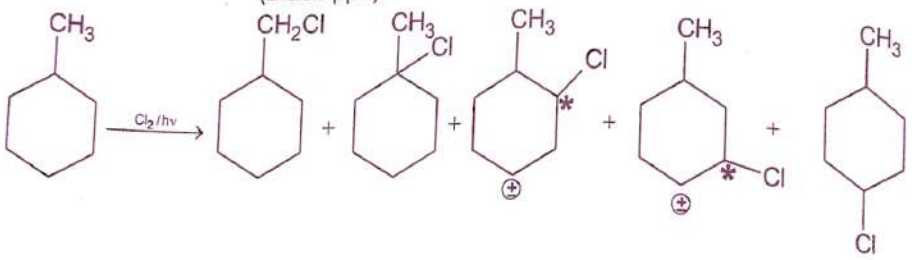
(i) $Na_2S_2O_3 + 2HCl \longrightarrow 2NaCl + H_2O + SO_2 +$ S (White turbidity)

(ii) $2Na_2S_2O_3$ (Sod. thiosulphate) $+ I_2 \longrightarrow$ $Na_2S_4O_6$ (Sod. tetrathionate) $+ 2NaI$

(iii) $Na_2S_2O_3 + 2AgNO_3 \longrightarrow$ $Ag_2S_2O_3\downarrow$ (Silver thiosulphate (White ppt.)) $+ 2NaNO_3$

$$Ag_2S_2O_3 + H_2O \longrightarrow \underset{\text{(Black ppt.)}}{As_2S\downarrow} + H_2SO_4$$

56.



58. 9

59. Due to presence of aromaticity.

62. 9

X:Y:Z = 1:5:3

MATRIX-MATCH THE FOLLOWING

12.